The `dfg-crc` class
User guide and implementation

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About this manual

This manual is typeset with the `dfg-crc` class itself. Executable examples show their source on the left and the rendered result on the right. Wide tables and plots put the result below the code. All examples use the fictional people, institutions, and projects embedded in the DTX.

Setup fragments, shell commands, and bibliography source records are shown as code only. Complete proposal and CV examples follow the user guide, before the implementation reference; see Chapters 4 and 5.

“Documentation must be regarded as an integral part of the process of design and coding.”(Hoare 1973, p. 195)

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1 Shared commands and first steps

The `dfg-crc` class is a starting point for a Collaborative Research Centre proposal and the accompanying CV section. Authors write the proposal chapters and maintain structured records for people, institutions, projects, work packages, and funding. The class uses those records to create project headings, references, tables, collaboration plots, Gantt charts, and totals that stay consistent across the document.

The supplied files include a complete example proposal, example CV, and customization file. Start with the generated examples to see how the source files fit together, then replace their content and metadata with those of your own centre. This chapter introduces the commands reused throughout the proposal. Chapter 2 then follows the proposal itself; Chapter 3 collects the complete key and settings reference. Applicants should still compare the result with the current DFG forms and programme rules.

1.1 Compiling and generated files

The maintained source of this bundle is `crc-proposal-documentation-and-code.dtx`. Files such as `local/dfg-crc.cls`, `README.md`, `CHANGELOG.md`, and the example files in `local/` are generated from that source. For template or documentation changes, edit the `.dtx` file and regenerate the derived files.

For a first local compile, use LuaLaTeX through `latexmk`:

First local compile

```
make files
cd local
latexmk -lualatex crc-proposal.tex
latexmk -lualatex crc-cv.tex
```

The class expects LuaLaTeX. Calling `LaTeX` with a capital `L` asks TeX for a format file named `LaTeX.fmt`; use `lualatex` or `latexmk -lualatex` instead.

If a change has no visible effect, check which generated class file is being loaded. Regenerate the class and local examples with `make files`, then compile the relevant example again.

To update a separately controlled proposal worktree, use `make update-class TARGET_DIR=/path/to/proposal-worktree` for the class alone. Use `make update-all TARGET_DIR=/path/to/proposal-worktree` to rebuild the manual and copy `dfg-crc.cls`, `crc-proposal-documentation-and-code.pdf`, `dfg-crc-proposal-example.pdf`, and `dfg-crc-cv-example.pdf`. Both targets require the destination to be the root of another Git worktree and report the affected files without committing or pushing them. They leave the destination's README, changelog, and licence unchanged.

1.2 How to read the examples

The examples use the same class commands as a proposal. In the template data, `per1` is Kim Sample, `per2` is Alex Example, and `A01` is their shared project. The manual has twelve PIs and fifteen active projects: `A01`–`A06`, `B01`–`B06`, and `Q01`–`Q03`. Research projects include both continuing and new projects; Q projects demonstrate service, research training, and administration. `A00` is a former, reference-only project and is excluded from active counts. Outputs are compiled from the displayed source.

Small examples place code beside its result; wide examples put the result below and allow it to continue onto another page. Headings inside live examples are unnumbered so they do not change the manual's outline. The complete shared metadata are listed once in Section 3.8. Finance examples show their additional input records at the point of use. Optional settings normally go immediately after the command, before its required arguments.

1.3 Declare records with \addEntry

\addEntry[⟨*key-set*⟩]{⟨*database*⟩}{⟨*key=value*⟩}

Adds an entry to the database ⟨*database*⟩ with the defined values from ⟨*key=value*⟩. An id is unique within its database. If a later call reuses the same id, the class issues a warning, removes all earlier rows with that id, and stores the later entry instead. The same id may still be used in a different database. Entries without an id, such as ordinary funding and staff rows, remain append-only.
See Example 1.1, Example 2.57.

A keyset defines the accepted fields; a database stores the resulting rows. Omit the optional keyset when it has the same name as the database, for example \addEntry{person}{...}. Give it explicitly for a separate collection, as below. Braces protect commas inside values. Use stable IDs rather than repeating names in the prose.

Example 1.1: Declare a record and retrieve its fields

```
\addEntry[person]{ManualContacts}{
  id        = {example-contact},
  firstname = {Kim},
  lastname  = {Sample}
}
\getEntry[ManualContacts]{example-contact
    }{firstname}\space
\getEntry[ManualContacts]{example-contact
    }{lastname}
```

Kim Sample

Store proposal records in metadata/dfg-metadata.tex, or in another metadata file loaded by the proposal. Tables share those records. The standard databases are meta (the CRC), person, project, institution, and affiliation; funding and staff tables use the named collections introduced with each table. Full field definitions are in Section 3.1.

1.4 Read, store, and compare fields

You can get any kind of information back from any database using \getEntry:

\getEntry[⟨*database*⟩]{⟨*value of id*⟩}{⟨*column*⟩}[⟨*alternative id-field*⟩]

Retrieve and typeset the value of ⟨*column*⟩ from the row whose id-cell equals ⟨*value of id*⟩. The optional ⟨*database*⟩ argument selects the datatool → CTAN database to query; it defaults to \tmpLocalDB, which is set by the surrounding context (typically by the table-generating command). The optional ⟨*alternative id-field*⟩ argument names the column that holds the row identifier and defaults to id; supply a different name when the row is keyed on something other than the standard id column (for example ref-project-id or number).
The retrieved value is wrapped in \dfgShowTag before being typeset, so when the showtags → P. 65 class option is active the output is annotated with a small overlay marker showing the originating database, id, and column. When the option is inactive, the value is typeset plain.
The command performs several checks and emits a warning to the log when something is wrong: a database-name-not-given warning if ⟨*database*⟩ is empty, a database-not-found warning if no database with the given name exists, and a missing-key warning if the row exists but the requested cell is empty or undefined. The last warning is suppressed during tabularray → CTAN's measuring phase to avoid duplicate log entries.
When the fastcompile-project class option is active, the lookup is skipped entirely. Project identifiers are typeset in bold black; other row identifiers use a quiet grey placeholder. Error diagnostics retain the configured error colour. Starred multi-field lookups initialise their destination macros to empty values without touching a database.

Examples:

Example 1.2: Field lookup and alternative identifier

```
\getEntry[person]{per9}{lastname}\par
\getEntry[project]{A06}{name}[number]\
    par
\def\tmpLocalDB{person}
\getEntry{per9}{firstname}
```

Navarro
Modelling microbial resilience and recovery
Leonie

When several columns from the same row are needed, prefer the starred form `\getEntry*`, which performs only a single row lookup.
See Example 1.2.

\getEntry* [⟨*database*⟩] {⟨*value of id*⟩} {⟨*assignments*⟩} [⟨*alternative id-field*⟩]

Starred variant of `\getEntry`. Retrieves multiple columns from the same row in a single lookup and assigns each value to the macro named on the left-hand side of the assignment. The ⟨*assignments*⟩ argument is a comma-separated list of ⟨|*macro*⟩ = ⟨*column*⟩ pairs. For each pair, the value of ⟨*column*⟩ for the row whose id-cell equals ⟨*value of id*⟩ is stored in ⟨|*macro*⟩. Unlike the unstarred form, this variant does not print anything and does not wrap results in `\dfgShowTag` – the caller is responsible for formatting the retrieved values.
Missing fields trigger the same `missing-key` warning as the unstarred form (suppressed during tabularray's measuring phase).

Example:

Example 1.3: Retrieve several fields at once

```
\getEntry*[person]{per8}{
  \exampleFirst = firstname,
  \exampleLast = lastname,
  \exampleDegree = academicdegree
}
\exampleDegree\space
\exampleFirst\space\exampleLast
```

Prof. Dr. rer. nat. Alexis-Marie Winters-Scholz

This is preferable over multiple `\getEntry` calls when several fields from the same row are needed: the row is located once instead of once per field.
See Example 1.3.

\tmpLocalDB {⟨*database*⟩}

Define the database locally (e.g. within a group) so you don't need to repeat it as [⟨*database*⟩] for e.g. `\getEntry` → P. 2.
See Example 1.2.

\compareEntry [⟨*database*⟩] {⟨*id*⟩} {⟨*field*⟩} {⟨*value*⟩} {⟨*true*⟩} {⟨*false*⟩}

Compare an entry with the ⟨*id*⟩ of ⟨*database*⟩ against ⟨*value*⟩ and show either ⟨*true*⟩ or ⟨*false*⟩.
See Example 1.4, Example 1.5.

For example, compare the surname of `per6` against an expected value.

Example 1.4: Compare a database entry against something.

```
\compareEntry[person]{per6}{lastname}{
    Hartmann}
  {The surname matches.}{The surname
    differs.}
```

The surname matches.

The next example prints an affiliation only when the institution refers to it.

Example 1.5: Conditionally print an affiliation

```
\compareEntry[institution]{ins3}{ref-
    affiliation-id}{aff3}
  {\getEntry[affiliation]{aff3}{name}}
  {Another university}
```

Northbridge University

1.5 Follow references between records

A `ref-institution-id` value links a person to institutions; each institution's `ref-affiliation-id` links it to a university or other parent organisation. Use one brace argument per step, ending with the field to print. Multiple IDs are followed automatically, so prose stays correct when affiliations change.

\get [⟨*database*⟩] {⟨*id*⟩} {⟨*field or reference*⟩} ...

Read a field directly, or follow metadata references before reading the final field. Each intermediate argument names a destination database and follows the current row's `ref-<database>-id` field. The source database defaults to `\tmpLocalDB`. No institution or affiliation ID needs to be repeated in the document text.

Example 1.6: Follow metadata references

```
\get[person]{per7}{lastname}\par
\get[person]{per7}{institution}{name}\
    par
\get[person]{per7}{institution}{
    affiliation}{name}\par
\get[project]{B04}{person}{lastname}\par
\get[RequestedFundingStaff]{per6}{person
    }[ref-person-id]{institution}{name}
```

Lindberg
Centre for Computational Microbiology,
Department of Experimental Immunology
Northbridge University
Miller, Sato
Centre for Computational Microbiology

The full reference-key spelling also works: `\get[person]{MM}{ref-institution-id}{ref-affiliation-id}{name}`. Traversal is generic: a `ref-other-unit-id` field points to rows of the `other-unit` database. The source may be any database populated by `\addEntry` → P. 2, including one using a separately named keyset. Each destination is matched by its `id` field. The lookup follows only the explicitly requested path; an affiliation reached through an institution therefore needs both steps.
A reference value may contain multiple comma-separated IDs. All branches are followed in stored order, and the resulting values are separated by commas. Repeated destination rows are printed once, in first-occurrence order. For example, two institutions of one university yield two institution names but only one affiliation name when the path ends at that university. Different rows with the same name are still distinct results.
Missing databases, rows, fields, and empty reference lists use the existing lookup warnings and contribute no value; valid branches still render without dangling separators. The final value retains its LaTeX formatting and the `showtags` → P. 65 annotation identifies the destination row and field. Focused fast compilation consumes the path but prints the source ID placeholder, just like `\getEntry` → P. 2.
With no following step, a full reference key returns its raw ID list: `\get[person]{MM}{ref-institution-id}`. An existing scalar field takes precedence over a shorthand relation name, so a following prose group is not absorbed. Use the explicit `ref-<database>-id` spelling if the names conflict. For a custom source identifier column, append the existing `[id-field]` argument after the first field/reference argument; destination rows still use `id`. The starred `\get*` form delegates to `\getEntry*` → P. 3 for assignments and does not traverse references.
See Example 1.6, Example 1.7.

Two institution IDs may lead to the same affiliation. The final destination is printed once, so this also works when a person's metadata changes later.

Example 1.7: Raw reference IDs and resolved organisation names

```
IDs: \get[person]{per7}{ref-institution-id
    }\par
Institutions: \get[person]{per7}{
    institution}{name}\par
Affiliation:
\get[person]{per7}{institution}{
    affiliation}{name}
```

IDs: ins3,ins4
Institutions: Centre for Computational Microbiology, Department of Experimental Immunology
Affiliation: Northbridge University

A lookup may branch at several steps: from a project to its PIs, then to their institutions and affiliations. Shared affiliations are still deduplicated.

Example 1.8: Find the affiliations behind a project

```
\getProject*{B04}\par
PIs: \get[project]{B04}{person}{lastname}\
    par
Affiliations:
\get[project]{B04}{person}{institution}{
    affiliation}{name}
```

B04
PIs: Miller, Sato
Affiliations: Partner University, Westhaven Research Centre

1.6 Names and references in prose

To quickly get a name of a person there is a dedicated command:

\getName{⟨*<id>*⟩}

Get last name of person that has the ⟨*id*⟩. This sets the database to `person`.
See Example 1.9.

\getFullName*{⟨*id*⟩}
\getFullName{⟨*id*⟩}

Get the full name of the person with the ⟨*id*⟩. The starred version adds the academic title in front.
See Example 1.9.

Example 1.9: Names with and without academic titles

```
\getName{per9}\par
\getFullName{per9}\par
\getFullName*{per9}
```

Navarro
Leonie Navarro
Prof. Dr. rer. nat. Leonie Navarro

PI and project commands also have all-lowercase aliases, with identical arguments, starred forms, formatting, and links. For example, `\getpiwithproject{per6}` is equivalent to `\getPIWithProject{per6}`, and `\getpiofproject*{B05}` is equivalent to `\getPIofProject*{B05}`. This also applies to `\getPI`, `\getPIsWithRole`, `\getProject`, `\getProjectColor`, `\getProjectCount`, `\getProjectWithPI`, `\getPIOfProjectWithProject`, and the legacy commands `\getPIProject` and `\getProjectPI`. Legacy aliases share the same once-per-command, per-LaTeX-run deprecation warning as their canonical spelling. The mixed spellings `\getPIwithProject`, `\getProjectwithPI`, and `\getPIOfProject` are accepted as well. Other capitalizations remain case-sensitive in TeX.

\getPI*{⟨*id-list*⟩}
\getPI{⟨*id-list*⟩}
\getPI{⟨*id-list*⟩}[⟨*text*⟩]

Reference one or multiple PIs from the `person` database. The ⟨*id-list*⟩ is a comma-separated list of person ids. By default the command prints the lastname; the starred version also prints the firstname. If multiple PI records share a lastname, unstarred references automatically

add the first given-name initial. For example, PI records for Kai Markus Schneider and Carolin V. Schneider are printed as **K. Schneider** and **C. Schneider**. If the initial is also shared, the full firstname is used instead. The comparison includes all records with role `PI` or `Co-PI`, independently of funding period or project selection. Project-first references inherit this distinction, for example **A04 (K. Schneider)** and **B09 (C. Schneider)**. With the optional ⟨*text*⟩ argument, ⟨*text*⟩ is printed instead, but the hyperlink still points to the PI. Set `PIReferenceFirstName=true` with `\changeSettings` → P. 8 to make the unstarred command include the firstname too. The starred form always includes it. This setting also applies to `\getPIProject` → P. 7 and `\getPIofProject` → P. 8. `\getProjectPI` → P. 7 keeps project PI names to lastnames with automatic disambiguation unless its starred form is used. Shared PI surnames use the initial rule even with `PIReferenceFirstName=true`; use the star to force full firstnames. Explicit display text bypasses automatic disambiguation. Fast project compilation retains the usual ID placeholder without reading the database.
The printed name is typeset bold, coloured with `referencePI`, and linked to the corresponding person's row in `\printTable{ProjectLeaders}`. Each row of that table provides the destination `pi:`⟨*id*⟩.
See Example 1.10.

Example 1.10: Reference one or multiple PIs.

```
\getPI{per6}\par
\getPI*{per6,per8}\par
\getPI{spokesperson}[the spokesperson]
```

Hartmann
Samira Hartmann and **Alexis-Marie Winters-Scholz**
the spokesperson

\getProject* { ⟨*id-list*⟩ } [⟨*text*⟩]

Reference one or multiple projects from the `project` database. The ⟨*id-list*⟩ is a comma-separated list of project ids. By default the command prints the project number; with the optional ⟨*text*⟩ argument, ⟨*text*⟩ is printed instead, but the hyperlink still points to the project. In CV mode the command prints the same text without a hyperlink. The starred form always uses this compact project-only representation, independently of `projectPIOncePerSection` → P. 67 and its first-reference state. Projects declared with `reference-only` → P. 56`=true` are never hyperlinked; instead, "former project" is appended to make the missing project section explicit.
The printed project reference is typeset bold and coloured according to the first letter of the project id. Multiple project references are joined as a natural-language list with commas and a final "and". With `projectPIOncePerSection` → P. 67`=true`, the command behaves like the unstarred `\getProjectPI` → P. 7: the first reference to each project in a section includes its PIs and later references use the compact project form. Generated tables and lists remain compact.
See Example 1.11.

Example 1.11: Reference one or multiple projects.

```
\changeSettings{projectPIOncePerSection=
    false}
\getProject{A03,B04,Q01}\par
\getProject{A06}[the modelling project]\
    par
\getProject*{A00}
```

A03, **B04**, and **Q01**
the modelling project
A00 (former project)

\getProjectCount [⟨*area*⟩]

Print the number of active projects. A project is active when its `project-status` → P. 56 is `C` or `N` and `consider-for-lists` → P. 56 is `true` and `reference-only` → P. 56 is `false`. With no optional argument all active projects are counted. Any explicit `project-area` → P. 56 value may be supplied to restrict the result; where that metadata is absent, the first character of the project ID is used. All area totals are computed together on first use and cached for subsequent calls.
See Example 1.12.

Example 1.12: Count active projects.

```
All: \getProjectCount\par
Area A: \getProjectCount[A]\par
Area B: \getProjectCount[B]\par
Area Q: \getProjectCount[Q]
```

All: 15
Area A: 6
Area B: 6
Area Q: 3

\CRC{⟨*crc-number*⟩}
\CRC*{⟨*crc-number*⟩}
\CRC*[⟨*crc-title*⟩]{⟨*crc-number*⟩}

Print a CRC reference. For the current CRC, identified by the `number` → P. 64 field of the `meta` entry with id `crc`, the unstarred command prints the short form stored in `\thisCRC`; the starred command prints the long form with the CRC title.
For a different CRC number, the unstarred command prints only the short reference, while the starred form prints a long reference. If ⟨*crc-title*⟩ is given, it is used as the title of the external CRC; otherwise the command falls back to the CRC number.
See Example 1.13.

\TRR{⟨*trr-number*⟩}
\TRR*{⟨*trr-number*⟩}
\TRR*[⟨*trr-title*⟩]{⟨*trr-number*⟩}

Print a CRC/TRR reference. The command follows the same rules as `\CRC`, but uses "CRC/TRR" as the printed programme type.
See Example 1.13.

Example 1.13: Reference current and external CRCs.

```
\CRC{9999}\par
\CRC*{9999}\par
\CRC*[External CRC title]{999}\par
\TRR*[External CRC/TRR title]{999}
```

CRC 9999
CRC 9999 *Mechanisms of Host–Microbiome Communication*
CRC 999 *External CRC title*
CRC/TRR 999 *External CRC/TRR title*

\getPIProject*{⟨*person-id-list*⟩}
\getPIProject{⟨*person-id-list*⟩}

Print one or multiple PIs together with their assigned projects. The ⟨*person-id-list*⟩ is a comma-separated list of ids from the `person` database. For each PI, the command reads `ref-project-id` → P. 64 from the person entry and prints the referenced project identifiers in parentheses using `\getProject*`, without repeating the projects' PI names.
The starred version prints firstnames in addition to lastnames.
See Example 1.14.

Example 1.14: Print PIs with their projects.

```
\changeSettings{projectPIOncePerSection=
    false}
\getPIProject{per6,per11}\par\medskip
\getPIProject*{per6,per11}
```

Hartmann (A02 / A03) and **Sato (B03 / B04)**

Samira Hartmann (A02 / A03) and **Mira Sato (B03 / B04)**

Use `\getPIWithProject` for the same person-first output. It takes person ids and prints each PI followed by their project list. The legacy `\getPIProject` remains available but warns that it is deprecated.

\getProjectPI*{⟨*project-id-list*⟩}
\getProjectPI{⟨*project-id-list*⟩}

Print one or multiple projects together with their assigned PIs. The ⟨*project-id-list*⟩ is a comma-separated list of ids from the `project` database. For each project, the command reads `ref-person-id`→P. 62 from the project entry and prints the referenced PIs in parentheses using `\getPI`→P. 5.
If a project id is identical to the current project established by `\printProjectHead`→P. 33, only the project itself is printed, just as with `\getProject`→P. 6; its PIs are omitted. In a list containing several projects, this exception applies only to the matching project.
By default, PI names are shown only for the first `\getProject`→P. 6 or `\getProjectPI`→P. 7 reference to each project within a section. Both commands share the same remembered state, and a new section resets it. Generated tables and lists remain compact. Set `projectPIOncePerSection`→P. 67`=false` with `\changeSettings` to make `\getProject`→P. 6 always compact and `\getProjectPI`→P. 7 always expanded.
The starred version prints firstnames in addition to lastnames.
See Example 1.15.

Example 1.15: Print projects with their PIs.

```
\changeSettings{projectPIOncePerSection=
    false}
\getProjectPI{A03,B04}\par\medskip
\getProjectPI*{A03,B04}
```

A03 (Hartmann / Lindberg) and **B04 (Miller / Sato)**

A03 (Samira Hartmann / Noah Lindberg) and **B04 (Kimmi Miller / Mira Sato)**

Use `\getProjectWithPI` for the same project-first output. It takes project ids and prints each project followed by its PI list. The legacy `\getProjectPI` remains available but warns that it is deprecated.

`\getPIOfProjectWithProject``*{`⟨*project-id-list*⟩`}`

Print the PIs of one or more projects, followed by each project's compact reference in parentheses. The argument is a comma-separated list of project ids, matching the project named after "OfProject" in the command. The starred form includes firstnames; the `PIReferenceFirstName=true` setting also enables firstnames in the unstarred form.

`\getPIofProject``*[`⟨*properties*⟩`]{`⟨*project-id*⟩`}[`⟨*delimiter*⟩`]`

Print the PIs assigned to a project. The command reads `ref-person-id`→P. 62 from the project entry and then prints selected person fields from the `person` database. Each name links to its row in the PI table and uses the same reference styling as `\getPI`→P. 5. The optional ⟨*properties*⟩ argument is a comma-separated list of fields to print and defaults to `lastname`. Typical values are `academicdegree`, `firstname`, and `lastname`. The optional ⟨*delimiter*⟩ separates multiple PIs and defaults to `\, /\,`.
The starred form defaults to `firstname,lastname` and joins names as "Alice Example, Bob Sample and Carol Demo". With two PIs it uses only "and"; with one PI it adds no separator. Explicit properties still override the default. In the starred form, an explicit trailing delimiter replaces only the final "and" separator; earlier names remain comma-separated.
See Example 1.16.

Example 1.16: Print the PIs of a project.

```
\getPIofProject{B05}\par
\getPIofProject[firstname,lastname]{B05}[;
    ]\par
\getPIofProject*{Q02}\par
\getPIofProject*{Q03}\par
\getPIofProject*[lastname]{Q03}
```

Redata / Keller
Friedhelm Redata; Robin Keller
Friedhelm Redata and **Leonie Navarro**
Hansi Meier and **Robin Keller**
Meier and **Keller**

1.7 Change shared settings

`\changeSettings``{`⟨*key=value*⟩`}`

Customize fonts, logo paths, colors, `thumbindex` → P. 67, and other class defaults. The former `\changeSetting` spelling remains available as a compatibility alias.
See Example 1.17.

Put proposal-wide settings in `preamble/dfg-customization.tex`. For example, `\changeSettings{salary@PhD=83700}` sets the annual salary used in funding calculations. Table, list, and plot options apply to a single call; shared settings establish defaults. The full list is in Section 3.2.

`\getValue`{⟨*customization key*⟩}

Read the current value of any key accepted by `\changeSettings` → P. 8. The command is expandable and follows local changes, so numeric values can also be used in `\num` and calculations. For example:

Using customization values in text

```
Housing costs are \num{\getValue{animalHousing@mouse}}\,EUR per mouse per week.
The annual postdoctoral rate is \num{\getValue{salary@postDoc}}\,EUR.
The configured font is \getValue{font}.
Thumb index enabled: \getValue{thumbindex}.
The reference color is \getValue{color@reference@PI}.
```

These examples read the configured values without changing them. With no override, `animalHousing@mouse` returns `0.75`; the getter uses the same stored rate as the animal-housing table. It returns a value without a currency or unit suffix; add those in the surrounding text.
Booleans return `true` or `false`; dimensions return points. Automatic thumb-index margins return `auto`, and orientation returns `inwards` or `outwards`. Text and formatting values retain their LaTeX commands. The compatibility keys `biblatexstyle` and `biblatexstyleFeature` read the current document mode's CV or proposal setting. Leading and trailing spaces around a key are ignored; unknown keys raise an error. This interface covers customization keys, not arbitrary commands or packages defined in the customization file.

`dfgLocation`=⟨*text*⟩
`dfgDate`=⟨*text*⟩
`dfgTitle`=⟨*text*⟩
`InfoTeX`=⟨*text*⟩

Override the matching public commands through `\changeSettings`, including from `preamble/dfg-customization.tex`. The defaults are `<Location>`, `<date>`, the quoted CRC name from metadata, and `LuaLaTeX (via Overleaf).`, respectively. The first three commands append `\xspace`; `InfoTeX` prints its value exactly. Values are stored without expansion, so references and formatting can be used in them. No fixed submission date is supplied. See Example 1.17.

Example 1.17: Set proposal text in the customization file

```
\changeSettings{
  dfgLocation = {Example City},
  dfgDate     = {30 September 2026},
  dfgTitle    = {Collaborative research
    programme},
  InfoTeX     = {LuaLaTeX, local build}
}
\dfgTitle\par
\dfgLocation, \dfgDate\par
\InfoTeX
```

Collaborative research programme
Example City, 30 September 2026
LuaLaTeX, local build

These macros already exist when the customization file is read, so direct overrides with `\renewcommand` work there too. For the three space-aware commands, include `\xspace` yourself when using `\renewcommand`. Definitions supplied before loading the class are preserved until explicitly overridden. If both interfaces assign a value, the last assignment wins.

1.8 Tables, lists, and plots: one calling convention

The dispatcher selects a named output and passes its arguments to it. Put settings in the leading square brackets. A simple target needs one pair of braces; a target with arguments uses an outer pair around the entire sequence: `\printTable{ProjectLeaders}`, `\printTable{{Projects}{A}{Area title}}`. The following chapter introduces each built-in target beside its result.

\printTable[⟨*options*⟩]{⟨*name*⟩}

Print one of the predefined template tables. The ⟨*name*⟩ is the table base name without the suffix `Table`; for example `ProjectLeaders` calls `\ProjectLeadersTable`. If ⟨*name*⟩ is a braced list, the first item is used as the table base name and the remaining items are passed as arguments to that table command. The gap above a table is set by `presep=6pt` in the shared `\SetTblrOuter` call below; adjust that value to change the gap above generated tables. Put options before the table specification. For example, `\printTable[allsexes]{GenderEqualityProjectLeaders}` selects the expanded gender overview for that call only. The command inserts `\par` before and after its output, so surrounding text always starts in separate paragraphs, even without blank source lines. Directly after a heading, it also inserts a zero-height box and a no-break penalty. This prevents the table from overlapping the heading at a page break; a manual `\null` before `\printTable` is unnecessary.
See Example 2.6, Example 2.57, Example 2.33.

\printList[⟨*options*⟩]{⟨*name*⟩}

Print one of the predefined template lists. The ⟨*name*⟩ is the list base name without the suffix `List`; for example `ParticipatingInstitutions` calls `\ParticipatingInstitutionsList`. The optional ⟨*options*⟩ are passed to the generated list command. Put sorting, formatting, database, and raster settings together in this leading option list, as with `\printPlot`. Both `{ParticipatingInstitutions}` and `{{ParticipatingInstitutions}}` are accepted as the list specification. Older trailing options remain supported, with leading settings taking precedence. A person's institutions use `\printList{{Institutions}{per1}}`; that form needs the person ID, while `\printList{ParticipatingInstitutions}` lists participating institutions across the proposal, with affiliation names in bold.
See Example 2.7, Example 2.9, Example 2.17.

`\printPlot` uses the same convention: `\printPlot[options]{{type}{project-id}}`. Within a project, omit the ID to use that project. The complete option and result examples follow the work-package declarations in Section 2.4.5.

2 Build the proposal in order

Follow the generated proposal from the title page through general information, finances, individual projects, bylaws, declarations, and back matter. Each output example pairs its source with its result. All sixteen built-in table targets, all three lists, and both plot types appear at their point of use. The shared metadata are documented once in Section 3.8; table-specific records are shown directly before their printing command.

2.1 Title page and applicant universities

\dfgApplicantAffiliationsTF {⟨*singular*⟩} {⟨*plural*⟩}

Use ⟨*singular*⟩ when `meta.ref-affiliation-id` contains one applicant affiliation and ⟨*plural*⟩ when it contains several. The title page, applicant-university section heading, bylaws, and declarations use this helper for their grammar. `\theCRCaffiliation` and `\theCRCaffiliationLocation` contain the complete human-readable name and location lists. The first id is the coordinating university and is exposed as `\theCRCCoordinatingAffiliation`; any remaining ids are exposed as `\theCRCOtherApplicantAffiliations`, matching the two separate fields on the DFG title page.
See Example 2.1.

Example 2.1: Choose applicant wording from metadata

```
\dfgApplicantAffiliationsTF
  {Applicant university:}
  {Applicant universities:}
\theCRCaffiliation.
```

Applicant university: Example University.

\crcLogo [⟨*graphics-options*⟩]

Print the CRC logo. The optional ⟨*graphics-options*⟩ are passed to `\includegraphics`. The actual file is configured with `crcLogoPath` → P. 66 through `\changeSettings` and remains available as `\crcLogoPath`.
See Example 2.3.

\affiliationLogo [⟨*graphics-options*⟩]

Print the affiliation logo. The optional ⟨*graphics-options*⟩ are passed to `\includegraphics`. The actual file is configured with `affiliationLogoPath` → P. 66 through `\changeSettings` and remains available as `\affiliationLogoPath`.
See Example 2.3.

\Hrule [⟨*key=value*⟩]

Print a configurable horizontal rule. The available keys are `color`, `above`, `below`, and `thickness`.
See Example 2.2.

Example 2.2: A horizontal rule with custom spacing

```
Scientific objectives
\Hrule[color=blue,thickness=.6pt,above=4pt
    ,below=6pt]
Experimental approach
```

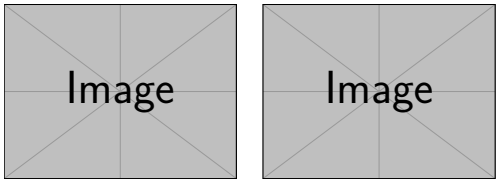
Scientific objectives

Experimental approach

Use your own logo files in a proposal. This example substitutes the bundled placeholder image so the size options can be tried without external files.

Example 2.3: Configure and print the title-page logos

```
\changeSettings{
  crcLogoPath=example-image,
  affiliationLogoPath=example-image
}
\crcLogo[width=.4\linewidth]\quad
\affiliationLogo[width=.4\linewidth]
```



2.2 General information

2.2.1 Key data: governing bodies and contact details

\getPersonInformation[⟨*database*⟩]{⟨*person-id-list*⟩}

Print full PI contact details, including institutions and contact fields. The database defaults to `person`. Use explicit person IDs for a selected pair; use `\getPIsWithRole` below to select people by their metadata role.
See Example 2.4.

Example 2.4: Contact details for two selected PIs

```
\getPersonInformation[person]{per1,per2}
```

Kim Sample, Prof. Dr. rer. nat.

0000-0000-0000-0001
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
(Germany)

Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City
(Germany)
+49 000 100-1000
kim.sample@example.org

Alex Example, Prof. Dr. phil.

0000-0000-0000-0002
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City
(Germany)
+49 000 200-2000
alex.example@example.org

\getPIsWithRole[⟨*options*⟩]{⟨*role*⟩}

Print persons with the given ⟨*role*⟩ in PI boxes. Put all options before the role, for example `database=person,raster columns=2`. The database defaults to `person`. Only persons belonging to the current funding period are shown. The default three-column raster has a 12 pt column gap. Rows of boxes and the content after the raster are separated by 20 pt. Raster settings in ⟨*options*⟩ are passed to the underlying `tcolorbox` CTAN raster and may override those defaults. Use `academicdegree-below` to place the academic degree on a separate line below the name, inside the same PI-box title. This applies only to the current call; omitting the option or setting `academicdegree-below=false` keeps the degree inline after the name. For example: `\getPIsWithRole[academicdegree-below]{steering-committee-member}`. The contact markers in each PI box are controlled globally with `\changeSettings{PIcontactStyle=icons}` (the default), `PIcontactStyle=text` for the labels “Phone:” and “E-Mail:”, or `PIcontactStyle=none` to suppress both markers. Older calls with a bare database name or trailing raster options remain supported; leading settings take precedence over trailing settings.
In both this raster and the two-column raster of `\getPersonInformation`, titles are ragged right and top-aligned. Each title reserves the height of the tallest title in its raster, keeping the contact information aligned when names or academic degrees wrap. Equal-height groups settle over successive LaTeX runs; a clean build can require three passes. Phone and email

each remain unbroken, but their one-em separating space permits a line break when both contacts do not fit on the same line.
See Example 2.5.

Example 2.5: Print all PIs with a given role.

```
\getPIsWithRole[raster columns=2]{steering-committee-member}
```

Alex Example, Prof. Dr. phil.
0000-0000-0000-0002
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City
(Germany)
+49 000 200-2000
alex.example@example.org

Samira Hartmann, Prof. Dr. rer. nat.
0000-0000-0000-0016
Centre for Computational Microbiology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)
+49 000 300-0006
samira@example.org

Hansi Meier, Dr. rer. nat.
0000-0000-0000-0003
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City
(Germany)
+49 000 200-3000
hansi.meier@example.org

Friedhelm Redata, Prof. Dr.-Ing.
0000-0000-0000-0005
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
(Germany)
+49 000 100-5000
friedhelm.redata@example.org

Morgan Sample, Prof. Dr. med.
0000-0000-0000-0006
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
(Germany)
+49 000 100-6000
morgan.sample@example.org

2.2.2 Project leaders

The table resolves each PI's institutions and affiliations from the shared metadata (Section 3.8). Declare those records once; changing a name or institution then updates every table and reference.

Example 2.6: Project leaders and their affiliations

```
\printTable{ProjectLeaders}
```

Project leader	Gende	Doctorate obtained in	Home institution, location	Project(s)
Example, Alex Prof. Dr. phil.	m	1999	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	**A01** **B01** **Q01**
Hartmann, Samira Prof. Dr. rer. nat.	f	2010	Centre for Computational Microbiology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)	**A02** **A03**

Continue on the following page ...

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Project leader	Gende	Doctorate obtained in	Home institution, location	Project(s)
Keller, Robin Prof. Dr. rer. nat.	d	2010	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany) Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	**B05** **B06** **Q03**
Lindberg, Noah Prof. Dr. rer. nat.	m	2010	Centre for Computational Microbiology Northbridge University Science Avenue 3, 34567 Northbridge (Germany) Department of Experimental Immunology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)	**A03** **A04** **Q01**
Meier, Hansi Dr. rer. nat.	f	1999	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	**B01** **B03** **Q03**
Miller, Kimmi Prof. Dr. rer. nat.	f	1999	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	**B02** **B04**
Navarro, Leonie Prof. Dr. rer. nat.	f	2010	Institute of Metabolic Medicine Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	**A05** **A06** **Q02**
Okafor, Daniel Prof. Dr. rer. nat.	m	2010	Institute of Metabolic Medicine Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany) Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	**A06** **B02** **B06**
Redata, Friedhelm Prof. Dr.-Ing.	m	1998	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany)	**B05** **Q02**
Sample, Kim Prof. Dr. rer. nat.	d	2000	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany) Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	**A01** **A02**
Sato, Mira Prof. Dr. rer. nat.	f	2010	Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	**B03** **B04**
Winters-Scholz, Alexis-Marie Prof. Dr. rer. nat.	d	2010	Department of Experimental Immunology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)	**A04** **A05**

2.2.3 Participating institutions

Use `columns=2` or `columns=3` to flow affiliation groups down balanced columns, keeping each affiliation with its institutions. Short groups follow one another without equal-height row padding. Reading order is down each column. The gap defaults to 12 pt; set `column sep=16pt` to change it. Flowing lists start a new paragraph with `\medskipamount` space above them.

Without `columns`, participating institutions use one raster cell per affiliation, including its institutions. The default is one column, with a 12 pt column gap and a 10 pt row gap. Pass `raster columns=2` or other `tcolorbox`→CTAN raster options through ⟨*options*⟩. Sorting defaults to `order=count`; `order=alpha` sorts affiliations alphabetically. The bare `count` and `alpha` options remain accepted.

Example 2.7: Flowing institution columns

```
\printList[columns=2]{ParticipatingInstitutions}
```

Example 2.7: Flowing institution columns (continued)

Northbridge University (Northbridge)
Centre for Computational Microbiology
Department of Experimental Immunology

Westhaven Research Centre (Westhaven)
Institute of Metabolic Medicine
Centre for Clinical Translation

Example University (Example City)
Institute of Microbial Systems Medicine

Partner University (Partner City)
Department of Translational Biosciences

Example 2.8: Three flowing institution columns

```
\printList[columns=3,column sep=16pt]{ParticipatingInstitutions}
```

Northbridge University (Northbridge)
Centre for Computational Microbiology
Department of Experimental Immunology

Westhaven Research Centre (Westhaven)
Institute of Metabolic Medicine
Centre for Clinical Translation

Example University (Example City)
Institute of Microbial Systems Medicine

Partner University (Partner City)
Department of Translational Biosciences

Example 2.9: Alphabetical institution raster

```
\printList[order=alpha,raster columns=2]{ParticipatingInstitutions}
```

Example University (Example City)
Institute of Microbial Systems Medicine

Partner University (Partner City)
Department of Translational Biosciences

Northbridge University (Northbridge)
Centre for Computational Microbiology
Department of Experimental Immunology

Westhaven Research Centre (Westhaven)
Institute of Metabolic Medicine
Centre for Clinical Translation

Example 2.10: Institution raster ordered by PI count

```
\printList[order=count,raster columns=3]{ParticipatingInstitutions}
```

Northbridge University (Northbridge)
Centre for Computational Microbiology
Department of Experimental Immunology

Partner University (Partner City)
Department of Translational Biosciences

Westhaven Research Centre (Westhaven)
Institute of Metabolic Medicine
Centre for Clinical Translation

Example University (Example City)
Institute of Microbial Systems Medicine

2.2.4 Project groups and projects

Projects are grouped by their area. The same command serves each area; the second argument supplies the heading. The shared dataset contains six A, six B, and three Q projects, declared once in Section 3.8.

Example 2.11: Project overview: area A

```
\printTable{{Projects}{A}{Microbiota and metabolic pathways}}
```

Project area A – Microbiota and metabolic pathways

Project		Title	Research area	Project leader(s), institute(s), location(s)
A01		Microbial signalling across epithelial barriers	2.21-02	Kim **Sample**, Institute of Microbial Systems Medicine, Example University; Department of Translational Biosciences, Partner University Alex **Example**, Department of Translational Biosciences, Partner University
A02	N	Metabolic control of tissue-resident immunity	2.21-05	Kim **Sample**, Institute of Microbial Systems Medicine, Example University; Department of Translational Biosciences, Partner University Samira **Hartmann**, Centre for Computational Microbiology, Northbridge University
A03		Microbiome-derived metabolites and immune memory	2.21-02	Samira **Hartmann**, Centre for Computational Microbiology, Northbridge University Noah **Lindberg**, Centre for Computational Microbiology, Northbridge University; Department of Experimental Immunology, Northbridge University
A04	N	Spatial niches of host–microbe interactions	2.21-02	Noah **Lindberg**, Centre for Computational Microbiology, Northbridge University; Department of Experimental Immunology, Northbridge University Alexis-Marie **Winters-Scholz**, Department of Experimental Immunology, Northbridge University
A05		Dietary control of microbial community dynamics	2.21-02	Alexis-Marie **Winters-Scholz**, Department of Experimental Immunology, Northbridge University Leonie **Navarro**, Institute of Metabolic Medicine, Westhaven Research Centre
A06	N	Modelling microbial resilience and recovery	2.21-02	Leonie **Navarro**, Institute of Metabolic Medicine, Westhaven Research Centre Daniel **Okafor**, Institute of Metabolic Medicine, Westhaven Research Centre; Centre for Clinical Translation, Westhaven Research Centre

Example 2.12: Project overview: area B

```
\printTable{{Projects}{B}{Immune mechanisms and clinical translation}}
```

Project area B – Immune mechanisms and clinical translation

Project		Title	Research area	Project leader(s), institute(s), location(s)
B01		Host defence during microbiome perturbation	2.21-03 2.21-05 2.22-15	Hansi **Meier**, Department of Translational Biosciences, Partner University Alex **Example**, Department of Translational Biosciences, Partner University
B02	N	Molecular signatures of tissue regeneration	2.22-15	Kimmi **Miller**, Department of Translational Biosciences, Partner University Daniel **Okafor**, Institute of Metabolic Medicine, Westhaven Research Centre; Centre for Clinical Translation, Westhaven Research Centre

Continue on the following page …

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Project		Title	Research area	Project leader(s), institute(s), location(s)
B03		Immune circuits linking intestine and liver	2.21-02	Hansi **Meier,** Department of Translational Biosciences, Partner University Mira **Sato,** Centre for Clinical Translation, Westhaven Research Centre
B04	N	Cross-organ consequences of inflammatory stress	2.21-02	Kimmi **Miller,** Department of Translational Biosciences, Partner University Mira **Sato,** Centre for Clinical Translation, Westhaven Research Centre
B05		Patient-derived models of metabolic disease	2.21-02	Friedhelm **Redata,** Institute of Microbial Systems Medicine, Example University Robin **Keller,** Institute of Microbial Systems Medicine, Example University; Centre for Clinical Translation, Westhaven Research Centre
B06	N	Translating microbial biomarkers into interventions	2.21-02	Daniel **Okafor,** Institute of Metabolic Medicine, Westhaven Research Centre; Centre for Clinical Translation, Westhaven Research Centre Robin **Keller,** Institute of Microbial Systems Medicine, Example University; Centre for Clinical Translation, Westhaven Research Centre

Example 2.13: Service, training, and administration projects

```
\printTable{{Projects}{Q}{Shared services, training, and coordination}}
```

Project area Q – Shared services, training, and coordination

Project	Title	Research area	Project leader(s), institute(s), location(s)
Q01	Integrated imaging and analysis services	2.21-02 2.22-15	Alex **Example,** Department of Translational Biosciences, Partner University Noah **Lindberg,** Centre for Computational Microbiology, Northbridge University; Department of Experimental Immunology, Northbridge University
Q02	Integrated Research Training Group		Friedhelm **Redata,** Institute of Microbial Systems Medicine, Example University Leonie **Navarro,** Institute of Metabolic Medicine, Westhaven Research Centre
Q03	Central coordination and data stewardship		Hansi **Meier,** Department of Translational Biosciences, Partner University Robin **Keller,** Institute of Microbial Systems Medicine, Example University; Centre for Clinical Translation, Westhaven Research Centre

The status legend can follow the project overview. The funding-period wording and research-area list are taken from metadata, so they stay consistent with the tables. The footnote remains in the normal page flow.

Example 2.14: Project-status legend and research areas

```
{\footnotesize
N: New projects started in the \theFundingPeriodText\space funding period;\space
E: Ending projects not continued in the \theFundingPeriodText\space funding period;\space
Research areas according to the German Research Foundation (2024--2028)%
\footnote{\raggedright
  \url{https://www.dfg.de/resource/blob/331950/fachsystematik-2024-2028-en.pdf}}:
\printList{ResearchAreasFromProjects}\par}
```

N: New projects started in the third funding period; E: Ending projects not continued in the third funding period; Research

areas according to the German Research Foundation (2024–2028)[1]: **2.21-02**: Microbial Ecology and Applied Microbiology, **2.21-03**: Medical Microbiology and Mycology, Hygiene, Molecular Infection Biology, **2.21-05**: Immunology, and **2.22-15**: Gastroenterology.

Example 2.15: Research areas excluding selected projects

```
\printList[A01,B01]{ResearchAreasFromProjects}
```

2.21-02: Microbial Ecology and Applied Microbiology, **2.21-05**: Immunology, and **2.22-15**: Gastroenterology.

The optional argument of `ResearchAreasFromProjects` excludes the listed project IDs from the collection of research areas.

Example 2.16: Look up research-area descriptions

```
\dfgResearchAreasList{2.21-02}\par
\dfgResearchAreasList{2.21-05}
```

2.21-02: Microbial Ecology and Applied Microbiology
2.21-05: Immunology

Person-specific institution lists Use `\printList{{Institutions}{per7}}` to print the institutions assigned to person `per7`. The person ID is required. For a non-default database, use `\printList[database=other]{{Institutions}{per7}}`; the database defaults to `person`. Put formatting options in the same leading list, as in `\printList[database=person,abbreviation,narrow]{{Institutions}{per7}}`. The older bare-database and trailing-formatting forms remain supported. The list reads `ref-institution-id`→P. 64 from the person entry and then resolves the referenced entries in the `institution` and `affiliation` databases. Options are `abbreviation`, `narrow`, `noaddress`, `nozip`, and `nolocation`.

In the standard layout, every institution prints the institution name, affiliation name, street, ZIP code, location, and country when these values are available. If a person references more than one institution, the entries are separated by a paragraph break with a small vertical gap. If an institution row has an empty `name`→P. 64, the output starts directly with the affiliation name.

`abbreviation` Use the institution's `abbreviation`→P. 58 instead of its full `name`→P. 64. If the abbreviation is absent, the full name is used for compatibility.

`narrow` Print a compact inline form. Fields inside one institution are separated by commas and multiple institutions by semicolons. This is useful in narrow CV/table cells.

`noaddress` Omit the street address.

`nozip` Omit the ZIP code.

`nolocation` Omit the location and country.

The former `\dfgInstitutionList` spelling remains as a compatibility alias.

1 https://www.dfg.de/resource/blob/331950/fachsystematik-2024-2028-en.pdf

Example 2.17: One person with multiple institutions

```
\get[person]{per7}{ref-institution-id}\par
\printList{{Institutions}{per7}}
\medskip
\printList[abbreviation,narrow,noaddress,
    nozip,nolocation]
  {{Institutions}{per7}}
```

ins3,ins4
Centre for Computational Microbiology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)

Department of Experimental Immunology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany) CCM, Northbridge University; DEI,

Northbridge University

2.2.5 Frame-text references

\printFrameBibliography [⟨*biblatex options*⟩]

Print the frame-text bibliography, forwarding the optional arguments to `\printbibliography` and preserving its normal font size. With `emphasizeEntries` →P. 70 enabled, a complete entry is bold when its `keywords` field contains `frametext-highlight`. Project highlight keywords do not affect this list; the frame keyword does not affect project lists or inline citations. Like project bibliographies, output is suppressed when `fastcompile-project-show-publications` →P. 65 is disabled.

Add the keyword to the publication's existing keywords, for example `keywords = {pabst-A,B06-highlight,frametext-highlight}`. The keyword controls formatting, not inclusion: cite the entry or add it with `\nocite` as usual. Enable `emphasizeEntries` →P. 70 in `\changeSettings`, then use:

```
\defbibnote{frame-references}{%
  \usekomafont{subsection}%
  \bfseries\sffamily References
}
\printFrameBibliography[
  prenote={frame-references},
  heading=none
]
```

Replace the plain `\printbibliography` call in existing chapter 1 files; template updates preserve those edited files.

2.2.6 Research profile: summaries and highlighted statements

`\begin{`**dfgBoxEnv**`}` [⟨*further specifications*⟩]
⟨*environment content*⟩
`\end{`**dfgBoxEnv**`}`

Box environment to be used for highlighting things To be used in `content.tex`. See Example 2.18.

Example 2.18: Highlight text with a box environment

```
\begin{dfgBoxEnv}[title={Research question
    }]
How do host and microbiome communicate?
\end{dfgBoxEnv}
```

Research question

How do host and microbiome communicate?

`\begin{`**dfgSummaryBox**`}` [⟨*further specifications, e.g. 'title=NN'*⟩]
⟨*environment content*⟩
`\end{`**dfgSummaryBox**`}`

Box environment to be used for e.g. summaries that are lists. Use `\item` for bulletpoints. To be used in `content.tex`
See Example 2.19.

Example 2.19: Example use of a summary box

```
\begin{dfgSummaryBox}[title={Next steps}]
  \item Validate biomarkers.
  \item Share the results.
\end{dfgSummaryBox}
```

Next steps

- Validate biomarkers.
- Share the results.

`\dfgBox[`⟨*further specifications*⟩`]`

Box environment to be used for anything. To be used anywhere in the text.
See Example 2.20.

Example 2.20: Example of a minimal box

```
\dfgBox[title={Research finding}]
  {Microbial signals shape immunity.}
```

Research finding

Microbial signals shape immunity.

2.2.7 Tables and figures in the scientific narrative

Use the following environments when the prose needs a table or a figure that is not one of the predefined metadata tables.

`\begin{TableEnv}{`⟨*column specification*⟩`}` `[`⟨*table options*⟩`]{`⟨*caption*⟩`}{`⟨*label*⟩`}`
⟨*environment content*⟩
`\end{TableEnv}`

Typeset a generic long table using the template's main table theme. The optional ⟨*table options*⟩ are inserted into the inner `tabularray` CTAN specification and default to an empty option list, so the square brackets may be omitted when no additional options are needed. Table text defaults to `\footnotesize` at theme scope. Consequently, cell-local declarations such as `font=\bfseries` or `font=\itshape` retain the default size; include an explicit size such as `font=\normalsize\bfseries` only when a table or cell should override it.
See Example 2.21, Example 2.36.

Example 2.21: TableEnv with column and row options

```
\begin{TableEnv}{X[2,l] X[1,r]}[
  row{1}={font=\bfseries},
  rowsep=3pt,
  column{2}={halign=r}
]{Projects by area}{tab:manual-area-counts
    }
Area & Projects \\
A & \getProjectCount[A] \\
B & \getProjectCount[B] \\
Q & \getProjectCount[Q]
\end{TableEnv}
```

Table 1: Projects by area

Area	Projects
A	6
B	6
Q	3

2.2.8 Figures

For inserting figures we highly recommend to use the command from the package `hvfloat`.

`\hvFloat*[`⟨*options*⟩`]+{`⟨*floattype*⟩`}{`⟨*floatingobject*⟩`}[`⟨*shortcaption*⟩`]{`⟨*longcaption*⟩`}{`⟨*label*⟩`}`

See Example 2.22.

Example 2.22: A figure with a caption

```
\hvFloat[capPos=bottom]{figure}
  {\includegraphics[width=\linewidth]{
    example-image}}
  {Example artwork and its caption.}
  {fig:manual-hvfloat}
```

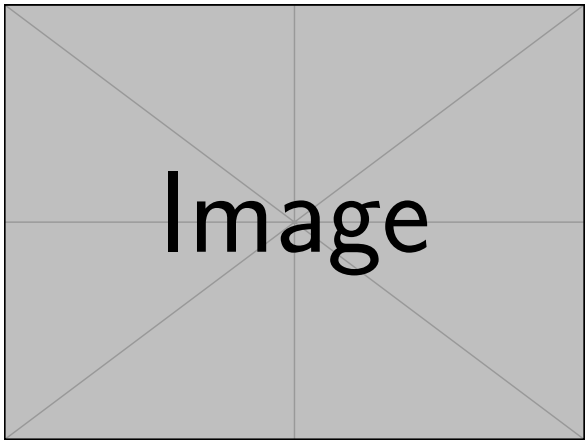


Fig. 1: Example artwork and its caption.

2.2.9 Early-career researchers and equality objectives

\EarlyCareerResearchersTable[⟨*database*⟩]

List doctoral candidates from `EarlyCareerResearcher`, using the `person` keyset. This is a named training roster, separate from the monetary staff request. `PhD` and `MD` select the row colours. An empty `periodEnd` prints an ongoing appointment as “since”; bold names can mark completed doctorates, as explained by the table legend.
See Example 2.23.

Example 2.23: Doctoral candidates: completed, ongoing, and medical projects

```
\addEntry[person]{EarlyCareerResearcher}{
  id             = {doc-phd-completed},
  ref-project-id = {A01},
  firstname      = {Robin},
  lastname       = {\textbf{Meyer}},
  financed-by    = {DFG},
  category       = {PhD},
  topic          = {Microbial metabolites and epithelial signalling},
  periodStart    = {2021-10},
  periodEnd      = {2025-09}
}
\addEntry[person]{EarlyCareerResearcher}{
  id             = {doc-phd-ongoing},
  ref-project-id = {B03},
  firstname      = {Alex},
  lastname       = {Reed},
  financed-by    = {University},
  category       = {PhD},
  topic          = {Tissue-resident immune-cell programming},
  periodStart    = {2025-04},
  periodEnd      = {}
}
\addEntry[person]{EarlyCareerResearcher}{
  id             = {doc-md},
  ref-project-id = {B04},
  firstname      = {Sam},
  lastname       = {Taylor},
  financed-by    = {DFG},
  category       = {MD},
  topic          = {Clinical markers of cross-organ inflammation},
  periodStart    = {2026-04},
  periodEnd      = {2026-09}
}
```

Example 2.23: Doctoral candidates: completed, ongoing, and medical projects (continued)

```
\printTable{EarlyCareerResearchers}
```

Project	Surname, first name	Type of funding	Topic	Duration
A01	**Meyer**, Robin	DFG	*Microbial metabolites and epithelial signalling*	2021-10–2025-09
B03	Reed, Alex	University	*Tissue-resident immune-cell programming*	since 2025-04
B04	Taylor, Sam	DFG	*Clinical markers of cross-organ inflammation*	2026-04–2026-09

Note: Row colours indicate category of doctoral candidates: Dr. rer. nat., Dr. med.. **RS** = Rising Star. Doctoral candidates with highlighted names have obtained the doctoral degree. MD denotes medical students supported by DFG stipends for up to 6 months.

A. Objectives for the participation of female researchers

This example supplies the previous objective, observed number and percentage, and future objective for doctoral and postdoctoral researchers. All figures are fictitious. Use the `GenderEquality` keyset for both equality tables.

Example 2.24: A. Objectives for the participation of female researchers

```
\addEntry[GenderEquality]{GenderEqualityStaff}{
  position                              = {PhD},
  previous-objective-percentage-female  = {40},
  status-quo-number-female              = {12},
  status-quo-percentage-female          = {50},
  future-objective-percentage-female    = {50}
}
\addEntry[GenderEquality]{GenderEqualityStaff}{
  position                              = {postDoc},
  previous-objective-percentage-female  = {40},
  status-quo-number-female              = {4},
  status-quo-percentage-female          = {40},
  future-objective-percentage-female    = {50}
}
\printTable{GenderEqualityStaff}
```

	Second funding period			Third funding period
	Objective [%]	Status Quo[a]		Objective [%]
		Number	[%]	f
Doctoral researchers	40	12	50	50
Postdoctoral researchers	40	4	40	50

[a] Reference date: proposal for third funding period

B. Project leaders

Here the records specify objectives for each career category. Status-quo counts and percentages are calculated from the `gender`, `position`, and funding-period membership of the `person` records shown in Section 2.2.4. Only proposals with `fundingPeriodNumeral = {3}` show the additional first-funding-period columns, using membership in period 1. By default, the table shows only female numbers and percentages. For a third-period proposal, it follows DFG form 60.200: periods 1 and 2 each have Number/Percentage columns; period 3 has its original objective and current Number/Percentage columns. The existing `future-objective-percentage-female` field supplies that period 3 objective. Include period 3 membership in the person records to calculate the proposed leadership composition.

Pass `[all-sexes]` before the table name in `\printTable` → p. 10 to select the expanded f/m/d comparison. For a third-period proposal, every period includes Number and Percentage columns for f/m/d, alongside the previous period's objective and the proposed period's future objective. All percentages use the complete f/m/d population in the same career category (or across the four categories in the total row). Do not enter these calculated values again in the objectives database; include historical PIs with the appropriate `period` membership.

```
\printTable{GenderEqualityProjectLeaders}            % default: female-only
\printTable[all-sexes]{GenderEqualityProjectLeaders} % f/m/d for all periods
```

all-sexes=true|false (initially `false`)

A per-table option for `\GenderEqualityProjectLeadersTable`. The bare flag `[all-sexes]` selects f/m/d counts and percentages, including period 3 in a third-period proposal, alongside the previous/future objectives. Omit the option (or use `[all-sexes=false]`) for the female-only DFG layout. Percentages always use all genders as their denominator. The option affects only the current table and does not change subsequent tables or prose commands. Use `[database=MyObjectives,all-sexes]` for a custom objectives database; the older bare database argument also remains supported.

Example 2.25: B. Project leaders: objectives and calculated participation

```
\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {postDoc},
  previous-objective-percentage-female  = {40},
  future-objective-percentage-female    = {50}
}
\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {groupLeader},
  previous-objective-percentage-female  = {40},
  future-objective-percentage-female    = {50}
}
\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {prof-W2},
  previous-objective-percentage-female  = {40},
  future-objective-percentage-female    = {50}
}
\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {prof-W3},
  previous-objective-percentage-female  = {40},
  future-objective-percentage-female    = {50}
}
\printTable{GenderEqualityProjectLeaders}
```

Position (female)	First funding period		Second funding period		Status Quo[a] Third funding period		
	Number	[%]	Number	[%]	Original objective [%]	Number	[%]
Postdoctoral researchers	0	0	1	100	50	1	100
Research group leaders, junior research group leaders, junior professors	0	0	0	0	50	0	0
Professors W2/C3	0	0	1	100	50	4	44
Professors W3/C4	0	0	0	0	50	0	0
Total	**0**	**0**	**2**	**50**		**5**	**42**

[a] Reference date: proposal for third funding period

Use the same statistics in prose

\getPositionGenderPhaseAmount [⟨*database*⟩]{⟨*position*⟩}{⟨*gender*⟩}{⟨*period*⟩} prints the count for one position/gender/period tuple. See Example 2.26.

\getPositionPhaseTotal [⟨*database*⟩] {⟨*position*⟩} {⟨*period*⟩} prints the all-gender total. See Example 2.26.

\getGenderPhaseTotal [⟨*database*⟩] {⟨*gender*⟩} {⟨*period*⟩} prints the all-position total. See Example 2.26.

\getPositionGenderPhasePercentage [⟨*database*⟩] {⟨*position*⟩} {⟨*gender*⟩} {⟨*period*⟩} prints the rounded integer percentage value within a position. See Example 2.26.

\getGenderPhasePercentage [⟨*database*⟩] {⟨*gender*⟩} {⟨*period*⟩} prints the rounded integer percentage value across all counted positions. See Example 2.26.

The period argument is the funding-period number, not a calendar year. Percentage commands return a number; append `\%` when using it in prose. The commands below retrieve the numbers printed in section 1.4.2 B for any funding period, independently of which proposal period is currently selected. Use `f`, `m`, or `d` for the gender. The supported table positions are `postDoc`, `groupLeader`, `prof-W2`, and `prof-W3`. The total-row commands sum these four positions and count each person with `role=PI` once per period; a person belonging to several periods is counted in each matching period. The optional database defaults to `person`. Percentages are rounded to the nearest integer and return zero if their denominator is zero. These are person statistics; calculating the share of projects with female leadership requires a separate calculation that counts each project once.

```
% First funding period: female W3 professors, number and percentage.
\getPositionGenderPhaseAmount{prof-W3}{f}{1}
\getPositionGenderPhasePercentage{prof-W3}{f}{1}\%
% All genders within the same career category.
\getPositionPhaseTotal{prof-W3}{1}
% First-period total row: female project leaders, number and percentage.
\getGenderPhaseTotal{f}{1}
\getGenderPhasePercentage{f}{1}\%
% Second-period total row: change only the final argument.
\getGenderPhaseTotal{f}{2}
\getGenderPhasePercentage{f}{2}\%
```

Example 2.26: Person counts and percentages for a funding period

```
W3 professors in period 3:
\getPositionPhaseTotal{prof-W3}{3}.\par
Women among them:
\getPositionGenderPhaseAmount{prof-W3}{f
    }{3}
(\getPositionGenderPhasePercentage{prof-W
    3}{f}{3}\%).\par
Women across counted positions:
\getGenderPhaseTotal{f}{3}
(\getGenderPhasePercentage{f}{3}\%).
```

W3 professors in period 3: 1.
Women among them: 0 (0%).
Women across counted positions: 5 (42%).

2.3 Existing and requested funds

The proposal-wide tables combine institutional resources and project requests. The class transfers project declarations through the auxiliary files, so compile with `latexmk` until the totals settle. The project-level declarations and their local tables follow in Section 2.4.6.

2.3.1 Finances: existing funds and resources

The following overviews precede the project chapters in a proposal. By default, current-period figures are restored from project declarations through the auxiliary file; compile again after changing those declarations. Custom databases in the next two examples isolate their inputs from the other demonstrations.

Existing funds for direct costs

The five historical rows use full EUR amounts. Current-period rows group funds by `ref-affiliation-id`: the applicant affiliation `aff1`, another participant `aff2`, and the special value `other-funds`. Funding-source text does not choose the column. In a proposal, declare the current rows in `RequestedAndExistingFundingDirectCosts` and use `\printTable{ExistingFundsDirectCosts}` to collect them automatically.

Example 2.27: Existing direct costs: previous and new funding periods

```
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year                = {1},
  applicant           = {2000},
  participant         = {1000},
  other-funds         = {500}
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year                = {2},
  applicant           = {2000},
  participant         = {1000},
  other-funds         = {500}
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year                = {3},
  applicant           = {2000},
  participant         = {1000},
  other-funds         = {500}
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year                = {4},
  applicant           = {2000},
  participant         = {1000},
  other-funds         = {500}
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year                = {5},
  applicant           = {2000},
  participant         = {1000},
  other-funds         = {500}
}
\addEntry[funding]{ManualExistingCosts}{
  ref-project-id      = {A01},
  ref-affiliation-id  = {aff1},
  financed-by         = {Example core or third-party funds},
  sum                 = {year1 = {4000}, year2 = {4000}, year3 = {4000}, year4 = {4000},
    year5 = {4000}}
}
\addEntry[funding]{ManualExistingCosts}{
  ref-project-id      = {A01},
  ref-affiliation-id  = {aff2},
  financed-by         = {Example core or third-party funds},
  sum                 = {year1 = {2000}, year2 = {2000}, year3 = {2000}, year4 = {2000},
    year5 = {2000}}
}
\addEntry[funding]{ManualExistingCosts}{
  ref-project-id      = {A01},
  ref-affiliation-id  = {other-funds},
  financed-by         = {Example core or third-party funds},
  sum                 = {year1 = {1000}, year2 = {1000}, year3 = {1000}, year4 = {1000},
    year5 = {1000}}
}
\printTable[ManualExistingCosts]{ExistingFundsDirectCosts}
```

Financial year/ funding period	Core support provided by applicant university	Core support provided by other participating institutions	Other funds	Total
2022|2	2,000	1,000	500	3,500

Continue on the following page ...

...continuing from previous page.

Financial year/ funding period	Core support provided by applicant university	Core support provided by other participating institutions	Other funds	Total
2023	2,000	1,000	500	3,500
2024	2,000	1,000	500	3,500
2025	2,000	1,000	500	3,500
2026|1	2,000	1,000	500	3,500
Ending Funding period	10,000	5,000	2,500	17,500
2027|2	4,000	2,000	1,000	7,000
2028	4,000	2,000	1,000	7,000
2029	4,000	2,000	1,000	7,000
2030	4,000	2,000	1,000	7,000
2031|1	4,000	2,000	1,000	7,000
New Funding period	20,000	10,000	5,000	35,000

Note: Figures in EUR

Existing staff

These records inherit institutions from the PI metadata already shown. The custom database supplies all columns read by the overview; empty organisation fields allow the person's institution references to be used. The overview counts existing staff at applicant and participating affiliations. In a proposal, add them to `RequestedFundingStaff` and use `\printTable{OverviewExistingStaff}` without a database override.

Example 2.28: Existing staff by affiliation and career category

```
\addEntry[staff]{ManualExistingStaff}{
  ref-project-id       = {A01},
  funding-status       = {existing},
  staff-type           = {research},
  ref-person-id        = {per1},
  lastname             = {Sample},
  category             = {prof-W3},
  ref-institution-id   = {},
  ref-affiliation-id   = {},
  field-of-research    = {Life sciences},
  commitment           = {8},
  funding-source       = {State funds}
}
\addEntry[staff]{ManualExistingStaff}{
  ref-project-id       = {A02},
  funding-status       = {existing},
  staff-type           = {research},
  ref-person-id        = {per6},
  lastname             = {Hartmann},
  category             = {prof-W2},
  ref-institution-id   = {},
  ref-affiliation-id   = {},
  field-of-research    = {Life sciences},
  commitment           = {8},
  funding-source       = {State funds}
}
\addEntry[staff]{ManualExistingStaff}{
  ref-project-id       = {B03},
  funding-status       = {existing},
  staff-type           = {research},
  ref-person-id        = {per11},
  lastname             = {Sato},
  category             = {prof-W2},
  ref-institution-id   = {},
  ref-affiliation-id   = {},
  field-of-research    = {Life sciences},
  commitment           = {8},
  funding-source       = {State funds}
```

Example 2.28: Existing staff by affiliation and career category (continued)

```
}
\printTable[ManualExistingStaff]{OverviewExistingStaff}
```

Category	Number of persons			
	at the applicant university	at other participating institutions		
	EU	PU	NU	WRC
Professors	1	1	1	1
Junior research group leaders	0	0	0	0
Postdoctoral researchers	0	0	0	0
Doctoral researchers	0	0	0	0
Other research staff	0	0	0	0
Non-research staff	0	0	0	0
Student and graduate assistants	0	0	0	0

Affiliations: EU: Example University; **PU**: Partner University; **NU**: Northbridge University; **WRC**: Westhaven Research Centre.

\ExistingInstrumentationTable[⟨*database*⟩]

List equipment already available to the projects. The database defaults to `existingInstrumentation` and uses the `instrumentation` keyset. The `cost` field is in thousands of euros: `250` means EUR 250,000. A comma-separated `ref-project-id` lists shared equipment once for several projects. Rows are sorted in ascending order by `ref-project-id`; shared equipment is ordered by its listed project IDs and remains a single row.
See Example 2.29.

Example 2.29: Existing instrumentation, including shared equipment

```
\addEntry[instrumentation]{existingInstrumentation}{
  ref-project-id = {A01},
  quantity       = {1},
  instrument     = {Confocal laser-scanning microscope},
  company        = {Example Instruments},
  year           = {2025},
  cost           = {250},
  financed-by    = {DFG}
}
\addEntry[instrumentation]{existingInstrumentation}{
  ref-project-id = {A01,B03},
  quantity       = {2},
  instrument     = {Anaerobic workstation},
  company        = {Example Laboratory Systems},
  year           = {2024},
  cost           = {80},
  financed-by    = {University funds}
}
\printTable{ExistingInstrumentation}
```

Project	Description of instrumentation	Year of purchase	Cost of purchase[a]	Source of funding
A01	1 × Confocal laser-scanning microscope (Example Instruments)	2025	250	DFG
A01 / B03	2 × Anaerobic workstation (Example Laboratory Systems)	2024	80	University funds

[a] Figures in thousands

2.3.2 Finances: previous funding and new requests

The next table combines five historical rows with the current requests declared in the funding examples. Its new-period amounts include the earlier A02 requests, the equipment records below, and the complete A06 request in Section 2.4.6. These records are collected from the auxiliary file even when their declarations follow the overview. All input amounts are full EUR; do not duplicate a project declaration to populate this overview.

Example 2.30: Previous funding and current requests across all five groups

```
\addEntry[funding]{PreviousRequestedFunding}{
  year                = {1},
  staff               = {100000},
  costs               = {20000},
  instrumentation     = {30000},
  fellowships         = {5000},
  globalfunds         = {10000}
}
\addEntry[funding]{PreviousRequestedFunding}{
  year                = {2},
  staff               = {100000},
  costs               = {20000},
  instrumentation     = {0},
  fellowships         = {5000},
  globalfunds         = {10000}
}
\addEntry[funding]{PreviousRequestedFunding}{
  year                = {3},
  staff               = {100000},
  costs               = {20000},
  instrumentation     = {0},
  fellowships         = {5000},
  globalfunds         = {10000}
}
\addEntry[funding]{PreviousRequestedFunding}{
  year                = {4},
  staff               = {100000},
  costs               = {20000},
  instrumentation     = {0},
  fellowships         = {5000},
  globalfunds         = {10000}
}
\addEntry[funding]{PreviousRequestedFunding}{
  year                = {5},
  staff               = {100000},
  costs               = {20000},
  instrumentation     = {0},
  fellowships         = {5000},
  globalfunds         = {10000}
}
\printTable{PreviousAndCurrentRequestedFunding}
```

Financial year/ funding period	Funding for					Total
	Staff	Direct costs	Instrumentation	Fellowships	Globalfunds	
2022|2	100,000	20,000	30,000	5,000	10,000	165,000
2023	100,000	20,000	0	5,000	10,000	135,000
2024	100,000	20,000	0	5,000	10,000	135,000
2025	100,000	20,000	0	5,000	10,000	135,000
2026|1	100,000	20,000	0	5,000	10,000	135,000
Ending Funding period	500,000	100,000	30,000	25,000	50,000	705,000
2027|2	54,404	2,000	170,000	3,000	1,500	230,904
2028	108,810	4,000	0	6,000	3,000	121,810
2029	108,810	4,000	45,000	6,000	3,000	166,810
2030	108,810	4,000	0	6,000	3,000	121,810
2031|1	54,404	2,000	0	3,000	1,500	60,904

Continue on the following page ...

...continuing from previous page.

Financial year/ funding period	Staff	Direct costs	Funding for Instrumentation	Fellowships	Globalfunds	Total
New Funding period	435,238	16,000	215,000	24,000	12,000	702,238

Note: Figures in EUR

The staff overview counts requested positions by category and year; salary percentages affect monetary costs, not the number of positions. This example uses a separate database, `ManualStaffRequest`, to show precisely these two records. Pass that database as the leading optional argument. In a proposal, use `RequestedFunding` for the records and omit the table's optional argument to collect requests across all projects. Projects without requested positions remain visible with empty-value markers.

Example 2.31: Requested staff positions by category and year

```
\addEntry[funding]{ManualStaffRequest}{
  ref-project-id = {B03},
  funding-type   = {staff},
  category       = {PhD},
  percentage     = {65},
  quantity       = {
    year1 = {1}, year2 = {1}, year3 = {2}, year4 = {2}, year5 = {1}
  }
}
\addEntry[funding]{ManualStaffRequest}{
  ref-project-id = {Q01},
  funding-type   = {staff},
  category       = {postDoc},
  percentage     = {100},
  quantity       = {
    year1 = {0}, year2 = {1}, year3 = {1}, year4 = {1}, year5 = {0}
  }
}
\printTable[ManualStaffRequest]{OverviewOfFundsRequestedForStaff}
```

Project	2027|2					2028					2029					2030					2031|1				
	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants
A01	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
A02	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
A03	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
A04	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
A05	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
A06	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
B01	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
B02	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
B03	×	1	×	×	×	×	1	×	×	×	×	2	×	×	×	×	2	×	×	×	×	1	×	×	×
B04	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
B05	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
B06	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Q01	×	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	×	×	×	×	×

Continue on the following page ...

...continuing from previous page.

Project	2027|2					2028					2029					2030					2031|1				
	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants
Q02	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Q03	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
Total	×	**1**	×	×	×	**1**	**1**	×	×	×	**1**	**2**	×	×	×	**1**	**2**	×	×	×	×	**1**	×	×	×

Requested equipment belongs in `RequestedFunding` with `funding-type={instrumentation}`. Unlike existing-equipment purchase costs, the yearly `sum` values are full EUR amounts. The overview collects these project declarations through the auxiliary file; two compilations may be needed after adding or changing them. Its result also includes the `A06` imaging workstation declared in the complete project-funding example in Section 2.4.6.

Example 2.32: Requested instrumentation by project and year

```
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q01},
  funding-type   = {instrumentation},
  category       = {Live-cell imaging system},
  sum            = {
    year1 = {150000},
    year2 = {0},
    year3 = {0},
    year4 = {0},
    year5 = {0}
  }
}
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {B03},
  funding-type   = {instrumentation},
  category       = {Automated sample preparation},
  sum            = {
    year1 = {0},
    year2 = {0},
    year3 = {45000},
    year4 = {0},
    year5 = {0}
  }
}
\printTable{OverviewFundsRequestedInstrumentation}
```

Project	Description of instrumentation	Funds requested for				
		2027|2	2028	2029	2030	2031|1
A06	Imaging workstation	20,000	0	0	0	0
B03	Automated sample preparation	0	0	45,000	0	0
Q01	Live-cell imaging system	150,000	0	0	0	0
Total		170,000	0	45,000	0	0

Note: Figures in EUR

Upkeep of laboratory animals

Each row specifies the animal count, average weeks kept, purchase cost per animal, and requested and existing funds in EUR. Housing rates come from the class settings; `mouse`, `rat`, and `mouse-gnotobiotic` select the species/rate.

Example 2.33: Laboratory animals: purchase, housing, and funding

```
\addEntry{UpkeepLaboratoryAnimals}{
  ref-project-id       = {A01},
  species              = {mouse},
  quantity             = {100},
  weeks-kept           = {12},
  purchasing-costs     = {25},
  requested-funds      = {9000},
  existing-funds       = {3000}
}
\addEntry{UpkeepLaboratoryAnimals}{
  ref-project-id       = {B03},
  species              = {rat},
  quantity             = {20},
  weeks-kept           = {8},
  purchasing-costs     = {40},
  requested-funds      = {1600},
  existing-funds       = {800}
}
\addEntry{UpkeepLaboratoryAnimals}{
  ref-project-id       = {A02},
  species              = {mouse-gnotobiotic},
  quantity             = {30},
  weeks-kept           = {10},
  purchasing-costs     = {50},
  requested-funds      = {6000},
  existing-funds       = {2500}
}
\printTable{UpkeepLaboratoryAnimals}
```

Project	Species	Quantity	Average number of weeks kept	Housing costs per animal per week[a]	Purchasing costs per animal[a]	Requested funds[a]	Existing funds[a]
A01	Mouse	100	12	0.75	25	9,000	3,000
A02	Mouse (gnotobiotic)	30	10	3.9	50	6,000	2,500
B03	Rat	20	8	1.4	40	1,600	800
Total		**Mice: 130** **Rats: 20**				16,600	6,300

[a] Figures in EUR

2.3.3 Refer to financial totals in prose

The near-identical funding names encode two different scopes. Commands using the singular name `RequestedFunding` use the proposal-wide `RequestedFundingOverview` database by default. Commands using the plural name `RequestedFunds` use the project-level `RequestedFunding` database and therefore require a project id. In the signatures below, ⟨*sum-fields*⟩ means one or more fields such as `sum-year1` through `sum-year5`.

`\getCountRequestedStaff` `[`⟨*project*⟩`]{`⟨*year*⟩`}{`⟨*category*⟩`}` prints the requested staff quantity, optionally restricted to one project. See Example 2.35.

`\getSumRequestedFunding` `[`⟨*database*⟩`]{`⟨*year*⟩`}{`⟨*type*⟩`}` prints the proposal-wide total for one funding type; use `year=all` for all five years. See Example 2.34.

\getTotalSumRequestedFunding [⟨*database*⟩] {⟨*year*⟩} {⟨*type*⟩} has the same proposal-wide scope and additionally accepts `type=all` for the grand total. See Example 2.34.

\getSumPreviousRequestedFunding [⟨*database*⟩] {⟨*year*⟩} {⟨*fields*⟩} totals the selected columns in the previous-period database; `year=all` includes every row.

\getSumRequestedFunds [⟨*database*⟩] {⟨*project*⟩} {⟨*type*⟩} {⟨*sum-fields*⟩} prints a project total for one funding type. See Example 2.34.

\getTotalSumRequestedFunds [⟨*database*⟩] {⟨*project*⟩} {⟨*sum-fields*⟩} prints a project total across all funding types. See Example 2.34.

\getSumRequestedFundsInstrumentation [⟨*project*⟩] {⟨*sum-fields*⟩} prints the proposal-wide instrumentation total or restricts it to the optional project.

\getSumRequestedFundingDirectCosts [⟨*database*⟩] {⟨*year*⟩} {⟨*group*⟩} totals direct costs for `applicant`, `participant`, `other-funds`, or `all`.

\getSumExistingFunds [⟨*database*⟩] {⟨*project*⟩} {⟨*sum-fields*⟩} prints existing direct-cost funding for one project.

For example:

Example 2.34: Project counts and funding totals

```
Area A projects: \getProjectCount[A]\par
Staff: \getSumRequestedFunding{all}{staff
    }\par
Total: \getTotalSumRequestedFunding{all}{
    all}\par
A02 staff, year 1:
\getSumRequestedFunds{A02}{staff}{sum-year
    1}\par
A02 total, year 1:
\getTotalSumRequestedFunds{A02}{sum-year1}
```

Area A projects: 6
Staff: 435,238
Total: 702,238
A02 staff, year 1: 27,202
A02 total, year 1: 28,202

Use the optional project argument to compare a project with the full request. Here `1` is the first funding-year column; `PhD` is the staff category.

Example 2.35: Requested positions and multi-year direct costs

```
A02 doctoral positions, year 1:
\getCountRequestedStaff[A02]{1}{PhD}.\par
All doctoral positions, year 1:
\getCountRequestedStaff{1}{PhD}.\par
A02 direct costs, years 1 and 2:
\getSumRequestedFunds{A02}{costs}{sum-year
    1,sum-year2}\,EUR.
```

A02 doctoral positions, year 1: 1.
All doctoral positions, year 1: 1.
A02 direct costs, years 1 and 2: 3,000 EUR.

2.4 Individual projects

2.4.1 Choose the structure from project status and type

Set `project-status` in the project's metadata: `N` for new, `C` for continuing, and `E` for ending. This is independent of `project-type`: `research`, `service`, `mgk` (integrated research training), or `admin` (central coordination). Changing metadata adjusts the generated heading and some tables; it does not rewrite the authored sections. Start from the matching generated content file.

Example 2.36: New, continuing, and ending project structures

```
\begin{TableEnv}{X[1,l] X[2,l] X[3,l]}[row{1}={font=\bfseries}]
  {Choose a project structure}{tab:manual-project-status}
```

Example 2.36: New, continuing, and ending project structures (continued)

```
Status & Background & Forward-looking content \\
N: new & Research rationale and preliminary work
  & Project plan, collaborations, and requested funding \\
C: continuing & Progress to date and current state of research
  & Project plan, collaborations, and requested funding \\
E: ending & Project history, report, and published results
  & Closing funding and staff from the ending period
\end{TableEnv}
```

Table 2: Choose a project structure

Status	Background	Forward-looking content
N: new	Research rationale and preliminary work	Project plan, collaborations, and requested funding

Continue on the following page ...

...continuing from previous page.

Status	Background	Forward-looking content
C: continuing	Progress to date and current state of research	Project plan, collaborations, and requested funding

Continue on the following page ...

...continuing from previous page.

Status	Background	Forward-looking content
E: ending	Project history, report, and published results	Closing funding and staff from the ending period

The starter files `03_project_A02.tex` and `03_project_A01.tex` demonstrate new and continuing research projects, respectively. Both have a summary, aims, work packages, a timeline, the relevance of sex/gender/diversity, the project's CRC role, differentiation from other grants, and project funding. The ending example `03_project_A00.tex` instead contains a report, three result categories, and closing funding information; it does not request a new period or present a new work programme.

For service projects, `03_project_Q01.tex` uses the continuing structure but focuses on services, access, capacity, and quality assurance. The training example `03_project_Q02.tex` and coordination example `03_project_Q03.tex` have dedicated content structures. Their headings, like ending-project headings, omit the research-area line. A `reference-only=true` project is a separate concept: it records a former project for references and is excluded from active counts. The manual's `A00` demonstrates this case.

2.4.2 Start and finish a project

\printProject{⟨*specification*⟩}

Use the unified project-boundary interface. Start a project with its id and finish the active project without repeating that id:

```
\printProject{{Head}{A01}}
\ldots
\printProject{{Tail}}
```

The `Head` form delegates to `\printProjectHead`; the `Tail` form delegates to `\printProjectTail` → P. 34. The original commands remain available as compatibility fallbacks.
See Example 2.37, Example 2.39.

\printProjectHead{⟨*project-id*⟩}

Start the section for one project. The command sets the current project, creates the project heading, resets project-local counters, starts a bibliography refsection, prints the project leaders, and prepares the project-specific thumbindex and headers. Active research and

service projects also receive the DFG research-areas heading from their `research-area` →P. 56 metadata; ending, MGK, and administrative projects do not. For every person id in the project's comma-separated `ref-person-id` →P. 62 value, all names stored in `name-alias` →P. 53 are registered as emphasized bibliography names. An empty alias value is ignored.
With `fastcompile-project` →P. 65, the current-project state, counters, labels, a compact running header containing the project ID, and the optional bibliography refsection are prepared. A light hook-based draft watermark is added to every shipped page; the calculated heading, project-leader cards, contract information, legal-issues table, thumb index, and table-of-contents entry are omitted.
See Example 2.38.

`\printProjectTail`

Finish the current project section. The command closes the bibliography refsection, resets bold bibliography names, restores the table-of-contents depth, and starts a new page. In `fastcompile-project` →P. 65, it only closes the optional publication refsection and does not force a page break.
See Example 2.38.

These commands manage complete pages and bibliography sections. The following code/result examples therefore show the beginning of the actual example proposal pages, rather than running a project wrapper inside the manual's own section. The complete proposal and inline examples share the same metadata. The omitted project body belongs between the two commands. Former project `A00` illustrates the ending-project structure; it remains excluded from active project counts and funding requests.

Example 2.37: New project: heading excerpt from the example proposal

```
% In project metadata: project-status = {N}
\printProject{{Head}{A02}}
% Summary, research rationale, plan, and funding.
\printProject{{Tail}}
```

3.1 General information about Project A02 N

3.1.1 Project title: Metabolic control of tissue-resident immunity

3.1.2 Research area: 2.21-05: Immunology

3.1.3 Project leaders

Kim Sample, Prof. Dr. rer. nat.

0000-0000-0000-0001
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
(Germany)

Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 100-1000
kim.sample@example.org

Samira Hartmann, Prof. Dr. rer. nat.

0000-0000-0000-0016
Centre for Computational Microbiology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)
+49 000 300-0006 samira@example.org

Example 2.38: Continuing project: heading excerpt from the example proposal

```
% In project metadata: project-status = {C}
\printProjectHead{A01}
% Summary, progress to date, plan, and funding.
\printProjectTail
```

Example 2.38: Continuing project: heading excerpt from the example proposal (continued)

3.1 General information about Project A01

3.1.1 Project title: Microbial signalling across epithelial barriers

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Kim Sample, Prof. Dr. rer. nat.

0000-0000-0000-0001
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City (Germany)

Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 100-1000
kim.sample@example.org

Alex Example, Prof. Dr. phil.

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Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-2000
alex.example@example.org

Example 2.39: Ending project: heading excerpt from the example proposal

```
% In project metadata: project-status = {E}
\printProject{{Head}{A00}}
% Project history, published results, closing funding.
\printProject{{Tail}}
```

3.1 General information about Project A00 (former project) E

3.1.1 Project title: Former project on microbial community dynamics

3.1.2 Project leaders

Kimmi Miller, Prof. Dr. rer. nat.

0000-0000-0000-0004
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
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kimmi.mueller@example.org

Daniel Okafor, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)

Centre for Clinical Translation
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0010 daniel@example.org

2.4.3 Publications for new and continuing projects

\projectPublicationsHeading

Print the standard project-publications subsection heading only when `fastcompile-project-show-publications` → P. 65 is enabled.
See Example 2.40.

\projectNocite{⟨*entry keys*⟩}

Add project publication entries with `\nocite` only when `fastcompile-project-show-publications` → P. 65 is enabled.
See Example 2.40.

\printProjectBibliography[⟨*biblatex options*⟩]

Print a project bibliography only when fastcompile-project-show-publications → P. 65 is enabled. Use these three project-specific commands together so disabling publication output also removes its heading and explicit bibliography work. Project bibliography entries are typeset in \footnotesize. With emphasizeEntries → P. 70 enabled, an entry carrying the current project's <project-id>-highlight keyword is printed in bold. A direct \printbibliography call bypasses this formatting, the font setting, and the publication-output switch.
See Example 2.40.

A local refsection keeps this example's bibliography independent of the manual. In an ordinary project, use the reference section established by the project wrapper. The leading options are passed to biblatex.

Example 2.40: Project publications with the standard heading

```
\begin{refsection}
\projectPublicationsHeading
\projectNocite{FAIR2016}
\printProjectBibliography[heading=none]
\end{refsection}
```

Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

2.4.4 Project plan: aims and work packages

For writing aims and workpackages we provide some special commands and features, including a gantt-chart.

\begin{**aims**}[⟨*options*⟩]
⟨*environment content*⟩
\end{**aims**}

Environment to be used in a project section to summarize in a condensed way the aims of the project. Use label={RDM Aims} to replace the default Aim prefix, keeping the item number and colon. The same text is used verbatim for singular and plural references through \cref, \Cref, and \getAim → P. 37, including references outside the list or before it. Compile twice after changing labels. Without this option, the defaults remain Aim and Aims. Other list options, such as start=3 and itemsep=2pt, are passed to enumitem.
Use resume to continue numbering from the previous aims list. Repeat the custom prefix, as in [label={RDM Aims},resume], to keep it in the numbered items and their cross-references. With [resume] alone, numbering continues but the prefix returns to the default Aim/Aims. Resuming uses the previous aims list regardless of its label text; custom labels do not create separate counters.
See Example 2.41.

\addAim{⟨*label*⟩}

Add an aim within the aims environment. You can refer to this aim by using \getAim → P. 37{⟨*label*⟩}.
See Example 2.41.

These aim and work-package examples use project A03 as their current project, as normally established by \printProjectHead.

Example 2.41: Aims and their references

```
\begin{aims}
  \addAim{manual-a} Identify pathways.
  \addAim{manual-b} Validate biomarkers.
  \addAim{manual-c} Test interventions.
\end{aims}
See \getAim{manual-a,manual-b,manual-c}.
\begin{aims}[label={RDM Aims}]
  \addAim{manual-rdm-a} Document research
    data.
  \addAim{manual-rdm-b} Publish reusable
    datasets.
\end{aims}
See \getAim{manual-rdm-a,manual-rdm-b}.
\begin{aims}[label={RDM Aims},resume]
  \addAim{manual-rdm-c} Archive the
    datasets.
\end{aims}
Continued as \getAim{manual-rdm-c}.
```

Aim 1: Identify pathways.

Aim 2: Validate biomarkers.

Aim 3: Test interventions.

See **Aims 1 to 3.**

RDM Aims 1: Document research data.

RDM Aims 2: Publish reusable datasets.

See **RDM Aims 1 and 2.**

RDM Aims 3: Archive the datasets.

Continued as **RDM Aims 3.**

\getAim{⟨*label*⟩}

Refer to an aim with the {⟨*label*⟩} in the `aims` → P. 36 environment. You can refer to multiple aims, too: `\getAim{A,B,C}`.
The default is referencing the aims of the same project. You can also refer to an aim of a different project: `\getAim`[⟨*project-id*⟩]{⟨*label*⟩}.
See Example 2.41, Example 2.44.

\begin{**workpackages**}
⟨*environment content*⟩
\end{**workpackages**}

Environment to be used in a project section when workpackages are written. This environment contains all descriptions of the workpackages that are introduced by `\addWP`.
See Example 2.42, Example 2.43.

\addWP{⟨*label*⟩}{⟨*key=values*⟩}

Add a workpackage with the {⟨*label*⟩} to the list. Further specifications are done using `name` → P. 64, `description` → P. 62 (text is used for the Gantt chart printed with `\printPlot` → P. 39), `periodStart` → P. 53, `periodEnd` → P. 53, `style` → P. 39. At the top level, the `name` → P. 64 value is printed as the bold work-package heading. Sub-workpackages do not print a `name` → P. 64; their authored prose starts directly after the list label. Immediate source whitespace, including a blank line after `\addWP`, is consumed without inserting another space.
See Example 2.42, Example 2.43.

Example 2.42: Work packages and their references

```
\begin{workpackages}
  \addWP{manual-wp1}{
    name={Identify pathways},
    description={Pathway discovery},
    periodStart=1, periodEnd=6,
    style={alternative}
  }
  Collect and analyse samples.
  \addWP{manual-wp2}{
    name={Validate biomarkers},
    description={Validation},
    periodStart=7, periodEnd=12,
    style={important}
  }
  Test independent cohorts.
\end{workpackages}
See \getWP{manual-wp1,manual-wp2}.
```

WP 1: Identify pathways Collect and analyse samples.

WP 2: Validate biomarkers Test independent cohorts.

See **WPs 1** and **2.**

Furthermore, PI, work-package, and aim references each have an independent color setting. They default to `black` and are independent of `color@reference@boldAuthor` → P. 68. Configure them with `\changeSettings`:

```
\changeSettings{
  color@reference@PI   = my-blue,
  color@reference@WP   = my-blue,
  color@reference@Aim = my-blue
}
```

\getWP{⟨*label*⟩}

Refer to a workpackage with the {⟨*label*⟩} in the `workpackages` → P. 37 environment. You can refer to multiple workpackages, too: `\getWP{wp2,wp3,wp4}`. Consecutive references under the same parent work package are compacted into a range. Its `to` connector uses the surrounding font weight while the `WPs` prefix and endpoint labels stay bold. A narrow, unbreakable space separates `WP` or `WPs` from its number. References under later parent work packages remain separate and are not discarded during compaction.
The default is referencing the workpackage of the same project. You can also refer to a workpackage of a different project: `\getWP[`⟨*project-id*⟩`]`{⟨*label*⟩}. Parent and nested work packages can be combined in one call, for example `\getWP{wp1a,wp1b,wp2,wp3a,wp3b}`. The parent `wp2` is retained alongside the nested items; sorting follows the hierarchy and ranges do not cross parent boundaries.
See Example 2.42, Example 2.43.

For this example the current project is `B04`. Parent and nested work packages can be referenced together; nested ranges stay within their parent.

Example 2.43: Nested work packages and mixed references

```
\begin{workpackages}
  \addWP{manual-nested-1}{name={Discovery}}
  \begin{workpackages}
    \addWP{manual-nested-1a}{} Recruit participants.
    \addWP{manual-nested-1b}{} Profile samples.
    \addWP{manual-nested-1c}{} Validate signals.
  \end{workpackages}
  \addWP{manual-nested-2}{name={Translation}}
  Test the selected intervention.
\end{workpackages}
See \getWP{manual-nested-1a,manual-nested-1b,manual-nested-1c,manual-nested-2}.
```

WP 1: Discovery
WP 1a: Recruit participants.
WP 1b: Profile samples.
WP 1c: Validate signals.
WP 2: Translation Test the selected intervention.
See **WPs 1a** to **1c** and **2**.

The next example is in `A03`. Its explicit `[B04]` argument points back to the work packages above, while `[A03]` selects the earlier aim example.

Example 2.44: References to another project's work packages

```
Exchange samples with
\getWP[B04]{manual-nested-1a,manual-nested
    -2}.\par
This supports \getAim[A03]{manual-a,manual
    -c}.
```

Exchange samples with **WPs 1a** and **2**.
This supports **Aims 1 and 3**.

While some keys have been described already, here is what is missing so far:

style=default|alternative|crosshatch|important|striped (initially: default)

Define the style of the bar of this workpackage in the gantt-chart.
For customizing the colors of the various styles have a look at: color@gantt@default → P. 69, color@gantt@alternative → P. 69, color@gantt@crosshatch → P. 69, color@gantt@important → P. 69, color@gantt@stripe-one → P. 69 and color@gantt@stripe-two → P. 69.
See Example 2.54, Example 2.55, and Example 2.56.

2.4.5 Timeline and collaboration plots

\printPlot[⟨*options*⟩]{⟨*plot specification*⟩}

Print a Gantt or collaboration plot. Use the lowercase type names gantt and collaborations in the braced specification. The second braced item selects a project, as in \printPlot{{gantt}{B06}} or \printPlot{{collaborations}{B06}}. Omit the project ID inside a project to use its current ID: \printPlot{{gantt}} or \printPlot{{collaborations}}. Without a current project, give the ID. Place plot options before the specification, for example \printPlot[arcs=true]{{collaborations}{B06}} or \printPlot[caption={Additional details}]{{gantt}}. Use the same leading caption option with an explicit ID, as in \printPlot[caption={Additional details}]{{gantt}{B06}}. Captions stay below plots; older trailing captions remain supported. For either plot type, an empty caption={} leaves the default caption intact and serves as a placeholder for additional project-specific text. The former type names Gantt and ProjectCollaboration and the singular lowercase collaboration remain accepted.
The collaboration arc diagram is the default. Select the heatmap with layout=heatmap. Heatmap columns use a shared dynamic box width based on the project area with the most projects, and inactive no-collaboration projects have white backgrounds. Project areas are read from the optional project-area field of each project entry; if it is absent, the first character of the project ID is used. Rows are created for every distinct area, ordered by area and project ID. The familiar A, B, and Q colors are retained; other areas use a rotating palette. Override individual colors with area-colors={C=orange,D=violet} or replace the fallback palette with area-palette={teal,brown,magenta} in the plot options. Use collaborationAreaColors → P. 67 and collaborationAreaPalette → P. 67 in \changeSettings → P. 8 to apply the same customization to every collaboration plot.
See Example 2.52, Example 2.47, Example 2.49. For bar colours and patterns, see Example 2.54 and Example 2.55.

Live plot variants For examples without an explicit ID, the current project is A03, as it would be after \printProjectHead{A03}. Specifying B04 overrides that context for that call. Each example starts with the default plot options.

Example 2.45: Collaborations of the current project

```
\printPlot{{collaborations}}
```

Example 2.45: Collaborations of the current project (continued)

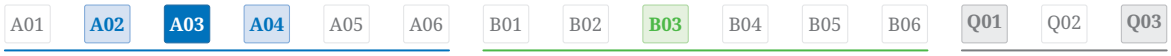


Fig. 2: Collaborations of **A03**.

Example 2.46: Explicit ID overrides the current project

```
\printPlot{{collaborations}{B04}}
```



Fig. 3: Collaborations of **B04**.

Example 2.47: Current-project connections as arcs

```
\printPlot[arcs=true]{{collaborations}}
```



Fig. 4: Collaborations of **A03**.

Example 2.48: Current-project heatmap

```
\printPlot[layout=heatmap]{{collaborations}}
```

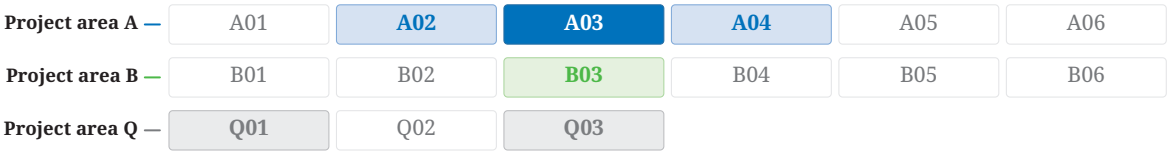


Fig. 5: Collaborations of **A03**.

Example 2.49: Explicit-project heatmap and caption

```
\printPlot[layout=heatmap,caption={Shared experimental platforms.}]
  {{collaborations}{B04}}
```

Example 2.49: Explicit-project heatmap and caption (continued)

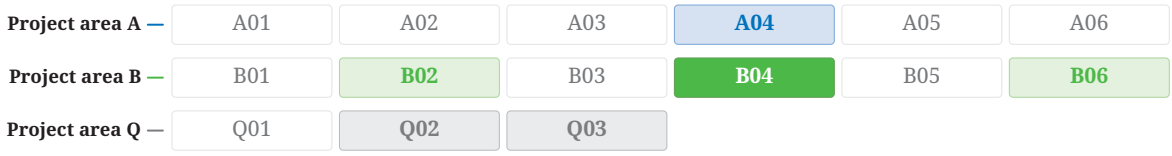


Fig. 6: Collaborations of **B04**. Shared experimental platforms.

Example 2.50: Custom project-area colours

```
\printPlot[layout=heatmap,area-colors={A=teal,B=orange,Q=violet}]
  {{collaborations}{A03}}
```

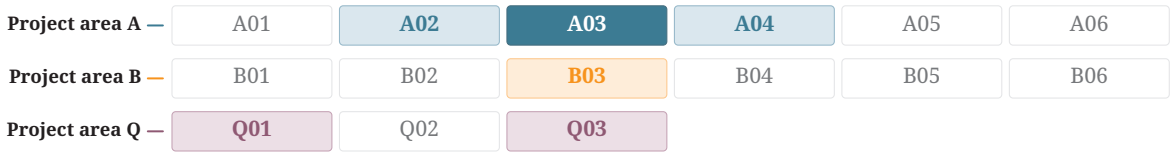


Fig. 7: Collaborations of **A03**.

Example 2.51: Layout aliases and additional caption

```
\printPlot[arc-diagram,caption={Partners of the imaging service.}]
  {{collaborations}{Q01}}
```



Fig. 8: Collaborations of **Q01**. Partners of the imaging service.

Example 2.52: Gantt chart of the current project

```
\printPlot[caption={Discovery followed by validation.}]{{gantt}}
```

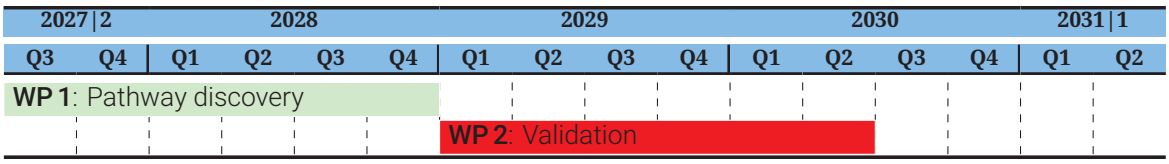


Fig. 9: Gantt chart of **A03**. Discovery followed by validation.

Example 2.53: Gantt chart with explicit project ID

```
\printPlot{{gantt}{A03}}
```

Example 2.53: Gantt chart with explicit project ID (continued)

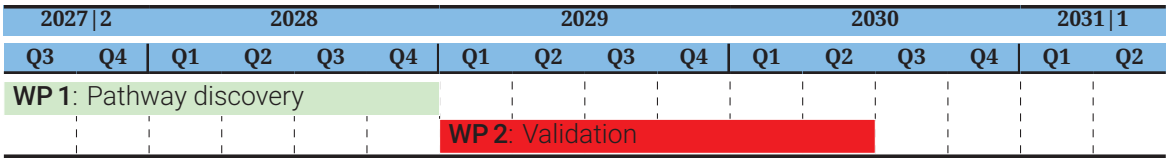


Fig. 10: Gantt chart of **A03**.

For older sources, `Gantt` means `gantt`; `ProjectCollaboration` and `collaboration` mean `collaborations`. These aliases render the same results and accept the same leading options.

Gantt bar styles and colours Choose a bar's appearance with the `style` → P. 39 key of `\addWP` → P. 37. The three solid fills are `default`, `alternative`, and `important`; `crosshatch` adds a hatch pattern, and `striped` alternates two colours. The following example defines one work package of each style in `B06`. Its five rows are reused in the palette examples below, so the colours can be compared without changing the work plan.

Example 2.54: Five Gantt bar styles

```
\begin{workpackages}
  \addWP{manual-colour-default}{
    name={Baseline}, description={Default},
    periodStart=1, periodEnd=6, style={default}
  }
  \addWP{manual-colour-alternative}{
    name={Parallel analysis}, description={Alternative},
    periodStart=3, periodEnd=9, style={alternative}
  }
  \addWP{manual-colour-important}{
    name={Validation}, description={Important},
    periodStart=7, periodEnd=12, style={important}
  }
  \addWP{manual-colour-crosshatch}{
    name={Integration}, description={Crosshatch},
    periodStart=9, periodEnd=14, style={crosshatch}
  }
  \addWP{manual-colour-striped}{
    name={Joint experiments}, description={Striped},
    periodStart=11, periodEnd=16, style={striped}
  }
\end{workpackages}
\printPlot[caption={One row for each built-in bar style.}]{gantt}{B06}
```

WP 1: Baseline

WP 2: Parallel analysis

WP 3: Validation

WP 4: Integration

WP 5: Joint experiments

Example 2.54: Five Gantt bar styles (continued)

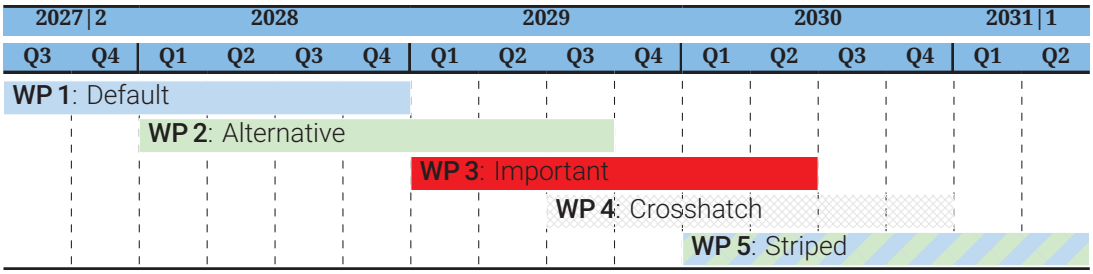


Fig. 11: Gantt chart of **B06**. One row for each built-in bar style.

For a document-wide palette, put `\changeSettings` in `preamble/dfg-customization.tex`. These colour settings are applied at `\begin{document}`. They accept colour names and xcolor mixes such as `teal!45`. The `frame` colour controls the outline of the bars, while `crosshatch` controls the hatch strokes and `stripe-one`/`stripe-two` control the two alternating stripe colours.

Document-wide Gantt palette: place in the customization file

```
\changeSettings{
  color@gantt@default     = {teal!45},
  color@gantt@alternative = {orange!50},
  color@gantt@important   = {violet!70},
  color@gantt@crosshatch  = {black!25},
  color@gantt@stripe-one  = {teal!20},
  color@gantt@stripe-two  = {teal!80},
  color@gantt@frame       = {black!50}
}
```

To recolour just one chart after the document has started, use `\colorlet` on the live colour names inside a group. The next example also defines an RGB colour; the surrounding group restores the previous palette afterwards. The colour names are set before `\printPlot` → P. 39; they are not plot options.

Example 2.55: Custom colours for one Gantt chart

```
\begingroup
\definecolor{ganttOcean}{RGB}{0,114,178}
\colorlet{color@gantt@default}{ganttOcean!45}
\colorlet{color@gantt@alternative}{teal!50}
\colorlet{color@gantt@important}{orange!90}
\colorlet{color@gantt@crosshatch}{violet!35}
\colorlet{color@gantt@stripe-one}{ganttOcean!25}
\colorlet{color@gantt@stripe-two}{ganttOcean!55}
\colorlet{color@gantt@frame}{black!50}
\printPlot[caption={Custom fills, coloured hatching, and two-tone stripes.}]
  {{gantt}{B06}}
\endgroup
```

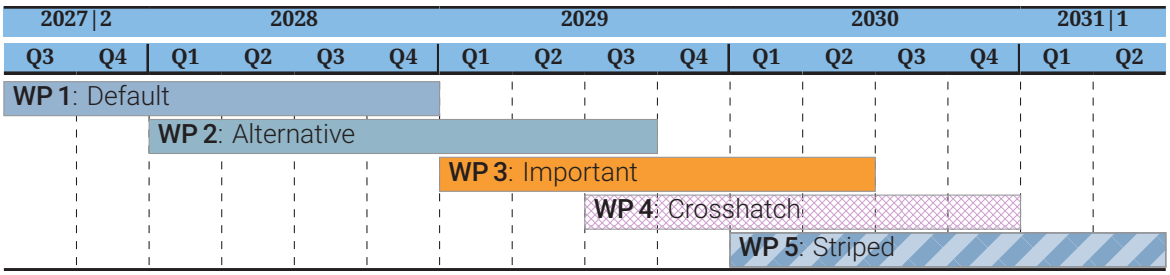


Fig. 12: Gantt chart of **B06**. Custom fills, coloured hatching, and two-tone stripes.

Example 2.55: Custom colours for one Gantt chart (continued)

For grayscale output, use different gray levels and retain the hatch and stripe patterns. The following chart uses the same work packages and dates; its black outlines make the bar boundaries explicit.

Example 2.56: Grayscale Gantt bars with patterns

```
\begingroup
\colorlet{color@gantt@default}{black!20}
\colorlet{color@gantt@alternative}{black!35}
\colorlet{color@gantt@important}{black!50}
\colorlet{color@gantt@crosshatch}{black!25}
\colorlet{color@gantt@stripe-one}{white}
\colorlet{color@gantt@stripe-two}{black!35}
\colorlet{color@gantt@frame}{black}
\printPlot[caption={Grayscale fills with hatching and stripes.}]{{gantt}{B06}}
\endgroup
```

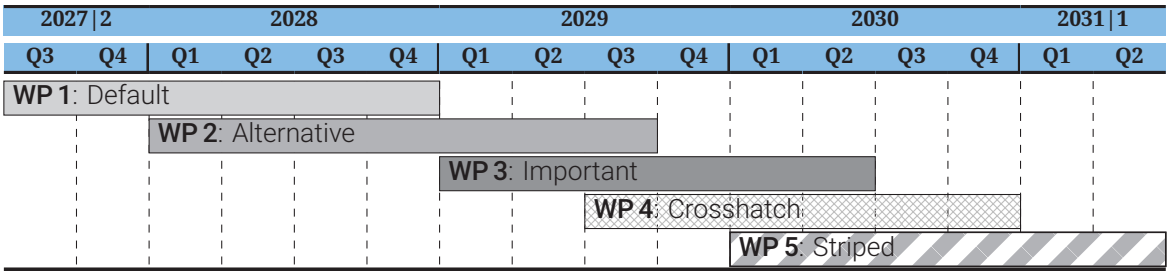


Fig. 13: Gantt chart of **B06**. Grayscale fills with hatching and stripes.

2.4.6 Individual project funding

The example below adds a funding request for `A06`, whose project metadata are already loaded. It covers all five funding groups: `staff`, `costs`, `fellowships`, `globalfunds`, and `instrumentation`. The doctoral position at 65% uses the configured annual salary; the other groups specify full EUR amounts directly, including a one-off equipment purchase. The year headings and partial-year factors come from `meta.fundingYears`. The earlier `A02` funding examples use separate records.

Example 2.57: Requested funding for a project

```
\changeSettings{salary@PhD=83700}
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A06},
  funding-type   = {staff},
  category       = {PhD},
  percentage     = {65},
  quantity       = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  }
}
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A06},
  funding-type   = {costs},
```

Example 2.57: Requested funding for a project (continued)

```
  category       = {consumables},
  sum            = {
    year1 = {1000},
    year2 = {2000},
    year3 = {2000},
    year4 = {2000},
    year5 = {1000}
  }
}
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A06},
  funding-type   = {fellowships},
  category       = {Short-term research fellowships},
  sum            = {
    year1 = {3000},
    year2 = {6000},
    year3 = {6000},
    year4 = {6000},
    year5 = {3000}
  }
}
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A06},
  funding-type   = {globalfunds},
  category       = {Training and networking},
  sum            = {
    year1 = {1500},
    year2 = {3000},
    year3 = {3000},
    year4 = {3000},
    year5 = {1500}
  }
}
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A06},
  funding-type   = {instrumentation},
  category       = {Imaging workstation},
  sum            = {
    year1 = {20000},
    year2 = {0},
    year3 = {0},
    year4 = {0},
    year5 = {0}
  }
}
\printTable{{RequestedFunding}{A06}}
```

Funding for	2027|2		2028		2029		2030		2031|1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Fellowships		Sum		Sum		Sum		Sum		Sum
Short-term research fellowships		3,000		6,000		6,000		6,000		3,000
Total		3,000		6,000		6,000		6,000		3,000
Globalfunds		Sum		Sum		Sum		Sum		Sum
Training and networking		1,500		3,000		3,000		3,000		1,500
Total		1,500		3,000		3,000		3,000		1,500
Instrumentation		Sum		Sum		Sum		Sum		Sum
Imaging workstation		20,000		0		0		0		0
Total		20,000		0		0		0		0
Total		52,702		65,405		65,405		65,405		32,702

Note: Figures in EUR

The project staff roster describes who does the work. It uses `staff` records in `RequestedFundingStaff`; monetary requests remain separate `funding` records. Existing staff may inherit their name and institution through `ref-person-id`. Use `lastname={TBA}` for an unnamed requested position. The `description` text is printed as a numbered job description below the table.

Example 2.58: Project staff: an existing investigator and a doctoral vacancy

```
\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id    = {B03},
  funding-status    = {existing},
  staff-type        = {research},
  ref-person-id     = {per11},
  field-of-research = {Immunology},
  commitment        = {8},
  funding-source    = {State funds},
  description       = {Supervises the experiments and data analysis.}
}
\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id     = {B03},
  funding-status     = {requested},
  staff-type         = {research},
  lastname           = {TBA},
  positionText       = {Doctoral researcher (65\%)},
  field-of-research  = {Immunology},
  ref-institution-id = {ins1},
  category           = {PhD},
  description        = {Performs longitudinal experiments and prepares the thesis.}
}
\printTable{{RequestedFundingStaff}{B03}}
```

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Existing staff							
Research staff	1.	**Sato**, Mira, Prof. Dr. rer. nat. (Professor of Systems Medicine)	Immunology	CCT	8		State funds

Continue on the following page ...

...continuing from previous page.

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Requested staff							
Research staff	2.	TBA (Doctoral researcher (65%))	Immunology	IMSM		Doctoral researcher	

Institutions: CCT: Centre for Clinical Translation (WRC); **IMSM**: Institute of Microbial Systems Medicine (EU). **Affiliations: WRC**: Westhaven Research Centre; **EU**: Example University.

Job descriptions of staff for the proposed funding period (supported through existing funds):
1. **Prof. Dr. rer. nat. Mira Sato** Supervises the experiments and data analysis.

Job descriptions of staff for the proposed funding period (requested funds):
2. **TBA** Performs longitudinal experiments and prepares the thesis.

Existing direct-cost funding is declared separately from new requests. The following records identify the participating affiliations and funding sources; the `sum` values are full EUR amounts for each funding-year column.

Example 2.59: Existing direct-cost funding from two sources

```
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {B03},
  ref-affiliation-id = {aff1},
  financed-by        = {University core funds},
  sum                = {
    year1 = {1000}, year2 = {2000}, year3 = {2000}, year4 = {2000}, year5 = {1000}
  }
}
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {B03},
  ref-affiliation-id = {aff2},
  financed-by        = {Partner university},
  sum                = {
    year1 = {500}, year2 = {1000}, year3 = {1000}, year4 = {1000}, year5 = {500}
  }
}
\printTable{{RequestedAndExistingFundingDirectCosts}{B03}}
```

	2027|2	2028	2029	2030	2031|1
EU: existing funds from University core funds	1,000	2,000	2,000	2,000	1,000
PU: existing funds from Partner university	500	1,000	1,000	1,000	500
Sum of existing funds	1,500	3,000	3,000	3,000	1,500
Sum of requested funds	0	0	0	0	0

Note: Figures in EUR

`\begin{`**`FundingDirectCostsTable`**`}[`⟨*heading*⟩`]`
⟨*environment content*⟩
`\end{`**`FundingDirectCostsTable`**`}`

Typeset a three-column breakdown of direct costs. Its optional heading and the table stay together when there is insufficient space left on a page. The currency column uses its natural width so that codes such as EUR fit in the selected font; the description column fills the remaining width.
See Example 2.60.

Example 2.60: Explain individual direct-cost items

```
\begin{FundingDirectCostsTable}[
    Consumables]
  Assay reagents & EUR & 1200 \\
  Culture media & EUR & 800 \\
\end{FundingDirectCostsTable}
```

Consumables

Assay reagents	EUR	1,200
Culture media	EUR	800

2.4.7 Report and results of an ending project

\printProjectResults{⟨*key-value list*⟩}

Print the three result categories required for an ending project. The keys `quality-assured`, `other`, and `patents` each accept a comma-separated list of bibliography entry keys. Use the literal value `none` when a category produced no results. An omitted or empty key retains a drafting instruction instead, so an unfinished list is not accidentally reported as empty.
Every non-empty category is placed in a separate biblatex reference segment within the reference section opened by `\printProjectHead`. Consequently, category membership is local to the current project. The command also places any pending floats before the result lists.
The combined result-key list is written to the project's auxiliary file. On the next normal LaTeX pass, a `projectReport` bibliography uses this list to omit report citations that are printed later as project results. This follows the usual biblatex compilation cycle and does not require the entry keys to be duplicated in the report.
See Example 2.61.

\begin{**projectReport**}
⟨*environment content*⟩
\end{**projectReport**}

Isolate all citations in the project report in their own reference segment and print that segment's bibliography at the end of the environment. Any entries assigned to `\printProjectResults` for the same project are omitted, because they are printed in the later result lists. Place the environment immediately after the report subsection heading and label.
See Example 2.61.

Example 2.61: Published results of an ending project

```
\begin{refsection}
\begin{projectReport}
  Follow the FAIR principles \cite{FAIR2016}
  and document software design \cite{Hoare1973}.
\end{projectReport}
\printProjectResults{
  quality-assured={FAIR2016},
  other=none,
  patents=none
}
\end{refsection}
```

Follow the FAIR principles Wilkinson et al. 2016 and document software design Hoare 1973.

References

Hoare, C. A. R. (1973). "Hints on programming language design". In: *Computer Systems Reliability*. Ed. by C. Bunyan. State of the Art Report 20, pp. 193–216. (Visited on 2018).

Publications with scientific quality assurance

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

Example 2.61: Published results of an ending project (continued)

Other publications and published results

None.

Patents

None.

For `project-status=E`, the staff table reuses `\printTable{{RequestedFundingStaff}{project-id}}`. Rows with `funding-status=requested` describe staff funded with approved grant money in the ending period, rather than a request for a new period. The complete B02 example shows this closing table together with its input records.

2.5 Bylaws and declarations

The bylaws are authored prose in `content/04_bylaws-of-the-crc.tex`. Use the applicant-wording helper introduced with the title page for singular or plural university names. The declaration file `content/05_declarations.tex` then prints the required signatures from the same person metadata.

`\printSignature``[`⟨*database*⟩`]{`⟨*id-description-pairs*⟩`}`

Print one signature block for each `{`⟨*id*⟩`}{`⟨*description*⟩`}` pair. The ⟨*database*⟩ defaults to `person`. Inside ⟨*description*⟩, `\theAffiliationOfRoleID` contains the person's affiliation id and can be used for further database lookups.
See Example 2.62.

`\printSignaturesWithRole``[`⟨*database*⟩`]{`⟨*role*⟩`}{`⟨*description*⟩`}`

Print a signature block for every current-period person with ⟨*role*⟩. This is used for the rector, chancellor, or president of every applicant university, so a proposal with several applicants does not require a hard-coded list of person ids. Inside ⟨*description*⟩, `\theAffiliationOfRoleID` refers to the affiliation of the current signatory.
See Example 2.63.

Example 2.62: Signature blocks

```
\printSignature{
  {spokesperson}{(Spokesperson of \thisCRC)}
  {rector}{(Rector of Example University)}
}
```

Example City
<date>

Prof. Dr. med. Morgan Sample
(Spokesperson of CRC 9999)

Example City
<date>

Prof. Dr. Robin Rector
(Rector of Example University)

The role-based form resolves the affiliation for each signatory; it avoids repeating names or maintaining a separate list of rectors in the content.

Example 2.63: Select signatories by role

```
\printSignaturesWithRole{rector}{%
  Rector of
  \get[affiliation]{\theAffiliationOfRoleID}{name}}
```

Example City
<date>

Prof. Dr. Casey President
Rector of Partner University

Example City
<date>

Prof. Dr. Robin Rector
Rector of Example University

2.6 Back matter: compact project overview

The closing overview reuses the project and PI records already used in general information. No separate database is needed.

Example 2.64: Compact project overview for the closing pages

```
\printTable{projectOverview}
```



Project	Title	Project leader(s)
A01	Microbial signalling across epithelial barriers	Kim **Sample** Alex **Example**
A02 N	Metabolic control of tissue-resident immunity	Kim **Sample** Samira **Hartmann**
A03	Microbiome-derived metabolites and immune memory	Samira **Hartmann** Noah **Lindberg**
A04 N	Spatial niches of host–microbe interactions	Noah **Lindberg** Alexis-Marie **Winters-Scholz**
A05	Dietary control of microbial community dynamics	Alexis-Marie **Winters-Scholz** Leonie **Navarro**
A06 N	Modelling microbial resilience and recovery	Leonie **Navarro** Daniel **Okafor**
B01	Host defence during microbiome perturbation	Hansi **Meier** Alex **Example**
B02 N	Molecular signatures of tissue regeneration	Kimmi **Miller** Daniel **Okafor**
B03	Immune circuits linking intestine and liver	Hansi **Meier** Mira **Sato**
B04 N	Cross-organ consequences of inflammatory stress	Kimmi **Miller** Mira **Sato**
B05	Patient-derived models of metabolic disease	Friedhelm **Redata** Robin **Keller**
B06 N	Translating microbial biomarkers into interventions	Daniel **Okafor** Robin **Keller**
Q01	Integrated imaging and analysis services	Alex **Example** Noah **Lindberg**
Q02	Integrated Research Training Group	Friedhelm **Redata** Leonie **Navarro**
Q03	Central coordination and data stewardship	Hansi **Meier** Robin **Keller**

3 Metadata and configuration reference

This chapter collects the detailed field definitions, class settings, CV and bibliography conventions, and maintenance notes. Read it as a reference while working through the preceding proposal examples.

3.1 Metadata, databases and keysets

This reference lists the accepted fields for each keyset. The walkthrough introduces the records at their point of use; consult this section when adding fields or choosing a non-default value. The class stores records with `datatool`[→CTAN].

Databases We have these databases that are used and picked up by several commands by default:

meta All the metainformation about the proposal

person Everything that is related to a person

project Anything that is related to a project

institution Info about an institution, that is linked to:

affiliation storing info about the affiliation

There are also more databases created on the fly but mainly used for project specific purposes. The examples below use one coherent sample dataset: `MM` and `KM` are persons, `A01` is a project, `IAIS` is an institution, and `RWTH` is the related affiliation.

3.1.1 Keyset person

Example keyset to describe a person-entity.

```
\addEntry[→P.2]{person}{
  id[→P.63]                    = {MM},
  firstname[→P.64]             = {Max S.},
  lastname[→P.64]              = {Doe},
  academicdegree[→P.64]        = {Prof. Dr. rer. nat.},
  gender[→P.64]                = {m},
  ref-institution-id[→P.64]    = {IAIS},
  ref-project-id[→P.64]        = {A01},
  role[→P.64]                  = {PI,spokesperson},
  position[→P.64]              = {prof-W3},
  positionText[→P.64]          = {Professor of Microbial Systems},
  period[→P.64]                = {1},
  name-alias[→P.53]            = {{Max Doe},{M. S. Doe}},
  orcid[→P.53]                 = {0000-0002-3555-7218},
  email[→P.52]                 = {max.doe@example.edu},
  phone[→P.52]                 = {+49 241 80 12345},
  url[→P.52]                   = {https://example.edu/mm},
  doctorateYear[→P.53]         = {1995},
  doctorate[→P.53]             = {Dr. rer. nat.},
  financed-by[→P.52]           = {RWTH},
  topic[→P.52]                 = {Host--microbiome interactions},
  category[→P.63]              = {prof},
  dateDefense[→P.53]           = {1995},
  periodStart[→P.53]           = {2025},
  periodEnd[→P.53]             = {2029},
  contract-information[→P.52] = {
    C1[→P.52]  = {yes},
    C2[→P.52]  = {no},
    C3[→P.52]  = {no},
    C4[→P.52]  = {yes},
  }
}
```

There are several keys you can and should use regarding a person. They are described further done as an example on how to use `\addEntry` → P. 2 etc.

email=⟨*string*⟩

Provide the email-address of the person.

url=⟨*url*⟩

URL of the persons academic/professional homepage (e.g. institute).

phone=⟨*string*⟩

Provide the phone number of the person.

financed-by=⟨*string*⟩

Describe the source of funding for the person's e.g. position etc.

topic=⟨*string*⟩

Especially for the early career researchers describe their topic of research.

category=`groupLeader|prof|postDoc|medDoc|PhD|research|nonResearch|student|assistant`

`category` → P. 63 is used in various databases. Mostly it is used to state the category of the person, but it is also used then for the early career researchers.

contract-information=⟨*key-values*⟩

Provide information about the person's contractual situation. The ⟨*key-value*⟩ is further specified with `C1`, `C2`, `C2a`, `C2b`, `C3`, `C3a`, `C4`, `C4a`. The generated wording for the yes/no keys `C1`–`C4` is singular/plural sensitive. It is adjusted automatically to match the number of project leaders and institutions represented in the proposal.

C1=`yes|no`

This key collects the answer to *Is the employment of the project leader(s) at the institution(s) indicated contractually secured for the duration of the proposed funding period?*

C2=`yes|no`

This key collects the answer to *Do any of the above mentioned persons hold fixed-term positions?*

C2a=⟨*YYYY-MM-DD*⟩

If key `C2` has been answered with ⟨*yes*⟩, this question *End date of fixed-term contract:*⟨*YYYY-MM-DD*⟩ requires ⟨*YYYY-MM-DD*⟩ as answer, too.

C2b⟨*YYYY-MM-DD*⟩

If key `C2` has been answered with ⟨*yes*⟩, this question *Further employment is planned until* ⟨*YYYY-MM-DD*⟩ requires ⟨*YYYY-MM-DD*⟩ as answer, too.

C3=`yes|no`

This key collects the answer to *Will one or more of the above-mentioned persons retire during the proposed funding period?*.

C3a=⟨*YYYY-MM-DD*⟩

If key `C3` has been answered with ⟨*yes*⟩, this question *anticipated retirement date:* ⟨*YYYY-MM-DD*⟩ requires ⟨*YYYY-MM-DD*⟩ as answer, too.

C4=`yes|no`

This key collects the answer to *Funding for the position(s) of the project leader(s) at the institution(s) indicated is covered by core support (state funds or similar):*

C4a=⟨*string*⟩

If key `C4` has been answered with ⟨*no*⟩, this key provides type and source of funding.

Following keys are mainly used for the CV-section but are closely tight to the person:

orcid=⟨*0000-0000-0000-0000*⟩

The individual string with the ORCiD. This value is also required for `\getORCiD` →P. 71.

name-alias=⟨*string*⟩

More versions of the person's name. The various aliases for `name-alias` are also described under `\addboldnames` →P. 73.

doctorateYear=⟨*YYYY*⟩
doctorate=⟨*string*⟩
degree-programme=⟨*string*⟩

Providing information about the doctorate.

3.1.2 Database entries for a cv

The header of the CV section is (partly) based on the information you parse to the document using the `\addEntry` →P. 2 command with the keyset **person**: Keys like `name-alias`, `orcid`, `doctorate`, `degree-programme`, `url` →P. 52 are used especially for the CV-section.

Let's assume we have a person with the following information:

Partly person-entity dataset focussing on its CV-entry.

```
\addEntry→P. 2{person}{
  id→P. 63                =MM},
  firstname→P. 64         ={Max S.},
  lastname→P. 64          ={Doe},
  academicdegree→P. 64    ={Prof. Dr. rer. nat.},
  role→P. 64              ={PI},
  \ldots
  name-alias        ={{Maxi Doe},{Maximilian Doe}},
  orcid             ={0000-0002-3555-7218},
  doctorateYear     ={1995},
  doctorate         ={Dr. rer. nat.},
  degree-programme  ={Biology},
  url→P. 52                 ={https://example.edu/mm},
}
```

This set of information is unique to a person and therefore only needs to be parsed once.

In the cv we need more dedicated information about the person. This is done with a separate keyset so it will not clutter the person-entry too much. You can have multiple of the following entries below for a person. Most important stages should be indicated with `cv-type=mandatory`. If you state `cv-type=supplement` you only need to use the key `description`, same goes for `cv-type=activities` which will then be in a different section, and last `cv-type=supervision`. There are more sections with further information about the PI. All the information go to the database ⟨*cv*⟩.

ref-person-id=⟨*id*⟩

Use the ⟨*id*⟩ of the person to whom this entry belongs to.

description=⟨*string*⟩

Further description of the current entry.

cv-type=mandatory|supplement|activities|supervision|
distinction|other

Specifying the type of one CV entry with additional information. Each CV of a person needs to have a ⟨*mandatory*⟩ entry (as part of *Qualification and Career*).

periodStart={⟨*date*⟩} (starting date)
periodEnd={⟨*date*⟩} (ending date)
dateDefense={⟨*date*⟩} (date of the defense of the thesis)

Specifying the time span of an event. The ⟨*date*⟩ should be given in an IS8601-form.

Some information comes from the block described above, but for the *Stages* you need to describe them separately using the keys `ref-person-id`→P. 62, `cv-type`, `periodStart`, `periodEnd`, `name`→P. 64, `position`→P. 64, `positionText`→P. 64, `description`→P. 62, `ref-institution-id`→P. 64.

Example dataset-entry for a mandatory person-cv-entity.

```
\addEntry→P. 2{cv}{
  ref-person-id→P. 62       ={MM},
  cv-type→P. 53          ={mandatory},
  periodStart→P. 53         ={1995},
  periodEnd→P. 53           ={1998},
  position→P. 64             ={prof-W3},
  positionText→P. 64            ={Professor of Microbial Systems},
  description→P. 62         ={Established an independent research group on ↩
    host--microbiome interactions.},
  ref-institution-id→P. 64 ={IAIS},
}
```

For each stage on the career path repeat the `\addEntry{`⟨*cv*⟩`}{`⟨ *..., type=mandatory, ...*⟩`}`-block.

There are five more blocks to which a researcher can provide information. For each block and for each entry within one block you need to create an entry.

supplement: Supplementary Career Information

activities: Activities in the Research System

supervision: Supervision of Researchers in Early Career Phases

distinction: Academic Distinctions

other: Other Information

Here is a more concrete example

Example dataset-entry for a common CV-entity

```
\addEntry→P. 2{cv}{
  ref-person-id→P. 62       ={MM},
  cv-type→P. 53             ={supplement},
  description→P. 62         ={Member of the university research council since 2024.},
}
```

`\dfgCV``{`⟨*person-id*⟩`}`

Print the complete CV for one person. The command expects a dedicated `cv-`⟨*person-id*⟩ database and combines the header, mandatory career stages, optional CV sections, selected publications, and the data privacy statement.
See Example 3.1.

`\dfgCVsection``[`⟨*database*⟩`]`

Print the CVs for all PIs in the current funding period. The ⟨*database*⟩ defaults to `person`; persons are sorted by lastname and firstname before their CVs are printed with `\dfgCV`.
See Example 3.1.

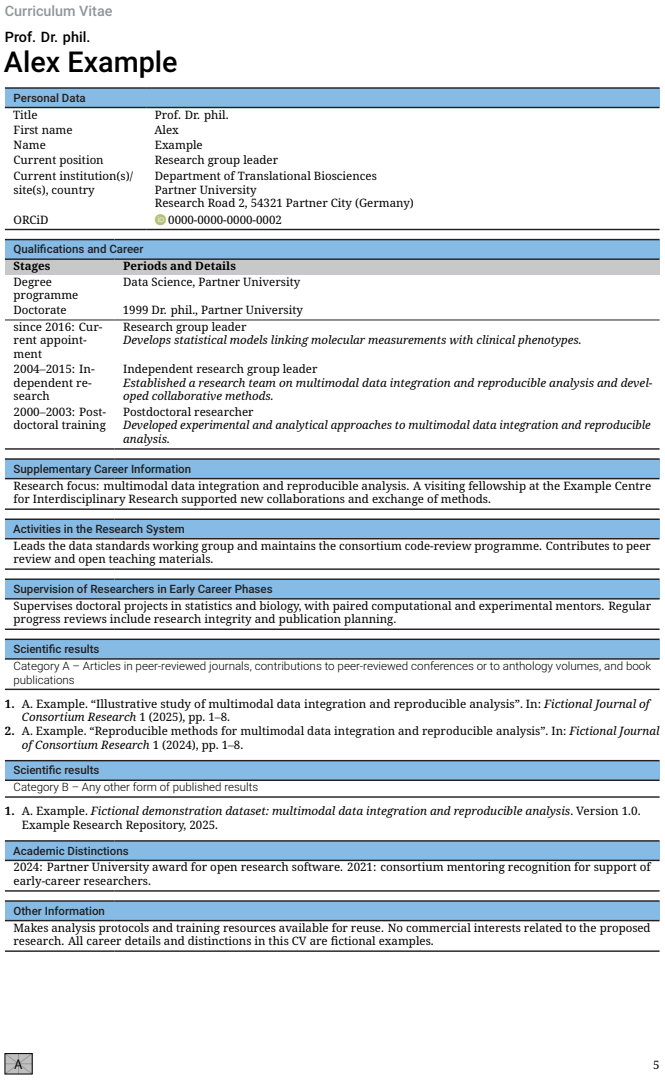
The starter CV loads person and career records from `metadata/dfg-metadata.tex` and `metadata/dfg-metadata-cv.tex`. The following example shows the first CV page produced by the complete appendix. Its first PI is `per2` (Alex Example); selecting just that person produces the same CV. The local proposal, CV appendix, and manual use the same twelve-PI consortium. All pages, including the privacy statement, follow in Chapter 5.

Example 3.1: Generate the CV appendix or select one person

```
\documentclass[english,cv,final=true]{dfg-crc}
\begin{document}
\dfgCVsection
% For one person, replace the line above with:
```

Example 3.1: Generate the CV appendix or select one person (continued)

```
% \dfgCV{per2}
\end{document}
```



Curriculum Vitae

Prof. Dr. phil.

Alex Example

Personal Data	
Title	Prof. Dr. phil.
First name	Alex
Name	Example
Current position	Research group leader
Current institution(s)/ site(s), country	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)
ORCiD	0000-0000-0000-0002

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Data Science, Partner University
Doctorate	1999 Dr. phil., Partner University
since 2016: Current appointment	Research group leader *Develops statistical models linking molecular measurements with clinical phenotypes.*
2004–2015: Independent research	Independent research group leader *Established a research team on multimodal data integration and reproducible analysis and developed collaborative methods.*
2000–2003: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to multimodal data integration and reproducible analysis.*

Supplementary Career Information

Research focus: multimodal data integration and reproducible analysis. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Leads the data standards working group and maintains the consortium code-review programme. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supervises doctoral projects in statistics and biology, with paired computational and experimental mentors. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. A. Example. “Illustrative study of multimodal data integration and reproducible analysis”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. A. Example. “Reproducible methods for multimodal data integration and reproducible analysis”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. A. Example. *Fictional demonstration dataset: multimodal data integration and reproducible analysis*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Partner University award for open research software. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

A

5

3.1.3 Keyset project

Following keys are required to describe a project.

Example dataset-entry to describe a project-entity.

```
\addEntry^{→P.2}{project}{
  id^{→P.63} = {A01},
  number^{→P.64} = {A01},
  name^{→P.64}   = {Microbial metabolism in barrier tissues},
  project-type^{→P.56}   = {research},
  project-status^{→P.56} = {N},
  project-area^{→P.56}   = {A},
  consider-for-lists^{→P.56} = {true},
```

```
    reference-only→P.56 = {false},
    research-area→P.56 = {1.11-01,1.11-03},
    ref-person-id→P.62 = {MM,KM},
    collaborations→P.57 = {A02,B01,Q01},
    legal-issues→P.57  = {
      L1→P.57  = {yes},
      L1a→P.57 = {yes},
      L2→P.57  = {no},
      L2a→P.57 = {},
      L3→P.57  = {yes},
      L4→P.57  = {no},
      L5→P.57  = {no},
      L5a→P.57 = {},
      L6→P.57  = {no},
      L7→P.57  = {no},
      L7a→P.57 = {},
    }
  }
```

Other required keys are `id`→P.63, `name`→P.64, `number`→P.64, `ref-person-id`→P.62.

Within an `\addEntry`→P.2`{project}{...}` entry, the `ref-person-id`→P.62 field is reserved exclusively for the person or persons who are PIs of that project. Every person listed in this field is interpreted as a project PI; no other person roles are inspected or considered. Persons associated with the project only in another role must therefore not be included in this field.

project-type=`research|service|mgk|admin`

Hereby you specify the type of the project. ⟨*research*⟩ stands for a classical research project, ⟨*service*⟩ is service project, ⟨*mgk*⟩ is an integrated graduate college, ⟨*admin*⟩ is an administrative project.

project-status=`N|E|C`

Define the status of the project, where ⟨*N*⟩ means a *new* project, ⟨*E*⟩ stands for *ending* and ⟨*C*⟩ for *continuing* project. The generated local proposal contains complete, separate source examples for all three statuses. Continuing projects use “Project progress to date”, new projects use “Research rationale”, and ending projects use the shorter “Project history”, “Published project results”, and “Funding” structure. In every case `\printProjectHead`→P.33 supplies the preceding general-information section from metadata.

project-area=⟨*area-id*⟩

Assign the project to a collaboration-plot area such as `A`, `B`, or `Q`. Any number of area identifiers is supported. If the key is omitted or empty, the first character of the project ID is used for compatibility with existing metadata (for example, `A01` belongs to area `A`).

consider-for-lists=`true|false`

Control whether the project participates in automatically generated project lists, project overview tables, research-area aggregates, and project counts. The default is `true`. Setting this key to `false` does not remove the database row, so the project remains available to explicit reference commands such as `\getProject`→P.6.

reference-only=`true|false`

Declare that the project exists only so historical or external project ids can be referenced. The default is `false`. A reference-only project is printed without a hyperlink because it has no project section that could provide a link target. In prose it receives the explicit “former project” indicator. Generated tables and lists instead append a compact superscript asterisk and add a local “Former project” legend when at least one such project occurs. This key is independent of `consider-for-lists`: an ending project with a real project section can be excluded from lists without becoming reference-only.

research-area=⟨*ticklist*⟩

Collect all research areas (according to the DFG list) in a comma-separated structure. Spaces around individual codes are ignored, e.g. `2.21-02, 2.22-31`.

collaborations=⟨*project-id-list*⟩

List the other CRC projects collaborating with this project. Collaborations are treated as undirected and duplicate declarations are collapsed, so the reverse relation does not need to be repeated. The project collaboration plot reads its edges from this field and its nodes from the same `project` database. Its default arc layout shows all active projects on one axis and highlights the focus project and its direct collaborators. Set `arcs=true` to draw the connecting arcs, or select the optional heatmap layout. Ending, reference-only, and `consider-for-lists=false` projects are omitted, as are edges to those intentionally excluded nodes. All arcs reach one common peak height, so their horizontal span does not imply collaboration strength or weight. Project identifiers are printed inside the nodes, and in the arc layout each project-area rule spans its complete contiguous group of nodes.

legal-issues=⟨*further key-values*⟩

Provide information about the project's legal issues. The ⟨*key-value*⟩ is further specified with `L1`, `L1a`, `L2`, `L2a`, `L3`, `L4`, `L5`, `L5a`. `L6`. `L7`. `L7a`.

L1=`yes|no`

This key records whether the project requires compliance with discipline-specific research ethics guidelines.

L1a=`yes|no`

If key `L1` has been answered with ⟨*yes*⟩, this statement records whether a statement from the responsible ethics committee is required and will be obtained before funding starts.

L2=`yes|no`

This key collects the answer to *clinical trials*

L2a=`yes|no`

If key `L2` has been answered with ⟨*yes*⟩, this statement *A copy of the studies' registration is included with the proposal* requires an answer, too.

L3=`yes|no`

This key collects the answer to *experiments involving vertebrates*

L4=`yes|no`

This key collects the answer to *experiments involving recombinant DNA*

L5=`yes|no`

This key collects the answer to *research involving human embryonic stem cells*

L5a=`yes|no`

If key `L5` has been answered with ⟨*yes*⟩, this statement *Legal authorization has been obtained* requires an answer, too.

L6=`yes|no`

This key collects the answer to *research concerning the Convention on Biological Diversity*

L7=`yes|no`

This key collects the answer to *investigations involving dual use research of concern*

L7a=`yes|no`

If key `L7` has been answered with ⟨*yes*⟩, this statement *A copy of the statement by the institution's research ethics committee is included with the proposal* requires an answer, too.

3.1.4 Keyset institution

An institution describes the organisational unit a person belongs to, for example an institute, department, centre, or clinic. It is linked to its parent affiliation through `ref-affiliation-id` →P. 64. The `abbreviation` →P. 58 is the compact, reader-facing label used in generated table cells; the internal `id` →P. 63 is never intended as display text. Full institution lists and legends continue to use `name` →P. 64. For compatibility, a missing `abbreviation` →P. 58 falls back to `name` →P. 64. The `name` →P. 64 key may be left empty when the affiliation itself is the relevant institution-level text; in that case institution-list commands start with the affiliation name.

Example dataset-entry to describe an institution-entity.

```
\addEntry →P. 2 {institution}{
  id →P. 63 = {IAIS},
  name →P. 64 = {Institute for Applied Immunology and Systems Biology},
  abbreviation = {IAIS},
  ref-affiliation-id →P. 64 = {RWTH},
}
```

Institution without a separate institution name.

```
\addEntry →P. 2 {institution}{
  id →P. 63 = {RWTH-main},
  name →P. 64 = {},
  abbreviation = {RWTH},
  ref-affiliation-id →P. 64 = {RWTH},
}
```

Person with multiple institutions.

```
\addEntry →P. 2 {person}{
  id →P. 63 = {MM},
  firstname →P. 64 = {Maria},
  lastname →P. 64 = {Sample},
  ref-institution-id →P. 64 = {IAIS,RWTH-main},
}
```

In standard institution lists, multiple institutions are printed as separate paragraphs with a small vertical gap. In `narrow` mode, they are joined inline with semicolons. The `abbreviation` option selects institution abbreviations (with a full-name fallback); generated compact tables enable this option automatically.

3.1.5 Keyset affiliation

Following keys are required to describe an affiliation.

Example dataset-entry to describe an affiliation-entity.

```
\addEntry →P. 2 {affiliation}{
  id →P. 63              = {RWTH},
  name →P. 64            = {Rheinisch-Westfälische Technische Hochschule Aachen},
  abbreviation    = {RWTH Aachen University},
  addressStreet   = {Templergraben 55},
  addressZIP →P. 59      = {52062},
  location →P. 59        = {Aachen},
  country →P. 59         = {Germany}
}
```

abbreviation=⟨*string*⟩

In addition to `name` →P. 64 provide an abbreviated version of this affiliation.

addressStreet=⟨*string*⟩

Specify the street of the affiliation.

addressZIP=⟨*string*⟩

Specify the ZIP code of the affiliation.

location=⟨*string*⟩

Specify the location (city) of the affiliation.

country=⟨*string*⟩

Specify the country the affiliation is located at.

3.1.6 Keyset funding

Each row represents one funding line. For a staff row, the class can derive the annual amount from the declared personnel category, percentage, and yearly quantity. For other funding types, annual values continue to be supplied with `sum` →P. 60.

Complete dataset for a funding entity. Not all keys are always required.

```
\addEntry →P. 2 [funding]{RequestedFunding}{
  id →P. 63                  = {A01-staff-prof},
  ref-project-id →P. 64      = {A01},
  ref-affiliation-id →P. 64  = {RWTH},
  financed-by →P. 52         = {DFG},
  funding-type       = {staff},
  percentage         = {100},
  category →P. 63            = {prof},
  quantity →P. 60            = {
    { year1 = {1} },
    { year2 = {1} },
    { year3 = {1} },
    { year4 = {1} },
    { year5 = {1} },
  },
  sum →P. 60 = {
    { year1 = {1000} },
    { year2 = {1000} },
    { year3 = {1000} },
    { year4 = {1000} },
    { year5 = {1000} },
  },
  year →P. 60                = {2026},
  applicant →P. 60           = {5000},
  participant →P. 60         = {2000},
  other-funds →P. 61         = {0},
  staff →P. 61               = {5000},
  costs →P. 61               = {1200},
  instrumentation →P. 61     = {0},
  fellowships →P. 61         = {0},
  globalfunds →P. 61         = {0}
}
```

Not all of the above mentioned keys are necessary for each funding table. We will provide examples for the various funding databases and their required keysets.

funding-type=`staff|costs|fellowships|globalfunds|instrumentation`

Define the type of this funding according to the predefined set. This is important since different features are activated depending on the value of this key.

percentage=⟨*number*⟩

This is only relevant when `funding-type` is set to `staff`. State the percentage of a full position, for example `100` for a full position or `65` for a 65% position. Integer and decimal values are accepted by the automatic calculation.

positionText=⟨*text*⟩

Optional replacement for the staff description in the requested-funding table. When supplied, it is printed instead of the translated `category` name. If omitted or empty, the category name is printed. The category still determines the salary calculation.

category=groupLeader|prof|prof-W2|prof-W3|postDoc|medDoc|PhD|research|nonResearch|student|assistant

This is only relevant when `funding-type` is set to `staff`. Describe the kind of the required person. Depending on this ⟨*category*⟩ the matching `salary@...` setting is used. The categories `prof`, `prof-W2`, and `prof-W3` all use `salary@prof`; the other category names correspond directly to their setting names below.

quantity=⟨*key*=*value*⟩

This is a key that enables further ⟨*key-values*⟩ with a predefined subset `year1` through `year5`. State how many positions of the `category` are required in each year. Decimal quantities are supported, for example `year1={0.5}` for half a position.

sum=⟨*key*=*value*⟩

This is a key that enables further ⟨*key-values*⟩ with a predefined subset for the individual years. State the sum for a person for a certain `category`. For `funding-type=staff` this key is optional: every omitted yearly sum is calculated from the salary setting for `category`, `percentage`, and that year's `quantity`. Because the five-year CRC period starts and ends with a half-year, automatically calculated amounts for `year1` and `year5` receive an additional factor of `0.5`; `year2` through `year4` use the complete annual rate. The result is rounded to the nearest euro. An explicit yearly sum remains an override, so existing declarations and exceptional personnel costs keep their stated value.

For every omitted staff sum, the class applies

$$\text{round}\left(\text{salary(category)} \times \frac{\text{percentage}}{100} \times \text{quantity(year)} \times \text{period-factor(year)}, 0\right).$$

Here, period-factor $= 0.5$ for `year1` and `year5` and 1 for `year2`–`year4`. Thus the two half-years together count as one complete salary year. The calculation is performed when the row is added and the result is stored in the normal `sum-year1` through `sum-year5` fields. Funding tables therefore use calculated and explicitly supplied amounts in exactly the same way. A missing salary setting, percentage, or valid quantity produces a warning instead of silently inventing an amount.

Staff funding with automatically calculated sums.

```
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A01},
  funding-type   = {staff},
  category       = {PhD},
  percentage     = {65},
  quantity       = {
    year1 = {1}, year2 = {1}, year3 = {1},
    year4 = {1}, year5 = {1}
  }
}
```

year=⟨*integer*⟩

State the year as integer. Actually only relevant in combination with `applicant`, `participant`, `other-funds`, `staff`, `costs`, `instrumentation`, `fellowships`, `globalfunds`.

applicant=⟨*integer*⟩

State the sum of direct costs for the applicant institution.

participant=⟨*integer*⟩

State the sum of direct costs for any other participating (i.e. not-applicant) institution.

other-funds=⟨*integer*⟩

State the sum of direct costs when neither the applicant nor a participating institution is covering the costs.

staff=⟨*integer*⟩

State the ⟨*sum*⟩ of staff for the previous funding phase. This is a fixed sum and therefore not necessary to calculate the sum like we do with `category`→P. 63 etc.

costs=⟨*integer*⟩

State the ⟨*sum*⟩ of direct costs for the previous funding phase. This is a fixed sum and therefore not necessary to calculate the sum like we do with `category`→P. 63 etc.

instrumentation=⟨*integer*⟩

State the ⟨*sum*⟩ of instrumentation for the previous funding phase. This is a fixed sum and therefore not necessary to calculate the sum like we do with `category`→P. 63 etc.

fellowships=⟨*integer*⟩

State the ⟨*sum*⟩ of staff for the previous funding phase. This is a fixed sum and therefore not necessary to calculate the sum like we do with `category`→P. 63 etc.

globalfunds=⟨*integer*⟩

State the ⟨*sum*⟩ of globalfunds for the previous funding phase. This is a fixed sum and therefore not necessary to calculate the sum like we do with `category`→P. 63 etc.

3.1.7 Keyset staff: requested-funding roster

The `staff` key family also supplies the project-level roster required by the DFG section “Requested funding for staff”. These records use the fixed database `RequestedFundingStaff` and are independent of the monetary `RequestedFunding` rows above. The public output call is `\printTable`→P. 10`{{RequestedFundingStaff}{A01}}`.

Complete staff-roster declaration.

```
\addEntry→P. 2[staff]{RequestedFundingStaff}{
  ref-project-id→P. 64      = {A01},
  funding-status      = {existing},
  staff-type          = {research},
  ref-person-id→P. 62       = {MM},
  field-of-research→P. 62   = {Microbiology},
  commitment→P. 63          = {8},
  funding-source→P. 62      = {State funds},
  description→P. 62         = {
    Leads the experimental work and supervises the research staff.
  }
}
```

ref-project-id=⟨*project-id*⟩

Required. Selects the project for which the roster row is printed. The value must match the project argument in the table call. For example, a row with `ref-project-id={A01}` is selected when `A01` is passed as the project argument to `\printTable`→P. 10.

funding-status=`existing|requested`
staff-type=`research|nonResearch`

These two required keys select one of the four official DFG groups. Empty subgroups are omitted and declaration order is retained within each group. In a project with `project-status=E`, `funding-status=requested` denotes staff funded with approved grant money in the ending funding period; the table changes that group label and its job-description heading automatically.

ref-person-id=⟨*string*⟩
firstname=⟨*string*⟩
lastname=⟨*string*⟩
academic-degree=⟨*string*⟩
positionText=⟨*string*⟩

A resolvable `ref-person-id` obtains the name, academic degree, position, category, and organisation from the central `person` fields `firstname`, `lastname`, `academicdegree`, `positionText`, `position`, `ref-institution-id`, and the compatibility fallback `ref-affiliation-id`. Any explicitly supplied roster field overrides the central value. An empty or omitted `academic-degree` is silently left out of both the roster and the job-description heading. For an unnamed vacancy, use `lastname={TBA}`; `firstname` may then also be omitted, and both the roster and the job-description lead-in print a clean "TBA" marker. Multiple unnamed vacancies are distinguished by their category, position, field of research, and institution when duplicate records are checked. Either a person reference or at least a last name is required. For existing staff resolved through `ref-person-id`, the lastname is printed in bold in the roster. The position is optional: omitting `positionText` inherits the central person's position, whereas an explicit `positionText={}` suppresses it. If the resulting position is empty, its line and parentheses are omitted without a missing-field warning. The former roster key `position` remains accepted as a compatibility alias for `positionText`.

field-of-research=⟨*string*⟩
funding-source=⟨*string*⟩
description=⟨*string*⟩

`field-of-research` supplies its table cell for every roster row. It may be empty or omitted for existing non-research staff; in that case the empty cell receives the `tableNoValue` background and no missing-field warning is issued. It remains an applicable, warning-checked value for the other three staff groups. `funding-source` applies only to existing staff; the corresponding cell for requested staff is marked as not applicable. `description` supplies the numbered text below the table and accepts ordinary paragraph material, citations, and other LaTeX commands. Each description is an item in the applicable existing- or requested-staff list. Its number is the flush-left item label; academic degree, first name, last name, and all continuation lines share a consistent hanging indent. The bold name and description are separated by a normal interword space. Missing applicable values are highlighted and reported in the log.

ref-institution-id=⟨*id-list*⟩

Resolve one or more comma-separated institution ids through the central `institution` database. Each institution's `abbreviation`→P. 58 is printed in the department/institution column; the full `name`→P. 64 is retained in the automatically generated legend together with its parent affiliation. Parent-affiliation abbreviations are checked and explained in first-use order. This key may be omitted when a resolvable `ref-person-id` already supplies `person.ref-institution-id`; a value on the staff row acts as an explicit override. When `ref-person-id` is empty or omitted, `ref-institution-id`→P. 64 must be supplied explicitly. The class warns at declaration time if both keys are empty, because the institution cannot be inherited and an existing staff member could not be assigned to an affiliation in `\OverviewExistingStaffTable`→P. 276.

department=⟨*string*⟩

Compatibility override for unstructured declarations. When present, this text is printed directly and takes precedence over institution lookup. New declarations should normally use `ref-institution-id`→P. 64.

ref-affiliation-id=⟨*affiliation-id*⟩

Compatibility fallback used only when neither `department` nor `ref-institution-id`→P. 64 is supplied. It prints the affiliation abbreviation and adds the affiliation to the legend. New declarations should use `ref-institution-id`→P. 64 so the organisational unit and its parent affiliation remain separate structured data. In particular, this compatibility key does not replace the required `ref-institution-id`→P. 64 on a row that has no `ref-person-id`.

commitment=⟨*number*⟩

Project commitment in hours per week for existing staff. The value may be a non-negative integer or decimal; no artificial upper limit is imposed. For requested staff the table cell is marked as not applicable.

category=prof|groupLeader|postDoc|medDoc|PhD|research|
nonResearch|student|assistant|prof-W2|prof-W3

Required for requested staff unless it can be inherited from `person.position` through `ref-person-id` →P.62. Only keys registered in `\g__dfg_dfg_category_names_prop` are accepted, and the table expands each key to the label stored in that property; for example, `postDoc` prints "Postdoctoral researcher". Existing staff inherit the category for the chapter-2 overview when possible, but may still supply an explicit override; their Category cell in the project-level roster is marked as not applicable.

The table traverses `RequestedFundingStaff` only once for the selected project, then assigns continuous sequence numbers in display order. The same numbers are reused below the table for the two job-description blocks. Missing values, unresolved references, and exact duplicate identities produce warnings and visible placeholders without stopping compilation.

Reuse in chapter 2: overview of existing staff The same declarations also populate the proposal-wide table in section 2.1.2; no separate totals database is needed. The call is `\printTable` →P.10`{OverviewExistingStaff}`. It selects only rows with `funding-status={existing}`, counts one person per distinct parent affiliation, and groups the values into the seven categories prescribed by the DFG form. An unnamed or otherwise unreferenced person therefore needs an explicit `ref-institution-id`; omitting both it and `ref-person-id` produces a warning when the roster row is declared. The mappings are `prof`, `prof-W2`, and `prof-W3` to Professors, `groupLeader` to Junior research group leaders, `postDoc` to Postdoctoral researchers, `PhD` to Doctoral researchers, `research` and `medDoc` to Other research staff, `nonResearch` to Non-research staff, and `student` and `assistant` to Student and graduate assistants.

The category column uses `X[3,l]`, reduced from `X[4,l]`; each affiliation column remains `X[1,c]`. The label reads "Junior research group" on its first line and "leaders" on its second line. This layout is maintained in the class source, so a proposal customization override is not needed after updating the generated class.

The proposal workspace contains a standalone layout test at `proposal-overleaf-com/__admin/scripts/test-overview-staff.tex`. It declares 85 complete fictional staff records across all seven categories and nine real affiliations in an isolated database, independently of project auxiliary files. The source lists the expected counts and build commands. Run LuaLaTeX twice from the proposal workspace; the PDF is written to `output/pdf/test-overview-staff.pdf`. The test PDF labels the counts as illustrative rather than proposal totals.

The applicant affiliation or affiliations are read as a comma-separated list from `meta.ref-affiliation-id` and printed first; other affiliations are obtained from each row's `ref-institution-id`. The group heading says enquoteapplicant university for one id and enquoteapplicant universities for several ids, spanning all corresponding columns. Multiple institutions belonging to the same affiliation count that person only once in that affiliation column. Since the project declarations normally occur after chapter 2, the class transfers the rows through the `.aux` file. The normal multi-pass `latexmk` build therefore fills the table automatically on the next pass.

3.1.8 Shared identifiers and CRC metadata

Following keys are available for the ⟨*person*⟩-keyset.

id=⟨*unique string*⟩

Any kind of a unique string that serves as an id, especially for commands like `\getEntry` →P.2. Almost all databases require this key.

firstname=⟨*string*⟩
lastname=⟨*string*⟩
academicdegree=⟨*string*⟩
gender=f|m|d
period=1|2|3 (multi choice!!)
role=PI|spokesperson|manager|steering-committee-member|rector
position=PhD|postDoc|groupLeader|research|nonResearch|student|prof|prof-W2|prof-W3
positionAt=⟨*string*⟩ (Specifying the position of the person, current or in the CV-section for the respective period.)

Personal and individual *required* information about a person.

ref-institution-id=⟨*string*⟩
ref-affiliation-id=⟨*string*⟩
ref-project-id=⟨*string*⟩

Referencing to an external database with its dedicated ⟨*id*⟩. **Note:** When you reference using `ref-institution-id`, there is no need to explicitly reference the affiliation with `ref-affiliation-id`.

The institution data is resolved through a two-step relationship. A person stores one or more institution ids in `ref-institution-id`; each institution stores the id of its parent affiliation in `ref-affiliation-id`. Commands such as `\printList` → P. 10 therefore follow this chain:

`person.ref-institution-id` ⟶ `institution.id` ⟶ `institution.ref-affiliation-id` ⟶ `affili`

The affiliation id usually does not need to be duplicated in the person entry.

For the database `meta` there are also dedicated keys. The database `meta` contains specific information on the metalevel about the CRC. For the `meta` row with `id=crc`, `ref-affiliation-id` accepts one or more comma-separated affiliation ids, for example `{aff1,aff2}`. These are the applicant or coordinating universities. Their names and locations are resolved in declaration order for the title page and related prose; applicant funding totals include every listed id. Besides `id` → P. 63, `ref-affiliation-id` there are these dedicated keys:

name=⟨*string*⟩ (applicable to other keysets, too.)
nameAddon=⟨*string*⟩
number=⟨*integer*⟩
fundingPeriodNumeral=1|2|3
fundedSince=⟨*YYYY*⟩
fundingYears=⟨*YYYY1,YYYY2,YYYY3,YYYY4,YYYY5*⟩
fundingYearsPrevious=⟨*YYYY1,YYYY2,YYYY3,YYYY4,YYYY5*⟩ (applicable to 2nd and 3rd phases)

Specify the core information of a CRC. The key `fundingPeriodNumeral` corresponds to `period`.

3.2 Adjustments and Customizations

3.2.1 Documentclass options

\documentclass{⟨*key=value*⟩}

Adjust on a metalevel various options (e.g. `showmarginalia` → P. 65, `fastcompile` → P. 65, `fastcompile-show-tags` → P. 65, `fastcompile-project` → P. 65, `fastcompile-project-show-images` → P. 65, `fastcompile-project-show-publications` → P. 65, `fastcompile-project-show-funding-tables` → P. 65, `showtags` → P. 65, `showtags-tables` → P. 66, `skip10pub` → P. 66, `cv` → P. 66, `final` → P. 66, `colorscheme` → P. 66) for the whole document. Not so much content wise but rather organisational.
See Example 3.1.

BCOR=⟨*dim*⟩ (initially `5mm`)

Reserve binding space on the inner edge of each page. The class passes this KOMA-Script option to geometry as its binding offset. With the default 30 mm nominal inner margin, `BCOR=5mm` gives 35 mm of inner clearance and a 160 mm text block; `BCOR=8mm` gives 38 mm and 157 mm.

showmarginalia=true|false (initially `false`.)

Show further information in the marginpar area, esp. total sum in funding tables.

fastcompile=true|false (initially `false`.)

Skip generated tables and lists, role-card grids, Gantt charts, and the CV publication block. Metadata and ordinary database-backed references remain available, making this the less aggressive compatibility mode.

fastcompile-show-tags=true|false (initially `false`.)

Show database provenance annotations during either fast-compilation mode. In focused project mode, annotations show the starting row and requested lookup path because metadata rows are skipped. This option has no effect outside fast compilation; final mode disables the annotations.

fastcompile-project=true|false (initially `false`.)

Compile authored project text with minimal calculated framing. This mode implies `fastcompile`, does not load metadata files unless `fastcompile-project-show-images` needs project rows, and executes only the three project-local funding entry types when `fastcompile-project-show-funding-tables` is enabled. It replaces the calculated project head with state-only setup, disables title/front-matter helpers when wrapped with `\dfgIfFastCompileProjectF`, and uses draft links. Graphics also default to draft placeholders unless `fastcompile-project-show-images` is enabled. Database references fall back to their IDs so prose remains readable. Apart from the optional project funding tables, generated tables and lists retain the same visible skipped-item notice as `fastcompile`.

fastcompile-project-show-images=true|false (initially `false`.)

Render images and generated plots encountered in the selected project while `fastcompile-project` is active. Project entries from the metadata file are retained for database-backed plots such as `\printPlot{{ProjectCollaboration}{B06}}`; other metadata entries, calculated headings, generated tables and the remaining fast-mode work stay disabled. With the default value, graphicx stays in draft mode, displays image placeholders, and `\printPlot` → P. 39 emits the usual skipped item notice.

fastcompile-project-show-publications=true|false (initially `false`.)

Show or suppress project publication headings, explicit project `\nocite` lists, project bibliographies, report references, and ending-project result lists while `fastcompile-project` is active. Ordinary and final builds always enable publication output. When this option is false in focused project mode, inline citation commands render compact bibliography-key markers without requesting biber.

fastcompile-project-show-funding-tables=true|false (initially `true`.)

Show or suppress `\RequestedFundingTable`, `\RequestedFundingStaffTable` → P. 305, and `\RequestedAndExistingFundingDirectCostsTable` while `fastcompile-project` is active. Focused mode processes only the corresponding project-local `RequestedFunding`, `RequestedFundingStaff`, and `RequestedAndExistingFundingDirectCosts` entries. Local staff-cost and total calculations remain available, including the per-year and full-period sums of existing direct-cost rows. Multiple rows accumulate in those local sums. Missing staff-table values use field-name placeholders such as `<firstname>` instead of “not provided” in fast compilation, since metadata needed for inherited values may have been skipped. Auxiliary-file restoration and proposal-wide overview caches stay disabled. This option has no effect on ordinary or final builds, where these tables are always rendered.

showtags=true|false (initially `false`.)

Annotate direct database values with their database, row identifier, and field, and calculated values with a concise calculation and source-database label. Successive labels alternate between two vertical levels to keep adjacent annotations legible. During fast compilation,

fastcompile-show-tags controls whether these annotations appear. In focused project mode, annotations identify the starting row and full requested field path because skipped metadata cannot supply destination row identifiers. Final mode disables annotations.

showtags-tables=true|false (initially true.)

Allow provenance tags inside generated table cells when showtags → P. 65 is enabled. Set this to false to retain tags in prose and other output while keeping tables uncluttered. It has no effect when showtags → P. 65 is disabled.

skip10pub=⟨*true*|*false*⟩ (initially false.)

Especially in the CV part, it might take longer to compile the reference list with the 10 most important publications. With this set to ⟨*true*⟩ the reference list will be skipped.

colorscheme=rgb|cmyk (initially cmyk.)

Set the colorscheme according to your needs. For a PDF that is meant for only being displayed on a screen, go with ⟨*rgb*⟩, for a printed version choose ⟨*cmyk*⟩.

cv=true|false (initially false.)

In the document which prints the CVs, set this option to ⟨*true*⟩, since it will show additional/needed text and modifications.

final=true|false (initially false.)

When your proposal has been finalised and is ready to be submitted, set this option to ⟨*true*⟩ for the final compilation. The capitalized alias Final is accepted for compatibility. This option has consequences for showmarginalia → P. 65, showtags → P. 65, fastcompile → P. 65, fastcompile-project → P. 65, skip10pub, fastcompile-project-show-publications → P. 65, colorscheme, and open-access markers in the bibliography. Open-access DOI API lookup is enabled only in final documents when doiAPIMail is non-empty and LuaSocket is available.

Example configuration for documentclass mit final-flag

```
\documentclass[
  final            =true,
% sets to false:
showmarginalia→P.65 =false,
  showtags→P.65        =false,
  fastcompile→P.65     =false,
  fastcompile-project→P.65=false,
  fastcompile-project-show-funding-tables→P.65=true,
  skip10pub       =false,
  fastcompile-project-show-publications→P.65=true,
  colorscheme     =cmyk
]{dfg-crc}
```

3.2.2 Change default settings

In the preamble/dfg-customization.tex you can overwrite the default settings of this documentclass.

Read these settings in text with \getValue → P. 9, for example \num{\getValue{animalHousing@mouse}}\,EU

The following keys are to be set or changed in \changeSettings → P. 8:

crcLogoPath=⟨*file path*⟩ (initially example-image-a)
affiliationLogoPath=⟨*file path*⟩ (initially example-image-b)

Select the image files used by \crcLogo and \affiliationLogo. As with \includegraphics, the file extension may be omitted.

doiAPIMail=⟨*email address*⟩ (initially empty)

Set the contact address required by the DOI API used for automatic open-access markers. Network lookup remains disabled unless this value is non-empty, the document uses final=true, and LuaLaTeX has socket support.

projectPIOncePerSection=true|false (initially true)

Show the PIs attached to a project only on that project's first shared \getProject→P.6 or \getProjectPI→P.7 reference in each section. Later references use the compact project form. Set this to false to keep \getProject→P.6 compact and repeat the PI names at every \getProjectPI→P.7 reference. References to the current project and references in generated tables and lists always remain compact.

PIReferenceFirstName=true|false (initially false)

Include the firstname in unstarred \getPI→P.5 references. The starred form always includes it. Set this through \changeSettings→P.8; it also affects person-first \getPIProject→P.7 and \getPIofProject→P.8. Project-first \getProjectPI→P.7 references keep lastnames unless the command is starred. Shared PI surnames are automatically disambiguated by the first given-name initial in unstarred references, regardless of this setting. If that initial is also shared, the full firstname is used. Use the starred command to request full firstnames explicitly.

projectsTableInstitutionAbbreviations=true|false (initially false)

Control the institution display in \ProjectsTable and \ProjectLeadersTable. With true, both tables use compact institution abbreviations; each projects table is followed by a tightly set legend containing only the abbreviations used in that table, sorted alphabetically by abbreviation. Reusing one abbreviation for different institution names raises a package warning. With false, full institution names are printed and no legend is added.

collaborationAreaColors=⟨*color list*⟩ (initially A=projectA, B=projectB, Q=projectQ)
collaborationAreaPalette=⟨*color list*⟩ (initially a rotating seven-color palette)

Set collaboration-area colors once for every collaboration plot. collaborationAreaColors is a key–value mapping such as {A=projectA, B=projectB, C=orange}. Areas absent from that mapping receive colors from collaborationAreaPalette in deterministic area order. The corresponding area-colors and area-palette plot options override these settings for one \printPlot→P.39 call.

salary@prof=⟨*euros per year*⟩ (initially 126600)
salary@groupLeader=⟨*euros per year*⟩ (initially 105000)
salary@postDoc=⟨*euros per year*⟩ (initially 90600)
salary@medDoc=⟨*euros per year*⟩ (initially 112200)
salary@PhD=⟨*euros per year*⟩ (initially 83700)
salary@research=⟨*euros per year*⟩ (initially 69000)
salary@nonResearch=⟨*euros per year*⟩ (initially 61500)
salary@student=⟨*euros per year*⟩ (initially 1)
salary@assistant=⟨*euros per year*⟩ (initially 1)

Set the annual salary rates used to calculate requested staff funding when a funding entry omits its explicit yearly sums. The professorial rate is shared by the prof, prof-W2, and prof-W3 categories.

animalHousing@mouse=⟨*euros per animal and week*⟩ (initially 0.75)
animalHousing@mouseGnotobiotic=⟨*euros per animal and week*⟩ (initially 3.9)
animalHousing@rat=⟨*euros per animal and week*⟩ (initially 1.4)

Set the weekly animal-housing rates used by the upkeep-of-laboratory-animals table. All salary and animal-housing settings accept integer or decimal numbers.

thumbindex=true|false (initially true)

Shows a thumbindex in the outer margin. This can be customized further.

Customize further all about the thumbindex. Modification done in dfg-customization.tex.

thumbindex@pattern=⟨*int*⟩ (initially 10)

Define how long the pattern of the thumbindex is until it starts again from the top of the page.

thumbindex@topmargin=⟨*dim*⟩|auto (initially auto)
thumbindex@bottommargin=⟨*dim*⟩|auto (initially auto)

With auto, the first tab starts at the top of the text block and the last tab ends at its bottom; the tabs divide the current \textheight into thumbindex@pattern equal slots. Set either key to a dimension to place that end of the tab stack the specified distance from the corresponding page edge. With the default auto values, the dimensions are recalculated for the current page layout, so changes to the text height also change the thumb-index spacing.

thumbindex@width=⟨*dim*⟩ (initially 1.2cm)

Define how far a thumb-index tab extends into the outer margin. The default 12 mm tab fits inside the 13 mm marginal column.

thumbindex@orientation=outwards|inwards (initially outwards)

Choose which way the thumb-index text faces on both left and right pages. With outwards, the tops of the letters point towards the outer page edge (the existing orientation). With inwards, they point towards the text block: the label is turned by 180 degrees. The tab's position, dimensions, and slot order stay the same. In both orientations the label is inset 2 mm from the tab center towards the text block on each page side. Set \changeSettings{thumbindex@orientation=inwards} in preamble/dfg-customization.tex to flip the labels.

thumbindex@font=⟨*ticklist*⟩ (initially \bfseries\Huge)

Define the ⟨*font*⟩ style of the text in the thumbindex.

color@thumbindex@bg=⟨*colorname*⟩ (initially black)
color@thumbindex@font=⟨*colorname*⟩ (initially white)

Define the color of the thumbindex: like the background color (⟨*bg*⟩) or the color of the ⟨*font*⟩

You can adjust any color in this template. In case you want to reuse a custom color of e.g. your corporate design, set this up first:

```
\definecolor{my-blue}{RGB}{0,84,159}
```

Afterwards you can use my-blue to any of the colors below.

color@reference@boldAuthor=⟨*colorname*⟩ (initially black)

Define the color of emphasized author names in bibliographies.

color@reference@PI=⟨*colorname*⟩ or current (initially black)

Define the color of references to principal investigators. Use current to inherit the surrounding text color at each reference instead of applying a fixed color. This applies to \getPI (including its starred and custom-text forms) and the project-reference commands that use it. Set this key in the preamble, for example:

```
\changeSettings{color@reference@PI=current}
% Or use a named xcolor color:
\changeSettings{color@reference@PI=blue}
```

color@reference@WP=⟨*colorname*⟩ (initially black)

Define the color of references to work packages.

color@reference@Aim=⟨*colorname*⟩ (initially black)

Define the color of references to aims.

color@table@headingsBG=⟨*colorname*⟩ (initially blue-50)

Define the background color of a table header (i.e. first row).

color@table@headingsBGalt=⟨*colorname*⟩ (initially blue-25)

Define an alternative background color of a table header (i.e. for subdividing row).

color@table@headingsFont=⟨*colorname*⟩ (initially black)

Define the color for the font in the header rows.

color@table@rows=⟨*colorname*⟩ (initially black-10)

Define the color for alternating rows.

color@table@lines=⟨*colorname*⟩ (initially black-10)

Define the color for inner table lines.

color@table@outerLines=⟨*colorname*⟩ (initially black)

Define the color for outer table lines and both the horizontal rules and dashed vertical grid lines in Gantt charts.

color@table@noValue=⟨*colorname*⟩ (initially black-10)

Define the color for cells that are supposed to contain no value.

color@table@firstColumn=⟨*colorname*⟩ (initially black-10)

Define the color for the first column. This is currently not supported.

color@table@firstColumn=⟨*colorname*⟩ (initially black)

Depending on the color of the table heading, we need to modify the color of the hyperlink since it can be the same as the colorlink from hyperref→CTAN.

color@table@firstColumn=⟨*colorname*⟩ (initially black)

Depending on the color of the table heading, we need to modify the color of the hyperlink since it can be the same as the colorlink from hyperref→CTAN.

color@tags@font=⟨*colorname*⟩ (initially red-100)
color@tags@line=⟨*colorname*⟩ (initially black-50)

Adjust the color of ⟨*font*⟩ or ⟨*line*⟩ when showmarginalia→P. 65 is activated.

color@item@i=⟨*colorname*⟩ (initially black)

Depending on the color of the table heading, we need to modify the color of the hyperlink since it can be the same as the colorlink from hyperref→CTAN.

color@font@alternative=⟨*colorname*⟩ (initially black-50)

Define the alternative color for headline fonts, especially in the CV section, cv→P. 66.

color@gantt@frame=⟨*colorname*⟩ (initially white)
color@gantt@default=⟨*colorname*⟩ (initially blue-25)
color@gantt@alternative=⟨*colorname*⟩ (initially green-25)
color@gantt@crosshatch=⟨*colorname*⟩ (initially black-10)
color@gantt@important=⟨*colorname*⟩ (initially red)
color@gantt@stripe-one=⟨*colorname*⟩ (initially blue-25)
color@gantt@stripe-two=⟨*colorname*⟩ (initially green-25)

Adjust the colors for the various styles of the gantt chart in the project section. The style striped is a alternating combination of color@gantt@stripe-one and color@gantt@stripe-two. Set these keys with \changeSettings→P. 8 in the customization file, before the document starts. For rendered examples and grouped overrides of an individual chart, see Example 2.54, Example 2.55, and Example 2.56.

biblatexstyleCV=⟨*style*⟩ (initially ext-numeric-comp)
biblatexstyleFeatureCV=⟨*clist*⟩ (initially sorting=none)
biblatexstyleProposal=⟨*style*⟩ (initially ext-authoryear-ecomp)
biblatexstyleFeatureProposal=⟨*clist*⟩ (initially empty)

Adjust the citation and bibliography style separately for standalone CVs and proposals. The feature values are comma-separated additional biblatex package options. Proposal features default to empty so changing styles does not pass unsupported options. Add style-specific options such as dashed=false explicitly when supported. Options shared by both modes, including backend=biber, autocite=inline, defernumbers=true, giveninits=true, the

citation-name limits, and `maxbibnames=99`, are supplied by the class. The class selects the CV pair when the document-class option `cv` → P. 66 is true and the proposal pair otherwise. The legacy keys `biblatexstyle` and `biblatexstyleFeature` remain available; each applies its value to both modes.

Configure CV and proposal bibliographies independently.

```
\changeSettings{
  biblatexstyleCV              = ext-numeric-comp,
  biblatexstyleFeatureCV       = {sorting=none},
  biblatexstyleProposal        = ext-authoryear-ecomp,
  biblatexstyleFeatureProposal = {},
}
```

Bibliography font and entry spacing The length `\bibitemsep` controls the extra vertical gap between bibliography entries, independently of the line spacing within an entry. A gap can already be present without a template override: the alphabetic styles inherit `\itemsep`, which is `4pt plus 2pt minus 1pt` in the default 10 pt setup. Other bibliography styles may supply different defaults.

To use footnote-sized text and remove this extra gap, add the following hook outside `\changeSettings` in `preamble/dfg-customization.tex`. The file is read before biblatex, so its lengths and font command must be changed after the package has loaded.

Compact bibliographies through the customization file.

```
\AddToHook{package/biblatex/after}{%
  \renewcommand*{\bibfont}{\normalfont\footnotesize}%
  \setlength{\bibitemsep}{0pt}%
}
```

This applies to ordinary `\printbibliography` calls and to `\printProjectBibliography` → P. 35 and `\printFrameBibliography` → P. 19. Replace `0pt` with `2pt`, for example, to keep a small extra gap. The CV publication environment `DFGlist` uses `nosep` already; to add space there, set `itemsep=2pt` after `nosep` in its list configuration. There is no `\changeSettings` key for entry spacing.

emphasizeAuthors=`true|false` (initially `true`.)

Emphasize the authors in a e.g. project section when they are authors of a referenced publication.

emphasizeEntries=`true|false` (initially `false`.)

Bold a complete entry in `\printProjectBibliography` when its `Keywords` field contains the current project ID followed by `-highlight`. For example, `B06-highlight` applies only to Project B06. In `\printFrameBibliography`, the keyword `frametext-highlight` selects complete entries instead. Each keyword applies only to its corresponding bibliography context. This setting is independent of `emphasizeAuthors`, which controls name highlighting. A `B06-project` keyword alone does not trigger bold type. To show only the keyword-selected full entries in bold, use `emphasizeAuthors=false` together with `emphasizeEntries=true`.

font=⟨*name*⟩ (initially `Noto Serif`)
sansfont=⟨*name*⟩ (initially `Roboto Light`)

Define the main font and the sansserif font.

fontspec=⟨*name*⟩ (initially `none`)
sansfontspec=⟨*name*⟩ (initially `ItalicFont=Roboto-LightItalic, BoldFont=Roboto-Medium, BoldItalicFont=Roboto-Medium-Italic`)

Customize font and sansserif font features.

3.2.3 Thumbindex

In your document you can use the following commands for displaying or switching off a thumbindex *locally*.

\printThumbIndex[⟨*color*⟩]{⟨*string*⟩}

Set the tab's ⟨*color*⟩ and displayed ⟨*string*⟩ for following pages. Place this command before a `\clearpage` if the new tab should appear on the next page.

\printThumbIndexOff

Remove the tab from following pages. This changes visibility only.

\resetThumbIndex

Reset the thumb-index slot counter. The next `\printThumbIndex` call starts at the first slot again. This changes only the position counter; use `\printThumbIndexOff` separately if the current tab should be hidden.

3.2.4 ORCiD

The ORCiD is foremost used in the CV section and stated in the entry of a person. So there is no need for a manual insertion of it. However you can do so using one of two following commands.

\ORCiD{⟨*number*⟩}

Display the ORCiD with the given ⟨*number*⟩. Adding the ORCiD logo in front.
See Example 3.2.

\getORCiD{⟨*id*⟩}

Display the ORCiD of the person with the given ⟨*id*⟩ If there is no `orcid`-entry in the database for this person, then a warning will be thrown and you can read about in the `.log`-file.
See Example 3.2.

Example 3.2: An ORCiD from metadata

```
\ORCiD{0000-0000-0000-0001}\par
\getORCiD{per1}\par
\getORCiDvalue{per1}
```

(0000-0000-0000-0001)
0000-0000-0000-0001

3.3 Bibliography/Publication

3.3.1 Equal contribution

In some cases it is very important to state that two or more authors have contributed equally to a publication. Take the following publication as example:

```
Schneider, K.M.*, C. Elfers*, A. Ghallab, C.V. Schneider,
E.J.C. Galvez, A. Mohs, W. Gui, L.S. Candels, T.H. Wirtz,
S. Zuehlke, M. Spiteller, M. Myllys, A. Roulet, A.
Ouzerdine, B. Lelouvier, K. Kilic, L. Liao, A. Nier, E.
Latz, I. Bergheim, C.A. Thaiss, J.G. Hengstler, T.
Strowig, and C. Trautwein. 2020. Intestinal Dysbiosis
Amplifies Acetaminophen-Induced Acute Liver Injury. Cell.
Mol. Gastroenterol. Hepatol. 11(4):909-933. *equal contribution
```

The first authors have `equal contribution`. In the bibtex-entry add *two* lines:

```
author+an       = {1=equal;2=equal},
addendum        = equal,
```

With `author+an` you annotate the authors, the numbers refer to the position of the author in the authors's list.

The complete entry of the mentioned example looks like:

```
@article{schneider2020a,
  author          = {Schneider, Kai Markus and Elfers, Carsten and Ghallab,
      Ahmed and Schneider, Carolin Victoria and Galvez, Eric J.C.
      and Mohs, Antje and Gui, Wenfang and Candels, Lena Susanna and
      Wirtz, Theresa Hildegard and Zuehlke, Sebastian and Spiteller,
      Michael and Myllys, Maiju and Roulet, Alain and Ouzerdine,
      Amirouche and Lelouvier, Benjamin and Kilic, Konrad and Liao,
      Lijun and Nier, Anika and Latz, Eicke and Bergheim, Ina and
      Thaiss, Christoph A. and Hengstler, Jan G. and Strowig, Till
      and Trautwein, Christian},
  author+an       = {1=equal;2=equal},
  date            = 2020,
  title           = {Intestinal Dysbiosis Amplifies Acetaminophen-Induced Acute
      Liver Injury},
  volume          = 11,
  pages           = {909--933},
  language        = {pt},
  journal         = {Cellular and Molecular Gastroenterology and Hepatology},
  doi             = {10.1016/j.jcmgh.2020.11.002},
  number          = 4,
  source          = {Crossref},
  url             = {https://doi.org/10.1016/j.jcmgh.2020.11.002},
  publisher       = {Elsevier BV},
  issn            = {2352-345X},
  date            = 2021,
  addendum        = equal,
}
```

3.3.2 Shared last authorship

Very similar to the `equal contribution`-part there is also a way to specify authors who `shared last authorship`.

In the bibtex-entry add *two* lines:

```
  author+an       = {1=sharedLast;2=sharedLast},
  addendum        = sharedLast,
```

Here is a complete example.

```
@Article{roessner2021a,
  Author          = {Roessner, P.M. and Cid, L.Llao and Lupar, E. and Roider,
      T. and Bordas, M. and Schifflers, C. and Arseni, L. and
      Gaupel, A.C. and Kilpert, F. and Krotschel, M. and
      Arnold, S.J. and Sellner, L. and Colomer, D. and
      Stilgenbauer, S. and Dietrich, S. and Lichter, P. and
      Izcue, A. and Seiffert, M.},
  Date            = {2021},
  author+an       = {17=sharedLast;18=sharedLast},
  Addendum        = sharedLast,
  Keywords        = {MM-A},
  Title           = {{EOMES} and {IL}-10 regulate antitumor activity of T
      regulatory type 1 {CD4(+)} T cells in chronic lymphocytic
      leukemia},
  Volume          = {35},
  Pages           = {2311--2324},
  Language        = {en},
  JournalTitle    = {Leukemia}
}
```

3.3.3 Emphasizing authors

You can have certain authors in bold letters in your bibliography section to highlight e.g. the core people of the proposal.

Notice: This feature is automatically triggered (unless switched off, see `emphasizeAuthors` → P. 70) for the CV section (see 3.1.2). We explain its function for additional use.

To add a person to the (sectioned) bibliography use `\addboldnames`.

`\addboldnames``{⟨{nn}, {nn}, {nn}⟩}`

Displaying authors in an emphasized way.

Notice that for each variation of a person's name you have to add one set of `{⟨nn⟩}`. For example you have an author with different spellings in your `.bib`-file, you have to list all of them to get them in bold letters (or adjust the entries in your `.bib`-file).

```
\addboldnames{
  {Max Doe},
  {M. Doe},
  {Doe, Max H.},
  {Doe, Max Heinrich}
}
```

After a (sectioned) bibliography and you do not want to display the before defined authors in bold letters anymore, you can use `\resetboldnames`:

`\resetboldnames`

Switch off showing the under `\addboldnames` defined names.

In case you do not need to have authors in bold letters you can disable the feature modification done in `dfg-customization.tex`.

3.3.4 Providing publication information

Providing the most relevant publications is done through the referenced `.bib`-file. Within a bibtex-entry you need to add a keyword with certain values to it preceeded with the ID of the person.

-A: Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications
-B: Category B – Any other form of published results

For example:

```
@article{...,
    keywords = {<ID>-A},
  }
```

To bold the complete bibliography entry only in Project B06, add its project-specific highlight keyword alongside any publication keywords:

```
@article{...,
  keywords = {pabst-A,B06-highlight},
}
```

This affects project bibliography lists, including project-report references and published-result lists. The same entry is not bolded in other projects unless their own `<project-id>-highlight` keyword is present. Set `emphasizeEntries=true` in `\changeSettings` to enable this feature. Only one highlight keyword per distinct project is needed, even when several PIs from that project coauthor the publication. For a publication to be bold in both B06 and Q04, use each keyword once:

```
@article{...,
  keywords = {pabst-A,B06-highlight,Q04-highlight},
}
```

The optional `B06-project` keyword can record project membership for other workflows, but it does not activate `emphasizeEntries` → P. 70. Existing CV-selection keywords such as `pabst-A` can stay in the same field.

For the frame-text bibliography in chapter 1, add `frametext-highlight` and use `\printFrameBibliography` → P. 19:

```
@article{...,
  keywords = {pabst-A,B06-highlight,frametext-highlight},
}
```

With `emphasizeEntries=true`, this entry is bold in both the B06 project bibliography and the frame-text bibliography. Neither keyword changes formatting in other bibliography contexts or in inline citations.

Of course several people can use the same reference for this section:

```
@article{...,
    keywords = {<ID1>-A, <ID2>-B, <ID3>-A},
  }
```

Open-access markers are handled by `biblatex-ext-oa`. If the API-based open-access request does not work for a publication, add the fallback annotation directly to the relevant `.bib` entry:

```
@article{...,
    options         = {openaccess=auto},
    doi+an:default  = {=openaccess},
  }
```

3.4 Automatic wording and list cardinality

Automatic singular, plural, and cardinality-dependent output The class derives the following visible wording and layout from the number of metadata items. These are the places where changing a one-item list to two or more items changes more than the retrieved data itself.

Project PIs `\printProjectHead` → P. 33 counts the comma-separated PI ids in the project's `ref-person-id` → P. 62 field. Exactly one id selects the singular branch; two or more ids select the plural branch. The field is required, so an empty list is invalid metadata; internally it falls back to the singular branch.

This PI count controls all of the following output:

• the heading "Project leader" or "Project leaders";

• the `\getPersonInformation` layout, which shows one PI in a single card and several PIs in a two-column raster of cards;

• contract question C1, which uses "project leader at the institution" or "project leaders at the institutions";

• contract question C2, which switches between "Does the above mentioned person hold a fixed-term position?" and "Do any of the above mentioned persons hold fixed-term positions?";

• contract question C3, which switches between "Will the above mentioned person" and "Will one or more of the above-mentioned persons";

• contract question C4, which again uses singular or plural "project leader(s)" and "institution(s)"; and

• contract answers, where PI surnames are appended in parentheses only when several PIs must be distinguished. Lists of dates or other follow-up values likewise append surnames when more than one matching PI contributes a value.

Research areas In `\printProjectHead` → P. 33, one value in the project's `research-area` → P. 56 field produces the heading "Research area"; two or more values produce "Research areas".

Applicant affiliations `\dfgApplicantAffiliationsTF` → P. 11 selects its first argument for one id in `meta.ref-affiliation-id` and its second argument for two or more ids. This helper controls:

• whether the title page prints an additional-applicant block;

• "applicant university" or "applicant universities" in the existing-direct-costs table;

• the Chapter 1 heading "Research profile of the applicant university/universities";

• "coordinating university/universities" in the Chapter 4 bylaws heading; and

• both the noun and verb in the Chapter 5 declaration: "The applicant university confirms" or "The applicant universities confirm".

On the title page, `\dfgOtherApplicantAffiliationsTF` applies the same rule after the first, coordinating affiliation has been removed: exactly two applicant affiliations in total produce "Other applicant university", while three or more produce "Other applicant universities".

Existing-staff overview `\OverviewExistingStaffTable` → P. 276 uses the number of applicant affiliations for its spanning group heading: one applicant id produces "applicant university", while several produce "applicant universities". Other affiliation columns do not affect this wording.

One-, two-, and longer-item lists Several formatters change punctuation rather than nouns. In prose, `\getPI` → P. 5, `\getProject` → P. 6, `\getProjectPI` → P. 7, and `\getPIProject` → P. 7 print one item without a separator, two items with "and", and three or more with commas plus a final "and". Project and PI references inside tables instead use compact slash separators. The complete applicant-name, applicant-location, and other-applicant lists exposed by `\theCRCaffiliation`, `\theCRCaffiliationLocation`, and `\theCRCOtherApplicantAffiliations` use the same natural-language rule. `\printList` → P. 10`{ResearchAreasFromProjects}` uses a comma for exactly two items and a final "and" for lists of three or more.

Fixed text written directly in project content is not altered automatically; authors must choose its singular or plural form themselves.

3.5 Project-area entries in the contents

`\dfgTOCprojectPart``{`⟨*letter*⟩`}[`⟨*label*⟩`]{`⟨*title*⟩`}`

Add a project-area heading to the table of contents. The ⟨*letter*⟩ identifies the project area, ⟨*label*⟩ defaults to `Project area`, and ⟨*title*⟩ is the text printed after the area label.

Wrap a project section.

```
\dfgTOCprojectPart{A}{Microbiota and metabolic pathways}
\printProjectHead{A01}
\ldots
\printProjectTail
```

3.6 Optional inputs and diagnostics

`\checkFile``{`⟨*file*⟩`}`

Input ⟨*file*⟩ if it exists. If the file is missing, compilation continues and a warning is written to the log. This is used for optional metadata, customization, and content files.

3.7 Troubleshooting

The class reports missing or inconsistent data in the terminal and the `.log` file. If your editor does not collect those messages, this command shows the class messages together:

Show DFG class messages in a log file

```
grep 'Package dfg' <filename>.log
```

`Unknown database name` The database has not been loaded, or the database name was misspelled. Check the relevant metadata file and whether it is input by the class.

Missing or empty key warnings The row exists, but the requested field is empty. This is intentional where the template needs to alert the user, for example for missing ORCiD values.

`Not allowed in LR mode` Vertical material such as `\vspace`, `\addvspace`, or a paragraph break was inserted in a table place that only accepts horizontal material. Prefer the documented table and list commands, or use options such as `narrow` for compact table output.

Stale output Regenerate the class with `make files`, and check whether the document uses the generated class from `local/`.

3.8 Complete shared metadata for the examples

The following records define the people, institutions, affiliations, and projects used throughout the live examples. They are extracted from the same DTX records loaded by this manual and generated into `local/metadata/dfg-metadata.tex` for the local proposal and CV. Matching career records are generated into `local/metadata/dfg-metadata-cv.tex`. The shared data are listed once here to avoid repeating the setup before every PI and project table. Adapt these declarations for your own consortium, then use the table and list commands from the walkthrough. The final command produces the project-leader table already shown in the general-information section.

Example 3.3: Shared input records for the proposal examples

```
\addEntry{meta}{
  id                   = {crc}, %<--- id (do not change)
  name                 = {Mechanisms of Host--Microbiome Communication},
  nameAddon            = {From Molecular Interactions to Organ Function},
  number               = {9999},
  fundingPeriodNumeral = {3}, % 1|2|3
  ref-affiliation-id   = {aff1}, % or {aff1,aff2} for two applicants
  fundedSince          = {2019-07-01},
  fundingYears         = {2027|2,2028,2029,2030,2031|1},
  fundingYearsPrevious = {2022|2,2023,2024,2025,2026|1}
}

\addEntry{institution}{
  id                 = {ins1},
  name               = {Institute of Microbial Systems Medicine},
  abbreviation       = {IMSM},
  ref-affiliation-id = {aff1}
}

\addEntry{institution}{
  id                 = {ins2},
  name               = {Department of Translational Biosciences},
  abbreviation       = {DTB},
  ref-affiliation-id = {aff2}
}

\addEntry{affiliation}{
  id            = {aff1},
  name          = {Example University},
  abbreviation  = {EU},
  addressStreet = {University Avenue 1},
```

Example 3.3: Shared input records for the proposal examples (continued)

```
  addressZIP    = {12345},
  location      = {Example City},
  country       = {Germany}
}

\addEntry{affiliation}{
  id            = {aff2},
  name          = {Partner University},
  abbreviation  = {PU},
  addressStreet = {Research Road 2},
  addressZIP    = {54321},
  location      = {Partner City},
  country       = {Germany}
}

\addEntry{person}{
  id                   = {per1},
  firstname            = {Kim},
  lastname             = {Sample},
  academicdegree       = {Prof. Dr. rer. nat.},
  period               = {1,2,3},
  role                 = {PI},
  gender               = {d},
  ref-institution-id   = {ins1,ins2},
  ref-project-id       = {A01,A02},
  name-alias           = {
    {Kim Sample},
    {Sample, Kim S.}
  },
  orcid                = {0000-0000-0000-0001},
  doctorateYear        = {2000},
  doctorate            = {Dr. rer. nat., Example University},
  degree-programme     = {Biology, Example University},
  position             = {prof-W3},
  positionText         = {Professor and Director},
  url                  = {https://www.example.org/kim-sample},
  phone                = {+49 000 100-1000},
  email                = {kim.sample@example.org},
  contract-information = {
    C1  = {no},
    C2  = {yes},
    C2a = {2028-12-31},
    C2b = {2029-01-03},
    C3  = {yes},
    C3a = {2054-03-10},
    C4  = {no},
    C4a = {DFG}
  }
}

\addEntry{person}{
  id                   = {per2},
  period               = {2,3},
  firstname            = {Alex},
  lastname             = {Example},
  academicdegree       = {Prof. Dr. phil.},
  position             = {groupLeader},
  positionText         = {Research group leader},
  role                 = {PI,steering-committee-member},
  doctorateYear        = {1999},
  doctorate            = {Dr. phil., Partner University},
  degree-programme     = {Data Science, Partner University},
  gender               = {m},
  ref-institution-id   = {ins2},
  ref-project-id       = {A01,B01,Q01},
  orcid                = {0000-0000-0000-0002},
  phone                = {+49 000 200-2000},
  email                = {alex.example@example.org},
  contract-information = {
    C1  = {no},
```

Example 3.3: Shared input records for the proposal examples (continued)

```
    C2  = {no},
    C2a = {},
    C2b = {},
    C3  = {no},
    C3a = {},
    C4  = {yes},
    C4a = {BMFTR}
  }
}

\addEntry{person}{
  id                   = {per3},
  period               = {2,3},
  firstname            = {Hansi},
  lastname             = {Meier},
  academicdegree       = {Dr. rer. nat.},
  position             = {postDoc},
  positionText         = {Senior research scientist},
  role                 = {PI,steering-committee-member},
  doctorateYear        = {1999},
  doctorate            = {Dr. rer. nat., Partner University},
  degree-programme     = {Microbiology, Partner University},
  gender               = {f},
  ref-institution-id   = {ins2},
  ref-project-id       = {B01,B03,Q03},
  orcid                = {0000-0000-0000-0003},
  phone                = {+49 000 200-3000},
  email                = {hansi.meier@example.org},
  contract-information = {
    C1  = {yes},
    C2  = {no},
    C2a = {},
    C2b = {},
    C3  = {no},
    C3a = {},
    C4  = {yes},
    C4a = {DFG}
  }
}

\addEntry{person}{
  id                   = {per4},
  period               = {2,3},
  firstname            = {Kimmi},
  lastname             = {Miller},
  academicdegree       = {Prof. Dr. rer. nat.},
  position             = {prof-W2},
  positionText         = {Professor of Molecular Medicine},
  role                 = {PI},
  doctorateYear        = {1999},
  doctorate            = {Dr. rer. nat., Partner University},
  degree-programme     = {Molecular Medicine, Partner University},
  gender               = {f},
  ref-institution-id   = {ins2},
  ref-project-id       = {B02,B04},
  orcid                = {0000-0000-0000-0004},
  phone                = {+49 000 200-4000},
  email                = {kimmi.mueller@example.org},
  contract-information = {
    C1  = {yes},
    C2  = {no},
    C2a = {},
    C2b = {},
    C3  = {no},
    C3a = {},
    C4  = {yes},
    C4a = {DFG}
  }
}
```

Example 3.3: Shared input records for the proposal examples (continued)

```
\addEntry{person}{
  id                   = {per5},
  period               = {3},
  firstname            = {Friedhelm},
  lastname             = {Redata},
  academicdegree       = {Prof. Dr.-Ing.},
  orcid                = {0000-0000-0000-0005},
  position             = {prof-W2},
  positionText         = {Professor of Research Training},
  role                 = {PI,steering-committee-member},
  doctorateYear        = {1998},
  doctorate            = {Dr.-Ing., Example University},
  degree-programme     = {Biomedical Engineering, Example University},
  gender               = {m},
  ref-institution-id   = {ins1},
  ref-project-id       = {B05,Q02},
  phone                = {+49 000 100-5000},
  email                = {friedhelm.redata@example.org},
  contract-information = {
    C1  = {yes},
    C2  = {yes},
    C2a = {2028-12-31},
    C2b = {2029-01-03},
    C3  = {yes},
    C3a = {2054-03-10},
    C4  = {no},
    C4a = {DFG}
  }
}

\addEntry{project}{
  id                 = {A01},
  project-area       = {A},
  number             = {A01},
  project-status     = {C},
  consider-for-lists = {true},
  reference-only     = {false},
  name               = {Microbial signalling across epithelial barriers},
  project-type       = {research},
  research-area      = {2.21-02},
  ref-person-id      = {per1,per2},
  collaborations     = {A02,A04,A06,B01,B03,B06,Q01},
  legal-issues       = {
      L1  = no,
      L1a = no,
      L2  = no,
      L2a = no,
      L3  = no,
      L4  = no,
      L5  = no,
      L5a = no,
      L6  = no,
      L7  = no,
      L7a = no
    }
}

\addEntry{project}{
  id                 = {A02},
  project-area       = {A},
  number             = {A02},
  project-status     = {N},
  consider-for-lists = {true},
  reference-only     = {false},
  project-type       = {research},
  name               = {Metabolic control of tissue-resident immunity},
  research-area      = {2.21-05},
  ref-person-id      = {per1,per6},
  collaborations     = {A01,A03,B02,Q02},
  legal-issues       = {
```

Example 3.3: Shared input records for the proposal examples (continued)

```
    L1  = no,
    L1a = no,
    L2  = yes,
    L2a = no,
    L3  = yes,
    L4  = yes,
    L5  = no,
    L5a = no,
    L6  = yes,
    L7  = no,
    L7a = yes
  }
}


\addEntry{project}{
  id                 = {B01},
  project-area       = {B},
  number             = {B01},
  project-status     = {C},
  consider-for-lists = {true},
  reference-only     = {false},
  project-type       = {research},
  research-area      = {2.21-03,2.21-05,2.22-15},
  name               = {Host defence during microbiome perturbation},
  ref-person-id      = {per3,per2},
  collaborations     = {A01,B02,B03,Q02},
  legal-issues       = {
    L1  = no,
    L1a = no,
    L2  = no,
    L2a = no,
    L3  = no,
    L4  = no,
    L5  = no,
    L5a = no,
    L6  = no,
    L7  = no,
    L7a = no
  }
}

\addEntry{project}{
  id                 = {B02},
  project-area       = {B},
  number             = {B02},
  project-status     = {N},
  consider-for-lists = {true},
  reference-only     = {false},
  project-type       = {research},
  research-area      = {2.22-15},
  name               = {Molecular signatures of tissue regeneration},
  ref-person-id      = {per4,per10},
  collaborations     = {A02,B01,B04,Q01},
  legal-issues       = {
      L1  = yes,
      L1a = no,
      L2  = yes,
      L2a = yes,
      L3  = yes,
      L4  = yes,
      L5  = yes,
      L5a = no,
      L6  = yes,
      L7  = yes,
      L7a = no
    }
}

\addEntry{project}{
```

Example 3.3: Shared input records for the proposal examples (continued)

```
  id                 = {Q01},
  project-area       = {Q},
  number             = {Q01},
  project-status     = {C},
  consider-for-lists = {true},
  reference-only     = {false},
  project-type       = {service},
  name               = {Integrated imaging and analysis services},
  research-area      = {2.21-02,2.22-15},
  ref-person-id      = {per2,per7},
  collaborations     = {A01,A03,A04,A06,B02,B03,B05},
  legal-issues       = {
    L1  = no,
    L1a = no,
    L2  = no,
    L2a = no,
    L3  = no,
    L4  = no,
    L5  = no,
    L5a = no,
    L6  = no,
    L7  = no,
    L7a = no
  }
}

\addEntry{project}{
  id                 = {Q02},
  project-area       = {Q},
  number             = {Q02},
  project-status     = {C},
  consider-for-lists = {true},
  reference-only     = {false},
  project-type       = {mgk},
  name               = {Integrated Research Training Group},
  research-area      = {},
  ref-person-id      = {per5,per9},
  collaborations     = {A02,A05,B01,B04,B06}
}

\addEntry{project}{
  id                 = {Q03},
  project-area       = {Q},
  number             = {Q03},
  project-status     = {C},
  consider-for-lists = {true},
  reference-only     = {false},
  project-type       = {admin},
  name               = {Central coordination and data stewardship},
  research-area      = {},
  ref-person-id      = {per3,per12},
  collaborations     = {A01,A02,A03,A04,A05,A06,B01,B02,B03,B04,B05,B06,Q01,Q02}
}




\addEntry{affiliation}{
  id                  = {aff3},
  name                = {Northbridge University},
  abbreviation        = {NU},
  addressStreet       = {Science Avenue 3},
  addressZIP          = {34567},
  location            = {Northbridge},
  country             = {Germany}
}

\addEntry{affiliation}{
  id                  = {aff4},
```

Example 3.3: Shared input records for the proposal examples (continued)

```
  name                = {Westhaven Research Centre},
  abbreviation        = {WRC},
  addressStreet       = {Science Avenue 3},
  addressZIP          = {34567},
  location            = {Westhaven},
  country             = {Germany}
}

\addEntry{institution}{
  id                  = {ins3},
  name                = {Centre for Computational Microbiology},
  abbreviation        = {CCM},
  ref-affiliation-id  = {aff3}
}

\addEntry{institution}{
  id                  = {ins4},
  name                = {Department of Experimental Immunology},
  abbreviation        = {DEI},
  ref-affiliation-id  = {aff3}
}

\addEntry{institution}{
  id                  = {ins5},
  name                = {Institute of Metabolic Medicine},
  abbreviation        = {IMM},
  ref-affiliation-id  = {aff4}
}

\addEntry{institution}{
  id                  = {ins6},
  name                = {Centre for Clinical Translation},
  abbreviation        = {CCT},
  ref-affiliation-id  = {aff4}
}

\addEntry{person}{
  id                  = {per6},
  orcid               = {0000-0000-0000-0016},
  firstname           = {Samira},
  lastname            = {Hartmann},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI,steering-committee-member},
  gender              = {f},
  ref-institution-id  = {ins3},
  ref-project-id      = {A02,A03},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0006},
  email               = {samira@example.org}
}

\addEntry{person}{
  id                  = {per7},
  firstname           = {Noah},
  lastname            = {Lindberg},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {m},
  ref-institution-id  = {ins3,ins4},
  ref-project-id      = {A03,A04,Q01},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
```

Example 3.3: Shared input records for the proposal examples (continued)

```
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0007},
  email               = {noah@example.org}
}

\addEntry{person}{
  id                  = {per8},
  firstname           = {Alexis-Marie},
  lastname            = {Winters-Scholz},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {d},
  ref-institution-id  = {ins4},
  ref-project-id      = {A04,A05},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0008},
  email               = {alexis-marie@example.org}
}

\addEntry{person}{
  id                  = {per9},
  firstname           = {Leonie},
  lastname            = {Navarro},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {f},
  ref-institution-id  = {ins5},
  ref-project-id      = {A05,A06,Q02},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0009},
  email               = {leonie@example.org}
}

\addEntry{person}{
  id                  = {per10},
  firstname           = {Daniel},
  lastname            = {Okafor},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {m},
  ref-institution-id  = {ins5,ins6},
  ref-project-id      = {A06,B02,B06},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0010},
  email               = {daniel@example.org}
}

\addEntry{person}{
  id                  = {per11},
  firstname           = {Mira},
  lastname            = {Sato},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {f},
```

Example 3.3: Shared input records for the proposal examples (continued)

```
  ref-institution-id  = {ins6},
  ref-project-id      = {B03,B04},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0011},
  email               = {mira@example.org}
}

\addEntry{person}{
  id                  = {per12},
  firstname           = {Robin},
  lastname            = {Keller},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {d},
  ref-institution-id  = {ins1,ins6},
  ref-project-id      = {B05,B06,Q03},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0012},
  email               = {robin@example.org}
}

\addEntry{project}{
  id                  = {A03},
  project-area        = {A},
  number              = {A03},
  project-status      = {C},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Microbiome-derived metabolites and immune memory},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per6,per7},
  collaborations      = {A02,A04,B03,Q01}
}

\addEntry{project}{
  id                  = {A04},
  project-area        = {A},
  number              = {A04},
  project-status      = {N},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Spatial niches of host--microbe interactions},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per7,per8},
  collaborations      = {A01,A03,A05,B04,Q01}
}

\addEntry{project}{
  id                  = {A05},
  project-area        = {A},
  number              = {A05},
  project-status      = {C},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Dietary control of microbial community dynamics},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per8,per9},
  collaborations      = {A04,A06,B05,Q02}
```

Example 3.3: Shared input records for the proposal examples (continued)

```
}

\addEntry{project}{
  id                  = {A06},
  project-area        = {A},
  number              = {A06},
  project-status      = {N},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Modelling microbial resilience and recovery},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per9,per10},
  collaborations      = {A01,A05,B06,Q01}
}

\addEntry{project}{
  id                  = {B03},
  project-area        = {B},
  number              = {B03},
  project-status      = {C},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Immune circuits linking intestine and liver},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per3,per11},
  collaborations      = {A01,A03,B01,B05,Q01}
}

\addEntry{project}{
  id                  = {B04},
  project-area        = {B},
  number              = {B04},
  project-status      = {N},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Cross-organ consequences of inflammatory stress},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per4,per11},
  collaborations      = {A04,B02,B06,Q02}
}

\addEntry{project}{
  id                  = {B05},
  project-area        = {B},
  number              = {B05},
  project-status      = {C},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Patient-derived models of metabolic disease},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per5,per12},
  collaborations      = {A05,B03,B06,Q01}
}

\addEntry{project}{
  id                  = {B06},
  project-area        = {B},
  number              = {B06},
  project-status      = {N},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Translating microbial biomarkers into interventions},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per10,per12},
  collaborations      = {A01,A06,B04,B05,Q02}
```

Example 3.3: Shared input records for the proposal examples (continued)

```
}


\printTable{ProjectLeaders}
```

3.9 Release workflow for maintainers

For a new release, update `\dfgVersion` and `\dfgVersionDate` at the top of `crc-proposal-documentation-and-code.dtx`, the release line in the README block, the changelog, and `dfg-crc.pkg`. Keep these values in one release commit.

Release and CTAN checks

```
make ctan-check
make check
% Optional online validation without submission:
make ctan-validate
```

The focused targets `make test-command-options` and `make test-participating-institutions` run 30 argument/option checks and 10 layout cases, respectively. The command-options fixture also checks 14 document-text cases: the generic date, early customization, later settings, direct redefinitions, deferred references, and local scope. Both targets are included in `make check`. The general regression fixture lives in `ci/test-template.tex`; its build copy is `local/test-template.tex`. The manual scratch file `local/test.tex` is neither generated nor compiled by these checks.

The first command builds the manual and the curated `output/ctan/dfg-crc.zip`, verifies its exact allow-list, extracts it in a clean temporary directory, runs the portable installer, and compiles a minimal document against the resulting class. The online target validates but does not submit the archive. Before committing, inspect `git status --short`, run `git diff --check`, and review the ZIP file list.

3.9.1 Maintaining executable examples

The driver uses `documentation=true` to keep the class typography and rendering commands while loading its fixtures explicitly. This option defaults to false; it suppresses automatic proposal customization, metadata, and bibliography inputs. It is intended for this manual, not for ordinary proposals. The fixtures and the manual driver are maintained in the DTX. A documentation build extracts them again, so the manual needs no private proposal files.

Add runnable examples with the manual's `dfgExample` environment. Its required argument is the title; add a stable `label={ex:manual-name}` option and refer to it from the relevant command description with `\dfgExampleRef{ex:manual-name}`. Add `dfg wide` to the options for full-width output below the code. Each example executes the displayed code in a local group. Floating output is set to non-floating inside the result panel. Use the fictional IDs `per1`, `per2`, `ins1`, `aff1`, and `A01` where possible. Use `dfg project=A03` on the example environment when an example needs a current-project context, and explain that context in the surrounding text. Keep data-only declarations, incomplete preamble fragments, full-document project wrappers, external commands, and BibTeX records in `dispListing`. The shared project and PI metadata are listed once with `\dfgTableSetup` in the reference chapter; live table examples and the generated local proposal use those same records. Maintain matching `cv-person-id` career records in the `meta-cv` block and fictional bibliography records with `person-id-A` or `person-id-B` keywords for every PI included in the example CV appendix. Project-boundary examples show labelled excerpts from the standalone proposal, because those commands change pages, headers, and bibliography scope. Running `make doc` checks that the PDFs contain all fifteen active project chapters, the ending-project example, and twelve PI CVs before compiling the manual. Inspect the PDF as well as the build log after changing a layout example.

4 Complete example proposal

The following pages show the complete fictional proposal generated from this DTX using the same metadata as the documentation examples: twelve PIs, six projects in area A, six in area B, and three in area Q. The title page, general information, financial overviews, all fifteen active project descriptions, the former project A00, bylaws, declarations, and back matter are included. The scientific text, people, figures, and funding values are illustrative.

The proposal is built with `final=true`, which suppresses draft tags and margin diagnostics. Each imported page is reduced to 82% and framed, with an “Example proposal” label above it. The proposal’s own page numbers, headers, and thumb index remain inside the frame; the manual page number appears below it, clearly separating the two documents. For the proposal’s own clickable references and bookmarks, open the accompanying standalone example proposal PDF.

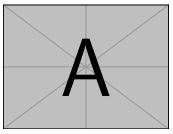
To regenerate the manual and both complete examples from the DTX, run `make doc` in the template repository. Use `make proposal-doc-example` to build only the standalone example. Both the local proposal and the final-mode documentation build use the same metadata and chapter inputs in `local/`. Its generated wrapper is:

```
\PassOptionsToClass{final=true}{dfg-crc}
\input{crc-proposal.tex}
```

Collaborative Research Centre 9999

Mechanisms of Host–Microbiome Communication

From Molecular Interactions to Organ Function



Example University

Funding Proposal

for the third Funding Period
2027 | 2–2028–2029–2030–2031 | 1



Proposal for the third Funding Period of
Collaborative Research Centre 9999

“Mechanisms of Host–Microbiome Communication: From Molecular Interactions to Organ Function”

funded since

2019-07-01

for

2027 | 2–2028–2029–2030–2031 | 1

Coordinating university:

Example University

Spokesperson of the Collaborative Research Centre:

Morgan Sample, Prof. Dr. med.
0000-0000-0000-0006
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City (Germany)
+49 000 100-6000
morgan.sample@example.org

Management or office of the Collaborative Research Centre:

Jamie Coordinator, Dr. rer. nat.
0000-0000-0000-0007
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-7000
jamie.coordinator@example.org

Example City
<date>

Prof. Dr. med. Morgan Sample
(Spokesperson of Collaborative Research Centre)

Example City
<date>

Prof. Dr. Casey President
(Rector, chancellor, or president of Partner University)

Example City
<date>

Prof. Dr. Robin Rector
(Rector, chancellor, or president of Example University)

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1 General information

1.1 Key data

1.1.1 Governing bodies of the Collaborative Research Centre

Steering Committee of the CRC 9999

Alex Example, Prof. Dr. phil.
0000-0000-0000-0002
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For the third funding period, the steering committee includes representatives of both research areas and the Integrated Research Training Group. The spokesperson and committee members are elected by the CRC general assembly in accordance with the bylaws. The composition is reviewed annually with regard to scientific breadth, career stage, and equitable participation.

1.1.2 Project leaders

Project leader	Gende	Doctorate obtained in	Home institution, location	Project(s)
Example, Alex Prof. Dr. phil.	m	1999	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	A01 B01 Q01
Hartmann, Samira Prof. Dr. rer. nat.	f	2010	Centre for Computational Microbiology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)	A02 A03
Keller, Robin Prof. Dr. rer. nat.	d	2010	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany) Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	B05 B06 Q03
Lindberg, Noah Prof. Dr. rer. nat.	m	2010	Centre for Computational Microbiology Northbridge University Science Avenue 3, 34567 Northbridge (Germany) Department of Experimental Immunology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)	A03 A04 Q01

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Project leader	Gende	Doctorate obtained in	Home institution, location	Project(s)
Meier, Hansi Dr. rer. nat.	f	1999	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	B01 B03 Q03
Miller, Kimmi Prof. Dr. rer. nat.	f	1999	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	B02 B04
Navarro, Leonie Prof. Dr. rer. nat.	f	2010	Institute of Metabolic Medicine Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	A05 A06 Q02
Okafor, Daniel Prof. Dr. rer. nat.	m	2010	Institute of Metabolic Medicine Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany) Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	A06 B02 B06
Redata, Friedhelm Prof. Dr.-Ing.	m	1998	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany)	B05 Q02
Sample, Kim Prof. Dr. rer. nat.	d	2000	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany) Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)	A01 A02
Sato, Mira Prof. Dr. rer. nat.	f	2010	Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)	B03 B04
Winters-Scholz, Alexis-Marie Prof. Dr. rer. nat.	d	2010	Department of Experimental Immunology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)	A04 A05

1.1.3 Participating institutions

Northbridge University (Northbridge)
Centre for Computational Microbiology
Department of Experimental Immunology

Westhaven Research Centre (Westhaven)
Institute of Metabolic Medicine
Centre for Clinical Translation

Example University (Example City)
Institute of Microbial Systems Medicine

Partner University (Partner City)
Department of Translational Biosciences

1.1.4 Project groups and projects

Project area A – Microbiota and metabolic pathways

Project		Title	Research area	Project leader(s), institute(s), location(s)
A01		Microbial signalling across epithelial barriers	2.21-02	Kim **Sample**, Institute of Microbial Systems Medicine, Example University; Department of Translational Biosciences, Partner University Alex **Example**, Department of Translational Biosciences, Partner University
A02	N	Metabolic control of tissue-resident immunity	2.21-05	Kim **Sample**, Institute of Microbial Systems Medicine, Example University; Department of Translational Biosciences, Partner University Samira **Hartmann**, Centre for Computational Microbiology, Northbridge University
A03		Microbiome-derived metabolites and immune memory	2.21-02	Samira **Hartmann**, Centre for Computational Microbiology, Northbridge University Noah **Lindberg**, Centre for Computational Microbiology, Northbridge University; Department of Experimental Immunology, Northbridge University
A04	N	Spatial niches of host–microbe interactions	2.21-02	Noah **Lindberg**, Centre for Computational Microbiology, Northbridge University; Department of Experimental Immunology, Northbridge University Alexis-Marie **Winters-Scholz**, Department of Experimental Immunology, Northbridge University
A05		Dietary control of microbial community dynamics	2.21-02	Alexis-Marie **Winters-Scholz**, Department of Experimental Immunology, Northbridge University Leonie **Navarro**, Institute of Metabolic Medicine, Westhaven Research Centre
A06	N	Modelling microbial resilience and recovery	2.21-02	Leonie **Navarro**, Institute of Metabolic Medicine, Westhaven Research Centre Daniel **Okafor**, Institute of Metabolic Medicine, Westhaven Research Centre; Centre for Clinical Translation, Westhaven Research Centre

Project area B – Immunological circuits

Project		Title	Research area	Project leader(s), institute(s), location(s)
B01		Host defence during microbiome perturbation	2.21-03 2.21-05 2.22-15	Hansi **Meier**, Department of Translational Biosciences, Partner University Alex **Example**, Department of Translational Biosciences, Partner University
B02	N	Molecular signatures of tissue regeneration	2.22-15	Kimmi **Miller**, Department of Translational Biosciences, Partner University Daniel **Okafor**, Institute of Metabolic Medicine, Westhaven Research Centre; Centre for Clinical Translation, Westhaven Research Centre
B03		Immune circuits linking intestine and liver	2.21-02	Hansi **Meier**, Department of Translational Biosciences, Partner University Mira **Sato**, Centre for Clinical Translation, Westhaven Research Centre
B04	N	Cross-organ consequences of inflammatory stress	2.21-02	Kimmi **Miller**, Department of Translational Biosciences, Partner University Mira **Sato**, Centre for Clinical Translation, Westhaven Research Centre
B05		Patient-derived models of metabolic disease	2.21-02	Friedhelm **Redata**, Institute of Microbial Systems Medicine, Example University Robin **Keller**, Institute of Microbial Systems Medicine, Example University; Centre for Clinical Translation, Westhaven Research Centre

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Project		Title	Research area	Project leader(s), institute(s), location(s)
B06	N	Translating microbial biomarkers into interventions	2.21-02	Daniel **Okafor,** Institute of Metabolic Medicine, Westhaven Research Centre; Centre for Clinical Translation, Westhaven Research Centre Robin **Keller,** Institute of Microbial Systems Medicine, Example University; Centre for Clinical Translation, Westhaven Research Centre

Project area Q – Central projects

Project	Title	Research area	Project leader(s), institute(s), location(s)
Q01	Integrated imaging and analysis services	2.21-02 2.22-15	Alex **Example,** Department of Translational Biosciences, Partner University Noah **Lindberg,** Centre for Computational Microbiology, Northbridge University; Department of Experimental Immunology, Northbridge University
Q02	Integrated Research Training Group		Friedhelm **Redata,** Institute of Microbial Systems Medicine, Example University Leonie **Navarro,** Institute of Metabolic Medicine, Westhaven Research Centre
Q03	Central coordination and data stewardship		Hansi **Meier,** Department of Translational Biosciences, Partner University Robin **Keller,** Institute of Microbial Systems Medicine, Example University; Centre for Clinical Translation, Westhaven Research Centre

N: New projects started in the third funding period; E: Ending projects not continued in the third funding period; Research areas according to the German Research Foundation (2024–2028)[1]: **2.21-02**: Microbial Ecology and Applied Microbiology, **2.21-03**: Medical Microbiology and Mycology, Hygiene, Molecular Infection Biology, **2.21-05**: Immunology, and **2.22-15**: Gastroenterology.

1 https://www.dfg.de/resource/blob/331950/fachsystematik-2024-2028-en.pdf

1.2 Research profile of the Collaborative Research Centre

Spotlights: Moving Forward

- Resolve causal mechanisms of communication between microbial communities, epithelial barriers, and tissue-resident immune cells.
- Combine spatial profiling, experimental perturbation, and predictive modelling across the participating projects.
- Translate shared methods into reproducible services, interoperable data, and structured training for researchers in early career phases.

1.2.1 Summary of the research programme

The CRC investigates how microbial and host-derived signals cross epithelial barriers and influence organ function. During the previous funding periods, the consortium established shared cohorts, experimental models, imaging workflows, and computational methods. The new programme links these resources in hypothesis-driven projects that test causality across molecular, cellular, and tissue scales.

1.2.2 Detailed presentation of the research programme

Project area A examines microbial metabolites and barrier signalling, whereas project area B focuses on immune circuits and tissue responses. The central projects provide imaging, analysis, training, coordination, and research-data services. Common samples, harmonised metadata, and reciprocal collaborations connect the individual work programmes and enable validation across models and participating institutions.

1.2.3 Positioning of the Collaborative Research Centre within its general research area

The programme combines mechanistic microbiome research with systems immunology and organ physiology. Its distinguishing feature is the coordinated validation of candidate signals from discovery cohorts through controlled experimental systems to tissue-level readouts, supported by shared technology and transparent data-management standards.

1.2.4 National and international cooperation and networking

The CRC maintains method-oriented collaborations with national infrastructure networks and international partner laboratories. Cooperation agreements define sample and data transfer, authorship, quality assurance, and access to specialised technologies. Visiting-researcher and exchange formats are coordinated through the Integrated Research Training Group.

1.3 Research profile of the applicant university

1.3.1 Strategy and planning

Example University has identified host–microbiome research as a strategic priority and supports the CRC through permanent scientific positions, shared technical platforms, research-data infrastructure, and central administration. Annual reviews align institutional investments with the scientific milestones of the consortium.

1.3.2 Staff situation

The participating units contribute principal investigators, research staff, technical personnel, data stewards, and administrative support. Recruitment planning prioritises timely appointments, transparent selection procedures, career development, and continuity of essential technical expertise.

1.3.3 Research infrastructure

The CRC has access to advanced microscopy, flow cytometry, sequencing, gnotobiotic facilities, secure data storage, and high-performance computing. Central projects coordinate access, training, maintenance, and quality control so that methods and datasets remain comparable across all projects.

1.4 Support structures

1.4.1 Researchers in early career phases

Researchers in early career phases receive structured scientific and career mentoring, transferable-skills training, annual progress reviews, and access to methods courses and exchange opportunities organised by Q02 (**Redata / Navarro**).

Project	Surname, first name	Type of funding	Topic	Duration
A01	Sample, Kimmi	DFG	*Epithelial sensing of microbial metabolites*	2025-04–2029-06
B01	Wunderlich, Kimmi	DFG	*Immune signatures during microbiome perturbation*	2025-04–2029-06

Note: Row colours indicate category of doctoral candidates: Dr. rer. nat., Dr. med.. **RS** = Rising Star. Doctoral candidates with highlighted names have obtained the doctoral degree. MD denotes medical students supported by DFG stipends for up to 6 months.

1.4.2 Promotion of equity and diversity

Recruitment, supervision, and committee appointments follow transparent criteria and are monitored by career stage and gender. Flexible working arrangements, mentoring, accessibility measures, and targeted career support are available through the central coordination project.

A. Objectives for the participation of female researchers

	Second funding period			Third funding period
	Objective [%]	Status Quo[a]		Objective [%]
		Number	[%]	f
Doctoral researchers	40	20	14	40
Postdoctoral researchers	22	34	44	34

[a] Reference date: proposal for third funding period

B. Project leaders

Position (female)	First funding period		Second funding period		Status Quo[a] Third funding period		
	Number	[%]	Number	[%]	Original objective [%]	Number	[%]
Postdoctoral researchers	0	0	1	100	34	1	100
Research group leaders, junior research group leaders, junior professors	0	0	0	0	34	0	0
Professors W2/C3	0	0	1	100	34	4	44
Professors W3/C4	0	0	0	0	34	0	0
Total	**0**	**0**	**2**	**50**		**5**	**42**

[a] Reference date: proposal for third funding period

1.4.3 Knowledge transfer and science communication

The CRC communicates results through open research resources, public lectures, school and patient-group activities, and dialogue formats with clinical and societal stakeholders. Transfer opportunities are assessed early, with clear procedures for intellectual property, responsible communication, and reusable research outputs.

1.5 Risks in international cooperation

International partnerships are reviewed for legal, ethical, export-control, data-protection, and research-security risks. Written agreements specify permitted data and material transfers, access rights, publication principles, and responsibilities. The participating universities reassess collaborations when partners, technologies, or regulatory conditions change.

1.6 Considerations on aspects of ecological sustainability in project planning and implementation

The CRC considers ecological sustainability when planning experiments, procurement, travel, computing, sample storage, and shared infrastructure. Projects consolidate measurement campaigns, share equipment and reference materials, select proportionate data-retention and computing strategies, and prefer lower-emission travel or remote participation where this is compatible with the scientific objective. The steering committee reviews these measures alongside scientific feasibility, research quality, legal obligations, animal welfare, and the responsible use of resources.

2 Existing funds and requested funds

2.1 Existing funds

2.1.1 Overview of existing funds for direct costs

Financial year/ funding period	Core support provided by applicant university	Core support provided by other participating institutions	Other funds	Total
2022 | 2	1,000	2,000	3,000	6,000
2023	1,000	2,000	3,000	6,000
2024	1,000	2,000	3,000	6,000
2025	1,000	2,000	3,000	6,000
2026 | 1	1,000	2,000	3,000	6,000
Ending Funding period	5,000	10,000	15,000	30,000
2027 | 2	13,600	3,200	0	16,800
2028	56,300	2,500	0	58,800
2029	23,600	23,100	0	46,700
2030	3,100	7,600	0	10,700
2031 | 1	40,800	5,400	0	46,200
New Funding period	137,400	41,800	0	179,200

Note: Figures in EUR

2.1.2 Overview of existing staff

Category	Number of persons	
	at the applicant university	at other participating institutions
	EU	PU
Professors	1	2
Junior research group leaders	0	0
Postdoctoral researchers	0	0
Doctoral researchers	0	0
Other research staff	0	0
Non-research staff	1	1
Student and graduate assistants	0	0

Affiliations: EU: Example University; **PU**: Partner University.

2.1.3 List of existing instrumentation

Project	Description of instrumentation	Year of purchase	Cost of purchase[a]	Source of funding
A01	1 × Confocal laser-scanning microscope (Example Instruments GmbH)	2025	250	DFG
A01 / B01	2 × Anaerobic workstation (Laboratory Systems AG)	2024	80	Institutional funds

[a] Figures in thousands

2.2 Previous and requested funds

2.2.1 Overview

Financial year/ funding period	Funding for					Total
	Staff	Direct costs	Instrumentation	Fellowships	Globalfunds	
2022 | 2	1,000	2,000	3,000	4,000	5,000	15,000
2023	1,000	2,000	3,000	4,000	5,000	15,000

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Financial year/ funding period	Staff	Direct costs	Funding for Instrumentation	Fellowships	Globalfunds	Total
2024	1,000	2,000	3,000	4,000	5,000	15,000
2025	1,000	2,000	3,000	4,000	5,000	15,000
2026|1	1,000	2,000	3,000	4,000	5,000	15,000
Ending Funding period	5,000	10,000	15,000	20,000	25,000	75,000
2027|2	734,158	19,112	6,000	1,000	6,000	766,270
2028	1,468,325	28,067	2,000	1,000	7,000	1,506,392
2029	1,468,325	39,843	3,000	1,000	8,000	1,520,168
2030	1,468,325	34,375	4,000	1,000	9,000	1,516,700
2031|1	734,158	25,983	5,000	1,000	10,000	776,141
New Funding period	5,873,291	147,380	20,000	5,000	40,000	6,085,671

Note: Figures in EUR

2.2.2 Overview of funds requested for staff

Project	2027|2					2028					2029					2030					2031|1				
	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants	Postdoctoral researchers	Doctoral researchers	Other research staff	Non-research staff	Student and graduate assistants
A01	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
A02	1	1	×	×	×	1	1	×	×	×	1	1	×	×	×	1	1	×	×	×	1	1	×	×	×
A03	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
A04	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
A05	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
A06	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
B01	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×
B02	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
B03	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
B04	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
B05	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
B06	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×
Q01	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×
Q02	×	1	×	1	×	×	1	×	1	×	×	1	×	1	×	×	1	×	1	×	×	1	×	1	×
Q03	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×	×	×	×	1	×
Total	**2**	**12**	×	**3**	×	**2**	**12**	×	**3**	×	**2**	**12**	×	**3**	×	**2**	**12**	×	**3**	×	**2**	**12**	×	**3**	×

2.2.3 Overview of funds requested for instrumentation

Project	Description of instrumentation	Funds requested for				
		2027|2	2028	2029	2030	2031|1
A01	Live-cell imaging system	1,000	2,000	3,000	4,000	5,000
A02	Specialised instrument	5,000	0	0	0	0
Total		6,000	2,000	3,000	4,000	5,000

Note: Figures in EUR

2.3 Upkeep of laboratory animals

Project	Species	Quantity	Average number of weeks kept	Housing costs per animal per week[a]	Purchasing costs per animal[a]	Requested funds[a]	Existing funds[a]
A01	Mouse	1,212	43	0.75	23	100	232
A01 / A02	Rat	2	3	1.4	24	1,230	232
B01	Mouse (gnotobiotic)	3	43	3.9	23	201	132
Total		**Mice: 1,215 Rats: 2**				1,531	596

[a] Figures in EUR

3 Project details

3.1 General information about Project A01

3.1.1 Project title: Microbial signalling across epithelial barriers

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Kim Sample, Prof. Dr. rer. nat.

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A01

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Sample), **no** (Example)
- Do any of the above mentioned persons hold fixed-term positions? **yes** (Sample), **no** (Example)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **yes** (Sample), **no** (Example)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Sample), **yes** (Example)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

Summarise the continuing project's central question, progress, planned work, methods, and contribution to the Collaborative Research Centre.

3.3 Project progress to date

3.3.1 Report and current state of research

Report the work and results from the previous funding periods, relate them to the current state of research, and explain how they motivate the new plan.

3.3.2 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Define the underlying mechanism.

Aim 2: Validate the mechanism experimentally.

Aim 3: Integrate the findings across the CRC.

Refer to one aim as **Aim 1**, or to several as **Aims 1 to 3**.

WP 1: Mechanistic analysis

Describe the work, methods, milestones, risks, and expected results of **WP 1**.

WP 1a: Establish and validate the experimental system.

WP 1b: Perform the mechanistic experiments and define decision points.

WP 2: Experimental validation

Describe the validation strategy, expected results, and contingency plan of **WP 2**.

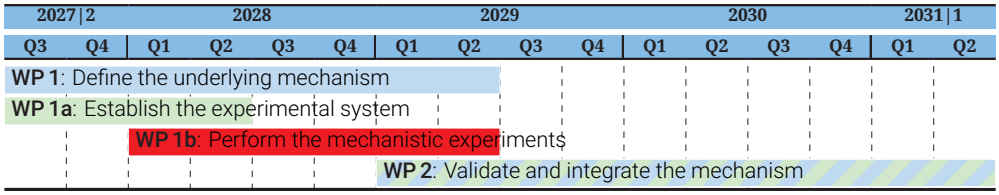


Fig. 1: Gantt chart of **A01**.

3.5 Relevance of sex, gender and/or diversity in the research project

Explain whether and how sex, gender, and diversity affect the research questions, design, analysis, and interpretation.

3.6 Role within the Collaborative Research Centre

Describe the project's contribution to the CRC and its principal collaborations.

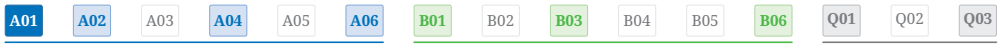


Fig. 2: Collaborations of **A01**.

3.7 Differentiation from other funded projects

Explain how this project differs from, and complements, other funded work of the project leaders.

3.8 Project funding

3.8.1 Previous funding

Funding of this project within the Collaborative Research Centre started in July 2019. Preparatory work was funded by Example University's seed-fund programme from January 2018 to June 2019 (reference number EU-EXAMPLE-01).

3.8.2 Requested funding

Funding for	**2027 | 2**		**2028**		**2029**		**2030**		**2031 | 1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Professor, 100 %	2	126,600	2	253,200	2	253,200	2	253,200	2	126,600
Doctoral researcher, 65 %	1	26,500	1	53,000	1	53,000	1	53,000	1	26,500
Total		153,100		306,200		306,200		306,200		153,100
Direct Costs		Sum		Sum		Sum		Sum		Sum
Laboratory consumables		2,112		67		9,843		2,375		983
Sequencing and data analysis		1,000		2,000		3,000		4,000		5,000
Total		3,112		2,067		12,843		6,375		5,983
Instrumentation		Sum		Sum		Sum		Sum		Sum
Live-cell imaging system		1,000		2,000		3,000		4,000		5,000
Total		1,000		2,000		3,000		4,000		5,000
Total		157,212		310,267		322,043		316,575		164,083

Note: Figures in EUR

Requested funding for staff for the new funding period

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Existing staff							
Research staff	1.	**Sample**, Kim, Prof. Dr. rer. nat. (Professor and Director)	Microbiology and systems medicine	IMSM, DTB	8		State funds
Non-research staff	2.	Example, Alex, M.Sc. (Technical assistant)	Laboratory management	IMSM	20		Core support
Requested staff							
Research staff	3.	TBA (Postdoctoral researcher)	Host–microbiome interactions	DTB		Postdoctoral researcher	
Non-research staff	4.	Demo, Taylor, B.Sc. (Research assistant)	Biobanking and data curation	DTB		Research assistant	

Institutions: IMSM: Institute of Microbial Systems Medicine (EU); **DTB**: Department of Translational Biosciences (PU).
Affiliations: EU: Example University; **PU**: Partner University.

Job descriptions of staff for the proposed funding period (supported through existing funds):

1. **Prof. Dr. rer. nat. Kim Sample** Leads the experimental programme and supervises the research staff assigned to the project.
2. **M.Sc. Alex Example** Maintains shared laboratory workflows and coordinates sample logistics.

Job descriptions of staff for the proposed funding period (requested funds):

3. **TBA** Performs the planned mechanistic experiments and integrates molecular and clinical datasets.
4. **B.Sc. Taylor Demo** Supports biobanking, quality control, and structured curation of the project datasets.

Requested funding for direct costs for the new funding period

	2027 | 2	2028	2029	2030	2031 | 1
PU: existing funds from Partner University core support	1,200	500	21,100	5,600	3,400
EU: existing funds from Example University core support	11,100	53,800	21,100	600	38,300
Sum of existing funds	12,300	54,300	42,200	6,200	41,700

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	2027 | 2	2028	2029	2030	2031 | 1
Sum of requested funds	3,112	2,067	12,843	6,375	5,983

Note: Figures in EUR

Text

Laboratory consumables and antibodies	EUR	15,000
Single-cell sequencing and spatial profiling	EUR	12,000
Research data storage and analysis	EUR	3,000

Requested funding for instrumentation for the new funding period

Text

Automated live-cell imaging system	EUR	95,000
Environmental control module	EUR	18,000
Analysis workstation and software	EUR	7,000

A01

3.1 General information about Project A02 N

3.1.1 Project title: Metabolic control of tissue-resident immunity

3.1.2 Research area: 2.21-05: Immunology

3.1.3 Project leaders

Kim Sample, Prof. Dr. rer. nat.

0000-0000-0000-0001
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
(Germany)

Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 100-1000
kim.sample@example.org

Samira Hartmann, Prof. Dr. rer. nat.

0000-0000-0000-0016
Centre for Computational Microbiology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)
+49 000 300-0006 samira@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Sample), **no** (Hartmann)
- Do any of the above mentioned persons hold fixed-term positions? **yes** (Sample), **no** (Hartmann)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **yes** (Sample), **no** (Hartmann)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Sample), **no** (Hartmann)

A02

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✓ yes
	A copy of the studies' registration is included with the proposal.	✗ no
3.	experiments involving vertebrates.	✓ yes
4.	experiments involving recombinant DNA.	✓ yes
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✓ yes
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

Summarise the new project's central hypothesis, methods, expected results, and contribution to the Collaborative Research Centre.

3.3 Research rationale

3.3.1 Current state of research and preliminary work

Describe the relevant state of research, preliminary work, open questions, and the scientific rationale for the proposed project.

3.3.2 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish the experimental and analytical framework.

Aim 2: Test the central hypothesis and validate the findings.

WP 1: Framework and pilot experiments

Describe the methods, milestones, expected results, and risks of **WP 1** and its contribution to **Aim 1**.

WP 2: Hypothesis testing and validation

Describe the validation strategy and contingency plan for **WP 2** and **Aim 2**.

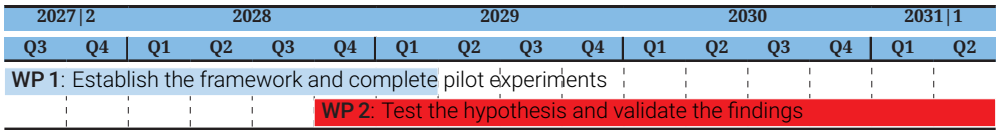


Fig. 1: Gantt chart of A02.

3.5 Relevance of sex, gender and/or diversity in the research project

Explain whether and how sex, gender, and diversity affect the research questions, design, analysis, and interpretation.

3.6 Role within the Collaborative Research Centre

Describe the project's contribution to the CRC and its principal collaborations.



Fig. 2: Collaborations of A02.

3.7 Differentiation from other funded projects of the project leader(s)

Explain how this new project differs from, and complements, other funded work of the project leaders.

3.8 Project funding

3.8.1 Previous funding

State that this is a new CRC project and list related previous funding of the project leaders where applicable.

3.8.2 Requested funding

Funding for	2027|2		2028		2029		2030		2031|1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Postdoctoral researcher, 100 %	1	45,300	1	90,600	1	90,600	1	90,600	1	45,300
Doctoral researcher, 65 %	1	26,500	1	53,000	1	53,000	1	53,000	1	26,500
Total		71,800		143,600		143,600		143,600		71,800
Direct Costs		Sum		Sum		Sum		Sum		Sum
Consumables		2,000		2,000		2,000		2,000		2,000

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Funding for	2027 | 2	2028	2029	2030	2031 | 1
Total	2,000	2,000	2,000	2,000	2,000
Instrumentation	Sum	Sum	Sum	Sum	Sum
Specialised instrument	5,000	0	0	0	0
Total	5,000	0	0	0	0
Total	78,800	145,600	145,600	145,600	73,800

Note: Figures in EUR

Requested funding for staff

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Requested staff							
Research staff	1.	TBA (Postdoctoral researcher)	Experimental systems medicine	IMSM		Postdoctoral researcher	

Institutions: IMSM: Institute of Microbial Systems Medicine (EU). **Affiliations: EU**: Example University.

Job descriptions of staff for the proposed funding period (requested funds):

1. **TBA** Establishes the new experimental platform, performs the mechanistic studies, and analyses the resulting datasets.

Provide the scientific and organisational justification for every requested staff position.

Requested funding of direct costs

	2027 | 2	2028	2029	2030	2031 | 1
EU: existing funds from State funds	500	500	500	500	500
Sum of existing funds	500	500	500	500	500
Sum of requested funds	2,000	2,000	2,000	2,000	2,000

Note: Figures in EUR

Text

Consumables for pilot experiments	EUR	10,000
External data analysis	EUR	5,000

Requested funding for instrumentation

Text

Specialised instrument including installation	EUR	5,000

3.1 General information about Project A03

3.1.1 Project title: Microbiome-derived metabolites and immune memory

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Samira Hartmann, Prof. Dr. rer. nat.

0000-0000-0000-0016
Centre for Computational Microbiology
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Science Avenue 3, 34567 Northbridge
(Germany)
+49 000 300-0006 samira@example.org

Noah Lindberg, Prof. Dr. rer. nat.

Centre for Computational Microbiology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)

Department of Experimental Immunology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)
+49 000 300-0007 noah@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Hartmann), **no** (Lindberg)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Hartmann), **no** (Lindberg)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Hartmann), **no** (Lindberg)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Hartmann), **no** (Lindberg)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

A03

3.2 Summary

This project investigates Microbiome-derived metabolites and immune memory. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Project progress to date

The preceding funding period established the experimental models and shared datasets. The next phase tests causal mechanisms and validates their relevance across the participating projects.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2**.

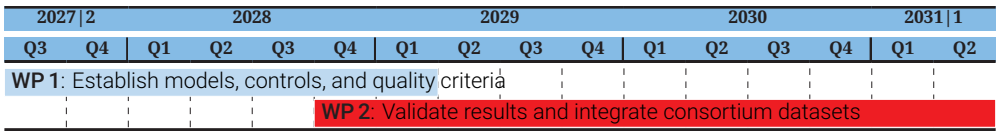


Fig. 1: Gantt chart of A03.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of A03.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project A04 N

3.1.1 Project title: Spatial niches of host–microbe interactions

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Noah Lindberg, Prof. Dr. rer. nat.

Centre for Computational Microbiology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)

Department of Experimental Immunology
Northbridge University
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(Germany)
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Alexis-Marie Winters-Scholz, Prof. Dr. rer. nat.

Department of Experimental Immunology
Northbridge University
Science Avenue 3, 34567 Northbridge
(Germany)
+49 000 300-0008
alexis-marie@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Lindberg), **no** (Winters-Scholz)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Lindberg), **no** (Winters-Scholz)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Lindberg), **no** (Winters-Scholz)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Lindberg), **no** (Winters-Scholz)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Spatial niches of host–microbe interactions. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Research rationale

Preliminary observations motivate a new testable hypothesis. The project establishes its experimental framework and independent validation strategy during the requested funding period.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1.**

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2.**

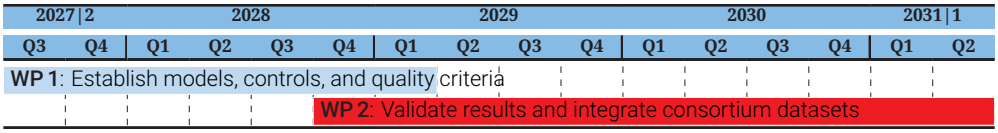


Fig. 1: Gantt chart of **A04**.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of **A04**.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	2027|2		2028		2029		2030		2031|1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project A05

3.1.1 Project title: Dietary control of microbial community dynamics

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Alexis-Marie Winters-Scholz, Prof. Dr. rer. nat.

Department of Experimental Immunology
Northbridge University
Science Avenue 3, 34567 Northbridge (Germany)
+49 000 300-0008
alexis-marie@example.org

Leonie Navarro, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0009 leonie@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Winters-Scholz), **no** (Navarro)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Winters-Scholz), **no** (Navarro)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Winters-Scholz), **no** (Navarro)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Winters-Scholz), **no** (Navarro)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Dietary control of microbial community dynamics. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Project progress to date

The preceding funding period established the experimental models and shared datasets. The next phase tests causal mechanisms and validates their relevance across the participating projects.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). “The FAIR Guiding Principles for scientific data management and stewardship”. In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2.**

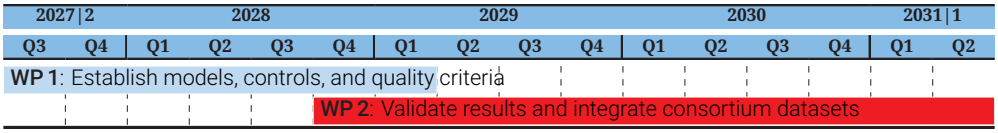


Fig. 1: Gantt chart of A05.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of A05.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project A06 N

3.1.1 Project title: Modelling microbial resilience and recovery

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Leonie Navarro, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
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+49 000 300-0009 leonie@example.org

Daniel Okafor, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)

Centre for Clinical Translation
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0010 daniel@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Navarro), **no** (Okafor)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Navarro), **no** (Okafor)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Navarro), **no** (Okafor)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Navarro), **no** (Okafor)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Modelling microbial resilience and recovery. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Research rationale

Preliminary observations motivate a new testable hypothesis. The project establishes its experimental framework and independent validation strategy during the requested funding period.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). “The FAIR Guiding Principles for scientific data management and stewardship”. In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: **Experimental framework** Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: **Validation and integration** Test robustness and share the validated results for **Aim 2**.

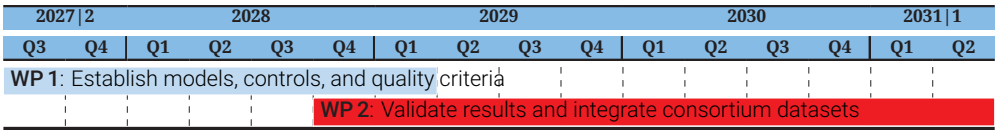


Fig. 1: Gantt chart of A06.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of A06.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	2027 | 2		2028		2029		2030		2031 | 1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project B01

3.1.1 Project title: Host defence during microbiome perturbation

3.1.2 Research areas: 2.21-03: Medical Microbiology and Mycology, Hygiene, Molecular Infection Biology, 2.21-05: Immunology, 2.22-15: Gastroenterology

3.1.3 Project leaders

Hansi Meier, Dr. rer. nat.
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Department of Translational Biosciences
Partner University
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hansi.meier@example.org

Alex Example, Prof. Dr. phil.
0000-0000-0000-0002
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-2000
alex.example@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **yes** (Meier), **no** (Example)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Meier), **no** (Example)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Meier), **no** (Example)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **yes** (Meier), **yes** (Example)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

Summarise the continuing project and the work proposed for the new funding period.

3.3 Project progress to date

3.3.1 Report and current state of research

Report the previous work, published and unpublished results, and the current state of research.

3.3.2 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Complete the mechanistic analysis.

Aim 2: Validate the findings with CRC partners.

WP 1: Mechanistic analysis Describe **WP 1**, including methods, milestones, and risks.

WP 2: Collaborative validation Describe **WP 2** and its links to **Aim 2**.

B01

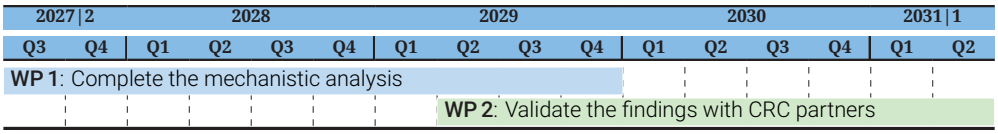


Fig. 1: Gantt chart of B01.

3.5 Relevance of sex, gender and/or diversity in the research project

Explain the relevance of sex, gender, and diversity, or justify why they do not affect this project's research questions and methods.

3.6 Role within the Collaborative Research Centre

Describe the role of the project and its reciprocal collaborations.



Fig. 2: Collaborations of B01.

3.7 Differentiation from other funded projects

Differentiate the project from other funded work of its project leaders.

3.8 Project funding

3.8.1 Previous funding

State when CRC funding began and list other relevant previous funding.

3.8.2 Requested funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
(Jr.) Research group leader, 100 %	2	105,000	2	210,000	2	210,000	2	210,000	2	105,000
Postdoctoral researcher, 100 %	1	45,000	1	90,000	1	90,000	1	90,000	1	45,000
Total		150,000		300,000		300,000		300,000		150,000
Globalfunds		Sum		Sum		Sum		Sum		Sum
Project-wide data stewardship		1,000		2,000		3,000		4,000		5,000
Total		1,000		2,000		3,000		4,000		5,000
Total		151,000		302,000		303,000		304,000		155,000

Note: Figures in EUR

Requested funding for staff for the new funding period

	No.	**Name, academic degree, position**	**Field of research**	**Departm. of (non-)univers. institution**	**Project commitm. (h/wk)**	**Category**	**Funding source**
Requested staff							
Research staff	1.	TBA (Research group leader)	Translational microbiology	DTB		(Jr.) Research group leader	

Institutions: DTB: Department of Translational Biosciences (PU). **Affiliations: PU**: Partner University.

B01

Job descriptions of staff for the proposed funding period (requested funds):
1. **TBA** Coordinates the experimental programme and the collaborative validation studies.

Requested funding for direct costs for the new funding period

	2027 | 2	2028	2029	2030	2031 | 1
PU: existing funds from Institutional funds	1,000	1,000	1,000	1,000	1,000
Sum of existing funds	1,000	1,000	1,000	1,000	1,000
Sum of requested funds	0	0	0	0	0

Note: Figures in EUR

Text

Experimental consumables	EUR	10,000
Data-processing services	EUR	5,000

Requested funding for instrumentation for the new funding period

Text

No instrumentation requested	EUR	0

B01

A

3.1 General information about Project B02 N

3.1.1 Project title: Molecular signatures of tissue regeneration

3.1.2 Research area: 2.22-15: Gastroenterology

3.1.3 Project leaders

Kimmi Miller, Prof. Dr. rer. nat.

0000-0000-0000-0004
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-4000
kimmi.mueller@example.org

Daniel Okafor, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)

Centre for Clinical Translation
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0010 daniel@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **yes** (Miller), **no** (Okafor)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Miller), **no** (Okafor)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Miller), **no** (Okafor)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **yes** (Miller), **no** (Okafor)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✓ yes
	A statement from the responsible ethics committee is required and will be obtained before funding starts.	✗ no
2.	clinical trials	✓ yes
	A copy of the studies' registration is included with the proposal.	✓ yes
3.	experiments involving vertebrates.	✓ yes
4.	experiments involving recombinant DNA.	✓ yes
5.	research involving human embryonic stem cells.	✓ yes
	Legal authorization has been obtained.	✗ no
6.	research concerning the Convention on Biological Diversity.	✓ yes
7.	investigations involving dual use research of concern.	✓ yes
	A copy of the statement by the institution's research ethics committee is included with the proposal.	✗ no

3.2 Summary

This project investigates Molecular signatures of tissue regeneration. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Research rationale

Preliminary observations motivate a new testable hypothesis. The project establishes its experimental framework and independent validation strategy during the requested funding period.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

B02

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2**.

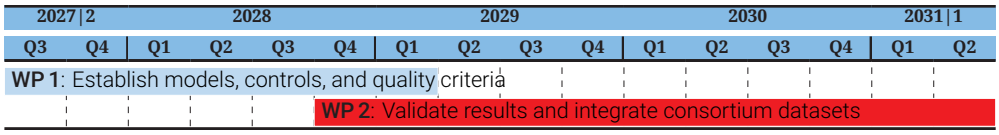


Fig. 1: Gantt chart of B02.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of B02.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project B03

3.1.1 Project title: Immune circuits linking intestine and liver

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

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Mira Sato, Prof. Dr. rer. nat.

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Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
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- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **yes** (Meier), **no** (Sato)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Meier), **no** (Sato)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Meier), **no** (Sato)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **yes** (Meier), **no** (Sato)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Immune circuits linking intestine and liver. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Project progress to date

The preceding funding period established the experimental models and shared datasets. The next phase tests causal mechanisms and validates their relevance across the participating projects.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). “The FAIR Guiding Principles for scientific data management and stewardship”. In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: **Experimental framework** Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: **Validation and integration** Test robustness and share the validated results for **Aim 2**.

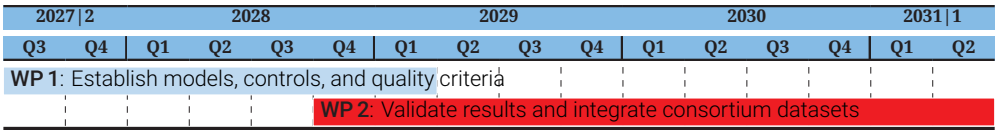


Fig. 1: Gantt chart of B03.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of B03.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	2027 | 2		2028		2029		2030		2031 | 1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project B04 N

3.1.1 Project title: Cross-organ consequences of inflammatory stress

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Kimmi Miller, Prof. Dr. rer. nat.

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Science Avenue 3, 34567 Westhaven (Germany)
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- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **yes** (Miller), **no** (Sato)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Miller), **no** (Sato)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Miller), **no** (Sato)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **yes** (Miller), **no** (Sato)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Cross-organ consequences of inflammatory stress. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Research rationale

Preliminary observations motivate a new testable hypothesis. The project establishes its experimental framework and independent validation strategy during the requested funding period.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2**.

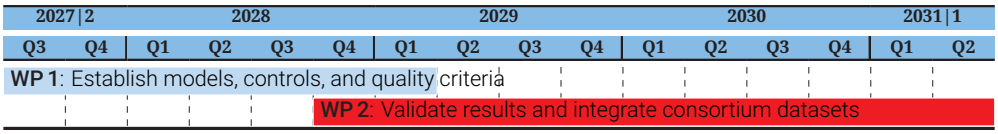


Fig. 1: Gantt chart of B04.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.



Fig. 2: Collaborations of B04.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	2027|2		2028		2029		2030		2031|1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project B05

3.1.1 Project title: Patient-derived models of metabolic disease

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Friedhelm Redata, Prof. Dr.-Ing.

0000-0000-0000-0005
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
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friedhelm.redata@example.org

Robin Keller, Prof. Dr. rer. nat.

Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City
(Germany)

Centre for Clinical Translation
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0012 robin@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **yes** (Redata), **no** (Keller)
- Do any of the above mentioned persons hold fixed-term positions? **yes** (Redata), **no** (Keller)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **yes** (Redata), **no** (Keller)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Redata), **no** (Keller)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Patient-derived models of metabolic disease. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Project progress to date

The preceding funding period established the experimental models and shared datasets. The next phase tests causal mechanisms and validates their relevance across the participating projects.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). “The FAIR Guiding Principles for scientific data management and stewardship”. In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2.**

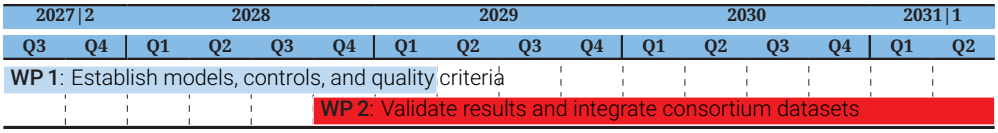


Fig. 1: Gantt chart of B05.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.

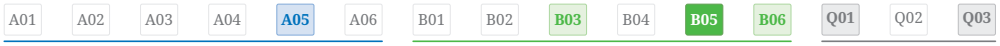


Fig. 2: Collaborations of B05.

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	**2027 | 2**		**2028**		**2029**		**2030**		**2031 | 1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project B06 N

3.1.1 Project title: Translating microbial biomarkers into interventions

3.1.2 Research area: 2.21-02: Microbial Ecology and Applied Microbiology

3.1.3 Project leaders

Daniel Okafor, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
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Centre for Clinical Translation
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Robin Keller, Prof. Dr. rer. nat.

Institute of Microbial Systems Medicine
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Centre for Clinical Translation
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Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0012 robin@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Okafor), **no** (Keller)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Okafor), **no** (Keller)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Okafor), **no** (Keller)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Okafor), **no** (Keller)

B06

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

3.2 Summary

This project investigates Translating microbial biomarkers into interventions. Its project leaders combine experimental and analytical approaches within the shared research programme of CRC 9999.

3.3 Research rationale

Preliminary observations motivate a new testable hypothesis. The project establishes its experimental framework and independent validation strategy during the requested funding period.

3.3.1 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Establish reproducible experimental and analytical workflows.

Aim 2: Validate the findings in complementary models and datasets.

WP 1: Experimental framework Define the controls, sampling strategy, and analysis workflow for **Aim 1**.

WP 2: Validation and integration Test robustness and share the validated results for **Aim 2.**

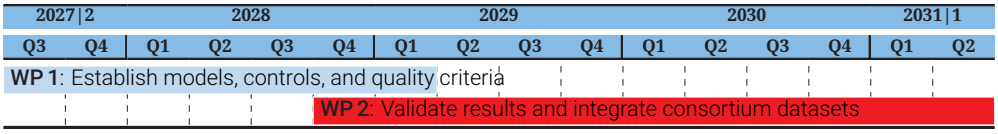


Fig. 1: Gantt chart of B06.

3.5 Relevance of sex, gender and/or diversity in the research project

Include relevant biological variables in sampling and analysis; document the rationale for each model and the limits of generalisation.

3.6 Role within the Collaborative Research Centre

Share methods, data, and validation resources with the linked projects.

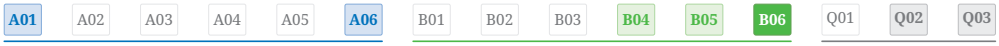


Fig. 2: Collaborations of B06.

B06

3.7 Differentiation from other funded projects

The proposed work addresses the CRC-specific hypothesis and does not duplicate personnel or experiments supported by other grants.

3.8 Project funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Total		27,202		54,405		54,405		54,405		27,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
consumables		1,000		2,000		2,000		2,000		1,000
Total		1,000		2,000		2,000		2,000		1,000
Total		28,202		56,405		56,405		56,405		28,202

Note: Figures in EUR

The doctoral researcher performs the experiments and integrates the data. Consumables cover sample preparation, assays, and independent validation.

3.1 General information about Project Q01

3.1.1 Project title: Integrated imaging and analysis services

3.1.2 Research areas: 2.21-02: Microbial Ecology and Applied Microbiology, 2.22-15: Gastroenterology

3.1.3 Project leaders

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Noah Lindberg, Prof. Dr. rer. nat.

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- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **no** (Example), **no** (Lindberg)
- Do any of the above mentioned persons hold fixed-term positions? **no** (Example), **no** (Lindberg)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **no** (Example), **no** (Lindberg)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **yes** (Example), **no** (Lindberg)

3.1.4 Legal issues

This project includes

1.	research requiring compliance with discipline-specific research ethics guidelines.	✗ no
2.	clinical trials	✗ no
3.	experiments involving vertebrates.	✗ no
4.	experiments involving recombinant DNA.	✗ no
5.	research involving human embryonic stem cells.	✗ no
6.	research concerning the Convention on Biological Diversity.	✗ no
7.	investigations involving dual use research of concern.	✗ no

Q01

3.2 Summary

Summarise the service portfolio, its scientific purpose, its users, and the developments planned for the new funding period.

3.3 Project progress to date

3.3.1 Report and current state of research

Report completed and ongoing services, usage by CRC projects, technical developments, quality assurance, and relevant service-project research.

3.3.2 Project- and subject-related list of publications

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

3.4 Project plan

Aim 1: Maintain and quality-assure the core service portfolio.

Aim 2: Extend the portfolio for new collaborative requirements.

WP 1: Core service provision Describe access, sample or data handover, turnaround, quality assurance, responsibilities, and capacity for **WP 1**.

WP 2: Service development Describe development milestones and the projects participating in **WP 2**.

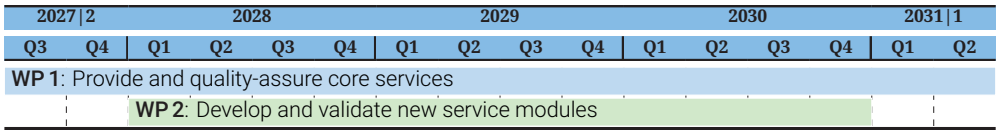


Fig. 1: Gantt chart of Q01.

3.5 Relevance of sex, gender and/or diversity in the research project

Explain relevance for the service methods and the studies supported by them.

3.6 Role within the Collaborative Research Centre

Describe which projects use the services and how priorities, access, and collaborations are coordinated.



Fig. 2: Collaborations of Q01.

3.7 Differentiation from other funded projects

Differentiate the CRC service from existing institutional infrastructure and other funded service activities.

3.8 Project funding

3.8.1 Previous funding

State when the service project began and describe relevant previous funding.

3.8.2 Requested funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Non-research staff, 65 %	1	19,988	1	39,975	1	39,975	1	39,975	1	19,988
Research assistant, 50 %	1	9,000	1	18,000	1	18,000	1	18,000	1	9,000
Total		28,988		57,975		57,975		57,975		28,988
Direct Costs		Sum		Sum		Sum		Sum		Sum
Invitation		1,000		2,000		3,000		4,000		5,000
Total		1,000		2,000		3,000		4,000		5,000
Total		29,988		59,975		60,975		61,975		33,988

Note: Figures in EUR

Requested funding for staff for the new funding period

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Requested staff							
Non-research staff	1.	Specialist, Service (Technical specialist)	Core-facility services	IMSM		Non-research staff	

Institutions: IMSM: Institute of Microbial Systems Medicine (EU). **Affiliations: EU**: Example University.

Job descriptions of staff for the proposed funding period (requested funds):

1. **Service Specialist** Operates the service platform, performs quality control, and supports participating projects.

Requested funding for direct costs for the new funding period

	2027 | 2	2028	2029	2030	2031 | 1
EU: existing funds from Core-facility funds	1,000	1,000	1,000	1,000	1,000
Sum of existing funds	1,000	1,000	1,000	1,000	1,000
Sum of requested funds	1,000	2,000	3,000	4,000	5,000

Note: Figures in EUR

Text

Service-specific consumables	EUR	10,000
Maintenance and quality assurance	EUR	5,000

Requested funding for instrumentation for the new funding period

Text

Service-platform upgrade	EUR	5,000

3.1 General information about Project Q02

3.1.1 Project title: Integrated Research Training Group

3.1.2 Project leaders

Friedhelm Redata, Prof. Dr.-Ing.

0000-0000-0000-0005
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City (Germany)
+49 000 100-5000
friedhelm.redata@example.org

Leonie Navarro, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0009 leonie@example.org

- Is the employment of the project leaders at the institutions indicated contractually secured for the duration of the proposed funding period? **yes** (Redata), **no** (Navarro)
- Do any of the above mentioned persons hold fixed-term positions? **yes** (Redata), **no** (Navarro)
- Will one or more of the above-mentioned persons retire during the proposed funding period? **yes** (Redata), **no** (Navarro)
- Funding for the position of the project leaders at the institutions indicated is covered by core support (state funds or similar): **no** (Redata), **no** (Navarro)

3.2 Summary

Summarise the objectives, development, and added value of the Integrated Research Training Group.

3.3 Tabular report

Provide the required tabular report of all researchers funded by the Integrated Research Training Group to date.

3.4 Qualification programme

Describe the qualification programme, workshops, courses, retreats, career development measures, and other training activities.

3.5 Management and supervision

Describe management responsibilities, recruitment, supervision arrangements, mentoring, conflict-resolution processes, and quality assurance.

3.6 Environment of the Integrated Research Training Group

Describe the scientific, institutional, interdisciplinary, and international training environment.

3.7 Interim assessment and outlook

Assess progress during the current funding period and describe priorities and measurable objectives for the next funding period.

3.8 Project funding

3.8.1 Previous funding

The Integrated Research Training Group has been funded within the Collaborative Research Centre since April 2018. Replace this date and add any relevant funding under another programme.

3.8.2 Existing funds for the new funding period

Describe the coordination, training, staff, and direct-cost resources made available to the Integrated Research Training Group by the participating institutions.

Existing staff

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Existing staff							
Non-research staff	1.	Trainer, Taylor (Training coordinator)	Structured doctoral training	DTB	20		Partner University core funds

Institutions: DTB: Department of Translational Biosciences (PU). **Affiliations: PU**: Partner University.

Job descriptions of staff for the proposed funding period (supported through existing funds):

1. **Taylor Trainer** Coordinates recruitment, course administration, progress monitoring, mentoring, and quality assurance for the qualification programme.

	2027|2	2028	2029	2030	2031|1
PU: existing funds from Graduate-school funds	1,000	1,000	1,000	1,000	1,000
Sum of existing funds	1,000	1,000	1,000	1,000	1,000
Sum of requested funds	2,000	2,000	2,000	2,000	2,000

Note: Figures in EUR

3.8.3 Requested funding

Funding for	**2027|2**		**2028**		**2029**		**2030**		**2031|1**	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum
Doctoral researcher, 65 %	1	27,202	1	54,405	1	54,405	1	54,405	1	27,202
Non-research staff, 50 %	1	15,000	1	30,000	1	30,000	1	30,000	1	15,000
Total		42,202		84,405		84,405		84,405		42,202
Direct Costs		Sum		Sum		Sum		Sum		Sum
Qualification programme		2,000		2,000		2,000		2,000		2,000
Total		2,000		2,000		2,000		2,000		2,000
Fellowships		Sum		Sum		Sum		Sum		Sum
Short-term fellowships		1,000		1,000		1,000		1,000		1,000
Total		1,000		1,000		1,000		1,000		1,000
Globalfunds		Sum		Sum		Sum		Sum		Sum
Training and networking		3,000		3,000		3,000		3,000		3,000
Total		3,000		3,000		3,000		3,000		3,000
Total		48,202		90,405		90,405		90,405		48,202

Note: Figures in EUR

3.8.4 Requested funding for staff

Text		
Doctoral-researcher position: interdisciplinary research and participation in the complete qualification programme	EUR	5,000

3.8.5 Requested funding for direct costs

Text		
Workshops and transferable-skills courses	EUR	10,000
Visiting researchers and networking	EUR	5,000

Q02

3.8.6 Requested funding for fellowships

Text		
Short-term research fellowships: state the type of fellowship, average number of fellows, average duration, monthly allowance per person, and justification	EUR	5,000

3.8.7 Requested global funds

Text		
Training and networking measures: explain the purpose, allocation rules, selection process, and administration	EUR	15,000

3.1 General information about Project Q03

3.1.1 Project title: Central coordination and data stewardship

3.1.2 Project leaders

Hansi Meier, Dr. rer. nat.

0000-0000-0000-0003
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-3000
hansi.meier@example.org

Robin Keller, Prof. Dr. rer. nat.

Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City (Germany)

Centre for Clinical Translation
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0012 robin@example.org

3.2 Documentation of the central activities of the Collaborative Research Centre

Document governance and decision-making, scientific coordination, reporting, internal communication, data stewardship, publication support, equal- opportunity measures, early-career support, outreach, and the interfaces with all applicant and participating institutions. Distinguish CRC-specific tasks from permanent institutional duties and state responsibilities, workflows, results from the current period, and priorities for the new funding period.

3.3 Project funding

3.3.1 Existing funds for the new funding period

Describe the central staff, facilities, and direct-cost resources contributed by the applicant and participating institutions.

Existing staff

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Existing staff							
Non-research staff	1.	**Coordinator**, Jamie, Dr. rer. nat. (Scientific manager)	Research management	DTB	20		Partner University core funds

Institutions: DTB: Department of Translational Biosciences (PU). **Affiliations: PU**: Partner University.

Job descriptions of staff for the proposed funding period (supported through existing funds):

1. **Dr. rer. nat. Jamie Coordinator** Coordinates governance, reporting, internal communication, central measures, and interfaces with applicant and participating institutions.

	2027|2	2028	2029	2030	2031|1
EU: existing funds from Central university funds	1,000	1,000	1,000	1,000	1,000
Sum of existing funds	1,000	1,000	1,000	1,000	1,000
Sum of requested funds	2,000	2,000	2,000	2,000	2,000

Note: Figures in EUR

3.3.2 Requested funding for the new funding period

Funding for	2027|2		2028		2029		2030		2031|1	
Staff	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum	Qty	Sum

Continue on the following page …

…continuing from previous page.

Funding for	2027 | 2	2028	2029	2030	2031 | 1
Non-research staff, 100 %	1 30,750	1 61,500	1 61,500	1 61,500	1 30,750
Research assistant, 50 %	1 12,500	1 25,000	1 25,000	1 25,000	1 12,500
Total	43,250	86,500	86,500	86,500	43,250
Direct Costs	Sum	Sum	Sum	Sum	Sum
Central coordination and communication	2,000	2,000	2,000	2,000	2,000
Total	2,000	2,000	2,000	2,000	2,000
Globalfunds	Sum	Sum	Sum	Sum	Sum
Central measures	2,000	2,000	2,000	2,000	2,000
Total	2,000	2,000	2,000	2,000	2,000
Total	47,250	90,500	90,500	90,500	47,250

Note: Figures in EUR

3.3.3 Requested funding for staff

Text

Scientific coordination: justify the category, percentage, period, and CRC-specific responsibilities	EUR	5,000

3.3.4 Requested funding for direct costs

Text

Governance meetings and scientific exchange	EUR	10,000
Communication and outreach	EUR	5,000

3.3.5 Requested global funds

Text

Central measures: explain the purpose, allocation rules, selection process, and administration	EUR	10,000

Q03

3.1 General information about Project A00 (former project) E

3.1.1 Project title: Former project on microbial community dynamics

3.1.2 Project leaders

Kimmi Miller, Prof. Dr. rer. nat.

0000-0000-0000-0004
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-4000
kimmi.mueller@example.org

Daniel Okafor, Prof. Dr. rer. nat.

Institute of Metabolic Medicine
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)

Centre for Clinical Translation
Westhaven Research Centre
Science Avenue 3, 34567 Westhaven (Germany)
+49 000 300-0010 daniel@example.org

3.2 Project history

3.2.1 Report

Describe the complete funding history, objectives, methods, results, and contribution of the ending project. Citations used only in the report are printed automatically at the end of this environment. This dummy citation also demonstrates the exclusion of a result listed below (Wilkinson et al. 2016).

References

3.2.2 Published project results

Publications with scientific quality assurance

Wilkinson, M. D. et al. (2016). "The FAIR Guiding Principles for scientific data management and stewardship". In: *Scientific Data* 3.1, p. 160018. DOI: 10.1038/sdata.2016.18. (Visited on 05/31/2021).

Other publications and published results

None.

Patents

None.

3.3 Funding

Funding of this project within the Collaborative Research Centre started in April 2018. The project will be completed by the end of the current funding period. Replace these dates and, where applicable, add previous funding under another DFG programme and its reference number.

3.3.1 Project staff in the ending funding period

	No.	Name, academic degree, position	Field of research	Departm. of (non-)univers. institution	Project commitm. (h/wk)	Category	Funding source
Existing staff							
Research staff	1.	**Miller**, Kimmi, Prof. Dr. rer. nat. (Professor of Molecular Medicine)	Molecular tissue regeneration	DTB	8		State funds
Staff funded with approved grant money							
Research staff	2.	TBA (Postdoctoral researcher)	Regenerative cell biology	DTB		Postdoctoral researcher	

Institutions: DTB: Department of Translational Biosciences (PU). **Affiliations: PU**: Partner University.

Job descriptions of staff (supported through existing funds):

1. **Prof. Dr. rer. nat. Kimmi Miller** Led the project, supervised the experimental work, and integrated its results across the CRC throughout the ending funding period.

Job descriptions of staff (funded with approved grant money):

2. **TBA** Completed the final validation experiments and data publication from April 2025 until the end of the current funding period.

4 Bylaws of the Collaborative Research Centre

§1 Name, coordinating university, and purpose of the Collaborative Research Centre

§1.1 The Collaborative Research Centre (CRC) “Mechanisms of Host–Microbiome Communication – From Molecular Interactions to Organ Function” is a legally dependent institution of Example University.

§1.2 The Collaborative Research Centre conducts interrelated research on microbial signalling, epithelial barriers, immune regulation, and organ function. It is divided into research project areas and central projects.

§1.3 The Collaborative Research Centre promotes interaction with other research institutions, supports researchers in early career phases, and fosters responsible international cooperation, equity, and equal opportunity.

§2 Membership and governing bodies

§2.1 Members are the project leaders and persons appointed to central scientific or administrative responsibilities by the general assembly.

§2.2 The governing bodies are the general assembly, the spokesperson, and the steering committee. The general assembly elects the spokesperson and steering committee for the applicable funding period.

§2.3 The steering committee prepares strategic and budgetary decisions, monitors milestones, and reports to the general assembly at least once per year.

§3 Decisions and allocation of shared resources

§3.1 Decisions are normally taken by simple majority of members present; changes to these bylaws require a two-thirds majority.

§3.2 Shared funds and infrastructure are allocated through documented, transparent procedures based on scientific need, feasibility, and equal access across participating projects.

§3.3 Members disclose conflicts of interest and do not participate in the corresponding deliberation or decision.

§4 Research integrity, equity, and conflict resolution

§4.1 All members follow the applicable rules of good research practice, data protection, animal welfare, research ethics, and occupational safety.

§4.2 The CRC implements equal-opportunity and accessibility measures and monitors participation in recruitment, training, and governance.

§4.3 Concerns may be raised confidentially with the spokesperson, the CRC management, or the competent independent offices of the participating institutions.

§5 Research-data governance

§5.1 Projects document data, code, samples, access rights, retention, and publication in project-specific data-management plans.

§5.2 Central data stewardship defines common metadata and quality standards, manages controlled access, and supports publication according to the FAIR principles and applicable legal requirements.

5 Declarations

I/We hereby confirm the accuracy of the information contained in the proposal, including the research profiles.

The *DFG Rules of Procedure for Dealing with Scientific Misconduct* (*known as VerfOwF*) apply to individuals with a high level of scientific/academic responsibility in connection with funding proposals submitted by universities. The applicant university confirms that all individuals with a high level of scientific/academic responsibility in the project proposal submitted by the university (cf. programme guidelines, DFG form 50.06) recognise the *DFG Rules of Procedure for Dealing with Scientific Misconduct* as legally binding by submitting a corresponding declaration of commitment.

In submitting this proposal, all project leaders undertake to comply with the relevant discipline-specific research ethics guidelines, to determine whether a statement from the responsible ethics committee is required and, where necessary, to obtain it in good time before the start of funding. They will inform the DFG immediately if an ethics committee expresses substantive concerns or issues a negative statement and will submit the statement if the DFG requests it.

The DFG takes the protection of your personal data very seriously. Please note the DFG's data protection notice on research funding, which can be viewed and downloaded at `www.dfg.de/privacy_policy`

Please forward this information to those individuals whose data will be processed by the DFG because they are mentioned in your proposal.

I/We have read the DFG's data protection notice.

I/We consent to the processing and transfer of special categories of personal data (including the data of all parties involved) for the purposes stated in the Privacy Policy if, after careful consideration, I/we provide such data either now or at a later date.

Example City
<date>

Prof. Dr. med. Morgan Sample
(Spokesperson of Collaborative Research Centre)

Example City
<date>

Prof. Dr. Casey President
(Rector, chancellor, or president of Partner University)

Example City
<date>

Prof. Dr. Robin Rector
(Rector, chancellor, or president of Example University)



Project	Title	Project leader(s)
A01	Microbial signalling across epithelial barriers	Kim **Sample** Alex **Example**
A02 N	Metabolic control of tissue-resident immunity	Kim **Sample** Samira **Hartmann**
A03	Microbiome-derived metabolites and immune memory	Samira **Hartmann** Noah **Lindberg**
A04 N	Spatial niches of host–microbe interactions	Noah **Lindberg** Alexis-Marie **Winters-Scholz**
A05	Dietary control of microbial community dynamics	Alexis-Marie **Winters-Scholz** Leonie **Navarro**
A06 N	Modelling microbial resilience and recovery	Leonie **Navarro** Daniel **Okafor**
B01	Host defence during microbiome perturbation	Hansi **Meier** Alex **Example**
B02 N	Molecular signatures of tissue regeneration	Kimmi **Miller** Daniel **Okafor**
B03	Immune circuits linking intestine and liver	Hansi **Meier** Mira **Sato**
B04 N	Cross-organ consequences of inflammatory stress	Kimmi **Miller** Mira **Sato**
B05	Patient-derived models of metabolic disease	Friedhelm **Redata** Robin **Keller**
B06 N	Translating microbial biomarkers into interventions	Daniel **Okafor** Robin **Keller**
Q01	Integrated imaging and analysis services	Alex **Example** Noah **Lindberg**
Q02	Integrated Research Training Group	Friedhelm **Redata** Leonie **Navarro**
Q03	Central coordination and data stewardship	Hansi **Meier** Robin **Keller**

5 Complete example CV appendix

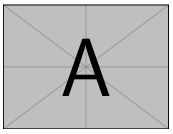
The following pages contain the complete fictional CV appendix generated from `crc-cv.tex` and the shared consortium metadata. It includes the cover, contents, and all twelve PI CVs. Each shows three career stages, supplementary research information, activities, supervision, distinctions, and further information, followed by two category-A articles, a category-B dataset, and the privacy statement. The bibliography records are explicitly fictional and use person-specific selection keywords such as `per12-A` and `per12-B`. Replace them with real publications and the appropriate keywords, as explained in Section 3.3.4. See Example 3.1 for the code that selects all PIs or one person.

As with the proposal, the CV is built with `final=true`. Every imported page is reduced to 82%, framed, and labelled “Example CV appendix”; the manual page number stays outside the frame. Open the accompanying standalone example CV PDF for its own clickable references and bookmarks. Run `make cv-doc-example` to regenerate just that PDF, or `make doc` to rebuild both examples and this manual from the DTX.

Collaborative Research Centre 9999

Mechanisms of Host–Microbiome Communication

From Molecular Interactions to Organ Function



Example University

Funding Proposal

for the third Funding Period
2027 | 2–2028–2029–2030–2031 | 1

Appendix: Research profiles



Proposal for the third Funding Period of
Collaborative Research Centre 9999

“Mechanisms of Host–Microbiome Communication: From Molecular Interactions to Organ Function”

funded since

2019-07-01

for

2027 | 2–2028–2029–2030–2031 | 1

Coordinating university:

Example University

Spokesperson of the Collaborative Research Centre:

Morgan Sample, Prof. Dr. med.
0000-0000-0000-0006
Institute of Microbial Systems Medicine
Example University
University Avenue 1, 12345 Example City (Germany)
+49 000 100-6000
morgan.sample@example.org

Management or office of the Collaborative Research Centre:

Jamie Coordinator, Dr. rer. nat.
0000-0000-0000-0007
Department of Translational Biosciences
Partner University
Research Road 2, 54321 Partner City (Germany)
+49 000 200-7000
jamie.coordinator@example.org

Example City
<date>

Prof. Dr. med. Morgan Sample
(Spokesperson of Collaborative Research Centre)

Example City
<date>

Prof. Dr. Casey President
(Rector, chancellor, or president of Partner University)

Example City
<date>

Prof. Dr. Robin Rector
(Rector, chancellor, or president of Example University)

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Okafor, Daniel — 19

Redata, Friedhelm — 21

Sample, Kim — 23

Sato, Mira — 25

Winters-Scholz, Alexis-Marie — 27

Curriculum Vitae

Prof. Dr. phil.

Alex Example

Personal Data	
Title	Prof. Dr. phil.
First name	Alex
Name	Example
Current position	Research group leader
Current institution(s)/ site(s), country	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)
ORCiD	0000-0000-0000-0002

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Data Science, Partner University
Doctorate	1999 Dr. phil., Partner University
since 2016: Current appointment	Research group leader *Develops statistical models linking molecular measurements with clinical phenotypes.*
2004–2015: Independent research	Independent research group leader *Established a research team on multimodal data integration and reproducible analysis and developed collaborative methods.*
2000–2003: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to multimodal data integration and reproducible analysis.*

Supplementary Career Information

Research focus: multimodal data integration and reproducible analysis. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Leads the data standards working group and maintains the consortium code-review programme. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supervises doctoral projects in statistics and biology, with paired computational and experimental mentors. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. A. Example. “Illustrative study of multimodal data integration and reproducible analysis”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. A. Example. “Reproducible methods for multimodal data integration and reproducible analysis”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. A. Example. *Fictional demonstration dataset: multimodal data integration and reproducible analysis*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Partner University award for open research software. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr. phil. Alex Example) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Samira Hartmann

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Samira
Name	Hartmann
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Centre for Computational Microbiology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)
ORCiD	0000-0000-0000-0016

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Combines microbial genetics with functional assays of host metabolic responses.*
2015–2017: Independent research	Independent research group leader *Established a research team on microbial metabolic pathways and immune adaptation and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to microbial metabolic pathways and immune adaptation.*

Supplementary Career Information

Research focus: microbial metabolic pathways and immune adaptation. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Coordinates the pathway-discovery workshop and joint protocol reviews across the consortium. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supervises paired microbiology and immunology projects, including laboratory exchanges for doctoral researchers. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. S. Hartmann. “Illustrative study of microbial metabolic pathways and immune adaptation”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. S. Hartmann. “Reproducible methods for microbial metabolic pathways and immune adaptation”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. S. Hartmann. *Fictional demonstration dataset: microbial metabolic pathways and immune adaptation*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example Foundation award for microbial systems research. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr. rer. nat. Samira Hartmann) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Robin Keller

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Robin
Name	Keller
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany) Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)
ORCiD	

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Combines patient-derived models with biomarker translation and stewardship of shared research data.*
2015–2017: Independent research	Independent research group leader *Established a research team on patient-derived models and biomarker translation and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to patient-derived models and biomarker translation.*

Supplementary Career Information

Research focus: patient-derived models and biomarker translation. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Chairs the consortium reproducibility forum and coordinates workshops on reusable clinical research data. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Co-supervises doctoral researchers across clinical and computational projects, with individual career-development plans. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. R. Keller. “Illustrative study of patient-derived models and biomarker translation”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. R. Keller. “Reproducible methods for patient-derived models and biomarker translation”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. R. Keller. *Fictional demonstration dataset: patient-derived models and biomarker translation*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example University award for reproducible translational medicine. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information
Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data

✗ I (Prof. Dr. rer. nat. Robin Keller) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court.

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Curriculum Vitae

Prof. Dr. rer. nat.

Noah Lindberg

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Noah
Name	Lindberg
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Centre for Computational Microbiology Northbridge University Science Avenue 3, 34567 Northbridge (Germany) Department of Experimental Immunology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)
ORCiD	

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Links spatial profiling to microscopy and quantitative descriptions of immune-cell neighbourhoods.*
2015–2017: Independent research	Independent research group leader *Established a research team on spatial immune niches and quantitative imaging and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to spatial immune niches and quantitative imaging.*

Supplementary Career Information

Research focus: spatial immune niches and quantitative imaging. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Leads the imaging quality-control forum and teaches the annual spatial-analysis course. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Mentors doctoral researchers in image analysis, with shared supervision across experimental and computational groups. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. N. Lindberg. “Illustrative study of spatial immune niches and quantitative imaging”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. N. Lindberg. “Reproducible methods for spatial immune niches and quantitative imaging”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. N. Lindberg. *Fictional demonstration dataset: spatial immune niches and quantitative imaging*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example Imaging Society early-career research prize. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information
Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data

✗ I (Prof. Dr. rer. nat. Noah Lindberg) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court.

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I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Dr. rer. nat.

Hansi Meier

Personal Data	
Title	Dr. rer. nat.
First name	Hansi
Name	Meier
Current position	Senior research scientist
Current institution(s)/ site(s), country	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)
ORCiD	0000-0000-0000-0003

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Microbiology, Partner University
Doctorate	1999 Dr. rer. nat., Partner University
since 2018: Current appointment	Senior research scientist *Coordinates experimental infection models and validation of immune mechanisms.*
2004–2017: Independent research	Independent research group leader *Established a research team on microbial sensing and innate immune memory and developed collaborative methods.*
2000–2003: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to microbial sensing and innate immune memory.*

Supplementary Career Information

Research focus: microbial sensing and innate immune memory. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Organises a practical course on host defence and serves on the laboratory methods committee. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Mentors early-career scientists in experimental design, biosafety and reproducible laboratory practice. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. H. Meier. “Illustrative study of microbial sensing and innate immune memory”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. H. Meier. “Reproducible methods for microbial sensing and innate immune memory”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. H. Meier. *Fictional demonstration dataset: microbial sensing and innate immune memory*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example Society prize for translational infection research. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Dr. rer. nat. Hansi Meier) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Kimmi Miller

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Kimmi
Name	Miller
Current position	Professor of Molecular Medicine
Current institution(s)/ site(s), country	Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)
ORCiD	0000-0000-0000-0004

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Molecular Medicine, Partner University
Doctorate	1999 Dr. rer. nat., Partner University
since 2015: Current appointment	Professor of Molecular Medicine *Connects organoid perturbation experiments with molecular measurements of tissue recovery.*
2004–2014: Independent research	Independent research group leader *Established a research team on epithelial repair and tissue regeneration and developed collaborative methods.*
2000–2003: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to epithelial repair and tissue regeneration.*

Supplementary Career Information

Research focus: epithelial repair and tissue regeneration. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Coordinates the organoid methods network and an interdisciplinary regeneration seminar. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supervises doctoral researchers in molecular medicine and provides mentoring for clinician scientists. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. K. Miller. “Illustrative study of epithelial repair and tissue regeneration”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. K. Miller. “Reproducible methods for epithelial repair and tissue regeneration”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. K. Miller. *Fictional demonstration dataset: epithelial repair and tissue regeneration*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Partner University award for collaborative tissue research. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr. rer. nat. Kimmi Miller) expressly consent to the processing of the voluntary (optional) information, including “special categories of personal data” in connection with the DFG’s review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit “special categories of personal data” relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG’s Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Leonie Navarro

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Leonie
Name	Navarro
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Institute of Metabolic Medicine Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)
ORCiD	

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Integrates dietary interventions with metabolic profiling and predictive modelling.*
2015–2017: Independent research	Independent research group leader *Established a research team on nutrient sensing and predictive metabolic models and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to nutrient sensing and predictive metabolic models.*

Supplementary Career Information

Research focus: nutrient sensing and predictive metabolic models. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Organises the metabolic modelling summer school and contributes to graduate admissions panels. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Mentors doctoral researchers in model validation, interdisciplinary communication and career planning. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. L. Navarro. “Illustrative study of nutrient sensing and predictive metabolic models”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. L. Navarro. “Reproducible methods for nutrient sensing and predictive metabolic models”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. L. Navarro. *Fictional demonstration dataset: nutrient sensing and predictive metabolic models*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example Society distinction for integrative metabolic research. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr. rer. nat. Leonie Navarro) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Daniel Okafor

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Daniel
Name	Okafor
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Institute of Metabolic Medicine Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany) Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)
ORCiD	

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Connects patient-derived measurements with biomarker validation and experimental tissue models.*
2015–2017: Independent research	Independent research group leader *Established a research team on microbial biomarkers and clinical tissue responses and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to microbial biomarkers and clinical tissue responses.*

Supplementary Career Information

Research focus: microbial biomarkers and clinical tissue responses. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Coordinates the translational data-analysis workshop and the clinical assay harmonisation group. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supervises clinician scientists and laboratory researchers in joint translational thesis projects. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. D. Okafor. “Illustrative study of microbial biomarkers and clinical tissue responses”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. D. Okafor. “Reproducible methods for microbial biomarkers and clinical tissue responses”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. D. Okafor. *Fictional demonstration dataset: microbial biomarkers and clinical tissue responses*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Westhaven Research Centre translational science award. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data

✗ I (Prof. Dr. rer. nat. Daniel Okafor) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court.

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I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr.-Ing.

Friedhelm Redata

Personal Data	
Title	Prof. Dr.-Ing.
First name	Friedhelm
Name	Redata
Current position	Professor of Research Training
Current institution(s)/ site(s), country	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany)
ORCiD	0000-0000-0000-0005

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Biomedical Engineering, Example University
Doctorate	1998 Dr.-Ing., Example University
since 2014: Current appointment	Professor of Research Training *Directs the graduate programme and evaluates interdisciplinary training formats.*
2003–2013: Independent research	Independent research group leader *Established a research team on structured doctoral training and research integrity and developed collaborative methods.*
1999–2002: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to structured doctoral training and research integrity.*

Supplementary Career Information

Research focus: structured doctoral training and research integrity. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Chairs the graduate curriculum committee and coordinates the annual doctoral retreat. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supports thesis committees, individual development plans and peer mentoring across all research areas. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. F. Redata. “Illustrative study of structured doctoral training and research integrity”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. F. Redata. “Reproducible methods for structured doctoral training and research integrity”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. F. Redata. *Fictional demonstration dataset: structured doctoral training and research integrity*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example University prize for doctoral education. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr.-Ing. Friedhelm Redata) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Kim Sample

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Kim
Name	Sample
Current position	Professor and Director
Current institution(s)/ site(s), country	Institute of Microbial Systems Medicine Example University University Avenue 1, 12345 Example City (Germany) Department of Translational Biosciences Partner University Research Road 2, 54321 Partner City (Germany)
ORCiD	0000-0000-0000-0001

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Biology, Example University
Doctorate	2000 Dr. rer. nat., Example University
since 2013: Current appointment	Professor and Director *Leads the mechanistic microbiome programme and coordinates the shared experimental strategy.*
2005–2012: Independent research	Independent research group leader *Established a research team on microbial metabolites and epithelial signalling and developed collaborative methods.*
2001–2004: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to microbial metabolites and epithelial signalling.*

Supplementary Career Information

Research focus: microbial metabolites and epithelial signalling. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Chairs the consortium steering committee and an annual host–microbiome symposium. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Mentors laboratory and clinical doctoral researchers through joint thesis committees. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. **K. Sample**. “Illustrative study of microbial metabolites and epithelial signalling”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. **K. Sample**. “Reproducible methods for microbial metabolites and epithelial signalling”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. **K. Sample**. *Fictional demonstration dataset: microbial metabolites and epithelial signalling*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example Academy fellowship for interdisciplinary microbiome research. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data

✗ I (Prof. Dr. rer. nat. Kim Sample) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court.

I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent).

I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Mira Sato

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Mira
Name	Sato
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Centre for Clinical Translation Westhaven Research Centre Science Avenue 3, 34567 Westhaven (Germany)
ORCiD	

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Coordinates tissue sampling and comparative analyses of inflammatory responses across organs.*
2015–2017: Independent research	Independent research group leader *Established a research team on inflammatory circuits between intestine and liver and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to inflammatory circuits between intestine and liver.*

Supplementary Career Information

Research focus: inflammatory circuits between intestine and liver. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Leads the clinical sample-coordination group and teaches a course on cross-organ immunology. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Mentors doctoral researchers in clinical cohort design, sample governance and mechanistic validation. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. M. Sato. “Illustrative study of inflammatory circuits between intestine and liver”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. M. Sato. “Reproducible methods for inflammatory circuits between intestine and liver”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. M. Sato. *Fictional demonstration dataset: inflammatory circuits between intestine and liver*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example Clinical Research Society award for collaborative immunology. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr. rer. nat. Mira Sato) expressly consent to the processing of the voluntary (optional) information, including "special categories of personal data" in connection with the DFG's review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit "special categories of personal data" relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG's Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

Curriculum Vitae

Prof. Dr. rer. nat.

Alexis-Marie Winters-Scholz

Personal Data	
Title	Prof. Dr. rer. nat.
First name	Alexis-Marie
Name	Winters-Scholz
Current position	Professor of Systems Medicine
Current institution(s)/ site(s), country	Department of Experimental Immunology Northbridge University Science Avenue 3, 34567 Northbridge (Germany)
ORCiD	

Qualifications and Career	
Stages	**Periods and Details**
Degree programme	Life Sciences, Example University
Doctorate	2010 Dr. rer. nat., Example University
since 2018: Current appointment	Professor of Systems Medicine *Develops experimental models of diet-dependent microbial colonisation and mucosal responses.*
2015–2017: Independent research	Independent research group leader *Established a research team on dietary perturbations and mucosal microbial communities and developed collaborative methods.*
2011–2014: Postdoctoral training	Postdoctoral researcher *Developed experimental and analytical approaches to dietary perturbations and mucosal microbial communities.*

Supplementary Career Information

Research focus: dietary perturbations and mucosal microbial communities. A visiting fellowship at the Example Centre for Interdisciplinary Research supported new collaborations and exchange of methods.

Activities in the Research System

Coordinates the experimental-models working group and a public seminar series on microbiome science. Contributes to peer review and open teaching materials.

Supervision of Researchers in Early Career Phases

Supervises doctoral projects in microbial ecology and supports exchange placements between partner laboratories. Regular progress reviews include research integrity and publication planning.

Scientific results

Category A – Articles in peer-reviewed journals, contributions to peer-reviewed conferences or to anthology volumes, and book publications

1. A.-M. Winters-Scholz. “Illustrative study of dietary perturbations and mucosal microbial communities”. In: *Fictional Journal of Consortium Research* 1 (2025), pp. 1–8.
2. A.-M. Winters-Scholz. “Reproducible methods for dietary perturbations and mucosal microbial communities”. In: *Fictional Journal of Consortium Research* 1 (2024), pp. 1–8.

Scientific results

Category B – Any other form of published results

1. A.-M. Winters-Scholz. *Fictional demonstration dataset: dietary perturbations and mucosal microbial communities*. Version 1.0. Example Research Repository, 2025.

Academic Distinctions

2024: Example University award for experimental microbiome research. 2021: consortium mentoring recognition for support of early-career researchers.

Other Information

Makes analysis protocols and training resources available for reuse. No commercial interests related to the proposed research. All career details and distinctions in this CV are fictional examples.

Data protection and consent to the processing of optional data	
✗	I (Prof. Dr. rer. nat. Alexis-Marie Winters-Scholz) expressly consent to the processing of the voluntary (optional) information, including “special categories of personal data” in connection with the DFG’s review and decision-making process regarding my proposal. This also includes forwarding my data to the external reviewers, committee members and, where applicable, foreign partner organisations who are involved in the decision-making process. To the extent that these recipients are located in a third country (outside the European Economic Area), I additionally consent to them being granted access to my data for the above-mentioned purposes, even though a level of data protection comparable to EU law may not be guaranteed. For this reason, compliance with the data protection principles of EU law is not guaranteed in such cases. In this respect, there may be a violation of my fundamental rights and freedoms and resulting damages. This may make it more difficult for me to assert my rights under the General Data Protection Regulation (e.g. information, rectification, erasure, compensation) and, if necessary, to enforce these rights with the help of authorities or in court. I may *revoke* my consent in whole or in part at any time – with effect for the future, freely and without giving reasons – vis-à-vis the DFG (postmaster@dfg.de). The lawfulness of the processing carried out up to that point remains unaffected. Insofar as I transmit “special categories of personal data” relating to third parties, I confirm that the necessary legitimation under data protection law exists (e.g. based on consent). I have taken note of the DFG’s Data Protection Notice relating to research funding, which I can access at `www.dfg.de/privacy_policy` and I will forward it to such persons whose data the DFG processes as a result of being mentioned in this CV.

6 Code documentation

The following part is meant for developers. Don't worry if you might not understand parts of the code. We try to use as much LaTeX3 as possible for a more granular data handling.

6.1 Documentclass

The class requires the June 2022 LaTeX kernel, whose native `\ProcessKeyOptions` interface processes class options through expl3 key families. `\ProvidesExplClass` declares the class to the kernel and simultaneously activates the expl3 (LaTeX3) syntax, so every line from here onwards may use the `:`, `_` and `@@` conventions of the L3 programming layer.

The ⟨*version*⟩ and ⟨*date*⟩ values come from the single definition at the very top of the `.dtx` file (`\def\dfgVersion` etc.), so a version bump only ever needs to be made in one place.

```
\NeedsTeXFormat{LaTeX2e}[2022-06-01]
\ProvidesExplClass
{dfg-crc}
{\dfgVersionDate}
{\dfgVersion}
{Create DFG Collaborative Research Centre proposals and CVs}
```

6.1.1 Message catalogue

All user-visible diagnostics are declared here once using `\msg_new:nnn` from the L3 message system. Centralising them serves several purposes:

1. The exact wording of every log line can be reviewed and updated in one place without searching through the entire code.
2. They are bound to the `dfg` module name, so a user can grep the `.log` with `grep "Package dfg"` to see a condensed overview.
3. Using the L3 message system instead of raw `\PackageWarning` gives consistent formatting, the fileline context via `\msg_line_context:`, and the ability to suppress or redirect messages at a global level with `\msg_redirect_module:nnn`.

The engine check fails early with a direct instruction instead of letting fontspec or a later `\directlua` call produce a less helpful error. The next message is emitted when an optional input file (e.g. a metadata or preamble fragment) cannot be found on the search path. Compilation is not aborted, but the user is strongly warned.

```
\msg_new:nnn { dfg } { unsupported-engine }
{
  The~dfg-crc~class~requires~LuaLaTeX.^^J
  Compile~this~document~with~lualatex~or~latexmk~-lualatex.
}
\sys_if_engine_luatex:F
  { \msg_fatal:nn { dfg } { unsupported-engine } }

\msg_new:nnn { dfg } { file-not-found }
{ File~'#1'~not~found.^^J
  ~You~can~proceed~without~it~
  but~it~is~not~recommended.
}
```

database-not-found Emitted by `\getEntry` → P. 2, `\compareEntry` → P. 3 and each table command when the database name passed by the user does not exist in the datatool register. The most common cause is a typo in the database name, or calling a retrieval command before the corresponding `\addEntry` → P. 2 block has been processed.

```
\msg_new:nnn { dfg } { database-not-found }
{ Unknown~ database~ name:~ #1^^J^^J
  Database~'#1'~not~found~\msg_line_context:.~
  Either~you~have~misspelled~it~or~
  ~the~database~has~not~yet~been~created.
}
```

database-name-not-given Fired when the optional ⟨*database*⟩ argument of `\getEntry` → P. 2 is empty because `\tmpLocalDB` has not been set and no explicit database name was supplied. The message text suggests two remedies: set `\tmpLocalDB` at the top of a section, or pass the database name inline.

```
\msg_new:nnn { dfg } { database-name-not-given }
{ Missing~ database~ name~ for~ entry:~ #1^^J ^^J
  You~forgot~to~pass~a~database~name~\msg_line_context:.~
  You~can~provide~a~name~with~ e.g.~ '\def\tmpLocalDB{<database>}'
  ~or~within~entries~like~
  '\getEntry[<database>]{<id>}{<field>}'.
}
```

biblatex-ext-oa-doiapi-unavailable Warns when biblatex-ext-oa's DOI API integration cannot be enabled because `socket.http` is not available in the current LuaTeX run. The package is still loaded, so manual BibTeX annotations continue to work.

```
\msg_new:nnn { dfg } { biblatex-ext-oa-doiapi-unavailable }
{ biblatex-ext-oa~ DOI~ API~ lookup~ disabled:~ LuaSocket~ module~
  'socket.http'~ is~ not~ available~ in~ this~ LuaTeX~ run.~
  Compile~ with~ '--socket'~ to~ enable~ DOI~ API~ requests.~
  Open-access~ markers~ still~ work~ from~ manual~ BibTeX~ annotations.
}
```

biblatex-ext-oa-doiapi-email-missing DOI lookups require a real contact address. An empty default keeps network access opt-in and prevents all users of the class from sharing a placeholder or maintainer address.

```
\msg_new:nnn { dfg } { biblatex-ext-oa-doiapi-email-missing }
{ biblatex-ext-oa~ DOI~ API~ lookup~ disabled:~ no~contact~address~was~set.^^J
  Set~'doiAPIMail=<address>'~with~\string\changeSettings~to~enable~lookups.
  Open-access~markers~still~work~from~manual~BibTeX~annotations.
}
```

missing-orcid Triggered specifically by `\getORCiD` → P. 71 when the `orcid` field is absent or empty for the requested person. The message body reproduces the exact `\addEntry` → P. 2 snippet the user needs to fill in, lowering the barrier to fixing the data.

```
\msg_new:nnn { dfg } { missing-orcid }
{DB:~'#1';~ID:~'#2';~null/empty:~'orcid'.~^^J^^J
  The~ORCiD~for~the~person~with~the~ID~'#2'~has~not~been~given.~
  You~asked~for~it~\msg_line_context: ~^^J
  If~the~person~has~an~ORCiD, ~go~to~where~you~have~written~^^J
  \addEntry{#1}{^^J
   id   =#2, ^^J
   orcid=<INSERT~ORCID~HERE>^^J
  }
}
```

missing-key Generic warning for any database field that is null or empty when `\getEntry` → P. 2 tries to retrieve it. The check is deliberately suppressed during the tabularray cell-measuring pass (`\lTblrMeasuringBool` is `true`) to prevent each cell from generating duplicate log entries — tabularray renders each cell twice for width calculation before the final output pass.

```
\msg_new:nnn { dfg } { missing-key }
{DB:~'#1';~ID:~'#2';~null/empty:~'#3'.^^J^^J
  In~ the~ database~ '#1', the~ key~ '#3'~ which~ belongs~ to~ the~ ID~ '#2'~
  is~ either~ null/not~ given~ or~ simply~ empty.~^^J
  You~ asked~ for~ it~ \msg_line_context:  }
```

item-skipped / \dfgItemSkipped Tables built from the datatool databases are the most expensive part of each compilation run because they involve full database queries. When either fastcompile mode is active, generated table/list commands call \dfgItemSkipped instead of rendering the real content. This prints a red placeholder in the PDF and issues an item-skipped note to the log. The colour red is overridden to white when final=true, so the placeholder is invisible in a submission build.

```
\msg_new:nnn { dfg } { item-skipped }
{Item~ #1~ skipped~ (compile-time).^^J
  To~ see~ the~ item,~ disable~ fast~ compilation^^J
  in~ the~ documentclass-options.^^J
  The~ item~ is~ \msg_line_context:  }
\NewDocumentCommand{\dfgItemSkipped}{m}{%
  \msg_note:nne {dfg}{item-skipped}{ #1 }
  \group_begin:
  \color{error}
  Item~ \emph{\normalcolor#1}~ skipped~ due~ to~ faster~ compilation.~\\
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  { To~ see~ the~ item,~ set~ \texttt{fastcompile-project=false}~(default)~ }
  { To~ see~ the~ item,~ set~ \texttt{fastcompile=false}~(default)~ }
  in~ the~ documentclass-options.
  \group_end:
}
```

research-area-not-found The property list \g_@@_dfg_research_areas_prop contains every DFG research area code from the official classification list. When a code stored in the project database does not appear in that map the formatted label cannot be generated and this warning is raised. The raw code is still printed so the proposal remains compilable.

```
\msg_new:nnn { dfg } { research-area-not-found }
{ Unknown~ research~ area:~ #1^^J^^J
  Doublecheck~ the~ research~ area~ in~ the~ database:~^^J
  \addEntry{project}{~^^J
   research-area~ =~ {#1}~^^J
  }
}
```

item-not-matched Raised as a generic fall-through in \str_case:VnF dispatchers where a string value is not matched by any of the expected branches.

```
\msg_new:nnn { dfg } { item-not-matched }
{ The~ string~ '#1'~does~not~match.~^^J^^J
  Doublecheck~ your~input!~^^J
  The~ item~ is~ \msg_line_context:
}
```

unknown-key and invalid-value These two messages are the default handlers wired into the unknown .code:n and \meta{choice} / unknown .code:n slots of every \keys_define:nn block. They give the user immediate feedback in the log whenever a key name is misspelled or a value is outside the permitted set, instead of silently ignoring the input or producing a cryptic low-level TeX error.

```
\msg_new:nnn { dfg } { unknown-key }
{
  Key~ '\l_keys_path_str'~ is~ unknown~ and~ being~ ignored.~^^J^^J
  You~passed~the~value~'#1'~ to~ that~ key.~^^J
  Check~the~key~spelling~and~the~keys~documented~for~this~family.~^^J
  The~value~was~supplied~\msg_line_context:.
}
```

Warning issued when a key receives a value outside its set of permitted alternatives. This message is the standard fallback for keys defined with the l3keys→CTAN .choices:nn property when their /unknown handler delegates to the dispatcher \person_questions_invalid:n. The two arguments are the rejected value and a comma-separated list of the values that would have been accepted. The full key path is taken from \l_keys_path_str, so the warning identifies not only the offending value but also which key was being set when the error occurred – useful when the same value name (such as yes or 1) is plausible for several keys but only valid for some of them. The warning does not stop processing; the database field is simply not created for

the rejected value, and subsequent keys in the same \addEntry→ P. 2 call continue to be parsed. Diagnostics list input values users can supply, not internal variable names. List-valued keys report each rejected item separately, preserving valid items. Format checks use invalid-format with examples; handlers that replace an invalid choice use invalid-value-fallback and name the replacement. Date warnings explicitly state that the input is retained.

```
\msg_new:nnn { dfg } { invalid-value }
{
  Invalid~value~'#1'~for~key~'\l_keys_path_str'.~^^J^^J
  Allowed~values~are:~'#2'.~^^J
  This~value~is~ignored.~^^J
  The~value~was~supplied~\msg_line_context:.
}
\msg_new:nnn { dfg } { invalid-value-fallback }
{
  Invalid~value~'#1'~for~key~'\l_keys_path_str'.~^^J^^J
  Allowed~values~are:~'#2'.~^^J
  Using~'#3'~instead.~^^J
  The~value~was~supplied~\msg_line_context:.
}
\msg_new:nnn { dfg } { invalid-format }
{
  Invalid~value~'#1'~for~key~'\l_keys_path_str'.~^^J^^J
  Expected~#2.~This~value~is~ignored.~^^J
  The~value~was~supplied~\msg_line_context:.
}
\msg_new:nnn { dfg } { unknown-name }
{
  Unknown~#1~'#2'.~^^J
  Allowed~values~are:~'#3'.~^^J
  A~mismatch~marker~is~printed.~^^J
  The~value~was~used~\msg_line_context:.
}
\msg_new:nnn { dfg } { unknown-institution-order }
{
  Invalid~order~'#1'~for~ParticipatingInstitutions.~^^J
  Allowed~values~are:~'alpha,~count'.~Using~'alpha'~instead.~^^J
  The~order~was~supplied~\msg_line_context:.
}
\msg_new:nnn { dfg } { invalid-project-count-area }
{
  No~active,~list-considered~projects~were~found~for~area~'#1'~in~
  \string\getProjectCount;~printing~0.
}
\msg_new:nnn { dfg } { collaboration-focus-not-found }
{
  Cannot~draw~the~project~collaboration~plot:~focus~project~'#1'~is~not~an~
  eligible~row~of~the~project~database.
}
\msg_new:nnn { dfg } { collaboration-target-not-found }
{
  Ignoring~collaboration~'#1--#2':~project~'#2'~is~not~an~eligible~row~of~
  the~project~database.
}
\msg_new:nnn { dfg } { plot-current-project-missing }
{
  Cannot~print~the~'#1'~plot:~no~current~project~has~been~set.~Start~a~
  project~first,~or~pass~an~explicit~project~ID.
}
```

Warning issued when a date-typed key receives a value that does not match the ISO-8601 calendar-date shape YYYY-MM-DD. Used by the contract-information helper \dfg_contract_date:nn for the fields C2a→ P. 52, C2b→ P. 52, and C3a→ P. 52, all of which are expected to hold dates. The two arguments are the field name and the offending value. The check is purely structural: it verifies that the value consists of four digits, a hyphen, two digits, a hyphen, and two digits, but it does not validate the resulting date as a real calendar entry (so 2025-13-45 passes the format check). When the warning fires, the supplied value is still recorded in the database so the typeset output makes the malformed entry visible; this trades log-cleanliness for ease of locating the source of the problem in the rendered document.

```
\msg_new:nnn { dfg } { invalid-date }
{
  Invalid~date~'#2'~for~key~'\l_keys_path_str'~(field~'#1').~^^J
  Expected~YYYY-MM-DD,~for~example~2026-10-11,~or~an~empty~value.~^^J
  The~supplied~value~is~retained;~correct~its~format~in~the~source.~^^J
  The~value~was~supplied~\msg_line_context:.
}
```

Requested-funding staff diagnostics The staff roster is deliberately permissive at output time: incomplete or unresolved records remain visible in the document while these messages make the underlying metadata problem searchable in the log.

```
\msg_new:nnn { dfg } { requested-staff-no-data }
{ No~RequestedFundingStaff~data~have~been~provided~for~project~'#1'. }
\msg_new:nnn { dfg } { requested-staff-required }
{ RequestedFundingStaff~entry~is~missing~required~metadata:~#1. }
\msg_new:nnn { dfg } { requested-staff-institution-required }
{
  RequestedFundingStaff~entry~for~project~'#1'~has~neither~ref-person-id~nor~
  ref-institution-id.~Provide~ref-institution-id~when~the~person~is~not~
  referenced;~otherwise~the~institution~cannot~be~resolved.
}
\msg_new:nnn { dfg } { requested-staff-missing-field }
{ Project~'#1':~requested-funding~staff~#2~is~missing. }
\msg_new:nnn { dfg } { requested-staff-unresolved-person }
{ Project~'#1':~person~reference~'#2'~cannot~be~resolved. }
\msg_new:nnn { dfg } { requested-staff-unresolved-institution }
{ Project~'#1':~institution~reference~'#2'~cannot~be~resolved. }
\msg_new:nnn { dfg } { requested-staff-unresolved-affiliation }
{ Project~'#1':~affiliation~reference~'#2'~cannot~be~resolved. }
\msg_new:nnn { dfg } { requested-staff-missing-abbreviation }
{ Affiliation~'#1'~has~no~abbreviation;~its~full~name~is~used. }
\msg_new:nnn { dfg } { duplicate-institution-abbreviation }
{
  Institution~abbreviation~'#1'~is~assigned~to~different~institutions:~^^J
  '#2'~and~'#3'~(institution~ids:~#4).~^^J
  Use~a~unique~abbreviation~for~each~institution.
}
\msg_new:nnn { dfg } { duplicate-entry-id }
{
  Duplicate~id~'#2'~in~database~'#1'.~^^J
  The~earlier~row(s)~are~removed;~the~later~entry~replaces~them.~^^J
  The~replacement~is~\msg_line_context:.
}
\msg_new:nnn { dfg } { requested-staff-duplicate }
{ Duplicate~RequestedFundingStaff~entry~for~#1. }
\msg_new:nnn { dfg } { funding-staff-cost-unavailable }
{ Automatic~staff-cost~calculation~is~not~available~for~#1. }
\msg_new:nnn { dfg } { existing-staff-no-data }
{ No~existing~staff~rows~were~found~in~database~'#1'. }
\msg_new:nnn { dfg } { existing-staff-no-organisation }
{ Existing~staff~entry~'#1'~cannot~be~assigned~to~an~affiliation. }
```

\StartTimer
\StopTimer

A simple pair of profiling helpers for measuring how long a stretch of LaTeX code takes to execute. Wrap the code under investigation between `\StartTimer` and `\StopTimer{`⟨*label*⟩`}`; the elapsed time in milliseconds is written to the terminal and the `.log` file, prefixed with the supplied ⟨*label*⟩ so several measurements in one run can be told apart.

Typical use:

```
\StartTimer
\ParticipatingInstitutionsList[count]
\StopTimer{ParticipatingInstitutionsList}
```

producing a line such as

```
TIMING ParticipatingInstitutionsList: 115 ms
```

in the log.

Only one measurement can be in progress at any one time, since the start time is held in a single shared integer variable. For nested or overlapping measurements, define additional counters or switch to the labelled-timer variant.

The implementation uses the kernel-native expl3 function `\sys_timer:`, which abstracts over the engine-specific timer and avoids an additional package dependency.

```
\int_new:N \l_@@_timer_int
```

Storage for the start timestamp. We use an `int` rather than a `tl` so the later subtraction in `\int_eval:n` is unambiguous.

```
\NewDocumentCommand { \StartTimer } { }
{
  \bool_if:NF \l_@@_class_fastcompile_project_bool
  { \int_set:Nn \l_@@_timer_int { \sys_timer: } }
```

```
}
```

`\sys_timer:` expands to the time since job start in *scaled seconds* – units of $1/65536$ s. We capture that raw value here and convert to milliseconds only at reporting time, to keep the start-up overhead minimal.

```
\NewDocumentCommand { \StopTimer } { m }
{
  \bool_if:NF \l_@@_class_fastcompile_project_bool
  {
    \iow_term:x
    {
      TIMING~#1:~
      \int_eval:n
      { ( \sys_timer: - \l_@@_timer_int ) * 1000 / 65536 }
      \c_space_tl ms
    }
  }
}
```

The arithmetic computes elapsed scaled-seconds first, then multiplies by 1000 before dividing by 65536. Doing the multiplication first preserves precision: a naive `(now - start) / 65536 * 1000` would truncate sub-second measurements to zero. The result is an integer count of milliseconds, accurate to roughly ± 1 ms.

`\iow_term:x` writes to the terminal *and* the log, fully expanding its argument once – which is what makes `\int_eval:n` produce a literal number rather than the unevaluated expression. `\c_space_tl` inserts a real space between the number and the `ms` unit; a literal space there would be consumed by `expl3`'s tokenisation rules.

Note that the reported time is wall-clock and includes any `\typeout`, file I/O, or shell-escape work performed inside the timed block. For tight measurements of pure computation, silence debugging output during the run, and take the second of two consecutive timings to avoid first-run effects such as font caching or `.aux` reads.

6.1.2 Documentclass Options

We have very few global options we use from the KOMAscript. But you can use any known option of the KOMAscript `scrbook`.

```
\PassOptionsToClass{
  fontsize  =10pt,
  twoside   =true,
  BCOR      =5mm,
  headings  =optiontoheadandtoc,
  parskip   =half-,
  numbers   =noenddot,
  toc       =indentunnumbered
}{scrbook}
```

6.1.3 Class options (`dfg/class` key family)

All class-level toggles live in the `dfg/class` key family and are processed via `\ProcessKeyOptions`. They control the two major compilation modes (`fastcompile` → P. 65, `fastcompile-project` → P. 65, `final` → P. 66) plus several debugging and output flags.

The general design principle is that every flag defaults to `false` or to a safe, submission-ready value. Authors enable the helpers they need during drafting and simply remove them (or set `final=true`) before submitting. Setting `final=true` enforces this by hard-resetting all drafting flags to `false` automatically.

```
\keys_define:nn { dfg / class }
{
```

The manual loads explicit example inputs, independent of proposal files.

```
  documentation .bool_set:N = \l_@@_class_documentation_bool,
  documentation .initial:n = false,
```

`final`→P. 66: Set to `true` immediately before submission. It forces `cmyk` colour model (required for print), silently sets `fastcompile=false`, `showtags=false`, `showmarginalia=false` and `skip10pub=false`, and turns the draft-green placeholder colour to white so it is invisible in the output.

```
226 final      .bool_set:N = \l_@@_class_final_bool,
227 final      .initial:n  = false,
228 Final      .meta:n     = { final = #1 },
229 Final      .default:n  = true,
```

`fastcompile`→P. 65: When `true`, generated tables and lists are replaced by a placeholder. It also suppresses role-card grids, the Gantt chart, and the ten-publications CV block while retaining metadata-backed prose references. Automatically disabled when `final=true`.

```
230 fastcompile   .bool_set:N = \l_@@_class_fastcompile_bool,
231 fastcompile   .initial:n  = false,
```

`fastcompile-project`→P. 65: Aggressive project-text drafting mode. It implies `fastcompile`→P. 65, but additionally prevents metadata rows from being loaded or restored, replaces calculated project headings with a state-only setup, uses draft rendering for graphics and links, and provides literal fallbacks for database-backed references. Authored sections, paragraphs, aims, work packages, labels, and inline citations remain. Automatically disabled when `final=true`.

```
232 fastcompile-project .bool_set:N = \l_@@_class_fastcompile_project_bool,
233 fastcompile-project .initial:n  = false,
```

`fastcompile-project-show-images`→P. 65: Opt back into real image rendering while aggressive project drafting is active. It has no effect on ordinary builds, where graphics are rendered normally. Final mode resets it to `false` together with the other drafting controls.

```
234 fastcompile-project-show-images .bool_set:N = \l_@@_class_fastcompile_project_show_images_bool,
235 fastcompile-project-show-images .initial:n  = false,
```

`fastcompile-project-show-publications`→P. 65: Controls the standard project publication heading, explicit project `\nocite` lists, project bibliographies, project-report references, and ending-project result lists during focused project drafting. Ordinary and final builds force it to `true`.

```
236 fastcompile-project-show-publications .bool_set:N = \l_@@_class_fastcompile_project_show_publications_bool,
237 fastcompile-project-show-publications .initial:n  = false,
```

`fastcompile-project-show-funding-tables`→P. 65: Retain the three project-local funding tables in aggressive project drafting mode. The option defaults to `true` and has no effect outside that mode.

```
238 fastcompile-project-show-funding-tables .bool_set:N = \l_@@_class_fastcompile_project_show_funding_tables_bool,
239 fastcompile-project-show-funding-tables .initial:n  = true,
```

`fastcompile-show-tags`→P. 65: Enables provenance overlays during fast compilation. Focused project lookups show their source row and requested field path because metadata rows are skipped.

```
240 fastcompile-show-tags .bool_set:N = \l_@@_class_fastcompile_show_tags_bool,
241 fastcompile-show-tags .initial:n  = false,
```

`showtags`→P. 65: Enables provenance overlays for direct database values and concise source labels for calculated values in ordinary builds. During fast compilation, `fastcompile-show-tags`→P. 65 controls the setting instead.

```
242 showtags   .bool_set:N = \l_@@_class_showtags_bool,
243 showtags   .initial:n  = false,
```

`showtags-tables`→P. 66: Controls only provenance overlays rendered in table cells. It defaults to `true`, but remains inert unless `showtags`→P. 65 is enabled. Keeping it separate lets authors debug prose without widening or visually overloading dense generated tables.

```
244 showtags-tables  .bool_set:N = \l_@@_class_showtags_tables_bool,
245 showtags-tables  .initial:n  = true,
```

showmarginalia→P. 65: Enables marginal notes that display running subtotals alongside the funding tables. Useful for cross-checking calculated sums during data entry. Disabled by final=true.

```
246  showmarginalia .bool_set:N = \l_@@_class_showmarginalia_bool,
247  showmarginalia .initial:n  = false,
```

skip10pub→P. 66: Suppresses the mandatory ten-publication block in each CV section. Because each bibliography list requires a full biber run and can take several seconds per PI, skipping it during early drafting significantly reduces compilation time. Automatically disabled by final=true.

```
248  skip10pub  .bool_set:N = \l_@@_class_skipTENpublications_bool,
249  skip10pub  .initial:n  = false,
```

colorscheme→P. 66: Passed directly to \selectcolormodel at \AtBeginDocument. Use cmyk for press-quality PDF output (DFG submission default) or RGB for on-screen review. Overridden to cmyk when final=true.

```
250  colorscheme .tl_set:N   =  \l_@@_class_color_model_tl,
251  colorscheme .initial:n  = cmyk,
```

cv→P. 66: Set to true in the standalone CV document (crc-cv.tex). When true, the class adjusts heading styles and suppresses proposal-specific front matter so the output is a self-contained curriculum vitae suitable for DFG submission.

```
252  cv   .bool_set:N = \l_@@_class_cv_bool,
253  cv   .initial:n  = false,
254  unknown .code:n = { \PassOptionsToClass{\CurrentOption}{scrbook} },
255  }
```

6.1.4 Conditional execution based on class options

The class option cv toggles whether the document is rendered as a CV (curriculum vitae) supplement. Internally this is stored in the expl3 boolean \l_@@_class_cv_bool. Since end-user document files should not be required to wrap conditional content in \ExplSyntaxOn/\ExplSyntaxOff, the following three commands expose the switch through a conventional LaTeX interface.

\dfgIfCV `\dfgIfCV{⟨true code⟩}{⟨false code⟩}`

Executes ⟨*true code*⟩ if the cv option is active, otherwise ⟨*false code*⟩. Use this when both branches need to produce content, for instance to switch between a full bibliography and a short selected-publications list:

```
\dfgIfCV{\printbibliography[heading=none]}{\printshortbib}
```

\dfgIfCVT `\dfgIfCVT{⟨true code⟩}`

Shorthand for the common case where only the true branch matters. Typical use is gating a section that exists only in the CV variant:

```
\dfgIfCVT{\par\textcolor{tableHeadingsFont}{Appendix:~Research~profiles}}
```

Equivalent to \dfgIfCV{⟨*true code*⟩}{} but avoids the empty second argument and reads more naturally in document source.

\dfgIfCVF `\dfgIfCVF{⟨false code⟩}`

Mirror image of \dfgIfCVT: executes ⟨*false code*⟩ only when the cv option is *not* active. Useful for content that belongs in the main proposal but should be suppressed in the CV variant.

None of these commands evaluates its arguments unless the relevant branch is taken, so wrapping expensive or side-effecting code is safe. All three are defined with \NewDocumentCommand, so their arguments are read as standard mandatory braced groups and may contain verbatim material only where the underlying typesetting context permits.

```
256  \NewDocumentCommand \dfgIfCV  { +m +m }
257    { \bool_if:NTF \l_@@_class_cv_bool {#1} {#2} }
258  \NewDocumentCommand \dfgIfCVT { +m }
259    { \bool_if:NT  \l_@@_class_cv_bool {#1} }
260  \NewDocumentCommand \dfgIfCVF { +m }
```

```
    { \bool_if:NF  \l_@@_class_cv_bool {#1} }

  \NewDocumentCommand \dfgIfFinalF { +m }
    { \bool_if:NF  \l_@@_class_final_bool {#1} }
  \NewDocumentCommand \dfgIfFastCompileProject { +m +m }
    { \bool_if:NTF \l_@@_class_fastcompile_project_bool {#1} {#2} }
  \NewDocumentCommand \dfgIfFastCompileProjectT { +m }
    { \bool_if:NT \l_@@_class_fastcompile_project_bool {#1} }
  \NewDocumentCommand \dfgIfFastCompileProjectF { +m }
    { \bool_if:NF \l_@@_class_fastcompile_project_bool {#1} }
  \NewDocumentCommand \dfgIfProjectPublications { +m +m }
    { \bool_if:NTF \l_@@_class_fastcompile_project_show_publications_bool {#1} {#2} }
```

Unknown options are forwarded by the key handler above, so ordinary KOMA-Script options remain available without processing the class options a second time. Finally process and execute all options and key-values.

```
  \ProcessKeyOptions[ dfg / class ]
  % Resolve mode precedence before any package, customisation, metadata, or
  % auxiliary-file work can observe the drafting booleans.
  \bool_if:NTF \l_@@_class_final_bool
  {
    \bool_set_false:N \l_@@_class_fastcompile_bool
    \bool_set_false:N \l_@@_class_fastcompile_project_bool
    \bool_set_false:N \l_@@_class_fastcompile_project_show_images_bool
    \bool_set_true:N  \l_@@_class_fastcompile_project_show_funding_tables_bool
    \bool_set_false:N \l_@@_class_showtags_bool
    \bool_set_false:N \l_@@_class_showmarginalia_bool
    \bool_set_false:N \l_@@_class_skipTENpublications_bool
    \bool_set_true:N  \l_@@_class_fastcompile_project_show_publications_bool
  }
  {
    \bool_if:NTF \l_@@_class_fastcompile_project_bool
    {
      \bool_set_true:N  \l_@@_class_fastcompile_bool
      \bool_set_false:N \l_@@_class_showmarginalia_bool
      \bool_set_true:N  \l_@@_class_skipTENpublications_bool
      \PassOptionsToPackage{draft}{hyperref}
      \PassOptionsToPackage{draft}{microtype}
    }
    { \bool_set_true:N \l_@@_class_fastcompile_project_show_publications_bool }
    \bool_if:NT \l_@@_class_fastcompile_bool
    {
      \bool_if:NTF \l_@@_class_fastcompile_show_tags_bool
        { \bool_set_true:N \l_@@_class_showtags_bool }
        { \bool_set_false:N \l_@@_class_showtags_bool }
    }
  }
  \LoadClass{scrbook}
  \PassOptionsToPackage{table}{xcolor} % getting rid of errors and conflicts.
  \RequirePackage{xcolor}
```

6.1.5 Colour palette

All colours follow the RWTH Aachen corporate-identity palette and are defined in all four models at once (`cmyk/RGB/HTML/gray`) using xcolor's multiple-model syntax. This allows the same symbolic name to be used regardless of the active colour model chosen via `colorscheme=`.

Each hue comes in five tints named `-100` (full) through `-10` (near white), matching the official RWTH tint steps. This naming scheme is intentional: table headers, zebra stripes and box backgrounds are all specified in terms of these canonical names (e.g. `blue-50`, `black-10`), so a complete rebranding only requires changing the definitions here.

The ORCiD brand green is also declared here since it is used by `\orcidIcon` throughout the document.

```
  \definecolor{blue-100}{cmyk/RGB/HTML/gray}{1,.50,0,0/0,84,159/00549F/1}% primary
  \definecolor{blue-75}{cmyk/RGB/HTML/gray}{.75,.38,0,0/64,127,183/407FB7/.75}%
  \definecolor{blue-50}{cmyk/RGB/HTML/gray}{.45,.14,0,0/142,186,229/8EBAE5/.5}% secondary
  \definecolor{blue-25}{cmyk/RGB/HTML/gray}{.23,.07,0,0/199,221,242/C7DDF2/.25}%
  \definecolor{blue-10}{cmyk/RGB/HTML/gray}{.09,.03,0,0/232,241,250/E8F1FA/.1}%

  \definecolor{black-100}{cmyk/RGB/HTML}{0,0,0,1/0,0,0/000000}%
  \definecolor{black-75}{cmyk/RGB/HTML}{0,0,0,.75/100,101,103/646567}%
  \definecolor{black-50}{cmyk/RGB/HTML}{0,0,0,.5/156,158,159/9C9E9F}%
  \definecolor{black-25}{cmyk/RGB/HTML}{0,0,0,.25/207,209,210/CFD1D2}%
  \definecolor{black-10}{cmyk/RGB/HTML}{0,0,0,.1/236,237,237/ECEDED}%

  \definecolor{green-100}{cmyk/RGB/HTML}{.7,0,1,0/87,171,39/57AB27}%
```

```
\definecolor{green-75}{cmyk/RGB/HTML}{.52,0,.75,0/141,192,96/8DC060}%
\definecolor{green-50}{cmyk/RGB/HTML}{.35,0,.5,0/184,214,152/B8D698}%
\definecolor{green-25}{cmyk/RGB/HTML}{.18,0,.25,0/221,235,206/DDEBCE}%
\definecolor{green-10}{cmyk/RGB/HTML}{.07,0,.1,0/242,247,236/F2F7EC}%

\definecolor{red-100}{cmyk/RGB/HTML}{.15,1,1,0/204,7,30/CC071E}%
\definecolor{red-75}{cmyk/RGB/HTML}{.11,.75,.75,0/216,92,65/D85C41}%
\definecolor{red-50}{cmyk/RGB/HTML}{.8,.5,.5,0/230,150,121/E69679}%
\definecolor{red-25}{cmyk/RGB/HTML}{.4,.25,.25,0/243,205,187/F3CDBB}%
\definecolor{red-10}{cmyk/RGB/HTML}{.2,.1,.1,0/250,235,227/FAEBE3}%

\definecolor{bordeaux-100}{cmyk/RGB/HTML}{.25,1,.70,.20/161,16,53/A11035}%
\definecolor{bordeaux-75}{cmyk/RGB/HTML}{.19,.75,.52,15/182,82,86/B65256}%
\definecolor{bordeaux-50}{cmyk/RGB/HTML}{.13,.5,.35,.1/205,139,135/CD8B87}%
\definecolor{bordeaux-25}{cmyk/RGB/HTML}{.06,.25,.18,.05/229,197,192/E5C5C0}%
\definecolor{bordeaux-10}{cmyk/RGB/HTML}{.03,.1,.07,.02/245,232,229/F5E8E5}%

\definecolor{orcid}{cmyk/RGB}{0.15,0.00,0.59,0.20/174,205,84}
```

Let's start with the common section on packages etc.

6.1.6 Customisation key family (`dfg/custom`)

We provide as much as possible all settings to customize to your desire and needs. The customization is done with key-values, which we set up below.

```
% Public document text exists before the customization file is read. Keep
% externally supplied definitions (including InfoTeX from the build command).
\providecommand{\dfgLocation}{
  \bool_if:NTF \l_@@_class_fastcompile_bool
    { \textcolor{error}{<Location>} }
    { <Location> }\xspace
}
\providecommand{\dfgDate}{
  \bool_if:NTF \l_@@_class_fastcompile_bool
    { \textcolor{error}{<date>} }
    { <date> }\xspace
}
\providecommand{\dfgTitle}{\enquote{\get[meta]{crc}{name}}\xspace}
\providecommand{\InfoTeX}{LuaLaTeX~ (via~ Overleaf).}
\bool_new:N \l_@@_custom_thumbindex_topmargin_set_bool
\bool_new:N \l_@@_custom_thumbindex_bottommargin_set_bool
\dim_new:N \l_@@_custom_thumbindex_topmargin_dim
\dim_new:N \l_@@_custom_thumbindex_bottommargin_dim
\int_new:N \l_@@_custom_thumbindex_rotation_int
\tl_new:N \l_@@_custom_pi_contact_style_tl
\keys_define:nn { dfg / custom  }
{
  % Assign without expansion; metadata references resolve when printed.
  dfgLocation .code:n = \cs_set:Npn \dfgLocation {#1\xspace},
  dfgDate     .code:n = \cs_set:Npn \dfgDate {#1\xspace},
  dfgTitle    .code:n = \cs_set:Npn \dfgTitle {#1\xspace},
  InfoTeX     .code:n = \cs_set:Npn \InfoTeX {#1},

  % You can pass any font you like (and |fontspec| will understand or is known to
  % your system). But since the DFG is requiring Arial, we set this as default.
  font                       .tl_set:N   = \l_@@_custom_font_tl,
  fontspec                   .tl_set:N   = \l_@@_custom_fontspec_tl,
  font                       .initial:n  = Noto Serif,
  sansfont                   .tl_set:N   = \l_@@_custom_sansfont_tl,
  sansfontspec               .tl_set:N   = \l_@@_custom_sansfontspec_tl,
  sansfont                   .initial:n  = Roboto Light,
  sansfontspec            .initial:n     = {
    ItalicFont=Roboto-LightItalic,
    BoldFont=Roboto-Medium,
    BoldItalicFont=Roboto-Medium-Italic},
  % \paragraph{Logo paths}
  crcLogoPath                 .tl_set:N   = \l_@@_custom_crc_logo_path_tl,
  crcLogoPath                 .initial:n  = example-image-a,
  affiliationLogoPath         .tl_set:N   = \l_@@_custom_affiliation_logo_path_tl,
  affiliationLogoPath         .initial:n  = example-image-b,
  % \paragraph{Project-reference repetition}
  projectPIOncePerSection      .bool_set:N =
    \l_@@_custom_project_pi_once_per_section_bool,
  projectPIOncePerSection      .initial:n  = true,
  PIReferenceFirstName .bool_set:N =
    \l_@@_custom_pi_reference_firstname_bool,
  PIReferenceFirstName .initial:n = false,
  % Project tables use full institution names by default. Set true for
  % compact abbreviations and a table-local full-name legend.
  projectsTableInstitutionAbbreviations .bool_set:N =
    \l_@@_custom_projects_table_institution_abbreviations_bool,
  projectsTableInstitutionAbbreviations .initial:n = false,
```

```
  % Contact labels in PI boxes: icons (default), text labels, or no labels.
  PIcontactStyle .tl_set:N = \l_@@_custom_pi_contact_style_tl,
  PIcontactStyle .initial:n = icons,
  % Collaboration-plot area colors can be set once for the complete proposal.
  % Per-plot area-colors and area-palette options may override these defaults.
  collaborationAreaColors      .tl_set:N =
    \l_@@_custom_collaboration_area_colors_tl,
  collaborationAreaColors      .initial:n =
    { A=projectA, B=projectB, Q=projectQ!75!black },
  collaborationAreaPalette     .code:n =
    {
      \clist_if_empty:nF {#1}
        { \tl_set:Nn \l_@@_custom_collaboration_area_palette_tl {#1} }
    },
  collaborationAreaPalette     .initial:n =
    { orange!80!black, violet!75!black, teal!70!black,
      brown!75!black, magenta!70!black, cyan!70!black, lime!60!black },
  % \paragraph{DFG salary and animal-housing rates}
  salary@prof                  .fp_set:N   = \l_@@_custom_salary_prof_fp,
  salary@prof                  .initial:n  = 126600,
  salary@groupLeader           .fp_set:N   = \l_@@_custom_salary_group_leader_fp,
  salary@groupLeader           .initial:n  = 105000,
  salary@postDoc               .fp_set:N   = \l_@@_custom_salary_postdoc_fp,
  salary@postDoc               .initial:n  = 90600,
  salary@medDoc                .fp_set:N   = \l_@@_custom_salary_meddoc_fp,
  salary@medDoc                .initial:n  = 112200,
  salary@PhD                   .fp_set:N   = \l_@@_custom_salary_phd_fp,
  salary@PhD                   .initial:n  = 83700,
  salary@research              .fp_set:N   = \l_@@_custom_salary_research_fp,
  salary@research              .initial:n  = 69000,
  salary@nonResearch           .fp_set:N   = \l_@@_custom_salary_nonresearch_fp,
  salary@nonResearch           .initial:n  = 61500,
  salary@student               .fp_set:N   = \l_@@_custom_salary_student_fp,
  salary@student               .initial:n  = 1,
  salary@assistant             .fp_set:N   = \l_@@_custom_salary_assistant_fp,
  salary@assistant             .initial:n  = 1,
  animalHousing@mouse          .fp_set:N   = \l_@@_custom_animal_mouse_fp,
  animalHousing@mouse          .initial:n  = 0.75,
  animalHousing@mouseGnotobiotic .fp_set:N =
    \l_@@_custom_animal_mouse_gnotobiotic_fp,
  animalHousing@mouseGnotobiotic .initial:n = 3.9,
  animalHousing@rat            .fp_set:N   = \l_@@_custom_animal_rat_fp,
  animalHousing@rat            .initial:n  = 1.4,
  % \paragraph{Thumb-index keys}
  % These keys control the decorative side tab on the outer margin.
  % The index divides the text block into \meta{pattern} equal-height slots by
  % default.  Explicit page-edge margins can override either end of the stack.
  % Each call to \cs{printThumbIndex}
  % advances the slot counter; when it reaches \meta{pattern} it wraps,
  % creating a repeating stripe aiding chapter navigation.
  %
  thumbindex                 .bool_set:N = \l_@@_custom_thumbindex_bool,
  thumbindex                 .initial:n  = true,
  %
  thumbindex@pattern         .int_set:N  = \l_@@_custom_thumbindexpattern_int,
  thumbindex@pattern         .initial:n  = 10,
  %
  thumbindex@topmargin .code:n =
    {
      \str_if_eq:nnTF {#1} {auto}
        { \bool_set_false:N \l_@@_custom_thumbindex_topmargin_set_bool }
        {
          \dim_set:Nn \l_@@_custom_thumbindex_topmargin_dim {#1}
          \bool_set_true:N \l_@@_custom_thumbindex_topmargin_set_bool
        }
    },
  thumbindex@topmargin .initial:n = auto,
  %
  thumbindex@bottommargin .code:n =
    {
      \str_if_eq:nnTF {#1} {auto}
        { \bool_set_false:N \l_@@_custom_thumbindex_bottommargin_set_bool }
        {
          \dim_set:Nn \l_@@_custom_thumbindex_bottommargin_dim {#1}
          \bool_set_true:N \l_@@_custom_thumbindex_bottommargin_set_bool
        }
    },
  thumbindex@bottommargin .initial:n = auto,
  %
  thumbindex@width     .dim_set:N  = \l_@@_custom_thumbindex_width_dim,
  thumbindex@width .initial:n  = 1.2cm,
  %
  thumbindex@orientation .choice:,
  thumbindex@orientation / outwards .code:n =
    \int_zero:N \l_@@_custom_thumbindex_rotation_int,
  thumbindex@orientation / inwards .code:n =
    \int_set:Nn \l_@@_custom_thumbindex_rotation_int {180},
```

```
    thumbindex@orientation .initial:n = outwards,
    thumbindex@orientation .value_required:n = true,
    %
    thumbindex@font .tl_set:N = \l_@@_custom_thumbindex_font_tl,
    thumbindex@font .initial:n = \bfseries\Huge,
    % \paragraph{Bibliography keys}
    % |emphasizeAuthors| controls the bold-name mechanism.  When true (the
    % default), it marks every PI's name in bold across all bibliography and
    % citation lists.
    % |emphasizeEntries| independently bolds entire bibliography entries with
    % the current project ID plus |-highlight| in project lists, or with
    % |frametext-highlight| in the frame-text list.
    % CV and proposal bibliography styles and features are stored separately.
    %
    doiAPIMail                  .tl_set:N  = \l_@@_custom_doi_api_mail_tl,
    doiAPIMail                  .initial:n = {},
    %
    emphasizeAuthors           .bool_set:N = \l_@@_custom_emphasize_authors_bool,
    emphasizeAuthors           .initial:n  = true,
    emphasizeEntries           .bool_set:N = \l_@@_custom_emphasize_entries_bool,
    emphasizeEntries           .initial:n  = false,
    %
    biblatexstyleCV                 .tl_set:N  = \l_@@_custom_biblatexstyle_cv_tl,
    biblatexstyleCV                 .initial:n = ext-numeric-comp,
    biblatexstyleFeatureCV          .tl_set:N  = \l_@@_custom_biblatexfeatures_cv_tl,
    biblatexstyleFeatureCV          .initial:n = {sorting=none},
    biblatexstyleProposal           .tl_set:N  = \l_@@_custom_biblatexstyle_proposal_tl,
    biblatexstyleProposal           .initial:n = ext-authoryear-ecomp,
    biblatexstyleFeatureProposal    .tl_set:N  = \l_@@_custom_biblatexfeatures_proposal_tl,
    biblatexstyleFeatureProposal    .initial:n = {},
    % Compatibility: the former keys apply the supplied value to both modes.
    biblatexstyle .code:n =
      {
        \tl_set:Nn \l_@@_custom_biblatexstyle_cv_tl       {#1}
        \tl_set:Nn \l_@@_custom_biblatexstyle_proposal_tl {#1}
      },
    biblatexstyleFeature .code:n =
      {
        \tl_set:Nn \l_@@_custom_biblatexfeatures_cv_tl       {#1}
        \tl_set:Nn \l_@@_custom_biblatexfeatures_proposal_tl {#1}
      },
    % \paragraph{Colour keys}
    % All colour keys store a colour \emph{name} referencing an already-defined
    % xcolor colour (e.g.\ |blue-50|, |black-10|).  Actual \cs{colorlet}
    % bindings happen in \cs{AtBeginDocument} so preamble overrides are
    % processed first.  Groups: |color@gantt@*| for the five bar styles;
    % |color@table@*| for all table chrome; |color@thumbindex@*| for the
    % thumb tab; |color@tags@*| for the showtags debug annotations.
    %
    color@reference@boldAuthor .tl_set:N   = \l_@@_custom_referenceBoldAuthor_tl,
    color@reference@boldAuthor .initial:n  = black,
    color@reference@PI .tl_set:N           = \l_@@_custom_referencePI_tl,
    color@reference@PI .initial:n          = black,
    color@reference@WP .tl_set:N           = \l_@@_custom_referenceWP_tl,
    color@reference@WP .initial:n          = black,
    color@reference@Aim .tl_set:N          = \l_@@_custom_referenceAim_tl,
    color@reference@Aim .initial:n         = black,

    color@gantt@frame      .tl_set:N    = \l_@@_custom_gantt_frame_tl,
    color@gantt@frame      .initial:n = {white},
    color@gantt@default    .tl_set:N    =\l_@@_custom_gantt_default_tl,
    color@gantt@default    .initial:n = {blue-25},
    color@gantt@alternative    .tl_set:N    =\l_@@_custom_gantt_alternative_tl,
    color@gantt@alternative    .initial:n = {green-25},
    color@gantt@crosshatch .tl_set:N    = \l_@@_custom_gantt_crosshatch_tl,
    color@gantt@crosshatch .initial:n = {black-10},
    color@gantt@important  .tl_set:N    = \l_@@_custom_gantt_important_tl,
    color@gantt@important  .initial:n = {red},
    color@gantt@stripe-one .tl_set:N    = \l_@@_custom_gantt_stripe_one_tl,
    color@gantt@stripe-one .initial:n = {blue-25},
    color@gantt@stripe-two .tl_set:N    = \l_@@_custom_gantt_stripe_two_tl,
    color@gantt@stripe-two .initial:n = {green-25},

    color@table@headingsBG     .tl_set:N   = \l_@@_custom_tableHeadingsBG_tl,
    color@table@headingsBG     .initial:n  = blue-50,
    color@table@headingsBGalt  .tl_set:N   = \l_@@_custom_tableHeadingsBGalt_tl,
    color@table@headingsBGalt  .initial:n  = black-25,
    color@table@headingsFont   .tl_set:N   = \l_@@_custom_tableHeadingsFont_tl,
    color@table@headingsFont   .initial:n  = black,
    color@table@rows           .tl_set:N   = \l_@@_custom_tableRows_tl,
    color@table@rows           .initial:n  = black-10,
    color@table@lines          .tl_set:N   = \l_@@_custom_tableLines_tl,
    color@table@lines          .initial:n  = black-10,
    color@table@outerLines     .tl_set:N   = \l_@@_custom_tableOuterLines_tl,
    color@table@outerLines     .initial:n  = black,
    color@table@noValue        .tl_set:N   = \l_@@_custom_tableNoValue_tl,
    color@table@noValue        .initial:n  = black-10,
```

```
    color@table@firstColumn     .tl_set:N   = \l_@@_custom_tableFirstColumn_tl,
    color@table@firstColumn     .initial:n  = black,
    color@table@link            .tl_set:N   = \l_@@_custom_tableLink_tl,
    color@table@link            .initial:n = black,
    color@table@contFootHead    .tl_set:N  = \l_@@_custom_tableContFootHead_tl,
    color@table@contFootHead    .initial:n = black,
    color@table@caption         .tl_set:N  = \l_@@_custom_tableCaption_tl,
    color@table@caption         .initial:n = black,
    color@thumbindex@bg         .tl_set:N  = \l_@@_custom_thumbIndexBG_tl,
    color@thumbindex@bg         .initial:n = black,
    color@thumbindex@font       .tl_set:N  = \l_@@_custom_thumbIndexFont_tl,
    color@thumbindex@font       .initial:n = white,
    color@tags@font             .tl_set:N  = \l_@@_custom_colorTagsFont_tl,
    color@tags@font             .initial:n = red-100,
    color@tags@line             .tl_set:N  = \l_@@_custom_colorTagsLine_tl,
    color@tags@line             .initial:n = black-50,
    % The alternative font color which is used e.g. for the CV headline.
    color@font@alternative      .tl_set:N  = \l_@@_custom_colorfontalternative_tl,
    color@font@alternative      .initial:n = black-50,

    color@item@i                .tl_set:N  = \l_@@_custom_color_item_i_tl,
    color@item@i                .initial:n = black,
    % That's it with the key-values.
  }
```

Bibliography package options must be fixed before biblatex is loaded. Consequently the customization file is read here, after all custom keys have been declared but before package loading. It is not read a second time at the end of the class.

```
\NewDocumentCommand{\changeSettings}{m}
  { \keys_set:nn { dfg / custom } {#1} }
% Compatibility alias for proposals created before the command was pluralised.
\NewDocumentCommand{\changeSetting}{m}
  { \changeSettings{#1} }
```

Read live storage, not cached input: typed values are normalised, grouped changes unwind, and internal overrides remain visible. Keep the registry in sync with all public keys in `dfg/custom`.

```
\prop_const_from_keyval:Nn \c_@@_custom_getters_prop
  {
    font = { \tl_use:N \l_@@_custom_font_tl },
    fontspec = { \tl_use:N \l_@@_custom_fontspec_tl },
    sansfont = { \tl_use:N \l_@@_custom_sansfont_tl },
    sansfontspec = { \tl_use:N \l_@@_custom_sansfontspec_tl },
    crcLogoPath = { \tl_use:N \l_@@_custom_crc_logo_path_tl },
    affiliationLogoPath = { \tl_use:N \l_@@_custom_affiliation_logo_path_tl },
    projectPIOncePerSection = { \bool_if:NTF \l_@@_custom_project_pi_once_per_section_bool {true} {false} },
    PIReferenceFirstName = { \bool_if:NTF \l_@@_custom_pi_reference_firstname_bool {true} {false} },
    projectsTableInstitutionAbbreviations = { \bool_if:NTF \l_@@_custom_projects_table_institution_abbreviations_bool
    PIcontactStyle = { \tl_use:N \l_@@_custom_pi_contact_style_tl },
    collaborationAreaColors = { \tl_use:N \l_@@_custom_collaboration_area_colors_tl },
    salary@prof = { \fp_use:N \l_@@_custom_salary_prof_fp },
    salary@groupLeader = { \fp_use:N \l_@@_custom_salary_group_leader_fp },
    salary@postDoc = { \fp_use:N \l_@@_custom_salary_postdoc_fp },
    salary@medDoc = { \fp_use:N \l_@@_custom_salary_meddoc_fp },
    salary@PhD = { \fp_use:N \l_@@_custom_salary_phd_fp },
    salary@research = { \fp_use:N \l_@@_custom_salary_research_fp },
    salary@nonResearch = { \fp_use:N \l_@@_custom_salary_nonresearch_fp },
    salary@student = { \fp_use:N \l_@@_custom_salary_student_fp },
    salary@assistant = { \fp_use:N \l_@@_custom_salary_assistant_fp },
    animalHousing@mouse = { \fp_use:N \l_@@_custom_animal_mouse_fp },
    animalHousing@mouseGnotobiotic = { \fp_use:N \l_@@_custom_animal_mouse_gnotobiotic_fp },
    animalHousing@rat = { \fp_use:N \l_@@_custom_animal_rat_fp },
    thumbindex = { \bool_if:NTF \l_@@_custom_thumbindex_bool {true} {false} },
    thumbindex@pattern = { \int_use:N \l_@@_custom_thumbindexpattern_int },
    thumbindex@width = { \dim_use:N \l_@@_custom_thumbindex_width_dim },
    thumbindex@font = { \tl_use:N \l_@@_custom_thumbindex_font_tl },
    doiAPIMail = { \tl_use:N \l_@@_custom_doi_api_mail_tl },
    emphasizeAuthors = { \bool_if:NTF \l_@@_custom_emphasize_authors_bool {true} {false} },
    emphasizeEntries = { \bool_if:NTF \l_@@_custom_emphasize_entries_bool {true} {false} },
    biblatexstyleCV = { \tl_use:N \l_@@_custom_biblatexstyle_cv_tl },
    biblatexstyleFeatureCV = { \tl_use:N \l_@@_custom_biblatexfeatures_cv_tl },
    biblatexstyleProposal = { \tl_use:N \l_@@_custom_biblatexstyle_proposal_tl },
    biblatexstyleFeatureProposal = { \tl_use:N \l_@@_custom_biblatexfeatures_proposal_tl },
    color@reference@boldAuthor = { \tl_use:N \l_@@_custom_referenceBoldAuthor_tl },
    color@reference@PI = { \tl_use:N \l_@@_custom_referencePI_tl },
    color@reference@WP = { \tl_use:N \l_@@_custom_referenceWP_tl },
    color@reference@Aim = { \tl_use:N \l_@@_custom_referenceAim_tl },
    color@gantt@frame = { \tl_use:N \l_@@_custom_gantt_frame_tl },
    color@gantt@default = { \tl_use:N \l_@@_custom_gantt_default_tl },
    color@gantt@alternative = { \tl_use:N \l_@@_custom_gantt_alternative_tl },
    color@gantt@crosshatch = { \tl_use:N \l_@@_custom_gantt_crosshatch_tl },
    color@gantt@important = { \tl_use:N \l_@@_custom_gantt_important_tl },
    color@gantt@stripe-one = { \tl_use:N \l_@@_custom_gantt_stripe_one_tl },
```

```
644       color@gantt@stripe-two = { \tl_use:N \l_@@_custom_gantt_stripe_two_tl },
645       color@table@headingsBG = { \tl_use:N \l_@@_custom_tableHeadingsBG_tl },
646       color@table@headingsBGalt = { \tl_use:N \l_@@_custom_tableHeadingsBGalt_tl },
647       color@table@headingsFont = { \tl_use:N \l_@@_custom_tableHeadingsFont_tl },
648       color@table@rows = { \tl_use:N \l_@@_custom_tableRows_tl },
649       color@table@lines = { \tl_use:N \l_@@_custom_tableLines_tl },
650       color@table@outerLines = { \tl_use:N \l_@@_custom_tableOuterLines_tl },
651       color@table@noValue = { \tl_use:N \l_@@_custom_tableNoValue_tl },
652       color@table@firstColumn = { \tl_use:N \l_@@_custom_tableFirstColumn_tl },
653       color@table@link = { \tl_use:N \l_@@_custom_tableLink_tl },
654       color@table@contFootHead = { \tl_use:N \l_@@_custom_tableContFootHead_tl },
655       color@table@caption = { \tl_use:N \l_@@_custom_tableCaption_tl },
656       color@thumbindex@bg = { \tl_use:N \l_@@_custom_thumbIndexBG_tl },
657       color@thumbindex@font = { \tl_use:N \l_@@_custom_thumbIndexFont_tl },
658       color@tags@font = { \tl_use:N \l_@@_custom_colorTagsFont_tl },
659       color@tags@line = { \tl_use:N \l_@@_custom_colorTagsLine_tl },
660       color@font@alternative = { \tl_use:N \l_@@_custom_colorfontalternative_tl },
661       color@item@i = { \tl_use:N \l_@@_custom_color_item_i_tl },
662       dfgLocation = { \dfgLocation },
663       dfgDate = { \dfgDate },
664       dfgTitle = { \dfgTitle },
665       InfoTeX = { \InfoTeX },
666       collaborationAreaPalette = { \tl_use:N \l_@@_custom_collaboration_area_palette_tl },
667       thumbindex@topmargin = { \bool_if:NTF \l_@@_custom_thumbindex_topmargin_set_bool { \dim_use:N \l_@@_custom_thumbi
668       thumbindex@bottommargin = { \bool_if:NTF \l_@@_custom_thumbindex_bottommargin_set_bool { \dim_use:N \l_@@_custom_
669       thumbindex@orientation = { \int_compare:nNnTF {\l_@@_custom_thumbindex_rotation_int} = {0} {outwards} {inwards} }
670       biblatexstyle = { \bool_if:NTF \l_@@_class_cv_bool {\tl_use:N \l_@@_custom_biblatexstyle_cv_tl} {\tl_use:N \l_@@_
671       biblatexstyleFeature = { \bool_if:NTF \l_@@_class_cv_bool {\tl_use:N \l_@@_custom_biblatexfeatures_cv_tl} {\tl_us
672     }
673   \msg_new:nnn { dfg } { unknown-customization-value }
674     { Unknown~customization~key~'#1'~in~\string\getValue.
675       ~Use~a~key~supported~by~\string\changeSettings. }
676   \cs_new:Npn \@@_get_custom_value:n #1
677     {
678       \prop_if_in:NnTF \c_@@_custom_getters_prop {#1}
679         { \exp_args:Ne \use:n { \prop_item:Nn \c_@@_custom_getters_prop {#1} } }
680         { \msg_expandable_error:nnn {dfg} {unknown-customization-value} {#1} }
681     }
682   \NewExpandableDocumentCommand \getValue { m }
683     { \exp_args:Ne \@@_get_custom_value:n { \tl_trim_spaces:n {#1} } }
684   % Compatibility accessors retained for the funding and animal-cost code.
685   \NewExpandableDocumentCommand{\dfgSalaryProf}{}
686     { \fp_use:N \l_@@_custom_salary_prof_fp }
687   \NewExpandableDocumentCommand{\dfgSalaryGroupLeader}{}
688     { \fp_use:N \l_@@_custom_salary_group_leader_fp }
689   \NewExpandableDocumentCommand{\dfgSalaryPostDoc}{}
690     { \fp_use:N \l_@@_custom_salary_postdoc_fp }
691   \NewExpandableDocumentCommand{\dfgSalaryMedDoc}{}
692     { \fp_use:N \l_@@_custom_salary_meddoc_fp }
693   \NewExpandableDocumentCommand{\dfgSalaryPhD}{}
694     { \fp_use:N \l_@@_custom_salary_phd_fp }
695   \NewExpandableDocumentCommand{\dfgSalaryResearch}{}
696     { \fp_use:N \l_@@_custom_salary_research_fp }
697   \NewExpandableDocumentCommand{\dfgSalaryNonResearch}{}
698     { \fp_use:N \l_@@_custom_salary_nonresearch_fp }
699   \NewExpandableDocumentCommand{\dfgSalaryStudent}{}
700     { \fp_use:N \l_@@_custom_salary_student_fp }
701   \NewExpandableDocumentCommand{\dfgSalaryAssistant}{}
702     { \fp_use:N \l_@@_custom_salary_assistant_fp }
703   \NewExpandableDocumentCommand{\dfgMouseHousing}{}
704     { \fp_use:N \l_@@_custom_animal_mouse_fp }
705   \NewExpandableDocumentCommand{\dfgMouseHousingGnotobiotic}{}
706     { \fp_use:N \l_@@_custom_animal_mouse_gnotobiotic_fp }
707   \NewExpandableDocumentCommand{\dfgRatHousing}{}
708     { \fp_use:N \l_@@_custom_animal_rat_fp }
709   \bool_if:NF \l_@@_class_documentation_bool
710   {
711   \file_if_exist:nTF {preamble/dfg-customization.tex}
712     { \ExplSyntaxOff \input{preamble/dfg-customization.tex} \ExplSyntaxOn }
713     { \msg_warning:nnn { dfg } { file-not-found } {preamble/dfg-customization.tex} }
714   }
715   \bool_if:NT \l_@@_class_fastcompile_project_bool
716   {
717     \bool_set_false:N \l_@@_custom_thumbindex_bool
718     \bool_set_false:N \l_@@_custom_emphasize_authors_bool
719   }
```

6.1.7 Colour binding at document start

The `\AtBeginDocument` hooks below transform the stored colour *names* from the `dfg/custom` key family into live `\colorlet` aliases that the rest of the document uses. Deferring this to `\AtBeginDocument` means the user's `\changeSettings` calls in `dfg-customization.tex` are already in force when the bindings run.

The second \AtBeginDocument block handles final-mode side effects: it resets all drafting flags (both fast modes, tags, marginalia, skip-pubs) to false, enables project publications, forces the colour model to cmyk, and switches the color@final@white-or-green alias from green to white so that any placeholder boxes are invisible in the submitted PDF.

Packages are loaded in an order that avoids common conflicts: xcolor is loaded first (with the table option forwarded beforehand) because almost every other package depends on it. tabularray and its siunitx library are set up before hyperref; reversing this order breaks tblr's PDF bookmark integration. hyperref and cleveref are always loaded last (standard LaTeX best practice).

The two-sided A4 layout uses a 25 mm inner margin plus the KOMA-Script BCOR binding correction (initially 5 mm), and a 20 mm outer margin. Geometry applies BCOR as its binding offset, leaving a 160 mm text block at the default setting. The outer margin accommodates a 13 mm note column and a 2 mm separation from the text, with 5 mm spare. On verso pages, the binding and outer margins, notes, and thumb index all switch sides. The top margin is 20 mm and the bottom margin 15 mm. This gives the text block 5 mm more height than the former 20 mm bottom margin while retaining the footer. Reserve 20 pt for two-line project running heads. The header remains outside the text block, so its height does not change the 20 mm top margin or the text height.

```
\RequirePackage{multicol}
% Defensive definition against lineno/multicol conflict
\AtBeginDocument{%
  \@ifpackageloaded{lineno}{%
   \providecommand*{\@LN@col}[1]{\relax}%
  }{}%
}
\RequirePackage{babel}
\RequirePackage{geometry}
\geometry{
  a4paper,
  includefoot,
  headheight     =20pt,
  top            =20mm,
  bottom         =15mm,
  inner          =25mm,
  outer          =20mm,
  bindingoffset  =\ta@bcor,
  marginparsep   =2mm,
  marginparwidth =13mm,
  % showframe
}
```

We provide a color which is used during the writing process. Since there are several steps that are calculated on the fly we want to be able to see if they match any criteria or if there has been a wrongly inserted number. The following colors can be overwritten in the dfg-customization.tex-file. Therefor we load them just before the document starts.

```
\AtBeginDocument{
  \selectcolormodel{\tl_use:N \l_@@_class_color_model_tl}
  \colorlet{color@item@i}{\l_@@_custom_color_item_i_tl}
  \colorlet{color@tags@font}{\l_@@_custom_colorTagsFont_tl}
  \colorlet{color@tags@line}{\l_@@_custom_colorTagsLine_tl}
  \colorlet{color@font@alternative}{\l_@@_custom_colorfontalternative_tl}
  \colorlet{color@reference@boldAuthor}{\l_@@_custom_referenceBoldAuthor_tl}
  \colorlet{tableHeadingsBG}{\l_@@_custom_tableHeadingsBG_tl}
  \colorlet{tableHeadingsBGalt}{\l_@@_custom_tableHeadingsBGalt_tl}
  \colorlet{tableHeadingsFont}{\l_@@_custom_tableHeadingsFont_tl}
  \colorlet{tableRows}{\l_@@_custom_tableRows_tl}
  \colorlet{tableLines}{\l_@@_custom_tableLines_tl}
  \colorlet{tableOuterLines}{\l_@@_custom_tableOuterLines_tl}
  \colorlet{tableNoValue}{\l_@@_custom_tableNoValue_tl}
  \colorlet{tableFirstColumn}{\l_@@_custom_tableFirstColumn_tl}
  \colorlet{tableLink}{\l_@@_custom_tableLink_tl}
  \colorlet{tableContFootHead}{\l_@@_custom_tableContFootHead_tl}
  \colorlet{tableCaption}{\l_@@_custom_tableCaption_tl}
  \colorlet{thumbIndexBG}{\l_@@_custom_thumbIndexBG_tl}
  \colorlet{thumbIndexFont}{\l_@@_custom_thumbIndexFont_tl}
  % all for the gantt chart
  \colorlet{color@gantt@default}{\l_@@_custom_gantt_default_tl}
  \colorlet{color@gantt@alternative}{\l_@@_custom_gantt_alternative_tl}
  \colorlet{color@gantt@frame}{\l_@@_custom_gantt_frame_tl}
  \colorlet{color@gantt@crosshatch}{\l_@@_custom_gantt_crosshatch_tl}
  \colorlet{color@gantt@important}{\l_@@_custom_gantt_important_tl}
  \colorlet{color@gantt@stripe-one}{\l_@@_custom_gantt_stripe_one_tl}
  \colorlet{color@gantt@stripe-two}{\l_@@_custom_gantt_stripe_two_tl}

  % NOTE: The aliases below shadow LaTeX/xcolor standard colour names
  % (black, red, green). Any subsequently loaded package relying on the
```

```
  % standard red=RGB(255,0,0) will receive bordeaux-100 instead.
  % Consider using dfg-prefixed aliases if you load additional packages.
  \colorlet{black}{black-100}
  \colorlet{blue}{blue-100}
  \colorlet{blueLight}{blue-50}
  \colorlet{green}{green-100}
  \colorlet{greenLight}{green-50}
  \colorlet{red}{bordeaux-100}
  \colorlet{redLight}{bordeaux-50}
  \colorlet{grey}{black-50}
  \colorlet{greyLight}{black-10}

  \colorlet{projectA}{blue}
  \colorlet{projectB}{green}
  \colorlet{projectQ}{grey}
  \colorlet{error}{red}
  % Inherited reference colors are resolved at the call site. Keep the
  % concrete alias available for PI tables and other fixed-color layouts.
  \tl_if_eq:NnTF \l_@@_custom_referencePI_tl {current}
    { \colorlet{referencePI}{black} }
    { \colorlet{referencePI}{\l_@@_custom_referencePI_tl} }
  \colorlet{referenceWP}{\l_@@_custom_referenceWP_tl}
  \colorlet{referenceAim}{\l_@@_custom_referenceAim_tl}
  % Now closing the AtBeginDocument-part:
}
```

Once you set the flag `final` the color (green) is replaced by white so it is ready for submission without indicating the doublechecking aspect. If it stays red you should not proceed to submit the proposal.

```
\AtBeginDocument{
  \bool_if:NTF \l_@@_class_final_bool
  {
   \colorlet{color@final@white-or-green}{white}
   \bool_set_false:N \l_@@_class_showmarginalia_bool
   \bool_set_false:N \l_@@_class_showtags_bool
    \bool_set_false:N \l_@@_class_fastcompile_bool
    \bool_set_false:N \l_@@_class_fastcompile_project_bool
    \bool_set_false:N \l_@@_class_fastcompile_project_show_images_bool
    \bool_set_true:N  \l_@@_class_fastcompile_project_show_funding_tables_bool
    \bool_set_false:N \l_@@_class_skipTENpublications_bool
    \bool_set_true:N \l_@@_class_fastcompile_project_show_publications_bool
   \selectcolormodel{cmyk}
  }
  {
   \colorlet{color@final@white-or-green}{green}
  }
}

\RequirePackage{xspace}
\RequirePackage{etoolbox}

\RequirePackage{ragged2e}
\raggedbottom
```

We are getting rid of having the chapter number added to the figure number, same goes for tables.

```
\counterwithout{figure}{chapter}
\counterwithout{table}{chapter}

\RequirePackage{qrcode}
\RequirePackage{graphicx}
\RequirePackage{mwe}% provides the default example-image-a/b logo placeholders
% Set the graphics mode after loading \pkg{graphicx}.  This explicitly clears
% a previously inherited draft setting when project images are requested and
% makes repeated false/true drafting builds deterministic.
\bool_if:NT \l_@@_class_fastcompile_project_bool
{
  \bool_if:NTF \l_@@_class_fastcompile_project_show_images_bool
    { \setkeys{Gin}{draft=false} }
    { \setkeys{Gin}{draft=true} }
}
\graphicspath{
  {figures/}
  {img/}
}
\RequirePackage{fontspec}
\AtBeginDocument{
  \setmainfont{\tl_use:N \l_@@_custom_font_tl}[
   \tl_use:N \l_@@_custom_fontspec_tl
  ]
  \setsansfont{\tl_use:N \l_@@_custom_sansfont_tl}[
   \tl_use:N \l_@@_custom_sansfontspec_tl
  ]
```

```
}

\RequirePackage{fontawesome5}
\RequirePackage{microtype}
\RequirePackage{enumitem}
% customizing bulletpoints
\newcommand{\dfgArrowHead}{\faCaretRight}
\newcommand{\dfgBulletPoint}{\faCaretRight}
\newcommand{\dfgTargetPoint}{\faBullseye}
\newcommand{\dfgDefaultPoint}{\faCaretRight}
\setlist[enumerate]{nosep}
\setlist[description]{
  %leftmargin=*,
  leftmargin=*
}
\setlist[itemize]{
  nosep,
  label={\textcolor{color@item@i}{\dfgDefaultPoint}},
}
\newlist{contractinformation}{itemize}{2}
\setlist[contractinformation,1]{
  nosep,
  leftmargin=*,
  label={\textcolor{color@item@i}{\dfgBulletPoint}},
}
\setlist[contractinformation,2]{
  nosep,
  leftmargin=*,
  label={\textcolor{color@item@i}{\dfgDefaultPoint}},
}
```

6.1.8 Project-body list environments

workpackages A custom enumitem list with `WP\,\arabic*:` labels at level 1 and `WP\,\meta{parent}\alph*:` at level 2. The `resume` option at level 1 means the work-package counter continues across multiple projects in the same document, giving each WP a globally unique number for cross-references. Labels are set via cleveref in `referenceWP` colour. Bold face is scoped to the prefix and label itself so range connectors remain at the surrounding font weight.

aims A special `enumerate` list that automatically wraps its contents in a `\dfgBoxEnv` frame (the blue highlighted aims box). The `before` hook also temporarily sets all reference colours to the table heading font colour so that PI/WP/Aim cross-references inside the aims box are visually consistent with the box style. The `after` hook closes the box.

```
\newlist{workpackages}{enumerate}{3}
\setlist[workpackages,1]{
  % wide=\parindent,
  leftmargin=0cm,
  labelindent=*,
  % leftmargin=*,
  % labelindent=-3em,
  % font=\normalfont,
  % first=\bfseries,
  topsep=3pt,
  font=\bfseries,
  % nosep,
  label={WP\,\arabic*:},
  ref = {\arabic*},
  resume
}

\setlist[workpackages,2]{
  ref={\theworkpackagesi\alph*},
  %wide=\parindent,
  leftmargin=0cm,
  labelindent=*,
  % font=\normalfont,
  % first=\bfseries,
  topsep=0pt,
  font=\bfseries,
  % nosep,
  label={WP\,\theworkpackagesi\alph*:}
}
```

bylaws A three-level enumitem list styled for the bylaws section of the CRC charter. Level 1 uses `§\arabic*` paragraph signs; level 2 uses `\thebylawsi.\arabic*` for sub-paragraphs. Both levels are set `nosep` to conserve vertical space.

```
\newlist{bylaws}{enumerate}{3}
\setlist[bylaws]{leftmargin=*}
\setlist[bylaws,1]{
  font=\bfseries,
  first=\bfseries,
  % after=\normalfont,
  label={§\arabic*},
  % itemsep=.5\baselineskip,
  nosep
}
\setlist[bylaws,2]{
  first=\normalfont,
  label={\thebylawsi.\arabic*},
  labelindent=0em,
}
```

A list for the aims within the projects

```
\newcommand{\addAim}[1]{\item\label{\l__tmp_current_project_tl-aim-#1}}
\newlist{aims}{enumerate}{1}
\setlist[aims,1]{
  before={\begin{dfgBoxEnv}%
    [%watermark text={Aims in }
                left=0mm,right=0pt,
         colupper=blue,
    ]
    \colorlet{referencePI}{tableHeadingsFont}
    \colorlet{referenceWP}{tableHeadingsFont}
    \colorlet{referenceAim}{tableHeadingsFont}},
  after={\end{dfgBoxEnv}},
  wide=\parindent,
  leftmargin=*,
  % font=\normalfont,
  % first=\bfseries,
  topsep=3pt,
  font=\bfseries,
  % nosep,
  label={Aim~ \arabic*:},
  ref = {\arabic*},
  %  resume
}
```

The public `label` option is a prefix, not enumitem's complete label format. Keep the original list and counter so existing list settings and numeric `\ref` values remain valid. A stable type per prefix prevents a later list from renaming earlier references. Store its name in the aux file so cleveref knows it before constructing formats and forward references on the next run. Other enumitem options pass through unchanged. Expand the label explicitly: older expl3 releases provide the hash function without its V variant.

```
\tl_new:N \l_@@_aim_label_tl
\tl_new:N \l_@@_aim_options_tl
\tl_new:N \l_@@_aim_type_tl
\keys_define:nn { dfg / aims }
{
  label .tl_set:N = \l_@@_aim_label_tl,
  label .value_required:n = true,
}
\NewCommandCopy\dfgOriginalAims\aims
\NewCommandCopy\dfgOriginalEndAims\endaims
\cs_new_protected:Npn \@@_aims_begin:n #1
  { \dfgOriginalAims[#1] }
\RenewDocumentEnvironment{aims}{ O{} }
{
  \tl_set:Nn \l_@@_aim_label_tl { \q_no_value }
  \keys_set_known:nnN { dfg / aims } {#1} \l_@@_aim_options_tl
  \quark_if_no_value:NF \l_@@_aim_label_tl
  {
    \tl_set:Ne \l_@@_aim_type_tl
      { dfg-aim-\exp_args:NV \str_mdfive_hash:n \l_@@_aim_label_tl }
    \exp_args:NnV \crefalias { aimsi } \l_@@_aim_type_tl
    \protected@write\@auxout{}
    {
      \string\dfgAimReferenceName
        {\l_@@_aim_type_tl}{\exp_not:V \l_@@_aim_label_tl}
    }
    \tl_put_left:Ne \l_@@_aim_options_tl
    {
```

```
      label={\exp_not:V \l_@@_aim_label_tl
        \exp_not:n {~\arabic*:}},
      \exp_not:n {ref={\arabic*},}
    }
  }
  \exp_args:NV \@@_aims_begin:n \l_@@_aim_options_tl
}
{ \dfgOriginalEndAims }
```

6.1.9 Page style (headers and footers)

The page style uses scrlayer-scrpage with a minimal layout: The outer footer shows the page number in bold sans-serif. The inner footer shows the CRC logo (`\crcLogo`) at 0.5 cm height — this places the logo on every page of the proposal for visual identification without cluttering the headers. Both header and footer use `\markcase=noupper` so section titles are not forced to uppercase in running heads. `\footheight` is set explicitly to suppress the scrlayer-scrpage warning about an undersized footer layer.

```
\RequirePackage[
  %headsepline=.5pt,
  markcase=noupper,
  %footsepline,
  autooneside=false, % use left and right marks with a onesided document
  %automark, % set \ leftmark and \ rightmark automatically by \ section and \ subsection
  draft=false,
]{scrlayer-scrpage}
```

First we clear any presettings so we can apply our own style.

```
\clearmainofpairofpagestyles
\pagestyle{scrheadings}
\setlength{\footheight}{17.5264pt}
\renewcommand{\chaptermarkformat}{}
\renewcommand{\sectionmarkformat}{}
\ofoot*{\pagemark}%Clear the outer foot
\ifoot*{\crcLogo[height=0.5cm]}%pagemark in the center only on plain pages
\addtokomafont{pagehead}{\normalfont\sffamily}
\addtokomafont{pagefoot}{\bfseries}
% Mark every shipped page of an aggressive project draft without adding a
% package-level watermark dependency.  The native background hook keeps the
% mark behind authored text and is registered only after final-mode precedence
% has been resolved.
\bool_if:NT \l_@@_class_fastcompile_project_bool
{
  \AddToHook{shipout/background}[dfg/fastcompile-project-watermark]
  {
    \put(0.5\paperwidth,-0.5\paperheight)
    {
      \makebox(0,0)
      {
        \rotatebox{45}
        {
          \textcolor{black!7}
          {
            \sffamily\bfseries\fontsize{100}{50}\selectfont
            FAST~COMPILE
          }
        }
      }
    }
  }
}
```

Remodelling the footnotes for more pleasant display

```
\deffootnote{0em}{0em}{\thefootnotemark\hskip .5em}
\renewcommand{\footnoterule}{%
  % \kern -4.5pt
  % \hrule width \textwidth height 0.5pt
  % \kern 4pt
}
```

6.1.10 Heading fonts (KOMA-Script customisation)

All heading sizes and weights are re-declared here to achieve a compact, dense layout that conserves space on the tightly page-budgeted DFG forms: `disposition` is reset to `\normalfont` (KOMA default is bold) so all heading levels start from the same base. Chapter headings (section

dividers) are set at 20 pt bold to stand out without the very large spacing of \Huge. Sections use 12 pt bold type; subsections and subsubsections use 10 pt type for a compact hierarchy. Section, subsection, and subsubsection headings have one baseline of space above them. Body text starts on the next line with no additional after-skip or first-line indent; zero after-skip does not make a run-in heading. Numbered section and subsection titles use the same 12 mm number column, so the title text begins at one horizontal position throughout the project information block. This also gives wrapped titles a matching hang indent. The TOC is configured to omit chapter page numbers (they would be meaningless given the DFG form structure) and to show section entries with a dash-style page separator. Project TOC entries deliberately use only their compact project number, so the fixed number column remains narrow and ordinary section titles are not pushed unnecessarily far to the right.

```
\setkomafont{title}{\Huge\bfseries}
% \setkomafont{disposition}{\normalfont}
\setkomafont{disposition}{\normalfont\sffamily\bfseries}
\setkomafont{descriptionlabel}{\normalfont\bfseries}
%\setkomafont{chapter}{\huge\bfseries}
%\setkomafont{section}{\normalsize\bfseries}
%\setkomafont{subsection}{\normalsize}
%setkomafont{subsubsection}{\normalsize}
\setkomafont{chapter}{\fontsize{20}{10}\selectfont\bfseries}
\setkomafont{section}{\fontsize{12}{10}\selectfont\bfseries}
\setkomafont{subsection}{\fontsize{10}{10}\selectfont}
\setkomafont{subsubsection}{\fontsize{10}{10}\selectfont}
\renewcommand*{\chapterformat}{\makebox[12mm][l]{\thechapter\autodot}}
\renewcommand*{\sectionformat}{\makebox[12mm][l]{\thesection\autodot}}
\renewcommand*{\subsectionformat}{\makebox[12mm][l]{\thesubsection\autodot}}
\setkomafont{footnotenumber}{\normalfont\sffamily}
\setkomafont{pageheadfoot}{\normalfont\footnotesize\sffamily}
```

Adjusting the style for the table of contents. First we want to have the normalfont for the chapter entries

```
\addtokomafont{chapterentry}{\normalfont}
```

```
\RedeclareSectionCommands[
  font=\usekomafont{subsection},
  % beforeskip=1pt,
  % afterskip=.5pt
]{subsubsection}

\RedeclareSectionCommands[
  tocraggedpagenumber,
  toclinefill=\tocpageseparator,
  tocindent=0em,
  tocnumwidth=4em,
  tocpagenumberbox=\tocpagenumberbox,% <- added
  % tocpagenumberformat=\textsf,
  beforeskip= 1\baselineskip,
  afterskip= 0pt,
  runin=false,
  afterindent=false
]{section,subsection,subsubsection}

\RedeclareSectionCommands[
  tocbeforeskip=2ex,
  tocentryformat=\bfseries\large,
]{section}

\newcommand\tocgobble[1]{}% <- added
\newcommand\tocpageseparator{\normalsize\bfseries\,\mbox{\color{\l_@@_custom_tableLines_tl}---}\,}
\newcommand\tocpagenumberbox[1]{\mbox{#1}}% <- added
% no page number for chapters in TOC
\RedeclareSectionCommands[
  tocraggedpagenumber,
  toclinefill           =\tocgobble,
  tocindent             =0em,
  tocnumwidth           =4em,
  tocpagenumberbox      =\tocgobble,% <- added
  % tocpagenumberformat =\textsf,
  beforeskip            = 0ex,
  afterskip             =0.1ex,
  tocentryformat        =\bfseries\Large\sffamily,
]{chapter}
```

the depth of chapter is configurable, so using this is instead of a absolute number is recommended

```
\setcounter{secnumdepth}{\subsectiontocdepth}
\setcounter{tocdepth}{\subsectiontocdepth}
```

6.1.11 ORCiD display commands

Two commands render ORCiD identifiers:

\ORCiD{⟨***0000-0000-0000-0000***⟩**}** Inline use: renders the ORCiD icon followed by the ID as a hyperlink. Intended for manual insertion in text.

\getORCiD[⟨***db***⟩**]{**⟨***id***⟩**}** Database-driven use: looks up the `orcid` field from the `person` database for the given person ID and renders `\ORCiD` automatically. If the field is missing, a `missing-orcid` warning is raised instead of silently omitting the identifier. The warning is suppressed during the tabularray measuring pass (`\lTblrMeasuringBool`) to avoid duplicates.

The ORCiD brand green (`orcid` colour) is defined in the colour palette section above. The icon uses the academicons `\aiOrcid` glyph.

```
\RequirePackage{pifont}
\RequirePackage{academicons}
% ============================================================================
% ORCiD commands
% ============================================================================

% Store the ORCiD value pulled from the database.
\providecommand{\theORCiDValue}{}

% --- Icon ------------------------------------------------------------------
\NewDocumentCommand{\orcidIcon}{}
{{\sffamily\textcolor{orcid}{\aiOrcid}}}

% --- Pure value-only lookup (no immediate output) ---------------------------
% Fetch the ORCiD value from the database into \theORCiDValue.
% Useful when the value has already been loaded by a previous multi-fetch step,
% or when formatting is needed later.
\NewDocumentCommand{\getORCiDvalue}{O{person} m}
{%
  % #1: Database (default is person)
  % #2: Person id
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  { \cs_set:Npn \theORCiDValue { } }
  { \DTLgetvalueforkey{\theORCiDValue}{orcid}{#1}{id}{#2} }%
}

% --- Formatting a preloaded ORCiD value -------------------------------------
% Expects \theORCiDValue to be set (for example via \getORCiDvalue or
% per \getEntry*[...]{id}{\theORCiDValue=orcid, ...}).
\NewDocumentCommand{\formatORCiD}{O{person} m}
{%
  % #1: Database (for warnings / tag overlay)
  % #2: Person id (for warnings / tag overlay)
  \DTLifnullorempty{\theORCiDValue}
  {%
   \bool_if:NF \lTblrMeasuringBool
   { \msg_warning:nnee {dfg}{missing-orcid}{#1}{#2} }%
   \mbox{~} % if no orcid present
  }
  {%
   \orcidIcon\,%
   \dfgShowTag{#2}{orcid}
   {\href{https://orcid.org/\theORCiDValue}{\theORCiDValue}}[#1]%
  }%
}

% --- Convenience command: fetch + format in a single call -------------------
% Keep the same API as the original \getORCiD so existing calls continue
% to work.
\NewDocumentCommand{\getORCiD}{O{person} m}
{%
  % #1: Database (default is person)
  % #2: Person id
  \bool_if:NF \l_@@_class_fastcompile_project_bool
  {
    \getORCiDvalue[#1]{#2}%
    \formatORCiD[#1]{#2}%
  }
}

% --- Inline variant for literal ORCiD values --------------------------------
% For direct values without database lookup.
\NewDocumentCommand{\ORCiD}{m}
{(\raisebox{-1pt}{\orcidIcon}\,\href{https://orcid.org/#1}{#1})}
```

6.1.12 siunitx configuration

siunitx options must be forwarded with `\PassOptionsToPackage` *before* the package is implicitly loaded by `\UseTblrLibrary{siunitx}`, otherwise they arrive too late.

Key choices: `group-minimum-digits=4`: numbers with 4+ digits get a thousands separator (so `90600` formats as `90,600` EUR). `group-separator={,}`: European style comma separator. `table-auto-round=true`: silently rounds values to fit the `table-format` without requiring `\num` wrappers everywhere. `drop-zero-decimal=true`: suppresses redundant decimal zeros in numeric and EUR cells while retaining genuinely fractional values such as explicit animal-housing rates. `mode=match`: inherits the surrounding text font, so numbers in tables match the `\footnotesize` row font automatically. `text-family-to-math=true`, `text-series-to-math=true`: pass the current font family and series into math mode, keeping bold/italic numbers consistent with their surrounding text.

```
\PassOptionsToPackage{
  mode =  match,
  propagate-math-font = true,
  % reset-math-version  = false,
  % reset-text-family   = false,
  % reset-text-series   = false,
  % reset-text-shape    = false,
  text-family-to-math = true,
  text-series-to-math = true,
  tight-spacing       = true,
  group-minimum-digits  = 4, % --> 1.234 instead of 1234
  group-separator        = {,},% for "12.345" instead of "12 345" %
  table-format          = 6,
  table-alignment-mode  = format,
  table-auto-round      = true,
  drop-zero-decimal     = true,
  locale                = UK,
  input-decimal-markers = {.},
  output-decimal-marker = {.},
  input-ignore          = {,},
  parse-numbers         = true
}{siunitx}
```

This is the package for our tables and we are using quite a bunch of additional libraries.

```
\RequirePackage{tabularray}
\UseTblrLibrary{booktabs}
\UseTblrLibrary{diagbox}
\UseTblrLibrary{siunitx}
\UseTblrLibrary{counter}
\UseTblrLibrary{functional}
```

6.1.13 tabularray setup and table themes

tabularray renders each cell *twice*: once in a hidden measuring pass to determine column widths, and once for final output. This means every database query, numeric calculation or warning inside a cell body would be executed twice. The global boolean `\g_@@_tblr_inside_bool` (toggled via environment hooks) lets code detect the measuring pass so it can skip expensive side effects (e.g. the `missing-key` warning is suppressed when `\lTblrMeasuringBool` is `true`).

```
\bool_new:N \g_@@_tblr_inside_bool
% -- Hook into the beginning and ending of the tblr environment ---------------
\AddToHook{env/tblr/begin}{\bool_set_true:N \g_@@_tblr_inside_bool}
\AddToHook{env/tblr/end}{\bool_set_false:N \g_@@_tblr_inside_bool}
```

6.1.14 Custom table definition

Base Table Theme Defines the shared templates and outer settings used by all table themes. The table font size is established as an ambient declaration here instead of through a `rows={font=...}` property. A more specific cell property such as `font=\bfseries` replaces a row font property, but it inherits an ambient `\footnotesize`; an explicit size in the cell property can still override the default.

```
% Reduce tabularray's default 1.5\bigskipamount above every generated table.
% Change presep here to adjust the gap before tables printed by \printTable.
\SetTblrOuter[tblr]{label=none,entry=none,presep=6pt,headsep=4pt}
% For long tables:
```

```
\cs_new_protected:Npn \mytblr_basecode: {
  \footnotesize
  % Continuous head/footer templates (use ~ for spaces)
  \DeclareTblrTemplate {conthead-text}{default}{%
   \normalfont\color{tableContFootHead}%
   \sffamily\footnotesize%
   \ldots~continuing~from~previous~page.%
  }
  \DeclareTblrTemplate {contfoot-text}{default}{%
   \normalfont\color{tableContFootHead}%
   \sffamily\footnotesize%
   Continue~on~the~following~page~\ldots%
  }
  % Caption continuation uses conthead
  \DeclareTblrTemplate {capcont}{default}{%
   \UseTblrTemplate{conthead-text}{default}%
  }
  % Keep all text rendered below a table at least at footnote size.
  \SetTblrStyle{note-tag}{font=\footnotesize}
  \SetTblrStyle{note-sep}{font=\footnotesize}
  \SetTblrStyle{note-text}{font=\footnotesize}
  \SetTblrStyle{remark-tag}{font=\footnotesize}
  \SetTblrStyle{remark-sep}{font=\footnotesize}
  \SetTblrStyle{remark-text}{font=\footnotesize}
  % Left-align tables
  \RenewDocumentCommand \TblrAlignLeft {} {\RaggedRight}%
  % Common outer settings
}
```

Table themes Three named tblr themes are provided. All inherit the shared `\mytblr_basecode`: helper (continuous head/footer templates, left alignment) via a direct call at theme definition time.

plain: minimal two-rule + alternating-row look used for simple supplementary tables (e.g. the direct-costs breakdown).

```
\NewTblrTheme{plain}{
  \mytblr_basecode:
  % plain-only inner tweaks:
  \SetTblrInner{
    % Keep row metrics independent of descenders such as p, q, f.
    stretch   = 1,
    rowsep    = .667pt,
   row{even}={bg=tableRows},
   width=\linewidth,
   hline{1,Z}={1pt,fg=tableOuterLines},
  }
}
```

main: the full-featured style used by all DFG data tables. Adds a styled first-row header (blue background, bold sans-serif text), a midrule below row 1, caption fonts, and row continuation labels for long tables that span multiple pages.

```
\RenewDocumentCommand\TblrAlignLeft{}{\RaggedRight}
\NewTblrTheme{main}{
  \mytblr_basecode:
  % main-only styles & inner grid:
  \SetTblrStyle{caption-tag}{font=\bfseries\footnotesize\sffamily,tableCaption}
  \SetTblrStyle{caption-text}{
   tableCaption,
   font=\footnotesize,
   halign=l
  }
  \SetTblrStyle{caption}{halign=l}
  \SetTblrTemplate{caption}{simple}
  \SetTblrInner{%
   row{odd}={bg=tableRows},
   width=\linewidth,
   hline{1,Z}={1pt,fg=tableOuterLines},
   rowhead      = 1,
   width        = \textwidth,
   hline{1,Z}   = {
    1pt,
    fg =tableOuterLines
   },
   hline{2} = {1pt, fg=tableOuterLines}, % midrule
    % Keep row metrics independent of descenders such as p, q, f.
    stretch   = .9,
    rowsep    = 1pt,
   row{1}       = {
    bg       = tableHeadingsBG,
    fg       = tableHeadingsFont,
        font     = \bfseries,
```

```
    halign   = {c},
    valign   = {t},
        abovesep = 3pt,
    guard
   },
  }%
  % \SetTblrInner{
  %   rows={ht=.5\baselineskip},
  %   rowsep       = 0pt,
  % }
}
```

requestedFundingStaff: caption styling for the staff roster without a table-wide row background. This lets the roster mark individual non-applicable cells with the `tableNoValue` background.

```
\NewTblrTheme{requestedFundingStaff}{
  \mytblr_basecode:
  \SetTblrStyle{caption-tag}{font=\bfseries\footnotesize\sffamily,tableCaption}
  \SetTblrStyle{caption-text}{tableCaption,\footnotesize}
}
```

A generic long-table environment. Its second argument supplies additional tabularray options and defaults to an empty option list. Let tabularray measure the caption normally; negative skips can pull it into the preceding paragraph or the top rule of the table.

```
\NewDocumentEnvironment{TableEnv}{ m O{} m m }
{
  % #1: colspec
  % #2: further options
  % #3: caption
  % #4: label
  \par
  \begin{tblr}[%
   %  expand=\tblrbody,
   theme=main,
   label  ={#4},
    entry  ={#3},
   caption={#3},
   long
  ]{%
   colspec ={#1},
        row{2-Z}={font=\RaggedRight},
   rowhead = 1, %<- first row is head
   #2 %<- in case we need further adjustments
  }
}
{\end{tblr}\par}
```

At the final part of projects the costs must be listed more granularly. Keep the optional heading and the unbreakable table in one minipage, so the heading cannot be left alone at a page boundary. Measure the currency column at its natural width: a fixed 0.5cm column is too narrow for EUR in some document fonts. The X description column absorbs the width difference while the amount column stays rightmost.

```
\NewDocumentEnvironment{FundingDirectCostsTable}{O{}}
{\par\noindent
  \begin{minipage}{\linewidth}
  \tl_if_blank:nF {#1}
    {\noindent\textbf{\sffamily#1}\par}
  \begin{tblr}[%
   theme=plain, label=none, entry=none]{
   colspec={
     X[t]
        Q[t,r]
   Q[t,si={table-format=9}]
    },
  }
}
{\end{tblr}
  \end{minipage}\par}


\RequirePackage{caption}
\captionsetup{%
  % format=llapx,
  hypcap=false,
  format=plain,
  singlelinecheck=off,
  labelsep={colon}, % saving space
  %justification=RaggedRight,
  font={footnotesize,sf},
```

```
    labelfont={
      sf,
      bf, % bold face
     }
}
% KOMA's caption name also applies to figures boxed by hvFloat.
\renewcaptionname{english}{\figurename}{Fig.}
\RequirePackage{pgfgantt}
\RequirePackage{placeins}
\RequirePackage{hvfloat} % load after tabularray!!
\hvFloatSet{capPos=bottom,capVPos=top,floatPos=!hb}% this are the default values.
\RequirePackage{marginalia}
\marginaliasetup{
  xsep=0pt,
  width=13mm,
  style=\raggedleft\footnotesize,
  pos=auto,
}
\bool_if:NTF \l_@@_class_cv_bool
  {
    \tl_set_eq:NN \l_@@_custom_biblatexstyle_active_tl
      \l_@@_custom_biblatexstyle_cv_tl
    \tl_set_eq:NN \l_@@_custom_biblatexfeatures_active_tl
      \l_@@_custom_biblatexfeatures_cv_tl
  }
  {
    \tl_set_eq:NN \l_@@_custom_biblatexstyle_active_tl
      \l_@@_custom_biblatexstyle_proposal_tl
    \tl_set_eq:NN \l_@@_custom_biblatexfeatures_active_tl
      \l_@@_custom_biblatexfeatures_proposal_tl
  }
\RequirePackage[
  backend=biber,
  % using the most common style and also saving space.
  % style=ext-numeric, % taken from biblatex-ext
  % just in case we want to switch the style.
  % style=ext-alphabetic,
  %style=ext-authornumber-ecomp,
  style=\l_@@_custom_biblatexstyle_active_tl,
  %introcite=label,
  autocite=inline,
  defernumbers=true, % reset numbered bibliography for each chapter
  giveninits=true, % abbreviate the first names
  maxcitenames=1,
  mincitenames=1,
  maxbibnames=99, % show up to 6 authors of an bib entry (et al. thereafter)
  %nohashothers=true,
  %uniquelist=true,
  %uniquename=allfull,
  %labeldatepartes=true
  \l_@@_custom_biblatexfeatures_active_tl
]{biblatex}
```

In final documents, biblatex-ext-oa can mark open-access publications directly in the bibliography. Draft documents do not load the package, so they avoid DOI API checks and related warnings. The DOI API backend is useful, but biblatex-ext-oa-doiapi and the `doiapi` option can only be used with LuaLaTeX. With LuaTeX binaries from version 1.17.0 onwards, the LuaSocket module `socket.http` is exposed only when LuaTeX is started with the command-line option `--socket`. If we always passed `doiapi=true` in final mode, a normal final compile without `--socket` would abort while loading biblatex-ext-oa-doiapi.

After a user opts in with a contact address, the guard asks LuaTeX whether `require("socket.http")` succeeds. The result is written back into the temporary expl3 string `\l_tmpa_str` as the literal text `true` or `false`. The Lua code is written without TeX-relevant whitespace, because expl3 ignores spaces while tokenising this part of the class source; semicolons keep the Lua syntax unambiguous after docstrip has generated the `.cls` file.

Automatic lookup is enabled only when LuaSocket is available and the user has supplied a non-empty `doiAPIMail` → P. 66. The address is passed to `\SetDOIAPIMail`; the class never sends a placeholder or maintainer address on behalf of its users. If either prerequisite is missing, the class emits a specific warning and loads biblatex-ext-oa with `doiapi=false` so manual BibTeX annotations such as `doi+an:default = {=openaccess}` continue to work.

The TikZ symbol package draws the open-access symbols. The `oanet` symbol is the open-access symbol from open-access.net (https://open-access.net/), logo by Medien + Design, Center for Digital

Systems, Competence Center for E-Learning and Multimedia, Freie Universität Berlin, CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/).

```
\cs_new_protected:Npn \@@_biblatex_oa_load_without_api:
  {
    \RequirePackage[
      doiapi=false,
      symbolpackage=tikz,
      symbol=oanet]{biblatex-ext-oa}
  }

\bool_if:NT \l_@@_class_final_bool
  {
    \tl_if_blank:VTF \l_@@_custom_doi_api_mail_tl
      {
        \msg_info:nn { dfg } { biblatex-ext-oa-doiapi-email-missing }
        \@@_biblatex_oa_load_without_api:
      }
      {
        \str_set:Nx \l_tmpa_str
          {
            \directlua
              {if(pcall(require,"socket.http"))then;tex.sprint("true");else;tex.sprint("false");end}
          }
        \str_if_eq:VnTF \l_tmpa_str { true }
          {
            \RequirePackage[
              doiapi=true,
              symbolpackage=tikz,
              symbol=oanet]{biblatex-ext-oa}
            \ExecuteBibliographyOptions{openaccess=doiapi}
            \exp_args:NV \SetDOIAPIMail \l_@@_custom_doi_api_mail_tl
          }
          {
            \msg_warning:nn { dfg } { biblatex-ext-oa-doiapi-unavailable }
            \@@_biblatex_oa_load_without_api:
          }
      }
  }
% A project-text build with publication processing disabled must not require a
% matching biber run merely to keep inline source references readable.  Replace
% the common single-citation commands with quiet key markers; enabling
% |fastcompile-project-show-publications| restores biblatex's normal commands unchanged.
\bool_if:nTF
  {
    \l_@@_class_fastcompile_project_bool
    && ! \l_@@_class_fastcompile_project_show_publications_bool
  }
  {
    % Do not register bibliography data sources or parse a stale full-build
    % .bbl.  Keep biblatex's ordinary initialisation and refsection setup, but
    % replace its data-file stage with the minimum empty-section state.
    \cs_set_protected:Npn \blx@bblfile
      {
        \blx@secinit
        \blx@generate@bbl@mdfivesum@notfound
        \global\togglefalse{blx@bbldone}
        \csnumgdef{blx@labelnumber@\the\c@refsection}{0}
      }
    \NewDocumentCommand{\dfgFastCitation}{m}
      { {\color{black-50}\scriptsize\ttfamily [#1]} }
    \RenewDocumentCommand{\autocite}{s o o m}
      { \dfgFastCitation{#4} }
    \RenewDocumentCommand{\cite}{s o o m}
      { \dfgFastCitation{#4} }
    \RenewDocumentCommand{\parencite}{s o o m}
      { \dfgFastCitation{#4} }
    \RenewDocumentCommand{\textcite}{s o o m}
      { \dfgFastCitation{#4} }
    \RenewDocumentCommand{\footcite}{s o o m}
      { \dfgFastCitation{#4} }
    \RenewDocumentCommand{\smartcite}{s o o m}
      { \dfgFastCitation{#4} }
    \RenewDocumentCommand{\nocite}{m} { }
    % Ignore citation requests restored from auxiliary files produced by a full
    % proposal build.  The fast source commands above do not write replacements.
    \cs_set_protected:Npn \abx@aux@cite   #1#2   { }
    \cs_set_protected:Npn \abx@aux@segm   #1#2#3 { }
    \cs_set_protected:Npn \abx@aux@nocite #1#2   { }
    % This mode deliberately has no bibliography work, so do not advertise a
    % biber rerun in the end-of-document logreq record.
    \AtEndDocument
    {
      \global\let\blx@rerun@biber\relax
      \global\togglefalse{blx@runbiber}
    }
  }
```

```
1457    {
1458      \bool_if:NF \l_@@_class_documentation_bool
1459      {
1460      \addbibresource{bib/dfg-base.bib}
1461      % The conversion workflow stores bibliography entries that still need
1462      % review separately.  Load that file when present without requiring it for
1463      % normal template installations.
1464      \IfFileExists{bib/conversion-missing.bib}
1465        {\addbibresource{bib/conversion-missing.bib}}
1466        {}
1467      \addbibresource{bib/dfg-strings.bib}
1468      }
1469    }
```

The `\AtEveryBibitem` hook strips fields that are irrelevant for a funding proposal (URL, month, note, ISSN/ISBN, language) from every entry, keeping the reference list compact. These fields are cleared during output only; the `.bib` source files are unchanged. A second part of the hook selects the full-entry bold face. The frame and project bibliography wrappers set a local keyword only while printing their list; biblatex groups each item, so the font change does not leak into the following entry. The hook runs after `\bibfont` and before the entry's fields are printed.

```
1470  \tl_new:N \l_@@_bibentry_highlight_keyword_tl
1471  \AtEveryBibitem{
1472    \clearfield{url}
1473    \clearfield{month}
1474    \clearfield{note}
1475    \clearfield{issn}
1476    \clearfield{isbn}
1477    \clearlist{language}
1478    \tl_if_empty:NF \l_@@_bibentry_highlight_keyword_tl
1479    {
1480      \exp_args:NV \ifkeyword \l_@@_bibentry_highlight_keyword_tl
1481        {\bfseries}{}
1482    }}
```

6.1.15 Shared authorship annotations

The `annote` field in `.bib` entries supports two special values: `equal` (for equal-first-author contributions) and `sharedlast` (for shared last authorship). When present, `\mkbibnamefamily` appends an asterisk `*` after the family name via `\addendumasterisk`. The asterisk is only printed inside bibliography environments (`\ifbibliography`), not in inline citations.

Use in `.bib`: add `addendum = equal` or `addendum = sharedlast` (no `{}`) to the affected author's annotation field.

```
1483  \newcommand{\addendumasterisk}{\ifbibliography{*}{}}
1484  \newcommand{\addendumhashtag}{\ifbibliography{\textsuperscript{\#}{}}}
1485  % There is also the possibility that authors share the last authorship.
1486  % with the bib-string `lastShared` we can get it work.
1487  \renewcommand*{\mkbibnamefamily}[1]{%
1488    \ifitemannotation{equal} % first check for equal contribution
1489    {#1\addendumasterisk}% add a * for those authors
1490    {
1491     % Then we check for "shared last authorship"
1492     \ifitemannotation{sharedlast}%
1493     {#1\addendumhashtag}% adding a symbol
1494     {#1}% do nothing for any one else.
1495    }}
```

```
1496  % \subsection{datatool: database backend}
1497  %
1498  % \pkg{datatool} provides the |DTL*| commands that are the actual in-memory
1499  % database engine behind the keyset-based metadata system.  \pkg{csquotes}
1500  % is loaded just before it because \cs{blockcquote} is used in the
1501  % documentation abstract and the class patches \cs{mkccitation} to suppress
1502  % the extra parentheses that csquotes adds around \cs{autocite} results.
1503  \RequirePackage{csquotes}
1504  \renewcommand{\mkccitation}[1]{ (#1)}
1505  \RequirePackage{datatool}
```

loading `tikz` at least for the thumb index.

```
1506  \RequirePackage{tikz}
1507  \ExplSyntaxOff
1508  \usetikzlibrary{calc}
1509  \usetikzlibrary{shapes.misc}
1510  \usetikzlibrary{patterns,patterns.meta}
1511  \ExplSyntaxOn
```

loading these two packages last

```
\RequirePackage{hyperref}
\RequirePackage{cleveref}
% enumitem creates a dedicated counter for each custom-list level but does not
% expose the parent relation to cleveref.  Register that relation so nested
% work-package references sort by parent first and ranges never cross parents.
\apptocmd{\cref@resetby}
  {\cref@ifstreq{#1}{workpackagesii}{\def#2{workpackagesi}}{}}
  {}
  {\PackageError{dfg}
    {Unable~to~register~the~nested~work-package~counter}
    {Nested~work-package~references~may~be~sorted~incorrectly.}}
% labelcref requires one reference type for mixed parent/child lists.
% Keep the parent-counter registration above for sorting and range boundaries.
\crefalias{workpackagesii}{workpackagesi}
\crefname{workpackagesi}{\color{referenceWP}\textbf{WP}}{\color{referenceWP}\textbf{WPs}}
\creflabelformat{workpackagesi}{\color{referenceWP}#2\textbf{#1}#3}
\crefname{workpackagesii}{\color{referenceWP}\textbf{WP}}{\color{referenceWP}\textbf{WPs}}
\creflabelformat{workpackagesii}{\color{referenceWP}#2\textbf{#1}#3}

\crefname{aimsi}{\bfseries\color{referenceAim}Aim}{\bfseries\color{referenceAim}Aims}
\creflabelformat{aimsi}{\color{referenceAim}#2\color{referenceAim}\bfseries#1\color{referenceAim}#3}
% Aux-file entry point: use the authored spelling for both reference forms.
\NewDocumentCommand{\dfgAimReferenceName}{m m}
{
  \crefname{#1}{\bfseries\color{referenceAim}#2}
    {\bfseries\color{referenceAim}#2}
  \Crefname{#1}{\bfseries\color{referenceAim}#2}
    {\bfseries\color{referenceAim}#2}
  \creflabelformat{#1}
    {\color{referenceAim}##2\bfseries##1##3}
  % The kernel reads aux files in a group, after cleveref's format setup.
  % Preserve only the names globally; cleveref's format builders already
  % make global definitions without leaking their temporary variables.
  \clist_map_inline:nn { cref,Cref }
  {
    \cs_gset_eq:cc {##1@#1@name} {##1@#1@name}
    \cs_gset_eq:cc {##1@#1@name@plural} {##1@#1@name@plural}
  }
  \@crefdefineallformats{#1}
}


% \subsection{Preamble end: global metadata extraction and hyperref}
%
% The \cs{AfterEndPreamble} hook fires after all user preamble code
% (including |dfg-customization.tex| and all \cs{addEntry} calls) but
% before \cs{begin}|{document}|.  This is the earliest point at which
% the |meta| database is guaranteed to be fully populated.
%
% The block extracts the five core CRC fields into named macros
% (\cs{theCRCname}, \cs{theCRCnumber}, \cs{theCRCaffiliation}, etc.)
% that are used in headers, footers, and the hyperref PDF metadata.  It
% treats |meta.ref-affiliation-id| as a comma-separated list: the public
% affiliation macros contain human-readable name and location lists, while
% \cs{theApplicantID} retains the normalized id list for database filters.  It
% also derives \cs{theFundingPeriodNumeralPrevious} by subtracting 1 from
% \cs{theFundingPeriodNumeral}, which is used in the gender equality tables
% to reference the previous funding period.
%
% The \cs{hypersetup} call here (rather than in the preamble) avoids
% needing the user to duplicate CRC metadata in a separate hypersetup block.
\clist_new:N \l_@@_applicant_affiliation_id_clist
\seq_new:N \l_@@_applicant_affiliation_name_seq
\seq_new:N \l_@@_applicant_affiliation_location_seq
\seq_new:N \l_@@_other_applicant_affiliation_name_seq
\tl_new:N \l_@@_applicant_affiliation_name_tl
\tl_new:N \l_@@_applicant_affiliation_location_tl

\cs_new_protected:Npn \@@_set_applicant_affiliation_metadata:
{
  \DTLifnullorempty{\theApplicantID}
    { \clist_clear:N \l_@@_applicant_affiliation_id_clist }
    {
      \clist_set:Nx \l_@@_applicant_affiliation_id_clist
        { \theApplicantID }
      \clist_remove_duplicates:N \l_@@_applicant_affiliation_id_clist
    }
  \tl_set:Nx \l_tmpa_tl
    { \clist_use:Nn \l_@@_applicant_affiliation_id_clist {,} }
  \cs_set:Npx \theApplicantID { \exp_not:V \l_tmpa_tl }
  \cs_set_eq:NN \theAffiliationValue \theApplicantID
  \seq_clear:N \l_@@_applicant_affiliation_name_seq
  \seq_clear:N \l_@@_applicant_affiliation_location_seq
  \clist_map_inline:Nn \l_@@_applicant_affiliation_id_clist
  {
```

```
    \DTLifdbexists{affiliation}
    {
      \DTLgetvalueforkey{\l_@@_applicant_affiliation_name_tl}
        {name}{affiliation}{id}{##1}
      \DTLgetvalueforkey{\l_@@_applicant_affiliation_location_tl}
        {location}{affiliation}{id}{##1}
    }
    {
      \tl_set:Nn \l_@@_applicant_affiliation_name_tl {##1}
      \tl_clear:N \l_@@_applicant_affiliation_location_tl
    }
    \DTLifnulloremptyempty{\l_@@_applicant_affiliation_name_tl}
      { \tl_set:Nn \l_@@_applicant_affiliation_name_tl {##1} }
      { }
    \tl_set:Ne \l_tmpb_tl
    {
      \exp_not:N \dfgShowTag{##1}{name}
        {\exp_not:V \l_@@_applicant_affiliation_name_tl}[affiliation]
    }
    \seq_put_right:NV \l_@@_applicant_affiliation_name_seq \l_tmpb_tl
    \DTLifnulloremptyempty{\l_@@_applicant_affiliation_location_tl}
      { }
      {
        \seq_if_in:NVF \l_@@_applicant_affiliation_location_seq
          \l_@@_applicant_affiliation_location_tl
        {
          \tl_set:Ne \l_tmpb_tl
          {
            \exp_not:N \dfgShowTag{##1}{location}
              {\exp_not:V \l_@@_applicant_affiliation_location_tl}
              [affiliation]
          }
          \seq_put_right:NV \l_@@_applicant_affiliation_location_seq
            \l_tmpb_tl
        }
      }
  }
  \cs_set:Npn \theCRCaffiliation
  {
    \seq_use:Nnnn \l_@@_applicant_affiliation_name_seq
      {~and~} {,~} {,~and~}
  }
  \cs_set:Npn \theCRCaffiliationLocation
  {
    \seq_use:Nnnn \l_@@_applicant_affiliation_location_seq
      {~and~} {,~} {,~and~}
  }
  \seq_set_eq:NN \l_@@_other_applicant_affiliation_name_seq
    \l_@@_applicant_affiliation_name_seq
  \seq_pop_left:NNF \l_@@_other_applicant_affiliation_name_seq \l_tmpa_tl
    { \tl_clear:N \l_tmpa_tl }
  \cs_set:Npn \theCRCCoordinatingAffiliation
    { \seq_item:Nn \l_@@_applicant_affiliation_name_seq {1} }
  \cs_set:Npn \theCRCOtherApplicantAffiliations
  {
    \seq_use:Nnnn \l_@@_other_applicant_affiliation_name_seq
      {~and~} {,~} {,~and~}
  }
}

\NewExpandableDocumentCommand{\dfgApplicantAffiliationsTF}{m m}
{
  \int_compare:nNnTF
    { \clist_count:N \l_@@_applicant_affiliation_id_clist } > {1}
    {#2}
    {#1}
}

\NewExpandableDocumentCommand{\dfgOtherApplicantAffiliationsTF}{m m}
{
  \int_compare:nNnTF
    { \clist_count:N \l_@@_applicant_affiliation_id_clist } > {2}
    {#2}
    {#1}
}

\NewDocumentCommand{\dfgCRCName}{}
  { \dfgShowTag{crc}{name}{\theCRCname}[meta] }
\NewDocumentCommand{\dfgCRCNameAddon}{}
  { \dfgShowTag{crc}{nameAddon}{\theCRCnameAddon}[meta] }
\NewDocumentCommand{\dfgCRCNumber}{}
  { \dfgShowTag{crc}{number}{\theCRCnumber}[meta] }
\NewDocumentCommand{\dfgFundingPeriodText}{}
  { \dfgShowTag{crc}{fundingPeriodText}{\theFundingPeriodText}[meta] }
\NewDocumentCommand{\dfgFundingPeriodNumeral}{}
  { \dfgShowTag{crc}{fundingPeriodNumeral}{\theFundingPeriodNumeral}[meta] }
\NewDocumentCommand{\dfgFundingYear}{m}
```

```
{
  \dfgShowTag{crc}{fundingYear#1}
  {
    \int_case:nnF {#1}
    {
      {1}{\tmpYearOne}
      {2}{\tmpYearTwo}
      {3}{\tmpYearThree}
      {4}{\tmpYearFour}
      {5}{\tmpYearFive}
    }
    {#1}
  }[meta]
}
\NewDocumentCommand{\dfgFundingYearPrevious}{m}
{
  \dfgShowTag{crc}{fundingYearPrevious#1}
  {
    \int_case:nnF {#1}
    {
      {1}{\tmpYearPreviousOne}
      {2}{\tmpYearPreviousTwo}
      {3}{\tmpYearPreviousThree}
      {4}{\tmpYearPreviousFour}
      {5}{\tmpYearPreviousFive}
    }
    {#1}
  }[meta]
}

\AfterEndPreamble{
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  {
    \clist_clear:N \l_@@_applicant_affiliation_id_clist
    \cs_set:Npn \theApplicantID                  { }
    \cs_set:Npn \theCRCname                      { CRC~project~draft }
    \cs_set:Npn \theCRCnameAddon                 { }
    \cs_set:Npn \theCRCnumber                    { draft }
    \cs_set:Npn \theFundingPeriodNumeral         { 1 }
    \cs_set:Npn \theFundingPeriodText            { draft }
    \cs_set:Npn \theAffiliationValue             { }
    \cs_set:Npn \theCRCaffiliation               { }
    \cs_set:Npn \theCRCaffiliationLocation       { }
    \cs_set:Npn \theCRCCoordinatingAffiliation   { }
    \cs_set:Npn \theCRCOtherApplicantAffiliations{ }
    \cs_set:Npn \tmpYearOne                      { Year~1 }
    \cs_set:Npn \tmpYearTwo                      { Year~2 }
    \cs_set:Npn \tmpYearThree                    { Year~3 }
    \cs_set:Npn \tmpYearFour                     { Year~4 }
    \cs_set:Npn \tmpYearFive                     { Year~5 }
    \cs_set:Npn \tmpYearPreviousOne              { Previous~year~1 }
    \cs_set:Npn \tmpYearPreviousTwo              { Previous~year~2 }
    \cs_set:Npn \tmpYearPreviousThree            { Previous~year~3 }
    \cs_set:Npn \tmpYearPreviousFour             { Previous~year~4 }
    \cs_set:Npn \tmpYearPreviousFive             { Previous~year~5 }
  }
  {
  \DTLgetvalueforkey{\theApplicantID}{ref-affiliation-id}{meta}{id}{crc}
  \DTLgetvalueforkey{\theCRCname}{name}{meta}{id}{crc}% getting Title of CRC
  \DTLgetvalueforkey{\theCRCnameAddon}{nameAddon}{meta}{id}{crc}% getting name addon of CRC
  \DTLgetvalueforkey{\theCRCnumber}{number}{meta}{id}{crc}% getting number of CRC
  \DTLgetvalueforkey{\theFundingPeriodNumeral}{fundingPeriodNumeral}{meta}{id}{crc}% getting period of CRC
  \DTLgetvalueforkey{\theFundingPeriodText}{fundingPeriodText}{meta}{id}{crc}% getting period of CRC
  \@@_set_applicant_affiliation_metadata:
  \DTLgetvalueforkey{\tmpYearOne}  {fundingYear1}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearTwo}  {fundingYear2}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearThree}{fundingYear3}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearFour} {fundingYear4}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearFive} {fundingYear5}{meta}{id}{crc}%

  \DTLgetvalueforkey{\tmpYearPreviousOne}  {fundingYearPrevious1}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearPreviousTwo}  {fundingYearPrevious2}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearPreviousThree}{fundingYearPrevious3}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearPreviousFour} {fundingYearPrevious4}{meta}{id}{crc}%
  \DTLgetvalueforkey{\tmpYearPreviousFive} {fundingYearPrevious5}{meta}{id}{crc}%
  }


  \providecommand{\thisCRC}
    {CRC\,\texorpdfstring
      {\dfgShowTag{crc}{number}{\theCRCnumber}[meta]}
      {\theCRCnumber}\xspace}
  % Shared implementation for the user-level \CRC and \TRR commands.
  % #1 is the star boolean, #2 the optional title, #3 the printed programme
  % type, and #4 the CRC/TRR number to print and compare with the current CRC.
  % Current CRCs use \thisCRC in the unstarred form; starred current references
  % include the CRC title from the metadata unless an explicit title is given.
```

```
1771    % External CRCs/TRRs always print the requested number and add the optional
1772    % title when one is supplied.
1773  \cs_new_protected:Npn \@@_crc_print:nnnn #1 #2 #3 #4
1774    {
1775      \tl_if_eq:VnTF \theCRCnumber {#4}
1776        {
1777          \IfBooleanTF{#1}
1778            {
1779              #3\,#4
1780              \IfValueTF{#2}
1781                { \ \textit{#2} }
1782                { \ \textit{\dfgShowTag{crc}{name}{\theCRCname}[meta]} }
1783            }
1784            {
1785              \thisCRC
1786              \IfValueT{#2}
1787                { \ \textit{#2} }
1788            }
1789        }
1790        {
1791          #3\,#4
1792          \IfValueT{#2}
1793            { \ \textit{#2} }
1794        }
1795    }
1796
1797    % User-level wrappers around \@@_crc_print:nnnn. They differ only in the
1798    % printed programme type.
1799  \NewDocumentCommand{\CRC}{s o m}
1800    {
1801      \@@_crc_print:nnnn {#1} {#2} {CRC} {#3}
1802    }
1803
1804  \NewDocumentCommand{\TRR}{s o m}
1805    {
1806      \@@_crc_print:nnnn {#1} {#2} {CRC/TRR} {#3}
1807    }
1808   %\DTLgetvalueforkey{\theNameSpokespersonPosition}{position}{VIP}{specialRole}{spokesperson}% getting Title of Spokes
1809    %\DTLgetvalueforkey{\theNameSpokespersonName}{name}{VIP}{specialRole}{spokesperson}% getting Title of CRC
1810    %\DTLgetvalueforkey{\theNameSpokespersonNameFirst}{nameFirst}{VIP}{specialRole}{spokesperson}% getting Title of CRC
1811    \newcommand{\theFundingPeriodNumeralPrevious}{%
1812     \number\numexpr\theFundingPeriodNumeral-1\relax%
1813    }
1814    % hyperref options
1815    \hypersetup{          % setup the hyperref-package options
1816     % unicode      = true,
1817     pdftitle     = {\theCRCname.\space\theCRCnameAddon},  %   - title (PDF meta)
1818     pdfsubject   = {\crc},%   - subject (PDF meta)
1819     pdfauthor    = {CRC\,\theCRCnumber~ \theCRCname},  %   - author (PDF meta)
1820     %   pdfauthortitle = {\theNameSpokespersonPosition},
1821     %   pdfcopyright = {Copyright (c) \thisCRC All rights reserved.},
1822     pdfhighlight = /N,
1823     pdfdisplaydoctitle = true,
1824     pdflang     = {en},
1825     %   pdfcaptionwriter = {\theNameSpokespersonNameFirst \theNameSpokespersonName},
1826     % pdfencoding = auto,
1827     pdfproducer = {LuaLaTeX},
1828     % pdflicenseurl = {http://creativecommons.org/licenses/by-nc-nd/4.0/},
1829     % plainpages  = false,      %   -
1830     colorlinks  = true, % -Colours links instead of ugly boxes
1831     urlcolor    = blue, %Colour for external hyperlinks
1832     linkcolor   = black,%rwthBlue, %Colour of internal links
1833     citecolor   = black, %Colour of citations
1834     % linkbordercolor=rwthRed,
1835     linktoc     = page,
1836     pdfborder   = {0 0 0},      %   -
1837     breaklinks  = true,      %   - allow line break inside links
1838     bookmarksnumbered  = true,     %
1839     bookmarksopenlevel = 1,
1840     bookmarksopen      = true,     %
1841     bookmarksdepth     = 3,
1842     %  final       = true,  % = true, only for web documents.
1843     %  hypertexnames = false,
1844     %  pageanchor=false,
1845     %https://tex.stackexchange.com/questions/623377/how-to-increase-the-setting-dest-names-size-using-
    lualatex-on-tl-2021
1846    }
1847  }
```

6.1.16 `\changeSettings`

`\changeSettings{`⟨*key=value,...*⟩`}` is the public command used to customise the document class beyond its defaults. It delegates directly to `\keys_set:nn{dfg/custom}{\meta{key=value,...}}`.

It should be called inside `preamble/dfg-customization.tex` so that all customisations are centralised. Examples:

Typical changeSettings call

```
%   \changeSettings{
%     font                          = Arial,
%     crcLogoPath                   = img/logo-crc,
%     affiliationLogoPath           = img/logo-affiliation,
%     doiAPIMail                    = researcher@example.org,
%     projectPIOncePerSection       = true,
%     collaborationAreaColors       = {A=projectA,B=projectB,Q=projectQ},
%     collaborationAreaPalette      = {orange,violet,teal,brown},
%     salary@postDoc                = 90000,
%     animalHousing@mouse           = 0.80,
%     thumbindex                    = false,
%     biblatexstyleCV               = ext-numeric-comp,
%     biblatexstyleFeatureCV        = {sorting=none},
%     biblatexstyleProposal         = ext-authoryear-ecomp,
%     biblatexstyleFeatureProposal= {},
%     color@table@rows              = blue-10,
%   }
%
```

The command is defined before biblatex is loaded so bibliography package options from the customization file take effect in time.

6.1.17 Thumb index implementation

The thumb index is a series of colour tabs printed on the outer margin of each page to help readers navigate between chapters or project sections. Each tab occupies one slot in a vertical grid that repeats with a period equal to `thumbindex@pattern` (default: 10).

Geometry calculation: `\thumb_compute_height:` calculates the slot height as `(\cs{paperheight} - topmargin - bottommargin) / pattern`. In the default `auto` mode, the margins are derived from the current text-block position and `\textheight`, so the slot height is `\textheight / pattern`. The calculation runs when the tab is drawn, allowing later layout changes to affect the thumb-index grid.

Placement: `\thumb_place_node:nnnn` draws an absolutely-positioned TikZ node on the current page using `remember picture,overlay`. Its reference point is `(current page.north west/east) -` followed by `(0, topmargin + height*(index-1/2))`, centering each tab in its slot. The rectangular tab has 7pt rounded corners and is shifted 7pt past the page boundary, clipping the outer corners while retaining the configured visible depth. Its label is rotated 90 degrees in opposite directions on facing pages so the tab stays within the marginal column. The default `thumbindex@orientation=outwards` points the tops of the letters towards the outer edge. Choosing `inwards` adds 180 degrees to the label rotation on both sides, without changing the tab geometry. The label is inset 2 mm towards the text block on either side; paired horizontal kerns leave the tab dimensions and placement unchanged.

Call pattern: `\printThumbIndex[`⟨*colour*⟩`]{`⟨***label***⟩`}` increments the slot counter, wrapping at `pattern`, then injects the node into both the header and footer layers via `\lehead`/`\lohead` so the tab appears on the page regardless of which scrlayer-scrpage style is active. The optional colour argument allows project-specific hues (typically the `projectA`, `projectB`, `projectQ` aliases).

`\printThumbIndexOff` clears the header/footer layers to suppress the tab on pages where no thumb index is desired (e.g. the bibliography). `\resetThumbIndex` resets the slot counter so that the next `\printThumbIndex` call starts at the first slot again. This is useful when a manually inserted section should begin a new thumb-index sequence.

```
1848  \int_new:N \l_thumb_index_int
1849  \int_zero:N \l_thumb_index_int
```

The effective margins follow the current text block unless overridden with explicit page-edge dimensions. Calculate them when drawing each tab rather than at class load time, when page geometry may not yet be final.

```
\dim_new:N \l_thumb_height_dim
\dim_new:N \l_thumb_topmargin_dim
\dim_new:N \l_thumb_bottommargin_dim
\cs_new_protected:Npn \thumb_compute_height:
{
  \bool_if:NTF \l_@@_custom_thumbindex_topmargin_set_bool
    { \dim_set_eq:NN \l_thumb_topmargin_dim
        \l_@@_custom_thumbindex_topmargin_dim }
    {
      \dim_set:Nn \l_thumb_topmargin_dim
        { 1in + \voffset + \topmargin + \headheight + \headsep }
    }
  \bool_if:NTF \l_@@_custom_thumbindex_bottommargin_set_bool
    { \dim_set_eq:NN \l_thumb_bottommargin_dim
        \l_@@_custom_thumbindex_bottommargin_dim }
    {
      \dim_set:Nn \l_thumb_bottommargin_dim
        {
          \paperheight
          - ( 1in + \voffset + \topmargin + \headheight + \headsep )
          - \textheight
        }
    }
  \dim_set:Nn \l_thumb_height_dim
    {
      ( \paperheight - \l_thumb_topmargin_dim
        - \l_thumb_bottommargin_dim )
      / \l_@@_custom_thumbindexpattern_int
    }
}
```

defining now the layout of the thumb index. Be aware of the stripped spaces in the LaTeX3 environment.

```
\tikzset
{
  thumb/.style=
   {
    fill   = thumbIndexBG ,      % colour may be changed later
    text   = thumbIndexFont,
    minimum~ height = \dim_use:N \l_thumb_height_dim,
  minimum~ width  = \dim_eval:n
    { \l_@@_custom_thumbindex_width_dim + 7pt },
  outer~ sep      = 0pt,
    font            = \tl_use:N \l_@@_custom_thumbindex_font_tl,
  inner~ sep      = 1mm,
   }
}
```

we need a couple of helper commands though. The <even/odd> macros are now tiny wrappers around a single protected function that does the arithmetic in expl3.

```
\cs_new_protected:Npn \thumb_place_node:nnnn #1#2#3#4
{
  %  #1 = west|east    (anchor direction)
  %  #2 = shift past the page edge
  %  #3 = label rotation
  %  #4 = node contents
  \thumb_compute_height:
  \begin{tikzpicture}[remember~ picture,overlay]
   \node[
    thumb,
    anchor  =#1,
  rectangle,
  rounded~ corners=7pt,
  xshift  =#2
   ]
   % ------------------------------------------------------------
   %   absolute position = (current page.north <west/east>) -
   %                       (0 , vertical offset)
   %   The whole thing must be inside $ … $.
   % ------------------------------------------------------------
   at
   ($ (current~page.north~ #1)~ --~
    (0,\dim_eval:n{
            \l_thumb_topmargin_dim
            + \l_thumb_height_dim * (\l_thumb_index_int - 1)
            + \l_thumb_height_dim / 2
    }) $)
 {
```

```
      % Inset the text by 2mm; opposite kerns preserve the tab dimensions.
      \str_if_eq:nnTF {#1}{west}
        {\kern2mm}{\kern-2mm}
      \rotatebox[origin=c]{#3}{#4}
      \str_if_eq:nnTF {#1}{west}
        {\kern-2mm}{\kern2mm}
    };
  \end{tikzpicture}
}

\NewDocumentCommand{\evenpageletterthumb}{m}
{
  \thumb_place_node:nnnn {west}{-7pt}
    {\int_eval:n {90 + \l_@@_custom_thumbindex_rotation_int}}{#1}
}
\NewDocumentCommand{\oddpageletterthumb}{m}
{
  \thumb_place_node:nnnn {east}{7pt}
    {\int_eval:n {-90 + \l_@@_custom_thumbindex_rotation_int}}{#1}
}
```

Now the key command `\printThumbIndex`.

```
\NewDocumentCommand{\printThumbIndex}{ O{thumbIndexBG} m }
{
  %   #1 (optional) - colour definition used when no project is given
  %   #2 (mandatory) - the letter (or any material) to be printed

  \bool_if:NT \l_@@_custom_thumbindex_bool
  {
   \colorlet{thumbIndexBG}{#1}

   % ---- reset index if we have reached the bottom of the column -----
   \int_compare:nNnT { \l_thumb_index_int } = { \l_@@_custom_thumbindexpattern_int }
   { \int_zero:N \l_thumb_index_int }

   % ---- step to the *next* position ----------------------------------
   \int_incr:N \l_thumb_index_int

   % ---- actually place the thumb on the current page ------------------
   %   (the same node is put into both the header and the foot, so it
   %   appears on the page no matter which page-style you use)
   \lehead[\evenpageletterthumb{#2}]{\evenpageletterthumb{#2}}
   \lohead[\oddpageletterthumb{#2}]{\oddpageletterthumb{#2}}
  }
}
\NewDocumentCommand{\resetThumbIndex}{}
{
  \int_zero:N \l_thumb_index_int
}
\NewDocumentCommand{\printThumbIndexOff}{}
{
  \lehead[]{}
  \lohead[]{}
}
```

6.1.18 Bold-author mechanism

The DFG requires that the names of all CRC project leaders are printed in bold in every bibliography list so reviewers can immediately identify the investigators' output. This is technically non-trivial because biblatex does not natively support singling out individual author names.

The implementation uses a hash-based approach:

1. For each PI, `\addboldnames` → P. 73 is called with their name as it appears in the `.bib` file (and all common aliases via `name-alias` → P. 53). Each name is written as a `\@misc` `dataonly` entry to an auxiliary `.bib` file.
2. During the `\AfterPreamble` pass, the cite command `\nhblx@getmethehash` processes each entry and extracts its biblatex internal hash (a fingerprint stable across style changes) into the list `\nhblx@boldhashes`.
3. The name wrapper `boldifhashinlist` applies `\mkboldifhashinlist` to every printed author name; if the author's hash appears in the list, the name is wrapped in `\mkbibbold` and coloured with `color@reference@boldAuthor`.

The mechanism is only activated when `emphasizeAuthors=true` (the default). Fallback no-op definitions of `\addboldnames` → P. 73 and `\resetboldnames` → P. 73 are provided via `\providecommand` so that content files compile correctly even if the feature is disabled.

\addboldnames → P. 73 should be called in the preamble of both the proposal document and the standalone CV document, typically inside preamble/dfg-customization.tex. For displaying certain authors in bold letters, emphasizeAuthors → P. 70 (for the switch) and \addboldnames → P. 73 (for the addition of authors).

```
\int_new:N  \g_@@_boldnames_int
\iow_new:N  \g_@@_boldnames_iow
\prop_new:N \g_@@_boldnames_hashes_prop
\tl_new:N   \l_@@_boldnames_name_tl

\bool_if:NT \l_@@_custom_emphasize_authors_bool
{
  % ---------- Auxiliary .bib file ----------------------------------

  \bool_if:NTF \l_@@_class_documentation_bool
    {\tl_const:Nn \c_@@_boldnames_bibfile_tl {\jobname-boldAuthors.bib}}
    {\tl_const:Nn \c_@@_boldnames_bibfile_tl {bib/\jobname-boldAuthors.bib}}
  \iow_open:Nn \g_@@_boldnames_iow { \c_@@_boldnames_bibfile_tl }
  \iow_now:Nn  \g_@@_boldnames_iow { @comment{Auto-generated~file} }
  \AtEndDocument { \iow_close:N \g_@@_boldnames_iow }
  \addbibresource{\c_@@_boldnames_bibfile_tl}

  % ---------- Writing a single name --------------------------------
  %
  % We need:
  %   - the entry KEY  (nhblx@name@<N>)  fully expanded
  %   - the AUTHOR field written verbatim (so \"a etc. survive)
  %
  % \iow_now:Nn  → no expansion
  % \iow_now:Nx  → full expansion (breaks accents in names)
  % \iow_now:Nn + \exp_not:n on the name = best of both via \iow_now:Nx

  \cs_new_protected:Npn \@@_boldnames_write:n #1
  {
   \int_gincr:N \g_@@_boldnames_int
   % Stash the name so it survives the x-expansion intact.
   \tl_set:Nn \l_@@_boldnames_name_tl { #1 }
   \iow_now:Nx \g_@@_boldnames_iow
   {
    @misc{nhblx@name@\int_use:N \g_@@_boldnames_int,
      author~=~{\exp_not:V \l_@@_boldnames_name_tl},
      options~=~{dataonly=true},
     }
   }
   % Schedule hash extraction after biblatex has loaded the entry.
   \exp_args:Nx \AfterPreamble
   {
    \exp_not:N \hbox_set:Nn \exp_not:N \l_tmpa_box
    { \exp_not:N \nhblx@getmethehash
     { nhblx@name@ \int_use:N \g_@@_boldnames_int } }
   }
  }

  % ---------- User-level commands ----------------------------------

  \NewDocumentCommand \addboldnames { m }
  { \clist_map_function:nN { #1 } \@@_boldnames_write:n }

  \NewDocumentCommand \resetboldnames { }
  { \prop_gclear:N \g_@@_boldnames_hashes_prop }

  % ---------- biblatex glue ----------------------------------------

  \DeclareNameFormat{nhblx@hashextract}
  {
   \prop_gput:Nen \g_@@_boldnames_hashes_prop { \thefield{hash} } { }
  }

  \DeclareCiteCommand{\nhblx@getmethehash}
  {}{ \printnames[nhblx@hashextract][1-999]{author} }{}{}

  \cs_new_protected:Npn \@@_boldnames_mark:n #1
  {
   \prop_if_in:NeTF \g_@@_boldnames_hashes_prop { \thefield{hash} }
   { \mkbibbold { \textcolor{color@reference@boldAuthor} { #1 } } }
   { #1 }
  }

  \cs_set_eq:NN \mkboldifhashinlist \@@_boldnames_mark:n

  \DeclareNameWrapperFormat{boldifhashinlist}
  { \renewcommand*{\mkbibcompletename}{\mkboldifhashinlist} #1 }

  \DeclareNameWrapperAlias{sortname}{default}
  \DeclareNameWrapperAlias{default}{boldifhashinlist}
}
```

```
2055
```

We need the fallback of the commands in case the functionality isnt used at all.

```
2056 \providecommand { \addboldnames   } [1] { }
2057 \providecommand { \resetboldnames }     { }
```

6.1.19 Database infrastructure and key families

All structured data in the proposal (people, projects, institutions, funding entries, ...) is stored in datatool databases at compile time. Each database is created lazily: `\addEntry` → P. 2 checks whether the named database already exists and creates it if not, eliminating the need for explicit initialisation calls.

The central mechanism relies on the L3 variable `\l_@@_database_name_tl`, which holds the name of the currently-active database while a `\keys_set:nn` call is in progress. Every `.code:n` handler in a keyset calls `\DTLnewdbentry` with that variable as the database name, so no explicit database name is needed inside the keyset definition itself.

The token list is reset in each call to `\addEntry` → P. 2 and is wrapped in a `\group_begin:/end:` pair to prevent it from leaking across independent `\addEntry` → P. 2 calls.

Input validation strategy Choice-valued keys (e.g. `gender` → P. 64, `project-type` → P. 56) use `.choices:nn` with an `unknown .code:n` handler wired to the `dfg/invalid-value` message. Free-text keys accept any input. Required keys are marked with `.value\_required:n = true` so that omitting them produces an immediate error rather than a silent empty database field.

```
2058 \tl_new:N \l_@@_database_name_tl
```

Keyset meta The `meta` database holds exactly one row with `id=crc` that describes the CRC itself. Its fields are read in `\AfterEndPreamble` to populate the global macros used in headers, footers, and PDF metadata (`\theCRCname`, `\theCRCnumber`, `\theCRCaffiliation`, etc.).

The `fundingYears` → P. 64 and `fundingYearsPrevious` → P. 64 keys accept a comma-separated list of calendar years (one per funding year) and expand them into individually named fields `fundingYear1` ...`fundingYear5`, which are used e.g. as column headers in the Gantt chart title row.

`fundingPeriodNumeral` → P. 64 (1, 2 or 3) drives all period-filtered queries: only personnel with a matching `period` entry are included in the project leader table and the gender equality analysis.

```
2059 \keys_define:nn { meta }{
2060   id              .code:n     =  \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
2061   id              .value_required:n = true,
2062   name      .code:n  = \DTLnewdbentry{\l_@@_database_name_tl}{name}{#1},
2063   nameAddon .code:n  = \DTLnewdbentry{\l_@@_database_name_tl}{nameAddon}{#1},
2064   number    .code:n  = \DTLnewdbentry{\l_@@_database_name_tl}{number}{#1},
2065   fundingPeriodNumeral .choices:nn = { 1 , 2 , 3 }
2066    {
2067     % store the numeral itself
2068     \DTLnewdbentry{\l_@@_database_name_tl}{fundingPeriodNumeral}{#1}
2069     % map numeral → text without a property list
2070     \int_case:nnF {#1}
2071     {
2072      { 1 } { \DTLnewdbentry{\l_@@_database_name_tl}{fundingPeriodText}{first} }
2073      { 2 } { \DTLnewdbentry{\l_@@_database_name_tl}{fundingPeriodText}{second} }
2074      { 3 } { \DTLnewdbentry{\l_@@_database_name_tl}{fundingPeriodText}{third} }
2075     }
2076     {
2077      % fallback (should not happen because of the .choices clause)
2078         \msg_warning:nnnn { dfg } { invalid-value }
2079      { #1 } { 1 , 2 , 3 }
2080     }
2081    } ,
2082   fundingPeriodNumeral / unknown .code:n =
2083    {
2084       \msg_warning:nnnn { dfg } { invalid-value }
2085     { #1 } { 1 , 2 , 3 }
2086    } ,
2087   fundedSince          .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{fundedSince}{#1},
```

```
2088    fundingYears .code:n =
2089     {
2090      \seq_set_from_clist:Nn \l_meta_funding_years_seq {#1}
2091      %--- create the individual entries -----------------
2092      \int_zero_new:N \l_tmpa_int
2093      \seq_map_inline:Nn \l_meta_funding_years_seq
2094      {
2095       \int_incr:N \l_tmpa_int
2096       \DTLnewdbentry
2097       {\l_@@_database_name_tl}
2098       {fundingYear\int_use:N \l_tmpa_int}
2099       {##1}
2100      }
2101     },
2102    fundingYearsPrevious .code:n =
2103     {
2104      \seq_set_from_clist:Nn \l_meta_funding_years_previous_seq {#1}
2105
2106      %--- create the individual entries -----------------
2107      \int_zero_new:N \l_tmpa_int
2108      \seq_map_inline:Nn \l_meta_funding_years_previous_seq
2109      {
2110       \int_incr:N \l_tmpa_int
2111       \DTLnewdbentry
2112       {\l_@@_database_name_tl}
2113       {fundingYearPrevious\int_use:N \l_tmpa_int}
2114       {##1}
2115      }
2116     },
2117    ref-affiliation-id  .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-affiliation-id}{#1},
2118  }
```

Keyset GenderEquality This keyset is used to populate the `\GenderEqualityStaffTable` and `\GenderEqualityProjectLeadersTable`. Each row stores one position-level target for the upcoming funding period alongside the previous period's objective for comparison.

The live counts (current numbers of female/male/diverse staff per position and period) are *computed* at render time from the `person` database — they are not entered manually. Only the percentage-female objective values need to be provided here.

```
2119  \keys_define:nn { GenderEquality }{
2120    position                              .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{position}{#1},
2121    previous-objective-percentage-female .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{previous-
      objective-percentage-female}{#1},
2122    status-quo-number-female              .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{status-quo-
      number-female}{#1},
2123    status-quo-percentage-female          .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{status-quo-
      percentage-female }{#1},
2124    future-objective-percentage-female    .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{future-
      objective-percentage-female}{#1},
2125  }
```

Multi-value helper: `\__dfg_set_filtered_clist:Nnn` Several keys (`role`, `period`) accept comma-separated lists rather than a single choice. The private helper `\__dfg_set_filtered_clist:Nnn{\meta{result}}{\meta{valid}}{\meta{input}}` iterates over the input list, drops any item that is not in the valid list (and emits an `invalid-value` warning for it), and stores the cleaned result in a clist variable. This is used instead of `.choices:nn` because the standard choices handler only accepts a single value.

```
2126  \clist_new:N \l_@@_person_tmp_period_clist
2127  \clist_new:N \l_@@_person_tmp_role_clist
2128  \clist_new:N \l_@@_keys_tmp_clist
2129  \cs_new_protected:Npn \_@@_set_filtered_clist:Nnn #1#2#3
2130  {
2131    % #1: clist variable in which valid values are stored
2132    % #2: clist of valid values
2133    % #3: clist of given values
2134    % function assumes none of the values contain braces or `#` tokens
2135    \clist_set:Nn \l_@@_keys_tmp_clist {#2}
2136    \clist_clear:N #1
2137    \clist_map_inline:nn {#3}
2138    {
2139     \clist_if_in:NnTF \l_@@_keys_tmp_clist {##1}
2140     { \clist_put_right:Nn #1 {##1} }
2141      { \msg_warning:nnne { dfg } { invalid-value } {##1}
2142        { \clist_use:Nn \l_@@_keys_tmp_clist {,~} } }
2143    }
2144  }
```

Keyset project One row per CRC sub-project. The `id`→P.63 is the short project letter (e.g. `A01`); `number`→P.64 is the DFG-assigned project number; `name`→P.64 is the full project title. `project-type`→P.56 and `project-status`→P.56 control how the project appears in tables and are choice-validated.

The `research-area`→P.56 key is a tick-list of DFG classification codes in the form `1.11-01` (see the property list in Section 6.1.23). Multiple codes are passed as a brace-grouped comma list.

The `ref-person-id`→P.62 and `collaborations`→P.57 values are comma lists. Whitespace surrounding individual IDs is ignored and the canonical, space-free list is stored in the database. Thus `{person-a,person-b,person-c}` and `{person-a, person-b, person-c}` are equivalent for every later consumer. The optional `project-area` key controls the area row in collaboration plots. When omitted, the first character of the project ID is used as a fallback.

The `legal-issues`→P.57 sub-keyset mirrors the questions from the DFG online portal's project-information form (`L1`→P.57–`L7`→P.57). Nested key-value parsing is triggered via a single `legal-issues={...}` block.

```
\cs_new_protected:Npn \@@_new_clist_dbentry:nnn #1#2#3
{
  \clist_set:Nn \l_tmpa_clist {#3}
  \exp_args:NnnV \DTLnewdbentry {#1} {#2} \l_tmpa_clist
}

\bool_new:N \l_@@_project_consider_for_lists_bool
\bool_new:N \l_@@_project_reference_only_bool
\bool_new:N \g_@@_project_count_ready_bool

\keys_define:nn { project }{
  id                       .code:n           =   \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  id                       .value_required:n = true,
  number                   .code:n           = \DTLnewdbentry{\l_@@_database_name_tl}{number}{#1},
  number                   .value_required:n = true,
  name                     .code:n           = \DTLnewdbentry{\l_@@_database_name_tl}{name}{#1},
  project-type             .choices:nn       =  { research,service,mgk,admin} {\DTLnewdbentry{\l_@@_database_name_tl}{project-type}{#1}},
  project-type / unknown   .code:n           = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{research,service,mgk,admin} },
  project-type             .value_required:n = true,
  project-status           .choices:nn       =  { N,E,C} {\DTLnewdbentry{\l_@@_database_name_tl}{project-status}{#1}},
  project-status / unknown .code:n           = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{N,E,C}},
  project-status           .value_required:n = true,
  consider-for-lists           .choices:nn       = { true,false }
    { \str_if_eq:nnTF {#1} {true}
        { \bool_set_true:N \l_@@_project_consider_for_lists_bool }
        { \bool_set_false:N \l_@@_project_consider_for_lists_bool } },
  consider-for-lists / unknown .code:n           =
    { \msg_warning:nnnn { dfg } { invalid-value } {#1} {true,false} },
  consider-for-lists           .value_required:n = true,
  reference-only           .choices:nn       = { true,false }
    { \str_if_eq:nnTF {#1} {true}
        { \bool_set_true:N \l_@@_project_reference_only_bool }
        { \bool_set_false:N \l_@@_project_reference_only_bool } },
  reference-only / unknown .code:n           =
    { \msg_warning:nnnn { dfg } { invalid-value } {#1} {true,false} },
  reference-only           .value_required:n = true,
  research-area            .code:n           = \DTLnewdbentry{\l_@@_database_name_tl}{research-area}{#1},
  project-area             .code:n           = \DTLnewdbentry{\l_@@_database_name_tl}{project-area}{#1},
  ref-person-id            .code:n           =
  \@@_new_clist_dbentry:nnn {\l_@@_database_name_tl} {ref-person-id} {#1},
  collaborations           .code:n           =
  \@@_new_clist_dbentry:nnn {\l_@@_database_name_tl} {collaborations} {#1},
  legal-issues             .code:n           = { \keys_set:nn { project / legal-issues  } { #1 } }
}
```

The `legal-issues` key in a `project` entry accepts a sub-keyset that maps each legal-issue question to a value. The questions fall into two groups, which are handled differently:

Main questions (`L1`→P.57, `L2`→P.57, `L3`→P.57, `L4`→P.57, `L5`→P.57, `L6`→P.57, `L7`→P.57) Each must be answered with exactly `yes` or `no`. Any other value triggers a warning via `\person_questions_invalid:n`, and no database entry is created.

Sub-questions (`L1a`→P.57, `L2a`→P.57, `L5a`→P.57, `L7a`→P.57) These are conditional follow-ups whose answer only matters when the corresponding main question was answered with `yes`. Sub-questions accept `yes`, `no`, or an empty value. An empty value records an empty cell in the

database, which the typesetting code treats as “not answered” and consequently does not render in the legal-issues table.

The asymmetry exists because forcing every sub-question to be answered would require authors to fill in irrelevant follow-ups whenever the parent question is `no` – for example, providing an answer to “has legal authorization been obtained” for a project that does not involve human embryonic stem cells in the first place. Allowing an empty value lets the data file mirror the structure of the questionnaire: only the relevant cells need to be filled in.

Internally, the main questions use the `l3keys` → CTAN `.choices:nn` property, which natively rejects values outside the allowed set. The sub-questions cannot use `.choices:nn` directly because that property does not accept the empty string as a choice; they instead dispatch through the helper `\dfg_legal_subq:nn`, which inspects the supplied value and creates a database entry when the value is `yes`, `no`, or empty, and warns otherwise.

Example of a complete `legal-issues` block in a project entry:

Example entries for `legal-issues`

```
% legal-issues = {
%   L1  = {yes},
%   L1a = {yes},       % L1 was true, follow-up answered
%   L2  = {no},
%   L2a = {},           % L2 was false, follow-up irrelevant
%   L3  = {yes},
%   L4  = {no},
%   L5  = {no},
%   L5a = {},
%   L6  = {yes},
%   L7  = {yes},
%   L7a = {yes}
% }
%
```

The empty values for `L2a` → P. 57 and `L5a` → P. 57 are valid: they correspond to follow-ups whose parent question was answered `no`, so no follow-up information is required or rendered.

```
\keys_define:nn { project / legal-issues }
{
  % main questions: must be yes/no
  L1  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L1} {#1} },
  L2  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L2} {#1} },
  L3  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L3} {#1} },
  L4  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L4} {#1} },
  L5  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L5} {#1} },
  L6  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L6} {#1} },
  L7  .choices:nn = { yes, no } { \DTLnewdbentry{\l_@@_database_name_tl}{L7} {#1} },
  %
  % sub-questions: yes / no / empty all allowed
  L1a .code:n = { \dfg_legal_subq:nn {L1a} {#1} },
  L2a .code:n = { \dfg_legal_subq:nn {L2a} {#1} },
  L5a .code:n = { \dfg_legal_subq:nn {L5a} {#1} },
  L7a .code:n = { \dfg_legal_subq:nn {L7a} {#1} },
  %
  % unknown handlers (only need them for the binary keys)
  L1  / unknown .code:n = { \person_questions_invalid:n {#1} },
  L2  / unknown .code:n = { \person_questions_invalid:n {#1} },
  L3  / unknown .code:n = { \person_questions_invalid:n {#1} },
  L4  / unknown .code:n = { \person_questions_invalid:n {#1} },
  L5  / unknown .code:n = { \person_questions_invalid:n {#1} },
  L6  / unknown .code:n = { \person_questions_invalid:n {#1} },
  L7  / unknown .code:n = { \person_questions_invalid:n {#1} },
}

% Sub-question handler: accepts yes, no, or empty.
\cs_new_protected:Npn \dfg_legal_subq:nn #1#2
{
  \tl_if_blank:nTF {#2}
  { \DTLnewdbentry{\l_@@_database_name_tl}{#1}{} }
  {
   \str_case:nnF {#2}
   {
    { yes  } { \DTLnewdbentry{\l_@@_database_name_tl}{#1}{yes}  }
    { no } { \DTLnewdbentry{\l_@@_database_name_tl}{#1}{no} }
   }
```

```
    { \msg_warning:nnnn { dfg } { invalid-value } {#2}
      { yes,~no,~or~an~empty~value } }
  }
}
```

Keyset person The person database is the central personnel registry for the proposal. Every PI, manager, steering committee member and early-career researcher who appears in any table must have a row here.

Key semantics: role→P.64: A filterable comma list of roles. The supported values (PI, spokesperson, manager, steering-committee-member, rector) are validated against a fixed list using the multi-value helper so that multiple roles per person are permitted. position→P.64: A single choice describing the employment category (prof-W2, prof-W3, postDoc, PhD, etc.). Used in the gender equality and staff cost tables. period→P.64: Comma list of funding periods (1, 2, 3) in which this person is active. All period-filtered queries (e.g. the project-leaders table) use this field. name-alias→P.53: Brace-grouped list of name variants as the author name appears in .bib files. These are passed to \addboldnames so all spelling variants are highlighted consistently. contract-information→P.52: Sub-keyset (C1→P.52–C4a→P.52) for the four employment-contract questions required in the DFG project header. Internally stored as fields Q1–Q4a.

```
\keys_define:nn { person }{
  unknown .code:n = { \msg_warning:nnn  { dfg } { unknown-key } {#1}},
  unknown .default:V = \c_novalue_tl,
  id                .code:n =   \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  id                .value_required:n = true,
  firstname        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{firstname}{#1},
  firstname .value_required:n = true,
  lastname         .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{lastname}{#1},
  lastname .value_required:n = true,
  academicdegree   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{academicdegree}{#1},
  gender .choices:nn =  { m , f , d } {\DTLnewdbentry{\l_@@_database_name_tl}{gender}{#1}},
  gender / unknown .code:n = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{ m,f,d }},
  gender .value_required:n = true,
  ref-affiliation-id   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-affiliation-id}{#1},
  ref-institution-id   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-institution-id}{#1},
  ref-project-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-project-id}{#1},
```

With role→P.64 we want to differentiate between various people so we easily can get the necessary person into the right table. Roles can be e.g. PI, service etc.

```
  role .code:n = {\_@@_set_filtered_clist:Nnn \l_@@_person_tmp_role_clist {
     PI,~ spokesperson,~ manager,~ steering-committee-member,~rector
    } {#1}
    \exp_args:Nnne \DTLnewdbentry{\l_@@_database_name_tl}{role}
                { \clist_use:N  \l_@@_person_tmp_role_clist}},
```

position is also bound to a value list

```
  position .choices:nn =  {
    PhD, postDoc, groupLeader, research,
    nonResearch, student, prof,
    prof-W2, prof-W3 }
   {\DTLnewdbentry{\l_@@_database_name_tl}{position}{#1}},
  position / unknown .code:n = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{
     PhD,~ postDoc,~ groupLeader,~ research,~
     nonResearch,~ student,~ prof,~ prof-W2,~ prof-W3 }},
  positionText .code:n =  \DTLnewdbentry{\l_@@_database_name_tl}{positionText}{#1},
```

we need to figure out in which funding period the person was active.

```
  period .code:n = {
    \_@@_set_filtered_clist:Nnn \l_@@_person_tmp_period_clist { 1, 2, 3 } {#1}
    \exp_args:Nnne \DTLnewdbentry{\l_@@_database_name_tl}{period}
                { \clist_use:N  \l_@@_person_tmp_period_clist }
   },
```

The following keys are used for the CV section (additionally position). We start with name-alias which is used to get the author's name in bold letters into the publications list. Write all aliases of the author according to their appearence in the .bib-file. Encapsulate each with one pair of {}.

```
  name-alias   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{name-alias}{{#1}},
  orcid        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{orcid}{#1},
  email        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{email}{#1},
  phone        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{phone}{#1},
  url          .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{url}{#1},
```

```
    doctorateYear .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{doctorateYear}{#1},
    doctorate    .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{doctorate}{#1},
    degree-programme   .code:n =
    \DTLnewdbentry{\l_@@_database_name_tl}{degree-programme}{#1},
```

Following part is for the early career researchers

```
    financed-by .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{financed-by}{#1},
    topic       .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{topic}{#1},
    category    .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{category}{#1},
    dateDefense .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{dateDefense}{#1},
    periodStart .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{periodStart}{#1},
    periodEnd .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{periodEnd}{#1},
```

For the PI we need to have information regarding their contract situation. Those questions are asked in the Header section of each project.

```
    contract-information .code:n = { \keys_set:nn { person / contract-information } { #1 } }
}
```

Helper: `\person_questions_invalid:n` Small private helper used as the `unknown .code:n` handler for every boolean choice key in the `contract-information` and `project/legal-issues` key families. It fires `dfg/invalid-value` with the allowed set hard-coded to `yes,no`.

```
\cs_new_protected:Npn \person_questions_invalid:n #1
{ \msg_warning:nnnn { dfg } { invalid-value } {#1} { yes , no } }


\keys_define:nn { person / contract-information }
{
  %--- yes/no questions ------------------------------------------------
  C1 .choices:nn = { yes , no }
   { \DTLnewdbentry{\l_@@_database_name_tl}{C1}{#1} },
  C1 / unknown .code:n = { \person_questions_invalid:n {#1} },

  C2 .choices:nn = { yes , no  }
   { \DTLnewdbentry{\l_@@_database_name_tl}{C2}{#1} },
  C2 / unknown .code:n = { \person_questions_invalid:n {#1} },

  C3 .choices:nn = { yes , no  }
   { \DTLnewdbentry{\l_@@_database_name_tl}{C3}{#1} },
  C3 / unknown .code:n = { \person_questions_invalid:n {#1} },

  C4 .choices:nn = {  yes , no  }
   { \DTLnewdbentry{\l_@@_database_name_tl}{C4}{#1} },
  C4 / unknown .code:n = { \person_questions_invalid:n {#1} },

  %--- free-form follow-ups (may be empty) ----------------------------
  C2a .code:n = { \dfg_contract_date:nn {C2a} {#1} },
  C2b .code:n = { \dfg_contract_date:nn {C2b} {#1} },
  C3a .code:n = { \dfg_contract_date:nn {C3a} {#1} },
  C4a .code:n = { \dfg_contract_text:nn {C4a} {#1} },
}

% Date follow-up: allows empty or ISO-8601 (YYYY-MM-DD).
\cs_new_protected:Npn \dfg_contract_date:nn #1#2
{
  \tl_if_blank:nTF {#2}
  { \DTLnewdbentry{\l_@@_database_name_tl}{#1}{} }
  {
   \dfg_is_iso_date:nTF {#2}
   { \DTLnewdbentry{\l_@@_database_name_tl}{#1}{#2} }
   {
    \msg_warning:nnnn { dfg } { invalid-date } {#1} {#2}
    \DTLnewdbentry{\l_@@_database_name_tl}{#1}{#2}
   }
  }
}

% Plain-text follow-up: allows empty or any string.
\cs_new_protected:Npn \dfg_contract_text:nn #1#2
{
  \DTLnewdbentry{\l_@@_database_name_tl}{#1}{#2}
}

% Format check: YYYY-MM-DD where Y/M/D are digits.
% Doesn't validate calendar correctness (e.g. Feb 30) -- just shape.
\prg_new_protected_conditional:Npnn \dfg_is_iso_date:n #1 { TF , T , F }
{
  \regex_match:nnTF { \A \d{4} - \d{2} - \d{2} \Z } {#1}
  { \prg_return_true: }
  { \prg_return_false: }
```

```
}
```

Keyset cv The `cv` database holds supplementary career information for each PI beyond what is already stored in the `person` row. Multiple entries per person are permitted; each entry is identified by `ref-person-id` and a `cv-type` → P. 53 that governs where in the printed CV the item appears.

```
\keys_define:nn { cv }{
  id           .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  ref-person-id.code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-person-id}{#1},
  cv-type .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{type}{#1},
  cv-type .choices:nn =  {mandatory, supplement,
    activities, supervision,
    distinction, other }
   {\DTLnewdbentry{\l_@@_database_name_tl}{type}{#1}},
  % Backward-compatibility alias for the former misspelling.
  cv-type / distinction .code:n = {   \DTLnewdbentry{\l_@@_database_name_tl}{type}{distinction}},
  cv-type / unknown .code:n = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{
     mandatory,~ supplement,~
     activities,~  supervision,~
     distinction,~ other }},
  positionText .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{positionText}{#1},
  position .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{position}{#1},
  ref-institution-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-institution-id}{#1},
  description .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{description}{#1},
  periodStart .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{periodStart}{#1},
  periodEnd .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{periodEnd}{#1},
}
```

Keysets affiliation and institution The template distinguishes between two levels of organisational unit: affiliation: A *university* or equivalent top-level organisation, described by its full name, abbreviation, and postal address. Referenced from `person` rows via `ref-affiliation-id`. institution: A *department, institute or centre* within an affiliation (e.g. `IT Center RWTH Aachen`). Referenced from `person` rows via `ref-institution-id` and from `affiliation` via `ref-affiliation-id`.

This two-level model allows the same address block to be reused across multiple people without duplicating data, and the same university to appear under different institute names for different projects.

```
\keys_define:nn { affiliation }
{
  id               .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  id               .value_required:n = true,
  name             .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{name}{#1},
  abbreviation  .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{abbreviation}{#1},
  addressStreet    .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{addressStreet}{#1},
  addressZIP   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{addressZIP}{#1},
  location         .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{location}{#1},
  country          .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{country}{#1},
}

\keys_define:nn { institution }
{
  id        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  id        .value_required:n = true,
  name      .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{name}{#1},
  abbreviation .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{abbreviation}{#1},
  ref-affiliation-id  .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{ref-affiliation-id}{#1},
}

% Resolve the compact reader-facing institution label once.  Older metadata
% without an abbreviation remains usable by falling back to the full name.
\tl_new:N \l_@@_institution_label_field_tl
\cs_new_protected:Npn \@@_institution_abbreviation_or_name:nN #1#2
{
  \tl_clear:N #2
  \tl_set:Nn \l_@@_institution_label_field_tl { abbreviation }
  \datatool_if_has_key:nnTF { institution } { abbreviation }
  {
    \DTLgetvalueforkey{#2}{abbreviation}{institution}{id}{#1}
  }
  { }
  \DTLifnullorempty{#2}
  {
```

```
      \tl_set:Nn \l_@@_institution_label_field_tl { name }
      \DTLgetvalueforkey{#2}{name}{institution}{id}{#1}
    }
    { }
  }
```

Keyset funding One `funding` row represents a single requested-funding line item, such as one FTE of PhD salary across five years, or a one-off equipment purchase. The same keyset is also used for*existing* and *previous* funding entries in the relevant tables.

`funding-type`: `staff`→P. 61: Personnel costs. Activates the restricted `category` validation (only staff-position names are accepted). `costs`→P. 61: Direct project costs (consumables, travel, etc.). `fellowships`→P. 61: Stipend payments. `globalfunds`→P. 61: Funds requested at the CRC level (coordination). `instrumentation`→P. 61: Equipment purchases.

`quantity`→P. 60 **and `sum`**→P. 60: both accept a `funding/year` sub-keyset `{year1={n},year2={n},...}`. The token-list variable `\l_funding_sum_quantity_tl` is set to the key name (`quantity` or `sum`) before the sub-parse so the `year\meta{n}` handlers can construct the correct database field name (`quantity-year1`, `sum-year2`, etc.) without extra parameters.

```
\str_new:N \l_@@_funding_tmp_type_str
\tl_new:N \l_funding_sum_quantity_tl
\clist_const:Nn \c_@@_funding_staff_categories_clist
  { prof, groupLeader, postDoc, medDoc, PhD, research,
    nonResearch, student, assistant }
\keys_define:nn { funding / year }
{
  year1 .code:n =
   {\DTLnewdbentry{\l_@@_database_name_tl}{\tl_use:N \l_funding_sum_quantity_tl-year1}{#1}},
  year2 .code:n =
   {\DTLnewdbentry{\l_@@_database_name_tl}{\tl_use:N \l_funding_sum_quantity_tl-year2}{#1}},
  year3 .code:n =
   {\DTLnewdbentry{\l_@@_database_name_tl}{\tl_use:N \l_funding_sum_quantity_tl-year3}{#1}},
  year4 .code:n =
   {\DTLnewdbentry{\l_@@_database_name_tl}{\tl_use:N \l_funding_sum_quantity_tl-year4}{#1}},
  year5 .code:n =
   {\DTLnewdbentry{\l_@@_database_name_tl}{\tl_use:N \l_funding_sum_quantity_tl-year5}{#1}},
}

\keys_define:nn { funding }
{
  id        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  id        .value_required:n = true,
  ref-project-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-project-id}{#1},
  ref-affiliation-id .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{ref-affiliation-id}{#1},
  financed-by .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{financed-by}{#1},
  funding-type .choices:nn =  {
    staff,~ costs,~ fellowships,~ globalfunds,~ instrumentation }
   {
    \str_set:Nn \l_@@_funding_tmp_type_str {#1}
    \DTLnewdbentry{\l_@@_database_name_tl}{type}{#1}
   },
  funding-type / unknown .code:n = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{
     staff,~ costs,~ fellowships,~ globalfunds,~ instrumentation }},
  funding-type        .value_required:n = true,
  percentage .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{percentage}{#1},
  positionText .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{positionText}{#1},
  quantity .code:n = {
    \tl_set:Nn \l_funding_sum_quantity_tl {quantity}
    \keys_set:nn { funding / year  } { #1 }
   },
  sum .code:n = {
    \tl_set:Nn \l_funding_sum_quantity_tl {sum}
    \keys_set:nn { funding / year  } { #1 }
   },
  category .code:n =
   {
    \str_if_eq:VnTF \l_@@_funding_tmp_type_str { staff }
    {% staff → restricted choices
        \clist_if_in:NnTF \c_@@_funding_staff_categories_clist {#1}
     {
      \DTLnewdbentry{\l_@@_database_name_tl}{category}{#1}
     }
     {
          \msg_warning:nnne { dfg } { invalid-value }{ #1 }
          { \clist_use:Nn \c_@@_funding_staff_categories_clist {,~} }
     }
```

```
    }
    {
     % non-staff → unrestricted
     \DTLnewdbentry{\l_@@_database_name_tl}{category}{#1}
    }
   },
  % following keys are especially for the previous fundings
  year            .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{year}{#1}},
  applicant       .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{applicant}{#1}},
  participant     .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{participant}{#1}},
  other-funds     .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{other-funds}{#1}},
  staff           .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{staff}{#1}},
  costs           .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{costs}{#1}},
  instrumentation .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{instrumentation}{#1}},
  fellowships     .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{fellowships}{#1}},
  globalfunds     .code:n = {\DTLnewdbentry{\l_@@_database_name_tl}{globalfunds}{#1}},
}
```

A keyset for instrumentation. The flat `cost` field belongs to the informational `existingInstrumentation` list, is expressed in thousands of EUR, and is never included in funding totals. The nested `sum` fields are requested-funding amounts expressed in full EUR.

```
\keys_define:nn { instrumentation }{
  ref-project-id  .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-project-id}{#1},
  quantity        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{quantity}{#1},
  instrument      .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{instrument}{#1},
  company         .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{company}{#1},
  year            .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{year}{#1},
  cost            .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{cost}{#1},
  financed-by     .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{financed-by}{#1},
  description     .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{description}{#1},
  sum .code:n = {
    \tl_set:Nn \l_funding_sum_quantity_tl {sum}
    \keys_set:nn { funding / year  } { #1 }
   },
}
```

Keyset staff The `staff` key family serves two related personnel data shapes. Its `year1`–`year5` subkeys store the headcount plan per position category for `\OverviewOfFundsRequestedForStaffTable`. The flat keys added below store the named and unnamed staff roster required by the DFG project template in the fixed `RequestedFundingStaff` database. All keys remain optional at the parser level so the older overview declarations continue to work unchanged; database-specific validation supplies useful warnings for roster entries.

The `position` →P. 64 sub-keyset holds named keys for each staff category (`postDoc`, `PhD`, `research`, etc.); each key's value is the number of FTEs for a specific year. `\l_staff_year_tl` is set to `year1`–`year5` before each sub-parse so the handlers can construct the correct field name (e.g. `postDoc-year2`) without extra parameters.

```
\tl_new:N \l_staff_year_tl          % will hold "year1", "year2", …
\keys_define:nn { staff / position }{
  postDoc      .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{postDoc-\tl_use:N \l_staff_year_tl}{#1},
  PhD          .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{PhD-\tl_use:N \l_staff_year_tl}{#1},
  research     .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{research-\tl_use:N \l_staff_year_tl}{#1},
  nonResearch  .code:n = \DTLnewdbentry{\l_@@_database_name_tl}
  {nonResearch-\tl_use:N \l_staff_year_tl}{#1},
  student      .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{student-\tl_use:N \l_staff_year_tl}{#1},
}

% Category warnings list the currently registered keys, so the diagnostic
% stays in sync with the category-name property list, including extensions.
\seq_new:N \l_@@_diagnostic_keys_seq
\cs_new_protected:Npn \@@_collect_diagnostic_keys:N #1
{
  \seq_clear:N \l_@@_diagnostic_keys_seq
  \prop_map_inline:Nn #1
    { \seq_put_right:Nn \l_@@_diagnostic_keys_seq {##1} }
}
\cs_new_protected:Npn \@@_warn_unknown_name:nnN #1#2#3
{
  \@@_collect_diagnostic_keys:N #3
  \msg_warning:nneee { dfg } { unknown-name } {#1} {#2}
    { \seq_use:Nn \l_@@_diagnostic_keys_seq {,~} }
}
\cs_new_protected:Npn \@@_staff_set_category:n #1
{
  \prop_if_in:NnTF \g_@@_dfg_category_names_prop {#1}
  { \DTLnewdbentry{\l_@@_database_name_tl}{category}{#1} }
  {
```

```
    \@@_collect_diagnostic_keys:N \g_@@_dfg_category_names_prop
    \msg_warning:nnne { dfg } { invalid-value } {#1}
      { \seq_use:Nn \l_@@_diagnostic_keys_seq {,~} }
  }
}

\keys_define:nn { staff }{
  unknown .code:n = { \msg_warning:nnn { dfg } { unknown-key } {#1} },
  unknown .default:V = \c_novalue_tl,
  ref-project-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-project-id}{#1},
  funding-status .choices:nn = { existing, requested }
  { \DTLnewdbentry{\l_@@_database_name_tl}{funding-status}{#1} },
  funding-status / unknown .code:n =
  { \msg_warning:nnnn { dfg } { invalid-value } {#1} { existing,~requested } },
  staff-type .choices:nn = { research, nonResearch }
  { \DTLnewdbentry{\l_@@_database_name_tl}{staff-type}{#1} },
  staff-type / unknown .code:n =
  { \msg_warning:nnnn { dfg } { invalid-value } {#1} { research,~nonResearch } },
  ref-person-id .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{ref-person-id}{#1},
  firstname .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{firstname}{#1},
  lastname .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{lastname}{#1},
  academic-degree .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{academic-degree}{#1},
  positionText .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{positionText}{#1},
  position .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{positionText}{#1},
  field-of-research .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{field-of-research}{#1},
  ref-institution-id .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{ref-institution-id}{#1},
  ref-affiliation-id .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{ref-affiliation-id}{#1},
  department .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{department}{#1},
  commitment .code:n =
  {
    \regex_match:nnTF { \A \d+ (?: \. \d+ )? \Z }
      {#1}
    { \DTLnewdbentry{\l_@@_database_name_tl}{commitment}{#1} }
    { \msg_warning:nnnn { dfg } { invalid-format } {#1}
      { a~non-negative~integer~or~decimal~using~a~decimal~point,
        ~for~example~0,~10,~or~10.5 } }
  },
  category .code:n = \@@_staff_set_category:n {#1},
  funding-source .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{funding-source}{#1},
  description .code:n =
  \DTLnewdbentry{\l_@@_database_name_tl}{description}{#1},
  year1 .code:n = {
    \tl_set:Nn \l_staff_year_tl {year1}
    \keys_set:nn { staff / position  } { #1 }
   },
  year2 .code:n = {
    \tl_set:Nn \l_staff_year_tl {year2}
    \keys_set:nn { staff / position  } { #1 }
   },
  year3 .code:n = {
    \tl_set:Nn \l_staff_year_tl {year3}
    \keys_set:nn { staff / position  } { #1 }
   },
  year4 .code:n = {
    \tl_set:Nn \l_staff_year_tl {year4}
    \keys_set:nn { staff / position  } { #1 }
   },
  year5 .code:n = {
    \tl_set:Nn \l_staff_year_tl {year5}
    \keys_set:nn { staff / position  } { #1 }
  },
}

% Pre-create every roster column once.  Datatool can otherwise report an
% unknown column when an optional key is omitted by every row in the database.
\bool_new:N \g_@@_requested_funding_staff_schema_ready_bool
\cs_new_protected:Npn \@@_requested_funding_staff_ensure_schema:
{
  \bool_if:NF \g_@@_requested_funding_staff_schema_ready_bool
  {
    \clist_map_inline:nn
    {
      ref-project-id, funding-status, staff-type, ref-person-id,
      firstname, lastname, academic-degree, positionText, field-of-research,
      ref-institution-id, ref-affiliation-id, department, commitment, category,
      funding-source, description
```

```
    }
    {
      \DTLifhaskey { RequestedFundingStaff } {##1}
      { }
      { \DTLaddcolumn { RequestedFundingStaff } {##1} }
    }
    \bool_gset_true:N \g_@@_requested_funding_staff_schema_ready_bool
  }
}

\tl_new:N \l_@@_requested_funding_staff_validate_project_tl
\tl_new:N \l_@@_requested_funding_staff_validate_status_tl
\tl_new:N \l_@@_requested_funding_staff_validate_type_tl
\tl_new:N \l_@@_requested_funding_staff_validate_person_tl
\tl_new:N \l_@@_requested_funding_staff_validate_lastname_tl
\tl_new:N \l_@@_requested_funding_staff_validate_institution_tl
\keys_define:nn { dfg / requested-funding-staff-validation }
{
  ref-project-id .tl_set:N =
    \l_@@_requested_funding_staff_validate_project_tl,
  funding-status .tl_set:N =
    \l_@@_requested_funding_staff_validate_status_tl,
  staff-type .tl_set:N =
    \l_@@_requested_funding_staff_validate_type_tl,
  ref-person-id .tl_set:N =
    \l_@@_requested_funding_staff_validate_person_tl,
  lastname .tl_set:N =
    \l_@@_requested_funding_staff_validate_lastname_tl,
  ref-institution-id .tl_set:N =
    \l_@@_requested_funding_staff_validate_institution_tl,
  unknown .code:n = { }
}

\cs_new_protected:Npn
  \@@_requested_funding_staff_warn_missing_institution:
{
  \msg_warning:nne { dfg } { requested-staff-institution-required }
    { \l_@@_requested_funding_staff_validate_project_tl }
}

\cs_new_protected:Npn \@@_requested_funding_staff_validate:n #1
{
  \tl_clear:N \l_@@_requested_funding_staff_validate_project_tl
  \tl_clear:N \l_@@_requested_funding_staff_validate_status_tl
  \tl_clear:N \l_@@_requested_funding_staff_validate_type_tl
  \tl_clear:N \l_@@_requested_funding_staff_validate_person_tl
  \tl_clear:N \l_@@_requested_funding_staff_validate_lastname_tl
  \tl_clear:N \l_@@_requested_funding_staff_validate_institution_tl
  \keys_set:nn { dfg / requested-funding-staff-validation } {#1}
  \tl_if_blank:VT \l_@@_requested_funding_staff_validate_project_tl
  { \msg_warning:nnn { dfg } { requested-staff-required }
    { ref-project-id } }
  \tl_if_blank:VT \l_@@_requested_funding_staff_validate_status_tl
  { \msg_warning:nnn { dfg } { requested-staff-required }
    { funding-status } }
  \tl_if_blank:VT \l_@@_requested_funding_staff_validate_type_tl
  { \msg_warning:nnn { dfg } { requested-staff-required }
    { staff-type } }
  \bool_lazy_all:nT
  {
    { \tl_if_blank_p:V \l_@@_requested_funding_staff_validate_person_tl }
    { \tl_if_blank_p:V \l_@@_requested_funding_staff_validate_lastname_tl }
  }
  { \msg_warning:nnn { dfg } { requested-staff-required }
    { ref-person-id~or~lastname } }
  \bool_lazy_all:nT
  {
    { \tl_if_blank_p:V \l_@@_requested_funding_staff_validate_person_tl }
    { \tl_if_blank_p:V \l_@@_requested_funding_staff_validate_institution_tl }
  }
  { \@@_requested_funding_staff_warn_missing_institution: }
}
```

A keyset for the upkeep of laboratory animals.

```
\keys_define:nn { UpkeepLaboratoryAnimals }{
  ref-project-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-project-id}{#1},
  species .choices:nn =  { mouse, mouse-gnotobiotic,rat}
   {\DTLnewdbentry{\l_@@_database_name_tl}{species}{#1}},
  species / unknown .code:n = {\msg_warning:nnnn { dfg } { invalid-value }{ #1 }{
        mouse,~ mouse-gnotobiotic,~ rat}},
  species   .value_required:n = true,
  quantity .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{quantity}{#1},
  weeks-kept .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{weeks-kept}{#1},
  purchasing-costs   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{purchasing-costs}{#1},
  requested-funds   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{requested-funds}{#1},
  existing-funds .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{existing-funds}{#1},
```

```
}
```

Keyset gantt Each row in a `gantt-\meta{project}` database describes one Gantt bar. Bars share their `id` → P. 63 with the corresponding work-package label, so `\addWP` automatically creates a matching Gantt entry when called.

`periodStart` → P. 53 and `periodEnd` → P. 53 are quarter numbers (1–16 for a standard 5-year CRC period). Setting both to the same value suppresses the bar, allowing a work-package item to exist in the list without occupying space in the chart.

`style` → P. 39 selects one of five visual styles for the bar. Unknown styles fall back to `default` with a warning, so the chart always compiles.

```
\keys_define:nn { gantt }{
  id          .code:n     =  \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  id          .value_required:n = true,
  name        .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{name}{#1},
  description .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{description}{#1},
  periodStart .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{periodStart}{#1},
  periodEnd   .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{periodEnd}{#1},
  style       .choices:nn =  { crosshatch,striped,important,alternative,default }
   {\DTLnewdbentry{\l_@@_database_name_tl}{style}{#1}},
  style / unknown .code:n = {
      \msg_warning:nnnnn { dfg } { invalid-value-fallback }{#1}
      {crosshatch,~striped,~important,~alternative,~default}{default}
    \DTLnewdbentry{\l_@@_database_name_tl}{style}{default}
   },
}
```

6.1.20 Core data-access commands

The following commands form the primary user-facing API for reading and writing the datatool databases. Together with the keyset definitions above they provide a strongly-typed, validated interface that prevents malformed data from silently entering the proposal.

`\dfgShowTag` and `\dfgShowCalculatedTag` Direct values use `db:id:field`; aggregates use the concise form `calc:command [db]`. Successive labels alternate between two vertical levels. The complete marker has zero width, height, and depth in both prose and tables, so annotations cannot change line spacing or table-row height. When `showtags-tables` → P. 66 is false, the wrapper returns its value unchanged while a `tblr` cell is being rendered. The begin-document specialisation keeps the normal `showtags=false` path as cheap as the unannotated output. In fast-project mode, skipped metadata prevents resolving the destination row of a reference chain; the annotation records its source row and full requested field path instead.

```
\NewDocumentCommand{\dfgShowTag}{ m m m o }
{%
  % #1: id
  % #2: field of database
  % #3: value from database field
  % #4: custom field (optional, e.g. database name)
  \bool_if:NTF \l_@@_class_showtags_bool
  { \texorpdfstring{\@@_show_tag_direct:nnnn {#1}{#2}{#3}{#4}}{#3} }
  {#3}%
}

\NewDocumentCommand{\dfgShowCalculatedTag}{ m m +m }
{%
  % #1: calculation command or label
  % #2: contributing database(s)
  % #3: calculated value
  \bool_if:NTF \l_@@_class_showtags_bool
  { \texorpdfstring{\@@_show_tag_value:nn { calc:#1~[#2] } {#3}}{#3} }
  {#3}%
}

\cs_new_protected:Npn \@@_show_tag_direct:nnnn #1#2#3#4
{
  \IfNoValueTF{#4}
    { \@@_show_tag_value:nn { #1:#2 } {#3} }
    { \@@_show_tag_value:nn { #4:#1:#2 } {#3} }
}

```

```
\cs_new_protected:Npn \@@_show_tag_value:nn #1#2
{
  \bool_if:nTF
    { \g_@@_tblr_inside_bool && ! \l_@@_class_showtags_tables_bool }
    {#2}
    { \@@_show_tag_overlay:nn {#1}{#2} }
}

\bool_new:N \g_@@_show_tag_upper_bool
\bool_new:N \l_@@_show_tag_leftward_bool
\dim_new:N \l_@@_show_tag_raise_dim

\cs_new_protected:Npn \@@_show_tag_overlay:nn #1#2
{
  % #1: preformatted provenance label
  % #2: displayed value
  \group_begin:
    \bool_if:NTF \g_@@_show_tag_upper_bool
      { \dim_set:Nn \l_@@_show_tag_raise_dim { 1.30em } }
      { \dim_set:Nn \l_@@_show_tag_raise_dim { .9em } }
    \bool_gset_inverse:N \g_@@_show_tag_upper_bool
    \leavevmode
    \smash{%
      \vbox to \z@ {%
   \vss
   \color{color@tags@line}%
      \bool_if:NTF \l_@@_show_tag_leftward_bool
      {%
        \llap{%
          \raise \l_@@_show_tag_raise_dim \hbox{%
            \fontsize{4}{0}\selectfont
            \ttfamily
            #1\,%
          }%
          \vrule
        }%
      }
      {%
        \rlap{%
          \vrule
          \raise \l_@@_show_tag_raise_dim \hbox{%
            \fontsize{4}{0}\selectfont
            \ttfamily
            \,%
            #1%
          }%
        }%
      }%
      }%
    }%
    \textcolor{color@tags@font}{#2}%
  \group_end:
}

% Aggregated monetary tables use full euro amounts.  Guarded tagged cells must
% invoke \cs{TblrNum} explicitly so they still use the current column's
% |table-format| for a common SI field width; outside tables \cs{num} provides
% ordinary inline output.  Marginalia temporarily clear the table-context
% boolean and therefore remain unpadded.  The informational
% existing-instrumentation cost and the animal-housing rate are formatted
% separately because their source units and precision differ from aggregated
% funding values.
\cs_new_protected:Npn \@@_show_table_number:n #1
{
  \bool_if:NTF \g_@@_tblr_inside_bool
    { \TblrNum{#1} }
    { \num{#1} }
}

\cs_new_protected:Npn \@@_show_eur:n #1
{
  \bool_if:NTF \g_@@_tblr_inside_bool
    { \TblrNum{#1} }
    {
      \num[
        round-mode = places,
        round-precision = 0
      ]{#1}
    }
}

% Existing-instrumentation costs are supplied in thousands of EUR and are not
% included in any funding aggregate.  Match the DFG table's one-decimal
% precision while still suppressing a redundant trailing zero.
\cs_new_protected:Npn \@@_show_existing_instrumentation_cost:n #1
{
  \bool_if:NTF \g_@@_tblr_inside_bool
```

```
2828      { \TblrNum{#1} }
2829      {
2830        \num[
2831          round-mode = places,
2832          round-precision = 1
2833        ]{#1}
2834      }
2835  }
2836
2837  \cs_new_protected:Npn \@@_show_calculated_number:nnN #1#2#3
2838  {
2839    \dfgShowCalculatedTag{#1}{#2}
2840    { \exp_args:NV \@@_show_eur:n #3 }
2841  }
2842
2843  \AtBeginDocument{%
2844    \bool_if:NTF \l_@@_class_showtags_bool
2845    {%
2846      \RenewDocumentCommand{\dfgShowTag}{ m m m o }
2847      { \texorpdfstring{\@@_show_tag_direct:nnnn {#1}{#2}{#3}{#4}}{#3} }%
2848      \RenewDocumentCommand{\dfgShowCalculatedTag}{ m m +m }
2849      { \texorpdfstring{\@@_show_tag_value:nn { calc:#1~[#2] } {#3}}{#3} }%
2850    }
2851    {%
2852      \RenewDocumentCommand{\dfgShowTag}{ m m m o }
2853      {#3}%
2854      \RenewDocumentCommand{\dfgShowCalculatedTag}{ m m +m }
2855      {#3}%
2856    }%
2857  }
2858
```

\addEntry `\addEntry [⟨keys⟩] {⟨database⟩} {⟨key-values⟩}` Adds a new row to the database named ⟨*database*⟩, populated by setting the keys defined under the key-path ⟨*keys*⟩ from ⟨*key-values*⟩. If ⟨*keys*⟩ is omitted, it defaults to the database name (⟨*database*⟩).

The `cv` database receives special treatment: entries are only recorded when the boolean `\l_dfg_class_cv_bool` is true (i.e. when a CV is actually being produced). Otherwise the entry is skipped and a warning is issued, so that bibliography-style CV entries can be left in the source unconditionally without polluting non-CV builds.

The internal helper `\__dfg_add_entry:nnn` performs the actual insertion and is shared between both branches to avoid code duplication.

Funding declarations for `RequestedFunding` may occur in project files, after the proposal-wide overview tables have already been printed. Such entries are additionally written to the current LaTeX auxiliary file and restored on the following run into `RequestedFundingOverview`. Entries encountered before `\begin{document}` are copied into that overview database immediately. Consequently the declarations remain next to their project text, while overview tables have access to the complete data set after the usual additional LaTeX run. A separate database prevents the restored rows from duplicating the live rows used by the project tables. The same forward-data mechanism copies `RequestedFundingStaff` declarations to `RequestedFundingStaffOverview` and `RequestedAndExistingFundingDirectCosts` declarations to `RequestedAndExistingFundingDirectCostsOverview`. Chapter 2 can therefore aggregate the existing staff and institutional direct-cost support declared later in the individual project chapters without a second user-maintained database.

Funding aggregate indexes The funding databases remain the authoritative row store. In addition, every funding declaration updates two LaTeX3 property lists: one for monetary sums grouped by database, project, funding type, and year field; and one for staff quantities grouped by database, project, category, and year field. Each index also stores the corresponding all-project and all-type aggregate where applicable.

Both live declarations and rows restored from auxiliary files pass through the same registration function. The indexes therefore remain correct as later project files add entries; there is no one-shot cache to invalidate. Summary cells perform a property lookup instead of traversing the growing database. Detailed project tables still read their source rows, but collect all five funding types in one project-filtered traversal rather than five.

Complete-proposal benchmark The 149-page complete proposal was measured on 2026-07-17 with LuaHBTeX from TeX Live 2026. Times below are for one final LuaLaTeX pass and compare the same proposal data before and after the aggregate-index refactor. Absolute values depend on the machine and document contents; the important result is that lookup time no longer grows through repeated full database sums.

Operation	Before	After	Reduction
Staff overview	135.745 s	6.975 s	94.9 %
Instrumentation overview	5.239 s	0.636 s	87.9 %
21 requested-funding tables	90.444 s	28.831 s	68.1 %
19 existing-funding tables	23.277 s	6.605 s	71.6 %
Combined	254.705 s	43.047 s	83.1 %

This saves approximately 211.7 seconds per LuaLaTeX pass. The monitored three-pass `make complete-proposal` release build completed in 33 minutes 21 seconds, so the funding changes account for approximately 10 minutes 35 seconds of avoided work across that build.

```
\tl_new:N \l_@@_add_entry_id_tl
\tl_new:N \l_@@_add_entry_existing_row_tl

\keys_define:nn { dfg / add-entry-id-probe }
{
  id      .tl_set:N = \l_@@_add_entry_id_tl,
  unknown .code:n   = { }
}

\cs_new_protected:Npn \@@_add_entry_replace_duplicate_id:n #1
{
  \tl_clear:N \l_@@_add_entry_id_tl
  \keys_set:nn { dfg / add-entry-id-probe } {#1}
  \tl_if_blank:VF \l_@@_add_entry_id_tl
  {
    \DTLifhaskey { \l_@@_database_name_tl } { id }
    {
      \DTLgetrowforkey
        { \l_@@_add_entry_existing_row_tl }
        { \l_@@_database_name_tl }
        { id }
        { \l_@@_add_entry_id_tl }
      \tl_if_blank:VF \l_@@_add_entry_existing_row_tl
      {
        \msg_warning:nnee { dfg } { duplicate-entry-id }
          { \l_@@_database_name_tl }
          { \l_@@_add_entry_id_tl }
        \DTLforeach { \l_@@_database_name_tl }
          { \l_tmpa_tl = id }
        {
          \str_if_eq:VVT \l_tmpa_tl \l_@@_add_entry_id_tl
            { \DTLremovecurrentrow }
        }
      }
    }
    { }
  }
}

\cs_new_protected:Npn \@@_add_entry:nnn #1#2#3
{
  % #1: keys     #2: database     #3: key-values
  \group_begin:
  \tl_set:Nn \l_@@_database_name_tl {#2}
  \DTLifdbexists {#2}
  { }
  { \DTLaction [ name = \l_@@_database_name_tl ] { new } }
  \@@_add_entry_replace_duplicate_id:n {#3}
  \bool_lazy_all:nT
  {
    { \str_if_eq_p:nn {#1} { staff } }
    { \str_if_eq_p:nn {#2} { RequestedFundingStaff } }
  }
  {
    \@@_requested_funding_staff_ensure_schema:
    \@@_requested_funding_staff_validate:n {#3}
  }
  \DTLaction [ name = \l_@@_database_name_tl ] { new~row }
  \str_if_eq:nnT {#1} { project }
  {
    \bool_set_true:N  \l_@@_project_consider_for_lists_bool
    \bool_set_false:N \l_@@_project_reference_only_bool
```

```
    }
    \exp_args:Ne \keys_set:nn {#1} {#3}
    \str_if_eq:nnT {#1} { project }
    {
      \bool_if:NTF \l_@@_project_consider_for_lists_bool
        { \DTLnewdbentry{\l_@@_database_name_tl}{consider-for-lists}{true} }
        { \DTLnewdbentry{\l_@@_database_name_tl}{consider-for-lists}{false} }
      \bool_if:NTF \l_@@_project_reference_only_bool
        { \DTLnewdbentry{\l_@@_database_name_tl}{reference-only}{true} }
        { \DTLnewdbentry{\l_@@_database_name_tl}{reference-only}{false} }
      \bool_gset_false:N \g_@@_project_count_ready_bool
    }
    \bool_lazy_all:nT
    {
      { \str_if_eq_p:nn {#1} { funding } }
      { \str_if_eq_p:Vn \l_@@_funding_tmp_type_str { staff } }
    }
    { \@@_funding_complete_staff_sums:n {#3} }
    \group_end:
  }

  % Funding tables repeatedly ask for the same sums, sometimes hundreds of
  % times while tabularray measures a table.  Keep the datatool databases as
  % the row store, but maintain compact aggregate indexes alongside them.
  % The key layout is
  %
  %   database / project-or-__ALL__ / type-or-__ALL__ / field
  %
  % for monetary sums and
  %
  %   database / project-or-__ALL__ / category / quantity-field
  %
  % for requested staff quantities.  Values are added when an entry is
  % declared, so a lookup never has to traverse a datatool database and the
  % index cannot become stale when later project files add more rows.
  \prop_new:N \g_@@_funding_sum_prop
  \prop_new:N \g_@@_funding_staff_quantity_prop
  \fp_new:N   \l_@@_funding_cache_fp
  \tl_new:N   \l_@@_funding_cache_key_tl
  \tl_new:N   \l_@@_funding_cache_value_tl
  \tl_new:N   \l_@@_funding_cache_project_tl
  \tl_new:N   \l_@@_funding_cache_type_tl
  \tl_new:N   \l_@@_funding_cache_category_tl
  \tl_new:N   \l_@@_funding_cache_percentage_tl
  \tl_new:N   \l_@@_funding_salary_command_tl
  \tl_new:N   \l_@@_funding_salary_value_tl
  \tl_new:N   \l_@@_funding_cache_sum_one_tl
  \tl_new:N   \l_@@_funding_cache_sum_two_tl
  \tl_new:N   \l_@@_funding_cache_sum_three_tl
  \tl_new:N   \l_@@_funding_cache_sum_four_tl
  \tl_new:N   \l_@@_funding_cache_sum_five_tl
  \tl_new:N   \l_@@_funding_cache_quantity_one_tl
  \tl_new:N   \l_@@_funding_cache_quantity_two_tl
  \tl_new:N   \l_@@_funding_cache_quantity_three_tl
  \tl_new:N   \l_@@_funding_cache_quantity_four_tl
  \tl_new:N   \l_@@_funding_cache_quantity_five_tl
  \bool_new:N \l_@@_funding_calculation_store_bool
  \bool_new:N \l_@@_funding_calculation_ready_bool
  \bool_new:N \l_@@_funding_calculation_needed_bool

  \prop_new:N \g_@@_funding_salary_commands_prop
  \prop_gset_from_keyval:Nn \g_@@_funding_salary_commands_prop
  {
    prof        = dfgSalaryProf,
    groupLeader = dfgSalaryGroupLeader,
    postDoc     = dfgSalaryPostDoc,
    medDoc      = dfgSalaryMedDoc,
    PhD         = dfgSalaryPhD,
    research    = dfgSalaryResearch,
    nonResearch = dfgSalaryNonResearch,
    student     = dfgSalaryStudent,
    assistant   = dfgSalaryAssistant,
    prof-W2     = dfgSalaryProf,
    prof-W3     = dfgSalaryProf
  }

  \keys_define:nn { dfg / funding-cache / sum }
  {
    year1 .tl_set:N = \l_@@_funding_cache_sum_one_tl,
    year2 .tl_set:N = \l_@@_funding_cache_sum_two_tl,
    year3 .tl_set:N = \l_@@_funding_cache_sum_three_tl,
    year4 .tl_set:N = \l_@@_funding_cache_sum_four_tl,
    year5 .tl_set:N = \l_@@_funding_cache_sum_five_tl,
    unknown .code:n = { }
  }

  \keys_define:nn { dfg / funding-cache / quantity }
```

```
{
  year1 .tl_set:N = \l_@@_funding_cache_quantity_one_tl,
  year2 .tl_set:N = \l_@@_funding_cache_quantity_two_tl,
  year3 .tl_set:N = \l_@@_funding_cache_quantity_three_tl,
  year4 .tl_set:N = \l_@@_funding_cache_quantity_four_tl,
  year5 .tl_set:N = \l_@@_funding_cache_quantity_five_tl,
  unknown .code:n = { }
}

\keys_define:nn { dfg / funding-cache }
{
  ref-project-id .tl_set:N = \l_@@_funding_cache_project_tl,
  funding-type   .tl_set:N = \l_@@_funding_cache_type_tl,
  category       .tl_set:N = \l_@@_funding_cache_category_tl,
  percentage     .tl_set:N = \l_@@_funding_cache_percentage_tl,
  sum .code:n =
    { \keys_set:nn { dfg / funding-cache / sum } {#1} },
  quantity .code:n =
    { \keys_set:nn { dfg / funding-cache / quantity } {#1} },
  unknown .code:n = { }
}

\cs_new_protected:Npn \@@_funding_capture:n #1
{
  \tl_clear:N \l_@@_funding_cache_project_tl
  \tl_clear:N \l_@@_funding_cache_type_tl
  \tl_clear:N \l_@@_funding_cache_category_tl
  \tl_clear:N \l_@@_funding_cache_percentage_tl
  \tl_clear:N \l_@@_funding_cache_sum_one_tl
  \tl_clear:N \l_@@_funding_cache_sum_two_tl
  \tl_clear:N \l_@@_funding_cache_sum_three_tl
  \tl_clear:N \l_@@_funding_cache_sum_four_tl
  \tl_clear:N \l_@@_funding_cache_sum_five_tl
  \tl_clear:N \l_@@_funding_cache_quantity_one_tl
  \tl_clear:N \l_@@_funding_cache_quantity_two_tl
  \tl_clear:N \l_@@_funding_cache_quantity_three_tl
  \tl_clear:N \l_@@_funding_cache_quantity_four_tl
  \tl_clear:N \l_@@_funding_cache_quantity_five_tl
  \keys_set:nn { dfg / funding-cache } {#1}
}

\cs_new_protected:Npn \@@_funding_calculation_needed:NN #1#2
{
  \bool_lazy_all:nT
  {
    { \tl_if_blank_p:V #1 }
    { ! \tl_if_blank_p:V #2 }
  }
  { \bool_set_true:N \l_@@_funding_calculation_needed_bool }
}

\cs_new_protected:Npn \@@_funding_calculate_one:NNnn #1#2#3#4
{
  % #1 sum tl, #2 quantity tl, #3 database field, #4 period factor
  \bool_lazy_all:nT
  {
    { \tl_if_blank_p:V #1 }
    { ! \tl_if_blank_p:V #2 }
    { \bool_if_p:N \l_@@_funding_calculation_ready_bool }
  }
  {
    \exp_args:NnV \regex_match:nnTF
      { \A \d+ (?: \. \d+ )? \Z } #2
    {
      \tl_set:Nx #1
      {
        \fp_eval:n
        {
          round(
            \l_@@_funding_salary_value_tl
            * \l_@@_funding_cache_percentage_tl / 100
            * #2
            * #4,
            0
          )
        }
      }
      \bool_if:NT \l_@@_funding_calculation_store_bool
      {
        \exp_args:Nnne \DTLnewdbentry
          {\l_@@_database_name_tl}{#3}{\tl_use:N #1}
      }
    }
    {
      \bool_if:NT \l_@@_funding_calculation_store_bool
      {
        \msg_warning:nne { dfg } { funding-staff-cost-unavailable }
```

```
3095          { project~'\l_@@_funding_cache_project_tl',~#3:~invalid~quantity~'#2' }
3096        }
3097      }
3098    }
3099  }
3100
3101  \cs_new_protected:Npn \@@_funding_calculate_staff_sums:
3102  {
3103    \str_if_eq:VnT \l_@@_funding_cache_type_tl { staff }
3104    {
3105      \bool_set_false:N \l_@@_funding_calculation_needed_bool
3106      \@@_funding_calculation_needed:NN
3107        \l_@@_funding_cache_sum_one_tl \l_@@_funding_cache_quantity_one_tl
3108      \@@_funding_calculation_needed:NN
3109        \l_@@_funding_cache_sum_two_tl \l_@@_funding_cache_quantity_two_tl
3110      \@@_funding_calculation_needed:NN
3111        \l_@@_funding_cache_sum_three_tl \l_@@_funding_cache_quantity_three_tl
3112      \@@_funding_calculation_needed:NN
3113        \l_@@_funding_cache_sum_four_tl \l_@@_funding_cache_quantity_four_tl
3114      \@@_funding_calculation_needed:NN
3115        \l_@@_funding_cache_sum_five_tl \l_@@_funding_cache_quantity_five_tl
3116      \bool_if:NT \l_@@_funding_calculation_needed_bool
3117      {
3118        \bool_set_true:N \l_@@_funding_calculation_ready_bool
3119        \prop_get:NVNTF \g_@@_funding_salary_commands_prop
3120          \l_@@_funding_cache_category_tl \l_@@_funding_salary_command_tl
3121        {
3122          \cs_if_exist:cTF { \l_@@_funding_salary_command_tl }
3123          { \tl_set:Nx \l_@@_funding_salary_value_tl
3124            { \use:c { \l_@@_funding_salary_command_tl } } }
3125          { \bool_set_false:N \l_@@_funding_calculation_ready_bool }
3126        }
3127        { \bool_set_false:N \l_@@_funding_calculation_ready_bool }
3128        \exp_args:NnV \regex_match:nnF
3129          { \A \d+ (?: \. \d+ )? \Z }
3130          \l_@@_funding_cache_percentage_tl
3131        { \bool_set_false:N \l_@@_funding_calculation_ready_bool }
3132        \exp_args:NnV \regex_match:nnF
3133          { \A \d+ (?: \. \d+ )? \Z }
3134          \l_@@_funding_salary_value_tl
3135        { \bool_set_false:N \l_@@_funding_calculation_ready_bool }
3136        \bool_lazy_all:nT
3137        {
3138          { \bool_if_p:N \l_@@_funding_calculation_store_bool }
3139          { ! \bool_if_p:N \l_@@_funding_calculation_ready_bool }
3140        }
3141        {
3142          \msg_warning:nne { dfg } { funding-staff-cost-unavailable }
3143          {
3144            project~'\l_@@_funding_cache_project_tl',~category~
3145            '\l_@@_funding_cache_category_tl',~percentage~
3146            '\l_@@_funding_cache_percentage_tl'
3147          }
3148        }
3149        \@@_funding_calculate_one:NNnn
3150          \l_@@_funding_cache_sum_one_tl
3151          \l_@@_funding_cache_quantity_one_tl { sum-year1 } { 0.5 }
3152        \@@_funding_calculate_one:NNnn
3153          \l_@@_funding_cache_sum_two_tl
3154          \l_@@_funding_cache_quantity_two_tl { sum-year2 } { 1 }
3155        \@@_funding_calculate_one:NNnn
3156          \l_@@_funding_cache_sum_three_tl
3157          \l_@@_funding_cache_quantity_three_tl { sum-year3 } { 1 }
3158        \@@_funding_calculate_one:NNnn
3159          \l_@@_funding_cache_sum_four_tl
3160          \l_@@_funding_cache_quantity_four_tl { sum-year4 } { 1 }
3161        \@@_funding_calculate_one:NNnn
3162          \l_@@_funding_cache_sum_five_tl
3163          \l_@@_funding_cache_quantity_five_tl { sum-year5 } { 0.5 }
3164      }
3165    }
3166  }
3167
3168  \cs_new_protected:Npn \@@_funding_complete_staff_sums:n #1
3169  {
3170    \bool_set_true:N \l_@@_funding_calculation_store_bool
3171    \@@_funding_capture:n {#1}
3172    \@@_funding_calculate_staff_sums:
3173  }
3174
3175  \cs_new_protected:Npn \@@_funding_cache_add:Nnn #1#2#3
3176  {
3177    \tl_set:Nx \l_@@_funding_cache_key_tl {#2}
3178    \tl_set:Nx \l_@@_funding_cache_value_tl {#3}
3179    \tl_if_blank:VF \l_@@_funding_cache_value_tl
3180    {
3181      \prop_get:NVNTF #1
```

```
      \l_@@_funding_cache_key_tl \l_@@_funding_cache_value_tl
    {
      \fp_set:Nn \l_@@_funding_cache_fp
        { \l_@@_funding_cache_value_tl + (#3) }
    }
    { \fp_set:Nn \l_@@_funding_cache_fp {#3} }
    \prop_gput:NVx #1 \l_@@_funding_cache_key_tl
      { \fp_to_decimal:N \l_@@_funding_cache_fp }
  }
}

\cs_new_protected:Npn \@@_funding_cache_sum_year:nnn #1#2#3
{
  % #1 database, #2 field, #3 value
  \tl_if_blank:VF \l_@@_funding_cache_type_tl
  {
    \@@_funding_cache_add:Nnn \g_@@_funding_sum_prop
      {#1/\l_@@_funding_cache_project_tl/
       \l_@@_funding_cache_type_tl/#2} {#3}
    \@@_funding_cache_add:Nnn \g_@@_funding_sum_prop
      {#1/__ALL__/\l_@@_funding_cache_type_tl/#2} {#3}
  }
  \@@_funding_cache_add:Nnn \g_@@_funding_sum_prop
    {#1/\l_@@_funding_cache_project_tl/__ALL__/#2} {#3}
  \@@_funding_cache_add:Nnn \g_@@_funding_sum_prop
    {#1/__ALL__/__ALL__/#2} {#3}
}

\cs_new_protected:Npn \@@_funding_cache_quantity_year:nnn #1#2#3
{
  % #1 database, #2 field, #3 value
  \@@_funding_cache_add:Nnn \g_@@_funding_staff_quantity_prop
    {#1/\l_@@_funding_cache_project_tl/
     \l_@@_funding_cache_category_tl/#2} {#3}
  \@@_funding_cache_add:Nnn \g_@@_funding_staff_quantity_prop
    {#1/__ALL__/\l_@@_funding_cache_category_tl/#2} {#3}
}

\cs_new_protected:Npn \@@_funding_cache_entry:nn #1#2
{
  \group_begin:
  \bool_set_false:N \l_@@_funding_calculation_store_bool
  \@@_funding_capture:n {#2}
  \@@_funding_calculate_staff_sums:
  \tl_if_blank:VF \l_@@_funding_cache_project_tl
  {
    \@@_funding_cache_sum_year:nnn {#1} {sum-year1}
      {\l_@@_funding_cache_sum_one_tl}
    \@@_funding_cache_sum_year:nnn {#1} {sum-year2}
      {\l_@@_funding_cache_sum_two_tl}
    \@@_funding_cache_sum_year:nnn {#1} {sum-year3}
      {\l_@@_funding_cache_sum_three_tl}
    \@@_funding_cache_sum_year:nnn {#1} {sum-year4}
      {\l_@@_funding_cache_sum_four_tl}
    \@@_funding_cache_sum_year:nnn {#1} {sum-year5}
      {\l_@@_funding_cache_sum_five_tl}
    \str_if_eq:VnT \l_@@_funding_cache_type_tl {staff}
    {
      \tl_if_blank:VF \l_@@_funding_cache_category_tl
      {
        \@@_funding_cache_quantity_year:nnn {#1} {quantity-year1}
          {\l_@@_funding_cache_quantity_one_tl}
        \@@_funding_cache_quantity_year:nnn {#1} {quantity-year2}
          {\l_@@_funding_cache_quantity_two_tl}
        \@@_funding_cache_quantity_year:nnn {#1} {quantity-year3}
          {\l_@@_funding_cache_quantity_three_tl}
        \@@_funding_cache_quantity_year:nnn {#1} {quantity-year4}
          {\l_@@_funding_cache_quantity_four_tl}
        \@@_funding_cache_quantity_year:nnn {#1} {quantity-year5}
          {\l_@@_funding_cache_quantity_five_tl}
      }
    }
  }
  \group_end:
}

\cs_new_protected:Npn \@@_funding_sum_set:Nnnnn #1#2#3#4#5
{
  % #1 result, #2 database, #3 project, #4 type, #5 field list
  \fp_zero:N \l_@@_funding_cache_fp
  \clist_map_inline:nn {#5}
  {
    \tl_set:Nx \l_@@_funding_cache_key_tl {#2/#3/#4/##1}
    \prop_get:NVNT \g_@@_funding_sum_prop
      \l_@@_funding_cache_key_tl \l_@@_funding_cache_value_tl
    {
      \fp_add:Nn \l_@@_funding_cache_fp
```

```
3269          { \l_@@_funding_cache_value_tl }
3270      }
3271    }
3272    \tl_set:Nx #1 { \fp_to_decimal:N \l_@@_funding_cache_fp }
3273  }
3274
3275  \bool_new:N \g_@@_document_started_bool
3276
3277  \AtBeginDocument
3278    { \bool_gset_true:N \g_@@_document_started_bool }
3279
3280  % This command is deliberately public and free of expl3 punctuation because
3281  % it is written to, and subsequently read from, LaTeX .aux files.
3282  \NewDocumentCommand \dfgRestoreRequestedFunding { +m }
3283  {
3284    \bool_if:NF \l_@@_class_fastcompile_project_bool
3285    {
3286      \@@_add_entry:nnn { funding } { RequestedFundingOverview } {#1}
3287      \@@_funding_cache_entry:nn { RequestedFundingOverview } {#1}
3288    }
3289  }
3290
3291  % This public command is written to the auxiliary file.  Keep its name free of
3292  % expl3 punctuation so LaTeX can read it before the document body starts.
3293  \cs_new_protected:Npn \@@_requested_funding_staff_overview_ensure_schema:
3294  {
3295    \DTLifdbexists { RequestedFundingStaffOverview }
3296    { }
3297    { \DTLaction [ name = RequestedFundingStaffOverview ] { new } }
3298    \clist_map_inline:nn
3299    {
3300      ref-project-id, funding-status, staff-type, ref-person-id,
3301      firstname, lastname, academic-degree, positionText, field-of-research,
3302      ref-institution-id, ref-affiliation-id, department, commitment, category,
3303      funding-source, description
3304    }
3305    {
3306      \DTLifhaskey { RequestedFundingStaffOverview } {##1}
3307      { }
3308      { \DTLaddcolumn { RequestedFundingStaffOverview } {##1} }
3309    }
3310  }
3311
3312  \NewDocumentCommand \dfgRestoreRequestedFundingStaff { +m }
3313  {
3314    \bool_if:NF \l_@@_class_fastcompile_project_bool
3315    {
3316      \@@_requested_funding_staff_overview_ensure_schema:
3317      \@@_add_entry:nnn { staff } { RequestedFundingStaffOverview } {#1}
3318    }
3319  }
3320
3321  \bool_if:NF \l_@@_class_fastcompile_project_bool
3322  { \@@_requested_funding_staff_overview_ensure_schema: }
3323
3324  % Existing direct-cost rows are normally declared in the individual project
3325  % descriptions, after section 2.1.1 has already been typeset.  Mirror them to
3326  % a dedicated overview database so that the project table keeps using its live
3327  % rows without counting the restored rows twice.
3328  \cs_new_protected:Npn
3329    \@@_requested_and_existing_direct_costs_overview_ensure_schema:
3330  {
3331    \DTLifdbexists { RequestedAndExistingFundingDirectCostsOverview }
3332    { }
3333    { \DTLaction [ name = RequestedAndExistingFundingDirectCostsOverview ] { new } }
3334    \clist_map_inline:nn
3335    {
3336      ref-project-id, ref-affiliation-id, financed-by,
3337      sum-year1, sum-year2, sum-year3, sum-year4, sum-year5
3338    }
3339    {
3340      \DTLifhaskey { RequestedAndExistingFundingDirectCostsOverview } {##1}
3341      { }
3342      { \DTLaddcolumn { RequestedAndExistingFundingDirectCostsOverview } {##1} }
3343    }
3344  }
3345
3346  % This public command is written to the auxiliary file.  Its name deliberately
3347  % contains no expl3 punctuation so that LaTeX can read it at the start of the
3348  % following run.
3349  \NewDocumentCommand \dfgRestoreRequestedAndExistingFundingDirectCosts { +m }
3350  {
3351    \bool_if:NF \l_@@_class_fastcompile_project_bool
3352    {
3353      \@@_requested_and_existing_direct_costs_overview_ensure_schema:
3354      \@@_add_entry:nnn { funding }
3355        { RequestedAndExistingFundingDirectCostsOverview } {#1}
```

```
  }
}

\bool_if:NF \l_@@_class_fastcompile_project_bool
{ \@@_requested_and_existing_direct_costs_overview_ensure_schema: }

% Keep the overview tables usable on the first run, before any project-local
% entries have reached the auxiliary files.  A newly created database has no
% columns, so establish the RequestedFunding schema explicitly: otherwise
% sorting or filtering an empty overview raises ``No sort columns'' and
% ``Unknown column key'' errors.  No placeholder row is required.
\bool_if:NF \l_@@_class_fastcompile_project_bool
{
  \DTLifdbexists { RequestedFundingOverview }
    { }
    {
      \DTLaction [ name = RequestedFundingOverview ] { new }
      \DTLaddcolumn { RequestedFundingOverview } { ref-project-id }
      \DTLaddcolumn { RequestedFundingOverview } { type }
      \DTLaddcolumn { RequestedFundingOverview } { category }
      \DTLaddcolumn { RequestedFundingOverview } { percentage }
      \DTLaddcolumn { RequestedFundingOverview } { quantity-year1 }
      \DTLaddcolumn { RequestedFundingOverview } { quantity-year2 }
      \DTLaddcolumn { RequestedFundingOverview } { quantity-year3 }
      \DTLaddcolumn { RequestedFundingOverview } { quantity-year4 }
      \DTLaddcolumn { RequestedFundingOverview } { quantity-year5 }
      \DTLaddcolumn { RequestedFundingOverview } { sum-year1 }
      \DTLaddcolumn { RequestedFundingOverview } { sum-year2 }
      \DTLaddcolumn { RequestedFundingOverview } { sum-year3 }
      \DTLaddcolumn { RequestedFundingOverview } { sum-year4 }
      \DTLaddcolumn { RequestedFundingOverview } { sum-year5 }
    }
}

\cs_new_protected:Npn \@@_cache_requested_funding:n #1
{
  \bool_if:NTF \g_@@_document_started_bool
  {
    \iow_now:Nx \@auxout
      {
        \token_to_str:N \dfgRestoreRequestedFunding
        { \exp_not:n {#1} }
      }
  }
  { \dfgRestoreRequestedFunding{#1} }
}

\cs_new_protected:Npn \@@_cache_requested_funding_staff:n #1
{
  \bool_if:NTF \g_@@_document_started_bool
  {
    \iow_now:Nx \@auxout
    {
      \token_to_str:N \dfgRestoreRequestedFundingStaff
      { \exp_not:n {#1} }
    }
  }
  { \dfgRestoreRequestedFundingStaff{#1} }
}

\cs_new_protected:Npn
  \@@_cache_requested_and_existing_direct_costs:n #1
{
  \bool_if:NTF \g_@@_document_started_bool
  {
    \iow_now:Nx \@auxout
    {
      \token_to_str:N \dfgRestoreRequestedAndExistingFundingDirectCosts
      { \exp_not:n {#1} }
    }
  }
  { \dfgRestoreRequestedAndExistingFundingDirectCosts{#1} }
}

% Focused project builds retain the project rows needed by generated plots when
% images are enabled, plus the rows needed by the three local funding tables.
% In particular, do not populate or restore any proposal-wide overview database
% through the auxiliary file.
\cs_new_protected:Npn \@@_fast_project_add_entry:nnn #1#2#3
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_images_bool
  {
    \str_if_eq:nnT {#1/#2} {/project}
      { \@@_add_entry:nnn {project} {project} {#3} }
    \str_if_eq:nnT {#1} {gantt}
      { \@@_add_entry:nnn {#1} {#2} {#3} }
  }
```

```
  \bool_if:NT \l_@@_class_fastcompile_project_show_funding_tables_bool
  {
    \str_case:nn {#1/#2}
    {
      {funding/RequestedFunding}
      {
        \@@_add_entry:nnn {#1} {#2} {#3}
        \@@_funding_cache_entry:nn {#2} {#3}
      }
      {staff/RequestedFundingStaff}
        { \@@_add_entry:nnn {#1} {#2} {#3} }
      {funding/RequestedAndExistingFundingDirectCosts}
      {
        \@@_add_entry:nnn {#1} {#2} {#3}
        \@@_funding_cache_entry:nn {#2} {#3}
      }
    }
  }

% Retain only the database-to-keyset mapping when focused builds skip rows.
% The get parser needs the schema to consume reference paths without values.
\prop_new:N \g_@@_database_keyset_prop
\NewDocumentCommand \addEntry { O{} m m }
{
  \tl_if_blank:nTF {#1}
    {\prop_gput:Nnn \g_@@_database_keyset_prop {#2}{#2}}
    {\prop_gput:Nnn \g_@@_database_keyset_prop {#2}{#1}}
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  { \@@_fast_project_add_entry:nnn {#1} {#2} {#3} }
  {
  \tl_if_eq:nnTF {#1} { cv }
  { \bool_if:NTF \l_@@_class_cv_bool
 { \@@_add_entry:nnn {#1} {#2} {#3} }
   { \msg_warning:nnn { dfg } { cv-entry-omitted } {#2} } }
  {% default behaviour: use #2 as family if #1 is empty
   \tl_if_empty:nTF {#1}
   { \@@_add_entry:nnn {#2} {#2} {#3} }
   {
      \@@_add_entry:nnn {#1} {#2} {#3}
      \str_if_eq:nnT {#1} { funding }
        { \@@_funding_cache_entry:nn {#2} {#3} }
      \bool_lazy_all:nT
        {
          { \str_if_eq_p:nn {#1} { funding } }
          { \str_if_eq_p:nn {#2} { RequestedFunding } }
        }
        { \@@_cache_requested_funding:n {#3} }
      \bool_lazy_all:nT
        {
          { \str_if_eq_p:nn {#1} { staff } }
          { \str_if_eq_p:nn {#2} { RequestedFundingStaff } }
        }
        { \@@_cache_requested_funding_staff:n {#3} }
      \bool_lazy_all:nT
        {
          { \str_if_eq_p:nn {#1} { funding } }
          { \str_if_eq_p:nn {#2} { RequestedAndExistingFundingDirectCosts } }
        }
        { \@@_cache_requested_and_existing_direct_costs:n {#3} }
    }
  }
  }
}
```

Issued by `\addEntry` when an entry is targeted at the `cv` database but no CV is being produced (`\l_dfg_class_cv_bool` is false). The omitted entry's key-value data is included in the message to aid debugging.

```
\msg_new:nnn { dfg } { cv-entry-omitted }
{
  CV~entry~omitted:~no~CV~is~being~created.^^J
  Entry~data:~'#1'.
}
```

There is a shortcut to define a database for e.g. a whole command to avoid all the time to write something like `\getEntry[⟨person⟩]{⟨MM⟩}{⟨lastname⟩}`. So you can define `\tmpLocalDB` with the database name.

```
\str_set:Nn \tmpLocalDB {}
```

\getEntry[→P. 2] The primary read command: \getEntry[⟨*db*⟩]{⟨*id-value*⟩}{⟨*field*⟩}[⟨*id-field*⟩].

1. #1: Database name (default: \tmpLocalDB).
2. #2: The value to look up in the id-field column.
3. #3: The field whose value is to be returned.
4. #4: The name of the ID column (default: id); change this to look up rows by a field other than the primary key.

The guard logic runs in order: If fastcompile-project[→P. 65]=true, replace the output with a literal identifier placeholder to skip the query entirely. If #1 is empty (\tmpLocalDB not set), warn and abort. If the database does not exist, warn and abort. Retrieve the value; if null/empty, warn (but only outside the tabularray measuring pass to avoid duplicate messages). Output via \dfgShowTag so that showtags[→P. 65] annotations are applied automatically.

PDF-bookmark note (FIXME): When #3 contains LaTeX commands, the value may produce garbage in PDF bookmarks. Wrapping the output in \texorpdfstring would fix this but requires knowing at definition time what the field contains, which is not possible. A workaround is to use \texorpdfstring at the call site for fields that appear in section headings.

```
%\getEntry [<DB>]{<id-value>}{<field>}                 [<id-column>]
%   single field, prints with \dfgShowTag wrapping
%
%\getEntry*[<DB>]{<id-value>}{<assignments>}            [<id-column>]
%   multi-field, assigns to macros, no printing
\NewDocumentCommand \dfgFastReference { m }
{ { \textcolor{error}{\texttt{<#1>}} } }

% Project IDs stay prominent when metadata lookups are skipped. Scope the
% style locally so following text and error diagnostics keep their colours.
\cs_new_protected:Npn \@@_fast_project_reference:n #1
{ \dfgFastReference{#1} }

\cs_new_protected:Npn \@@_fast_clear_assignment:w #1 = #2 \q_stop
{ \cs_set:Npn #1 { } }

\cs_new_protected:Npn \@@_fast_clear_assignments:n #1
{
  \clist_map_inline:nn {#1}
  { \@@_fast_clear_assignment:w ##1 \q_stop }
}

\NewDocumentCommand{\getEntry}{s O{\tmpLocalDB} m m O{id}}
{%
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  {
    \IfBooleanTF{#1}
    { \@@_fast_clear_assignments:n {#4} }
    {
      \dfgShowTag {#3}{#4}
        {
          \str_if_eq:eeTF {#2} {project}
            { \@@_fast_project_reference:n {#3} }
            { \dfgFastReference{#3} }
        }
        [#2]
    }
  }
  {
    \IfBooleanTF{#1}
    { \@@_get_entry_multi:nnnn  {#2}{#3}{#4}{#5} }
    { \@@_get_entry_single:nnnn {#2}{#3}{#4}{#5} }%
  }
}

```

Reference-aware lookup (\get) Parse relation steps before rendering, so metadata containing braces cannot become part of the path. The walker carries its path in arguments rather than shared scratch variables; this also permits branching reference lists.

```
\tl_new:N \l_@@_get_path_db_tl
\tl_new:N \l_@@_get_path_id_tl
\tl_new:N \l_@@_get_path_id_field_tl
\tl_new:N \l_@@_get_path_scan_db_tl
\tl_new:N \l_@@_get_path_field_tl
\tl_new:N \l_@@_get_path_tl
\tl_new:N \l_@@_get_path_keyset_tl
\tl_new:N \l_@@_get_path_target_tl
```

```
\tl_new:N \l_@@_get_path_value_tl
\seq_new:N \l_@@_get_path_values_seq
\seq_new:N \l_@@_get_path_annotation_seq
\prop_new:N \l_@@_get_path_seen_prop

\NewDocumentCommand{\get}{s O{\tmpLocalDB} m m O{id}}
{
  \IfBooleanTF{#1}
    {\getEntry*[#2]{#3}{#4}[#5]}
    {
      \group_begin:
      \tl_set:Ne \l_@@_get_path_db_tl {#2}
      \tl_set:Ne \l_@@_get_path_id_tl {#3}
      \tl_set:Nn \l_@@_get_path_id_field_tl {#5}
      \tl_set_eq:NN \l_@@_get_path_scan_db_tl \l_@@_get_path_db_tl
      \tl_clear:N \l_@@_get_path_tl
      \seq_clear:N \l_@@_get_path_annotation_seq
      \@@_get_path_scan:n {#4}
    }
}

\prg_new_protected_conditional:Npnn \@@_get_path_if_has_key:nn #1#2 {TF}
{
  \datatool_if_has_key:nnTF {#1}{#2}
    {\prg_return_true:}
    {
      \prop_get:NnNF \g_@@_database_keyset_prop {#1}
        \l_@@_get_path_keyset_tl
        {\tl_set:Nn \l_@@_get_path_keyset_tl {#1}}
      \exp_args:NV \keys_if_exist:nnTF \l_@@_get_path_keyset_tl {#2}
        {\prg_return_true:}{\prg_return_false:}
    }
}
\prg_generate_conditional_variant:Nnn \@@_get_path_if_has_key:nn {Vn} {TF}

\cs_new_protected:Npn \@@_get_path_scan:n #1
{
  \tl_put_right:Nn \l_@@_get_path_tl {{#1}}
  \seq_put_right:Nn \l_@@_get_path_annotation_seq {#1}
  \tl_set:Nn \l_@@_get_path_field_tl {#1}
  \tl_set:Nn \l_@@_get_path_target_tl {#1}
  \regex_match:nnTF {\A ref-.+-id \Z} {#1}
    {
      \regex_replace_once:nnN {\A ref-(.+)-id \Z} {\1}
        \l_@@_get_path_target_tl
      \@@_get_path_next:
    }
    {
      % An actual scalar field takes precedence over a database shorthand.
      \@@_get_path_if_has_key:VnTF \l_@@_get_path_scan_db_tl {#1}
        {\@@_get_path_finish:}
        {
          \@@_get_path_if_has_key:VnTF \l_@@_get_path_scan_db_tl {ref-#1-id}
            {\@@_get_path_next:}
            {
              \DTLifdbexists{#1}
                {\@@_get_path_next:}
                {\@@_get_path_finish:}
            }
        }
    }
}
% xparse preserves following prose when no optional braced step is present.
\NewDocumentCommand{\@@_get_path_next:}{g}
{
  \IfNoValueTF{#1}
    {\@@_get_path_finish:}
    {
      \tl_set_eq:NN \l_@@_get_path_scan_db_tl \l_@@_get_path_target_tl
      \@@_get_path_scan:n {#1}
    }
}

\cs_new_protected:Npn \@@_get_path_missing:nnn #1#2#3
{
  \tl_clear:N \l_@@_get_path_value_tl
  \bool_if:NF \lTblrMeasuringBool
    {\msg_warning:nnnnn {dfg}{missing-key}{#1}{#2}{#3}}
}
\cs_new_protected:Npn \@@_get_path_read:nnnn #1#2#3#4
{
  \tl_clear:N \l_@@_get_path_value_tl
  \tl_if_blank:nTF {#1}
    {\msg_warning:nnn {dfg}{database-name-not-given}{#2-/-#3}}
    {
      \DTLifdbexists{#1}
        {
```

```
            \datatool_if_has_key:nnTF {#1}{#3}
              {
                \datatool_if_has_key:nnTF {#1}{#4}
                  {
                    \DTLgetvalueforkey{\l_@@_get_path_value_tl}{#3}{#1}{#4}{#2}
                    \DTLifnullorempty{\l_@@_get_path_value_tl}
                      {\@@_get_path_missing:nnn {#1}{#2}{#3}} {}
                  }
                  {\@@_get_path_missing:nnn {#1}{#2}{#4}}
              }
              {\@@_get_path_missing:nnn {#1}{#2}{#3}}
          }
          {\msg_warning:nnn {dfg}{database-not-found}{#1}}
      }
  }

  \cs_new_protected:Npn \@@_get_path_walk:nnnw #1#2#3#4#5\q_stop
  {
    \tl_if_eq:nnTF {#5}{\q_nil}
      {
        \prop_if_in:NnF \l_@@_get_path_seen_prop {{#1}{#2}{#4}}
          {
            \prop_put:Nnn \l_@@_get_path_seen_prop {{#1}{#2}{#4}} {}
            \@@_get_path_read:nnnn {#1}{#2}{#4}{#3}
            \tl_if_empty:NF \l_@@_get_path_value_tl
              {
                \seq_put_right:Ne \l_@@_get_path_values_seq
                  {
                    \exp_not:N \dfgShowTag {#2}{#4}
                      {\exp_not:V \l_@@_get_path_value_tl}[#1]
                  }
              }
          }
      }
      {
        \tl_set:Nn \l_@@_get_path_target_tl {#4}
        \regex_replace_once:nnN {\A ref-(.+)-id \Z} {\1}
          \l_@@_get_path_target_tl
        \use:e
          {
            \exp_not:N \@@_get_path_read:nnnn {#1}{#2}
              {ref-\l_@@_get_path_target_tl-id}{#3}
          }
        \tl_if_empty:NF \l_@@_get_path_value_tl
          {
            % Freeze both the destination database and the complete ID list
            % before recursion changes the local lookup variables.
            \use:e
              {
                \exp_not:N \@@_get_path_branch:nnn
                  {\l_@@_get_path_target_tl}
                  {\exp_not:V \l_@@_get_path_value_tl}
                  {\exp_not:n {#5}}
              }
          }
      }
  }
  \cs_new_protected:Npn \@@_get_path_branch:nnn #1#2#3
  {
    \clist_map_inline:nn {#2}
      {\@@_get_path_walk:nnnw {#1}{##1}{id}#3\q_stop}
  }
  \cs_new_protected:Npn \@@_get_path_finish:
  {
    \bool_if:NTF \l_@@_class_fastcompile_project_bool
      {
        \tl_set:Ne \l_tmpa_tl
          {\seq_use:Nn \l_@@_get_path_annotation_seq {/}}
        \use:e
          {
            \exp_not:N \dfgShowTag
              {\exp_not:V \l_@@_get_path_id_tl}
              {\exp_not:V \l_tmpa_tl}
              {
                \exp_not:n
                  {
                    \str_if_eq:VnTF \l_@@_get_path_db_tl {project}
                      {\@@_fast_project_reference:n {\l_@@_get_path_id_tl}}
                      {\dfgFastReference{\l_@@_get_path_id_tl}}
                  }
              }
              [\exp_not:V \l_@@_get_path_db_tl]
          }
      }
      {
        \seq_clear:N \l_@@_get_path_values_seq
        \prop_clear:N \l_@@_get_path_seen_prop
```

```
      \use:e
        {
          \exp_not:N \@@_get_path_walk:nnnw
            {\l_@@_get_path_db_tl}{\l_@@_get_path_id_tl}
            {\l_@@_get_path_id_field_tl}
            \exp_not:V \l_@@_get_path_tl \exp_not:N \q_nil \exp_not:N \q_stop
        }
      \seq_use:Nn \l_@@_get_path_values_seq {,~}
    }
  \group_end:
}

% --- single-field (your existing behaviour, lightly cleaned) ----------------
\cs_new_protected:Nn \@@_get_entry_single:nnnn
{
  % #1 db, #2 id-value, #3 field, #4 id-column
  \tl_if_blank:nTF {#1}
  { \msg_warning:nne {dfg}{database-name-not-given}{#2-/-#3} }
  {%
   \DTLifdbexists{#1}
   {%
    \DTLgetvalueforkey{\thevalue}{#3}{#1}{#4}{#2}%
    \DTLifnullorempty{\thevalue}
    {%
     \bool_if:NF \lTblrMeasuringBool
     { \msg_warning:nneee {dfg}{missing-key}{#1}{#2}{#3} }%
    }
    { \dfgShowTag{#2}{#3}{\thevalue}[#1] }%
   }
   { \msg_warning:nne {dfg}{database-not-found}{#1} }%
  }%
}

% --- multi-field: hoists checks out of per-field loop -----------------------
\cs_new_protected:Nn \@@_get_entry_multi:nnnn
{
  \tl_if_blank:nTF {#1}
  { \msg_warning:nne {dfg}{database-name-not-given}{#2-/-multi} }
  {%
   \tl_set:Nx \l_tmpa_tl {#1}%
   \tl_set:Nx \l_tmpb_tl {#2}%
   \DTLifdbexists{\l_tmpa_tl}
   {%
    % \typeout{DEBUG:~about~to~assign~from~\l_tmpa_tl~for~id~\l_tmpb_tl}%
    \use:x
    {
     \exp_not:N \DTLassignfirstmatch
     { \exp_not:V \l_tmpa_tl }
     { \exp_not:n {#4} }
     { \exp_not:V \l_tmpb_tl }
     { \exp_not:n {#3} }
    }%
    % \typeout{DEBUG:~done.~theLastName~=~\theLastName}%
   }
   { \msg_warning:nne {dfg}{database-not-found}{\l_tmpa_tl} }%
  }%
}

\cs_new_protected:Npn \@@_check_empty:w #1 = #2 \q_stop #3#4
{
  % #1: target macro tokens, #2: field name, #3: db, #4: id
  \exp_args:NV \DTLifnullorempty #1
  { \msg_warning:nneee {dfg}{missing-key}{#3}{#4}{#2} }
  { }%
}

\cs_new_protected:Nn \@@_check_empty:nnn
{
  % #1 db, #2 id, #3 single "\macro = field" pair
  \@@_check_empty:w #3 \q_stop {#1}{#2}%
}

```

Sometimes you need to get the last name of a person in the text. Here is how to do it, \getName → P. 5.

```
\NewDocumentCommand{\getName}{m}{\getEntry[person]{#1}{lastname}}
```

Sometimes you need to get the full name of a person in the text. Here is how to do it, \getFullName → P. 5.

```
\NewDocumentCommand{\getFullName}{s m}
{%
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  { \dfgFastReference{#2} }
  {
```

```
3818    % #1: starred -> include academic degree
3819    % #2: person id
3820    \getEntry*[person]{#2}{
3821     \theAcademicDegree = academicdegree,
3822     \theFirstName      = firstname,
3823     \theLastName       = lastname
3824    }%
3825    \IfBooleanT{#1}
3826    {
3827      \DTLifnullorempty{\theAcademicDegree}{}
3828        {\dfgShowTag{#2}{academicdegree}{\theAcademicDegree}[person]\space}
3829    }%
3830    \dfgShowTag{#2}{firstname}{\theFirstName}[person]\space
3831    \dfgShowTag{#2}{lastname}{\theLastName}[person]
3832    }
3833  }
```

\compareEntry P. 3 `\compareEntry` P. 3 `[`⟨*db*⟩`]{`⟨*id*⟩`}{`⟨*field*⟩`}{`⟨*value*⟩`}{`⟨*true*⟩`}{`⟨*false*⟩`}` reads ⟨*field*⟩ from the row with the given ⟨*id*⟩ and executes ⟨*yes*⟩ if it matches ⟨*value*⟩, otherwise ⟨*no*⟩. It is used throughout the tables to branch on e.g. project type or status without repeating the full `\DTLgetvalueforkey`/`\if` pattern.

The internal implementation `\dfg_compare_entry:nnnnnn` runs the same three guard checks as `\getEntry` P. 2 (empty DB name, non-existent DB), then uses `\exp_args:Nx` to fully expand `\thevalue` before the `\tl_if_eq:nnTF` test, ensuring that stored macro values are resolved correctly.

```
3834  \cs_new_protected:Npn \dfg_compare_entry:nnnnnn #1#2#3#4#5#6
3835  {
3836    \bool_if:NTF \l_@@_class_fastcompile_project_bool
3837    {#6}
3838    {
3839    \tl_if_blank:nTF {#1}
3840    { \msg_warning:nnn { dfg } { database-name-not-given } {#1} }
3841    {
3842     \DTLifdbexists{#1}
3843     {
3844      \DTLgetvalueforkey{\thevalue}{#3}{#1}{id}{#2}%
3845      \str_if_eq:eeTF { \thevalue } {#4}
3846      {#5}
3847      {#6}
3848     }
3849     { \msg_warning:nnn { dfg } { database-not-found } {#1} }
3850    }
3851    }
3852  }
3853
3854  \NewDocumentCommand \compareEntry { O{\tmpLocalDB} m m m m m }
3855  {
3856    \dfg_compare_entry:nnnnnn {#1}{#2}{#3}{#4}{#5}{#6}
3857  }
```

6.1.21 Table and list dispatch (**\printTable**, **\printList**)

Rather than requiring the user to call `\ProjectLeadersTable`, `\OverviewOfFundsRequestedForStaffTab` etc. directly, two thin dispatchers are provided: `\printTable[`⟨*opts*⟩`]{`⟨*name*⟩`}` and `\printList[`⟨*opts*⟩`]{`⟨*name*⟩`}`.

They synthesise the command name by appending `Table` or `List` to the ⟨*name*⟩ argument (using `\cs:w`…`\cs_end:`), then pass the optional argument through if present. This pattern means new tables can be added to the class without any change to user-facing call sites.

All three commands check `fastcompile` first and emit a `\dfgItemSkipped` placeholder instead of running the real command when compilation speed is prioritised. Focused project mode makes narrow exceptions when `fastcompile-project-show-funding-tables=true`: the three project-local funding tables are rendered from their retained local rows; when `fastcompile-project-show-images=true`, plots are rendered from retained project metadata. During real rendering the commands also establish the compact former-project context: reference-only projects receive an asterisk, and a local legend is added only if at least one such project was encountered. A `\rowNumber` counter is reset before each table for use in zebra-stripe logic inside individual table implementations.

```
3858
```

```
\tl_new:N \l_tblr_body_tl                % token list that will hold the rows

\bool_new:N \l_@@_former_project_compact_bool
\bool_new:N \g_@@_former_project_seen_bool

\cs_new_protected:Npn \@@_former_project_note:
{
  \smallskip
  \noindent
  {\normalfont\normalcolor\footnotesize
    \textsuperscript{*}\,Former~project}\par
}

\cs_new_protected:Npn \@@_former_project_context:n #1
{
  \bool_set_true:N \l_@@_former_project_compact_bool
  % Table cells and list implementations introduce their own groups, so the
  % occurrence flag must be latched globally for the duration of this context.
  \bool_gset_false:N \g_@@_former_project_seen_bool
  #1
  \bool_if:NT \g_@@_former_project_seen_bool
    { \@@_former_project_note: }
  \bool_set_false:N \l_@@_former_project_compact_bool
  \bool_gset_false:N \g_@@_former_project_seen_bool
}

\cs_new_protected:Npn \@@_print_thing:nnnn #1#2#3#4 {
  \bool_if:NTF \l_@@_class_fastcompile_bool
  { \@@_print_thing_skipped:nn {#1} {#3} }
  {%
    \@@_former_project_context:n
    {
      #4%
      \StartTimer
      \@@_print_thing_call:nnn {#1} {#2} {#3}%
      \StopTimer { \@@_print_thing_label:n {#3} }%
    }
  }
}

\cs_new_protected:Npn \@@_print_thing_call:nnn #1#2#3 {
  \tl_if_head_is_group:nTF {#3}
  { \@@_print_thing_split:nnw {#1} {#2} #3 \empty \q_stop }
  { \cs:w #3 #1 \cs_end: #2 }
}

\cs_new_protected:Npn \@@_print_thing_split:nnw #1#2#3#4 \q_stop {
  \cs:w #3 #1 \cs_end: #2 #4
}

\cs_new:Npn \@@_print_thing_label:n #1 {
  \tl_if_head_is_group:nTF {#1}
  { \@@_print_thing_label_aux:w #1 \empty \q_stop }
  { #1 }
}

\cs_new:Npn \@@_print_thing_label_aux:w #1#2 \q_stop {
  #1 \tl_if_blank:nF {#2} { - \@@_print_thing_label_tail:n {#2} }
}

\cs_new:Npn \@@_print_thing_label_tail:n #1 {
  \tl_if_head_eq_charcode:nNTF {#1} [
    { \@@_print_thing_label_skip_optional:w #1 \q_stop }
    { \tl_head:n {#1} }
}

\cs_new:Npn \@@_print_thing_label_skip_optional:w [#1]#2 \q_stop {
  \tl_head:n {#2}
}

\cs_new_protected:Npn \@@_print_thing_skipped:nn #1#2 {
  \dfgItemSkipped { \@@_print_thing_label:n {#2} #1 }
}

\cs_new:Npn \@@_print_table_name:n #1
{
  \tl_if_head_is_group:nTF {#1}
    { \@@_print_table_name_aux:w #1 \q_stop }
    {#1}
}

\cs_new:Npn \@@_print_table_name_aux:w #1#2 \q_stop {#1}

\prg_new_protected_conditional:Npnn
  \@@_print_table_if_fast_project_funding:n #1 { TF }
{
  \bool_lazy_all:nTF
```

```
    {
      { \l_@@_class_fastcompile_project_bool }
      { \l_@@_class_fastcompile_project_show_funding_tables_bool }
    }
    {
      \str_case_e:nnF { \@@_print_table_name:n {#1} }
      {
        {RequestedFunding} { \prg_return_true: }
        {RequestedFundingStaff} { \prg_return_true: }
        {RequestedAndExistingFundingDirectCosts} { \prg_return_true: }
      }
      { \prg_return_false: }
    }
    { \prg_return_false: }
  }

  \cs_new_protected:Npn \@@_print_table_dispatch:nn #1#2
  {
    \@@_print_table_if_fast_project_funding:nTF {#2}
    {
      \@@_former_project_context:n
      {
        \setcounter{rowNumber}{0}
        \@@_print_thing_call:nnn {Table} {#1} {#2}
      }
    }
    { \@@_print_thing:nnnn {Table} {#1} {#2} {\setcounter{rowNumber}{0}} }
  }

  \NewDocumentCommand{\printTable}{ o +m }{%
    % Finish the preceding paragraph before a table theme changes the font.
    \par
    % Anchor the table after a heading before tabularray calculates page space.
    % Otherwise its page-start topskip correction can overlap the heading.
    \if@nobreak
      \null\nobreak
    \fi
    \IfNoValueTF {#1}
      { \@@_print_table_dispatch:nn { } {#2} }
      { \@@_print_table_dispatch:nn { [#1] } {#2} }
    \par
  }

  % Person-specific institution lists also run inside tables and PI cards.
  % Call their formatter directly so a nested list does not reset the outer
  % former-project context; this also preserves its focused-build output.
  \cs_new_protected:Npn \@@_print_list_dispatch:nn #1#2
  {
    \str_if_eq:eeTF { \@@_print_table_name:n {#2} } { Institutions }
      { \@@_print_thing_call:nnn { List } {#1} {#2} }
      { \@@_print_thing:nnnn { List } {#1} {#2} { } }
  }

  \NewDocumentCommand{\printList}{ o +m o }{%
    % Accept old trailing options, but let the leading options take precedence.
    \IfNoValueTF {#3}
    {
      \IfNoValueTF {#1}
        { \@@_print_list_dispatch:nn { } {#2} }
        { \@@_print_list_dispatch:nn { [#1] } {#2} }
    }
    {
      \IfNoValueTF {#1}
        { \@@_print_list_dispatch:nn { [#3] } {#2} }
        { \@@_print_list_dispatch:nn { [#3,#1] } {#2} }
    }
  }

  % Database-aware list commands share one leading option list. Bare existing
  % database names remain valid for compatibility with the previous interface.
  \tl_new:N \l_@@_list_database_tl
  \tl_new:N \l_@@_list_options_tl
  \keys_define:nn { @@ / list-options }
  {
    database .tl_set:N = \l_@@_list_database_tl,
  }
  \cs_new_protected:Npn \@@_list_options:n #1
  {
    \tl_set:Nn \l_@@_list_database_tl {person}
    \tl_clear:N \l_@@_list_options_tl
    \tl_if_blank:nF {#1}
    {
      \DTLifdbexists{#1}
        { \tl_set:Nn \l_@@_list_database_tl {#1} }
        {
          \keys_set_known:nnN { @@ / list-options } {#1}
            \l_@@_list_options_tl
```

```
      }
    }
  }

  \cs_new_protected:Npn \@@_print_plot_dispatch:nn #1#2
  {
    \bool_lazy_all:nTF
    {
      { \l_@@_class_fastcompile_project_bool }
      { \l_@@_class_fastcompile_project_show_images_bool }
    }
    {
      \@@_former_project_context:n
      { \@@_print_thing_call:nnn {Plot} {#1} {#2} }
    }
    { \@@_print_thing:nnnn {Plot} {#1} {#2} { } }
  }

  \NewDocumentCommand{\printPlot}{ o +m }{%
    \IfNoValueTF {#1}
    { \@@_print_plot_dispatch:nn { } {#2} }
    { \@@_print_plot_dispatch:nn { [#1] } {#2} }%
  }

  \newcounter{rowNumber}
```

6.1.22 Data tables

All tables follow the same three-phase pattern: **Body accumulation:** A `\DTLforeach*` loop iterates over the relevant database (filtered and sorted as required) and appends one tblr row to the token list `\l_tblr_body_tl` via `\tl_gput_right:Nx`. Using a global token list and `:Nx` (full expansion) ensures that database values are resolved at loop time rather than at tblr rendering time. **Table opening:** `\begin{tblr}` with the `main` theme, `expand=\l_tblr_body_tl`, `long` (for page-breaking), and a column spec tailored to the specific table. **Header row + body:** A literal first row provides column headings; `\l_tblr_body_tl` follows as the data rows.

ProjectLeadersTable Lists all PIs who are active in the current funding period (`\theFundingPeriodNumeral`), sorted by last name. The filter applies two conditions simultaneously: `period`$^{\rightarrow \text{P. } 64}$ must contain the current numeral and `role`$^{\rightarrow \text{P. } 64}$ must contain `PI`. The first cell of every row defines the hyperlink destination `pi:<id>` used by `\getPI`.

```

  % Use the same institution formatter as the CV section.  The public
  % \InstitutionsList maps institution ids through \_@@_one_institution:n
  % and uses \par\medskip between multiple institutions in the standard layout.
  % Keep the institution label in sync with the ProjectsTable display setting.
  \NewDocumentCommand{\dfgPLInstitutionList}{m}{%
    \bool_if:NTF \l_@@_custom_projects_table_institution_abbreviations_bool
      { \printList[abbreviation]{{Institutions}{#1}} }
      { \printList{{Institutions}{#1}} }
  }

  \int_new:N \l_@@_pl_iter_int

  % Keep the hyperref command protected while tabularray expands the assembled
  % table body.  It is executed only when the corresponding name cell is typeset.
  \cs_new_protected:Npn \@@_pi_target:n #1
    { \hypertarget{pi:#1}{} }

  \NewDocumentCommand{\ProjectLeadersTable}{}{%
    \def\tblrbody{}%

    \StartTimer
    \DTLsort*{lastname}{person}%
    \StopTimer{ProjectLeaderTable-sort}

    \int_zero:N \l_@@_pl_iter_int

    \StartTimer
    \DTLforeach*[
     \DTLisinlist{\theFundingPeriodNumeral}{\period}
     \and
     \DTLisinlist{PI}{\role}
    ]{person}{
     \id              = id,
     \role            = role,
     \lastname        = lastname,
     \firstname       = firstname,
```

```
    \academicdegree  = academicdegree,
    \period          = period,
    \refProjectId    = ref-project-id,
    \refInstitutionId = ref-institution-id,
    \doctorateYear   = doctorateYear,
    \gender          = gender
   }{%
    \int_incr:N \l_@@_pl_iter_int
    \protected@edef\tblrbody{\tblrbody
   {\noexpand\@@_pi_target:n{\id}%
    {\noexpand{\color{referencePI}\bfseries
         \noexpand\dfgShowTag{\id}{lastname}{\lastname}[person]},~
         \noexpand\dfgShowTag{\id}{firstname}{\firstname}[person]}
         \noexpand\\
         \noexpand\dfgShowTag{\id}{academicdegree}{\academicdegree}[person]}
       \noexpand& \noexpand\dfgShowTag{\id}{gender}{\gender}[person]
       \noexpand& \noexpand\dfgShowTag{\id}{doctorateYear}{\doctorateYear}[person]
     \noexpand& {\noexpand\dfgPLInstitutionList{\id}}
     \noexpand& {\noexpand\getProject*{\refProjectId}[\\]}
     \noexpand\\
    }%
   }
   \StopTimer{PL-loop}
   \iow_term:x { PL-iterations:~ \int_use:N \l_@@_pl_iter_int }

   \StartTimer
   \begin{tblr}[
     theme  = main,
     expand = \tblrbody,
     long
    ]{
     colspec   = {t{3.5cm}@{}t{.7cm}t{1.6cm}X[1.2,t]t{1.3cm}},
     column{1} = { fg = referencePI, halign = l, valign = m },
     column{2} = { halign = c, valign = m },
     column{3} = { halign = c, valign = m },
     column{4} = { halign = l, valign = m },
     column{5} = { halign = c, valign = m },
     rowhead   = 1,
     row{1}    = {
                  bg     = tableHeadingsBG,
                  fg     = tableHeadingsFont,
                  font   = \bfseries,
                  guard
                 },
    }
    Project~ leader
     & \mbox{Gender}
     & {Doctorate          \\ obtained~ in}
     & {Home~ institution, \\ location}
     & Project(s)                          \\
    \tblrbody
   \end{tblr}
   \StopTimer{PL-render}
 }
```

After the list of the PIs we need to have an aggregated list of the participating institutions.

```
\prop_new:N \l_@@_inst_to_aff_prop
\prop_new:N \l_@@_inst_name_prop
\prop_new:N \l_@@_inst_seen_prop
\prop_new:N \l_@@_aff_seen_prop
\prop_new:N \l_@@_aff_to_insts_prop  % values are seq names (tl)
\prop_new:N \l_@@_aff_name_prop
\seq_new:N  \l_@@_affiliations_seq
\seq_new:N  \l_@@_aff_sort_seq
\seq_new:N  \l_@@_inst_sort_seq
\tl_new:N   \l_@@_output_tl
\tl_new:N   \l_@@_aff_label_tl
\int_new:N  \l_@@_seq_counter_int
\cs_generate_variant:Nn \seq_set_from_clist:Nn { Ne , NV }

% Append an institution name to the per-affiliation seq.
% Lazily creates the seq on first use and stores its name in the prop.
\cs_new_protected:Npn \@@_append_inst:nn #1#2 {
  \prop_get:NnNTF \l_@@_aff_to_insts_prop {#1} \l_tmpd_tl
  {%
   % \l_tmpd_tl now holds the seq's csname; append to it
   \seq_put_right:cn { \l_tmpd_tl } {#2}%
  }
  {%
   \int_incr:N \l_@@_seq_counter_int
   \tl_set:Nx \l_tmpd_tl
   { l_@@_inst_seq_ \int_use:N \l_@@_seq_counter_int _seq }%
    \seq_clear_new:c { \l_tmpd_tl }%
   \seq_put_right:cn { \l_tmpd_tl } {#2}%
   \prop_put:NnV \l_@@_aff_to_insts_prop {#1} \l_tmpd_tl
  }
```

```
}
\cs_generate_variant:Nn \@@_append_inst:nn { VV }

\cs_new_protected:Npn \@@_collect_pi_institution:n #1
{
  \tl_if_blank:nF {#1}
  {
   \prop_if_in:NnF \l_@@_inst_seen_prop {#1}
   {%
    \prop_put:Nnn \l_@@_inst_seen_prop {#1} {1}%
    \prop_get:NnNT \l_@@_inst_to_aff_prop {#1} \l_tmpb_tl
    {%
     \prop_get:NnNT \l_@@_inst_name_prop {#1} \l_tmpc_tl
        {
          \tl_set:Ne \l_tmpe_tl
          {
            \exp_not:N \dfgShowTag {#1}{name}
              {\exp_not:V \l_tmpc_tl}[institution]
          }
          \@@_append_inst:VV \l_tmpb_tl \l_tmpe_tl
        }%
     \prop_if_in:NVF \l_@@_aff_seen_prop \l_tmpb_tl
     {%
      \prop_put:NVn \l_@@_aff_seen_prop \l_tmpb_tl {1}%
      \seq_put_right:NV \l_@@_affiliations_seq \l_tmpb_tl
     }%
    }%
   }%
  }
}

\int_new:N \l_@@_participating_columns_int
\dim_new:N \l_@@_participating_column_sep_dim
\tl_new:N \l_@@_participating_order_tl
\tl_new:N \l_@@_participating_raster_options_tl
\keys_define:nn { @@ / participating-institutions }
{
  columns .int_set:N = \l_@@_participating_columns_int,
  columns .value_required:n = true,
  column~sep .dim_set:N = \l_@@_participating_column_sep_dim,
  column~sep .value_required:n = true,
  order .choice:,
  order / alpha .code:n =
    { \tl_set:Nn \l_@@_participating_order_tl {alpha} },
  order / count .code:n =
    { \tl_set:Nn \l_@@_participating_order_tl {count} },
  order / unknown .code:n =
    {
      \msg_warning:nnn { dfg } { unknown-institution-order } {#1}
      \tl_set:Nn \l_@@_participating_order_tl {alpha}
    },
  alpha .meta:n = { order=alpha },
  count .meta:n = { order=count },
}

\cs_new_protected:Npn \@@_participating_institutions_raster:n #1
{
  \begin{tcbraster}[
    raster~ columns=1,
    raster~ equal~ height=rows,
    raster~ column~ skip=12pt,
    raster~ row~ skip=10pt,
    #1]
    \tl_use:N \l_@@_output_tl
  \end{tcbraster}
}

% Flow whole affiliation boxes down balanced columns without equal-height rows.
\cs_new_protected:Npn \@@_participating_institutions_columns:
{
  \par\addvspace{\medskipamount}
  \setlength{\columnsep}{\l_@@_participating_column_sep_dim}
  \setlength{\multicolsep}{0pt}
  \int_compare:nNnTF {\l_@@_participating_columns_int} > {1}
  {
    \begin{multicols}{\int_use:N \l_@@_participating_columns_int}
      \raggedcolumns
      \tl_use:N \l_@@_output_tl
    \end{multicols}
  }
  { \tl_use:N \l_@@_output_tl }
}

\NewDocumentCommand{\ParticipatingInstitutionsList}{ O{count} }
{%
  \group_begin:
  % Zero selects the existing raster; columns=1,2,... selects flowing columns.
```

```
\int_zero:N \l_@@_participating_columns_int
\dim_set:Nn \l_@@_participating_column_sep_dim {12pt}
% Accept the legacy alpha/count options as well as order and raster keys.
\tl_set:Nn \l_@@_participating_order_tl {count}
\keys_set_known:nnN { @@ / participating-institutions } {#1}
  \l_@@_participating_raster_options_tl
\prop_clear:N \l_@@_inst_to_aff_prop
\prop_clear:N \l_@@_inst_name_prop
\prop_clear:N \l_@@_inst_seen_prop
\prop_clear:N \l_@@_aff_seen_prop
\prop_clear:N \l_@@_aff_to_insts_prop
\seq_clear:N  \l_@@_affiliations_seq
\tl_clear:N   \l_@@_output_tl
\int_zero:N   \l_@@_seq_counter_int

% --- 1. One pass over institution: id -> aff-id, id -> name ---------
\DTLforeach*{institution}{
 \l_tmpa_tl = id,
 \l_tmpb_tl = ref-affiliation-id,
 \l_tmpc_tl = name
}{%
 \prop_put:NVV \l_@@_inst_to_aff_prop \l_tmpa_tl \l_tmpb_tl
 \prop_put:NVV \l_@@_inst_name_prop   \l_tmpa_tl \l_tmpc_tl
}

% --- 2. One pass over person: collect active PI institutions --------
\DTLforeach*[
 \DTLisinlist{\theFundingPeriodNumeral}{\pPeriod} \and
 \DTLisinlist{PI}{\pRole}
]{person}{ \l_tmpa_tl = ref-institution-id ,\pPeriod=period,\pRole=role}{%
 \seq_set_from_clist:NV \l_tmpa_seq \l_tmpa_tl
 \seq_map_inline:Nn \l_tmpa_seq
 { \@@_collect_pi_institution:n {##1} }%
}

% --- 3a. Precompute affiliation display names -----------------------
\prop_clear:N \l_@@_aff_name_prop
\seq_map_inline:Nn \l_@@_affiliations_seq {%
 \getEntry*[affiliation]{##1}{ \l_tmpa_tl = name }%
 \prop_put:NnV \l_@@_aff_name_prop {##1} \l_tmpa_tl
}

% --- 3b. Sort affiliations according to requested order ------------
\str_case:VnF \l_@@_participating_order_tl
{
 { alpha }
 {%
  \seq_sort:Nn \l_@@_affiliations_seq
  {%
   \prop_get:NnN \l_@@_aff_name_prop {##1} \l_tmpa_tl
   \prop_get:NnN \l_@@_aff_name_prop {##2} \l_tmpb_tl
   \str_compare:eNeTF { \l_tmpa_tl } < { \l_tmpb_tl }
    { \sort_return_same: }
    { \sort_return_swapped: }%
  }%
 }
 { count }
 {%
  \seq_sort:Nn \l_@@_affiliations_seq
  {%
   \prop_get:NnN \l_@@_aff_to_insts_prop {##1} \l_tmpc_tl
   \prop_get:NnN \l_@@_aff_to_insts_prop {##2} \l_tmpd_tl
   \int_compare:nNnTF
   { \seq_count:c { \l_tmpc_tl } } > { \seq_count:c { \l_tmpd_tl } }
    { \sort_return_same: }
    {%
     \int_compare:nNnTF
     { \seq_count:c { \l_tmpc_tl } } < { \seq_count:c { \l_tmpd_tl } }
      { \sort_return_swapped: }
      {%
       % equal counts → alphabetical tiebreak
       \prop_get:NnN \l_@@_aff_name_prop {##1} \l_tmpa_tl
       \prop_get:NnN \l_@@_aff_name_prop {##2} \l_tmpb_tl
       \str_compare:eNeTF { \l_tmpa_tl } < { \l_tmpb_tl }
        { \sort_return_same: }
        { \sort_return_swapped: }%
      }%
    }%
  }%
 }
}
{%
  \msg_warning:nne { dfg } { unknown-institution-order }
  { \l_@@_participating_order_tl }%
 \seq_sort:Nn \l_@@_affiliations_seq
 {%
  \prop_get:NnN \l_@@_aff_name_prop {##1} \l_tmpa_tl
```

```
      \prop_get:NnN \l_@@_aff_name_prop {##2} \l_tmpb_tl
      \str_compare:eNeTF { \l_tmpa_tl } < { \l_tmpb_tl }
       { \sort_return_same: }
       { \sort_return_swapped: }%
     }%
    }

    % --- 4. Emit output ------------------------------------------------
    \seq_map_inline:Nn \l_@@_affiliations_seq {%
      \int_compare:nNnTF {\l_@@_participating_columns_int} > {0}
      {
        \tl_put_right:Nn \l_@@_output_tl
        {
          \begin{tcolorbox}[blankest,before~upper=\RaggedRight,
            before={\par\noindent},after={\par\addvspace{10pt}}]
        }
      }
      {
        % Preserve the raster's own before/after hooks for row placement.
        \tl_put_right:Nn \l_@@_output_tl
          {\begin{tcolorbox}[blankest,before~upper=\RaggedRight]}
      }%
     \getEntry*[affiliation]{##1}{
      \theAffName     = name,
      \theAffLocation = location
     }%
     \prop_get:NnN \l_@@_aff_to_insts_prop {##1} \l_tmpd_tl
     \seq_set_eq:Nc \l_@@_inst_sort_seq { \l_tmpd_tl }%
     \seq_sort:Nn \l_@@_inst_sort_seq
     {%
      \str_compare:nNnTF {####1} < {####2}
       { \sort_return_same: }
       { \sort_return_swapped: }%
     }%
     \tl_set:Nx \l_@@_aff_label_tl { \seq_use:Nn \l_@@_inst_sort_seq {\\} }%
     \tl_put_right:Nx \l_@@_output_tl {%
      { \exp_not:N \usekomafont{subsubsection}
          \exp_not:N \textbf{%
            \exp_not:N \dfgShowTag {##1}{name}
              {\exp_not:V \theAffName}[affiliation]}
          \exp_not:n {\space(}
          \exp_not:N \dfgShowTag {##1}{location}
            {\exp_not:V \theAffLocation}[affiliation]
          \exp_not:n {)\\} }%
      \exp_not:V \l_@@_aff_label_tl
     }%
      \tl_put_right:Nn \l_@@_output_tl {\end{tcolorbox}}%
    }
    \seq_if_empty:NF \l_@@_affiliations_seq
    {
      \int_compare:nNnTF {\l_@@_participating_columns_int} > {0}
        { \@@_participating_institutions_columns: }
        {
          \exp_args:NV \@@_participating_institutions_raster:n
            \l_@@_participating_raster_options_tl
        }
    }
    \group_end:
}
```

Very similar to the table for the PIs we need one table for the projects. First we start with a helper command to get for each referenced person the required data.

```
\NewDocumentCommand{\getPersonInformation}{O{person} m}
{%
  % #1: database
  % #2: comma-separated list of person ids
  \def\tmpLocalDB{#1}%
  \seq_clear:N \l_tmpa_seq
  \seq_set_from_clist:Ne \l_tmpa_seq {#2}%
  \bool_if:NTF \l_@@_project_one_PI_bool
  {%
   % Single-PI mode: no boxing, just emit each entry inline
   \bool_set_true:N \l_tmpa_bool
   \seq_map_inline:Nn \l_tmpa_seq
   {%
    \bool_if:NTF \l_tmpa_bool
    { \bool_set_false:N \l_tmpa_bool }
    { ~ }%
    \__getperson_one:n {##1}%
   }%
  }
  {%
   % Multi-PI mode: lay out as a 2-column raster
   \begin{tcbraster}[
```

```
            raster~ valign=top,
      raster~ columns=2,
      raster~ equal~ height=rows,
            raster~ column~ skip=12pt,
      raster~ row~ skip=10pt,
      PIbox
     ]
     \seq_map_inline:Nn \l_tmpa_seq
     {%
      \__getperson_one:n {##1}%
     }%
    \end{tcbraster}
   }%
}

\bool_new:N \l_@@_pi_academicdegree_below_bool

\cs_new_protected:Npn \__getperson_one:n #1
{%
  \getEntry*[\tmpLocalDB]{#1}{
   \theFirstName      = firstname,
   \theLastName       = lastname,
   \theAcademicDegree = academicdegree,
   \thePhone          = phone,
   \theEmail          = email,
    \theInstitution    = ref-institution-id,
   \theORCiD         = orcid
  }%
  % Match the tallest title in this raster; equal-height groups settle over
  % successive LaTeX runs. Keep title text at the top of the reserved space.
  \begin{tcolorbox}[PIbox,title={
      \tcbox[blankest,tcbox~ width=forced~ left,
        equal~ height~ group={dfg-PI-titles-\thetcbraster}]{%
        \parbox[t]{\linewidth}{%
          \RaggedRight\bfseries
          \dfgShowTag{#1}{firstname}{\theFirstName}[\tmpLocalDB]\space
          \color{referencePI}
          \dfgShowTag{#1}{lastname}{\theLastName}[\tmpLocalDB]
          \normalfont\normalcolor
          \bool_if:NTF \l_@@_pi_academicdegree_below_bool
            {\newline}
            {,\space}
          \dfgShowTag{#1}{academicdegree}{\theAcademicDegree}[\tmpLocalDB]}}}]
   \ifdefempty{\theORCiD}{\mbox{~}\\[.75\baselineskip]}{\getORCiD{#1}\\}
    \ifdefempty{\theInstitution}{}{\printList{{Institutions}{#1}}\newline}
    \ifdefempty{\thePhone}{}
      {\str_case:Vn \l_@@_custom_pi_contact_style_tl
        {
          {icons}
            {\mbox{\faPhone\space
             \dfgShowTag{#1}{phone}{\thePhone}[\tmpLocalDB]}\quad}
          {text}
            {\mbox{Phone:\space
             \dfgShowTag{#1}{phone}{\thePhone}[\tmpLocalDB]}\quad}
          {none}
            {\mbox{\dfgShowTag{#1}{phone}{\thePhone}[\tmpLocalDB]}\quad}
        }}
    \ifdefempty{\theEmail}{}
      {\str_case:Vn \l_@@_custom_pi_contact_style_tl
        {
          {icons}
            {\mbox{\faEnvelope\space
              \dfgShowTag{#1}{email}{\theEmail}[\tmpLocalDB]}}
          {text}
            {\mbox{E-Mail:\space
              \dfgShowTag{#1}{email}{\theEmail}[\tmpLocalDB]}}
          {none}
            {\mbox{\dfgShowTag{#1}{email}{\theEmail}[\tmpLocalDB]}}
        }}
  \end{tcolorbox}
}

% ============================================================
%  Precompute compact project-table data per person
% ============================================================

% --- Shared allocations outside each function ----------------------
\prop_new:N \g_@@_proj_pi_prop
\bool_new:N \g_@@_proj_pi_built_bool
\prop_new:N \g_@@_aff_narrow_prop
\prop_new:N \g_@@_inst_narrow_prop
\prop_new:N \g_@@_proj_pi_institution_ids_prop
\prop_new:N \g_@@_projects_table_institution_abbreviation_prop
\prop_new:N \g_@@_projects_table_institution_name_prop
\prop_new:N \g_@@_projects_table_abbreviation_owner_id_prop
\prop_new:N \g_@@_projects_table_abbreviation_owner_name_prop
\prop_new:N \g_@@_projects_table_abbreviation_warned_prop
```

```
\tl_new:N \l_@@_projects_table_legend_tl
\seq_new:N \l_@@_projects_table_institution_ids_seq
\seq_new:N \l_@@_projects_table_sorted_institution_ids_seq
\clist_new:N \l_@@_projects_table_institution_ids_clist
\tl_new:N \l_@@_projects_table_institution_abbreviation_tl

\cs_new_protected:Npn \@@_projects_table_collect_person_institutions:n #1
{
  \prop_get:NnNT \g_@@_proj_pi_institution_ids_prop {#1} \l_tmpa_tl
  {
    \clist_set:NV \l_@@_projects_table_institution_ids_clist \l_tmpa_tl
    \clist_map_inline:Nn \l_@@_projects_table_institution_ids_clist
    {
      \seq_if_in:NnF \l_@@_projects_table_institution_ids_seq {##1}
        { \seq_put_right:Nn \l_@@_projects_table_institution_ids_seq {##1} }
    }
  }
}

\cs_new_protected:Npn \@@_projects_table_collect_institutions:n #1
{
  \bool_if:NT \l_@@_custom_projects_table_institution_abbreviations_bool
  {
    \clist_map_function:nN {#1}
      \@@_projects_table_collect_person_institutions:n
  }
}

\cs_new_protected:Npn \@@_projects_table_institution_legend:
{
  \bool_if:NT \l_@@_custom_projects_table_institution_abbreviations_bool
  {
    \tl_clear:N \l_@@_projects_table_legend_tl
    \seq_set_eq:NN \l_@@_projects_table_sorted_institution_ids_seq
      \l_@@_projects_table_institution_ids_seq
    \seq_sort:Nn \l_@@_projects_table_sorted_institution_ids_seq
    {
      \prop_get:NnN \g_@@_projects_table_institution_abbreviation_prop
        {##1} \l_tmpa_tl
      \prop_get:NnN \g_@@_projects_table_institution_abbreviation_prop
        {##2} \l_tmpb_tl
      \str_compare:eNeTF
        { \str_casefold:V \l_tmpa_tl }
        >
        { \str_casefold:V \l_tmpb_tl }
        { \sort_return_swapped: }
        { \sort_return_same: }
    }
    \seq_map_inline:Nn \l_@@_projects_table_sorted_institution_ids_seq
    {
      \prop_get:NnN \g_@@_projects_table_institution_abbreviation_prop
        {##1} \l_tmpa_tl
      \prop_get:NnN \g_@@_projects_table_institution_name_prop
        {##1} \l_tmpb_tl
      \DTLifnullorempty{\l_tmpa_tl}{}
      {
        \tl_if_blank:VF \l_@@_projects_table_legend_tl
          { \tl_put_right:Nn \l_@@_projects_table_legend_tl { ;~ } }
        \tl_put_right:Nx \l_@@_projects_table_legend_tl
        {
          \exp_not:N \textbf { \exp_not:V \l_tmpa_tl }
          \c_space_tl = \c_space_tl
          \exp_not:V \l_tmpb_tl
        }
      }
    }
    \tl_if_blank:VF \l_@@_projects_table_legend_tl
    {
      \par\vspace{.15\baselineskip}
      {\tiny\RaggedRight\setlength{\parskip}{0pt}\noindent
        \textit{Institution~abbreviations:}\space
        \tl_use:N \l_@@_projects_table_legend_tl
        \par
      }
    }
  }
}


% Helper for render-time formatting
\NewDocumentCommand{\dfgPIformat}{mmm}{%
  % #1 = person id, #2 = firstname, #3 = lastname
  \dfgShowTag{#1}{firstname}{#2}[person]\space
  {\color{referencePI}\bfseries
    \dfgShowTag{#1}{lastname}{#3}[person],}\space
}

```

```
% Precompute once: store only the tokens, not \color & \bfseries directly
\cs_new_protected:Npn \@@_proj_pi_build: {
  \bool_if:NF \g_@@_proj_pi_built_bool {
   \bool_gset_true:N \g_@@_proj_pi_built_bool

   % Pass 1+2: precompute institution strings (as before)
   \prop_new:N \l_@@_aff_narrow_prop
   \DTLforeach*{affiliation}{
    \l_tmpa_tl = id,
    \l_tmpb_tl = name
   }{%
    \prop_put:NVV \l_@@_aff_narrow_prop \l_tmpa_tl \l_tmpb_tl
   }

  \prop_new:N \l_@@_inst_narrow_prop
  \DTLforeach*{institution}{
    \l_tmpa_tl = id,
    \l_tmpb_tl = name,
  \l_tmpc_tl = ref-affiliation-id,
      \l_@@_projects_table_institution_abbreviation_tl = abbreviation
   }{%
  \prop_gput:NVV \g_@@_projects_table_institution_abbreviation_prop
      \l_tmpa_tl \l_@@_projects_table_institution_abbreviation_tl
  \prop_gput:NVV \g_@@_projects_table_institution_name_prop
      \l_tmpa_tl \l_tmpb_tl
  \DTLifnulloremptyf{\l_@@_projects_table_institution_abbreviation_tl}{}
    {
      \prop_get:NVNTF \g_@@_projects_table_abbreviation_owner_name_prop
        \l_@@_projects_table_institution_abbreviation_tl \l_tmpe_tl
      {
        \tl_if_eq:NNF \l_tmpe_tl \l_tmpb_tl
        {
          \prop_if_in:NVF \g_@@_projects_table_abbreviation_warned_prop
            \l_@@_projects_table_institution_abbreviation_tl
          {
            \prop_get:NVN \g_@@_projects_table_abbreviation_owner_id_prop
              \l_@@_projects_table_institution_abbreviation_tl \l_tmpf_tl
            \msg_warning:nneeee { dfg } { duplicate-institution-abbreviation }
              { \l_@@_projects_table_institution_abbreviation_tl }
              { \l_tmpe_tl }
              { \l_tmpb_tl }
              { \l_tmpf_tl,~\l_tmpa_tl }
            \prop_gput:NVn \g_@@_projects_table_abbreviation_warned_prop
              \l_@@_projects_table_institution_abbreviation_tl { 1 }
          }
        }
      }
      {
        \prop_gput:NVV \g_@@_projects_table_abbreviation_owner_id_prop
          \l_@@_projects_table_institution_abbreviation_tl \l_tmpa_tl
        \prop_gput:NVV \g_@@_projects_table_abbreviation_owner_name_prop
          \l_@@_projects_table_institution_abbreviation_tl \l_tmpb_tl
      }
    }
  \bool_if:NTF
    \l_@@_custom_projects_table_institution_abbreviations_bool
  {
    \@@_institution_abbreviation_or_name:nN
      { \l_tmpa_tl } \l_tmpf_tl
  }
  {
    \tl_set:Nn \l_@@_institution_label_field_tl { name }
    \DTLgetvalueforkey{\l_tmpf_tl}{name}{institution}{id}{\l_tmpa_tl}
  }
    \prop_get:NVNT \l_@@_aff_narrow_prop \l_tmpc_tl \l_tmpd_tl {
          \tl_if_blank:VTF \l_tmpf_tl
          {
            \tl_set:Nx \l_tmpe_tl
            {
              \exp_not:N \dfgShowTag
                {\exp_not:V \l_tmpc_tl}{name}
                {\exp_not:V \l_tmpd_tl}[affiliation]
            }
          }
          {
            \tl_set:Nx \l_tmpe_tl
            {
              \exp_not:N \dfgShowTag
                {\exp_not:V \l_tmpa_tl}
                {\exp_not:V \l_@@_institution_label_field_tl}
                {\exp_not:V \l_tmpf_tl}[institution]
              ,~
              \exp_not:N \dfgShowTag
                {\exp_not:V \l_tmpc_tl}{name}
                {\exp_not:V \l_tmpd_tl}[affiliation]
            }
          }
```

```
      \prop_put:NVV \l_@@_inst_narrow_prop \l_tmpa_tl \l_tmpe_tl
     }
    }

    % Pass 3: per Person die ganze Zelle vorberechnen, mit \dfgPIformat
    \DTLforeach*{person}{
     \l_tmpa_tl = id,
     \l_tmpb_tl = firstname,
     \l_tmpc_tl = lastname,
     \l_tmpd_tl = ref-institution-id
    }{%
   \prop_gput:NVV \g_@@_proj_pi_institution_ids_prop
       \l_tmpa_tl \l_tmpd_tl
     \tl_clear:N \l_tmpe_tl
     \tl_put_right:Nx \l_tmpe_tl {%
      \exp_not:N \dfgPIformat
         { \exp_not:V \l_tmpa_tl }
      { \exp_not:V \l_tmpb_tl }
      { \exp_not:V \l_tmpc_tl }%
     }%
     % Institutions, separator ";~"
     \seq_set_from_clist:NV \l_tmpa_seq \l_tmpd_tl
     \bool_set_true:N \l_tmpa_bool
     \seq_map_inline:Nn \l_tmpa_seq {%
      \bool_if:NTF \l_tmpa_bool
      { \bool_set_false:N \l_tmpa_bool }
      { \tl_put_right:Nn \l_tmpe_tl { ;~ } }%
      \prop_get:NnNT \l_@@_inst_narrow_prop {##1} \l_tmpf_tl
      { \tl_put_right:NV \l_tmpe_tl \l_tmpf_tl }%
     }
     \prop_gput:NVV \g_@@_proj_pi_prop \l_tmpa_tl \l_tmpe_tl
    }
   }
}

% Schneller Drop-in
\NewDocumentCommand{\dfgFastPIinProjectsTable}{m}{%
  \@@_proj_pi_build:
  \seq_set_from_clist:Nn \l_tmpa_seq {#1}%
  \bool_set_true:N \l_tmpa_bool
  \seq_map_inline:Nn \l_tmpa_seq {%
   \bool_if:NTF \l_tmpa_bool
   { \bool_set_false:N \l_tmpa_bool }
   { \newline }%
   \prop_get:NnNT \g_@@_proj_pi_prop {##1} \l_tmpb_tl
   { \tl_use:N \l_tmpb_tl }%
  }
}

\NewDocumentCommand{\dfgPIinProjectsTable}{O{person} m}
{%
  % #1: database
  % #2: comma-separated list of person ids
  \def\tmpLocalDB{#1}%
  \seq_clear:N \l_tmpa_seq
  \seq_set_from_clist:Nn \l_tmpa_seq {#2}%
  \bool_set_true:N \l_tmpa_bool
  \seq_map_inline:Nn \l_tmpa_seq
  {%
   \bool_if:NTF \l_tmpa_bool
   { \bool_set_false:N \l_tmpa_bool }
   { \newline }%
   \__pi_in_projects_one:n {##1}%
  }%
}

\cs_new_protected:Npn \__pi_in_projects_one:n #1
{%
  \getEntry*[\tmpLocalDB]{#1}{
   \theAcademicDegree = academicdegree,
   \theFirstName = firstname,
   \theLastName  = lastname
  }%
  \group_begin:
  %%\theAcademicDegree\space%
  \dfgShowTag{#1}{firstname}{\theFirstName}[\tmpLocalDB]\space
  \color{referencePI}\bfseries
  \dfgShowTag{#1}{lastname}{\theLastName}[\tmpLocalDB],\space
  \normalfont\normalcolor
  \printList[abbreviation,noaddress,nolocation,nozip,narrow]{{Institutions}{#1}}%
  \group_end:
}
\NewDocumentCommand{\ProjectsTable}{mm}
{%
  \def\tblrbody{}%
  \clist_clear:N \l_@@_research_area_clist
  \tl_clear:N    \l_tmpa_tl
```

```
4783    \clist_clear:N \l_tmpa_clist
4784    \seq_clear:N   \l_tmpb_seq
4785    \seq_clear:N   \l_tmpa_seq
4786    \seq_clear:N   \l_@@_projects_table_institution_ids_seq
4787    \@@_proj_pi_build:
4788
4789    \StartTimer
4790    \DTLsort{number}{project}%
4791    \StopTimer{Proj-#1-sort}
4792
4793    \int_zero:N \l_tmpa_int
4794
4795    \StartTimer
4796    \DTLforeach*[
4797     \DTLisPrefix{\pNumber}{#1}
4798      \and
4799      \not\DTLiseq{\pConsiderForLists}{false}
4800    ]{project}{
4801     \pID          = id,
4802     \pNumber      = number,
4803     \pName        = name,
4804     \pStatus      = project-status,
4805     \pType        = project-type,
4806      \pConsiderForLists = consider-for-lists,
4807     \l_tmpa_clist = research-area,
4808     \pRefPersonId = ref-person-id
4809    }{%
4810     \int_incr:N \l_tmpa_int
4811      \exp_args:NV \@@_projects_table_collect_institutions:n \pRefPersonId
4812     % --- reset ALL flags each iteration --------------------------------
4813     \bool_set_false:N \l_@@_project_type_research_bool
4814     \bool_set_false:N \l_@@_project_type_service_bool
4815     \bool_set_false:N \l_@@_project_type_mgk_bool
4816     \bool_set_false:N \l_@@_project_type_admin_bool
4817
4818     \bool_set_false:N \l_@@_project_status_N_bool
4819     \bool_set_false:N \l_@@_project_status_E_bool
4820     \bool_set_false:N \l_@@_project_status_C_bool
4821
4822
4823     \str_case:Vn {\pStatus}
4824     {
4825      {N}{ \bool_set_true:N \l_@@_project_status_N_bool }
4826      {E}{ \bool_set_true:N \l_@@_project_status_E_bool }
4827      {C}{ \bool_set_true:N \l_@@_project_status_C_bool }
4828     }
4829
4830     \str_case:Vn {\pType}{
4831      {research}{ \bool_set_true:N \l_@@_project_type_research_bool }
4832      {service} { \bool_set_true:N \l_@@_project_type_service_bool  }
4833      {mgk}     { \bool_set_true:N \l_@@_project_type_mgk_bool      }
4834      {admin}   { \bool_set_true:N \l_@@_project_type_admin_bool    }
4835     }
4836
4837     \bool_if:nT
4838     { ( \l_@@_project_type_research_bool || \l_@@_project_type_service_bool ) && ! \l_@@_project_status_E_bool }
4839     {%
4840      \clist_set:NV \l_tmpa_clist \l_tmpa_clist
4841      \clist_sort:Nn \l_tmpa_clist
4842      { \fp_compare:nTF {##1 < ##2} {\sort_return_swapped:} {\sort_return_same:} }%
4843      \seq_set_from_clist:NN \l_tmpa_seq \l_tmpa_clist
4844      \seq_set_map:NNn \l_tmpb_seq \l_tmpa_seq
4845      {%
4846       \prop_if_in:NnTF \g_@@_dfg_research_areas_prop {##1}
4847       { ##1 }
4848       { \textcolor{error}{\textbf{##1}} }%
4849      }%
4850     }
4851
4852     \protected@edef\tblrbody{\tblrbody
4853        \noexpand\getProject{\pID}
4854        & \bool_if:NF \l_@@_project_status_C_bool
4855            { \noexpand\dfgShowTag{\pID}{project-status}{\pStatus}[project] }
4856        & \noexpand\dfgShowTag{\pID}{name}{\pName}[project]
4857      &
4858      \bool_if:nT
4859      { ( \l_@@_project_type_research_bool || \l_@@_project_type_service_bool ) && ! \l_@@_project_status_E_bool }
4860        {{\noexpand\dfgShowTag{\pID}{research-area}
4861            {\seq_use:Nn \l_tmpb_seq {\\}}[project]}}
4862      & \dfgFastPIinProjectsTable{\pRefPersonId}
4863      \\
4864     }%
4865    }
4866    \StopTimer{Proj-#1-loop}
4867    \iow_term:x { Proj-#1-iterations:~ \int_use:N \l_tmpa_int }
4868
4869    \StartTimer
```

```
4870    \begin{tblr}[
4871      theme   = main,
4872      expand  = \tblrbody,
4873      caption = {Project~area~#1~---~#2},
4874      long
4875     ]{
4876      colspec     = {m{1.1cm}@{}m{0.2cm}X[m]m{1.1cm}X[1.3,m]},
4877      rowhead     = 1,
4878          rows        = {halign=l,font=\raggedright},
4879      column{2,4} = {halign=c},
4880    % column{1}   = {fg=tableFirstColumn},
4881      row{1}      = {
4882                     bg     = tableHeadingsBG,
4883                     fg     = tableHeadingsFont,
4884                     font   = \bfseries,
4885                     % halign = c,
4886                     guard
4887                    },
4888     }
4889     Project
4890      & %{Status}
4891      & Title
4892      & {\mbox{Research}                                                  \\area}
4893      & Project\space leader(s),\space institute(s),\space location(s)         \\
4894   \tblrbody
4895    \end{tblr}
4896    \@@_projects_table_institution_legend:
4897    \StopTimer{Proj-#1-render}
4898  }
```

Project overview table `\projectOverviewTable` produces the compact project list used on the proposal's final page. Current and new projects remain in project-number order; ending projects (status `E`) are moved to the end. The status marker is omitted for current projects and printed after the project number for all other statuses.

Project-leader names are resolved while the table body is assembled. This avoids repeating database lookups during tabularray's measuring and rendering passes. All three headings use the canonical dark blue at a larger size; the cell-specific rule ensures that the first heading ("Project") is not overridden by the first-column styling. The final-page source can therefore remain deliberately small: `\pagestyle{empty}` followed by `\printTable{projectOverview}`.

```
4899  \tl_new:N  \l_@@_project_overview_body_tl
4900  \tl_new:N  \l_@@_project_overview_ending_body_tl
4901  \tl_new:N  \l_@@_project_overview_id_tl
4902  \tl_new:N  \l_@@_project_overview_number_tl
4903  \tl_new:N  \l_@@_project_overview_name_tl
4904  \tl_new:N  \l_@@_project_overview_status_tl
4905  \tl_new:N  \l_@@_project_overview_consider_for_lists_tl
4906  \tl_new:N  \l_@@_project_overview_person_ids_tl
4907  \tl_new:N  \l_@@_project_overview_leaders_tl
4908  \tl_new:N  \l_@@_project_overview_project_cell_tl
4909  \seq_new:N \l_@@_project_overview_person_seq
4910
4911  \cs_new_protected:Npn \@@_project_overview_set_leaders:n #1
4912  {
4913    \tl_clear:N \l_@@_project_overview_leaders_tl
4914    \seq_set_from_clist:Nn \l_@@_project_overview_person_seq {#1}
4915    \seq_map_indexed_inline:Nn \l_@@_project_overview_person_seq
4916    {
4917      \DTLgetvalueforkey{\l_tmpa_tl}{firstname}{person}{id}{##2}
4918      \DTLgetvalueforkey{\l_tmpb_tl}{lastname}{person}{id}{##2}
4919      \int_compare:nNnT {##1} > {1}
4920        { \tl_put_right:Nn \l_@@_project_overview_leaders_tl { \newline } }
4921      \tl_if_blank:VF \l_tmpa_tl
4922      {
4923        \tl_put_right:Ne \l_@@_project_overview_leaders_tl
4924        {
4925          \exp_not:N \dfgShowTag {##2}{firstname}
4926            {\exp_not:V \l_tmpa_tl}[person]
4927          \exp_not:N \space
4928        }
4929      }
4930      \tl_put_right:Ne \l_@@_project_overview_leaders_tl
4931      {
4932        \exp_not:N \hyperlink { pi:\exp_not:n {##2} }
4933        {
4934          \exp_not:N \textbf
4935          {
4936            \exp_not:N \color { referencePI }
4937            \exp_not:N \dfgShowTag {##2}{lastname}
4938              {\exp_not:V \l_tmpb_tl}[person]
```

```
        }
      }
    }
  }
}

\cs_new_protected:Npn \@@_project_overview_append_row:N #1
{
  \exp_args:NV \@@_project_overview_set_leaders:n
    \l_@@_project_overview_person_ids_tl
  \tl_set:Ne \l_@@_project_overview_project_cell_tl
  {
    \exp_not:N \getProject
      { \exp_not:V \l_@@_project_overview_id_tl }
  }
  \str_if_eq:VnF \l_@@_project_overview_status_tl { C }
  {
    \tl_if_blank:VF \l_@@_project_overview_status_tl
    {
      \tl_put_right:Ne \l_@@_project_overview_project_cell_tl
      {
        \exp_not:N \,
        \exp_not:N \dfgShowTag
          {\exp_not:V \l_@@_project_overview_id_tl}{project-status}
          {\exp_not:V \l_@@_project_overview_status_tl}[project]
      }
    }
  }
  \tl_put_right:Ne #1
  {
    { \exp_not:V \l_@@_project_overview_project_cell_tl }
    & {
      \exp_not:N \dfgShowTag
        {\exp_not:V \l_@@_project_overview_id_tl}{name}
        {\exp_not:V \l_@@_project_overview_name_tl}[project]
    }
    & { \exp_not:V \l_@@_project_overview_leaders_tl }
    \exp_not:N \\
  }
}

\cs_new_protected:Npn \@@_project_overview_collect:
{
  \DTLsort{number}{project}
  \DTLforeach*{project}
  {
    \l_@@_project_overview_id_tl         = id,
    \l_@@_project_overview_number_tl     = number,
    \l_@@_project_overview_name_tl       = name,
    \l_@@_project_overview_status_tl     = project-status,
    \l_@@_project_overview_consider_for_lists_tl = consider-for-lists,
    \l_@@_project_overview_person_ids_tl = ref-person-id
  }
  {
    \str_if_eq:VnF \l_@@_project_overview_consider_for_lists_tl { false }
    {
      \str_if_eq:VnTF \l_@@_project_overview_status_tl { E }
        { \@@_project_overview_append_row:N \l_@@_project_overview_ending_body_tl }
        { \@@_project_overview_append_row:N \l_@@_project_overview_body_tl }
    }
  }
  \tl_put_right:NV \l_@@_project_overview_body_tl
    \l_@@_project_overview_ending_body_tl
}

\cs_new_protected:Npn \@@_project_overview_typeset:
{
  \par\vspace*{-5\baselineskip}
  \begin{center}
    \crcLogo[height=1cm]
  \end{center}
  \enlargethispage{3\baselineskip}
  \begin{tblr}[
    theme  = main,
    expand = \l_@@_project_overview_body_tl
  ]{
    vspan       = even,
    rowsep      = 1pt,
    rows        = {ht=2\baselineskip,valign=m},
    colspec     = {m{1.5cm}X[m]m{4cm}},
    rowhead     = 1,
 column{1}   = {font=\Large\bfseries,valign=m, fg=tableFirstColumn},
    column{2,3} = {font=\raggedright},
    row{1} = {font=\large\bfseries,fg=tableHeadingsFont},
  }
    Project
    & Title
```

```
    & Project\space leader(s) \\
    \l_@@_project_overview_body_tl
  \end{tblr}
}

\NewDocumentCommand{\projectOverviewTable}{}
{
  \group_begin:
  \DTLifdbexists{project}
  {
    \tl_clear:N \l_@@_project_overview_body_tl
    \tl_clear:N \l_@@_project_overview_ending_body_tl
    \@@_project_overview_collect:
    \@@_project_overview_typeset:
  }
  { \msg_warning:nnn { dfg } { database-not-found } { project } }
  \group_end:
}
```

We turn now to the part to get the research areas aggregated.

```
\clist_new:N \l_@@_research_area_clist   % new clist name
\clist_new:N \l_@@_research_area_final_clist   % new clist name
\NewDocumentCommand{\ResearchAreasFromProjectsList}{o}{
  % #1: IDs of project to be excluded
  \clist_clear:N \l_@@_research_area_clist
  \tl_clear:N \l_tmpa_tl

  \DTLforeach[
   % exclude mentioned projects (for whatsoever reason)
   \not\DTLisinlist{\pNumber}{#1}% 1st condition
    \and
    \not\DTLiseq{\pConsiderForLists}{false}
  ]
  {project}{%
   \pNumber       =number,
   \pType         =project-type,
    \pConsiderForLists=consider-for-lists,
   \l_tmpa_tl     =research-area%
  }{%
   % Quite heavy but it works to skip the research area for non-research projects
   \bool_set_false:N \l_@@_project_type_research_bool
   \bool_set_false:N \l_@@_project_type_service_bool
   \bool_set_false:N \l_@@_project_type_mgk_bool
   \bool_set_false:N \l_@@_project_type_admin_bool

   \str_case:Vn {\pType}{
    {research}{ \bool_set_true:N \l_@@_project_type_research_bool }
    {service} { \bool_set_true:N \l_@@_project_type_service_bool  }
    {mgk}     { \bool_set_true:N \l_@@_project_type_mgk_bool      }
    {admin}   { \bool_set_true:N \l_@@_project_type_admin_bool    }
   }
   \bool_lazy_any:nT {{ \l_@@_project_type_research_bool}}
   {\clist_put_right:NV \l_@@_research_area_clist { \l_tmpa_tl }}%
  }
  \clist_remove_duplicates:N \l_@@_research_area_clist
  % Sort the fixed-width classification code numerically after removing
  % punctuation; treating `2.21-02` as an fp expression would parse the
  % hyphen as subtraction.
  \clist_sort:Nn \l_@@_research_area_clist
  {
    \tl_set:Nn \l_tmpa_tl {##1}
    \tl_set:Nn \l_tmpb_tl {##2}
    \regex_replace_all:nnN { [-.] } {} \l_tmpa_tl
    \regex_replace_all:nnN { [-.] } {} \l_tmpb_tl
    \exp_args:NnV \regex_match:nnTF { \A \d+ \Z } \l_tmpa_tl
      {
        \exp_args:NnV \regex_match:nnTF { \A \d+ \Z } \l_tmpb_tl
          {
            \int_compare:nNnTF { \l_tmpa_tl } < { \l_tmpb_tl }
              { \sort_return_same: }
              { \sort_return_swapped: }
          }
          { \sort_return_same: }
      }
      {
        \exp_args:NnV \regex_match:nnTF { \A \d+ \Z } \l_tmpb_tl
          { \sort_return_swapped: }
          { \sort_return_same: }
      }
  }

  \clist_map_inline:Nn \l_@@_research_area_clist {
   \clist_put_right:NV \l_@@_research_area_final_clist { \dfgResearchAreasList{##1}}
  }

  \dfgShowCalculatedTag{ResearchAreasFromProjectsList}{project}
```

```
    {
      \clist_use:Nnnn \l_@@_research_area_final_clist {,~}{,~}{,~and~}%
      .%final dot.
    }
  }
```

6.1.23 DFG research area classification

The DFG publishes an official hierarchical subject classification list; a project must declare which areas its work belongs to. The global property list `\g_@@_dfg_research_areas_prop` maps every classification code (e.g. `2.11-01`) to its English label.

Usage: supply one or more codes as a comma-separated value in the `research-area` key of an `\addEntry` → p. 2 `{project}` block. `\printResearchAreas` then looks up and formats them as a natural-language list.

Sorting note: the codes are numeric strings with a hyphen (`2.11-01`), which cannot be compared as plain integers. The sort in `\printResearchAreas` uses `\fp_compare:nTF` after stripping the hyphen suffix — a known workaround for the absence of a lexicographic sort in expl3 for this mixed format.

```
\prop_new:N \g_@@_dfg_research_areas_prop
\prop_gset_from_keyval:Nn \g_@@_dfg_research_areas_prop
{
  1.11-01 = {Prehistory~and~World~Archaeology},
  1.11-02 = {Greek~and~Latin~Philology},
  1.11-03 = {Ancient~History},
  1.11-04 = {Classical,~Roman,~Christian~and~Islamic~Archaeology},
  1.11-05 = {Egyptology~and~Ancient~Near~Eastern~Studies},

  1.12-01 = {Medieval~History},
  1.12-02 = {Early~Modern~History},
  1.12-03 = {Modern~and~Contemporary~History},
  1.12-04 = {History~of~Science},

  1.13-01 = {Art~History},
  1.13-02 = {Musicology},
  1.13-03 = {Theatre~and~Media~Studies},

  1.14-01 = {General~and~Comparative~Linguistics,~Experimental~Linguistics,~Typology,~Non-
European~Languages},
  1.14-02 = {Individual~Linguistics,~Historical~Linguistics},
  1.14-03 = {Applied~Linguistics,~Computational~Linguistics},

  1.15-01 = {German~Medieval~Studies~(Medieval~German~Literature)},
  1.15-02 = {German~Literary~and~Cultural~Studies~(Modern~German~Literature)},
  1.15-03 = {European~and~American~Literary~and~Cultural~Studies},
  1.15-04 = {General~and~Comparative~Literary~Studies;~Cultural~Studies},

  1.16-01 = {Social~and~Cultural~Anthropology~and~Ethnology},
  1.16-02 = {Asian~Studies},
  1.16-03 = {African,~American~and~Oceania~Studies},
  1.16-04 = {Islamic~Studies,~Arabian~Studies,~Semitic~Studies},
  1.16-05 = {Religious~Studies~and~Jewish~Studies},

  1.17-01 = {Protestant~Theology},
  1.17-02 = {Roman~Catholic~Theology},

  1.18-01 = {History~of~Philosophy},
  1.18-02 = {Theoretical~Philosophy},
  1.18-03 = {Practical~Philosophy},

  1.21-01 = {General~Education~and~History~of~Education},
  1.21-02 = {General~and~Domain-Specific~Teaching~and~Learning},
  1.21-03 = {Education~Systems~and~Educational~Institutions},
  1.21-04 = {Educational~Research~on~Socialisation,~Welfare~and~Professionalism},

  1.22-01 = {General,~Cognitive~and~Mathematical~Psychology},
  1.22-02 = {Biological~Psychology~and~Cognitive~Neuroscience},
  1.22-03 = {Developmental~and~Educational~Psychology},
  1.22-04 = {Social~Psychology,~Industrial~and~Organisational~Psychology},
  1.22-05 = {Personality~Psychology,~Clinical~and~Medical~Psychology,~Methodology},

  1.23-01 = {Sociological~Theory},
  1.23-02 = {Empirical~Social~Research},
  1.23-03 = {Communication~Sciences},
  1.23-04 = {Political~Science},

  1.24-01 = {Economic~Theory},
  1.24-02 = {Economic~Policy,~Applied~Economics},
  1.24-03 = {Accounting~and~Finance},
```

```
    1.24-04 = {Management~and~Marketing},
    1.24-05 = {Operations~Management~and~Computer~Science~for~Business~Administration},
    1.24-06 = {Statistics~and~Econometrics},
    1.24-07 = {Economic~and~Social~History},

    1.25-01 = {Principles~of~Law~and~Jurisprudence},
    1.25-02 = {Private~Law},
    1.25-03 = {Public~Law},
    1.25-04 = {Criminal~Law},
    1.25-05 = {Criminology},

    2.11-01 = {Biochemistry},
    2.11-02 = {Biophysics},
    2.11-03 = {Cell~Biology},
    2.11-04 = {Structural~Biology},
    2.11-05 = {General~Genetics~and~Functional~Genome~Biology},
    2.11-06 = {Developmental~Biology},
    2.11-07 = {Bioinformatics~and~Theoretical~Biology},

    2.12-01 = {Evolution~and~Systematics~of~Plants~and~Fungi},
    2.12-02 = {Ecology~and~Biodiversity~of~Plants~and~Ecosystems},
    2.12-03 = {Organismic~Interactions,~Chemical~Ecology~and~Microbiomes~of~Plant~Systems},
    2.12-04 = {Plant~Physiology},
    2.12-05 = {Plant~Biochemistry~and~Biophysics},
    2.12-06 = {Plant~Cell~and~Developmental~Biology},
    2.12-07 = {Plant~Genetics~and~Genomics},

    2.13-01 = {Systematics~and~Morphology~(Zoology)},
    2.13-02 = {Evolution,~Anthropology},
    2.13-03 = {Ecology~and~Biodiversity~of~Animals~and~Ecosystems,~Organismic~Interactions},
    2.13-04 = {Sensory~and~Behavioural~Biology},
    2.13-05 = {Animal~Physiology~and~Biochemistry},
    2.13-06 = {Evolutionary~Cell~and~Developmental~Biology~(Zoology)},

    2.21-01 = {Metabolism,~Biochemistry~and~Genetics~of~Microorganisms},
    2.21-02 = {Microbial~Ecology~and~Applied~Microbiology},
    2.21-03 = {Medical~Microbiology~and~Mycology,~Hygiene,~Molecular~Infection~Biology},
    2.21-04 = {Virology},
    2.21-05 = {Immunology},
    2.21-06 = {Parasitology~and~Biology~of~Tropical~Infectious~Disease~Pathogens},

    2.22-01 = {Epidemiology~and~Medical~Biometry~/~Statistics},
    2.22-02 = {Public~Health,~Healthcare~Research,~Social~and~Occupational~Medicine},
    2.22-03 = {Human~Genetics},
    2.22-04 = {Anatomy~and~Physiology},
    2.22-05 = {Nutritional~Sciences},
    2.22-06 = {Pathology},
    2.22-07 = {Medical~Informatics~and~Medical~Bioinformatics},
    2.22-08 = {Pharmacy},
    2.22-09 = {Pharmacology},
    2.22-10 = {Toxicology,~Laboratory~Medicine},
    2.22-11 = {Anaesthesiology},
    2.22-12 = {Cardiology,~Angiology},
    2.22-13 = {Pneumology,~Thoracic~Surgery},
    2.22-14 = {Hematology,~Oncology},
    2.22-15 = {Gastroenterology},
    2.22-16 = {Nephrology},
    2.22-17 = {Endocrinology,~Diabetology,~Metabolism},
    2.22-18 = {Rheumatology},
    2.22-19 = {Dermatology},
    2.22-20 = {Pediatric~and~Adolescent~Medicine},
    2.22-21 = {Gynaecology~and~Obstetrics},
    2.22-22 = {Clinical~Immunology~and~Allergology},
    2.22-23 = {Reproductive~Medicine,~Urology},
    2.22-24 = {Biogerontology~and~Geriatric~Medicine},
    2.22-25 = {General~and~Visceral~Surgery},
    2.22-26 = {Cardiac~and~Vascular~Surgery},
    2.22-27 = {Orthopaedics,~Traumatology,~Reconstructive~Surgery},
    2.22-28 = {Dentistry,~Oral~Surgery},
    2.22-29 = {Otolaryngology,~Phoniatrics~and~Audiology},
    2.22-30 = {Radiology},
    2.22-31 = {Clinical~Infectiology~and~Tropical~Medicine},
    2.22-32 = {Medical~Physics,~Biomedical~Technology},
    2.22-33 = {Nuclear~Medicine,~Radiotherapy,~Radiobiology},

    2.23-01 = {Developmental~Neurobiology},
    2.23-02 = {Molecular~Biology~and~Physiology~of~Neurons~and~Glial~Cells},
    2.23-03 = {Experimental~and~Theoretical~Network~Neuroscience},
    2.23-04 = {Cognitive,~Systems~and~Behavioural~Neurobiology},
    2.23-05 = {Experimental~Models~for~the~Understanding~of~Nervous~System~Diseases},
    2.23-06 = {Molecular~and~Cellular~Neurology~and~Neuropathology},
    2.23-07 = {Clinical~Neurology;~Neurosurgery~and~Neuroradiology},
    2.23-08 = {Human~Cognitive~and~Systems~Neuroscience},
    2.23-09 = {Biological~Psychiatry},
    2.23-10 = {Clinical~Psychiatry,~Psychotherapy,~Child~and~Adolescent~Psychiatry},
    2.23-11 = {Ophthalmology},

```

```
5261    2.31-01 = {Soil~Sciences},
5262    2.31-02 = {Plant~Breeding~and~Plant~Pathology},
5263    2.31-03 = {Plant~Cultivation,~Plant~Nutrition,~Agricultural~Technology},
5264    2.31-04 = {Ecology~of~Land~Use},
5265    2.31-05 = {Agricultural~Economics,~Agricultural~Policy,~Agricultural~Sociology},
5266    2.31-06 = {Forestry},
5267    2.31-07 = {Animal~Breeding,~Animal~Nutrition,~Animal~Husbandry},
5268
5269    3.11-01 = {Inorganic~Molecular~Chemistry~-~Synthesis~and~Characterisation},
5270    3.11-02 = {Organic~Molecular~Chemistry~-~Synthesis~and~Characterisation},
5271
5272    3.12-01 = {Solid~State~and~Surface~Chemistry,~Material~Synthesis},
5273    3.12-02 = {Physical~Chemistry~of~Solids~and~Surfaces,~Material~Characterisation},
5274
5275    3.13-01 = {Physical~Chemistry~of~Molecules,~Liquids~and~Interfaces,~Biophysical~Chemistry},
5276
5277    3.14-01 = {Analytical~Chemistry},
5278
5279    3.15-01 = {Biological~and~Biomimetic~Chemistry},
5280    3.15-02 = {Food~Chemistry},
5281
5282    3.16-01 = {Preparatory~and~Physical~Chemistry~of~Polymers},
5283    3.16-02 = {Experimental~and~Theoretical~Physics~of~Polymers},
5284    3.16-03 = {Polymer~Materials},
5285
5286    3.17-01 = {Theoretical~Chemistry:~Electron~Structure,~Dynamics,~Simulation},
5287    3.17-02 = {Theoretical~Chemistry:~Molecules,~Materials,~Surfaces},
5288
5289    3.21-01 = {Experimental~Condensed~Matter~Physics},
5290    3.21-02 = {Theoretical~Condensed~Matter~Physics},
5291
5292    3.22-01 = {Statistical~Physics,~Nonlinear~Dynamics,~Complex~Systems,~Soft~and~Fluid~Matter,~Biological~Physics},
5293
5294    3.23-01 = {Optics,~Quantum~Optics~and~Physics~of~Atoms,~Molecules~and~Plasmas},
5295
5296    3.24-01 = {Nuclear~and~Elementary~Particle~Physics,~Quantum~Mechanics,~Relativity,~Fields},
5297
5298    3.25-01 = {Astrophysics~and~Astronomy},
5299
5300    3.31-01 = {Mathematics},
5301
5302    3.41-01 = {Atmospheric~Science},
5303    3.41-02 = {Oceanography},
5304
5305    3.42-01 = {Geology},
5306    3.42-02 = {Palaeontology},
5307
5308    3.43-01 = {Geophysics},
5309    3.43-02 = {Geodesy,~Photogrammetry,~Remote~Sensing,~Geoinformatics,~Cartography},
5310
5311    3.44-01 = {Mineralogy,~Petrology~and~Geochemistry},
5312
5313    3.45-01 = {Physical~Geography},
5314    3.45-02 = {Human~Geography},
5315
5316    3.46-01 = {Hydrogeology,~Hydrology,~Limnology,~Urban~Water~Management,~Water~Chemistry,~Integrated~Water~Resources~
5317
5318    4.11-01 = {Metal-Cutting~and~Abrasive~Manufacturing~Engineering},
5319    4.11-02 = {Primary~Shaping~and~Reshaping~Technology,~Additive~Manufacturing},
5320    4.11-03 = {Joining~and~Separation~Technology},
5321    4.11-04 = {Plastics~Engineering},
5322    4.11-05 = {Production~Systems,~Operations~Management,~Quality~Management~and~Factory~Planning},
5323    4.11-06 = {Production~Automation~and~Assembly~Technology},
5324
5325    4.12-01 = {Engineering~Design,~Machine~Elements,~Product~Development},
5326    4.12-02 = {Mechanics},
5327    4.12-03 = {Lightweight~Construction,~Textile~Technology},
5328    4.12-04 = {Acoustics},
5329
5330    4.21-01 = {Chemical~and~Thermal~Process~Engineering},
5331    4.21-02 = {Technical~Chemistry},
5332    4.21-03 = {Mechanical~Process~Engineering},
5333    4.21-04 = {Biological~Process~Engineering},
5334    4.21-05 = {Systems~Process~Engineering},
5335
5336    4.22-01 = {Energy~Process~Engineering},
5337    4.22-02 = {Technical~Thermodynamics},
5338    4.22-03 = {Fluid~Mechanics},
5339    4.22-04 = {Hydraulic~and~Turbo~Engines~and~Piston~Engines},
5340
5341    4.31-01 = {Metallurgical,~Thermal~and~Thermomechanical~Treatment~of~Materials},
5342    4.31-02 = {Materials~in~Sintering~Processes~and~Generative~Manufacturing~Processes},
5343    4.31-03 = {Coating~and~Surface~Technology},
5344    4.31-04 = {Mechanical~Properties~of~Metallic~Materials~and~their~Microstructural~Origins},
5345    4.31-05 = {Glass,~Ceramics~and~Derived~Composites},
5346    4.31-06 = {Polymeric~and~Biogenic~Materials~and~Derived~Composites},
5347
```

```
  4.32-01 = {Synthesis~and~Properties~of~Functional~Materials},
  4.32-02 = {Biomaterials},
  4.32-03 = {Thermodynamics~and~Kinetics~as~well~as~Properties~of~Phases~and~Microstructure~of~Materials},
  4.32-04 = {Computer-Aided~Design~of~Materials~and~Simulation~of~Materials~Behaviour~from~Atomic~to~Microscopic~Scal

  4.41-01 = {Automation,~Mechatronics,~Control~Systems,~Intelligent~Technical~Systems,~Robotics},
  4.41-02 = {Measurement~Systems},
  4.41-03 = {Microsystems},
  4.41-04 = {Traffic~and~Transport~Systems,~Intelligent~and~Automated~Traffic},
  4.41-05 = {Human~Factors,~Ergonomics,~Human-Machine~Systems},
  4.41-06 = {Biomedical~Systems~Technology},

  4.42-01 = {Electronic~Semiconductors,~Components~and~Circuits,~Integrated~Systems,~Sensor~Technology,~Theoretical~E
  4.42-02 = {Communication~Technology~and~Networks,~High-Frequency~Technology~and~Photonic~Systems,~Signal~Processing
  4.42-03 = {Electrical~Energy~Systems,~Power~Management,~Power~Electronics,~Electrical~Machines~and~Drives},
  4.42-04 = {Hardware~Systems~and~Architectures~for~Information~Technology~and~Artificial~Intelligence,~Quantum~Engin

  4.43-01 = {Theoretical~Computer~Science},
  4.43-02 = {Software~Engineering~and~Programming~Languages},
  4.43-03 = {Security~and~Dependability,~Operating,~Communication~and~Distributed~Systems},
  4.43-04 = {Artificial~Intelligence~and~Machine~Learning~Methods},
  4.43-05 = {Image~and~Language~Processing,~Computer~Graphics~and~Visualisation,~Human~Computer~Interaction,~Ubiquito
  4.43-06 = {Data~Management,~Data-Intensive~Systems,~Computer~Science~Methods~in~Business~Informatics},
  4.43-07 = {Computer~Architecture,~Embedded~and~Massively~Parallel~Systems},

  4.51-01 = {Architecture,~Building~and~Construction~History,~Construction~Research,~Sustainable~Building~Technology}
  4.51-02 = {Urbanism,~Spatial~Planning,~Transportation~and~Infrastructure~Planning,~Landscape~Planning},
  4.51-03 = {Construction~Material~Sciences,~Chemistry,~Building~Physics},
  4.51-04 = {Structural~Engineering,~Building~Informatics~and~Construction~Operation},
  4.51-05 = {Applied~Mechanics,~Statics~and~Dynamics},
  4.51-06 = {Geotechnics,~Hydraulic~Engineering},

}

\NewDocumentCommand{\dfgResearchAreasList}{m}{
  \exp_args:Ne \@@_research_area_print:n { \tl_trim_spaces:n {#1} }
}
\cs_new_protected:Npn \@@_research_area_print:n #1
{
  % #1: a single code, with surrounding spaces removed
  \prop_if_in:NnTF \g_@@_dfg_research_areas_prop {#1}
  {
   % key exists → fetch the description and typeset it
   \prop_get:NnN \g_@@_dfg_research_areas_prop {#1} \l_tmpa_tl
   \textbf{#1}:~\l_tmpa_tl
  }
  {
   % key missing → same fallback as your original macro
   \textcolor{error}{\textbf{#1}:~ research~ area~ not~ found}
   \msg_warning:nne { dfg } { research-area-not-found }{#1}
  }
}
```

We turn to the early career phases

```
\str_new:N \l_@@_row_color_str
\NewDocumentCommand{\EarlyCareerResearchersTable}{ O{EarlyCareerResearcher} }{%
  \def\dfgLocalDB{#1}
  \DTLifdbexists{\dfgLocalDB}{%
   \DTLsort*{ref-project-id,lastname}{\dfgLocalDB}%
   \def\tblrbody{}%
   \DTLforeach*{\dfgLocalDB}{%
    \pRefProjectID=ref-project-id,%
    \pLastname=lastname,%
    \pFirstname=firstname,%
    \pFinancedBy=financed-by,%
    \pTopic=topic,%
    \pCategory=category,%
    \pPeriodStart=periodStart,%
    \pPeriodEnd=periodEnd%
   }{%
    \str_case:en {\pCategory}{
     {MD}{\str_set:Nn \l_@@_row_color_str {green!10} }
     {PhD}{\str_set:Nn \l_@@_row_color_str {blue!10} }
    }
    \DTLgetvalueforkey{\theProjectNumber}{number}{project}{id}{\pRefProjectID}
    \protected@edef\tblrbody{\tblrbody
     \SetRow{\l_@@_row_color_str}
           \datatool_if_null:NTF {\theProjectNumber}{\pRefProjectID}{\getProject{\pRefProjectID}}
      & \noexpand\dfgShowTag{\pRefProjectID/\pLastname}{lastname}
          {\pLastname}[\dfgLocalDB],~
        \noexpand\dfgShowTag{\pRefProjectID/\pLastname}{firstname}
          {\pFirstname}[\dfgLocalDB]
      & \noexpand\dfgShowTag{\pRefProjectID/\pLastname}{financed-by}
          {\pFinancedBy}[\dfgLocalDB]
      & \noexpand\dfgShowTag{\pRefProjectID/\pLastname}{topic}
          {\pTopic}[\dfgLocalDB]
```

```
      &
      {
       \ifdefempty{\pPeriodEnd}{since~}{}
           \noexpand\dfgShowTag{\pRefProjectID/\pLastname}{periodStart}
             {\pPeriodStart}[\dfgLocalDB]
           \ifdefempty{\pPeriodEnd}{}{--\\
             \noexpand\dfgShowTag{\pRefProjectID/\pLastname}{periodEnd}
               {\pPeriodEnd}[\dfgLocalDB]}}
      \\
     }}%
    \begin{tblr}[%
     theme=main,
     expand=\tblrbody,
     remark{Note}={Row~colours~indicate~category~of~doctoral~candidates:~
                   \textcolor{blue}{Dr.\,rer.\,nat.},~
                   \textcolor{green}{Dr.\,med.}.~
                   \textbf{RS}~=~Rising~Star.~
                   Doctoral~candidates~with~highlighted~names~have~obtained~the~doctoral~degree.~
                   MD~denotes~medical~students~supported~by~DFG~stipends~for~up~to~6~months.
                  },
     long
    ]{
     colspec = {t{.8cm}t{3cm}t{1.2cm}X[t]t{1.8cm}},
     rowhead = 1,
     column{1} = {
                  font={\bfseries},
                  fg = {tableFirstColumn}
                 },
     % row{even}={bg=tableRows},
     column{4} = {font={\footnotesize\itshape}},
     row{1}       = {
                     bg       = tableHeadingsBG,
                     fg       = tableHeadingsFont,
                     font     = \footnotesize\bfseries,
                     halign   = {c},
                     valign   = {t},
                     guard
                    },
    }
    Project %
     & Surname,~ first~ name             %
     & {Type~ of~            \\ funding} %
     & Topic                             %
     & Duration                          \\
    \tblrbody
   \end{tblr}}
  {
   Database~ \texttt{#1}~ missing.
   \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}
```

Following now the section about diversity and equity. Before that there is a command to get proper position names based on a property list.

```

\prop_new:N \g_@@_dfg_position_names_prop
\prop_gset_from_keyval:Nn \g_@@_dfg_position_names_prop
{
  PhD         = {Doctoral~researchers},
  postDoc     = {Postdoctoral~researchers},
  groupLeader = {Research~ group~ leaders,~
    junior~ research~ group~ leaders,~
    junior~ professors},
  research    = {Other~ research~ staff},
  nonResearch = {Non-research~ staff},
  student     = {Student~ and~ graduate~ assistants},
  prof        = {Professors},
  prof-W2     = {Professors~ W2/C3},
  prof-W3     = {Professors~ W3/C4}
}
\prop_new:N \g_@@_dfg_category_names_prop
\prop_gset_from_keyval:Nn \g_@@_dfg_category_names_prop
{
  PhD         = {Doctoral~researcher},
  postDoc     = {Postdoctoral~researcher},
  medDoc      = {Medical~doctoral~researcher},
  groupLeader = {(Jr.)~ Research~ group~ leader},
  research    = {Other~ research~ staff},
  nonResearch = {Non-research~ staff},
  student     = {Student/graduate~ assistant},
  assistant   = {Research~assistant},
  prof        = {Professor},
  prof-W2     = {Professor~ (W2/C3)},
  prof-W3     = {Professor~ (W3/C4)},
}
```

```
\cs_generate_variant:Nn \prop_if_in:NnTF { NeTF }

\NewDocumentCommand{\dfgPositionName}{m}{%
  \prop_if_in:NeTF \g_@@_dfg_position_names_prop {#1}
  {
   \prop_get:NeN \g_@@_dfg_position_names_prop {#1} \l_tmpa_tl
   \l_tmpa_tl
  }
  {
   \textcolor{error}{mismatch:~#1}
    \@@_warn_unknown_name:nnN {position~in~\string\dfgPositionName} {#1}
      \g_@@_dfg_position_names_prop
  }
}
\NewDocumentCommand{\dfgCategoryName}{m}{
  \prop_if_in:NeTF \g_@@_dfg_category_names_prop {#1}
  {
   \prop_get:NeN \g_@@_dfg_category_names_prop {#1} \l_tmpa_tl
   \l_tmpa_tl
  }
  {
   \textcolor{error}{mismatch:~#1}
    \@@_warn_unknown_name:nnN {category~in~\string\dfgCategoryName} {#1}
      \g_@@_dfg_category_names_prop
  }
}

\NewDocumentCommand{\GenderEqualityStaffTable}{ O{GenderEqualityStaff} }{%
  \def\dfgLocalDB{#1}
  \DTLifdbexists{\dfgLocalDB}{%
   \tl_clear:N \l_tblr_body_tl
   \DTLforeach*[
    \DTLiseq{PhD}{\pPosition}
    \OR
    \DTLiseq{postDoc}{\pPosition}
   ]{\dfgLocalDB}{%
    \pPosition=position,%
    \pPrevObjPctFemale=previous-objective-percentage-female,%
    \pStatusQuoNumFemale=status-quo-number-female,%
    \pStatusQuoPctFemale=status-quo-percentage-female,%
    \pFutureObjPctFemale=future-objective-percentage-female%
   }{
    \tl_gput_right:Nx \l_tblr_body_tl
    {
      \dfgShowTag{\pPosition}{position}
        {\dfgPositionName{\pPosition}}[\dfgLocalDB] &
      \dfgShowTag{\pPosition}{previous-objective-percentage-female}
        {\pPrevObjPctFemale}[\dfgLocalDB] &
      \dfgShowTag{\pPosition}{status-quo-number-female}
        {\pStatusQuoNumFemale}[\dfgLocalDB] &
      \dfgShowTag{\pPosition}{status-quo-percentage-female}
        {\pStatusQuoPctFemale}[\dfgLocalDB] &
      \dfgShowTag{\pPosition}{future-objective-percentage-female}
        {\pFutureObjPctFemale}[\dfgLocalDB] \\
    }
   }
   \begin{tblr}[%
     theme=main,
   note{a}={Reference~ date:~ proposal~ for~ \dfgFundingPeriodText\space~ funding~ period},
     expand=\l_tblr_body_tl,
     long
    ]{
     vspan=even,
     %row{Z}= {font={\footnotesize\bfseries}},
     colspec = {
       X[m] % position
       m{2cm} % proportion
       m{2cm} % number
       m{2cm} % female
       m{3cm}
      },
     hspan=minimal, % !!!!! https://tex.stackexchange.com/a/637391/98739
     vline{3} = {3-3}{1pt,solid,fg={tableLines}},
     vline{5} = {1-3}{1pt,solid,fg={tableLines}},
     hline{4} = {1pt,solid,fg={tableLines}},
     % hline{Y} = {1pt,solid,fg={tableOuterLines}},
     hline{2} = {wd=0pt},
     rowhead = 3,
     column{2-5}={halign={c}},
     %  column{1} = {font={\bfseries},
     %    fg = {rwthBlue}},
     row{2-3} ={
                guard,
                valign={t},
                font={\footnotesize\bfseries},
                fg={tableHeadingsFont},
                bg={white}},
```

```
      }
      \SetCell[r=3]{c} % empty
       & \SetCell[c=3]{c} Second~ funding~ period  %
       &                                          %
       &                                          %
       & Third~ funding~ period                   \\
      % empty
       & {Objective~ [\unit{\percent}]}
       & \SetCell[c=2]{c}{Status~ Quo\TblrNote{a}}
       &
       & {Objective~ [\unit{\percent}]}            \\
      % empty
       &                                          % empty
       & Number
       & [\unit{\percent}]
       & f                                        \\
      \tl_use:N \l_tblr_body_tl
      % Total
      %  &                                          % objective
      %  &                                          % number
      %  &                                          % percentage
      %  &                                          % objective \\
     \end{tblr}}
    {
     Database~ \texttt{#1}~ missing.
     \msg_warning:nne { dfg } { database-not-found }{ #1 }
    }
  }

  \int_new:N \l_@@_count_int
  \int_new:N \l_@@_total_int
  \int_new:N \l_@@_tmp_int
  \int_new:N \l_@@_tmp_total_int
  \int_new:N \l_@@_tmp_position_int
  \int_new:N \l_@@_tmp_gender_int
  \int_new:N \l_@@_percent_int
```

Cached counting infrastructure: A single pass over the `person` database fills a property list with one count per (position, gender, phase) triple. All user-facing commands then read from this prop list in O(1) instead of doing a fresh full-table scan per call. This replaces 100 full database walks per gender table with 1.

```
  \prop_new:N \g_@@_counts_prop
  \tl_new:N   \l_@@_count_key_tl
  \tl_new:N   \g_@@_count_phase_tl
  \tl_new:N   \l_@@_count_db_tl
```

Keep the cache phase global, like the counts themselves: a nested group may switch periods, and subsequent calls must see the matching phase. Compute counts for ALL (position, gender) combinations in one pass. Stores them in `\g_@@_counts_prop` with key "<position>/<gender>". `\dfg_compute_counts:nn {<db>} {<phase>}`.

```
  \cs_new_protected:Npn \dfg_compute_counts:nn #1#2
  {
    % Skip if already computed for this (db, phase).
    \tl_set:Nx \l_tmpa_tl { #1 | #2 }
    \tl_if_eq:NNF \l_tmpa_tl \g_@@_count_phase_tl
    {
      \tl_gset:Nx \g_@@_count_phase_tl { #1 | #2 }
     \prop_gclear:N \g_@@_counts_prop
     \DTLforeach*[
      \DTLisinlist{#2}{\pPeriod}
      \and
      \DTLisinlist{PI}{\pRole}
     ]{#1}{
      \pPosition = position,
      \pGender   = gender,
      \pPeriod   = period,
      \pRole     = role
     }{
      \tl_set:Nx \l_@@_count_key_tl { \pPosition | \pGender }
      \prop_get:NVNTF \g_@@_counts_prop \l_@@_count_key_tl \l_tmpb_tl
      {
       \int_set:Nn \l_@@_count_int { \l_tmpb_tl + 1 }
       \prop_gput:NVx \g_@@_counts_prop \l_@@_count_key_tl
       { \int_use:N \l_@@_count_int }
      }
      {
       \prop_gput:NVn \g_@@_counts_prop \l_@@_count_key_tl { 1 }
      }
     }
    }
```

```
}
% O(1) lookup. Returns 0 for unknown keys.
% \dfg_count_lookup:nn {<position>} {<gender>}  -> integer literal
\cs_new:Npn \dfg_count_lookup:nn #1#2
{
  \prop_if_in:NnTF \g_@@_counts_prop { #1 | #2 }
  { \prop_item:Nn \g_@@_counts_prop { #1 | #2 } }
  { 0 }
}

% Set <int> to count for (position, gender) - uses cache.
% \dfg_count_position_set:Nnnnn <int> {<db>} {<position>} {<gender>} {<phase>}
\cs_new_protected:Npn \dfg_count_position_set:Nnnnn #1#2#3#4#5
{
  \prop_if_in:NnTF \g_@@_dfg_position_names_prop {#3}
  {
   \dfg_compute_counts:nn {#2} {#5}
   \int_set:Nn #1 { \dfg_count_lookup:nn {#3} {#4} }
  }
  {
   \textcolor{error}{mismatch:~#3}
    \@@_warn_unknown_name:nnN {staff~position~for~counts} {#3}
      \g_@@_dfg_position_names_prop
  }
}

% Sum across all genders for one position.
% \dfg_total_position_set:Nnnn <int> {<db>} {<position>} {<phase>}
\cs_new_protected:Npn \dfg_total_position_set:Nnnn #1#2#3#4
{
  \dfg_compute_counts:nn {#2} {#4}
  \int_set:Nn #1
  {
   \dfg_count_lookup:nn {#3} {f}
   + \dfg_count_lookup:nn {#3} {m}
   + \dfg_count_lookup:nn {#3} {d}
  }
}

% Sum across all positions for one gender.
% \dfg_total_gender_set:Nnnn <int> {<db>} {<gender>} {<phase>}
\cs_new_protected:Npn \dfg_total_gender_set:Nnnn #1#2#3#4
{
  \dfg_compute_counts:nn {#2} {#4}
  \int_set:Nn #1
  {
   \dfg_count_lookup:nn {postDoc}     {#3}
   + \dfg_count_lookup:nn {groupLeader} {#3}
   + \dfg_count_lookup:nn {prof-W2}    {#3}
   + \dfg_count_lookup:nn {prof-W3}    {#3}
  }
}

% Grand total across all positions and genders.
% \dfg_total_position_gender_set:Nnn <int> {<db>} {<phase>}
\cs_new_protected:Npn \dfg_total_position_gender_set:Nnn #1#2#3
{
  \dfg_compute_counts:nn {#2} {#3}
  \int_zero:N #1
  \clist_map_inline:nn {postDoc, groupLeader, prof-W2, prof-W3}
  {
   \clist_map_inline:nn { f , m , d }
   {
    \int_add:Nn #1 { \dfg_count_lookup:nn {##1} {####1} }
   }
  }
}
```

You can get the count of PIs of a certain position, gender and funding phase.

```
\NewDocumentCommand{\getPositionGenderPhaseAmount}{ O{person} m m m }
{
  % #1: database
  % #2: position
  % #3: gender
  % #4: phase
  \dfg_count_position_set:Nnnnn \l_@@_tmp_int {#1} {#2} {#3} {#4}
  \dfgShowCalculatedTag{getPositionGenderPhaseAmount}{#1}
    { \int_use:N \l_@@_tmp_int }
}
```

This command sums up a certain gender column based on a funding phase.

```
\NewDocumentCommand{\getPositionPhaseTotal}{ O{person} m m }
{
  % #1: database
```

```
5749    % #2: position
5750    % #3: phase
5751
5752    \dfg_total_position_set:Nnnn \l_@@_total_int {#1} {#2} {#3}
5753
5754    % no macro supplied → print the total
5755    \dfgShowCalculatedTag{getPositionPhaseTotal}{#1}
5756      { \int_use:N \l_@@_total_int }
5757  }
```

This command sums up a certain gender column based on a funding phase.

```
5758  \NewDocumentCommand{\getGenderPhaseTotal}{ O{person} m m  }
5759  {
5760    % #1: database
5761    % #2: gender
5762    % #3: phase
5763    \dfg_total_gender_set:Nnnn \l_@@_total_int {#1} {#2} {#3}
5764
5765    % % no macro supplied → print the total
5766    \dfgShowCalculatedTag{getGenderPhaseTotal}{#1}
5767      { \int_use:N \l_@@_total_int }
5768  }
```

This will get you the percentage of a certain position, gender and phase. Mainly for the table below, but can be used anywhere.

```
5769  \NewDocumentCommand{\getPositionGenderPhasePercentage}{ O{person} m m m}
5770  {
5771    % #1: database
5772    % #2: position
5773    % #3: gender
5774    % #4: phase
5775
5776    \dfg_total_position_set:Nnnn \l_@@_total_int {#1} {#2} {#4}
5777
5778    % --- number of the requested gender ---------------------------
5779    \dfg_count_position_set:Nnnnn \l_@@_tmp_int {#1}{#2}{#3}{#4}
5780
5781    % --- avoid division by zero ----------------------------------
5782    \dfgShowCalculatedTag{getPositionGenderPhasePercentage}{#1}
5783    {
5784      \int_compare:nNnTF { \l_@@_total_int } = { 0 }
5785      { 0 }
5786      {
5787        % --- integer percentage (rounded) ------------------------
5788        \int_set:Nn \l_@@_percent_int
5789        { \fp_eval:n { round( 100 * \l_@@_tmp_int / \l_@@_total_int ) } }
5790
5791        % print the integer together with the percent sign
5792        \int_use:N \l_@@_percent_int
5793      }
5794    }
5795  }
```

The following command is meant to get the overall percentage of a certain gender. It goes into the final total row of the table, but it also can be used elsewhere with any funding phase you want.

```
5796  \NewDocumentCommand{\getGenderPhasePercentage}{ O{person} m m  }
5797  {
5798    % #1: database
5799    % #2: gender
5800    % #3: phase
5801
5802    \int_zero:N \l_@@_percent_int
5803    \int_zero:N \l_@@_tmp_int
5804    \int_zero:N \l_@@_tmp_total_int
5805
5806    \dfg_total_position_gender_set:Nnn \l_@@_tmp_total_int {#1} {#3}
5807
5808
5809    % % --- number of the requested gender ---------------------------
5810    \dfg_total_gender_set:Nnnn \l_@@_tmp_gender_int {#1}{#2}{#3}
5811
5812    % --- avoid division by zero ----------------------------------
5813    \dfgShowCalculatedTag{getGenderPhasePercentage}{#1}
5814    {
5815      \int_compare:nNnTF { \l_@@_tmp_total_int } = { 0 }
5816      { 0 }
5817      {
5818        % --- integer percentage (rounded) ------------------------
5819        \int_set:Nn \l_@@_percent_int
5820        { \fp_eval:n { round( 100 * \l_@@_tmp_gender_int / \l_@@_tmp_total_int ) } }
5821
```

```
      % print the integer together with the percent sign
      \int_use:N \l_@@_percent_int
    }
  }
  % \par Gender: #2--
  % Phase: #3--
  % Count: \int_use:N \l_@@_tmp_gender_int --
  % Count Total:\int_use:N \l_@@_tmp_total_int\par------------
}
```

Finally the table which is mostly calculated automatically based on the person-database.

```
\bool_new:N \l_@@_gender_leaders_all_sexes_bool
\tl_new:N \l_@@_gender_leaders_database_tl
\tl_new:N \l_@@_gender_leaders_unknown_options_tl
\clist_new:N \l_@@_gender_leaders_unknown_options_clist
\keys_define:nn {dfg/gender-leaders}
{
  all-sexes .bool_set:N = \l_@@_gender_leaders_all_sexes_bool,
  all-sexes .default:n = true,
  database .tl_set:N = \l_@@_gender_leaders_database_tl,
}
\NewDocumentCommand{\GenderEqualityProjectLeadersTable}{ O{} }
{
  \group_begin:
  \bool_set_false:N \l_@@_gender_leaders_all_sexes_bool
  \tl_set:Nn \l_@@_gender_leaders_database_tl {GenderEqualityProjectLeaders}
  \keys_set_known:nnN {dfg/gender-leaders} {#1}
    \l_@@_gender_leaders_unknown_options_tl
  \tl_if_blank:VF \l_@@_gender_leaders_unknown_options_tl
  {
    % Preserve the old bare database argument, including when combined with
    % a new display flag. Unknown key=value pairs still receive key errors.
    \clist_set:NV \l_@@_gender_leaders_unknown_options_clist
      \l_@@_gender_leaders_unknown_options_tl
    \int_compare:nNnTF
      {\clist_count:N \l_@@_gender_leaders_unknown_options_clist} = {1}
    {
      \exp_args:NV \tl_if_in:nnTF \l_@@_gender_leaders_unknown_options_tl {=}
      {
        \exp_args:NnV \keys_set:nn {dfg/gender-leaders}
          \l_@@_gender_leaders_unknown_options_tl
      }
      {
        \tl_set:Ne \l_@@_gender_leaders_database_tl
          {\clist_item:Nn \l_@@_gender_leaders_unknown_options_clist {1}}
      }
    }
    {
      \exp_args:NnV \keys_set:nn {dfg/gender-leaders}
        \l_@@_gender_leaders_unknown_options_tl
    }
  }
  \bool_if:NTF \l_@@_gender_leaders_all_sexes_bool
    { \exp_args:NV \@@_gender_leaders_all:n \l_@@_gender_leaders_database_tl }
    { \exp_args:NV \@@_gender_leaders_female:n \l_@@_gender_leaders_database_tl }
  \group_end:
}
\cs_new_protected:Npn \@@_gender_leaders_all:n #1
{%
  \def\dfgLocalDB{#1}
  \DTLifdbexists{\dfgLocalDB}{%
    % Invalidate once per table. Both counts and their phase are global, so
    % cell groups and prose commands always agree on the cached period.
    \tl_gclear:N \g_@@_count_phase_tl
   \tl_clear:N \l_tblr_body_tl
   \DTLforeach*[
    % just making absolutely sure we only have the data sets we need.
    \DTLiseq{postDoc}{\pPosition}
    \OR
    \DTLiseq{groupLeader}{\pPosition}
    \OR
    \DTLiseq{prof-W2}{\pPosition}
    \OR
    \DTLiseq{prof-W3}{\pPosition}
   ]{\dfgLocalDB}{%
    \pPosition=position,%
    \pPrevObjPctFemale=previous-objective-percentage-female,%
    \pFutureObjPctFemale=future-objective-percentage-female%
   }{

    \tl_gput_right:Nx \l_tblr_body_tl
    {
      \dfgShowTag{\pPosition}{position}
        {\dfgPositionName{\pPosition}}[\dfgLocalDB] % position
      \int_compare:nNnT {\theFundingPeriodNumeral} = {3}
      {
```

```
5906        & \getPositionGenderPhaseAmount{\pPosition}{f}{1}
5907        & \getPositionGenderPhaseAmount{\pPosition}{m}{1}
5908        & \getPositionGenderPhaseAmount{\pPosition}{d}{1}
5909        & \getPositionGenderPhasePercentage{\pPosition}{f}{1}
5910        & \getPositionGenderPhasePercentage{\pPosition}{m}{1}
5911        & \getPositionGenderPhasePercentage{\pPosition}{d}{1}
5912        }
5913        & \dfgShowTag{\pPosition}{previous-objective-percentage-female}
5914            {\pPrevObjPctFemale}[\dfgLocalDB] % objective
5915       & \getPositionGenderPhaseAmount{\pPosition}{f}{\theFundingPeriodNumeralPrevious} % f number
5916       & \getPositionGenderPhaseAmount{\pPosition}{m}{\theFundingPeriodNumeralPrevious} % m number
5917       & \getPositionGenderPhaseAmount{\pPosition}{d}{\theFundingPeriodNumeralPrevious}% d number
5918       &  \getPositionGenderPhasePercentage{\pPosition}{f}{\theFundingPeriodNumeralPrevious}% f percentage
5919       &  \getPositionGenderPhasePercentage{\pPosition}{m}{\theFundingPeriodNumeralPrevious}% m percentage
5920       &  \getPositionGenderPhasePercentage{\pPosition}{d}{\theFundingPeriodNumeralPrevious}% d percentage
5921        & \dfgShowTag{\pPosition}{future-objective-percentage-female}
5922            {\pFutureObjPctFemale}[\dfgLocalDB] % f objective
5923        \int_compare:nNnT {\theFundingPeriodNumeral} = {3}
5924        {
5925          & \getPositionGenderPhaseAmount{\pPosition}{f}{3}
5926          & \getPositionGenderPhaseAmount{\pPosition}{m}{3}
5927          & \getPositionGenderPhaseAmount{\pPosition}{d}{3}
5928          & \getPositionGenderPhasePercentage{\pPosition}{f}{3}
5929          & \getPositionGenderPhasePercentage{\pPosition}{m}{3}
5930          & \getPositionGenderPhasePercentage{\pPosition}{d}{3}
5931        }
5932       \\
5933      }
5934     }
5935      \tl_gput_right:Nx \l_tblr_body_tl
5936      {
5937        \exp_not:n {Total\printMarginalia{
5938            Total~ no.:~
5939            \dfg_total_position_gender_set:Nnn \l_@@_tmp_total_int
5940            {person} % database
5941            {\theFundingPeriodNumeralPrevious} % phase
5942               \num{\int_use:N \l_@@_tmp_total_int}
5943
5944        }}
5945        \int_compare:nNnT {\theFundingPeriodNumeral} = {3}
5946        {
5947        & \getGenderPhaseTotal{f}{1}
5948        & \getGenderPhaseTotal{m}{1}
5949        & \getGenderPhaseTotal{d}{1}
5950        & \getGenderPhasePercentage{f}{1}
5951        & \getGenderPhasePercentage{m}{1}
5952        & \getGenderPhasePercentage{d}{1}
5953        }
5954       &                                                                  % objective
5955       & \getGenderPhaseTotal{f}{\theFundingPeriodNumeralPrevious}      % number f
5956       & \getGenderPhaseTotal{m}{\theFundingPeriodNumeralPrevious}      % number m
5957       & \getGenderPhaseTotal{d}{\theFundingPeriodNumeralPrevious}      % number d
5958       & \getGenderPhasePercentage{f}{\theFundingPeriodNumeralPrevious} % percentage f
5959       & \getGenderPhasePercentage{m}{\theFundingPeriodNumeralPrevious} % percentage m
5960       & \getGenderPhasePercentage{d}{\theFundingPeriodNumeralPrevious} % percentage d
5961        & % current-period objective
5962        \int_compare:nNnT {\theFundingPeriodNumeral} = {3}
5963        {
5964          & \getGenderPhaseTotal{f}{3}
5965          & \getGenderPhaseTotal{m}{3}
5966          & \getGenderPhaseTotal{d}{3}
5967          & \getGenderPhasePercentage{f}{3}
5968          & \getGenderPhasePercentage{m}{3}
5969          & \getGenderPhasePercentage{d}{3}
5970        }
5971        \\
5972      }
5973      \int_compare:nNnTF {\theFundingPeriodNumeral} = {3}
5974      {
5975        \@@_gender_leaders_render:nn
5976        {
5977          vline{2,8,15}={1-4}{1pt,solid,tableLines},
5978          vline{5,12,19}={3-4}{1pt,solid,tableLines},
5979          vline{9,16}={3-4}{1pt,solid,tableLines},
5980          colspec={X[m] *{6}{m{0.5cm}} m{1.3cm}
5981                        *{6}{m{0.5cm}} m{1.3cm} *{6}{m{0.5cm}}},
5982          colsep=1pt,
5983          column{1}={halign=l},
5984          column{2-21}={halign=c},
5985          cell{2}{8,15}={font=\scriptsize\bfseries},
5986        }
5987        {
5988          \SetCell[r=4]{l} Position
5989          & \SetCell[c=6]{c} First~ funding~ period & & & & &
5990          & \SetCell[c=7]{c} Second~ funding~ period & & & & & &
5991          & \SetCell[c=7]{c} Third~ funding~ period & & & & & & \\
5992          & \SetCell[c=6]{c} Status~ Quo & & & & &
```

```
        & {Objective~ [\unit{\percent}]}
        & \SetCell[c=6]{c} Status~ Quo\TblrNote{a} & & & & &
        & {Objective~ [\unit{\percent}]}
        & \SetCell[c=6]{c} Status~ Quo\TblrNote{a} & & & & & \\
        & \SetCell[c=3]{c} Number & &
        & \SetCell[c=3]{c} [\unit{\percent}] & &
        &
        & \SetCell[c=3]{c} Number & &
        & \SetCell[c=3]{c} [\unit{\percent}] & &
        &
        & \SetCell[c=3]{c} Number & &
        & \SetCell[c=3]{c} [\unit{\percent}] & & \\
        & f & m & d & f & m & d
        &
        & f & m & d & f & m & d
        & f
        & f & m & d & f & m & d \\
      }
    }
    {
      \@@_gender_leaders_render:nn
      {
        vline{3}={3-4}{1pt,solid,tableLines},
        vline{6}={4-4}{1pt,solid,tableLines},
        vline{9}={1-4}{1pt,solid,tableLines},
        colspec={X[m] m{2cm} *{6}{m{0.5cm}} m{3cm}},
        column{2-9}={halign=c},
      }
      {
        \SetCell[r=4]{l} Position
        & \SetCell[c=7]{c}
          \int_case:nnF {\theFundingPeriodNumeralPrevious}
            {{1}{First}{2}{Second}} {Previous}~ funding~ period & & & & & &
        & \dfgFundingPeriodText\space funding~ period \\
        & {Objective~ [\unit{\percent}]}
        & \SetCell[c=6]{c} Status~ Quo\TblrNote{a} & & & & &
        & {Objective~ [\unit{\percent}]} \\
        &
        & \SetCell[c=3]{c} Number & &
        & \SetCell[c=3]{c} [\unit{\percent}] & &
        & \\
        & & f & m & d & f & m & d & f \\
      }
    }
  }
  {
    Database~ \texttt{#1}~ missing.
    \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}
```

Female-only view: the third-period layout follows DFG form 60.200, including the current period's number/percentage next to its original target. The existing future-objective field supplies that target, preserving metadata.

```
\cs_new:Npn \@@_gender_leaders_female_cells:nn #1#2
{
  & \getPositionGenderPhaseAmount{#1}{f}{#2}
  & \getPositionGenderPhasePercentage{#1}{f}{#2}
}
\cs_new_protected:Npn \@@_gender_leaders_female:n #1
{
  \def\dfgLocalDB{#1}
  \DTLifdbexists{\dfgLocalDB}
  {
    \tl_gclear:N \g_@@_count_phase_tl
    \tl_clear:N \l_tblr_body_tl
    \DTLforeach*[
      \DTLiseq{postDoc}{\pPosition}\OR
      \DTLiseq{groupLeader}{\pPosition}\OR
      \DTLiseq{prof-W2}{\pPosition}\OR
      \DTLiseq{prof-W3}{\pPosition}
    ]{\dfgLocalDB}{
      \pPosition=position,
      \pPrevObjPctFemale=previous-objective-percentage-female,
      \pFutureObjPctFemale=future-objective-percentage-female
    }
    {
      \tl_gput_right:Nx \l_tblr_body_tl
      {
        \dfgShowTag{\pPosition}{position}
          {\dfgPositionName{\pPosition}}[\dfgLocalDB]
        \int_compare:nNnTF {\theFundingPeriodNumeral} = {3}
        {
          \@@_gender_leaders_female_cells:nn {\pPosition}{1}
          \@@_gender_leaders_female_cells:nn {\pPosition}{2}
```

```
          & \dfgShowTag{\pPosition}{future-objective-percentage-female}
              {\pFutureObjPctFemale}[\dfgLocalDB]
          \@@_gender_leaders_female_cells:nn {\pPosition}{3}
        }
        {
          & \dfgShowTag{\pPosition}{previous-objective-percentage-female}
              {\pPrevObjPctFemale}[\dfgLocalDB]
          \@@_gender_leaders_female_cells:nn
            {\pPosition}{\theFundingPeriodNumeralPrevious}
          & \dfgShowTag{\pPosition}{future-objective-percentage-female}
              {\pFutureObjPctFemale}[\dfgLocalDB]
        }
        \\
      }
    }
    \tl_gput_right:Nx \l_tblr_body_tl
    {
      Total
      \int_compare:nNnTF {\theFundingPeriodNumeral} = {3}
      {
        & \getGenderPhaseTotal{f}{1} & \getGenderPhasePercentage{f}{1}
        & \getGenderPhaseTotal{f}{2} & \getGenderPhasePercentage{f}{2}
        &
        & \getGenderPhaseTotal{f}{3} & \getGenderPhasePercentage{f}{3}
      }
      {
        &
        & \getGenderPhaseTotal{f}{\theFundingPeriodNumeralPrevious}
        & \getGenderPhasePercentage{f}{\theFundingPeriodNumeralPrevious}
        &
      }
      \\
    }
    \int_compare:nNnTF {\theFundingPeriodNumeral} = {3}
    {
      \@@_gender_leaders_render:nnnn
      {
        colspec={X[m] m{1.2cm} m{.8cm} m{1.2cm} m{.8cm}
                 m{2cm} m{1.2cm} m{.8cm}},
        vline{2,4,6}={1-2}{1pt,solid,tableLines},
        column{2-8}={halign=c},
      }
      {
        \SetCell[r=2]{l} Position~(female)
        & \SetCell[c=2]{c} First~ funding~ period &
        & \SetCell[c=2]{c} Second~ funding~ period &
        & \SetCell[c=3]{c} {Status~ Quo\TblrNote{a}\\Third~ funding~ period} & & \\
        & Number & [\unit{\percent}]
        & Number & [\unit{\percent}]
        & {Original~ objective~[\unit{\percent}]} & Number & [\unit{\percent}] \\
      }
      {2}{3}
    }
    {
      \@@_gender_leaders_render:nnnn
      {
        colspec={X[m] m{2cm} m{1.5cm} m{1.5cm} m{3cm}},
        vline{3}={2-2}{1pt,solid,tableLines},
        vline{5}={1-2}{1pt,solid,tableLines},
        column{2-5}={halign=c},
      }
      {
        \SetCell[r=2]{l} Position~(female)
        & \SetCell[c=3]{c}
          \int_case:nnF {\theFundingPeriodNumeralPrevious}
            {{1}{First}{2}{Second}} {Previous}~ funding~ period & &
        & \dfgFundingPeriodText\space funding~ period \\
        & {Original~ objective~[\unit{\percent}]}
        & {Number\TblrNote{a}} & [\unit{\percent}]
        & {Objective~[\unit{\percent}]} \\
      }
      {2}{3}
    }
  }
  {
    Database~ \texttt{#1}~ missing.
    \msg_warning:nne {dfg}{database-not-found}{#1}
  }
}
```

Share the table styling between the two-period and three-period layouts. The complete four-row header repeats if the table crosses a page boundary.

```
\cs_new_protected:Npn \@@_gender_leaders_render:nn #1#2
  { \@@_gender_leaders_render:nnnn {#1}{#2}{4}{5} }
```

```
\cs_new_protected:Npn \@@_gender_leaders_render:nnnn #1#2#3#4
{
  \begin{tblr}[
    theme=main,
    note{a}={Reference~ date:~ proposal~ for~ \dfgFundingPeriodText\space funding~ period},
    expand=\l_tblr_body_tl,
    long
  ]{
    vspan=even,
    hspan=minimal,
    row{Z}={font={\footnotesize\bfseries}},
    hline{#4}={1pt,solid,tableLines},
    hline{Y}={1pt,solid,fg={tableOuterLines}},
    hline{2}={wd=0pt},
    row{odd}={bg=white},
    row{even}={bg=tableRows},
    row{1-#3}={bg=tableHeadingsBG},
    rowhead=#3,
    row{2-#3}={guard,valign=t,font={\footnotesize\bfseries},
               fg={tableHeadingsFont},bg=white},
    #1
  }
    #2
    \l_tblr_body_tl
  \end{tblr}
}
```

We continue with the list of existing instrumentation

```
\NewDocumentCommand{\ExistingInstrumentationTable}{ O{existingInstrumentation} }{%
  \def\dfgLocalDB{#1}
  \def\tblrbody{}
  \DTLifdbexists{\dfgLocalDB}{%
    \DTLsort{ref-project-id}{\dfgLocalDB}%
   \DTLforeach*{\dfgLocalDB}{%
    \instProject=ref-project-id,%
    \instInstrument=instrument,%
    \instCompany=company,%
    \instQuantity=quantity,%
    \instYear=year,%
    \instCost=cost,%
    \instSource=financed-by%
   }{%
  \protected@edef\tblrbody{\tblrbody
     \getProject{\instProject}
      & \ifdefempty{\instQuantity}{}{
          \noexpand\dfgShowTag{\instProject/\instInstrument}{quantity}
            {\instQuantity}[\dfgLocalDB]\,\times\,}
        \noexpand\dfgShowTag{\instProject/\instInstrument}{instrument}
          {\instInstrument}[\dfgLocalDB]
      \ifdefempty{\instCompany}{}{\space(
        \noexpand\dfgShowTag{\instProject/\instInstrument}{company}
          {\instCompany}[\dfgLocalDB])}
      & \noexpand\dfgShowTag{\instProject/\instInstrument}{year}
          {\instYear}[\dfgLocalDB]
      & \noexpand\dfgShowTag{\instProject/\instInstrument}{cost}
          {\noexpand\@@_show_existing_instrumentation_cost:n{\instCost}}[\dfgLocalDB]
      & \noexpand\dfgShowTag{\instProject/\instInstrument}{financed-by}
          {\instSource}[\dfgLocalDB]
      \\
    }}%
   \begin{tblr}[%
     theme=main,
     expand=\tblrbody,
     long,
   note{a}={Figures~ in~ thousands},
    ]{%
     %rowsep=1pt,
     vspan=even,
   colspec = {X[-.8,m]X[4.5,m]X[m]X[-1,c,si={table-format=4.1}]X[1.7,m]},
     rowhead = 1,
     column{3}={halign={c}},
   column{4}={halign={r},guard},
     column{5}={halign={l}},
     % cell{1}{4}={valign={m}},
     cell{1}{1-5} = {cmd={\hypersetup{linkcolor=tableLink}}},
     row{1}={valign=t, halign={c},guard},
     % row{1} = {font={\bfseries},fg={white}}
    }
    Project %
     & Description~ of~ instrumentation                          %
     & {Year~ of~                      \\ purchase}
     & \SetCell{c}{{{Cost~ of~          \\purchase}}}\TblrNote{a}
     & {Source~ of~                     \\ funding}              \\
    \tblrbody
   \end{tblr}
  }
```

```
   {
    Database~ \texttt{#1}~ missing.
    \msg_warning:nne { dfg } { database-not-found }{ #1 }
    a }
}
```

We turn to the laboratory animals

```
\NewDocumentCommand{\UpkeepLaboratoryAnimalsTable}{ O{UpkeepLaboratoryAnimals} }{%
  \def\tmpLocalDB{#1}%
  \DTLifdbexists{\tmpLocalDB}{%
   \DTLsort{ref-project-id}{\tmpLocalDB}%
   \DTLsumcolumn{\tmpLocalDB}{quantity}{\totalQuantity}%
   \DTLsumcolumn{\tmpLocalDB}{requested-funds}{\totalRequestedFunds}%
   \DTLsumcolumn{\tmpLocalDB}{existing-funds}{\totalExistingFunds}%
   \DTLconverttodecimal{\totalRequestedFunds}{\totalRequestedFunds}%
   \DTLconverttodecimal{\totalExistingFunds}{\totalExistingFunds}%
   \def\tblrbody{}%
    \def\totalMouseQuantity{0}%
    \def\totalRatQuantity{0}%
   \DTLforeach*{\tmpLocalDB}{%
    \pProject=ref-project-id,%
    \pSpecies=species,%
    \pQuantity=quantity,
    \pWeeksKept=weeks-kept,%
    \pPurchasingCosts=purchasing-costs,%
    \pRequestedFunds=requested-funds,%
    \pExistingFunds=existing-funds%
   }{
    \str_case:en {\pSpecies}{

     {mouse}{%
      \protected@edef\crcSpecies{Mouse}%
      \protected@edef\crcHousingCosts{\dfgMouseHousing}%
      \DTLgadd{\totalMouseQuantity}{\totalMouseQuantity}{\pQuantity}}%
     {mouse-gnotobiotic}{%
      \protected@edef\crcSpecies{Mouse~ {\footnotesize(gnotobiotic)}}%
      \protected@edef\crcHousingCosts{\dfgMouseHousingGnotobiotic}%
      \DTLgadd{\totalMouseQuantity}{\totalMouseQuantity}{\pQuantity}}%
     {rat}{%
      \protected@edef\crcSpecies{Rat}%
      \protected@edef\crcHousingCosts{\dfgRatHousing}%
      \DTLgadd{\totalRatQuantity}{\totalRatQuantity}{\pQuantity}}
    }
    \protected@edef\tblrbody{\tblrbody
     \getProject{\pProject}
      & \noexpand\dfgShowTag{\pProject/\pSpecies}{species}
          {\crcSpecies}[\tmpLocalDB]
      & \noexpand\dfgShowTag{\pProject/\pSpecies}{quantity}
          {\noexpand\@@_show_table_number:n{\pQuantity}}[\tmpLocalDB]
      & \noexpand\dfgShowTag{\pProject/\pSpecies}{weeks-kept}
          {\noexpand\@@_show_table_number:n{\pWeeksKept}}[\tmpLocalDB]
      & \noexpand\dfgShowCalculatedTag{animalHousingRate}
          {settings,\tmpLocalDB}
          {\noexpand\@@_show_table_number:n{\crcHousingCosts}}
      & \noexpand\dfgShowTag{\pProject/\pSpecies}{purchasing-costs}
          {\noexpand\@@_show_eur:n{\pPurchasingCosts}}[\tmpLocalDB]
      & \noexpand\dfgShowTag{\pProject/\pSpecies}{requested-funds}
          {\noexpand\@@_show_eur:n{\pRequestedFunds}}[\tmpLocalDB]
      & \noexpand\dfgShowTag{\pProject/\pSpecies}{existing-funds}
          {\noexpand\@@_show_eur:n{\pExistingFunds}}[\tmpLocalDB] \\
    }   % \DTLconverttodecimal{\totalMouseQuantity}{\totalMouseQuantity}%
   }%
   \begin{tblr}[%
     theme=main,
     expand=\tblrbody,
     long,
   note{a}={Figures~ in~ EUR},
    ]{%
     vspan=even,
      colspec ={*{1}{Q[l]}
        *{1}{X[2,l]}
        *{2}{X[1.3,c,si={table-format=3}]}
        *{1}{X[-1,c,si={table-format=2.2,table-auto-round=false}]}
        *{1}{X[-1,c,si={table-format=5}]}
        *{2}{X[-1,c,si={table-format=6}]}
       },
     rowhead={1},
   column{3-8}={guard},
     %rowfoot = 1,
     hline{Y}={%
               wd=1pt,
               tableLines},
     row{Z}={%
             bg=white,
             fg=black,
             font=\footnotesize\bfseries,
```

```
              halign={c},
              valign={t}},
       % cell{Y}{3}={halign={l}},
       row{1}={%
              guard,
              halign={c},
              cmd={\hypersetup{linkcolor=tableLink}},
             },
     }
     Project
      & Species
      & Quantity
      & {Average~                                                                    \\number~ of~\\weeks~ kept}
      & {Housing~ costs~                                                            \\per~ animal~\\per~ week\TblrNo
      & {Purchasing~                                                                \\costs~ \\per~ animal\TblrNote{
      & {Requested~                                                                 \\funds\TblrNote{a}}
      & {Existing~                                                                  \\funds\TblrNote{a}}
     \tblrbody
     Total
      &
        & \SetCell{halign={l},font={\bfseries\footnotesize}}
         {Mice:~\dfgShowCalculatedTag{UpkeepLaboratoryAnimalsTable}{\tmpLocalDB}
             {\totalMouseQuantity}
          \\Rats:~\dfgShowCalculatedTag{UpkeepLaboratoryAnimalsTable}{\tmpLocalDB}
             {\totalRatQuantity}}
      & \SetCell{tableNoValue}
      & \SetCell{tableNoValue}
      & \SetCell{tableNoValue}
        & \dfgShowCalculatedTag{UpkeepLaboratoryAnimalsTable}{\tmpLocalDB}
           {\@@_show_eur:n{\totalRequestedFunds}}
        & \dfgShowCalculatedTag{UpkeepLaboratoryAnimalsTable}{\tmpLocalDB}
           {\@@_show_eur:n{\totalExistingFunds}}
    \end{tblr}}
   {
    Database~ \texttt{#1}~ missing.
    \msg_warning:nne { dfg } { database-not-found }{ #1 }
   }
}

\newcommand{\dfgRotate}[1]{\rotatebox{90}{\dfgPositionName{#1}}}
```

`\getCountRequestedStaff` with aggregate indexing Funding declarations update a property-list index when they are read. Consequently this command only performs one property lookup; it never traverses a datatool database. This matters especially for tabularray, which evaluates table cells more than once while measuring and typesetting them. Missing tuples and explicit zero values both print the faded multiplication sign used elsewhere in funding tables, preserving the display of the former database sum.

`\getCountRequestedStaff`[⟨*project*⟩]{⟨*year*⟩}{⟨*category*⟩}

Sum of the requested staff `quantity-year`⟨*year*⟩ values in the currently selected requested-funding database, normally `RequestedFunding`, filtered by ⟨*category*⟩ and (when ⟨*project*⟩ is given) by ⟨*project*⟩. Display follows the usual table conventions: an absent or zero sum prints a faded `\texttimes`, and any other value prints verbatim.

```
\tl_new:N \l_@@_requested_staff_database_tl
\tl_set:Nn \l_@@_requested_staff_database_tl { RequestedFunding }

\NewDocumentCommand{\getCountRequestedStaff}{omm}{%
  \tl_set:Nx \l_@@_funding_cache_key_tl
  {
    \IfNoValueTF{#1}
      {\tl_use:N \l_@@_requested_staff_database_tl/
       __ALL__/#3/quantity-year#2}
      {\tl_use:N \l_@@_requested_staff_database_tl/
       #1/#3/quantity-year#2}
  }
  \dfgShowCalculatedTag{getCountRequestedStaff}
    {\l_@@_requested_staff_database_tl}
  {
    \prop_get:NVNTF \g_@@_funding_staff_quantity_prop
      \l_@@_funding_cache_key_tl \l_@@_funding_cache_value_tl
    {
      \fp_compare:nNnTF {\l_@@_funding_cache_value_tl} = {0}
        {\textcolor{tableNoValue!60!black}{\texttimes}}
        {\l_@@_funding_cache_value_tl}
    }
    {\textcolor{tableNoValue!60!black}{\texttimes}}
  }
}
```

\OverviewExistingStaffTable → P. 276 The chapter-2 overview reuses the project-level staff roster instead of maintaining a second set of totals. Only records with `funding-status=existing` are counted. Each record contributes one person to every distinct parent affiliation reached through its `ref-institution-id` list; when that field is absent, both organisation and category are inherited through `ref-person-id` from the central `person` row. Multiple institutions at the same affiliation are counted once. The comma-separated applicant affiliations come from `meta` and occupy the first numeric columns. The heading is singular for one applicant affiliation and plural for several. Other affiliations are added in first-use order.

Because project declarations normally occur after chapter 2, the default database is `RequestedFundingStaffOverview`, restored from the preceding LaTeX run's auxiliary file. Passing `RequestedFundingStaff` explicitly is useful in focused tests or documents that load all rows before the table.

The DFG overview has seven fixed category rows. The more detailed roster categories `medDoc` and `assistant` are folded into enquoteOther research staff and enquoteStudent and graduate assistants, respectively.

\OverviewExistingStaffTable[⟨*database*⟩]

Print the DFG enquoteOverview of existing staff. The optional database defaults to `RequestedFundingStaffOverview`. Use the public call `\printTable` → P. 10 `{OverviewExistingStaff}` in chapter 2. Direct calls also emit the standard skipped-item notice instead of querying unavailable metadata when either fast-compilation mode is active.

```
\seq_new:N  \l_@@_oes_affiliation_seq
\prop_new:N \l_@@_oes_affiliation_name_prop
\prop_new:N \l_@@_oes_affiliation_abbreviation_prop
\prop_new:N \l_@@_oes_count_prop
\prop_new:N \l_@@_oes_row_affiliation_seen_prop
\tl_new:N   \l_@@_oes_applicant_affiliation_tl
\tl_new:N   \l_@@_oes_person_tl
\tl_new:N   \l_@@_oes_category_tl
\tl_new:N   \l_@@_oes_institutions_tl
\tl_new:N   \l_@@_oes_affiliation_tl
\tl_new:N   \l_@@_oes_candidate_affiliation_tl
\tl_new:N   \l_@@_oes_registered_affiliation_tl
\tl_new:N   \l_@@_oes_affiliation_name_tl
\tl_new:N   \l_@@_oes_affiliation_abbreviation_tl
\tl_new:N   \l_@@_oes_lookup_key_tl
\tl_new:N   \l_@@_oes_database_tl
\tl_new:N   \l_@@_oes_table_body_tl
\tl_new:N   \l_@@_oes_colspec_tl
\int_new:N  \l_@@_oes_affiliation_count_int
\int_new:N  \l_@@_oes_applicant_affiliation_count_int
\int_new:N  \l_@@_oes_other_affiliation_count_int
\int_new:N  \l_@@_oes_record_count_int
\clist_new:N \l_@@_oes_applicant_affiliation_clist

\cs_new_protected:Npn \@@_oes_resolve_person_fallbacks:
{
  \tl_if_blank:VF \l_@@_oes_person_tl
  {
    \DTLifdbexists{person}
    {
      \DTLgetvalueforkey{\l_tmpa_tl}{id}{person}{id}
        {\l_@@_oes_person_tl}
      \DTLifnullorempty{\l_tmpa_tl}
      {
        \msg_warning:nnee { dfg } { requested-staff-unresolved-person }
          { \pOESProject } { \l_@@_oes_person_tl }
      }
      {
        \tl_if_blank:VT \l_@@_oes_category_tl
        {
          \DTLgetvalueforkey{\l_tmpa_tl}{position}{person}{id}
            {\l_@@_oes_person_tl}
          \DTLifnullorempty{\l_tmpa_tl}
            { }
            { \tl_set:NV \l_@@_oes_category_tl \l_tmpa_tl }
        }
        \bool_lazy_all:nT
        {
          { \tl_if_blank_p:V \l_@@_oes_institutions_tl }
          { \tl_if_blank_p:V \l_@@_oes_affiliation_tl }
        }
        {
          \DTLgetvalueforkey{\l_tmpa_tl}{ref-institution-id}{person}{id}
            {\l_@@_oes_person_tl}
```

```
6442            \DTLifnullorempty{\l_tmpa_tl}
6443              { }
6444              { \tl_set:NV \l_@@_oes_institutions_tl \l_tmpa_tl }
6445            \tl_if_blank:VT \l_@@_oes_institutions_tl
6446            {
6447              \DTLgetvalueforkey{\l_tmpa_tl}{ref-affiliation-id}{person}{id}
6448                {\l_@@_oes_person_tl}
6449              \DTLifnullorempty{\l_tmpa_tl}
6450                { }
6451                { \tl_set:NV \l_@@_oes_affiliation_tl \l_tmpa_tl }
6452            }
6453          }
6454        }
6455      }
6456      {
6457        \msg_warning:nnee { dfg } { requested-staff-unresolved-person }
6458          { \pOESProject } { \l_@@_oes_person_tl }
6459      }
6460    }
6461  }
6462
6463  \cs_new_protected:Npn \@@_oes_register_affiliation:n #1
6464  {
6465    \tl_set:Ne \l_@@_oes_registered_affiliation_tl {#1}
6466    \tl_trim_spaces:N \l_@@_oes_registered_affiliation_tl
6467    \tl_if_blank:VF \l_@@_oes_registered_affiliation_tl
6468    {
6469      \seq_if_in:NVF \l_@@_oes_affiliation_seq
6470        \l_@@_oes_registered_affiliation_tl
6471      {
6472        \tl_clear:N \l_@@_oes_affiliation_name_tl
6473        \tl_clear:N \l_@@_oes_affiliation_abbreviation_tl
6474        \DTLifdbexists{affiliation}
6475        {
6476          \DTLgetvalueforkey{\l_@@_oes_affiliation_name_tl}
6477            {name}{affiliation}{id}{\l_@@_oes_registered_affiliation_tl}
6478          \DTLifnullorempty{\l_@@_oes_affiliation_name_tl}
6479          {
6480            \msg_warning:nnee { dfg } { requested-staff-unresolved-affiliation }
6481              { OverviewExistingStaff }
6482              { \l_@@_oes_registered_affiliation_tl }
6483            \tl_set:NV \l_@@_oes_affiliation_name_tl
6484              \l_@@_oes_registered_affiliation_tl
6485            \tl_set:NV \l_@@_oes_affiliation_abbreviation_tl
6486              \l_@@_oes_registered_affiliation_tl
6487          }
6488          {
6489            \DTLgetvalueforkey{\l_@@_oes_affiliation_abbreviation_tl}
6490              {abbreviation}{affiliation}{id}
6491              {\l_@@_oes_registered_affiliation_tl}
6492            \DTLifnullorempty{\l_@@_oes_affiliation_abbreviation_tl}
6493            {
6494              \msg_warning:nne { dfg } { requested-staff-missing-abbreviation }
6495                { \l_@@_oes_registered_affiliation_tl }
6496              \tl_set:NV \l_@@_oes_affiliation_abbreviation_tl
6497                \l_@@_oes_affiliation_name_tl
6498            }
6499            { }
6500          }
6501        }
6502        {
6503          \msg_warning:nnee { dfg } { requested-staff-unresolved-affiliation }
6504            { OverviewExistingStaff }
6505            { \l_@@_oes_registered_affiliation_tl }
6506          \tl_set:NV \l_@@_oes_affiliation_name_tl
6507            \l_@@_oes_registered_affiliation_tl
6508          \tl_set:NV \l_@@_oes_affiliation_abbreviation_tl
6509            \l_@@_oes_registered_affiliation_tl
6510        }
6511        \seq_put_right:NV \l_@@_oes_affiliation_seq
6512          \l_@@_oes_registered_affiliation_tl
6513        \prop_put:NVV \l_@@_oes_affiliation_name_prop
6514          \l_@@_oes_registered_affiliation_tl \l_@@_oes_affiliation_name_tl
6515        \prop_put:NVV \l_@@_oes_affiliation_abbreviation_prop
6516          \l_@@_oes_registered_affiliation_tl
6517          \l_@@_oes_affiliation_abbreviation_tl
6518      }
6519    }
6520  }
6521
6522  \cs_new_protected:Npn \@@_oes_count_affiliation:n #1
6523  {
6524    \tl_set:Ne \l_@@_oes_candidate_affiliation_tl {#1}
6525    \tl_trim_spaces:N \l_@@_oes_candidate_affiliation_tl
6526    \tl_if_blank:VF \l_@@_oes_candidate_affiliation_tl
6527    {
6528      \prop_if_in:NVF \l_@@_oes_row_affiliation_seen_prop
```

```
      \l_@@_oes_candidate_affiliation_tl
    {
      \prop_put:NVn \l_@@_oes_row_affiliation_seen_prop
        \l_@@_oes_candidate_affiliation_tl {1}
      \@@_oes_register_affiliation:n
        { \l_@@_oes_candidate_affiliation_tl }
      \tl_set:Nx \l_tmpb_tl
        { \l_@@_oes_category_tl / \l_@@_oes_candidate_affiliation_tl }
      \prop_get:NVNTF \l_@@_oes_count_prop \l_tmpb_tl \l_tmpc_tl
      { \int_set:Nn \l_tmpa_int { \l_tmpc_tl + 1 } }
      { \int_set:Nn \l_tmpa_int { 1 } }
      \prop_put:NVx \l_@@_oes_count_prop \l_tmpb_tl
        { \int_use:N \l_tmpa_int }
    }
  }
}

\cs_new_protected:Npn \@@_oes_collect_current:
{
  \DTLifnullorempty{\pOESPerson}
    { \tl_clear:N \l_@@_oes_person_tl }
    { \tl_set:Ne \l_@@_oes_person_tl {\pOESPerson} }
  \DTLifnullorempty{\pOESCategory}
    { \tl_clear:N \l_@@_oes_category_tl }
    { \tl_set:Ne \l_@@_oes_category_tl {\pOESCategory} }
  \DTLifnullorempty{\pOESInstitutions}
    { \tl_clear:N \l_@@_oes_institutions_tl }
    { \tl_set:Ne \l_@@_oes_institutions_tl {\pOESInstitutions} }
  \DTLifnullorempty{\pOESAffiliation}
    { \tl_clear:N \l_@@_oes_affiliation_tl }
    { \tl_set:Ne \l_@@_oes_affiliation_tl {\pOESAffiliation} }
  \@@_oes_resolve_person_fallbacks:
  \str_case:VnF \l_@@_oes_category_tl
  {
    { prof }        { \tl_set:Nn \l_@@_oes_category_tl { prof } }
    { prof-W2 }     { \tl_set:Nn \l_@@_oes_category_tl { prof } }
    { prof-W3 }     { \tl_set:Nn \l_@@_oes_category_tl { prof } }
    { groupLeader } { \tl_set:Nn \l_@@_oes_category_tl { groupLeader } }
    { postDoc }     { \tl_set:Nn \l_@@_oes_category_tl { postDoc } }
    { PhD }         { \tl_set:Nn \l_@@_oes_category_tl { PhD } }
    { medDoc }      { \tl_set:Nn \l_@@_oes_category_tl { research } }
    { research }    { \tl_set:Nn \l_@@_oes_category_tl { research } }
    { nonResearch } { \tl_set:Nn \l_@@_oes_category_tl { nonResearch } }
    { student }     { \tl_set:Nn \l_@@_oes_category_tl { student } }
    { assistant }   { \tl_set:Nn \l_@@_oes_category_tl { student } }
  }
  { \tl_clear:N \l_@@_oes_category_tl }
  \tl_if_blank:VF \l_@@_oes_category_tl
  {
    \int_incr:N \l_@@_oes_record_count_int
    \prop_clear:N \l_@@_oes_row_affiliation_seen_prop
    \tl_if_blank:VTF \l_@@_oes_institutions_tl
    {
      \tl_if_blank:VTF \l_@@_oes_affiliation_tl
      {
        \msg_warning:nne { dfg } { existing-staff-no-organisation }
          { \pOESProject / \pOESLastname }
      }
      { \@@_oes_count_affiliation:n { \l_@@_oes_affiliation_tl } }
    }
    {
      \clist_set:NV \l_tmpa_clist \l_@@_oes_institutions_tl
      \clist_map_inline:Nn \l_tmpa_clist
      {
        \tl_clear:N \l_tmpa_tl
        \DTLifdbexists{institution}
        {
          \DTLgetvalueforkey{\l_tmpa_tl}
            {ref-affiliation-id}{institution}{id}{##1}
        }
        { }
        \DTLifnullorempty{\l_tmpa_tl}
        {
          \msg_warning:nnee { dfg } { requested-staff-unresolved-institution }
            { OverviewExistingStaff } {##1}
        }
        { \@@_oes_count_affiliation:n { \l_tmpa_tl } }
      }
    }
  }
}

\cs_new_protected:Npn \@@_oes_append_empty_cells:n #1
{
  \int_step_inline:nn {#1}
    { \tl_put_right:Nn \l_@@_oes_table_body_tl { & } }
}
```

```
\cs_new_protected:Npn \@@_oes_append_category:nn #1#2
{
  \tl_put_right:Nn \l_@@_oes_table_body_tl {#2}
  \seq_map_inline:Nn \l_@@_oes_affiliation_seq
  {
    \tl_set:Ne \l_@@_oes_lookup_key_tl { #1 / ##1 }
    \prop_get:NVNTF \l_@@_oes_count_prop
      \l_@@_oes_lookup_key_tl \l_tmpc_tl
    {
      \tl_put_right:Ne \l_@@_oes_table_body_tl
      {
        & \exp_not:N \dfgShowCalculatedTag{OverviewExistingStaffTable}
          {\exp_not:V \l_@@_oes_database_tl,person,institution,affiliation}
          {\exp_not:V \l_tmpc_tl}
      }
    }
    {
      \tl_put_right:Ne \l_@@_oes_table_body_tl
      {
        & \exp_not:N \dfgShowCalculatedTag{OverviewExistingStaffTable}
          {\exp_not:V \l_@@_oes_database_tl,person,institution,affiliation}{0}
      }
    }
  }
  \tl_put_right:Nn \l_@@_oes_table_body_tl { \\ }
}

\cs_new_protected:Npn \@@_oes_build_table_body:
{
  \int_set:Nn \l_@@_oes_affiliation_count_int
    { \seq_count:N \l_@@_oes_affiliation_seq }
  \int_set:Nn \l_@@_oes_other_affiliation_count_int
    {
      \l_@@_oes_affiliation_count_int
      - \l_@@_oes_applicant_affiliation_count_int
    }
  \tl_set:Nx \l_@@_oes_colspec_tl
    {
      X[3,l]~*{\int_use:N \l_@@_oes_affiliation_count_int}{X[1,c]}
    }
  \tl_clear:N \l_@@_oes_table_body_tl
  \tl_put_right:Nx \l_@@_oes_table_body_tl
  {
    \exp_not:N \SetCell [r=3] {c,m} Category
    & \exp_not:N \SetCell
      [c=\int_use:N \l_@@_oes_affiliation_count_int] {c,m}
      {Number~of~persons}
  }
  \@@_oes_append_empty_cells:n
    { \l_@@_oes_affiliation_count_int - 1 }
  \tl_put_right:Nn \l_@@_oes_table_body_tl { \\ & }
  \int_compare:nNnTF
    { \l_@@_oes_applicant_affiliation_count_int } = {1}
  {
    \tl_put_right:Nn \l_@@_oes_table_body_tl
      { \SetCell{c,m} {at~the~applicant\\university} }
  }
  {
    \tl_put_right:Nx \l_@@_oes_table_body_tl
    {
      \exp_not:N \SetCell
        [c=\int_use:N \l_@@_oes_applicant_affiliation_count_int] {c,m}
        {at~the~applicant\\universities}
    }
    \@@_oes_append_empty_cells:n
      { \l_@@_oes_applicant_affiliation_count_int - 1 }
  }
  \int_compare:nNnT { \l_@@_oes_other_affiliation_count_int } > {0}
  {
    \tl_put_right:Nx \l_@@_oes_table_body_tl
    {
      & \exp_not:N \SetCell
        [c=\int_use:N \l_@@_oes_other_affiliation_count_int] {c,m}
        {at~other~participating\\institutions}
    }
    \@@_oes_append_empty_cells:n
      { \l_@@_oes_other_affiliation_count_int - 1 }
  }
  \tl_put_right:Nn \l_@@_oes_table_body_tl { \\ }
  \tl_put_right:Nn \l_@@_oes_table_body_tl { & }
  \seq_map_indexed_inline:Nn \l_@@_oes_affiliation_seq
  {
    \int_compare:nNnF {##1} = {1}
      { \tl_put_right:Nn \l_@@_oes_table_body_tl { & } }
    \tl_set:Nn \l_tmpa_tl {##2}
    \prop_get:NVN \l_@@_oes_affiliation_abbreviation_prop
```

```
6703        \l_tmpa_tl \l_tmpb_tl
6704      \tl_put_right:Ne \l_@@_oes_table_body_tl
6705      {
6706        \exp_not:N \dfgShowCalculatedTag{affiliationLabel}{affiliation}
6707          {\exp_not:V \l_tmpb_tl}
6708      }
6709    }
6710    \tl_put_right:Nn \l_@@_oes_table_body_tl { \\ }
6711    \@@_oes_append_category:nn { prof } { Professors }
6712    \@@_oes_append_category:nn { groupLeader }
6713      { {Junior~research~group\\leaders} }
6714    \@@_oes_append_category:nn { postDoc } { Postdoctoral~researchers }
6715    \@@_oes_append_category:nn { PhD } { Doctoral~researchers }
6716    \@@_oes_append_category:nn { research } { Other~research~staff }
6717    \@@_oes_append_category:nn { nonResearch } { Non-research~staff }
6718    \@@_oes_append_category:nn { student }
6719      { Student~and~graduate~assistants }
6720  }
6721
6722  \cs_new_protected:Npn \@@_oes_print_legend:
6723  {
6724    \vspace*{-1.5\baselineskip}%
6725    %\par\noindent
6726    \group_begin:
6727    \footnotesize\textbf{Affiliations:}\space
6728    \seq_map_indexed_inline:Nn \l_@@_oes_affiliation_seq
6729    {
6730      \int_compare:nNnF {##1} = {1} {;\space}
6731      \tl_set:Nn \l_tmpa_tl {##2}
6732      \prop_get:NVN \l_@@_oes_affiliation_abbreviation_prop
6733        \l_tmpa_tl \l_tmpb_tl
6734      \prop_get:NVN \l_@@_oes_affiliation_name_prop
6735        \l_tmpa_tl \l_tmpc_tl
6736      \textbf{\dfgShowCalculatedTag{affiliationLabel}{affiliation}
6737        {\l_tmpb_tl}}:~
6738      \dfgShowTag{##2}{name}{\l_tmpc_tl}[affiliation]
6739    }
6740    .\par
6741    \group_end:
6742  }
6743
6744  \cs_new_protected:Npn \@@_oes_typeset:n #1
6745  {
6746    \begin{tblr}[
6747      theme=main,
6748      long,
6749      expand=\l_@@_oes_table_body_tl
6750    ]{
6751      colspec={#1},
6752      rowhead=3,
6753      rows={valign=m},
6754      column{2-Z}={halign=c},
6755      row{1-3}={font=\footnotesize\bfseries,
6756        fg=tableHeadingsFont,bg=tableHeadingsBG},
6757      hline{4}={wd=1pt,tableLines},
6758      vline{2}={wd=.75pt,tableLines}
6759    }
6760      \l_@@_oes_table_body_tl
6761    \end{tblr}
6762  }
6763
6764  \cs_new_protected:Npn \@@_overview_existing_staff:n #1
6765  {
6766    \group_begin:
6767    \tl_set:Nn \l_@@_oes_database_tl {#1}
6768    \seq_clear:N \l_@@_oes_affiliation_seq
6769    \prop_clear:N \l_@@_oes_affiliation_name_prop
6770    \prop_clear:N \l_@@_oes_affiliation_abbreviation_prop
6771    \prop_clear:N \l_@@_oes_count_prop
6772    \int_zero:N \l_@@_oes_record_count_int
6773    \tl_clear:N \l_@@_oes_applicant_affiliation_tl
6774    \DTLifdbexists{meta}
6775    {
6776      \DTLgetvalueforkey{\l_@@_oes_applicant_affiliation_tl}
6777        {ref-affiliation-id}{meta}{id}{crc}
6778    }
6779    { }
6780    \DTLifnullorempty{\l_@@_oes_applicant_affiliation_tl}
6781      { \tl_set:Nn \l_@@_oes_applicant_affiliation_tl { applicant } }
6782      { }
6783    \clist_set:NV \l_@@_oes_applicant_affiliation_clist
6784      \l_@@_oes_applicant_affiliation_tl
6785    \clist_map_inline:Nn \l_@@_oes_applicant_affiliation_clist
6786      { \@@_oes_register_affiliation:n {##1} }
6787    \int_set:Nn \l_@@_oes_applicant_affiliation_count_int
6788      { \seq_count:N \l_@@_oes_affiliation_seq }
6789    \DTLifdbexists{#1}
```

```
    {
      \DTLforeach*{#1}
      {
        \pOESProject=ref-project-id,
        \pOESStatus=funding-status,
        \pOESLastname=lastname,
        \pOESPerson=ref-person-id,
        \pOESCategory=category,
        \pOESInstitutions=ref-institution-id,
        \pOESAffiliation=ref-affiliation-id
      }
      {
        \DTLifnullorempty{\pOESStatus}
        { }
        {
          \tl_set:Ne \l_tmpa_tl { \pOESStatus }
          \str_if_eq:VnT \l_tmpa_tl { existing }
            { \@@_oes_collect_current: }
        }
      }
    }
    { \msg_warning:nnn { dfg } { database-not-found } {#1} }
    \int_compare:nNnT { \l_@@_oes_record_count_int } = {0}
      { \msg_warning:nnn { dfg } { existing-staff-no-data } {#1} }
    \@@_oes_build_table_body:
    \exp_args:NV \@@_oes_typeset:n \l_@@_oes_colspec_tl
    \@@_oes_print_legend:
    \group_end:
  }

  \NewDocumentCommand{\OverviewExistingStaffTable}
    { O{RequestedFundingStaffOverview} }
  {
    \bool_if:NTF \l_@@_class_fastcompile_bool
      { \dfgItemSkipped{OverviewExistingStaffTable} }
      { \@@_overview_existing_staff:n {#1} }
  }
```

\OverviewOfFundsRequestedForStaffTable Each body row is assembled by `\tl_put_right:Nx` into `\l_tblr_body_tl` and contains twenty-five `\getCountRequestedStaff` → P. 275 calls (five years × five categories) plus the leading `\getProject{`⟨*project*⟩`}`; the foot row issues another twenty-five calls for the column totals.

No indexing logic appears here: `\addEntry` → P. 2 maintains the aggregate property list and `\getCountRequestedStaff` → P. 275 encapsulates its lookup. The speed-up therefore applies transparently to every other consumer of the macro as well (marginalia, individual project tables, etc.).

\OverviewOfFundsRequestedForStaffTable`[`⟨*database*⟩`]`

Emit the overview of requested staff funding. ⟨*database*⟩ defaults to `RequestedFundingOverview`, which combines funding declared before the document with project-local declarations restored from the preceding run's auxiliary files. The table contains five year bands of five category columns (`postDoc`, `PhD`, `research`, `nonResearch`, `student`), one body row per project, and a "Total" foot row.

```
 \NewDocumentCommand{\OverviewOfFundsRequestedForStaffTable}{ O{RequestedFundingOverview} }{%
  \def\tmpLocalDB{#1}
   \tl_set:Nn \l_@@_requested_staff_database_tl {#1}
   \DTLifdbexists{\tmpLocalDB}{%
  \tl_clear:N \l_tblr_body_tl
  \DTLsort{number}{project}
     \DTLforeach*[
       \not\DTLiseq{\tmpProjectStatus}{E}
       \and
       \not\DTLiseq{\tmpProjectConsiderForLists}{false}
     ]{project}{%
   \pProject               = number,
      \tmpProjectStatus       = project-status,
      \tmpProjectType         = project-type,
       \tmpProjectConsiderForLists = consider-for-lists,
  }{%
  \tl_put_right:Nx \l_tblr_body_tl
   {
    \getProject{\pProject}
    & \getCountRequestedStaff[\pProject]{1}{postDoc}
    & \getCountRequestedStaff[\pProject]{1}{PhD}
    & \getCountRequestedStaff[\pProject]{1}{research}
    & \getCountRequestedStaff[\pProject]{1}{nonResearch}
```

```
6850     & \getCountRequestedStaff[\pProject]{1}{student}
6851
6852     & \getCountRequestedStaff[\pProject]{2}{postDoc}
6853     & \getCountRequestedStaff[\pProject]{2}{PhD}
6854     & \getCountRequestedStaff[\pProject]{2}{research}
6855     & \getCountRequestedStaff[\pProject]{2}{nonResearch}
6856     & \getCountRequestedStaff[\pProject]{2}{student}
6857
6858     & \getCountRequestedStaff[\pProject]{3}{postDoc}
6859     & \getCountRequestedStaff[\pProject]{3}{PhD}
6860     & \getCountRequestedStaff[\pProject]{3}{research}
6861     & \getCountRequestedStaff[\pProject]{3}{nonResearch}
6862     & \getCountRequestedStaff[\pProject]{3}{student}
6863
6864     & \getCountRequestedStaff[\pProject]{4}{postDoc}
6865     & \getCountRequestedStaff[\pProject]{4}{PhD}
6866     & \getCountRequestedStaff[\pProject]{4}{research}
6867     & \getCountRequestedStaff[\pProject]{4}{nonResearch}
6868     & \getCountRequestedStaff[\pProject]{4}{student}
6869
6870     & \getCountRequestedStaff[\pProject]{5}{postDoc}
6871     & \getCountRequestedStaff[\pProject]{5}{PhD}
6872     & \getCountRequestedStaff[\pProject]{5}{research}
6873     & \getCountRequestedStaff[\pProject]{5}{nonResearch}
6874     & \getCountRequestedStaff[\pProject]{5}{student} \\
6875    }
6876       }
6877
6878    \begin{tblr}[%
6879      theme=main,
6880      expand=\l_tblr_body_tl,
6881      long,
6882     ]{%
6883      vspan=even,
6884      colspec = {*{1}{X[2,m]}
6885         *{24}{Q[c]}
6886       },
6887      rowhead = 2,
6888      colsep=4.5pt,
6889      vline{2,7,12,17,22}={tableLines,solid,wd=.75pt},
6890      vline{2,7,12,17,22}={3-Z}{tableLines,solid,wd=.75pt},
6891      hline{2}={wd=0pt},
6892      hline{3}={wd=1pt,tableLines},
6893      hline{Y}={wd=1pt,tableOuterLines},
6894      row{2} = {
6895                fg={tableHeadingsFont},
6896                bg={tableHeadingsBG},
6897                halign={c}
6898               },
6899      row{Z}= {font={\footnotesize\bfseries}}
6900     }
6901      &
6902     \SetCell[c=5]{c} \dfgFundingYear{1}
6903      & & & & &
6904     \SetCell[c=5]{c} \dfgFundingYear{2}
6905      & & & & &
6906     \SetCell[c=5]{c} \dfgFundingYear{3}
6907      & & & & &
6908     \SetCell[c=5]{c} \dfgFundingYear{4}
6909      & & & & &
6910     \SetCell[c=5]{c} \dfgFundingYear{5}
6911      & & & &
6912     \\
6913     \rotatebox{90}{Project} %
6914      & \dfgRotate{postDoc}
6915      & \dfgRotate{PhD}
6916      & \dfgRotate{research}
6917      & \dfgRotate{nonResearch}
6918      & \dfgRotate{student}
6919      & \dfgRotate{postDoc}
6920      & \dfgRotate{PhD}
6921      & \dfgRotate{research}
6922      & \dfgRotate{nonResearch}
6923      & \dfgRotate{student}
6924      & \dfgRotate{postDoc}
6925      & \dfgRotate{PhD}
6926      & \dfgRotate{research}
6927      & \dfgRotate{nonResearch}
6928      & \dfgRotate{student}
6929      & \dfgRotate{postDoc}
6930      & \dfgRotate{PhD}
6931      & \dfgRotate{research}
6932      & \dfgRotate{nonResearch}
6933      & \dfgRotate{student}
6934      & \dfgRotate{postDoc}
6935      & \dfgRotate{PhD}
6936      & \dfgRotate{research}
```

```
    & \dfgRotate{nonResearch}
    & \dfgRotate{student}
   \\
   \l_tblr_body_tl
   Total
    & \getCountRequestedStaff{1}{postDoc}
    & \getCountRequestedStaff{1}{PhD}
    & \getCountRequestedStaff{1}{research}
    & \getCountRequestedStaff{1}{nonResearch}
    & \getCountRequestedStaff{1}{student}
    & \getCountRequestedStaff{2}{postDoc}
    & \getCountRequestedStaff{2}{PhD}
    & \getCountRequestedStaff{2}{research}
    & \getCountRequestedStaff{2}{nonResearch}
    & \getCountRequestedStaff{2}{student}
    & \getCountRequestedStaff{3}{postDoc}
    & \getCountRequestedStaff{3}{PhD}
    & \getCountRequestedStaff{3}{research}
    & \getCountRequestedStaff{3}{nonResearch}
    & \getCountRequestedStaff{3}{student}
    & \getCountRequestedStaff{4}{postDoc}
    & \getCountRequestedStaff{4}{PhD}
    & \getCountRequestedStaff{4}{research}
    & \getCountRequestedStaff{4}{nonResearch}
    & \getCountRequestedStaff{4}{student}
    & \getCountRequestedStaff{5}{postDoc}
    & \getCountRequestedStaff{5}{PhD}
    & \getCountRequestedStaff{5}{research}
    & \getCountRequestedStaff{5}{nonResearch}
    & \getCountRequestedStaff{5}{student}
  \end{tblr}

 }
   {
  Database~ \texttt{#1}~ missing.
  \msg_warning:nne { dfg } { database-not-found }{ #1 }
   }
   \tl_set:Nn \l_@@_requested_staff_database_tl { RequestedFunding }
 }

 \NewDocumentCommand{\getSumRequestedFundsInstrumentation}{som}{
   % #2: project
   % #3: year
   \IfNoValueTF{#2}
     { \@@_funding_sum_set:Nnnnn \SumFundsRequestedInstrumentation
         {RequestedFundingOverview}{__ALL__}{instrumentation}{#3} }
     { \@@_funding_sum_set:Nnnnn \SumFundsRequestedInstrumentation
         {RequestedFundingOverview}{#2}{instrumentation}{#3} }
   \@@_show_calculated_number:nnN
     {getSumRequestedFundsInstrumentation}
     {RequestedFundingOverview}
     \SumFundsRequestedInstrumentation
 }
```

Requested instrumentation overview The overview filters its database for `funding-type=instrumentation`. When the filter yields no rows, the table is suppressed and a sentence names the proposed funding-period number instead. This also covers a freshly created overview database on the first compilation pass.

\OverviewFundsRequestedInstrumentationTable[⟨*database*⟩]

Print the requested-instrumentation overview from ⟨*database*⟩, or state that no instrumentation funds are requested for the current funding period. The database defaults to `RequestedFundingOverview`.

```
 \NewDocumentCommand{\OverviewFundsRequestedInstrumentationTable} { O{RequestedFundingOverview} }{%
 \def\tmpLocalDB{#1}
   \DTLifdbexists{\tmpLocalDB}{%
  \tl_clear:N \l_tblr_body_tl
  \DTLsort{ref-project-id}{\tmpLocalDB}
  \DTLforeach*[\DTLiseq{\pFundingType}{instrumentation}]{\tmpLocalDB}{%
   \pProject=ref-project-id,
         \pFundingType=type,
   \pDescription=category,
   \pSumYearOne=sum-year1,
   \pSumYearTwo=sum-year2,
   \pSumYearThree=sum-year3,
   \pSumYearFour=sum-year4,
   \pSumYearFive=sum-year5
  }{%
  \tl_put_right:Nx \l_tblr_body_tl
```

```
7005    {
7006     \getProject{\pProject}
7007     \printMarginalia{
7008      \getSumRequestedFundsInstrumentation[\pProject]{%
7009       sum-year1,
7010       sum-year2,
7011       sum-year3,
7012       sum-year4,
7013       sum-year5
7014      }
7015     }
7016        & \dfgShowTag{\pProject}{category}{\pDescription}[\tmpLocalDB]
7017        & \dfgShowTag{\pProject}{sum-year1}
7018            {\exp_not:N \@@_show_eur:n {\pSumYearOne}}[\tmpLocalDB]
7019        & \dfgShowTag{\pProject}{sum-year2}
7020            {\exp_not:N \@@_show_eur:n {\pSumYearTwo}}[\tmpLocalDB]
7021        & \dfgShowTag{\pProject}{sum-year3}
7022            {\exp_not:N \@@_show_eur:n {\pSumYearThree}}[\tmpLocalDB]
7023        & \dfgShowTag{\pProject}{sum-year4}
7024            {\exp_not:N \@@_show_eur:n {\pSumYearFour}}[\tmpLocalDB]
7025        & \dfgShowTag{\pProject}{sum-year5}
7026            {\exp_not:N \@@_show_eur:n {\pSumYearFive}}[\tmpLocalDB] \\
7027    }
7028       }
7029  \tl_if_empty:VTF \l_tblr_body_tl
7030  {
7031   \par\noindent
7032   No~funds~for~instrumentation~are~requested~for~funding~period~
7033   \dfgFundingPeriodNumeral.
7034   \par
7035  }
7036  {
7037  \begin{tblr}[%
7038     theme=main,
7039     expand=\l_tblr_body_tl,
7040     long,
7041     remark{Note} = {Figures~ in~ EUR}
7042    ]{%
7043     vspan=even,
7044     row{Z} = {guard,font={\footnotesize\bfseries}},
7045     colspec = {
7046       *{1}{X[-1,t]}
7047        *{1}{X[2,t]}
7048        *{5}{X[-1,c,si={table-format=8}]}
7049      },
7050     rowhead = 2,
7051     column{3-7}={guard},
7052     %colsep=4.5pt,
7053     hline{2}={wd=0pt},
7054     hline{3}={wd=1pt,tableOuterLines},
7055     hline{Y}={wd=1pt,tableOuterLines},
7056     row{2} = {
7057               fg={tableHeadingsFont},
7058               bg={tableHeadingsBG},
7059               halign={c},
7060               guard,
7061              },
7062    }
7063    Project
7064     & {Description~of                                    \\ instrumentation}
7065     & \SetCell[c=5]{c,guard} Funds~ requested~ for
7066     &                                                                    % year 2
7067     &                                                                    % year 3
7068     &                                                                    % year 4
7069     &                                                                    % year 5
7070    \\
7071     &
7072     & \dfgFundingYear{1}
7073     & \dfgFundingYear{2}
7074     & \dfgFundingYear{3}
7075     & \dfgFundingYear{4}
7076     & \dfgFundingYear{5}   \\
7077
7078    \l_tblr_body_tl
7079    \SetCell[c=2]{l}Total
7080    \printMarginalia{
7081     \getSumRequestedFundsInstrumentation{%
7082      sum-year1,
7083      sum-year2,
7084      sum-year3,
7085      sum-year4,
7086      sum-year5
7087     }
7088    }
7089     &
7090     & \getSumRequestedFundsInstrumentation{sum-year1}
7091     & \getSumRequestedFundsInstrumentation{sum-year2}
```

```
    & \getSumRequestedFundsInstrumentation{sum-year3}
    & \getSumRequestedFundsInstrumentation{sum-year4}
    & \getSumRequestedFundsInstrumentation{sum-year5}   \\
 \end{tblr}
}
}
  {
 Database~ \texttt{#1}~ missing.
 \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}

```

Generic memoized `\DTLsumforkeys`

```
\prop_new:N \g_@@_dtlsum_cache_prop
```

Memoized form WITH filter: `\@@_dtlsum_filtered:nnnnN {db}{filter-key}{filter-value}{cols}{result-cs}` This computes `\DTLsumforkeys[\DTLiseq{\pTmp}{value}][\pTmp=key]{d`

```

\cs_new_protected:Npn \@@_dtlsum_filtered:nnnnN #1#2#3#4#5 {
  \tl_set:Nx \l_tmpa_tl
  { #1 || \tl_to_str:n {#2} || \tl_to_str:n {#3} || \tl_to_str:n {#4} }
  \prop_get:NVNTF \g_@@_dtlsum_cache_prop \l_tmpa_tl \l_tmpb_tl
  { \cs_set_eq:NN #5 \l_tmpb_tl }
  {%
   \DTLsumforkeys[\DTLiseq{\pCacheKey}{#3}][\pCacheKey=#2]{#1}{#4}{#5}%
   \prop_gput:NVV \g_@@_dtlsum_cache_prop \l_tmpa_tl #5
  }
}

% Memoized form WITHOUT filter:
%   \@@_dtlsum_all:nnN {db}{cols}{result-cs}
\cs_new_protected:Npn \@@_dtlsum_all:nnN #1#2#3 {
  \tl_set:Nx \l_tmpa_tl
  { #1 || NOFILTER || \tl_to_str:n {#2} }
  \prop_get:NVNTF \g_@@_dtlsum_cache_prop \l_tmpa_tl \l_tmpb_tl
  { \cs_set_eq:NN #3 \l_tmpb_tl }
  {%
   \DTLsumforkeys{#1}{#2}{#3}%
   \prop_gput:NVV \g_@@_dtlsum_cache_prop \l_tmpa_tl #3
  }
}

% ============================================================
%  Drop-in replacements
% ============================================================

\providecommand{\SumPreviousRequestedFunding}{0}
\providecommand{\SumRequestedFunding}{0}

\NewDocumentCommand{\getSumPreviousRequestedFunding}{O{PreviousRequestedFunding}mm}{%
  \DTLifdbexists{#1}{%
   \str_if_eq:nnTF {#2} {all}
   { \@@_dtlsum_all:nnN {#1} {#3} \SumPreviousRequestedFunding }
   { \@@_dtlsum_filtered:nnnnN {#1} {year} {#2} {#3} \SumPreviousRequestedFunding }%
    \@@_show_calculated_number:nnN
      {getSumPreviousRequestedFunding}{#1}
      \SumPreviousRequestedFunding
  }{%
   \texttt{\color{error}XXX}%
   \msg_warning:nne { dfg } { database-not-found } { #1 }%
  }%
}

\NewDocumentCommand{\getSumRequestedFunding}{O{RequestedFundingOverview}mm}{%
  \DTLifdbexists{#1}{%
   \str_if_eq:nnTF {#2} {all}
    { \@@_funding_sum_set:Nnnnn \SumRequestedFunding {#1}
      {__ALL__}{#3}{sum-year1,sum-year2,sum-year3,sum-year4,sum-year5} }
    { \@@_funding_sum_set:Nnnnn \SumRequestedFunding {#1}
      {__ALL__}{#3}{sum-year#2} }%
    \@@_show_calculated_number:nnN
      {getSumRequestedFunding}{#1}
      \SumRequestedFunding
  }{%
   \texttt{\color{error}XXX}%
   \msg_warning:nne { dfg } { database-not-found } { #1 }%
  }%
}

\NewDocumentCommand{\getTotalSumRequestedFunding}{O{RequestedFundingOverview}mm}{%
  \DTLifdbexists{#1}{%
    \tl_set:Nn \l_tmpa_tl {#3}
```

```
7170      \str_if_eq:nnT {#3} {all} { \tl_set:Nn \l_tmpa_tl {__ALL__} }
7171      \str_if_eq:nnTF {#2} {all}
7172        { \@@_funding_sum_set:Nnnnn \SumRequestedFunding {#1}
7173            {__ALL__}{\l_tmpa_tl}
7174            {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5} }
7175        { \@@_funding_sum_set:Nnnnn \SumRequestedFunding {#1}
7176            {__ALL__}{\l_tmpa_tl}{sum-year#2} }%
7177      \@@_show_calculated_number:nnN
7178        {getTotalSumRequestedFunding}{#1}
7179        \SumRequestedFunding
7180    }{%
7181     \texttt{\color{error}XXX}%
7182     \msg_warning:nne { dfg } { database-not-found } { #1 }%
7183    }%
7184  }
7185
7186  \NewDocumentCommand{\PreviousAndCurrentRequestedFundingTable}{O{PreviousRequestedFunding}}
7187  {
7188    \def\tmpLocalDB{#1}
7189    \begin{tblr}[%
7190      theme=main,
7191      expand=\l_tblr_body_tl,
7192      long,
7193    remark{Note} = {Figures~ in~ EUR}
7194     ]{%
7195      row{8,Z} = {guard,font={\footnotesize\bfseries}},
7196      colspec = {
7197          *{1}{t{3.5cm}}@{}
7198          *{6}{X[-1,c,si={table-format=8}]}
7199        },
7200      rowhead = 2,
7201    column{2-7}={guard},
7202      %colsep=4.5pt,
7203      hline{2}={wd=0pt},
7204      hline{3}={wd=1pt,tableOuterLines},
7205      % hline{Y}={wd=1pt,tableOuterLines},
7206      row{1,2} = {
7207                  fg={tableHeadingsFont},
7208                  bg={tableHeadingsBG},
7209                  halign={c},
7210                  guard,
7211                 },
7212     }
7213     \SetCell[r=2]{b}{Financial~year/                          \\funding~period}
7214      & \SetCell[c=5]{c} Funding~ for                          % staff
7215      &                                                       % costs
7216      &                                                       % instrumentation
7217      &                                                       % fellowships
7218      &                                                       % globalfunds
7219      & \SetCell[r=2]{b} Total                                \\
7220      &
7221     Staff
7222      & Direct~ costs
7223      & Instrumentation
7224      & Fellowships
7225      & Globalfunds
7226      &                                                       \\
7227   \dfgFundingYearPrevious{1}
7228      \printMarginalia{%
7229        \getSumPreviousRequestedFunding{1}{%
7230          staff,costs,instrumentation,fellowships,globalfunds}%
7231      }
7232      & \getSumPreviousRequestedFunding{1}{staff}             % staff
7233      & \getSumPreviousRequestedFunding{1}{costs}             % costs
7234      & \getSumPreviousRequestedFunding{1}{instrumentation}   % instrumentation
7235      & \getSumPreviousRequestedFunding{1}{fellowships}       % fellowships
7236      & \getSumPreviousRequestedFunding{1}{globalfunds}       % globalfunds
7237      &
7238     \getSumPreviousRequestedFunding{1}{staff,costs,instrumentation,
7239     fellowships,globalfunds}
7240     \\ % Total
7241
7242   \dfgFundingYearPrevious{2}
7243      \printMarginalia{%
7244        \getSumPreviousRequestedFunding{2}{%
7245          staff,costs,instrumentation,fellowships,globalfunds}%
7246      }
7247      & \getSumPreviousRequestedFunding{2}{staff}             % staff
7248      & \getSumPreviousRequestedFunding{2}{costs}             % costs
7249      & \getSumPreviousRequestedFunding{2}{instrumentation}   % instrumentation
7250      & \getSumPreviousRequestedFunding{2}{fellowships}       % fellowships
7251      & \getSumPreviousRequestedFunding{2}{globalfunds}       % globalfunds
7252      &
7253     \getSumPreviousRequestedFunding{2}{staff,costs,instrumentation,
7254     fellowships,globalfunds}
7255     \\ % Total
7256   \dfgFundingYearPrevious{3}
```

```
7257      \printMarginalia{%
7258        \getSumPreviousRequestedFunding{3}{%
7259          staff,costs,instrumentation,fellowships,globalfunds}%
7260      }
7261      & \getSumPreviousRequestedFunding{3}{staff}             % staff
7262      & \getSumPreviousRequestedFunding{3}{costs}             % costs
7263      & \getSumPreviousRequestedFunding{3}{instrumentation}   % instrumentation
7264      & \getSumPreviousRequestedFunding{3}{fellowships}       % fellowships
7265      & \getSumPreviousRequestedFunding{3}{globalfunds}       % globalfunds
7266      &
7267     \getSumPreviousRequestedFunding{3}{staff,costs,instrumentation,
7268      fellowships,globalfunds}
7269     \\ % Total
7270
7271   \dfgFundingYearPrevious{4}
7272      \printMarginalia{%
7273        \getSumPreviousRequestedFunding{4}{%
7274          staff,costs,instrumentation,fellowships,globalfunds}%
7275      }
7276      & \getSumPreviousRequestedFunding{4}{staff}             % staff
7277      & \getSumPreviousRequestedFunding{4}{costs}             % costs
7278      & \getSumPreviousRequestedFunding{4}{instrumentation}   % instrumentation
7279      & \getSumPreviousRequestedFunding{4}{fellowships}       % fellowships
7280      & \getSumPreviousRequestedFunding{4}{globalfunds}       % globalfunds
7281      &
7282     \getSumPreviousRequestedFunding{4}{staff,costs,instrumentation,
7283      fellowships,globalfunds}
7284     \\ % Total
7285
7286   \dfgFundingYearPrevious{5}
7287      \printMarginalia{%
7288        \getSumPreviousRequestedFunding{5}{%
7289          staff,costs,instrumentation,fellowships,globalfunds}%
7290      }
7291      & \getSumPreviousRequestedFunding{5}{staff}             % staff
7292      & \getSumPreviousRequestedFunding{5}{costs}             % costs
7293      & \getSumPreviousRequestedFunding{5}{instrumentation}   % instrumentation
7294      & \getSumPreviousRequestedFunding{5}{fellowships}       % fellowships
7295      & \getSumPreviousRequestedFunding{5}{globalfunds}       % globalfunds
7296      &
7297     \getSumPreviousRequestedFunding{5}{staff,costs,instrumentation,
7298      fellowships,globalfunds}
7299     \\ % Total
7300     \SetHline{-}{tableOuterLines,0.5pt}
7301     \SetRow{belowsep=5pt}
7302      Ending~Funding~period
7303      \printMarginalia{%
7304        \getSumPreviousRequestedFunding{all}{%
7305          staff,costs,instrumentation,fellowships,globalfunds}%
7306      }
7307      & \getSumPreviousRequestedFunding{all}{staff}           % staff
7308      & \getSumPreviousRequestedFunding{all}{costs}           % costs
7309      & \getSumPreviousRequestedFunding{all}{instrumentation} % instrumentation
7310      & \getSumPreviousRequestedFunding{all}{fellowships}     % fellowships
7311      & \getSumPreviousRequestedFunding{all}{globalfunds}     % globalfunds
7312      &
7313     \getSumPreviousRequestedFunding{all}{staff,costs,instrumentation,
7314      fellowships,globalfunds}
7315     \\ % Total
7316
7317     \SetHline{-}{tableOuterLines,1pt}
7318   \dfgFundingYear{1}
7319      \printMarginalia{\getTotalSumRequestedFunding{1}{all}}
7320      & \getSumRequestedFunding{1}{staff}
7321      & \getSumRequestedFunding{1}{costs}
7322      & \getSumRequestedFunding{1}{instrumentation}
7323      & \getSumRequestedFunding{1}{fellowships}
7324      & \getSumRequestedFunding{1}{globalfunds}
7325      & \getTotalSumRequestedFunding{1}{all}
7326     \\
7327   \dfgFundingYear{2}
7328      \printMarginalia{\getTotalSumRequestedFunding{2}{all}}
7329      & \getSumRequestedFunding{2}{staff}
7330      & \getSumRequestedFunding{2}{costs}
7331      & \getSumRequestedFunding{2}{instrumentation}
7332      & \getSumRequestedFunding{2}{fellowships}
7333      & \getSumRequestedFunding{2}{globalfunds}
7334      & \getTotalSumRequestedFunding{2}{all}
7335     \\
7336
7337   \dfgFundingYear{3}
7338      \printMarginalia{\getTotalSumRequestedFunding{3}{all}}
7339      & \getSumRequestedFunding{3}{staff}
7340      & \getSumRequestedFunding{3}{costs}
7341      & \getSumRequestedFunding{3}{instrumentation}
7342      & \getSumRequestedFunding{3}{fellowships}
7343      & \getSumRequestedFunding{3}{globalfunds}
```

```
     & \getTotalSumRequestedFunding{3}{all}
    \\
  \dfgFundingYear{4}
     \printMarginalia{\getTotalSumRequestedFunding{4}{all}}
     & \getSumRequestedFunding{4}{staff}
     & \getSumRequestedFunding{4}{costs}
     & \getSumRequestedFunding{4}{instrumentation}
     & \getSumRequestedFunding{4}{fellowships}
     & \getSumRequestedFunding{4}{globalfunds}
     & \getTotalSumRequestedFunding{4}{all}
    \\
  \dfgFundingYear{5}
     \printMarginalia{\getTotalSumRequestedFunding{5}{all}}
     & \getSumRequestedFunding{5}{staff}
     & \getSumRequestedFunding{5}{costs}
     & \getSumRequestedFunding{5}{instrumentation}
     & \getSumRequestedFunding{5}{fellowships}
     & \getSumRequestedFunding{5}{globalfunds}
     & \getTotalSumRequestedFunding{5}{all}
    \\

    \SetHline{-}{tableOuterLines,0.5pt}
    New~ Funding~period
     \printMarginalia{\getTotalSumRequestedFunding{all}{all}}
     & \getTotalSumRequestedFunding{all}{staff}              % staff
     & \getTotalSumRequestedFunding{all}{costs}              % costs
     & \getTotalSumRequestedFunding{all}{instrumentation}    % instrumentation
     & \getTotalSumRequestedFunding{all}{fellowships}        % fellowships
     & \getTotalSumRequestedFunding{all}{globalfunds}        % globalfunds
     &
    \getTotalSumRequestedFunding{all}{all}
    \\ % Total
   \end{tblr}
}
```

Section 2.1.1 reuses the project-level existing-funds declarations. Both the stored values and the overview display use full euros. The shared euro formatter adds thousands separators and rounds display values to whole euros (for example, `5000` is printed as `5,000`).

`ref-affiliation-id` determines the column: every id listed in `meta/crc.ref-affiliation-id` is applicant core support, the sentinel `other-funds` selects other funding sources, and every other affiliation id is treated as another participating institution. The descriptive `financed-by` value does not perform this classification.

```
\cs_new_protected:Npn
  \@@_get_sum_previous_requested_funding_eur:nnn #1#2#3
{
  \DTLifdbexists{#1}
  {
    \str_if_eq:nnTF {#2} {all}
      { \@@_dtlsum_all:nnN {#1} {#3} \SumPreviousRequestedFunding }
      { \@@_dtlsum_filtered:nnnnN
          {#1} {year} {#2} {#3} \SumPreviousRequestedFunding }
    \@@_show_calculated_number:nnN
      {getSumPreviousRequestedFunding}{#1}
      \SumPreviousRequestedFunding
  }
  {
    \texttt{\color{error}XXX}
    \msg_warning:nne { dfg } { database-not-found } {#1}
  }
}

\NewDocumentCommand{\getSumRequestedFundingDirectCosts}{
  O{RequestedAndExistingFundingDirectCostsOverview}
  m m }
{
  % #1: database
  % #2: year
  % #3: affiliation

  \str_if_eq:nnTF {#2}{all}
  {
   \str_case:nn {#3}
   {
  {applicant} {\DTLsumforkeys[\DTLisinlist{\pAffiliation}{\theApplicantID}]
     [\pAffiliation=ref-affiliation-id]{#1}
     {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
     {\SumRequestedFundingDirectCosts}}

    {participant}{
     \DTLsumforkeys[
    \not\DTLisinlist{\pAffiliation}{\theApplicantID}
      \and
```

```
7418        \not\DTLiseq{\pAffiliation}{other-funds}
7419       ]
7420       [\pAffiliation=ref-affiliation-id]{#1}
7421       {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
7422       {\SumRequestedFundingDirectCosts}}
7423
7424      {other-funds}{\DTLsumforkeys[\DTLiseq{\pAffiliation}{other-funds}]
7425       [\pAffiliation=ref-affiliation-id]{#1}
7426       {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
7427       {\SumRequestedFundingDirectCosts}}
7428
7429      {all}{\DTLsumforkeys{#1}
7430       {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
7431       {\SumRequestedFundingDirectCosts}}
7432     }}
7433    {
7434     \str_case:nn {#3}
7435     {
7436    {applicant} {\DTLsumforkeys[\DTLisinlist{\pAffiliation}{\theApplicantID}]
7437       [\pAffiliation=ref-affiliation-id]{#1}
7438       {sum-year#2}
7439       {\SumRequestedFundingDirectCosts}}
7440
7441      {participant}{
7442       \DTLsumforkeys[
7443      \not\DTLisinlist{\pAffiliation}{\theApplicantID}
7444        \and
7445        \not\DTLiseq{\pAffiliation}{other-funds}
7446       ]
7447       [\pAffiliation=ref-affiliation-id]{#1}
7448       {sum-year#2}
7449       {\SumRequestedFundingDirectCosts}}
7450
7451      {other-funds}{\DTLsumforkeys[\DTLiseq{\pAffiliation}{other-funds}]
7452       [\pAffiliation=ref-affiliation-id]{#1}
7453       {sum-year#2}
7454       {\SumRequestedFundingDirectCosts}}
7455
7456      {all}{\DTLsumforkeys{#1}
7457       {sum-year#2}
7458       {\SumRequestedFundingDirectCosts}}
7459     }}
7460
7461
7462    \@@_show_calculated_number:nnN
7463      {getSumRequestedFundingDirectCosts}{#1}
7464      \SumRequestedFundingDirectCosts
7465  }
7466
7467  \NewDocumentCommand{\ExistingFundsDirectCostsTable}
7468    {O{RequestedAndExistingFundingDirectCostsOverview}}
7469  {
7470    \DTLifdbexists{#1}{
7471     \begin{tblr}[%
7472       theme=main,
7473       % expand=\l_tblr_body_tl,
7474       long,
7475     remark{Note} = {Figures~ in~ EUR}
7476      ]{%
7477       row{7,Z} = {guard,font={\footnotesize\bfseries}},
7478     colspec = {
7479          *{1}{t{3.5cm}}
7480          *{4}{X[-1,c,si={table-format=9}]}
7481        },
7482       % rowhead = 1,
7483     column{2-5}={guard},
7484       %colsep=4.5pt,
7485       % hline{3}={wd=1pt,tableOuterLines},
7486       % hline{Y}={wd=1pt,tableOuterLines},
7487       row{1} = {
7488                  fg={tableHeadingsFont},
7489                  bg={tableHeadingsBG},
7490                  halign={c},
7491                  guard,
7492                 },
7493      }
7494     {Financial~year/                                                                    \\funding~period}
7495     & {Core~ support~ provided~                                                        \\by~ applicant~
7496       \dfgApplicantAffiliationsTF{university}{universities}}
7497
7498       & {Core~ support~ provided~                                                         \\by~
7499          other~ participating~ institutions}
7500       & Other~ funds
7501       & Total
7502
7503    \dfgFundingYearPrevious{1}
7504        \printMarginalia{%
```

```
7505          \@@_get_sum_previous_requested_funding_eur:nnn
7506            {PreviousExistingFundsDirectCosts}{1}{%
7507            applicant,participant,other-funds}%
7508        }
7509     & {\@@_get_sum_previous_requested_funding_eur:nnn
7510          {PreviousExistingFundsDirectCosts}{1}{applicant}}
7511     & {\@@_get_sum_previous_requested_funding_eur:nnn
7512          {PreviousExistingFundsDirectCosts}{1}{participant}}
7513     & {\@@_get_sum_previous_requested_funding_eur:nnn
7514          {PreviousExistingFundsDirectCosts}{1}{other-funds}}
7515     &
7516    {\@@_get_sum_previous_requested_funding_eur:nnn
7517        {PreviousExistingFundsDirectCosts}{1}{
7518      applicant,
7519      participant,
7520      other-funds}}                                                  \\
7521
7522    \dfgFundingYearPrevious{2}
7523        \printMarginalia{%
7524          \@@_get_sum_previous_requested_funding_eur:nnn
7525            {PreviousExistingFundsDirectCosts}{2}{%
7526            applicant,participant,other-funds}%
7527        }
7528     & {\@@_get_sum_previous_requested_funding_eur:nnn
7529          {PreviousExistingFundsDirectCosts}{2}{applicant}}
7530     & {\@@_get_sum_previous_requested_funding_eur:nnn
7531          {PreviousExistingFundsDirectCosts}{2}{participant}}
7532     & {\@@_get_sum_previous_requested_funding_eur:nnn
7533          {PreviousExistingFundsDirectCosts}{2}{other-funds}}
7534     & {\@@_get_sum_previous_requested_funding_eur:nnn
7535          {PreviousExistingFundsDirectCosts}{2}{
7536        applicant,
7537        participant,
7538        other-funds}}                                                \\
7539
7540    \dfgFundingYearPrevious{3}
7541        \printMarginalia{%
7542          \@@_get_sum_previous_requested_funding_eur:nnn
7543            {PreviousExistingFundsDirectCosts}{3}{%
7544            applicant,participant,other-funds}%
7545        }
7546     & {\@@_get_sum_previous_requested_funding_eur:nnn
7547          {PreviousExistingFundsDirectCosts}{3}{applicant}}
7548     & {\@@_get_sum_previous_requested_funding_eur:nnn
7549          {PreviousExistingFundsDirectCosts}{3}{participant}}
7550     & {\@@_get_sum_previous_requested_funding_eur:nnn
7551          {PreviousExistingFundsDirectCosts}{3}{other-funds}}
7552     & {\@@_get_sum_previous_requested_funding_eur:nnn
7553          {PreviousExistingFundsDirectCosts}{3}{
7554        applicant,
7555        participant,
7556        other-funds}}                                                \\
7557
7558    \dfgFundingYearPrevious{4}
7559        \printMarginalia{%
7560          \@@_get_sum_previous_requested_funding_eur:nnn
7561            {PreviousExistingFundsDirectCosts}{4}{%
7562            applicant,participant,other-funds}%
7563        }
7564     & {\@@_get_sum_previous_requested_funding_eur:nnn
7565          {PreviousExistingFundsDirectCosts}{4}{applicant}}
7566     & {\@@_get_sum_previous_requested_funding_eur:nnn
7567          {PreviousExistingFundsDirectCosts}{4}{participant}}
7568     & {\@@_get_sum_previous_requested_funding_eur:nnn
7569          {PreviousExistingFundsDirectCosts}{4}{other-funds}}
7570     & {\@@_get_sum_previous_requested_funding_eur:nnn
7571          {PreviousExistingFundsDirectCosts}{4}{
7572        applicant,
7573        participant,
7574        other-funds}}                                                \\
7575
7576    \dfgFundingYearPrevious{5}
7577        \printMarginalia{%
7578          \@@_get_sum_previous_requested_funding_eur:nnn
7579            {PreviousExistingFundsDirectCosts}{5}{%
7580            applicant,participant,other-funds}%
7581        }
7582     & {\@@_get_sum_previous_requested_funding_eur:nnn
7583          {PreviousExistingFundsDirectCosts}{5}{applicant}}
7584     & {\@@_get_sum_previous_requested_funding_eur:nnn
7585          {PreviousExistingFundsDirectCosts}{5}{participant}}
7586     & {\@@_get_sum_previous_requested_funding_eur:nnn
7587          {PreviousExistingFundsDirectCosts}{5}{other-funds}}
7588     & {\@@_get_sum_previous_requested_funding_eur:nnn
7589          {PreviousExistingFundsDirectCosts}{5}{
7590        applicant,
7591        participant,
```

```
        other-funds}}                                                 \\

     \SetHline{-}{tableOuterLines,0.5pt}
     \SetRow{belowsep=5pt}
       Ending~Funding~period
       \printMarginalia{%
         \@@_get_sum_previous_requested_funding_eur:nnn
           {PreviousExistingFundsDirectCosts}{all}{%
           applicant,participant,other-funds}%
       }
    & {\@@_get_sum_previous_requested_funding_eur:nnn
         {PreviousExistingFundsDirectCosts}{all}{applicant}}
    & {\@@_get_sum_previous_requested_funding_eur:nnn
         {PreviousExistingFundsDirectCosts}{all}{participant}}
    & {\@@_get_sum_previous_requested_funding_eur:nnn
         {PreviousExistingFundsDirectCosts}{all}{other-funds}}
    & {\@@_get_sum_previous_requested_funding_eur:nnn
         {PreviousExistingFundsDirectCosts}{all}{
        applicant,
        participant,
        other-funds}}                                             \\

     \SetHline{-}{tableOuterLines,1pt}
  \dfgFundingYear{1}
      \printMarginalia{\getSumRequestedFundingDirectCosts[#1]{1}{all}}
    & \getSumRequestedFundingDirectCosts[#1]{1}{applicant}
    & \getSumRequestedFundingDirectCosts[#1]{1}{participant}
    & \getSumRequestedFundingDirectCosts[#1]{1}{other-funds}
    & \getSumRequestedFundingDirectCosts[#1]{1}{all}
  \dfgFundingYear{2}
      \printMarginalia{\getSumRequestedFundingDirectCosts[#1]{2}{all}}
    & \getSumRequestedFundingDirectCosts[#1]{2}{applicant}
    & \getSumRequestedFundingDirectCosts[#1]{2}{participant}
    & \getSumRequestedFundingDirectCosts[#1]{2}{other-funds}
    & \getSumRequestedFundingDirectCosts[#1]{2}{all}
  \dfgFundingYear{3}
      \printMarginalia{\getSumRequestedFundingDirectCosts[#1]{3}{all}}
    & \getSumRequestedFundingDirectCosts[#1]{3}{applicant}
    & \getSumRequestedFundingDirectCosts[#1]{3}{participant}
    & \getSumRequestedFundingDirectCosts[#1]{3}{other-funds}
    & \getSumRequestedFundingDirectCosts[#1]{3}{all}
  \dfgFundingYear{4}
      \printMarginalia{\getSumRequestedFundingDirectCosts[#1]{4}{all}}
    & \getSumRequestedFundingDirectCosts[#1]{4}{applicant}
    & \getSumRequestedFundingDirectCosts[#1]{4}{participant}
    & \getSumRequestedFundingDirectCosts[#1]{4}{other-funds}
    & \getSumRequestedFundingDirectCosts[#1]{4}{all}
  \dfgFundingYear{5}
      \printMarginalia{\getSumRequestedFundingDirectCosts[#1]{5}{all}}
    & \getSumRequestedFundingDirectCosts[#1]{5}{applicant}
    & \getSumRequestedFundingDirectCosts[#1]{5}{participant}
    & \getSumRequestedFundingDirectCosts[#1]{5}{other-funds}
    & \getSumRequestedFundingDirectCosts[#1]{5}{all}


     \SetHline{-}{tableOuterLines,0.5pt}
  New~ Funding~period
      \printMarginalia{\getSumRequestedFundingDirectCosts[#1]{all}{all}}
    & \getSumRequestedFundingDirectCosts[#1]{all}{applicant}
    & \getSumRequestedFundingDirectCosts[#1]{all}{participant}
    & \getSumRequestedFundingDirectCosts[#1]{all}{other-funds}
    & \getSumRequestedFundingDirectCosts[#1]{all}{all}
    \end{tblr}
  }{
    \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}
```

The core part is definitely all about the projects.

```
\clist_new:N \l_tmp_research_area_clist
\bool_new:N  \l_@@_project_one_PI_bool
\bool_new:N  \g_@@_fast_project_started_bool

\NewDocumentCommand{\printProjectHead}{m}
{%
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  {
    \bool_if:NF \g_@@_fast_project_started_bool
    {
      \setcounter{page}{1}% Ignore page checkpoints restored by excluded files.
      \bool_gset_true:N \g_@@_fast_project_started_bool
    }
    \def\tmpLocalDB{project}%
    \setcounter{section}{0}%
    \setcounter{figure}{0}%
    \setcounter{table}{0}%
```

```
7676      \setcounter{secnumdepth}{2}%
7677      \restartlist{workpackages}%
7678      \tl_set:Nn \l__tmp_current_project_tl {#1}%
7679      \cs_set:Npn \tmpProjectNumber       {#1}%
7680      \cs_set:Npn \tmpProjectTitle        { }%
7681      \cs_set:Npn \tmpProjectStatus       { }%
7682      \cs_set:Npn \tmpProjectType         { }%
7683      \cs_set:Npn \tmpProjectResearchArea { }%
7684      \cs_set:Npn \tmpProjectPIids        { }%
7685      \bool_set_true:N \l_@@_project_one_PI_bool
7686      \renewcommand*{\thesection}      {\arabic{section}}%
7687      \renewcommand*{\thesubsection}   {\thesection.\arabic{subsection}}%
7688      \renewcommand*{\thesubsubsection}{\thesubsection.\arabic{subsubsection}}%
7689      \renewcommand*{\thefigure}       {\arabic{figure}}%
7690      \renewcommand*{\thetable}        {\arabic{table}}%
7691      \renewcommand*{\theHpart}        {\thepart.#1}%
7692      \renewcommand*{\theHchapter}     {\thechapter.#1}%
7693      \renewcommand*{\theHsection}     {\thesection.#1}%
7694      \renewcommand*{\theHsubsection}  {\thesubsection.#1}%
7695      \renewcommand*{\theHsubsubsection}{\thesubsubsection.#1}%
7696      \renewcommand*{\theHfigure}      {\thefigure.#1}%
7697      \renewcommand*{\theHtable}       {\thetable.#1}%
7698      \renewcommand*{\theHfootnote}    {\thefootnote.#1}%
7699      \chead*{\textbf{Project~#1}\enspace|\enspace FAST~COMPILE~-~DRAFT}%
7700      \printThumbIndexOff
7701      \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
7702      {
7703        \@@_project_results_prepare:n {#1}%
7704        \citereset
7705        \begin{refsection}
7706      }
7707    }
7708    {
7709    % #1: project id
7710    \cleardoublepage
7711  \phantomsection
7712    \def\tmpLocalDB{project}%
7713    \setcounter{section}{0}%
7714    \setcounter{figure}{0}%
7715    \setcounter{table}{0}%
7716    \setcounter{tocdepth}{0}%
7717    \restartlist{workpackages}%
7718    \tl_set:Nn \l__tmp_current_project_tl {#1}%
7719    \@@_project_results_prepare:n {#1}%
7720
7721    \renewcommand*{\theHpart}        {\thepart.#1}%
7722    \renewcommand*{\theHchapter}     {\thechapter.#1}%
7723    \renewcommand*{\theHsection}     {\thesection.#1}%
7724    \renewcommand*{\theHsubsection}  {\thesubsection.#1}%
7725    \renewcommand*{\theHsubsubsection}{\thesubsubsection.#1}%
7726    \renewcommand*{\theHfigure}      {\thefigure.#1}%
7727    \renewcommand*{\theHtable}       {\thetable.#1}%
7728    \renewcommand*{\theHfootnote}    {\thefootnote.#1}%
7729
7730    % --- single row scan for all project columns we need --------------------
7731    \getEntry*[\tmpLocalDB]{#1}{
7732     \tmpProjectNumber       = number,
7733     \tmpProjectTitle        = name,
7734     \tmpProjectStatus       = project-status,
7735     \tmpProjectType         = project-type,
7736     \tmpProjectResearchArea = research-area,
7737     \tmpProjectPIids        = ref-person-id
7738    }%
7739
7740   % Register the bibliography aliases of every PI assigned to this project.
7741   % The mapping parameter is doubled because this loop occurs inside the
7742   % definition of \printProjectHead.
7743   \datatool_if_has_key:nnT { person } { name-alias }
7744   {
7745    \clist_map_inline:Nn \tmpProjectPIids
7746    {
7747     \DTLgetvalueforkey { \theAliases } { name-alias } { person } { id } { ##1 }
7748     \DTLifnullorempty { \theAliases }
7749     { }
7750     { \exp_args:No \addboldnames { \theAliases } }
7751    }
7752     }
7753
7754    % --- precompute the few derived booleans we actually use ----------------
7755    \int_compare:nNnTF { \clist_count:N \tmpProjectPIids } > {1}
7756     { \bool_set_false:N \l_@@_project_one_PI_bool }
7757     { \bool_set_true:N  \l_@@_project_one_PI_bool }%
7758
7759    \citereset
7760    \hypertarget{project:#1}{}%
7761
7762    \use:x
```

```
7763    {
7764     \exp_not:N \printThumbIndex
7765     [ project \tl_head:n {#1} ]
7766     { \exp_not:n {#1} }
7767    }%
7768
7769    \chead{\getPIofProject{#1}[\space\cdot\space]%
7770      \space|\space
7771      \dfgShowTag{#1}{name}{\tmpProjectTitle}[project]}%
7772
7773
7774
7775   % --- toc entry ------------------------------------------------------
7776    \addtocentrydefault{section}{
7777      \texorpdfstring
7778        { \dfgShowTag{#1}{number}{\tmpProjectNumber}[project] }
7779        { \tmpProjectNumber }
7780      \str_if_eq:eeF {\tmpProjectStatus} {C}
7781        {
7782          \space
7783          \texorpdfstring
7784            {\dfgShowTag{#1}{project-status}{\tmpProjectStatus}[project]}
7785            {\tmpProjectStatus}
7786        }%
7787    }
7788    {
7789      \texorpdfstring{\dfgShowTag{#1}{name}{\tmpProjectTitle}[project]}{}%
7790    }%
7791    \setcounter{secnumdepth}{2}%
7792    \addtocontents{toc}{\protect\setcounter{tocdepth}{0}}%
7793    \hypersetup{bookmarksdepth=0}%
7794    \begin{refsection}
7795
7796    \section{%
7797      General~information~about~Project~%
7798      \texorpdfstring
7799      {
7800        \getProject{#1}%
7801        \str_if_eq:eeF {\tmpProjectStatus} {C}
7802          { \c_space_tl
7803            \dfgShowTag{#1}{project-status}{\tmpProjectStatus}[project] }%
7804      }
7805      {\tmpProjectNumber}%
7806     }
7807   \subsection{Project~title:~
7808     \texorpdfstring
7809       {\dfgShowTag{#1}{name}{\tmpProjectTitle}[project]}
7810       {\tmpProjectTitle}}
7811
7812    % --- research areas (for active research and service projects) ----------
7813    \str_if_eq:eeF {\tmpProjectStatus} {E}
7814    {
7815     \bool_lazy_or:nnT
7816     { \str_if_eq_p:ee {\tmpProjectType} {research} }
7817     { \str_if_eq_p:ee {\tmpProjectType} {service}  }
7818     {%
7819     \clist_clear:N \l_tmp_research_area_clist
7820      \clist_set:NV \tmpProjectResearchArea \tmpProjectResearchArea
7821     \clist_sort:Nn \tmpProjectResearchArea
7822     { \fp_compare:nTF {##1 < ##2} {\sort_return_swapped:} {\sort_return_same:} }%
7823     \clist_map_inline:Nn \tmpProjectResearchArea
7824     { \clist_put_right:Nn \l_tmp_research_area_clist { \dfgResearchAreasList{##1} } }%
7825      \subsection{Research~area
7826        \int_compare:nNnT { \clist_count:N \tmpProjectResearchArea } > {1}
7827          {s}
7828        :~
7829        \texorpdfstring
7830          {\dfgShowTag{#1}{research-area}
7831            {\clist_use:Nn \l_tmp_research_area_clist {,~}}[project]}
7832          {\clist_use:Nn \l_tmp_research_area_clist {,~}}}%
7833     }
7834    }
7835
7836    \bool_if:NTF \l_@@_project_one_PI_bool
7837    { \subsection{Project~leader}  }
7838    { \subsection{Project~leaders} }
7839
7840    \getPersonInformation{\tmpProjectPIids}\par
7841    \vspace*{1em}
7842
7843    % --- contract information (skip for admin / ending) ---------------------
7844    \bool_lazy_or:nnF
7845    { \str_if_eq_p:ee {\tmpProjectType}   {admin} }
7846    { \str_if_eq_p:ee {\tmpProjectStatus} {E}     }
7847    {%
7848     \begin{contractinformation}
7849      \clist_map_inline:nn {C1,C2,C3,C4}
```

```
     { \dfgPIcontractInformation{##1} }%
    \end{contractinformation}
   }

   % --- legal issues (skip for admin / mgk / ending) -----------------------
   \bool_lazy_any:nF
   {
    { \str_if_eq_p:ee {\tmpProjectType}   {admin} }
    { \str_if_eq_p:ee {\tmpProjectType}   {mgk}   }
    { \str_if_eq_p:ee {\tmpProjectStatus} {E}     }
   }
   {%
    \subsection{Legal~issues}%
    \def\tmpLocalDB{project}%
  {\footnotesize This~project~includes}\\

    \tl_clear:N \l_mytblr_body_tl
    \tl_put_right:Ne \l_mytblr_body_tl
  { 1. & research~requiring~compliance~with~discipline-specific~research~ethics~guidelines. & \checkYesNo{L1} \\ }%
    \compareEntry{#1}{L1}{yes}
    {
     \tl_put_right:Ne \l_mytblr_body_tl
  { & A~statement~from~the~responsible~ethics~committee~is~required~and~will~be~obtained~before~funding~starts. & \che
    }
    {}%
    \tl_put_right:Ne \l_mytblr_body_tl
    { 2. & clinical~trials & \checkYesNo{L2} \\ }%
    \compareEntry{#1}{L2}{yes}
    {
     \tl_put_right:Ne \l_mytblr_body_tl
     { & A~copy~of~the~studies'~registration~is~included~with~the~proposal. & \checkYesNo{L2a} \\ }%
    }
    {}%
    \tl_put_right:Ne \l_mytblr_body_tl
    {
     3. & experiments~involving~vertebrates. & \checkYesNo{L3} \\
     4. & experiments~involving~recombinant~DNA. & \checkYesNo{L4} \\
     5. & research~involving~human~embryonic~stem~cells. & \checkYesNo{L5} \\
    }
    \compareEntry{#1}{L5}{yes}
    {
     \tl_put_right:Ne \l_mytblr_body_tl
     { & Legal~authorization~has~been~obtained. & \checkYesNo{L5a} \\ }%
    }
    {}%
    \tl_put_right:Ne \l_mytblr_body_tl
    {
   6. & research~concerning~the~Convention~on~Biological~Diversity. & \checkYesNo{L6} \\
     7. & investigations~involving~dual~use~research~of~concern. & \checkYesNo{L7} \\
    }
    \compareEntry{#1}{L7}{yes}
    {
     \tl_put_right:Ne \l_mytblr_body_tl
     { & A~copy~of~the~statement~by~the~institution's~research~ethics~committee~is~included~with~the~proposal. & \chec
    }
    {}%
    \begin{tblr}[
      theme  = plain,
      expand = \l_mytblr_body_tl,
     ]{
      rowhead   = 0,
    rowsep    = 0pt,
      hline{2}  = {0pt},
    rows      = {bg=white, halign=l,font=\footnotesize},
    cells     = {font=\footnotesize},
      column{1} = {halign=r},
      colspec   = {t{.25cm}X[t]t{.75cm}}
     }
     \tl_use:N \l_mytblr_body_tl
    \end{tblr}%
   }
   }
 }

 \NewDocumentCommand{\checkYesNo}{O{\tmpProjectNumber} m}
 {%
   \compareEntry{#1}%        current project ID
   {#2}%                    which cell
   {yes}%                  if cell value is 'yes'
   {\dfgShowTag{#1}{#2}{{\bfseries\color{green}\faCheck}\,yes}[project]}%
   {\dfgShowTag{#1}{#2}{{\bfseries\color{red}\faTimes}\space no}[project]}%
 }
```

Another helper command to retrieve the information about a PI and his/her/their involvement within the project.

```
% ------------------------------------------------------------------------
% PI contract information
% ------------------------------------------------------------------------

\seq_new:N   \l_@@_contract_pi_ids_seq
\seq_new:N   \l_@@_contract_true_pi_ids_seq
\seq_new:N   \l_@@_contract_false_pi_ids_seq
\clist_new:N \l_@@_contract_answers_clist
\tl_new:N    \l_@@_contract_answer_tl
\tl_new:N    \l_@@_contract_field_list_tl

\NewDocumentCommand{\dfgPIcontractInformation}{ o m }
 {
  % #1: optional comma-separated PI ids
  %     If omitted, use the current value of \tmpProjectPIids.
  % #2: question id, e.g. C1 / C2 / C3 / C4
  \IfNoValueTF {#1}
   {
    \@@_pi_contract_information:Vn \tmpProjectPIids {#2}
   }
   {
    \@@_pi_contract_information:nn {#1} {#2}
   }
 }

\cs_new_protected:Npn \@@_pi_contract_information:nn #1#2
 {
  % #1: comma-separated PI ids
  % #2: question id, e.g. C1 / C2 / C3 / C4

  \def\tmpLocalDB{person}%
  \seq_set_from_clist:Nn \l_@@_contract_pi_ids_seq {#1}

  \str_case:nn {#2}
   {
    {C1}
     {
      \item
      Is~the~employment~of~the~
      \bool_if:NTF \l_@@_project_one_PI_bool
       { project~leader~at~the~institution~ }
       { project~leaders~at~the~institutions~ }%
      indicated~contractually~secured~for~the~duration~of~
      the~proposed~funding~period?\space
      \@@_pi_contract_yes_no:n {#2}%
     }

    {C2}
     {
      \item
      \bool_if:NTF \l_@@_project_one_PI_bool
       { Does~the~above~mentioned~person~hold~a~fixed-term~position? }
       { Do~any~of~the~above~mentioned~persons~hold~fixed-term~positions? }%
      \space
      \@@_pi_contract_yes_no:n {#2}%

      \clist_if_in:NnT \l_@@_contract_answers_clist {yes}
       {
        \begin{contractinformation}
          \item End~date~of~fixed-term~contract:\space
          \@@_pi_contract_field_list:nN {C2a} \l_@@_contract_true_pi_ids_seq

          \item Further~employment~is~planned~until:\space
          \@@_pi_contract_field_list:nN {C2b} \l_@@_contract_true_pi_ids_seq
        \end{contractinformation}
       }
     }

    {C3}
     {
      \item
      \bool_if:NTF \l_@@_project_one_PI_bool
       { Will~the~above~mentioned~person~ }
       { Will~one~or~more~of~the~above-mentioned~persons~ }%
      retire~during~the~proposed~funding~period?\space
      \@@_pi_contract_yes_no:n {#2}%

      \clist_if_in:NnT \l_@@_contract_answers_clist {yes}
       {
        \begin{contractinformation}
          \item Anticipated~retirement~date:\space
          \@@_pi_contract_field_list:nN {C3a} \l_@@_contract_true_pi_ids_seq
        \end{contractinformation}
       }
     }

    {C4}
```

```
      {
       \item
       Funding~for~the~position~of~the~
       \bool_if:NTF \l_@@_project_one_PI_bool
        { project~leader~at~the~institution~ }
        { project~leaders~at~the~institutions~ }%
       indicated~is~covered~by~core~support~(state~funds~or~similar):\space
       \@@_pi_contract_yes_no:n {#2}%

       \clist_if_in:NnT \l_@@_contract_answers_clist {no}
        {
         \begin{contractinformation}
           \item Type~and~source~of~funding:\space
           \@@_pi_contract_field_list:nN {C4a} \l_@@_contract_false_pi_ids_seq
         \end{contractinformation}
        }
      }
    }
  }

 \cs_generate_variant:Nn \@@_pi_contract_information:nn { Vn }


 \cs_new_protected:Npn \@@_pi_contract_yes_no:n #1
  {
   % #1: question id, e.g. C1 / C2 / C3 / C4

   \clist_clear:N \l_@@_contract_answers_clist
   \seq_clear:N   \l_@@_contract_true_pi_ids_seq
   \seq_clear:N   \l_@@_contract_false_pi_ids_seq
   \tl_clear:N    \l_@@_contract_answer_tl

   \bool_set_true:N \l_tmpa_bool

   \seq_map_inline:Nn \l_@@_contract_pi_ids_seq
    {
     \bool_if:NTF \l_tmpa_bool
      { \bool_set_false:N \l_tmpa_bool }
      { \tl_put_right:Nn \l_@@_contract_answer_tl {,~} }%

     \DTLgetvalueforkey{\theLastName}{lastname}{person}{id}{##1}%

     \compareEntry{##1}{#1}{yes}
      {
       \tl_put_right:Nn \l_@@_contract_answer_tl
         { \dfgShowTag{##1}{#1}{\textbf{yes}}[person] }%
       \clist_put_right:Nn \l_@@_contract_answers_clist {true}%
       \seq_put_right:Nn   \l_@@_contract_true_pi_ids_seq {##1}%
      }
      {
       \tl_put_right:Nn \l_@@_contract_answer_tl
         { \dfgShowTag{##1}{#1}{\textbf{no}}[person] }%
       \clist_put_right:Nn \l_@@_contract_answers_clist {false}%
       \seq_put_right:Nn   \l_@@_contract_false_pi_ids_seq {##1}%
      }

     \bool_if:NF \l_@@_project_one_PI_bool
      {
       \tl_put_right:Nx \l_@@_contract_answer_tl
        {
         \exp_not:n {~(}
         \exp_not:N \dfgShowTag{##1}{lastname}
           {\exp_not:V \theLastName}[person]
         \exp_not:n {)}
        }%
      }
    }

   \tl_use:N \l_@@_contract_answer_tl
  }


 \cs_new_protected:Npn \@@_pi_contract_field_list:nN #1#2
  {
   % #1: field id, e.g. C2a / C2b / C3a / C4a
   % #2: sequence variable containing the PI ids for which the field is relevant

   \tl_clear:N \l_@@_contract_field_list_tl
   \bool_set_true:N \l_tmpa_bool

   \seq_map_inline:Nn #2
    {
     \bool_if:NTF \l_tmpa_bool
      { \bool_set_false:N \l_tmpa_bool }
      { \tl_put_right:Nn \l_@@_contract_field_list_tl {,~} }%

     \getEntry*[person]{##1}
```

```
      {
       \theFieldValue = #1,
       \theLastName   = lastname
      }%

     \tl_put_right:Nx \l_@@_contract_field_list_tl
      {
       \exp_not:N \textbf
       {
         \exp_not:N \dfgShowTag{##1}{#1}
           {\exp_not:V \theFieldValue}[person]
       }
       \int_compare:nNnT { \seq_count:N #2 } > {1}
        {
         \exp_not:n {~(}
         \exp_not:N \dfgShowTag{##1}{lastname}
           {\exp_not:V \theLastName}[person]
         \exp_not:n {)}
        }
      }%
    }

   \tl_use:N \l_@@_contract_field_list_tl
  }


 \cs_new_protected:Npn \__pi_question_field_list:nn #1#2
 {%
   \tl_clear:N \l_tmpa_tl
   \bool_set_true:N \l_tmpa_bool   % first-iteration flag
   \seq_map_inline:Nn #2
   {%
    \bool_if:NTF \l_tmpa_bool
    { \bool_set_false:N \l_tmpa_bool }
    { \tl_put_right:Nn \l_tmpa_tl {,~} }%
    \getEntry*[person]{##1}{
     \theFieldValue = #1,
     \theLastName   = lastname
    }%
    \tl_put_right:Nx \l_tmpa_tl
    {
   \exp_not:N \textbf
     {
       \exp_not:N \dfgShowTag{##1}{#1}
         {\exp_not:V \theFieldValue}[person]
     }
     \int_compare:nNnT { \seq_count:N #2 } > {1}
       {
         \exp_not:n {\space(}
         \exp_not:N \dfgShowTag{##1}{lastname}
           {\exp_not:V \theLastName}[person]
         \exp_not:n {)}
       }
    }%
   }%
   \tl_use:N \l_tmpa_tl
 }

 \tl_new:N \l_@@_yes_names_tl
 \tl_new:N \l_@@_no_names_tl

 \cs_new_protected:Npn \tmp_person_question_yes_no:n #1
 {%
   \clist_clear:N \l_tmpc_clist
   \seq_clear:N   \l_tmpd_seq
   \seq_clear:N   \l_tmpe_seq
   \tl_clear:N    \l_tmpa_tl
   \bool_set_true:N \l_tmpa_bool   % first-iteration flag
   \seq_map_inline:Nn \l_tmpa_seq
   {%
    \bool_if:NTF \l_tmpa_bool
    { \bool_set_false:N \l_tmpa_bool }
    { \tl_put_right:Nn \l_tmpa_tl {,~} }%
    \DTLgetvalueforkey{\theLastName}{lastname}{person}{id}{##1}%
    \compareEntry{##1}{#1}{yes}
    {%
     \tl_put_right:Nn \l_tmpa_tl
       { \dfgShowTag{##1}{#1}{\textbf{yes}}[person] }%
     \clist_put_right:Nn \l_tmpc_clist {yes}%
     \seq_put_right:Nn   \l_tmpd_seq   {##1}%
    }
    {%
     \tl_put_right:Nn \l_tmpa_tl
       { \dfgShowTag{##1}{#1}{\textbf{no}}[person] }%
     \clist_put_right:Nn \l_tmpc_clist {false}%
     \seq_put_right:Nn   \l_tmpe_seq   {##1}%
    }%
```

```
    \bool_if:NF \l_@@_project_one_PI_bool
    {%
     \tl_put_right:Nx \l_tmpa_tl
     {
       \exp_not:n {\space(}
       \exp_not:N \dfgShowTag{##1}{lastname}
         {\exp_not:V \theLastName}[person]
       \exp_not:n {)}
     }%
    }%
   }%
   \tl_use:N \l_tmpa_tl
}
```

Ending projects report three distinct kinds of published results. Each populated list is cited in its own reference segment. The combined list is persisted in the auxiliary file so a preceding project-report bibliography can omit entries that are printed again in the result lists.

```
\clist_new:N \l_@@_project_results_quality_assured_clist
\clist_new:N \l_@@_project_results_other_clist
\clist_new:N \l_@@_project_results_patents_clist
\clist_new:N \l_@@_project_results_all_clist
\clist_new:N \l_@@_project_results_excluded_clist
\prop_new:N  \g_@@_project_results_aux_prop

\NewDocumentCommand{\dfgProjectResultsAux}{mm}
{
  \prop_gput:Nnn \g_@@_project_results_aux_prop {#1} {#2}
}

\cs_new_protected:Npn \@@_project_results_prepare:n #1
{
  \clist_clear:N \l_@@_project_results_excluded_clist
  \prop_get:NnNT \g_@@_project_results_aux_prop {#1} \l_tmpa_tl
  {
    \clist_set:NV \l_@@_project_results_excluded_clist \l_tmpa_tl
  }
}

\cs_new_protected:Npn \@@_project_results_collect:N #1
{
  \clist_map_inline:Nn #1
  {
    \str_if_eq:nnF {##1} {none}
    { \clist_put_right:Nn \l_@@_project_results_all_clist {##1} }
  }
}

\cs_new_protected:Npn \@@_project_results_register:
{
  \clist_clear:N \l_@@_project_results_all_clist
  \@@_project_results_collect:N \l_@@_project_results_quality_assured_clist
  \@@_project_results_collect:N \l_@@_project_results_other_clist
  \@@_project_results_collect:N \l_@@_project_results_patents_clist
  \clist_remove_duplicates:N \l_@@_project_results_all_clist
  \clist_set_eq:NN
    \l_@@_project_results_excluded_clist
    \l_@@_project_results_all_clist
  \iow_now:Ne \@auxout
  {
    \exp_not:N \dfgProjectResultsAux
    { \exp_not:V \l__tmp_current_project_tl }
    { \exp_not:V \l_@@_project_results_all_clist }
  }
}

\cs_new_protected:Npn \@@_project_results_skip_if_listed:
{
  \clist_map_inline:Nn \l_@@_project_results_excluded_clist
  {
    \str_if_eq:eeT {##1} { \strfield{entrykey} }
    {
      \skipentry
      \clist_map_break:
    }
  }
}

\defbibcheck{dfg-project-report}
{ \@@_project_results_skip_if_listed: }

\defbibheading{dfg-project-report-references}
{ \subsubsection{References} }

\NewDocumentCommand{\projectPublicationsHeading}{}
{
```

```
  \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
  { \subsection{Project-~and~subject-related~list~of~publications} }
}

\NewDocumentCommand{\projectNocite}{m}
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
  { \nocite{#1} }
}

% The frame wrapper retains the configured bibliography font. Clear the
% local selector first so disabled emphasis cannot inherit an outer selector.
\NewDocumentCommand{\printFrameBibliography}{o}
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
  {
    \group_begin:
    \tl_clear:N \l_@@_bibentry_highlight_keyword_tl
    \bool_if:NT \l_@@_custom_emphasize_entries_bool
      { \tl_set:Nn \l_@@_bibentry_highlight_keyword_tl {frametext-highlight} }
    \IfNoValueTF{#1}
    { \printbibliography }
    { \printbibliography[#1] }
    \group_end:
  }
}

\NewDocumentCommand{\printProjectBibliography}{o}
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
  {
    \group_begin:
    % biblatex applies \bibfont when it starts the bibliography list.
    \renewcommand*{\bibfont}{\normalfont\footnotesize}%
    \bool_if:NT \l_@@_custom_emphasize_entries_bool
    {
      \tl_if_blank:VF \l__tmp_current_project_tl
      {
        \tl_set:Ne \l_@@_bibentry_highlight_keyword_tl
          {\l__tmp_current_project_tl-highlight}
      }
    }
    \IfNoValueTF{#1}
    { \printbibliography }
    { \printbibliography[#1] }
    \group_end:
  }
}

\NewDocumentEnvironment{projectReport}{}
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
  { \begin{refsegment} }
}
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
  {
    \printProjectBibliography[
      segment = \therefsegment,
      check   = dfg-project-report,
      heading = dfg-project-report-references
    ]
    \end{refsegment}
  }
}

\keys_define:nn { dfg / project-results }
{
  quality-assured .clist_set:N = \l_@@_project_results_quality_assured_clist,
  quality-assured .value_required:n = true,
  other .clist_set:N = \l_@@_project_results_other_clist,
  other .value_required:n = true,
  patents .clist_set:N = \l_@@_project_results_patents_clist,
  patents .value_required:n = true,
  unknown .code:n = { \msg_warning:nnn { dfg } { unknown-key } {#1} }
}

\cs_new_protected:Npn \@@_project_results_print:Nn #1#2
{
  \clist_if_empty:NTF #1
  {#2}
  {
    \str_if_eq:eeTF { \clist_use:Nn #1 {,} } { none }
    { None. }
    {
      \begin{refsegment}
        \exp_args:Ne \nocite { \clist_use:Nn #1 {,} }
```

```
8360          \printProjectBibliography[segment=\therefsegment, heading=none]
8361        \end{refsegment}
8362      }
8363    }
8364  }
8365
8366  \NewDocumentCommand{\printProjectResults}{m}
8367  {
8368    \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
8369    {
8370    \FloatBarrier
8371    \group_begin:
8372      \clist_clear:N \l_@@_project_results_quality_assured_clist
8373      \clist_clear:N \l_@@_project_results_other_clist
8374      \clist_clear:N \l_@@_project_results_patents_clist
8375      \keys_set:nn { dfg / project-results } {#1}
8376      \@@_project_results_register:
8377      \setbibpreamble{}
8378
8379      \subsubsection{Publications~with~scientific~quality~assurance}
8380      \@@_project_results_print:Nn
8381        \l_@@_project_results_quality_assured_clist
8382        { List~publications~with~scientific~quality~assurance. }
8383
8384      \subsubsection{Other~publications~and~published~results}
8385      \@@_project_results_print:Nn
8386        \l_@@_project_results_other_clist
8387        { List~datasets,~software,~preprints,~protocols,~or~other~relevant~
8388          published~results~that~are~not~included~above. }
8389
8390      \subsubsection{Patents}
8391      \@@_project_results_print:Nn
8392        \l_@@_project_results_patents_clist
8393        { List~patents~and~patent~applications,~or~state~that~none~resulted~
8394          from~the~project. }
8395    \group_end:
8396    }
8397  }

8398  \NewDocumentCommand{\printProjectTail}{}{
8399    \bool_if:NTF \l_@@_class_fastcompile_project_bool
8400    {
8401      \bool_if:NT \l_@@_class_fastcompile_project_show_publications_bool
8402      {
8403        \end{refsection}
8404        \resetboldnames
8405      }
8406    }
8407    {
8408      \end{refsection}
8409      \resetboldnames%
8410      \clearpage
8411      \addtocontents{toc}{\protect\setcounter{tocdepth}{1}}%
8412      \hypersetup{bookmarksdepth = 3}
8413      \colorlet{thumbIndexBG}{\l_@@_custom_thumbIndexBG_tl}
8414    }
8415  }
8416
8417  \cs_new_protected:Npn \@@_print_project_head:n #1
8418  { \printProjectHead{#1} }
8419
8420  \cs_new_protected:Npn \@@_print_project_error:n #1
8421  {
8422    \PackageError{dfg-crc}
8423      {Invalid~project~part~'#1'}
8424      {Use~\string\printProject{{Head}{project-id}}~or~\string\printProject{{Tail}}.}
8425  }
8426
8427  \cs_new_protected:Npn \@@_print_project_dispatch:w #1#2 \q_stop
8428  {
8429    \str_case:nnF {#1}
8430    {
8431      {Head}
8432      {
8433        \tl_if_blank:nTF {#2}
8434          { \@@_print_project_error:n {Head} }
8435          { \@@_print_project_head:n {#2} }
8436      }
8437      {Tail}
8438      {
8439        \tl_if_blank:nTF {#2}
8440          { \printProjectTail }
8441          { \@@_print_project_error:n {Tail} }
8442      }
8443    }
8444    { \@@_print_project_error:n {#1} }
8445  }
```

```
\NewDocumentCommand{\printProject}{m}
{ \@@_print_project_dispatch:w #1 \q_stop }
```

To get the interim sum printed in the margin area we use a custom command. Table rows are denser than the package's default collision distance, so these notes use fixed placement with no collision padding. This keeps every total on its row anchor instead of accumulating a downward displacement. The leader is placed before the value in the right margin and after the value in the left margin, keeping it adjacent to the table on both page sides. You can switch the notes off using the customization section.

```
\NewDocumentCommand{\printMarginalia}{O{\hrulefill}m}
{
  \bool_if:NT \l_@@_class_showmarginalia_bool
  {\bool_set_false:N \g_@@_tblr_inside_bool
 \bool_set_true:N \l_@@_show_tag_leftward_bool
    \marginalia[type=fixed,ysep=0pt]
    {
      \int_if_odd:nTF { \marginaliapage }
        { #1#2 }
        { #2#1 }
    }
 \bool_set_false:N \l_@@_show_tag_leftward_bool
   \bool_set_true:N \g_@@_tblr_inside_bool
  }
}

\tl_new:N \l__tmp_body_staff_tl
\tl_new:N \l__tmp_body_costs_tl
\tl_new:N \l__tmp_body_fellowships_tl
\tl_new:N \l__tmp_body_globalfunds_tl
\tl_new:N \l__tmp_body_instrumentation_tl
\cs_new_protected:Npn \@@_requested_funding_append_nonstaff:N #1
{
  \DTLaddall{\tmpHorizontalSum}{
    \pSumYearOne,
    \pSumYearTwo,
    \pSumYearThree,
    \pSumYearFour,
    \pSumYearFive
  }
  \tl_put_right:Ne #1
  {
    \dfgShowTag{\pProject/\pCategory}{category}{\pCategory}[\tmpLocalDB]
    \printMarginalia{\dfgShowCalculatedTag{RequestedFundingTable}{\tmpLocalDB}
      {\@@_show_eur:n{\tmpHorizontalSum}}}
    & & \dfgShowTag{\pProject/\pCategory}{sum-year1}
          {\@@_show_eur:n{\pSumYearOne}}[\tmpLocalDB]
    & & \dfgShowTag{\pProject/\pCategory}{sum-year2}
          {\@@_show_eur:n{\pSumYearTwo}}[\tmpLocalDB]
    & & \dfgShowTag{\pProject/\pCategory}{sum-year3}
          {\@@_show_eur:n{\pSumYearThree}}[\tmpLocalDB]
    & & \dfgShowTag{\pProject/\pCategory}{sum-year4}
          {\@@_show_eur:n{\pSumYearFour}}[\tmpLocalDB]
    & & \dfgShowTag{\pProject/\pCategory}{sum-year5}
          {\@@_show_eur:n{\pSumYearFive}}[\tmpLocalDB] \\
  }
}
\NewDocumentCommand{\RequestedFundingTable}{m}
{
  \group_begin:
  \sisetup{drop-zero-decimal=true}
  % #1: project
  \def\tmpLocalDB{RequestedFunding}%
  \DTLifdbexists{\tmpLocalDB}{%
  % Older funding entries may omit positionText; keep it an optional column.
  \DTLifhaskey{\tmpLocalDB}{positionText}{}{\DTLaddcolumn{\tmpLocalDB}{positionText}}
   \tl_clear:N \l__tmp_body_staff_tl
   \tl_clear:N \l__tmp_body_costs_tl
   \tl_clear:N \l__tmp_body_fellowships_tl
   \tl_clear:N \l__tmp_body_globalfunds_tl
   \tl_clear:N \l__tmp_body_instrumentation_tl
   \tl_put_right:Nn \l__tmp_body_staff_tl {

    \SetRow{tableHeadingsBGalt}
    \SetCell{font=\footnotesize\bfseries,halign=l}
  Staff
  & Qty & \SetCell{c} Sum
  & Qty & \SetCell{c} Sum
  & Qty & \SetCell{c} Sum
  & Qty & \SetCell{c} Sum
  & Qty & \SetCell{c} Sum \\
   }
```

```
8520
8521     \tl_put_right:Nn \l__tmp_body_costs_tl {
8522      \SetHline{-}{tableOuterLines,0.5pt}
8523      \SetRow{tableHeadingsBGalt}
8524      \SetCell{font=\footnotesize\bfseries,halign=l}
8525    Direct~ Costs
8526    & & \SetCell{c} Sum
8527    & & \SetCell{c} Sum
8528    & & \SetCell{c} Sum
8529    & & \SetCell{c} Sum
8530    & & \SetCell{c} Sum  \\
8531     }
8532     \tl_put_right:Nn \l__tmp_body_fellowships_tl {
8533      \SetHline{-}{tableOuterLines,0.5pt}
8534      \SetRow{tableHeadingsBGalt}
8535      \SetCell{font=\footnotesize\bfseries,halign=l}
8536    Fellowships
8537    & & \SetCell{c} Sum
8538    & & \SetCell{c} Sum
8539    & & \SetCell{c} Sum
8540    & & \SetCell{c} Sum
8541    & & \SetCell{c} Sum \\
8542     }
8543     \tl_put_right:Nn \l__tmp_body_globalfunds_tl {
8544      \SetHline{-}{tableOuterLines,0.5pt}
8545      \SetRow{tableHeadingsBGalt}
8546      \SetCell{font=\footnotesize\bfseries,halign=l}
8547    Globalfunds
8548    & & \SetCell{c} Sum
8549    & & \SetCell{c} Sum
8550    & & \SetCell{c} Sum
8551    & & \SetCell{c} Sum
8552    & & \SetCell{c} Sum  \\
8553     }
8554     \tl_put_right:Nn \l__tmp_body_instrumentation_tl {
8555      \SetHline{-}{tableOuterLines,0.5pt}
8556      \SetRow{tableHeadingsBGalt}
8557      \SetCell{font=\footnotesize\bfseries,halign=l}
8558    Instrumentation
8559    & & \SetCell{c} Sum
8560    & & \SetCell{c} Sum
8561    & & \SetCell{c} Sum
8562    & & \SetCell{c} Sum
8563    & & \SetCell{c} Sum  \\
8564     }
8565
8566
8567   % One filtered traversal is enough: dispatch each matching row to the
8568   % token list for its funding type.  The previous implementation traversed
8569   % the complete, growing database once for every type.
8570   \DTLforeach[\DTLiseq{#1}{\pProject}]{\tmpLocalDB}{
8571       \pProject           =ref-project-id,
8572       \pType              =type,
8573       \pCategory          =category,
8574       \pPercentage        =percentage,
8575       \pPositionText      =positionText,
8576       \pQuantityYearOne   =quantity-year1,
8577       \pQuantityYearTwo   =quantity-year2,
8578       \pQuantityYearThree =quantity-year3,
8579       \pQuantityYearFour  =quantity-year4,
8580       \pQuantityYearFive  =quantity-year5,
8581       \pSumYearOne        =sum-year1,
8582       \pSumYearTwo        =sum-year2,
8583       \pSumYearThree      =sum-year3,
8584       \pSumYearFour       =sum-year4,
8585       \pSumYearFive       =sum-year5
8586      }{
8587
8588       \str_case:enF {\pType}
8589       {
8590      {staff} {
8591            % Use the category only when no custom position label is supplied.
8592            \DTLifnullorempty{\pPositionText}
8593              {
8594                \tl_set:Ne \l_tmpa_tl
8595                  { \dfgShowTag{\pProject/\pCategory}{category}
8596                      {\dfgCategoryName{\pCategory}}[\tmpLocalDB] }
8597              }
8598              {
8599                \tl_set:Ne \l_tmpa_tl
8600                  { \dfgShowTag{\pProject/\pCategory}{positionText}
8601                      {\exp_not:V \pPositionText}[\tmpLocalDB] }
8602              }
8603       \DTLaddall{\tmpHorizontalSum}{
8604          \pSumYearOne,
8605          \pSumYearTwo,
8606          \pSumYearThree,
```

```
8607          \pSumYearFour,
8608          \pSumYearFive
8609         }
8610         \tl_put_right:Ne \l__tmp_body_staff_tl
8611         {
8612
8613              \exp_not:V \l_tmpa_tl
8614              ,~
8615              \dfgShowTag{\pProject/\pCategory}{percentage}
8616                {\qty{\pPercentage}{\percent}}[\tmpLocalDB]
8617        \printMarginalia{\dfgShowCalculatedTag{RequestedFundingTable}
8618                {\tmpLocalDB}{\@@_show_eur:n{\tmpHorizontalSum}}}
8619        & \dfgShowTag{\pProject/\pCategory}{quantity-year1}
8620                  {\@@_show_table_number:n{\pQuantityYearOne}}[\tmpLocalDB]
8621        & \dfgShowTag{\pProject/\pCategory}{sum-year1}
8622                  {\@@_show_eur:n{\pSumYearOne}}[\tmpLocalDB]
8623        & \dfgShowTag{\pProject/\pCategory}{quantity-year2}
8624                  {\@@_show_table_number:n{\pQuantityYearTwo}}[\tmpLocalDB] % year2
8625        & \dfgShowTag{\pProject/\pCategory}{sum-year2}
8626                  {\@@_show_eur:n{\pSumYearTwo}}[\tmpLocalDB]
8627        & \dfgShowTag{\pProject/\pCategory}{quantity-year3}
8628                  {\@@_show_table_number:n{\pQuantityYearThree}}[\tmpLocalDB] % year3
8629        & \dfgShowTag{\pProject/\pCategory}{sum-year3}
8630                  {\@@_show_eur:n{\pSumYearThree}}[\tmpLocalDB]
8631        & \dfgShowTag{\pProject/\pCategory}{quantity-year4}
8632                  {\@@_show_table_number:n{\pQuantityYearFour}}[\tmpLocalDB] % year4
8633        & \dfgShowTag{\pProject/\pCategory}{sum-year4}
8634                  {\@@_show_eur:n{\pSumYearFour}}[\tmpLocalDB]
8635        & \dfgShowTag{\pProject/\pCategory}{quantity-year5}
8636                  {\@@_show_table_number:n{\pQuantityYearFive}}[\tmpLocalDB] % year5
8637        & \dfgShowTag{\pProject/\pCategory}{sum-year5}
8638                  {\@@_show_eur:n{\pSumYearFive}}[\tmpLocalDB]
8639          \\
8640         }
8641     }
8642         {costs}
8643           {\@@_requested_funding_append_nonstaff:N \l__tmp_body_costs_tl}
8644         {fellowships}
8645           {\@@_requested_funding_append_nonstaff:N \l__tmp_body_fellowships_tl}
8646         {globalfunds}
8647           {\@@_requested_funding_append_nonstaff:N \l__tmp_body_globalfunds_tl}
8648         {instrumentation}
8649           {\@@_requested_funding_append_nonstaff:N \l__tmp_body_instrumentation_tl}
8650    }
8651       { }
8652  }
8653    \tl_put_right:Ne \l__tmp_body_staff_tl
8654    {
8655     \SetRow{belowsep=5pt}
8656     \SetHline{-}{tableOuterLines,0.5pt}
8657     Total~
8658     \printMarginalia{%
8659      \getSumRequestedFunds{#1}{staff}
8660      {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
8661     }
8662     & \SetCell{bg=tableNoValue}
8663     & \getSumRequestedFunds{#1}{staff}{sum-year1}
8664     & \SetCell{bg=tableNoValue}
8665     & \getSumRequestedFunds{#1}{staff}{sum-year2}
8666     & \SetCell{bg=tableNoValue}
8667     & \getSumRequestedFunds{#1}{staff}{sum-year3}
8668     & \SetCell{bg=tableNoValue}
8669     & \getSumRequestedFunds{#1}{staff}{sum-year4}
8670     & \SetCell{bg=tableNoValue}
8671     & \getSumRequestedFunds{#1}{staff}{sum-year5}
8672     \\
8673    }
8674
8675    \clist_map_inline:nn {
8676     costs,fellowships,globalfunds,instrumentation
8677    }{
8678
8679     \tl_put_right:ce { l__tmp_body_##1_tl }
8680     {
8681      \SetRow{belowsep=5pt}
8682      \SetHline{-}{tableLines,0.5pt}
8683      Total
8684      \printMarginalia{%
8685       \getSumRequestedFunds{#1}{##1}
8686       {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
8687      }
8688      & &
8689      \getSumRequestedFunds{#1}{##1}{sum-year1}
8690      & &
8691      \getSumRequestedFunds{#1}{##1}{sum-year2}
8692      & &
8693      \getSumRequestedFunds{#1}{##1}{sum-year3}
```

```
      & &
      \getSumRequestedFunds{#1}{##1}{sum-year4}
      & &
      \getSumRequestedFunds{#1}{##1}{sum-year5}
      \\
     }}

    % delete when zero
    \dfgCheckIfZeroShowCategory{#1}{staff}
    \dfgCheckIfZeroShowCategory{#1}{costs}
    \dfgCheckIfZeroShowCategory{#1}{fellowships}
    \dfgCheckIfZeroShowCategory{#1}{globalfunds}
    \dfgCheckIfZeroShowCategory{#1}{instrumentation}

    \begin{tblr}[
      theme=main,
      long,
      expand=
      \l__tmp_body_staff_tl
      \l__tmp_body_costs_tl
      \l__tmp_body_fellowships_tl
      \l__tmp_body_globalfunds_tl
      \l__tmp_body_instrumentation_tl,
    remark{Note}={Figures~ in~ EUR},
     ]{
      colsep=3pt,
    colspec = {
      *{1}{X[3.1,l,t]}
      *{5}{
       @{}X[0.75,c,si={table-format=1}]
       @{}X[1.2,c,si={table-format=7}]
      }
        % *{1}{X[2,t]}
        % *{10}{@{}X[c, si={table-format=4}]}
       },
      rows = {bg=white},
    column{2-Z}={guard},
      row{1}= {bg=tableHeadingsBG},
      row{1,2}= {guard},
     }
   Funding~for
    & \SetCell[c=2]{c} \dfgFundingYear{1}
      &
    & \SetCell[c=2]{c} \dfgFundingYear{2}
      &
    & \SetCell[c=2]{c} \dfgFundingYear{3}
      &
    & \SetCell[c=2]{c} \dfgFundingYear{4}
      &
    & \SetCell[c=2]{c} \dfgFundingYear{5}
      &                                          \\

     \l__tmp_body_staff_tl
     \l__tmp_body_costs_tl
     \l__tmp_body_fellowships_tl
     \l__tmp_body_globalfunds_tl
     \l__tmp_body_instrumentation_tl

     \SetHline{-}{tableOuterLines,.75pt}
     \SetCell{font=\bfseries\footnotesize}
     Total
     \printMarginalia{%
      \getTotalSumRequestedFunds{#1}
      {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
     }
      &
      & \getTotalSumRequestedFunds{#1}{sum-year1}
      &
      & \getTotalSumRequestedFunds{#1}{sum-year2}
      &
      & \getTotalSumRequestedFunds{#1}{sum-year3}
      &
      & \getTotalSumRequestedFunds{#1}{sum-year4}
      &
      & \getTotalSumRequestedFunds{#1}{sum-year5} \\
    \end{tblr}
   }{
    Database~ \texttt{#1}~ missing.
    \msg_warning:nne { dfg } { database-not-found }{ #1 }
   }
   \group_end:
}
```

\RequestedFundingStaffTable → P. 305 The DFG staff roster uses one database and one project-filtered traversal. Rows are first dispatched, in declaration order, to four sequences. The sequences are then rendered in the official order (existing research, existing non-research, requested research, requested non-research), which also assigns the continuous sequence numbers used by the descriptions. When both staff types occur within a funding-status block, a horizontal rule separates the research and non-research groups without repeating their row-spanning labels. Category values and the emphasized job-description headings are left aligned for easier scanning. Person, institution, and affiliation references are resolved during this assembly phase, before tabularray measures any cells. Thus neither measuring passes nor page breaks repeat Datatool queries.

\RequestedFundingStaffTable{⟨*project*⟩}

Print the official DFG requested-funding staff roster, the institution and parent-affiliation legend, and both applicable job-description blocks for ⟨*project*⟩. The data source is always `RequestedFundingStaff`. For an ending project (`project-status=E`), rows declared with `funding-status=requested` are automatically labelled as staff funded with approved grant money; this matches the terminology of the ending-project report while retaining the same two-value data model.

```
\seq_new:N \l_@@_rfs_existing_research_seq
\seq_new:N \l_@@_rfs_existing_nonresearch_seq
\seq_new:N \l_@@_rfs_requested_research_seq
\seq_new:N \l_@@_rfs_requested_nonresearch_seq
\seq_new:N \l_@@_rfs_institution_seq
\seq_new:N \l_@@_rfs_affiliation_seq
\prop_new:N \l_@@_rfs_duplicate_prop
\prop_new:N \l_@@_rfs_institution_name_prop
\prop_new:N \l_@@_rfs_institution_abbreviation_prop
\prop_new:N \l_@@_rfs_institution_affiliation_abbreviation_prop
\prop_new:N \l_@@_rfs_affiliation_abbreviation_prop
\prop_new:N \l_@@_rfs_affiliation_name_prop
\tl_new:N \l_@@_rfs_table_body_tl
\tl_new:N \l_@@_rfs_existing_descriptions_tl
\tl_new:N \l_@@_rfs_requested_descriptions_tl
\tl_new:N \l_@@_rfs_project_tl
\tl_new:N \l_@@_rfs_project_status_tl
\tl_new:N \l_@@_rfs_existing_group_label_tl
\tl_new:N \l_@@_rfs_requested_group_label_tl
\tl_new:N \l_@@_rfs_existing_description_heading_tl
\tl_new:N \l_@@_rfs_requested_description_heading_tl
\tl_new:N \l_@@_rfs_ref_person_tl
\tl_new:N \l_@@_rfs_firstname_tl
\tl_new:N \l_@@_rfs_lastname_tl
\tl_new:N \l_@@_rfs_degree_tl
\tl_new:N \l_@@_rfs_position_tl
\tl_new:N \l_@@_rfs_field_tl
\tl_new:N \l_@@_rfs_ref_institution_tl
\tl_new:N \l_@@_rfs_ref_affiliation_tl
\tl_new:N \l_@@_rfs_department_tl
\tl_new:N \l_@@_rfs_funding_status_tl
\tl_new:N \l_@@_rfs_staff_type_tl
\tl_new:N \l_@@_rfs_commitment_tl
\tl_new:N \l_@@_rfs_category_tl
\tl_new:N \l_@@_rfs_funding_source_tl
\tl_new:N \l_@@_rfs_description_tl
\tl_new:N \l_@@_rfs_name_cell_tl
\tl_new:N \l_@@_rfs_plain_name_tl
\tl_new:N \l_@@_rfs_description_name_tl
\tl_new:N \l_@@_rfs_field_cell_tl
\tl_new:N \l_@@_rfs_department_cell_tl
\tl_new:N \l_@@_rfs_commitment_cell_tl
\tl_new:N \l_@@_rfs_category_cell_tl
\tl_new:N \l_@@_rfs_funding_source_cell_tl
\tl_new:N \l_@@_rfs_group_cell_tl
\tl_new:N \l_@@_rfs_institution_name_tl
\tl_new:N \l_@@_rfs_institution_abbreviation_tl
\tl_new:N \l_@@_rfs_institution_affiliation_tl
\tl_new:N \l_@@_rfs_affiliation_name_tl
\tl_new:N \l_@@_rfs_affiliation_abbreviation_tl
\int_new:N \l_@@_rfs_display_number_int
\int_new:N \l_@@_rfs_record_count_int
\bool_new:N \l_@@_rfs_person_resolved_bool
\bool_new:N \l_@@_rfs_name_incomplete_bool

\keys_define:nn { dfg / requested-funding-staff-row }
{
  ref-person-id .tl_set:N = \l_@@_rfs_ref_person_tl,
  firstname .tl_set:N = \l_@@_rfs_firstname_tl,
  lastname .tl_set:N = \l_@@_rfs_lastname_tl,
```

```
8836    academic-degree .tl_set:N = \l_@@_rfs_degree_tl,
8837    positionText .tl_set:N = \l_@@_rfs_position_tl,
8838    position .tl_set:N = \l_@@_rfs_position_tl,
8839    field-of-research .tl_set:N = \l_@@_rfs_field_tl,
8840    ref-institution-id .tl_set:N = \l_@@_rfs_ref_institution_tl,
8841    ref-affiliation-id .tl_set:N = \l_@@_rfs_ref_affiliation_tl,
8842    department .tl_set:N = \l_@@_rfs_department_tl,
8843    funding-status .tl_set:N = \l_@@_rfs_funding_status_tl,
8844    staff-type .tl_set:N = \l_@@_rfs_staff_type_tl,
8845    commitment .tl_set:N = \l_@@_rfs_commitment_tl,
8846    category .tl_set:N = \l_@@_rfs_category_tl,
8847    funding-source .tl_set:N = \l_@@_rfs_funding_source_tl,
8848    description .tl_set:N = \l_@@_rfs_description_tl,
8849    unknown .code:n = { }
8850  }
8851
8852  \cs_new:Npn \@@_rfs_missing_text:n #1
8853  {
8854    \bool_if:NTF \l_@@_class_fastcompile_bool
8855      { \@@_rfs_error_text:n {#1} }
8856      { \textcolor{tableNoValue!55!black}{\emph{not~provided}} }
8857  }
8858
8859  \cs_new:Npn \@@_rfs_error_text:n #1
8860  { \textcolor{error}{\texttt{<#1>}} }
8861
8862  \cs_new_protected:Npn \@@_rfs_warn_missing:n #1
8863  {
8864    \msg_warning:nnee { dfg } { requested-staff-missing-field }
8865      { \l_@@_rfs_project_tl }
8866      { no.~\int_use:N \l_@@_rfs_display_number_int \c_colon_str\space #1 }
8867  }
8868
8869  \cs_new_protected:Npn \@@_rfs_row_clear:
8870  {
8871    \tl_clear:N \l_@@_rfs_ref_person_tl
8872    \tl_clear:N \l_@@_rfs_firstname_tl
8873    \tl_clear:N \l_@@_rfs_lastname_tl
8874    \tl_clear:N \l_@@_rfs_degree_tl
8875    \tl_clear:N \l_@@_rfs_position_tl
8876    \tl_clear:N \l_@@_rfs_field_tl
8877    \tl_clear:N \l_@@_rfs_ref_institution_tl
8878    \tl_clear:N \l_@@_rfs_ref_affiliation_tl
8879    \tl_clear:N \l_@@_rfs_department_tl
8880    \tl_clear:N \l_@@_rfs_funding_status_tl
8881    \tl_clear:N \l_@@_rfs_staff_type_tl
8882    \tl_clear:N \l_@@_rfs_commitment_tl
8883    \tl_clear:N \l_@@_rfs_category_tl
8884    \tl_clear:N \l_@@_rfs_funding_source_tl
8885    \tl_clear:N \l_@@_rfs_description_tl
8886  }
8887
8888  \cs_new_protected:Npn \@@_rfs_store:N #1
8889  {
8890    \seq_put_right:Ne #1
8891    {
8892      ref-person-id = { \exp_not:V \pRFSRefPerson },
8893      firstname = { \exp_not:V \pRFSFirstname },
8894      lastname = { \exp_not:V \pRFSLastname },
8895      academic-degree = { \exp_not:V \pRFSDegree },
8896      positionText = { \exp_not:V \pRFSPosition },
8897      field-of-research = { \exp_not:V \pRFSField },
8898      ref-institution-id = { \exp_not:V \pRFSRefInstitution },
8899      ref-affiliation-id = { \exp_not:V \pRFSRefAffiliation },
8900      department = { \exp_not:V \pRFSDepartment },
8901      funding-status = { \exp_not:V \pRFSFundingStatus },
8902      staff-type = { \exp_not:V \pRFSStaffType },
8903      commitment = { \exp_not:V \pRFSCommitment },
8904      category = { \exp_not:V \pRFSCategory },
8905      funding-source = { \exp_not:V \pRFSFundingSource },
8906      description = { \exp_not:V \pRFSDescription }
8907    }
8908  }
8909
8910  \cs_new_protected:Npn \@@_rfs_collect_current:
8911  {
8912    % Preserve the null marker for an omitted position until person lookup.
8913    \clist_map_inline:nn
8914    {
8915      \pRFSRefPerson, \pRFSFirstname, \pRFSLastname, \pRFSDegree,
8916      \pRFSField, \pRFSRefInstitution,
8917      \pRFSRefAffiliation, \pRFSDepartment, \pRFSCommitment,
8918      \pRFSCategory, \pRFSFundingSource, \pRFSDescription
8919    }
8920    { \DTLifnullorempty{##1}{\cs_set:Npn ##1 {}}{} }
8921    \tl_if_blank:VTF \pRFSRefPerson
8922    {
```

```
      \str_if_eq:VnTF \pRFSLastname { TBA }
      {
        \tl_set:Ne \l_tmpa_tl
        {
          \exp_not:V \pRFSProject / \exp_not:V \pRFSFundingStatus /
          \exp_not:V \pRFSStaffType / vacancy /
          \exp_not:V \pRFSCategory / \exp_not:V \pRFSPosition /
          \exp_not:V \pRFSField / \exp_not:V \pRFSRefInstitution /
          \exp_not:V \pRFSDepartment
        }
      }
      {
        \tl_set:Ne \l_tmpa_tl
        {
          \exp_not:V \pRFSProject / \exp_not:V \pRFSFundingStatus /
          \exp_not:V \pRFSStaffType / name /
          \exp_not:V \pRFSLastname / \exp_not:V \pRFSFirstname
        }
      }
    }
    {
      \tl_set:Ne \l_tmpa_tl
      {
        \exp_not:V \pRFSProject / \exp_not:V \pRFSFundingStatus /
        \exp_not:V \pRFSStaffType / person / \exp_not:V \pRFSRefPerson
      }
    }
    \prop_get:NVNTF \l_@@_rfs_duplicate_prop \l_tmpa_tl \l_tmpb_tl
    { \msg_warning:nne { dfg } { requested-staff-duplicate } { \l_tmpa_tl } }
    { \prop_put:NVn \l_@@_rfs_duplicate_prop \l_tmpa_tl { 1 } }
    \tl_set:Ne \l_tmpa_tl
      { \pRFSFundingStatus / \pRFSStaffType }
    \str_case:VnF \l_tmpa_tl
    {
      { existing/research }
        { \@@_rfs_store:N \l_@@_rfs_existing_research_seq }
      { existing/nonResearch }
        { \@@_rfs_store:N \l_@@_rfs_existing_nonresearch_seq }
      { requested/research }
        { \@@_rfs_store:N \l_@@_rfs_requested_research_seq }
      { requested/nonResearch }
        { \@@_rfs_store:N \l_@@_rfs_requested_nonresearch_seq }
    }
    { }
  }

  \cs_new_protected:Npn \@@_rfs_use_fallback:NN #1#2
  {
    \tl_if_blank:VT #1
    { \DTLifnulloremptyempty{#2}{ }{ \tl_set:NV #1 #2 } }
  }

  \cs_new_protected:Npn \@@_rfs_resolve_person:
  {
    \bool_set_false:N \l_@@_rfs_person_resolved_bool
    \tl_if_blank:VF \l_@@_rfs_ref_person_tl
    {
      \DTLifdbexists{person}
      {
        \DTLgetvalueforkey{\l_tmpa_tl}{lastname}{person}{id}
          {\l_@@_rfs_ref_person_tl}
        \DTLifnulloremptyempty{\l_tmpa_tl}
        {
          \msg_warning:nnee { dfg } { requested-staff-unresolved-person }
            { \l_@@_rfs_project_tl } { \l_@@_rfs_ref_person_tl }
        }
        {
          \bool_set_true:N \l_@@_rfs_person_resolved_bool
          \@@_rfs_use_fallback:NN \l_@@_rfs_lastname_tl \l_tmpa_tl
          \DTLgetvalueforkey{\l_tmpa_tl}{firstname}{person}{id}
            {\l_@@_rfs_ref_person_tl}
          \@@_rfs_use_fallback:NN \l_@@_rfs_firstname_tl \l_tmpa_tl
          \DTLgetvalueforkey{\l_tmpa_tl}{academicdegree}{person}{id}
            {\l_@@_rfs_ref_person_tl}
          \@@_rfs_use_fallback:NN \l_@@_rfs_degree_tl \l_tmpa_tl
          % Only a missing database value inherits; an explicit empty value wins.
          \DTLifnull{\l_@@_rfs_position_tl}
          {
            \DTLgetvalueforkey{\l_tmpa_tl}{positionText}{person}{id}
              {\l_@@_rfs_ref_person_tl}
            \tl_set:NV \l_@@_rfs_position_tl \l_tmpa_tl
          }
          { }
          \DTLgetvalueforkey{\l_tmpa_tl}{position}{person}{id}
            {\l_@@_rfs_ref_person_tl}
          \@@_rfs_use_fallback:NN \l_@@_rfs_category_tl \l_tmpa_tl
          \bool_lazy_all:nT
```

```
        {
          { \tl_if_blank_p:V \l_@@_rfs_department_tl }
          { \tl_if_blank_p:V \l_@@_rfs_ref_institution_tl }
          { \tl_if_blank_p:V \l_@@_rfs_ref_affiliation_tl }
        }
        {
          \DTLgetvalueforkey{\l_tmpa_tl}{ref-institution-id}{person}{id}
            {\l_@@_rfs_ref_person_tl}
          \@@_rfs_use_fallback:NN \l_@@_rfs_ref_institution_tl \l_tmpa_tl
          \tl_if_blank:VT \l_@@_rfs_ref_institution_tl
          {
            \DTLgetvalueforkey{\l_tmpa_tl}{ref-affiliation-id}{person}{id}
              {\l_@@_rfs_ref_person_tl}
            \@@_rfs_use_fallback:NN \l_@@_rfs_ref_affiliation_tl \l_tmpa_tl
          }
        }
      }
    }
    {
      \msg_warning:nnee { dfg } { requested-staff-unresolved-person }
        { \l_@@_rfs_project_tl } { \l_@@_rfs_ref_person_tl }
    }
  }
  \DTLifnull{\l_@@_rfs_position_tl}
    { \tl_clear:N \l_@@_rfs_position_tl }{ }
}

\cs_new_protected:Npn \@@_rfs_register_affiliation:n #1
{
  \tl_set:Ne \l_tmpa_tl {#1}
  \tl_clear:N \l_@@_rfs_affiliation_name_tl
  \tl_clear:N \l_@@_rfs_affiliation_abbreviation_tl
  \DTLifdbexists{affiliation}
  {
    \DTLgetvalueforkey{\l_@@_rfs_affiliation_name_tl}
      {name}{affiliation}{id}{\l_tmpa_tl}
    \DTLifnulloremptyl{\l_@@_rfs_affiliation_name_tl}
    {
      \msg_warning:nnee { dfg } { requested-staff-unresolved-affiliation }
        { \l_@@_rfs_project_tl } { \l_tmpa_tl }
      \tl_clear:N \l_@@_rfs_affiliation_name_tl
    }
    {
      \DTLgetvalueforkey{\l_@@_rfs_affiliation_abbreviation_tl}
        {abbreviation}{affiliation}{id}{\l_tmpa_tl}
      \DTLifnulloremptyl{\l_@@_rfs_affiliation_abbreviation_tl}
      {
        \msg_warning:nne { dfg } { requested-staff-missing-abbreviation }
          { \l_tmpa_tl }
        \tl_set:NV \l_@@_rfs_affiliation_abbreviation_tl
          \l_@@_rfs_affiliation_name_tl
      }
      { }
      \seq_if_in:NVF \l_@@_rfs_affiliation_seq \l_tmpa_tl
      {
        \seq_put_right:NV \l_@@_rfs_affiliation_seq \l_tmpa_tl
        \prop_put:NVV \l_@@_rfs_affiliation_abbreviation_prop
          \l_tmpa_tl \l_@@_rfs_affiliation_abbreviation_tl
        \prop_put:NVV \l_@@_rfs_affiliation_name_prop
          \l_tmpa_tl \l_@@_rfs_affiliation_name_tl
      }
    }
  }
  {
    \msg_warning:nnee { dfg } { requested-staff-unresolved-affiliation }
      { \l_@@_rfs_project_tl } { \l_tmpa_tl }
  }
}

\cs_new_protected:Npn \@@_rfs_resolve_organisation:
{
  \tl_if_blank:VT \l_@@_rfs_department_tl
  {
    \tl_if_blank:VTF \l_@@_rfs_ref_institution_tl
    {
      % Compatibility fallback for roster declarations written before the
      % institution-level key was introduced.
      \tl_if_blank:VTF \l_@@_rfs_ref_affiliation_tl
      {
        \@@_rfs_warn_missing:n { department~or~ref-institution-id }
        \tl_set:Nn \l_@@_rfs_department_tl
          { \SetCell{bg=tableNoValue}\@@_rfs_missing_text:n {department} }
      }
      {
        \@@_rfs_register_affiliation:n { \l_@@_rfs_ref_affiliation_tl }
        \tl_if_blank:VTF \l_@@_rfs_affiliation_name_tl
        {
```

```
            \tl_set:Ne \l_@@_rfs_department_tl
            {
              \exp_not:N \SetCell { bg=tableNoValue }
              \exp_not:N \@@_rfs_error_text:n
                { \exp_not:V \l_@@_rfs_ref_affiliation_tl }
            }
          }
          {
            \tl_set:NV \l_@@_rfs_department_tl
              \l_@@_rfs_affiliation_abbreviation_tl
          }
        }
      }
      {
        \tl_clear:N \l_@@_rfs_department_tl
        \DTLifdbexists{institution}
        {
          \clist_set:NV \l_tmpa_clist \l_@@_rfs_ref_institution_tl
          \bool_set_true:N \l_tmpa_bool
          \clist_map_inline:Nn \l_tmpa_clist
          {
            \bool_if:NTF \l_tmpa_bool
              { \bool_set_false:N \l_tmpa_bool }
              { \tl_put_right:Nn \l_@@_rfs_department_tl {,~} }
            \DTLgetvalueforkey{\l_@@_rfs_institution_affiliation_tl}
              {ref-affiliation-id}{institution}{id}{##1}
            \DTLifnulloremptyl{\l_@@_rfs_institution_affiliation_tl}
            {
              \msg_warning:nnee { dfg } { requested-staff-unresolved-institution }
                { \l_@@_rfs_project_tl } {##1}
              \tl_put_right:Ne \l_@@_rfs_department_tl
                { \exp_not:N \@@_rfs_error_text:n { \exp_not:n {##1} } }
            }
            {
              \DTLgetvalueforkey{\l_@@_rfs_institution_name_tl}
                {name}{institution}{id}{##1}
              \@@_institution_abbreviation_or_name:nN
                {##1} \l_@@_rfs_institution_abbreviation_tl
              \@@_rfs_register_affiliation:n
                { \l_@@_rfs_institution_affiliation_tl }
              \DTLifnulloremptyl{\l_@@_rfs_institution_name_tl}
              {
                \tl_set:NV \l_@@_rfs_institution_name_tl
                  \l_@@_rfs_affiliation_name_tl
              }
              { }
              \DTLifnulloremptyl{\l_@@_rfs_institution_abbreviation_tl}
              {
                \tl_set:NV \l_@@_rfs_institution_abbreviation_tl
                  \l_@@_rfs_affiliation_abbreviation_tl
              }
              { }
              \tl_put_right:NV \l_@@_rfs_department_tl
                \l_@@_rfs_institution_abbreviation_tl
              \tl_set:Nn \l_tmpa_tl {##1}
              \seq_if_in:NVF \l_@@_rfs_institution_seq \l_tmpa_tl
              {
                \seq_put_right:NV \l_@@_rfs_institution_seq \l_tmpa_tl
                \prop_put:NVV \l_@@_rfs_institution_name_prop
                  \l_tmpa_tl \l_@@_rfs_institution_name_tl
                \prop_put:NVV \l_@@_rfs_institution_abbreviation_prop
                  \l_tmpa_tl \l_@@_rfs_institution_abbreviation_tl
                \prop_put:NVV
                  \l_@@_rfs_institution_affiliation_abbreviation_prop
                  \l_tmpa_tl \l_@@_rfs_affiliation_abbreviation_tl
              }
            }
          }
        }
        {
          \msg_warning:nnee { dfg } { requested-staff-unresolved-institution }
            { \l_@@_rfs_project_tl } { \l_@@_rfs_ref_institution_tl }
          \tl_set:Ne \l_@@_rfs_department_tl
          {
            \exp_not:N \SetCell { bg=tableNoValue }
            \exp_not:N \@@_rfs_error_text:n
              { \exp_not:V \l_@@_rfs_ref_institution_tl }
          }
        }
      }
    }
  }

  \cs_new_protected:Npn \@@_rfs_make_cell:NnN #1#2#3
  {
    \tl_if_blank:VTF #1
    {
```

```
    \@@_rfs_warn_missing:n {#2}
    \tl_set:Nn #3 { \SetCell{bg=tableNoValue}\@@_rfs_missing_text:n {#2} }
  }
  { \tl_set:NV #3 #1 }
}

\cs_new_protected:Npn \@@_rfs_make_field_cell:
{
  \bool_lazy_all:nTF
  {
    { \str_if_eq_p:Vn \l_@@_rfs_funding_status_tl { existing } }
    { \str_if_eq_p:Vn \l_@@_rfs_staff_type_tl { nonResearch } }
    { \tl_if_blank_p:V \l_@@_rfs_field_tl }
  }
  { \tl_set:Nn \l_@@_rfs_field_cell_tl { \SetCell{bg=tableNoValue} } }
  {
    \@@_rfs_make_cell:NnN \l_@@_rfs_field_tl
      { field-of-research } \l_@@_rfs_field_cell_tl
  }
}

\cs_new_protected:Npn \@@_rfs_make_name_cell:
{
  \bool_set_false:N \l_@@_rfs_name_incomplete_bool
  \str_if_eq:VnTF \l_@@_rfs_lastname_tl { TBA }
  {
    \tl_set:Nn \l_@@_rfs_plain_name_tl { TBA }
    \tl_set:Nn \l_@@_rfs_description_name_tl { TBA }
    \tl_set:Nn \l_tmpa_tl { TBA }
  }
  {
    \tl_if_blank:VT \l_@@_rfs_lastname_tl
    {
      \@@_rfs_warn_missing:n { lastname~or~resolvable~ref-person-id }
      \tl_set:Nn \l_@@_rfs_lastname_tl
      {
        \bool_if:NTF \l_@@_class_fastcompile_bool
          { \@@_rfs_error_text:n {lastname} }
          { \@@_rfs_error_text:n {name} }
      }
      \bool_set_true:N \l_@@_rfs_name_incomplete_bool
    }
    \tl_if_blank:VT \l_@@_rfs_firstname_tl
    {
      \@@_rfs_warn_missing:n { firstname }
      \tl_set:Nn \l_@@_rfs_firstname_tl { \@@_rfs_missing_text:n {firstname} }
      \bool_set_true:N \l_@@_rfs_name_incomplete_bool
    }
    \tl_set:Ne \l_@@_rfs_plain_name_tl
      {
        \exp_not:V \l_@@_rfs_lastname_tl,\exp_not:N \space
        \exp_not:V \l_@@_rfs_firstname_tl
      }
    \tl_clear:N \l_@@_rfs_description_name_tl
    \tl_if_blank:VF \l_@@_rfs_degree_tl
      {
        \tl_put_right:Ne \l_@@_rfs_description_name_tl
          { \exp_not:V \l_@@_rfs_degree_tl \exp_not:N \space }
      }
    \tl_put_right:Ne \l_@@_rfs_description_name_tl
    {
      \exp_not:V \l_@@_rfs_firstname_tl \exp_not:N \space
      \exp_not:V \l_@@_rfs_lastname_tl
    }
    \tl_set:NV \l_tmpa_tl \l_@@_rfs_plain_name_tl
    \tl_if_blank:VF \l_@@_rfs_degree_tl
    {
      \tl_put_right:Ne \l_tmpa_tl
        { ,\exp_not:N \space \exp_not:V \l_@@_rfs_degree_tl }
    }
    \bool_if:NT \l_@@_rfs_person_resolved_bool
    {
      \str_if_eq:VnT \l_@@_rfs_funding_status_tl { existing }
      {
        \tl_set:Ne \l_tmpa_tl
        {
          \exp_not:N \textbf { \exp_not:V \l_@@_rfs_lastname_tl },
          \exp_not:N \space \exp_not:V \l_@@_rfs_firstname_tl
        }
        \tl_if_blank:VF \l_@@_rfs_degree_tl
        {
          \tl_put_right:Ne \l_tmpa_tl
            { ,\exp_not:N \space \exp_not:V \l_@@_rfs_degree_tl }
        }
      }
    }
  }
```

```
    \tl_if_blank:VF \l_@@_rfs_position_tl
    {
      \tl_put_right:Ne \l_tmpa_tl
      {
        \exp_not:N \newline
        (\exp_not:V \l_@@_rfs_position_tl)
      }
    }
    \bool_if:NF \l_@@_rfs_person_resolved_bool
    {
      \tl_if_blank:VF \l_@@_rfs_ref_person_tl
      {
        \bool_set_true:N \l_@@_rfs_name_incomplete_bool
        \tl_put_right:Ne \l_tmpa_tl
        {
          \exp_not:N \space
          \exp_not:N \@@_rfs_error_text:n
            { \exp_not:V \l_@@_rfs_ref_person_tl }
        }
      }
    }
    \tl_clear:N \l_@@_rfs_name_cell_tl
    \bool_if:NT \l_@@_rfs_name_incomplete_bool
    { \tl_put_right:Nn \l_@@_rfs_name_cell_tl { \SetCell{bg=tableNoValue} } }
    \bool_if:NTF \l_@@_rfs_person_resolved_bool
    {
      \tl_put_right:Ne \l_@@_rfs_name_cell_tl
      {
        \exp_not:N \hyperlink { pi:\exp_not:V \l_@@_rfs_ref_person_tl }
        {
          \exp_not:N \textcolor { referencePI }
          {
            \exp_not:N \dfgShowCalculatedTag{RequestedFundingStaffTable}
              {RequestedFundingStaff,person}
              {\exp_not:V \l_tmpa_tl}
          }
        }
      }
    }
    {
      \tl_put_right:Ne \l_@@_rfs_name_cell_tl
      {
        \exp_not:N \dfgShowCalculatedTag{RequestedFundingStaffTable}
          {RequestedFundingStaff,person}
          {\exp_not:V \l_tmpa_tl}
      }
    }
  }

  \cs_new_protected:Npn \@@_rfs_render_row:nnnN #1#2#3#4
  {
    \int_incr:N \l_@@_rfs_display_number_int
    \@@_rfs_resolve_person:
    \@@_rfs_resolve_organisation:
    \@@_rfs_make_name_cell:
    \@@_rfs_make_field_cell:
    \tl_if_blank:VF \l_@@_rfs_field_tl
    {
      \tl_set:Ne \l_@@_rfs_field_cell_tl
      {
        \exp_not:N \dfgShowTag
          {\exp_not:V \l_@@_rfs_project_tl/
           \int_use:N \l_@@_rfs_display_number_int}
          {field-of-research}{\exp_not:V \l_@@_rfs_field_cell_tl}
          [RequestedFundingStaff]
      }
    }
    \tl_set:NV \l_@@_rfs_department_cell_tl \l_@@_rfs_department_tl
    \tl_if_in:VnF \l_@@_rfs_department_cell_tl { \SetCell }
    {
      \tl_set:Ne \l_@@_rfs_department_cell_tl
      {
        \exp_not:N \dfgShowCalculatedTag{RequestedFundingStaffTable}
          {RequestedFundingStaff,person,institution,affiliation}
          {\exp_not:V \l_@@_rfs_department_cell_tl}
      }
    }
    \str_if_eq:VnTF \l_@@_rfs_funding_status_tl { requested }
    {
      \tl_set:Nn \l_@@_rfs_commitment_cell_tl
        { \SetCell{bg=tableNoValue} }
      \tl_set:Nn \l_@@_rfs_funding_source_cell_tl
        { \SetCell{bg=tableNoValue} }
    }
    {
      \@@_rfs_make_cell:NnN \l_@@_rfs_commitment_tl
        { commitment } \l_@@_rfs_commitment_cell_tl
```

```
      \@@_rfs_make_cell:NnN \l_@@_rfs_funding_source_tl
        { funding-source } \l_@@_rfs_funding_source_cell_tl
      \tl_if_blank:VF \l_@@_rfs_commitment_tl
      {
        \tl_set:Ne \l_@@_rfs_commitment_cell_tl
        {
          \exp_not:N \dfgShowTag
            {\exp_not:V \l_@@_rfs_project_tl/
             \int_use:N \l_@@_rfs_display_number_int}
            {commitment}{\exp_not:V \l_@@_rfs_commitment_cell_tl}
            [RequestedFundingStaff]
        }
      }
      \tl_if_blank:VF \l_@@_rfs_funding_source_tl
      {
        \tl_set:Ne \l_@@_rfs_funding_source_cell_tl
        {
          \exp_not:N \dfgShowTag
            {\exp_not:V \l_@@_rfs_project_tl/
             \int_use:N \l_@@_rfs_display_number_int}
            {funding-source}{\exp_not:V \l_@@_rfs_funding_source_cell_tl}
            [RequestedFundingStaff]
        }
      }
    }
    \str_if_eq:VnTF \l_@@_rfs_funding_status_tl { existing }
    {
      \tl_set:Nn \l_@@_rfs_category_cell_tl
        { \SetCell{bg=tableNoValue} }
    }
    {
      \tl_if_blank:VTF \l_@@_rfs_category_tl
      {
        \@@_rfs_warn_missing:n { category }
        \tl_set:Nn \l_@@_rfs_category_cell_tl
          { \SetCell{bg=tableNoValue}\@@_rfs_missing_text:n {category} }
      }
      {
        \prop_get:NVNTF \g_@@_dfg_category_names_prop
          \l_@@_rfs_category_tl \l_@@_rfs_category_cell_tl
        { }
        {
          \tl_set:Ne \l_@@_rfs_category_cell_tl
            { \exp_not:N \@@_rfs_error_text:n { \exp_not:V \l_@@_rfs_category_tl } }
        }
      }
    }
    \tl_if_in:VnF \l_@@_rfs_category_cell_tl { \SetCell }
    {
      \tl_set:Ne \l_@@_rfs_category_cell_tl
      {
        \exp_not:N \dfgShowCalculatedTag{RequestedFundingStaffTable}
          {RequestedFundingStaff,person}
          {\exp_not:V \l_@@_rfs_category_cell_tl}
      }
    }
    \int_compare:nNnTF {#1} = {1}
    {
      \tl_set:Ne \l_@@_rfs_group_cell_tl
      {
        \exp_not:N \SetCell [ r=#2 ] { l }
        { \exp_not:n {#3} } &
      }
    }
    { \tl_set:Nn \l_@@_rfs_group_cell_tl { & } }
    \tl_put_right:Ne \l_@@_rfs_table_body_tl
    {
      \exp_not:V \l_@@_rfs_group_cell_tl
      \int_use:N \l_@@_rfs_display_number_int .
      & \exp_not:V \l_@@_rfs_name_cell_tl
      & \exp_not:V \l_@@_rfs_field_cell_tl
      & \exp_not:V \l_@@_rfs_department_cell_tl
      & \exp_not:V \l_@@_rfs_commitment_cell_tl
      & \exp_not:V \l_@@_rfs_category_cell_tl
      & \exp_not:V \l_@@_rfs_funding_source_cell_tl
      \\
    }
    \tl_if_blank:VT \l_@@_rfs_description_tl
    {
      \@@_rfs_warn_missing:n { description }
      \tl_set:Nn \l_@@_rfs_description_tl
        { \textcolor{error}{Description~not~provided.} }
    }
    \tl_put_right:Ne #4
    {
      \exp_not:N \item
        [\exp_not:N \textbf
```

```
9445          { \int_use:N \l_@@_rfs_display_number_int. }]
9446      \exp_not:N \textbf
9447        { \exp_not:N \dfgShowCalculatedTag{RequestedFundingStaffTable}
9448            {RequestedFundingStaff,person}
9449            {\exp_not:V \l_@@_rfs_description_name_tl} }
9450      \space
9451      \exp_not:N \dfgShowTag
9452        {\exp_not:V \l_@@_rfs_project_tl/
9453         \int_use:N \l_@@_rfs_display_number_int}
9454        {description}{\exp_not:V \l_@@_rfs_description_tl}
9455        [RequestedFundingStaff]
9456    }
9457  }
9458
9459  \cs_new_protected:Npn \@@_rfs_render_group:NnN #1#2#3
9460  {
9461    \seq_map_indexed_inline:Nn #1
9462    {
9463      \@@_rfs_row_clear:
9464      \keys_set:nn { dfg / requested-funding-staff-row } {##2}
9465      \@@_rfs_render_row:nnnN {##1} { \seq_count:N #1 } {#2} #3
9466    }
9467  }
9468
9469  \cs_new_protected:Npn \@@_rfs_render_status:NNnN #1#2#3#4
9470  {
9471    \int_compare:nNnT { \seq_count:N #1 + \seq_count:N #2 } > {0}
9472    {
9473      \tl_put_right:Nn \l_@@_rfs_table_body_tl
9474      {
9475        \SetRow{bg=tableHeadingsBGalt,font=\footnotesize\bfseries}
9476        \SetCell[c=8]{l} #3 & & & & & & & \\*
9477      }
9478      \@@_rfs_render_group:NnN #1 { Research~staff } #4
9479      \seq_if_empty:NF #1
9480      {
9481        \seq_if_empty:NF #2
9482        {
9483          \tl_put_right:Nn \l_@@_rfs_table_body_tl
9484            { \SetHline{-}{tableOuterLines,0.5pt} }
9485        }
9486      }
9487      \@@_rfs_render_group:NnN #2 { Non-research~staff } #4
9488    }
9489  }
9490
9491  \cs_new_protected:Npn \@@_rfs_print_legend:
9492  {
9493    \bool_lazy_or:nnT
9494      { ! \seq_if_empty_p:N \l_@@_rfs_institution_seq }
9495      { ! \seq_if_empty_p:N \l_@@_rfs_affiliation_seq }
9496    {
9497      \par\smallskip\noindent
9498      \group_begin:
9499      \footnotesize
9500      \seq_if_empty:NF \l_@@_rfs_institution_seq
9501      {
9502        \textbf{Institutions:}\space
9503        \seq_map_indexed_inline:Nn \l_@@_rfs_institution_seq
9504        {
9505          \int_compare:nNnF {##1} = {1} {;\space}
9506          \tl_set:Nn \l_tmpa_tl {##2}
9507          \prop_get:NVN \l_@@_rfs_institution_name_prop
9508            \l_tmpa_tl \l_tmpb_tl
9509          \prop_get:NVN \l_@@_rfs_institution_abbreviation_prop
9510            \l_tmpa_tl \l_tmpd_tl
9511          \prop_get:NVN \l_@@_rfs_institution_affiliation_abbreviation_prop
9512            \l_tmpa_tl \l_tmpc_tl
9513          \textbf{\dfgShowCalculatedTag{institutionLabel}{institution}
9514            {\l_tmpd_tl}}:~
9515          \dfgShowTag{##2}{name}{\l_tmpb_tl}[institution]\space
9516          (\dfgShowCalculatedTag{affiliationLabel}{affiliation}
9517            {\l_tmpc_tl})
9518        }
9519        .\space
9520      }
9521      \seq_if_empty:NF \l_@@_rfs_affiliation_seq
9522      {
9523        \textbf{Affiliations:}\space
9524        \seq_map_indexed_inline:Nn \l_@@_rfs_affiliation_seq
9525        {
9526          \int_compare:nNnF {##1} = {1} {;\space}
9527          \tl_set:Nn \l_tmpa_tl {##2}
9528          \prop_get:NVN \l_@@_rfs_affiliation_abbreviation_prop
9529            \l_tmpa_tl \l_tmpb_tl
9530          \prop_get:NVN \l_@@_rfs_affiliation_name_prop
9531            \l_tmpa_tl \l_tmpc_tl
```

```
9532          \textbf{\dfgShowCalculatedTag{affiliationLabel}{affiliation}
9533            {\l_tmpb_tl}}:~
9534          \dfgShowTag{##2}{name}{\l_tmpc_tl}[affiliation]
9535        }
9536        .
9537      }
9538      \par
9539      \group_end:
9540    }
9541  }
9542
9543  \cs_new_protected:Npn \@@_rfs_no_data:n #1
9544  {
9545    \msg_warning:nnn { dfg } { requested-staff-no-data } {#1}
9546    \par\noindent
9547    \textcolor{tableNoValue!55!black}
9548      {No~staff~data~have~been~provided~for~this~project.}
9549    \par
9550  }
9551
9552  \NewDocumentCommand{\RequestedFundingStaffTable}{m}
9553  {
9554    \group_begin:
9555    \tl_set:Nn \l_@@_rfs_project_tl {#1}
9556    \tl_clear:N \l_@@_rfs_project_status_tl
9557    \DTLifdbexists{project}
9558    {
9559      \DTLgetvalueforkey{\l_@@_rfs_project_status_tl}
9560        {project-status}{project}{id}{#1}
9561    }
9562    { }
9563    \str_if_eq:VnTF \l_@@_rfs_project_status_tl { E }
9564    {
9565      \tl_set:Nn \l_@@_rfs_existing_group_label_tl {Existing~staff}
9566      \tl_set:Nn \l_@@_rfs_requested_group_label_tl
9567        {Staff~funded~with~approved~grant~money}
9568      \tl_set:Nn \l_@@_rfs_existing_description_heading_tl
9569        {Job~descriptions~of~staff~(supported~through~existing~funds):}
9570      \tl_set:Nn \l_@@_rfs_requested_description_heading_tl
9571        {Job~descriptions~of~staff~(funded~with~approved~grant~money):}
9572    }
9573    {
9574      \tl_set:Nn \l_@@_rfs_existing_group_label_tl {Existing~staff}
9575      \tl_set:Nn \l_@@_rfs_requested_group_label_tl {Requested~staff}
9576      \tl_set:Nn \l_@@_rfs_existing_description_heading_tl
9577        {Job~descriptions~of~staff~for~the~proposed~funding~period~
9578          (supported~through~existing~funds):}
9579      \tl_set:Nn \l_@@_rfs_requested_description_heading_tl
9580        {Job~descriptions~of~staff~for~the~proposed~funding~period~
9581          (requested~funds):}
9582    }
9583    \seq_clear:N \l_@@_rfs_existing_research_seq
9584    \seq_clear:N \l_@@_rfs_existing_nonresearch_seq
9585    \seq_clear:N \l_@@_rfs_requested_research_seq
9586    \seq_clear:N \l_@@_rfs_requested_nonresearch_seq
9587    \seq_clear:N \l_@@_rfs_institution_seq
9588    \seq_clear:N \l_@@_rfs_affiliation_seq
9589    \prop_clear:N \l_@@_rfs_duplicate_prop
9590    \prop_clear:N \l_@@_rfs_institution_name_prop
9591    \prop_clear:N \l_@@_rfs_institution_abbreviation_prop
9592    \prop_clear:N \l_@@_rfs_institution_affiliation_abbreviation_prop
9593    \prop_clear:N \l_@@_rfs_affiliation_abbreviation_prop
9594    \prop_clear:N \l_@@_rfs_affiliation_name_prop
9595    \tl_clear:N \l_@@_rfs_table_body_tl
9596    \tl_clear:N \l_@@_rfs_existing_descriptions_tl
9597    \tl_clear:N \l_@@_rfs_requested_descriptions_tl
9598    \DTLifdbexists{RequestedFundingStaff}
9599    {
9600      \DTLforeach[\DTLiseq{#1}{\pRFSProject}]
9601        {RequestedFundingStaff}
9602      {
9603        \pRFSProject=ref-project-id,
9604        \pRFSFundingStatus=funding-status,
9605        \pRFSStaffType=staff-type,
9606        \pRFSRefPerson=ref-person-id,
9607        \pRFSFirstname=firstname,
9608        \pRFSLastname=lastname,
9609        \pRFSDegree=academic-degree,
9610        \pRFSPosition=positionText,
9611        \pRFSField=field-of-research,
9612        \pRFSRefInstitution=ref-institution-id,
9613        \pRFSRefAffiliation=ref-affiliation-id,
9614        \pRFSDepartment=department,
9615        \pRFSCommitment=commitment,
9616        \pRFSCategory=category,
9617        \pRFSFundingSource=funding-source,
9618        \pRFSDescription=description
```

```
    }
    { \@@_rfs_collect_current: }
    \int_set:Nn \l_@@_rfs_record_count_int
    {
      \seq_count:N \l_@@_rfs_existing_research_seq
      + \seq_count:N \l_@@_rfs_existing_nonresearch_seq
      + \seq_count:N \l_@@_rfs_requested_research_seq
      + \seq_count:N \l_@@_rfs_requested_nonresearch_seq
    }
    \int_compare:nNnTF { \l_@@_rfs_record_count_int } = {0}
    { \@@_rfs_no_data:n {#1} }
    {
      \int_zero:N \l_@@_rfs_display_number_int
      \@@_rfs_render_status:NNnN
        \l_@@_rfs_existing_research_seq
        \l_@@_rfs_existing_nonresearch_seq
        {\tl_use:N \l_@@_rfs_existing_group_label_tl}
        \l_@@_rfs_existing_descriptions_tl
      \@@_rfs_render_status:NNnN
        \l_@@_rfs_requested_research_seq
        \l_@@_rfs_requested_nonresearch_seq
        {\tl_use:N \l_@@_rfs_requested_group_label_tl}
        \l_@@_rfs_requested_descriptions_tl
      \begin{tblr}[
        theme=requestedFundingStaff,
        long,
        postsep=0pt,
        expand=\l_@@_rfs_table_body_tl
      ]{
        colsep=2pt,
        width=\textwidth,
        hline{1,Z}={1pt,fg=tableOuterLines},
        hline{2}={1pt,fg=tableOuterLines},
        stretch=.8,
        rowsep=.500pt,
        colspec={X[1.1,l]
                 X[.42,c] % no
                 X[2.18,l] % Name
                 X[1.4,l] % field
                 X[1.5,c] % department
                 X[1.05,c] % proj com.
                 X[1.1,c] % category
                 X[.95,c] % funding
                 },
        column{1}={font=\footnotesize\bfseries,valign=m},
        column{2}={halign=c},
        rows={font=\footnotesize,valign=t},
        rowhead=1,
        row{1}={bg=tableHeadingsBG,fg=tableHeadingsFont,
          font=\scriptsize\bfseries,guard},
        cell{1}{1-8}={halign=c,valign=m}
      }
        & {No.}
        & {Name,~academic\\degree,~position}
        & {Field~of\\research}
        & {Departm.~of\\(non-)univers.\\institution}
        & {Project\\commitm.~(h/wk)}
        & Category
        & {Funding\\source} \\
        \l_@@_rfs_table_body_tl
      \end{tblr}
      \@@_rfs_print_legend:
      \tl_if_empty:NF \l_@@_rfs_existing_descriptions_tl
      {
        \par\medskip\noindent
        {\tl_use:N \l_@@_rfs_existing_description_heading_tl}\par\nobreak
        {\normalfont\normalsize\setlength{\parskip}{0pt}
          \begin{itemize}[
            label={},
            labelindent=0pt,
            labelwidth=1.5em,
            labelsep=.5em,
            leftmargin=2em,
            align=left,
            beginpenalty=10000]
            \tl_use:N \l_@@_rfs_existing_descriptions_tl
          \end{itemize}}
      }
      \tl_if_empty:NF \l_@@_rfs_requested_descriptions_tl
      {
        %\par\medskip\noindent
        {\tl_use:N \l_@@_rfs_requested_description_heading_tl}\par\nobreak
        {\normalfont\normalsize\setlength{\parskip}{0pt}
          \begin{itemize}[
            label={},
            labelindent=0pt,
            labelwidth=1.5em,
```

```
            labelsep=.5em,
            leftmargin=2em,
            align=left,
            beginpenalty=10000]
            \tl_use:N \l_@@_rfs_requested_descriptions_tl
          \end{itemize}}
      }
    }
  }
  { \@@_rfs_no_data:n {#1} }
  \group_end:
}


\NewDocumentCommand{\getTotalSumRequestedFunds}{O{RequestedFunding}mm}{
  % #1: database
  % #2: project
  % #3: year
  \def\tmpLocalDB{#1}%
  \DTLifdbexists{\tmpLocalDB}{%
    \@@_funding_sum_set:Nnnnn \TotalSumRequestedFunds
      {#1}{#2}{__ALL__}{#3}
    \@@_show_calculated_number:nnN
      {getTotalSumRequestedFunds}{#1}
      \TotalSumRequestedFunds
  }{
   \texttt{\color{error}XXX}
   \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}


\NewDocumentCommand{\getSumExistingFunds}{O{RequestedAndExistingFundingDirectCosts}mm}{
  % #1: database
  % #2: project
  % #3: year
  \def\tmpLocalDB{#1}%
  \DTLifdbexists{\tmpLocalDB}{
    \@@_funding_sum_set:Nnnnn \SumExistingFunds
      {#1}{#2}{__ALL__}{#3}

    \@@_show_calculated_number:nnN
      {getSumExistingFunds}{#1}
      \SumExistingFunds
  }{
   \texttt{\color{error}XXX}
   \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}


\providecommand{\SumRequestedFunds}{0}

\NewDocumentCommand{\getSumRequestedFunds}{ O{RequestedFunding} m m m }{%
  \DTLifdbexists{#1}{%
    \@@_funding_sum_set:Nnnnn \SumRequestedFunds {#1}{#2}{#3}{#4}%
    \@@_show_calculated_number:nnN
      {getSumRequestedFunds}{#1}
      \SumRequestedFunds
  }{%
   \texttt{\color{error}XXX}%
   \msg_warning:nne { dfg } { database-not-found } { #1 }%
  }%
}
\NewDocumentCommand{\dfgCheckIfZeroShowCategory}{ O{RequestedFunding} m m }
{
  % #1: database
  % #2: project
  % #3: category/type
  \def\tmpLocalDB{#1}
  \DTLifdbexists{\tmpLocalDB}{
    \@@_funding_sum_set:Nnnnn \SumRequestedFunds {#1}{#2}{#3}
      {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
   \fp_compare:nNnT { \SumRequestedFunds } = {0}
   { \tl_clear:c {l__tmp_body_#3_tl }}
  }
  {
   \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }
}
```

This is for `Requested funding for direct costs fo the new funding period` and is partially data from new key-values and partially it has to be reused.

```
\NewDocumentCommand{\RequestedAndExistingFundingDirectCostsTable}{m}
```

```
{
  % #1: project
  \def\tmpLocalDB{RequestedAndExistingFundingDirectCosts}%
  \DTLifdbexists{\tmpLocalDB}{%
   \tl_clear:N \l__tmp_body_costs_tl
   \DTLforeach[\DTLiseq{\pProject}{#1}]{\tmpLocalDB}{
    \pProject=ref-project-id,
    \pAffiliation=ref-affiliation-id,
    \pFinancedBy=financed-by,
    \pSumYearOne=sum-year1,
    \pSumYearTwo=sum-year2,
    \pSumYearThree=sum-year3,
    \pSumYearFour=sum-year4,
    \pSumYearFive=sum-year5
   }{
  \DTLaddall{\tmpHorizontalSum}{
     \pSumYearOne,
     \pSumYearTwo,
     \pSumYearThree,
     \pSumYearFour,
     \pSumYearFive
    }

    \tl_put_right:Ne \l__tmp_body_costs_tl
    {
     \str_if_eq:VnTF { \pAffiliation }  { other-funds }
     { Other }
      { \getEntry[affiliation]{\pAffiliation}{abbreviation} : } ~
      existing~ funds~ from~
      % Freeze the database value without expanding commands such as \RWTH,
      % whose trailing \xspace must only run when the completed cell is typeset.
      \dfgShowTag{\pProject/\pAffiliation}{financed-by}
        {\exp_not:V \pFinancedBy}[\tmpLocalDB]
      \printMarginalia{\dfgShowCalculatedTag
        {RequestedAndExistingFundingDirectCostsTable}{\tmpLocalDB}
        {\@@_show_eur:n{\tmpHorizontalSum}}}
      & \dfgShowTag{\pProject/\pAffiliation}{sum-year1}
          {\@@_show_eur:n{\pSumYearOne}}[\tmpLocalDB]
      & \dfgShowTag{\pProject/\pAffiliation}{sum-year2}
          {\@@_show_eur:n{\pSumYearTwo}}[\tmpLocalDB]
      & \dfgShowTag{\pProject/\pAffiliation}{sum-year3}
          {\@@_show_eur:n{\pSumYearThree}}[\tmpLocalDB]
      & \dfgShowTag{\pProject/\pAffiliation}{sum-year4}
          {\@@_show_eur:n{\pSumYearFour}}[\tmpLocalDB]
      & \dfgShowTag{\pProject/\pAffiliation}{sum-year5}
          {\@@_show_eur:n{\pSumYearFive}}[\tmpLocalDB] \\
    }
   }
   \tl_put_right:Ne \l__tmp_body_costs_tl
   {
    \SetHline{-}{tableOuterLines,0.5pt}
    \SetRow{belowsep=5pt}
    Sum~ of~ existing~ funds
    \printMarginalia{%
     \getSumExistingFunds{#1}
     {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
    }
    & \getSumExistingFunds{#1}{sum-year1}
    & \getSumExistingFunds{#1}{sum-year2}
    & \getSumExistingFunds{#1}{sum-year3}
    & \getSumExistingFunds{#1}{sum-year4}
    & \getSumExistingFunds{#1}{sum-year5} \\
   }

   \begin{tblr}[
     theme=main,
     long,
     expand=
     \l__tmp_body_costs_tl,
   remark{Note}={Figures~ in~ EUR},
    ]{
     % colsep=3pt,
     colspec = {
       *{1}{X[4,t]}
        *{5}{
       @{}X[c,si={table-format=8}]
        }},
     rows = {bg=white},
   column{2-6}={guard},
     row{1}= {guard,bg=tableHeadingsBG},
     row{Y,Z}= {
                 %bg=tableHeadingsBGalt,
                 font={\footnotesize\bfseries}},
    }

   & \dfgFundingYear{1}
   & \dfgFundingYear{2}
```

```
    & \dfgFundingYear{3}
    & \dfgFundingYear{4}
    & \dfgFundingYear{5}                                    \\

     \l__tmp_body_costs_tl

     \SetHline{-}{tableOuterLines,.75pt}
     %\SetCell{font=\bfseries\footnotesize}
     Sum~ of~ requested~ funds
     \printMarginalia{%
      \getSumRequestedFunds{#1}{costs}
      {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
     }
      & \getSumRequestedFunds{#1}{costs}{sum-year1}
      & \getSumRequestedFunds{#1}{costs}{sum-year2}
      & \getSumRequestedFunds{#1}{costs}{sum-year3}
      & \getSumRequestedFunds{#1}{costs}{sum-year4}
      & \getSumRequestedFunds{#1}{costs}{sum-year5} \\
    \end{tblr}
   }{
    Database~ \texttt{#1}~ missing.
    \msg_warning:nne { dfg } { database-not-found }{ #1 }
   }
}
```

6.1.24 Boxes

```
\RequirePackage[breakable,fitting,skins,most]{tcolorbox}

\tcbset{
  PIbox/.style={
    %enhanced, breakable,
    colback=white, colframe=white,
    coltitle=black,
    boxrule=0pt, %arc=2pt,
    before~ skip=0pt, after~ skip=7pt,
  before~ title=\raggedright,
  before~ upper=\raggedright,
    boxsep=0pt,
    left=0pt, right=0pt, top=7pt, bottom=0pt,
   },
}

\bool_if:NTF \l_@@_class_final_bool
{\newcommand{\dfgcomment}[2][]{}}
{\NewTotalTColorBox{\dfgcomment}{ O{} m }
  {%
   nobeforeafter,
   colupper=black!50!white,
   coltitle=black!75!white,
   fonttitle=\fontsize{7}{9}\selectfont\bfseries,
   colback=black!5!white,
   colframe=black!15!white,
   width=\linewidth,
   %  text height=4cm,
   breakable,
   enhanced,
   left=0pt,right=0pt,
   top=0pt,bottom=0pt,
   % text width=\linewidth,
   fontupper=\fontsize{7}{9}\selectfont,
   title={Anmerkung~der~DFG},
   #1
  }{#2}}
```

defining standard for boxes

```
\tcbset{
  dfgStyle/.style={%
    enhanced~ jigsaw,
    colback       =white,
    colbacktitle  =tableHeadingsBG,
    coltitle      =tableHeadingsFont,
    colframe      =tableOuterLines,
    %colupper      =tableHeadingsFont,
    fontupper     =\sffamily,
    fonttitle     =\bfseries,
    % after skip   =1em,
    before        ={\pagebreak[0]},
    after         ={\nopagebreak},
    % fonttitle    =\bfseries,
    %boxrule      =0.5pt,
    sharp~ corners,
    toprule      =1pt,
    rightrule    =0mm,
    leftrule     =0mm,
```

```
    % drop shadow =rwthGrey,
    %drop large lifted shadow,
    width         =\linewidth,
    bottomrule    =1pt,
    % draft,
    left          =2mm,
    bottom        =0mm,
    top           =0mm,
    capture       =minipage,
   }}

\NewTotalTColorBox{\dfgBox}{ O{} m }
{%
  dfgStyle,
  nobeforeafter,
  left=0pt,right=0pt,
  top=0pt,bottom=0pt,
  #1
}{#2}
```

A handy box to aggregate important points, `dfgSummaryBox` → P. 19.

```
\newtcolorbox{dfgSummaryBox}[1][]{
  dfgStyle,
  top=1mm,
  left=1mm,
  bottom=1mm,
  before={\par},
  after={\par},
  before~ upper={\begin{itemize}[leftmargin=.5cm]%
    \renewcommand{\labelitemi}{\dfgTargetPoint}%
    \hypersetup{linkcolor=tableHeadingsFont}%
    \colorlet{referencePI}{tableHeadingsFont}
    \colorlet{referenceWP}{tableHeadingsFont}
    \colorlet{referenceAim}{tableHeadingsFont}
   },
  after~ upper={\end{itemize}},
  %  height=2.5cm,
  fontupper=\rmfamily,
  % ------ title below.
  width=\linewidth,
  rounded~ corners,
  toprule=1pt,
  rightrule=1pt,
  leftrule=1pt,
  bottomrule=1pt,
  % attach boxed title to bottom center,
  boxed~ title~ style={empty,boxrule=1pt},
  varwidth~ boxed~ title=0.5\linewidth,
  #1
}
```

Box for anything like the aims section.

```
\newtcolorbox{dfgBoxEnv}[1][]{
  dfgStyle,
  % top=-1.5mm,
  % left=1mm,
  before={\par},
  #1
}
```

6.1.25 New Commands

Providing further commands that can and will be used throughout the document

```
\providecommand{\crc}{Collaborative~ Research~ Centre\xspace}

% getting the main logo of the CRC
% the filename needs to be adjusted or make this more generic.
\providecommand{\crcLogoPath}{\tl_use:N \l_@@_custom_crc_logo_path_tl}
\providecommand{\affiliationLogoPath}
  {\tl_use:N \l_@@_custom_affiliation_logo_path_tl}

\NewDocumentCommand{\crcLogo}{O{1cm}}{%
  \includegraphics[#1]{\crcLogoPath}}

% same for the affiliation.
\NewDocumentCommand{\affiliationLogo}{O{1cm}}{%
  \includegraphics[#1]{\affiliationLogoPath}}

\keys_define:nn { hrule }
{
  color     .tl_set:N   = \l__hrule_color_tl,
  color     .initial:n  = color@tags@line,
```

```
  above     .dim_set:N  = \l__hrule_above_dim,
  above     .initial:n  = 0pt,
  below     .dim_set:N  = \l__hrule_below_dim,
  below     .initial:n  = 0pt,
  thickness .dim_set:N  = \l__hrule_thickness_dim,
  thickness .initial:n  = 0.4pt,
}

\ProvideDocumentCommand{\Hrule}{O{}}
{%
  \group_begin:
  \keys_set:nn { hrule } {#1}%
  \par
  \addvspace{ \l__hrule_above_dim }%
  {\color{ \l__hrule_color_tl }\hrule height \l__hrule_thickness_dim}%
  \addvspace{ \l__hrule_below_dim }%
  \group_end:
}

% remodeling for the titlepage where we use some center-environments
\AtBeginEnvironment{center}{
  % \vspace*{-1em}
}

\seq_new:N \l_@@_sig_pairs_seq
\tl_new:N  \l_@@_sig_id_tl
\tl_new:N  \l_@@_sig_desc_tl
```

\printSignature `\printSignature[<db>]{ {<id>}{<desc>} {<id>}{<desc>} ... }` – one signature block per `{id}{desc}` pair. `<db>` defaults to `person`. Inside `<desc>` you may use `\theAffiliationOfRoleID`, which is set to the person's `ref-institution-id` before the description is typeset.

```
\NewDocumentCommand{\printSignature}{ O{person} m }
{
  \bool_if:NF \l_@@_class_fastcompile_project_bool
  {
    \par
    \seq_set_split:Nnn \l_@@_sig_pairs_seq { } {#2}
    \int_compare:nNnT { \seq_count:N \l_@@_sig_pairs_seq } > { 0 }
    { \@@_signature_consume_pairs:n {#1} }
  }
}

\tl_new:N \l_@@_signature_role_id_tl
\tl_new:N \l_@@_signature_role_tl
\tl_new:N \l_@@_signature_role_period_tl

\NewDocumentCommand{\printSignaturesWithRole}{ O{person} m m }
{
  \bool_if:NF \l_@@_class_fastcompile_project_bool
  {
    \DTLifdbexists{#1}
    {
      \DTLforeach*[
        \DTLisinlist{\theFundingPeriodNumeral}{\l_@@_signature_role_period_tl}
        \and
        \DTLisinlist{#2}{\l_@@_signature_role_tl}
      ]{#1}{
        \l_@@_signature_role_id_tl     = id,
        \l_@@_signature_role_tl        = role,
        \l_@@_signature_role_period_tl = period
      }{
        \@@_signature_block:nnn
          {#1}{\l_@@_signature_role_id_tl}{#3}
      }
    }
    { \msg_warning:nne { dfg } { database-not-found } {#1} }
  }
}
```

Pops `{id}{desc}` pairs off `\l_@@_sig_pairs_seq` until empty, typesetting one block per pair.

```
\cs_new_protected:Npn \@@_signature_consume_pairs:n #1
{
  \seq_pop_left:NN \l_@@_sig_pairs_seq \l_@@_sig_id_tl
  \seq_pop_left:NNTF \l_@@_sig_pairs_seq \l_@@_sig_desc_tl
  {
   \@@_signature_block:nVV {#1} \l_@@_sig_id_tl \l_@@_sig_desc_tl
   \seq_if_empty:NF \l_@@_sig_pairs_seq
   { \@@_signature_consume_pairs:n {#1} }
  }
  {
```

```
    \msg_warning:nnn { crc-proposal } { signature-odd-args } {#1}
   }
 }
 \msg_new:nnn { crc-proposal } { signature-odd-args }
 { \printSignature\ in\ db\ '#1':\ odd\ number\ of\ tokens,\ %
   expected\ {id}{desc}\ pairs. }
```

Typesets a single signature block for person `#2` in db `#1` with description `#3`. Sets `\theAffiliationOfRoleID` from the row's `ref-institution-id` before expanding `#3`.

```
 \cs_new_protected:Npn \@@_signature_block:nnn #1 #2 #3
 {
   \DTLgetvalueforkey{\theAffiliationOfRoleID}{ref-affiliation-id}{#1}{id}{#2}
   \DTLgetvalueforkey{\@@tmpDegree}   {academicdegree}{#1}{id}{#2}
   \DTLgetvalueforkey{\@@tmpFirst}    {firstname}     {#1}{id}{#2}
   \DTLgetvalueforkey{\@@tmpLast}     {lastname}      {#1}{id}{#2}
   \begin{tblr}{
                width=\linewidth,
                colspec={@{}b{3cm}b{1cm}X[b]@{}},
               }
               {\theCRCaffiliationLocation \\ \dfgDate} &  & \faFileSignature                                                                \\
               \SetHline{3-3}{tableOuterLines,1pt}
                                                          &  & {
                                                               \dfgShowTag{#2}{academicdegree}{\@@tmpDegree}[#1]\space
                                                                 \textcolor{referencePI}{
                                                                 \dfgShowTag{#2}{firstname}{\@@tmpFirst}[#1]\space
                                                                 \dfgShowTag{#2}{lastname}{\@@tmpLast}[#1]} \\
                                                                #3 } \\
   \end{tblr}
   \par\bigskip
 }
 \cs_generate_variant:Nn \@@_signature_block:nnn { nVV }
```

If there are multiple institutions mentioned we want to have a list of all of them with their affiliation. We are getting the possible list of institutions into one command Loop over your list and build the sequence.1

```
 \bool_new:N \l_@@_narrow_bool
 \bool_new:N \l_@@_institution_abbreviation_bool
 \bool_new:N \l_@@_noaddr_bool
 \bool_new:N \l_@@_nozip_bool
 \bool_new:N \l_@@_noloc_bool
 \bool_new:N \l_@@_has_country_bool
 \seq_new:N  \l_@@_inst_ids_seq
 \tl_new:N   \l_@@_sep_tl
 \tl_new:N   \l_@@_inst_sep_tl

 \cs_generate_variant:Nn \seq_set_from_clist:Nn { NV }

 \keys_define:nn { @@ / institution-list }
 {
   abbreviation .bool_set:N = \l_@@_institution_abbreviation_bool,
   narrow .bool_set:N = \l_@@_narrow_bool,
   noaddress .bool_set:N = \l_@@_noaddr_bool,
   nozip .bool_set:N = \l_@@_nozip_bool,
   nolocation .bool_set:N = \l_@@_noloc_bool,
 }

 \NewDocumentCommand{\InstitutionsList}{O{} m O{}}
 {
   \group_begin:
     \@@_list_options:n {#1}
     \tl_put_left:Nn \l_@@_list_options_tl {#3,}
     \exp_args:NVnV \@@_institutions_list:nnn
       \l_@@_list_database_tl {#2} \l_@@_list_options_tl
   \group_end:
 }

 \cs_new_protected:Npn \@@_institutions_list:nnn #1#2#3
 {
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  { \dfgFastReference{#2} }
  {
  \DTLifdbexists{#1}{
   % Parse key names rather than comparing tokens: options forwarded by
   % keys_set_known may have string catcodes. Boolean values also allow a
   % leading setting to override an older trailing flag.
   \bool_set_false:N \l_@@_institution_abbreviation_bool
   \bool_set_false:N \l_@@_narrow_bool
   \bool_set_false:N \l_@@_noaddr_bool
   \bool_set_false:N \l_@@_nozip_bool
   \bool_set_false:N \l_@@_noloc_bool
   \keys_set:nn { @@ / institution-list } {#3}
   \bool_if:NTF \l_@@_narrow_bool
```

```
  {
    \tl_set:Nn \l_@@_sep_tl {,~}
    \tl_set:Nn \l_@@_inst_sep_tl {;~}
  }
  {
    \tl_set:Nn \l_@@_sep_tl {\newline}
    \tl_set:Nn \l_@@_inst_sep_tl {\par\medskip}
  }
  \datatool_if_has_key:nnTF {affiliation}{country}
  { \bool_set_true:N \l_@@_has_country_bool }
  { \bool_set_false:N \l_@@_has_country_bool }

  % --- get list of institution ids for this person --------------------
  \DTLgetvalueforkey{\theInstitutionListValue}{ref-institution-id}{#1}{id}{#2}
  \seq_set_from_clist:NV \l_@@_inst_ids_seq \theInstitutionListValue

  % --- iterate ---------------------------------------------------------
  \bool_set_true:N \l_tmpa_bool
  \seq_map_inline:Nn \l_@@_inst_ids_seq
  {
   \bool_if:NTF \l_tmpa_bool
   { \bool_set_false:N \l_tmpa_bool }
   { \tl_use:N \l_@@_inst_sep_tl }
   \_@@_one_institution:n {##1}
  }
}
   {
     Database~ \texttt{#1}~ missing.
     \msg_warning:nne { dfg } { database-not-found }{ #1 }
   }
 }
}

% Compatibility spelling for older proposal sources.
\cs_new_eq:NN \dfgInstitutionList \InstitutionsList

\cs_new_protected:Npn \_@@_one_institution:n #1
{
 \DTLifdbexists{institution}{ \DTLgetvalueforkey{\theAffiliationIDValue}{ref-affiliation-
id}{institution}{id}{#1}

  \bool_if:NTF \l_@@_institution_abbreviation_bool
  {
    \@@_institution_abbreviation_or_name:nN
      {#1} \l_tmpa_tl
    \cs_set:Npx \theInstitutionNameValue { \exp_not:V \l_tmpa_tl }
  }
  { \DTLgetvalueforkey{\theInstitutionNameValue}{name}{institution}{id}{#1} }
  \DTLifnullorempty{\theInstitutionNameValue}{}
  {
    \dfgShowTag{#1}{\l_@@_institution_label_field_tl}
      {\theInstitutionNameValue}[institution]
   \tl_use:N \l_@@_sep_tl
  }

  \DTLgetvalueforkey{\theAffiliationNameValue}{name}{affiliation}{id}{\theAffiliationIDValue}
  \dfgShowTag{\theAffiliationIDValue}{name}
    {\theAffiliationNameValue}[affiliation]

  \bool_if:NF \l_@@_noaddr_bool
  {
   \tl_use:N \l_@@_sep_tl
   \DTLgetvalueforkey{\theAffiliationStreetValue}{addressStreet}{affiliation}{id}{\theAffiliationIDValue}
    \dfgShowTag{\theAffiliationIDValue}{addressStreet}
      {\theAffiliationStreetValue}[affiliation]
  }

  \bool_if:NF \l_@@_nozip_bool
  {
   ,~
   \DTLgetvalueforkey{\theAffiliationZIPValue}{addressZIP}{affiliation}{id}{\theAffiliationIDValue}
    \dfgShowTag{\theAffiliationIDValue}{addressZIP}
      {\theAffiliationZIPValue}[affiliation]
   \bool_if:NF \l_@@_noloc_bool { ~ }
  }

  \bool_if:NF \l_@@_noloc_bool
  {
   \bool_if:NT \l_@@_nozip_bool { ,~ }
   \DTLgetvalueforkey{\theAffiliationLocationValue}{location}{affiliation}{id}{\theAffiliationIDValue}
    \dfgShowTag{\theAffiliationIDValue}{location}
      {\theAffiliationLocationValue}[affiliation]
   \bool_if:NT \l_@@_has_country_bool
   {
    \DTLgetvalueforkey{\theAffiliationCountryValue}{country}{affiliation}{id}{\theAffiliationIDValue}
    \DTLifnullorempty{\theAffiliationCountryValue}{}
      { ~(\dfgShowTag{\theAffiliationIDValue}{country}
```

```
            {\theAffiliationCountryValue}[affiliation]) }
    }
   }
 }
    {
      Database~ \texttt{institution}~ missing.
      \msg_warning:nne { dfg } { database-not-found }{ institution }
    }
 }
```

get all the PIs who have a certain role in the crc this can be e.g. 'spokesperson', 'steering-committee-member'. The leading options include database (person by default) and raster settings. Trailing raster options remain supported for older documents.

```
\keys_define:nn { @@ / pi-role-list }
{
  academicdegree-below .bool_set:N = \l_@@_pi_academicdegree_below_bool,
}

\NewDocumentCommand{\getPIsWithRole}{O{} m O{}}
{
  \group_begin:
    \bool_set_false:N \l_@@_pi_academicdegree_below_bool
    \@@_list_options:n {#1}
    \tl_put_left:Nn \l_@@_list_options_tl {#3,}
    \exp_args:NnV \keys_set_known:nnN { @@ / pi-role-list }
      \l_@@_list_options_tl \l_@@_list_options_tl
    \exp_args:NVnV \@@_get_pis_with_role:nnn
      \l_@@_list_database_tl {#2} \l_@@_list_options_tl
  \group_end:
}

\cs_new_protected:Npn \@@_get_pis_with_role:nnn #1#2#3
{%
  \bool_if:NTF \l_@@_class_fastcompile_bool
  { \dfgItemSkipped{dfgPIsWithRole:~#2} }
  {%
    \DTLifdbexists{#1}
    {
      \def\tmpLocalDB{#1}%
      \DTLsort*{lastname}{#1}%
      \begin{tcbraster}[
          raster~ valign=top,
          raster~ columns=3,
          raster~ equal~ height=rows,
          raster~ column~ skip=12pt,
          raster~ row~ skip=20pt,
          raster~ after~ skip=20pt,
          PIbox,
          #3
        ]
        \DTLforeach*[
          \DTLisinlist{\theFundingPeriodNumeral}{\period}
          \and
          \DTLisinlist{#2}{\role}
        ]{#1}{
          \id     = id,
          \role   = role,
          \period = period
        }{%
          \__getperson_one:n {\id}%
        }%
      \end{tcbraster}
    }
    {
      Database~ \texttt{#1}~ missing.
      \msg_warning:nne { dfg } { database-not-found }{ #1 }
    }
  }
}
```

In the text you can reference to a PI with the following command. The name is highlighted and referenced.

```
\cs_new_protected:Npn \@@_pi_reference_seq_use:N #1
{
  \bool_if:NTF \g_@@_tblr_inside_bool
    { \seq_use:Nn #1 {\addnbthinspace/\addthinspace} }
    { \seq_use:Nnnn #1 {~and~} {,~} {,~and~} }
}

\NewDocumentCommand{\getPI}{s m o} {
  % #1: starred version -> firstname
  % #2: id of PI / person
```

```
  % #3: optional text, but reference linked to PI
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  { \IfNoValueTF{#3}{\dfgFastReference{#2}}{#3} }
  {
  \seq_clear:N \l_tmpa_seq
  \seq_set_from_clist:Nn \l_tmpa_seq {#2}%
  \seq_set_map:NNn \l_tmpb_seq \l_tmpa_seq {
   \exp_not:n {
    \group_begin:
    \DTLgetvalueforkey{\thePersonName}{lastname}{person}{id}{##1}
    \DTLgetvalueforkey{\thePersonFirstName}{firstname}{person}{id}{##1}
    \datatool_if_null:NTF {\thePersonName}
    { \getEntry[person]{##1}{lastname} %<- just to get the error message
     \textcolor{error}{\texttt{<##1>}}
    }
    {
          \tl_if_eq:NnF \l_@@_custom_referencePI_tl {current}
            { \color{referencePI} }
          \bfseries
     \IfNoValueTF{#3}%

               {\hyperlink{pi:##1}{\@@_pi_reference_name:nn {#1}{##1}}}
     {\hyperlink{pi:##1}{#3}}}
    \group_end:
   }}
  \@@_pi_reference_seq_use:N \l_tmpb_seq
  }
}
```

Unstarred PI references disambiguate shared surnames across all PI records with the first given-name initial, for example K. Schneider and C. Schneider. If that initial is shared as well, use the full given name. Starred references and explicit display text retain their existing behavior, as do unique names. Project-first references inherit this formatting through `\getPI`. Recompute from the current metadata so additions are reflected immediately.

```
\tl_new:N \l_@@_pi_lastname_tl
\tl_new:N \l_@@_pi_firstname_tl
\tl_new:N \l_@@_pi_initial_tl
\tl_new:N \l_@@_pi_candidate_lastname_tl
\tl_new:N \l_@@_pi_candidate_firstname_tl
\tl_new:N \l_@@_pi_candidate_initial_tl
\clist_new:N \l_@@_pi_candidate_roles_clist
\bool_new:N \l_@@_pi_candidate_is_pi_bool
\int_new:N \l_@@_pi_same_lastname_int
\int_new:N \l_@@_pi_same_initial_int

% Count matching PI surnames and first initials across the current metadata.
\cs_new_protected:Npn \@@_pi_count_name_matches:
  {
    \int_zero:N \l_@@_pi_same_lastname_int
    \int_zero:N \l_@@_pi_same_initial_int
    \DTLforeach*{person}
      {
        \dfgDisambiguationName=lastname,
        \dfgDisambiguationFirstname=firstname,
        \dfgDisambiguationRoles=role
      }
      {
        \clist_set:Nx \l_@@_pi_candidate_roles_clist {\dfgDisambiguationRoles}
        \bool_set_false:N \l_@@_pi_candidate_is_pi_bool
        \clist_if_in:NnT \l_@@_pi_candidate_roles_clist {PI}
          {\bool_set_true:N \l_@@_pi_candidate_is_pi_bool}
        \clist_if_in:NnT \l_@@_pi_candidate_roles_clist {Co-PI}
          {\bool_set_true:N \l_@@_pi_candidate_is_pi_bool}
        \bool_if:NT \l_@@_pi_candidate_is_pi_bool
          {
            \tl_set:Nx \l_@@_pi_candidate_lastname_tl {\dfgDisambiguationName}
            \tl_if_eq:NNT \l_@@_pi_lastname_tl \l_@@_pi_candidate_lastname_tl
              {
                \int_incr:N \l_@@_pi_same_lastname_int
                \tl_set:Nx \l_@@_pi_candidate_firstname_tl {\dfgDisambiguationFirstname}
                \tl_trim_spaces:N \l_@@_pi_candidate_firstname_tl
                \tl_set:Nx \l_@@_pi_candidate_initial_tl
                  {\tl_head:V \l_@@_pi_candidate_firstname_tl}
                \tl_if_eq:NNT \l_@@_pi_initial_tl \l_@@_pi_candidate_initial_tl
                  {\int_incr:N \l_@@_pi_same_initial_int}
              }
          }
      }
  }

% Render the name inside the PI hyperlink. #1 is the star, #2 the person ID.
% The caller has already fetched and validated the name fields.
\cs_new_protected:Npn \@@_pi_reference_name:nn #1#2
  {
```

```
    \group_begin:
    \IfBooleanTF{#1}
      {\dfgShowTag{#2}{firstname}{\thePersonFirstName}[person]\space}
      {
        \tl_set:Nx \l_@@_pi_lastname_tl {\thePersonName}
        \tl_set:Nx \l_@@_pi_firstname_tl {\thePersonFirstName}
        \tl_trim_spaces:N \l_@@_pi_firstname_tl
        \tl_set:Nx \l_@@_pi_initial_tl {\tl_head:V \l_@@_pi_firstname_tl}
        \@@_pi_count_name_matches:
        \int_compare:nNnTF {\l_@@_pi_same_lastname_int} > {1}
          {
            \int_compare:nNnTF {\l_@@_pi_same_initial_int} > {1}
              {\dfgShowTag{#2}{firstname}{\thePersonFirstName}[person]}
              {\tl_use:N \l_@@_pi_initial_tl.}
            \nobreakspace
          }
          {
            \bool_if:NT \l_@@_custom_pi_reference_firstname_bool
              {\dfgShowTag{#2}{firstname}{\thePersonFirstName}[person]\space}
          }
      }
    \dfgShowTag{#2}{lastname}{\thePersonName}[person]
    \group_end:
  }
```

Very similarly, we want to get a list of the projects.

```
\NewDocumentCommand \getProjectColor { m }
{
  \exp_args:Nx \color { project \tl_head:n { #1 } }
}
\tl_new:N \l_@@_project_reference_only_tl

\prg_new_protected_conditional:Npnn \@@_project_if_reference_only:n #1
  { T,F,TF }
{
  \DTLgetvalueforkey{\l_@@_project_reference_only_tl}
    {reference-only}{project}{id}{#1}
  \str_if_eq:VnTF \l_@@_project_reference_only_tl { true }
    { \prg_return_true: }
    { \prg_return_false: }
}

\cs_new_protected:Npn \@@_project_reference_only_indicator:
{
  \bool_if:NTF \l_@@_former_project_compact_bool
  {
    \bool_gset_true:N \g_@@_former_project_seen_bool
    {\normalfont\normalcolor\textsuperscript{*}}
  }
  {
    \nobreakspace
    {\normalfont\normalcolor(former~project)}
  }
}

% Project references read naturally in prose but need a compact delimiter in
% narrow table cells.  Keep the two container variants centralized so
% \cs{getProject}, \cs{getProjectPI}, and focused-project fallbacks cannot
% diverge.
\bool_new:N \l_@@_project_newline_separator_bool
\bool_new:N \l_@@_project_custom_text_bool
\cs_new_protected:Npn \@@_project_reference_seq_use:N #1
{
  \bool_if:NTF \l_@@_project_newline_separator_bool
    { \seq_use:Nn #1 {\\} }
    {
      \bool_if:NTF \g_@@_tblr_inside_bool
        { \seq_use:Nn #1 {\addnbthinspace/\addthinspace} }
        { \seq_use:Nnnn #1 {~and~} {,~} {,~and~} }
    }
}

\cs_new_protected:Npn \@@_project_reference_clist_use:N #1
{
  \bool_if:NTF \g_@@_tblr_inside_bool
    { \clist_use:Nn #1 {\addnbthinspace/\addthinspace} }
    { \clist_use:Nnnn #1 {~and~} {,~} {,~and~} }
}

\cs_new_protected:Npn \@@_get_project_compact:nn #1#2 {
  % #1: id of project
  % #2: optional text, or \newline/\\ for a stacked project list
  \bool_set_false:N \l_@@_project_newline_separator_bool
  \bool_set_false:N \l_@@_project_custom_text_bool
  \IfNoValueF{#2}
    {
```

```
      \bool_set_true:N \l_@@_project_custom_text_bool
      \tl_if_eq:nnT {#2} {\newline}
        { \bool_set_false:N \l_@@_project_custom_text_bool
          \bool_set_true:N \l_@@_project_newline_separator_bool }
      \tl_if_eq:nnT {#2} {\\}
        { \bool_set_false:N \l_@@_project_custom_text_bool
          \bool_set_true:N \l_@@_project_newline_separator_bool }
    }
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  {
    \seq_set_from_clist:Nn \l_tmpa_seq {#1}
    \seq_set_map:NNn \l_tmpb_seq \l_tmpa_seq
    {
      \exp_not:n
      {
        \bool_if:NTF \l_@@_project_custom_text_bool
          {#2}
          {
            \dfgShowTag {##1}{number}
              {\@@_fast_project_reference:n {##1}}[project]
          }
      }
    }
    \@@_project_reference_seq_use:N \l_tmpb_seq
  }
  {
  \seq_clear:N \l_tmpa_seq
  \seq_set_from_clist:Nn \l_tmpa_seq {#1}%
  \seq_set_map:NNn \l_tmpb_seq \l_tmpa_seq {
   \exp_not:n {
    \group_begin:
    \DTLgetvalueforkey{\theProjectNumber}{number}{project}{id}{##1}
    \datatool_if_null:NTF {\theProjectNumber}
    { \getEntry[project]{##1}{number} %<- just to get the error message
     \textcolor{error}{\texttt{<##1>}}
    }{
     \bfseries%
     \getProjectColor{##1}%
        \@@_project_if_reference_only:nTF {##1}
        {
          \bool_if:NTF \l_@@_project_custom_text_bool
            {#2}
            {\dfgShowTag{##1}{number}{\theProjectNumber}[project]}%
          \@@_project_reference_only_indicator:
        }
        {
          \bool_if:NTF \l_@@_project_custom_text_bool
          {
            \bool_if:NTF \l_@@_class_cv_bool
              { #2 }
              { \hyperlink{project:##1}{#2} }
          }
          {
            \bool_if:NTF \l_@@_class_cv_bool
              { \dfgShowTag{##1}{number}{\theProjectNumber}[project] }
              { \hyperlink{project:##1}
                  {\dfgShowTag{##1}{number}{\theProjectNumber}[project]} }
          }
        }
    }
    \group_end:
   }}%
  \@@_project_reference_seq_use:N \l_tmpb_seq
  }
}

\NewDocumentCommand{\getProject}{s m o}
{
  \IfBooleanTF{#1}
    { \@@_get_project_compact:nn {#2} {#3} }
    {
      \bool_if:NTF \l_@@_custom_project_pi_once_per_section_bool
        { \@@_get_project_with_people:nnn {\BooleanFalse} {#2} {#3} }
        { \@@_get_project_compact:nn {#2} {#3} }
    }
}

\int_new:N \g_@@_project_count_all_int
\prop_new:N \g_@@_project_count_area_prop
\tl_new:N  \l_@@_project_count_id_tl
\tl_new:N  \l_@@_project_count_area_tl
\tl_new:N  \l_@@_project_count_status_tl
\tl_new:N  \l_@@_project_count_consider_for_lists_tl
\tl_new:N  \l_@@_project_count_reference_only_tl

\cs_new_protected:Npn \@@_project_count_increment:n #1
{
```

```
  \int_gincr:N \g_@@_project_count_all_int
  \tl_set:Nx \l_tmpa_tl { \tl_upper_case:n {#1} }
  \prop_get:NVNTF \g_@@_project_count_area_prop
    \l_tmpa_tl \l_tmpb_tl
    {
      \prop_gput:NVx \g_@@_project_count_area_prop \l_tmpa_tl
        { \int_eval:n { \l_tmpb_tl + 1 } }
    }
    { \prop_gput:NVn \g_@@_project_count_area_prop \l_tmpa_tl { 1 } }
}

\cs_new_protected:Npn \@@_project_count_build:
{
  \int_gzero:N \g_@@_project_count_all_int
  \prop_gclear:N \g_@@_project_count_area_prop
  \DTLifdbexists{project}
  {
    \DTLifhaskey { project } { project-area }
      { }
      { \DTLaddcolumn { project } { project-area } }
    \DTLforeach*{project}
    {
      \l_@@_project_count_id_tl                = id,
      \l_@@_project_count_area_tl              = project-area,
      \l_@@_project_count_status_tl            = project-status,
      \l_@@_project_count_consider_for_lists_tl = consider-for-lists,
      \l_@@_project_count_reference_only_tl    = reference-only
    }
    {
      \bool_lazy_all:nT
      {
        { \str_if_eq_p:Vn \l_@@_project_count_consider_for_lists_tl { true } }
        { \str_if_eq_p:Vn \l_@@_project_count_reference_only_tl { false } }
        {
          \bool_lazy_or_p:nn
            { \str_if_eq_p:Vn \l_@@_project_count_status_tl { C } }
            { \str_if_eq_p:Vn \l_@@_project_count_status_tl { N } }
        }
      }
      {
        \DTLifnullorempty{\l_@@_project_count_area_tl}
          {
            \tl_set:Nx \l_@@_project_count_area_tl
              { \tl_head:V \l_@@_project_count_id_tl }
          }
          { }
        \tl_trim_spaces:N \l_@@_project_count_area_tl
        \exp_args:NV \@@_project_count_increment:n
          \l_@@_project_count_area_tl
      }
    }
  }
  { \msg_warning:nnn { dfg } { database-not-found } { project } }
  \bool_gset_true:N \g_@@_project_count_ready_bool
}

\NewDocumentCommand{\getProjectCount}{o}
{
  \bool_if:NF \g_@@_project_count_ready_bool
    { \@@_project_count_build: }
  \dfgShowCalculatedTag{getProjectCount}{project}
  {
    \IfNoValueTF{#1}
      { \int_use:N \g_@@_project_count_all_int }
      {
        \tl_set:Nx \l_tmpa_tl { \tl_upper_case:n {#1} }
        \prop_get:NVNTF \g_@@_project_count_area_prop
          \l_tmpa_tl \l_tmpb_tl
          { \tl_use:N \l_tmpb_tl }
        {
          \msg_warning:nnn { dfg } { invalid-project-count-area } {#1}
          0
        }
      }
  }
}
\clist_new:N \l_@@_person_project_clist
\clist_new:N \l_@@_project_id_clist
\clist_new:N \l_@@_project_number_clist
\bool_new:N \l_@@_suppress_deprecated_project_pi_warning_bool
% Global flags keep repeated calls and table measurement groups quiet.
\bool_new:N \g_@@_warned_get_pi_project_bool
\bool_new:N \g_@@_warned_get_project_pi_bool
\seq_new:N \l_@@_project_pi_with_project_seq
\msg_new:nnn {dfg} {deprecated-get-pi-project}
  {Command~\string\getPIProject~ ~is~deprecated;~use~\string\getPIWithProject.}
\msg_new:nnn {dfg} {deprecated-get-project-pi}
```

```
10671   {Command~\string\getProjectPI~ ~is~deprecated;~use~\string\getProjectWithPI.}
10672
10673 \NewDocumentCommand{\getPIProject}{s m}{
10674   \bool_if:NF \l_@@_suppress_deprecated_project_pi_warning_bool
10675     {
10676       \bool_if:NF \g_@@_warned_get_pi_project_bool
10677         {
10678           \bool_gset_true:N \g_@@_warned_get_pi_project_bool
10679           \msg_warning:nn {dfg}{deprecated-get-pi-project}
10680         }
10681     }
10682   \bool_if:NTF \l_@@_class_fastcompile_project_bool
10683   { \dfgFastReference{#2} }
10684   {
10685   %
10686   %  • Argument #1  (optional *-star)
10687   %      - if present, the PI's *first name* is printed before the last name;
10688   %        if the star is omitted only the last name is printed.
10689   %
10690   %  • Argument #2  (mandatory)
10691   %      - a comma-separated list of *person-IDs* (the field "id" in your
10692   %        "person" datatool table).
10693   %        Example:  `\getPIProject*{per1,per2}`.
10694   %
10695   %  What the macro does (high-level description)
10696   %  ------------------------------------------------------------------
10697   %  1. Turns the comma list #2 into a clist:   \l_@@_person_id_clist
10698   %  2. For **each** person-ID in that clist it builds an
10699   %     "author block" consisting of
10700   %        • a coloured, bold hyperlink to the PI's entry, and
10701   %        • the list of projects (formatted by \getProject*) that are
10702   %          attached to that PI.
10703   %  3. All author blocks are collected in a second clist
10704   %     \l_@@_person_project_clist.
10705   %  4. Finally the blocks are printed in an *Oxford-comma* style
10706   %     using \clist_use:Nnnn.
10707   %
10708   %  Internal containers (all declared elsewhere with \clist_new:N)
10709   %  ------------------------------------------------------------------
10710   %    \l_@@_person_id_clist          - list of the IDs passed in #2
10711   %    \l_@@_person_project_clist     - the final list of "author blocks"
10712   %    \l_@@_project_id_clist         - temporary clist that stores the raw
10713   %                                     CSV string returned by the DB field
10714   %                                     `ref-project-id` for the current PI.
10715   %    \l_@@_project_number_clist      - coloured project numbers for the
10716   %                                     current PI (built by the inner loop).
10717   %    \l_@@_project_id_clist (re-used) - holds the CSV string before it is
10718   %                                     split into \l_@@_project_id_clist.
10719   %    \l_tmpa_clist                  - loop variable used by
10720   %                                     \clist_map_variable:NNn (holds a *single*
10721   %                                     project-ID during the inner loop).
10722   %
10723   %---------------------------------------------------------------------
10724   %  Detailed walk-through of the code
10725   %---------------------------------------------------------------------
10726   %-------------------------------------------------------------
10727   %   0.  Initialise the containers
10728   %-------------------------------------------------------------
10729   % #2 is a comma-separated list of person IDs → turn it into a clist
10730   \clist_set:Nn \l_@@_person_id_clist {#2}
10731   % containers that will be filled later
10732   \clist_clear:N \l_@@_person_project_clist
10733   \clist_clear:N \l_@@_project_id_clist   % (used as a *scratch* clist)
10734
10735   %-------------------------------------------------------------
10736   %   1.  Outer loop - one iteration per PI
10737   %-------------------------------------------------------------
10738   \clist_map_inline:Nn \l_@@_person_id_clist
10739   {               % ##1 = the current person ID
10740    %-----------------------------------------------------------
10741    %   a)  Build the *author name* (hyperlinked, coloured,
10742    %       bold).  We wrap everything in \exp_not:n so that the
10743    %       whole fragment is **not expanded now** - it will be
10744    %       expanded later when the final clist is printed.
10745    %-----------------------------------------------------------
10746    \clist_put_right:Nn \l_@@_person_project_clist {
10747     \clist_clear:N   \l_@@_project_number_clist
10748     \IfBooleanTF{#1}
10749     {\getPI*{##1}}
10750     {\getPI{##1}}
10751     %-----------------------------------------------------------
10752     %   b)  Pull the raw CSV string that contains *all* project
10753     %       IDs belonging to this PI.
10754     %-----------------------------------------------------------
10755     \DTLgetvalueforkey{\l_@@_project_id_clist}
10756     {ref-project-id}
10757     {person}{id}{##1}
```

```
10758
10759       %-------------------------------------------------------
10760       %   c)  Print the project list inside parentheses.
10761       %       The raw CSV string is *still* a token list; we turn it
10762       %       into a clist by using the special "variable" version of
10763       %       \clist_map_… which gives us a temporary clist variable
10764       %       (\l_tmpa_clist) that holds *one* project ID at a time.
10765       %-------------------------------------------------------
10766       \space(
10767       \clist_map_variable:NNn \l_@@_project_id_clist \l_tmpa_clist {
10768        % ##1 (here called \l_tmpa_clist) is a single project ID.
10769           % Keep project references compact: the PI name is already printed.
10770        \clist_put_right:Ne \l_@@_project_number_clist {
10771             \getProject*{\l_tmpa_clist}
10772        }
10773       }
10774       % finally print the list of coloured project numbers, separated
10775       % by a thin space + comma
10776       \clist_use:Nn \l_@@_project_number_clist  {\addnbthinspace/\addthinspace}
10777       )
10778      }   % end of \clist_put_right:Nn  (one author block)
10779    } % end of outer \clist_map_inline:Nn
10780
10781    %-------------------------------------------------------------
10782    %   2.  Typeset the collected author blocks.
10783    %       \clist_use:Nnnn joins the items with
10784    %          "~and~"   for the *last* separator,
10785    %          ",~"      for the *middle* separators,
10786    %          ",~and~"  for the *penultimate* separator
10787    %-------------------------------------------------------------
10788    \clist_use:Nnnn \l_@@_person_project_clist {~and~} {,~} {,~and~}%
10789    }
10790  }
10791
10792  \clist_new:N \l_@@_person_clist
10793
10794  % Track which projects have already had their PI names expanded by either
10795  % \cs{getProject} or \cs{getProjectPI} in the current section. Chapter
10796  % boundaries also clear the state so references before the first section of a
10797  % new chapter cannot inherit the preceding section's state.
10798  \prop_new:N \g_@@_project_pi_seen_prop
10799  \cs_new_protected:Npn \@@_project_pi_seen_clear:
10800    { \prop_gclear:N \g_@@_project_pi_seen_prop }
10801  \AddToHook{cmd/chapter/before}[dfg/project-pi-once-per-section]
10802    { \@@_project_pi_seen_clear: }
10803  \AddToHook{cmd/section/before}[dfg/project-pi-once-per-section]
10804    { \@@_project_pi_seen_clear: }
10805
10806  % PI details are never appended to reference-only projects, the project
10807  % currently being typeset, or references inside generated tables and lists.
10808  % Otherwise the customization setting selects either the former always-expanded
10809  % \cs{getProjectPI} behaviour or one shared expansion per project and section.
10810  \prg_new_protected_conditional:Npnn \@@_project_pi_if_show_people:n #1
10811    { T,F,TF }
10812  {
10813    \@@_project_if_reference_only:nTF {#1}
10814      { \prg_return_false: }
10815      {
10816        \tl_if_eq:VnTF \l__tmp_current_project_tl {#1}
10817          { \prg_return_false: }
10818          {
10819            \bool_if:nTF
10820              {
10821                \g_@@_tblr_inside_bool
10822                || \l_@@_former_project_compact_bool
10823              }
10824              { \prg_return_false: }
10825              {
10826                \bool_if:NTF \l_@@_custom_project_pi_once_per_section_bool
10827                {
10828                  \prop_if_in:NnTF \g_@@_project_pi_seen_prop {#1}
10829                    { \prg_return_false: }
10830                    {
10831                      \prop_gput:Nnn \g_@@_project_pi_seen_prop {#1} { }
10832                      \prg_return_true:
10833                    }
10834                }
10835                { \prg_return_true: }
10836              }
10837          }
10838      }
10839  }
10840
10841  \cs_new_protected:Npn \@@_get_project_with_people:nnn #1#2#3
10842  {
10843    \bool_if:NTF \l_@@_class_fastcompile_project_bool
10844    { \@@_get_project_compact:nn {#2} {#3} }
```

```
10845    {
10846    \group_begin:
10847    % Project-first references use lastnames by default regardless of the
10848    % global PIReferenceFirstName setting. The explicit star still selects full
10849    % names through the starred \getPI call below.
10850    \bool_set_false:N \l_@@_custom_pi_reference_firstname_bool
10851    %-----------------------------------------------------------------
10852    % 0.  Initialise the containers we are going to use.
10853    %-----------------------------------------------------------------
10854    %   #2 is a comma-separated list of *project* IDs → turn it into a clist.
10855    \clist_set:Nn \l_@@_project_id_clist {#2}
10856    %   These three clists will be filled during the loops.
10857    \clist_clear:N \l_@@_person_project_clist   % final result: "project → PI list"
10858    \clist_clear:N \l_@@_person_id_clist        % temporary list of PIs for one project
10859    \clist_clear:N \l_@@_person_clist           % temporary list of formatted PI names
10860
10861    %-----------------------------------------------------------------
10862    % 1.  Outer loop - one iteration per *project* ID.
10863    %-----------------------------------------------------------------
10864    \clist_map_inline:Nn \l_@@_project_id_clist
10865    {
10866     % ##1 = the current project ID
10867     %-------------------------------------------------------------
10868     %   a)  Put the *project number* (coloured) at the beginning
10869     %       of the block that will be stored in
10870     %       \l_@@_person_project_clist.
10871     %-------------------------------------------------------------
10872     \clist_put_right:Nn \l_@@_person_project_clist
10873     {%
10874      % Reset the per-project temporaries before we start filling them.
10875      \clist_clear:N \l_@@_person_clist
10876      \clist_clear:N \l_@@_person_id_clist
10877
10878    % The coloured project number (e.g. "\color{blue}{12}")
10879    \@@_get_project_compact:nn {##1} {#3}
10880
10881    % Reference-only projects and the current project remain compact. For
10882    % other projects, the customization setting determines whether PI names
10883    % are always shown or only on the first reference in this section.
10884      \@@_project_pi_if_show_people:nT {##1}
10885      {
10886    %---------------------------------------------------------
10887    %   b)  Retrieve the *raw* CSV string that contains all
10888      %       person IDs attached to this project.
10889      %---------------------------------------------------------
10890      \DTLgetvalueforkey
10891      {\l_@@_person_id_clist}      % ← will receive something like "per1,per3"
10892      {ref-person-id}              % field name in the "project" table
10893      {project}{id}{##1}           % look it up for the current project
10894
10895      %---------------------------------------------------------
10896      %   c)  Build the *list of PI names* for this project.
10897      %       We use the "variable" version of \clist_map which
10898      %       gives us a temporary clist variable (\l_tmpa_clist)
10899      %       that holds a *single* person ID at each iteration.
10900      %---------------------------------------------------------
10901      \space(%
10902      \clist_map_variable:NNn \l_@@_person_id_clist \l_tmpa_clist
10903      {
10904       % \l_tmpa_clist now expands to one person ID.
10905       % We format it with \getPI (or \getPI* if the star was given).
10906       \clist_put_right:Ne \l_@@_person_clist
10907       {
10908        \IfBooleanTF{#1}
10909        {\getPI*{\l_tmpa_clist}}
10910        {\getPI{\l_tmpa_clist}}
10911       }
10912      }
10913      %---------------------------------------------------------
10914      %   d)  Print the formatted PI list, separating consecutive
10915      %       names with a thin slash (the macro \addnbthinspace and
10916      %       \addthinspace are assumed to be defined elsewhere).
10917      %---------------------------------------------------------
10918    \clist_use:Nn \l_@@_person_clist {\addnbthinspace/\addthinspace}
10919      ) % end of the parentheses that surround the PI list
10920     }
10921    }   % end of \clist_put_right:Nn   (one *project block*)
10922     } % end of outer \clist_map_inline:Nn
10923
10924    %-----------------------------------------------------------------
10925    % 2.  Typeset the collected project blocks.
10926    %     \clist_use:Nnnn joins the items with
10927    %        "~and~"   for the *last* separator,
10928    %        ",~"      for the *middle* separators,
10929    %        ",~and~"  for the *penultimate* separator.
10930    %-----------------------------------------------------------------
10931    \@@_project_reference_clist_use:N \l_@@_person_project_clist
```

```
  \group_end:
  }
}

\NewDocumentCommand{\getPIWithProject}{s m}
{
  \group_begin:
  \bool_set_true:N \l_@@_suppress_deprecated_project_pi_warning_bool
  \IfBooleanTF{#1}{\getPIProject*{#2}}{\getPIProject{#2}}
  \group_end:
}

\NewDocumentCommand{\getProjectPI}{s m}
  {
    \bool_if:NF \l_@@_suppress_deprecated_project_pi_warning_bool
      {
        \bool_if:NF \g_@@_warned_get_project_pi_bool
          {
            \bool_gset_true:N \g_@@_warned_get_project_pi_bool
            \msg_warning:nn {dfg}{deprecated-get-project-pi}
          }
      }
    \exp_args:NnnV \@@_get_project_with_people:nnn
      {#1} {#2} \c_novalue_tl
  }
\NewDocumentCommand{\getProjectWithPI}{s m}
{
  \group_begin:
  \bool_set_true:N \l_@@_suppress_deprecated_project_pi_warning_bool
  \IfBooleanTF{#1}{\getProjectPI*{#2}}{\getProjectPI{#2}}
  \group_end:
}

\NewDocumentCommand{\getPIOfProjectWithProject}{s m}
{
  \seq_clear:N \l_@@_project_pi_with_project_seq
  \clist_map_inline:nn {#2}
  {
    \seq_put_right:Nn \l_@@_project_pi_with_project_seq
    {
      \IfBooleanTF{#1}
        {\getPIofProject*{##1}}
        {\getPIofProject{##1}}
      \space(\getProject*{##1})
    }
  }
  \@@_project_reference_seq_use:N \l_@@_project_pi_with_project_seq
}
```

We also need a way to extract all PIs of a project.

```
% The star defaults to full names and a prose list; explicit properties still
% override the default. A starred delimiter overrides only the final join.
\clist_new:N \l_@@_project_pi_properties_clist
\tl_new:N \l_@@_project_pi_name_tl
\NewDocumentCommand{\getPIofProject}{s o m o}
{
  \group_begin:
  \IfNoValueTF{#2}
    {
      \bool_if:NTF \l_@@_custom_pi_reference_firstname_bool
        {\clist_set:Nn \l_@@_project_pi_properties_clist {firstname,lastname}}
        {
          \IfBooleanTF{#1}
            {\clist_set:Nn \l_@@_project_pi_properties_clist {firstname,lastname}}
            {\clist_set:Nn \l_@@_project_pi_properties_clist {lastname}}
        }
    }
    {\clist_set:Nn \l_@@_project_pi_properties_clist {#2}}
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
    {\@@_fast_project_reference:n {#3}}
    {
      \clist_clear:N \l_@@_person_clist
      \DTLgetvalueforkey{\l_@@_person_id_clist}
        {ref-person-id}{project}{id}{#3}
      \DTLifnullorempty{\l_@@_person_id_clist}
        {\clist_clear:N \l_@@_person_id_clist}{}
      \clist_map_variable:NNn \l_@@_person_id_clist \l_tmpa_clist
        {
          \DTLgetvalueforkey{\tmpPIacademicdegree}{academicdegree}{person}{id}{\l_tmpa_clist}
          \DTLgetvalueforkey{\tmpPIfirstname}{firstname}{person}{id}{\l_tmpa_clist}
          \DTLgetvalueforkey{\tmpPIlastname}{lastname}{person}{id}{\l_tmpa_clist}
          \tl_clear:N \l_@@_project_pi_name_tl
          \clist_if_in:NnT \l_@@_project_pi_properties_clist {academicdegree}
            {
              \tl_put_right:Ne \l_@@_project_pi_name_tl
                {\dfgShowTag{\l_tmpa_clist}{academicdegree}{\tmpPIacademicdegree}[person]\addnbthinspace}
```

```
            }
          \clist_if_in:NnT \l_@@_project_pi_properties_clist {firstname}
            {
              \tl_put_right:Ne \l_@@_project_pi_name_tl
                {\dfgShowTag{\l_tmpa_clist}{firstname}{\tmpPIfirstname}[person]\space}
            }
          \clist_if_in:NnT \l_@@_project_pi_properties_clist {lastname}
            {
              \tl_put_right:Ne \l_@@_project_pi_name_tl
                {\dfgShowTag{\l_tmpa_clist}{lastname}{\tmpPIlastname}[person]}
            }
          % Reuse getPI's target, reference styling, and missing-person handling.
          % Freeze the ID and selected fields before advancing the metadata loop.
          \tl_if_empty:NF \l_@@_project_pi_name_tl
            {
              \clist_put_right:Ne \l_@@_person_clist
                {
                  \exp_not:N \getPI {\exp_not:V \l_tmpa_clist}
                    [\exp_not:V \l_@@_project_pi_name_tl]
                }
            }
        }
      \IfBooleanTF{#1}
        {
          \IfNoValueTF{#4}
            {\clist_use:Nnnn \l_@@_person_clist {~and~} {,~} {~and~}}
            {\clist_use:Nnnn \l_@@_person_clist {#4} {,~} {#4}}
        }
        {
          \IfNoValueTF{#4}
            {\clist_use:Nn \l_@@_person_clist {\,/\,}}
            {\clist_use:Nn \l_@@_person_clist {#4}}
        }
    }
  \group_end:
}
```

PI and project references also accept an all-lowercase command spelling. Copy the complete command so stars, optional arguments, and legacy warnings follow the canonical implementation without a separate argument parser.

```
\clist_map_inline:nn
  {
    getPI, getPIsWithRole, getProject, getProjectColor, getProjectCount,
    getPIWithProject, getProjectWithPI, getPIOfProjectWithProject,
    getPIofProject, getPIProject, getProjectPI
  }
  { \cs_new_eq:cc { \str_lowercase:n {#1} } {#1} }
% Common mixed-case spellings retained alongside the lowercase aliases.
\cs_new_eq:NN \getPIwithProject \getPIWithProject
\cs_new_eq:NN \getProjectwithPI \getProjectWithPI
\cs_new_eq:NN \getPIOfProject \getPIofProject
```

For referencing Aims we need to define a new macro.

```
\clist_new:N \l_@@_aim_refs_clist
\NewDocumentCommand{\getAim}{O{\l__tmp_current_project_tl} m}
{
  % #1: current project
  % #2: id of aim(s)
  \clist_clear:N \l_@@_aim_refs_clist
  \clist_map_inline:nn { #2 }
  { \clist_put_right:Nn \l_@@_aim_refs_clist { #1-aim-##1 } }
  \exp_args:NV \cref \l_@@_aim_refs_clist
}
```

For work-package references, `\labelcref` keeps cleveref's sorting and ranges while we provide a narrow, unbreakable space after `WP` or `WPs`.

```
\tl_new:N \l__tmp_current_project_tl
\clist_new:N \l_@@_wp_refs_clist
\NewDocumentCommand{\getWP}{O{\l__tmp_current_project_tl} m}
{
  % #1: current project
  % #2: id of workpackage(s)
  \clist_clear:N \l_@@_wp_refs_clist
  \clist_map_inline:nn { #2 }
  { \clist_put_right:Nn \l_@@_wp_refs_clist { #1-wp-##1 } }
  \textcolor{referenceWP}
    {\textbf{WP\int_compare:nNnF
      { \clist_count:N \l_@@_wp_refs_clist } = { 1 } { s }}}
  \addnbthinspace
  \exp_args:NV \labelcref \l_@@_wp_refs_clist
}
```

The following command must be called within the environment `workpackages`.

```
\tl_new:N \l_@@_fast_workpackage_name_tl
\keys_define:nn { dfg / fast-workpackage }
{
  name .tl_set:N = \l_@@_fast_workpackage_name_tl,
  unknown .code:n = { }
}

% A nested workpackage starts directly with its authored body; its list label
% already supplies the required separation.  Remove only the immediately
% following spaces and paragraph tokens, without adding interword space.  Later
% paragraph breaks in the authored body remain untouched.
\cs_new_protected:Npn \@@_start_subworkpackage_body:
{
  \peek_meaning_remove_ignore_spaces:NTF \par
    { \@@_start_subworkpackage_body: }
    { \ignorespaces }
}

% The visible sub-workpackage label contains its parent number, but enumitem's
% child counter restarts for every parent.  Cleveref otherwise sees, for
% example, both WP 1b and WP 2b as child counter 2 and can discard the latter
% while sorting or compress a range across two parents.  Store the parent
% counter as Cleveref's numeric prefix so sorting and range detection respect
% the work-package hierarchy.
\NewDocumentCommand{\addWP}{ m m }{
  % #1: workpackage-id
  % #2: key-set
  \bool_if:NTF \l_@@_class_fastcompile_project_bool
  {
    \tl_clear:N \l_@@_fast_workpackage_name_tl
    \keys_set:nn { dfg / fast-workpackage } {#2}
    \bool_if:NT \l_@@_class_fastcompile_project_show_images_bool
      { \addEntry[gantt]{gantt-\l__tmp_current_project_tl}{id=#1,#2} }
  }
  {
    \addEntry[gantt]{gantt-\l__tmp_current_project_tl}{id=#1,#2}
  }
  \str_if_eq:eeTF { \@enumctr } { workpackagesi }
  {
    \bool_if:NTF \l_@@_class_fastcompile_project_bool
    {
      \item \textbf{
        \tl_if_blank:VTF \l_@@_fast_workpackage_name_tl
        { WP~#1 }
        { \tl_use:N \l_@@_fast_workpackage_name_tl }
      }%
    }
    {
      \item \textbf{\getEntry[gantt-\l__tmp_current_project_tl]{#1}{name}}%
    }
  }
  { \item }
  \label{\l__tmp_current_project_tl-wp-#1}%
  \str_if_eq:eeF { \@enumctr } { workpackagesi }
    { \@@_start_subworkpackage_body: }
}
```

Project collaborations are stored with each project as an undirected list. The default plot is a compact focus heatmap containing every active project; the focus and its direct collaborators are highlighted. An arc layout uses the same ordered nodes, draws only the focus project's edges, and gives every edge the same peak height. It prints identifiers inside uniformly sized nodes and places unlabelled project-area rules directly below them. Heatmap rows and colors are generated dynamically from `project-area` (with the project-ID prefix as fallback); a rotating palette supports any number of areas. Boxes use a shared dynamic column width determined by the widest project area, so the grid fills the complete text block after its area labels. Both layouts are deterministic, are generated entirely with TikZ, and omit ending, reference-only, and list-excluded projects. The built-in area palette cycles when more areas are present than palette entries; existing A, B, and Q projects retain their familiar colors.

```
\dim_new:N  \l_@@_collaboration_plot_x_dim
\dim_new:N  \l_@@_collaboration_plot_y_dim
\dim_new:N  \l_@@_collaboration_plot_x_step_dim
\dim_new:N  \l_@@_collaboration_plot_box_width_dim
\dim_new:N  \l_@@_collaboration_plot_box_gap_dim
\dim_new:N  \l_@@_collaboration_plot_area_label_width_dim
\dim_new:N  \l_@@_collaboration_plot_content_start_dim
\dim_new:N  \l_@@_collaboration_plot_width_dim
\dim_new:N  \l_@@_collaboration_plot_arc_height_dim
\box_new:N  \l_@@_collaboration_plot_area_label_box
```

```
\seq_new:N  \l_@@_collaboration_plot_nodes_seq
\seq_new:N  \l_@@_collaboration_plot_areas_seq
\seq_new:N  \l_@@_collaboration_plot_area_palette_seq
\seq_new:N  \l_@@_collaboration_plot_edges_seq
\prop_new:N \l_@@_collaboration_plot_all_status_prop
\prop_new:N \l_@@_collaboration_plot_status_prop
\prop_new:N \l_@@_collaboration_plot_lists_prop
\prop_new:N \l_@@_collaboration_plot_edges_prop
\prop_new:N \l_@@_collaboration_plot_visible_prop
\prop_new:N \l_@@_collaboration_plot_area_prop
\prop_new:N \l_@@_collaboration_plot_area_nodes_prop
\prop_new:N \l_@@_collaboration_plot_area_color_prop
\prop_new:N \l_@@_collaboration_plot_area_color_override_prop
\tl_new:N   \l_@@_collaboration_plot_focus_tl
\tl_new:N   \l_@@_collaboration_plot_layout_tl
\tl_new:N   \l_@@_collaboration_plot_id_tl
\tl_new:N   \l_@@_collaboration_plot_status_tl
\tl_new:N   \l_@@_collaboration_plot_consider_tl
\tl_new:N   \l_@@_collaboration_plot_reference_only_tl
\tl_new:N   \l_@@_collaboration_plot_collaborations_tl
\tl_new:N   \l_@@_collaboration_plot_area_tl
\tl_new:N   \l_@@_collaboration_plot_source_tl
\tl_new:N   \l_@@_collaboration_plot_target_tl
\tl_new:N   \l_@@_collaboration_plot_edge_tl
\tl_new:N   \l_@@_collaboration_plot_edge_left_tl
\tl_new:N   \l_@@_collaboration_plot_edge_right_tl
\tl_new:N   \l_@@_collaboration_plot_label_tl
\tl_new:N   \l_@@_collaboration_plot_color_tl
\tl_new:N   \l_@@_collaboration_plot_caption_tl
\int_new:N  \l_@@_collaboration_plot_group_index_int
\int_new:N  \l_@@_collaboration_plot_row_index_int
\int_new:N  \l_@@_collaboration_plot_node_count_int
\int_new:N  \l_@@_collaboration_plot_area_index_int
\bool_new:N \l_@@_collaboration_plot_arcs_bool

\cs_new_protected:Npn \@@_collaboration_plot_merge_area_colors:n #1
{
  \prop_set_from_keyval:Nn \l_tmpa_prop {#1}
  \prop_map_inline:Nn \l_tmpa_prop
    { \prop_put:Nnn \l_@@_collaboration_plot_area_color_override_prop
        {##1} {##2} }
}

\keys_define:nn { dfg / project-collaboration-plot }
{
  layout .choice:,
  layout / heatmap .code:n =
    { \tl_set:Nn \l_@@_collaboration_plot_layout_tl { heatmap } },
  layout / arc .code:n =
    { \tl_set:Nn \l_@@_collaboration_plot_layout_tl { arc } },
  layout / arc-diagram .meta:n = { layout = arc },
  heatmap .meta:n = { layout = heatmap },
  arc .meta:n = { layout = arc },
  arc-diagram .meta:n = { layout = arc },
  arcs .bool_set:N = \l_@@_collaboration_plot_arcs_bool,
  arcs .default:n = true,
  no-arcs .meta:n = { arcs = false },
  area-colors .code:n =
    { \@@_collaboration_plot_merge_area_colors:n {#1} },
  area-colours .meta:n = { area-colors = {#1} },
  area-palette .code:n =
    {
      \clist_if_empty:nF {#1}
        { \seq_set_from_clist:Nn \l_@@_collaboration_plot_area_palette_seq {#1} }
    },
  caption .tl_set:N = \l_@@_collaboration_plot_caption_tl,
  unknown .code:n = { \msg_warning:nnn { dfg } { unknown-key } {#1} }
}

\cs_new_protected:Npn \@@_collaboration_plot_collect_nodes:
{
  \seq_clear:N  \l_@@_collaboration_plot_nodes_seq
  \prop_clear:N \l_@@_collaboration_plot_all_status_prop
  \prop_clear:N \l_@@_collaboration_plot_status_prop
  \prop_clear:N \l_@@_collaboration_plot_lists_prop
  \prop_clear:N \l_@@_collaboration_plot_edges_prop
  \prop_clear:N \l_@@_collaboration_plot_area_prop
  % DATATOOL requires an iterated key to exist as a database column.  Older
  % proposals do not have project-area yet, so create the optional column and
  % let the ID-prefix fallback below handle its empty values.
  \DTLifhaskey { project } { project-area }
    { }
    { \DTLaddcolumn { project } { project-area } }
  \DTLifhaskey { project } { collaborations }
    { }
    { \DTLaddcolumn { project } { collaborations } }
  \DTLforeach*{project}
```

```
    {
      \l_@@_collaboration_plot_id_tl = id,
      \l_@@_collaboration_plot_status_tl = project-status,
      \l_@@_collaboration_plot_consider_tl = consider-for-lists,
      \l_@@_collaboration_plot_reference_only_tl = reference-only,
      \l_@@_collaboration_plot_area_tl = project-area,
      \l_@@_collaboration_plot_collaborations_tl = collaborations
    }
    {
      \prop_put:NVV \l_@@_collaboration_plot_all_status_prop
        \l_@@_collaboration_plot_id_tl
        \l_@@_collaboration_plot_status_tl
      \bool_lazy_all:nT
      {
        { \str_if_eq_p:Vn \l_@@_collaboration_plot_consider_tl { true } }
        { \str_if_eq_p:Vn \l_@@_collaboration_plot_reference_only_tl { false } }
        {
          \bool_lazy_or_p:nn
            { \str_if_eq_p:Vn \l_@@_collaboration_plot_status_tl { C } }
            { \str_if_eq_p:Vn \l_@@_collaboration_plot_status_tl { N } }
        }
      }
      {
        \DTLifnullorempty{\l_@@_collaboration_plot_collaborations_tl}
          { \tl_clear:N \l_@@_collaboration_plot_collaborations_tl }
          { }
        \seq_put_right:NV \l_@@_collaboration_plot_nodes_seq
          \l_@@_collaboration_plot_id_tl
        \DTLifnullorempty{\l_@@_collaboration_plot_area_tl}
          { \tl_set:Nx \l_@@_collaboration_plot_area_tl
              { \tl_head:V \l_@@_collaboration_plot_id_tl } }
          { }
        \tl_trim_spaces:N \l_@@_collaboration_plot_area_tl
        \prop_put:NVV \l_@@_collaboration_plot_area_prop
          \l_@@_collaboration_plot_id_tl \l_@@_collaboration_plot_area_tl
        \prop_put:NVV \l_@@_collaboration_plot_status_prop
          \l_@@_collaboration_plot_id_tl
          \l_@@_collaboration_plot_status_tl
        \prop_put:NVV \l_@@_collaboration_plot_lists_prop
          \l_@@_collaboration_plot_id_tl
          \l_@@_collaboration_plot_collaborations_tl
      }
    }
    \seq_sort:Nn \l_@@_collaboration_plot_nodes_seq
    {
      \prop_get:NnN \l_@@_collaboration_plot_area_prop {##1} \l_tmpa_tl
      \prop_get:NnN \l_@@_collaboration_plot_area_prop {##2} \l_tmpb_tl
      \str_compare:eNeTF
        { \tl_use:N \l_tmpa_tl / ##1 }
        <
        { \tl_use:N \l_tmpb_tl / ##2 }
        { \sort_return_same: }
        { \sort_return_swapped: }
    }
  }

  \cs_new_protected:Npn \@@_collaboration_plot_collect_edge:nn #1#2
  {
    \tl_set:Nn \l_@@_collaboration_plot_source_tl {#1}
    \tl_set:Nn \l_@@_collaboration_plot_target_tl {#2}
    \tl_trim_spaces:N \l_@@_collaboration_plot_source_tl
    \tl_trim_spaces:N \l_@@_collaboration_plot_target_tl
    \tl_if_blank:VF \l_@@_collaboration_plot_target_tl
    {
      \prop_if_in:NVTF \l_@@_collaboration_plot_status_prop
        \l_@@_collaboration_plot_target_tl
      {
        \str_if_eq:VVF
          \l_@@_collaboration_plot_source_tl
          \l_@@_collaboration_plot_target_tl
        {
          \str_compare:eNeTF
            { \tl_use:N \l_@@_collaboration_plot_source_tl }
            <
            { \tl_use:N \l_@@_collaboration_plot_target_tl }
            {
              \tl_set:Nx \l_@@_collaboration_plot_edge_tl
                {
                  \tl_use:N \l_@@_collaboration_plot_source_tl
                  /\tl_use:N \l_@@_collaboration_plot_target_tl
                }
            }
            {
              \tl_set:Nx \l_@@_collaboration_plot_edge_tl
                {
                  \tl_use:N \l_@@_collaboration_plot_target_tl
                  /\tl_use:N \l_@@_collaboration_plot_source_tl
```

```
11328               }
11329             }
11330           \prop_put:NVn \l_@@_collaboration_plot_edges_prop
11331             \l_@@_collaboration_plot_edge_tl { 1 }
11332         }
11333       }
11334       {
11335         \prop_if_in:NVF \l_@@_collaboration_plot_all_status_prop
11336           \l_@@_collaboration_plot_target_tl
11337         {
11338           \msg_warning:nnee { dfg } { collaboration-target-not-found }
11339             { \l_@@_collaboration_plot_source_tl }
11340             { \l_@@_collaboration_plot_target_tl }
11341         }
11342       }
11343     }
11344   }
11345
11346   \cs_new_protected:Npn \@@_collaboration_plot_collect_edges:
11347   {
11348     \seq_clear:N \l_@@_collaboration_plot_edges_seq
11349     \prop_map_inline:Nn \l_@@_collaboration_plot_lists_prop
11350     {
11351       \clist_map_inline:nn {##2}
11352         { \@@_collaboration_plot_collect_edge:nn {##1} {####1} }
11353     }
11354     \prop_map_inline:Nn \l_@@_collaboration_plot_edges_prop
11355       { \seq_put_right:Nn \l_@@_collaboration_plot_edges_seq {##1} }
11356     \seq_sort:Nn \l_@@_collaboration_plot_edges_seq
11357       {
11358         \str_compare:eNeTF {##1} < {##2}
11359           { \sort_return_same: }
11360           { \sort_return_swapped: }
11361       }
11362   }
11363
11364   \cs_new_protected:Npn \@@_collaboration_plot_unpack_edge:n #1
11365   {
11366     \seq_set_split:Nnn \l_tmpa_seq {/} {#1}
11367     \seq_pop_left:NN \l_tmpa_seq \l_@@_collaboration_plot_edge_left_tl
11368     \seq_pop_left:NN \l_tmpa_seq \l_@@_collaboration_plot_edge_right_tl
11369   }
11370
11371   \cs_new_protected:Npn \@@_collaboration_plot_prepare:
11372   {
11373     \prop_clear:N \l_@@_collaboration_plot_visible_prop
11374     \seq_clear:N \l_@@_collaboration_plot_areas_seq
11375     \prop_clear:N \l_@@_collaboration_plot_area_nodes_prop
11376     \prop_clear:N \l_@@_collaboration_plot_area_color_prop
11377     \int_zero:N \l_@@_collaboration_plot_area_index_int
11378     \prop_put:NVn \l_@@_collaboration_plot_visible_prop
11379       \l_@@_collaboration_plot_focus_tl { 1 }
11380     \seq_map_inline:Nn \l_@@_collaboration_plot_edges_seq
11381     {
11382       \@@_collaboration_plot_unpack_edge:n {##1}
11383       \bool_lazy_or:nnT
11384       {
11385         \str_if_eq_p:VV
11386           \l_@@_collaboration_plot_edge_left_tl
11387           \l_@@_collaboration_plot_focus_tl
11388       }
11389       {
11390         \str_if_eq_p:VV
11391           \l_@@_collaboration_plot_edge_right_tl
11392           \l_@@_collaboration_plot_focus_tl
11393       }
11394       {
11395         \prop_put:NVn \l_@@_collaboration_plot_visible_prop
11396           \l_@@_collaboration_plot_edge_left_tl { 1 }
11397         \prop_put:NVn \l_@@_collaboration_plot_visible_prop
11398           \l_@@_collaboration_plot_edge_right_tl { 1 }
11399       }
11400     }
11401     \seq_map_inline:Nn \l_@@_collaboration_plot_nodes_seq
11402     {
11403       \prop_get:NnN \l_@@_collaboration_plot_area_prop {##1}
11404         \l_@@_collaboration_plot_area_tl
11405       \prop_if_in:NVF \l_@@_collaboration_plot_area_nodes_prop
11406         \l_@@_collaboration_plot_area_tl
11407       {
11408         \seq_put_right:NV \l_@@_collaboration_plot_areas_seq
11409           \l_@@_collaboration_plot_area_tl
11410         \prop_get:NVNTF \l_@@_collaboration_plot_area_color_override_prop
11411           \l_@@_collaboration_plot_area_tl
11412           \l_@@_collaboration_plot_color_tl
11413         {
11414           \prop_put:NVV \l_@@_collaboration_plot_area_color_prop
```

```
            \l_@@_collaboration_plot_area_tl
            \l_@@_collaboration_plot_color_tl
        }
        {
          \int_incr:N \l_@@_collaboration_plot_area_index_int
          \int_set:Nn \l_tmpa_int
            {
              1 + \int_mod:nn
                { \l_@@_collaboration_plot_area_index_int - 1 }
                { \seq_count:N \l_@@_collaboration_plot_area_palette_seq }
            }
          \prop_put:NVx \l_@@_collaboration_plot_area_color_prop
            \l_@@_collaboration_plot_area_tl
            {
              \seq_item:Nn \l_@@_collaboration_plot_area_palette_seq
                { \l_tmpa_int }
            }
        }
        \prop_put:NVn \l_@@_collaboration_plot_area_nodes_prop
          \l_@@_collaboration_plot_area_tl { }
      }
      \prop_get:NVN \l_@@_collaboration_plot_area_nodes_prop
        \l_@@_collaboration_plot_area_tl \l_tmpa_tl
      \tl_if_blank:VTF \l_tmpa_tl
        { \tl_set:Nn \l_tmpa_tl {##1} }
        { \tl_put_right:Nn \l_tmpa_tl { , ##1 } }
      \prop_put:NVV \l_@@_collaboration_plot_area_nodes_prop
        \l_@@_collaboration_plot_area_tl \l_tmpa_tl
    }
  }

  \cs_new_protected:Npn \@@_collaboration_plot_set_color:n #1
  {
    \tl_set:Nx \l_@@_collaboration_plot_label_tl {#1}
    \prop_get:NnNTF \l_@@_collaboration_plot_area_prop
      {#1} \l_@@_collaboration_plot_area_tl
      { \prop_get:NVN \l_@@_collaboration_plot_area_color_prop
          \l_@@_collaboration_plot_area_tl
          \l_@@_collaboration_plot_color_tl }
      { \tl_set:Nn \l_@@_collaboration_plot_color_tl { grey } }
  }

  \cs_new_protected:Npn \@@_collaboration_plot_heatmap_cell:n #1
  {
    \@@_collaboration_plot_set_color:n {#1}
    \str_if_eq:VVTF
      \l_@@_collaboration_plot_focus_tl
      \l_@@_collaboration_plot_label_tl
    {
      \node[
        anchor=center,
        minimum~width=\dim_use:N \l_@@_collaboration_plot_box_width_dim,
        minimum~height=5.2mm,
        inner~sep=0pt,
        rounded~corners=1.4pt,
        draw=\tl_use:N \l_@@_collaboration_plot_color_tl!78!black,
        fill=\tl_use:N \l_@@_collaboration_plot_color_tl,
        text=white,
        font=\footnotesize\bfseries
      ] at
        (\dim_use:N \l_@@_collaboration_plot_x_dim,
         \dim_use:N \l_@@_collaboration_plot_y_dim)
        { \strut\tl_use:N \l_@@_collaboration_plot_label_tl };
    }
    {
      \prop_if_in:NVTF \l_@@_collaboration_plot_visible_prop
        \l_@@_collaboration_plot_label_tl
      {
        \node[
          anchor=center,
          minimum~width=\dim_use:N \l_@@_collaboration_plot_box_width_dim,
          minimum~height=5.2mm,
          inner~sep=0pt,
          rounded~corners=1.4pt,
          draw=\tl_use:N \l_@@_collaboration_plot_color_tl!48,
          fill=\tl_use:N \l_@@_collaboration_plot_color_tl!14,
          text=\tl_use:N \l_@@_collaboration_plot_color_tl,
          font=\footnotesize\bfseries
        ] at
          (\dim_use:N \l_@@_collaboration_plot_x_dim,
           \dim_use:N \l_@@_collaboration_plot_y_dim)
          { \strut\tl_use:N \l_@@_collaboration_plot_label_tl };
      }
      {
        \node[
          anchor=center,
          minimum~width=\dim_use:N \l_@@_collaboration_plot_box_width_dim,
```

```
        minimum~height=5.2mm,
        inner~sep=0pt,
        rounded~corners=1.4pt,
        draw=grey!22,
        fill=white,
        text=grey!70!black,
        font=\footnotesize
      ] at
        (\dim_use:N \l_@@_collaboration_plot_x_dim,
         \dim_use:N \l_@@_collaboration_plot_y_dim)
        { \strut\tl_use:N \l_@@_collaboration_plot_label_tl };
    }
  }
}

\cs_new_protected:Npn \@@_collaboration_plot_heatmap_row:nnn #1#2#3
{
  \clist_if_empty:nF {#1}
  {
    \int_incr:N \l_@@_collaboration_plot_row_index_int
    \dim_set:Nn \l_@@_collaboration_plot_y_dim
    {
      -7mm *
      \int_eval:n { \l_@@_collaboration_plot_row_index_int - 1 }
    }
    \node[
      anchor=east,
      inner~sep=0pt,
      font=\scriptsize\bfseries
    ] at
      (\dim_use:N \l_@@_collaboration_plot_area_label_width_dim,
       \dim_use:N \l_@@_collaboration_plot_y_dim)
      {Project~area~#3};
    \draw[draw=#2,line~width=.75pt,line~cap=round]
      (\dim_eval:n
         { \l_@@_collaboration_plot_area_label_width_dim + .8mm },
       \dim_use:N \l_@@_collaboration_plot_y_dim)
      --
      (\dim_eval:n
         { \l_@@_collaboration_plot_area_label_width_dim + 2.8mm },
       \dim_use:N \l_@@_collaboration_plot_y_dim);
    \int_zero:N \l_@@_collaboration_plot_group_index_int
    \clist_map_inline:nn {#1}
    {
      \int_incr:N \l_@@_collaboration_plot_group_index_int
      \dim_set:Nn \l_@@_collaboration_plot_x_dim
      {
        \l_@@_collaboration_plot_content_start_dim
        + \l_@@_collaboration_plot_box_width_dim / 2
        + \l_@@_collaboration_plot_x_step_dim *
        \int_eval:n { \l_@@_collaboration_plot_group_index_int - 1 }
      }
      \@@_collaboration_plot_heatmap_cell:n {##1}
    }
  }
}
\cs_generate_variant:Nn \@@_collaboration_plot_heatmap_row:nnn { V V n }

\cs_new_protected:Npn \@@_collaboration_plot_heatmap_picture:
{
    % Use the widest project area as the shared column grid.  Each column gets
    % a dynamic box width, and the final cell reaches the right edge of the
    % text block instead of leaving a large unused strip before the caption.
    \dim_zero:N \l_@@_collaboration_plot_area_label_width_dim
    \seq_map_inline:Nn \l_@@_collaboration_plot_areas_seq
      {
        \hbox_set:Nn \l_@@_collaboration_plot_area_label_box
          { \scriptsize\bfseries Project~area~##1 }
        \dim_compare:nNnT
          { \box_wd:N \l_@@_collaboration_plot_area_label_box }
          >
          { \l_@@_collaboration_plot_area_label_width_dim }
          {
            \dim_set:Nn \l_@@_collaboration_plot_area_label_width_dim
              { \box_wd:N \l_@@_collaboration_plot_area_label_box }
          }
      }
    \dim_set:Nn \l_@@_collaboration_plot_content_start_dim
      { \l_@@_collaboration_plot_area_label_width_dim + 4mm }
  \begin{tikzpicture}
    \int_zero:N \l_@@_collaboration_plot_node_count_int
    \seq_map_inline:Nn \l_@@_collaboration_plot_areas_seq
      {
        \prop_get:NnN \l_@@_collaboration_plot_area_nodes_prop {##1}
          \l_tmpa_tl
        \clist_set:NV \l_tmpa_clist \l_tmpa_tl
        \int_set:Nn \l_tmpa_int { \clist_count:N \l_tmpa_clist }
```

```
11589         \int_set:Nn \l_@@_collaboration_plot_node_count_int
11590           { \int_max:nn { \l_@@_collaboration_plot_node_count_int }
11591             { \l_tmpa_int } }
11592       }
11593     \dim_set:Nn \l_@@_collaboration_plot_box_gap_dim { 1mm }
11594     \int_compare:nNnTF
11595       { \l_@@_collaboration_plot_node_count_int } > { 0 }
11596       {
11597         \dim_set:Nn \l_@@_collaboration_plot_box_width_dim
11598         {
11599           ( \l_@@_collaboration_plot_width_dim
11600             - \l_@@_collaboration_plot_content_start_dim
11601             - \l_@@_collaboration_plot_box_gap_dim *
11602               \int_eval:n
11603                 { \l_@@_collaboration_plot_node_count_int - 1 }
11604           ) / \l_@@_collaboration_plot_node_count_int
11605         }
11606         \dim_set:Nn \l_@@_collaboration_plot_x_step_dim
11607         {
11608           \l_@@_collaboration_plot_box_width_dim
11609           + \l_@@_collaboration_plot_box_gap_dim
11610         }
11611       }
11612       {
11613         \dim_zero:N \l_@@_collaboration_plot_box_width_dim
11614         \dim_zero:N \l_@@_collaboration_plot_x_step_dim
11615       }
11616     \int_zero:N \l_@@_collaboration_plot_row_index_int
11617     \seq_map_inline:Nn \l_@@_collaboration_plot_areas_seq
11618       {
11619         \prop_get:NnN \l_@@_collaboration_plot_area_nodes_prop {##1}
11620           \l_tmpa_tl
11621         \prop_get:NnN \l_@@_collaboration_plot_area_color_prop {##1}
11622           \l_@@_collaboration_plot_color_tl
11623         \@@_collaboration_plot_heatmap_row:VVn
11624           \l_tmpa_tl \l_@@_collaboration_plot_color_tl {##1}
11625       }
11626   \end{tikzpicture}
11627 }
11628
11629 \cs_new_protected:Npn \@@_collaboration_plot_arc_place_nodes:
11630 {
11631   \int_set:Nn \l_@@_collaboration_plot_node_count_int
11632     { \seq_count:N \l_@@_collaboration_plot_nodes_seq }
11633   \int_compare:nNnTF
11634     { \l_@@_collaboration_plot_node_count_int } > { 1 }
11635     {
11636       \dim_set:Nn \l_@@_collaboration_plot_x_step_dim
11637       {
11638         ( \l_@@_collaboration_plot_width_dim - 6mm ) /
11639         \int_eval:n { \l_@@_collaboration_plot_node_count_int - 1 }
11640       }
11641     }
11642     { \dim_zero:N \l_@@_collaboration_plot_x_step_dim }
11643   \int_zero:N \l_@@_collaboration_plot_group_index_int
11644   \seq_map_inline:Nn \l_@@_collaboration_plot_nodes_seq
11645   {
11646     \dim_set:Nn \l_@@_collaboration_plot_x_dim
11647     {
11648       3mm + \l_@@_collaboration_plot_x_step_dim
11649       * \l_@@_collaboration_plot_group_index_int
11650     }
11651     \coordinate (dfg-collaboration-arc-##1) at
11652       (\dim_use:N \l_@@_collaboration_plot_x_dim,0);
11653     \int_incr:N \l_@@_collaboration_plot_group_index_int
11654   }
11655 }
11656
11657 \cs_new_protected:Npn \@@_collaboration_plot_draw_focus_arc:n #1
11658 {
11659   \@@_collaboration_plot_unpack_edge:n {#1}
11660   \tl_clear:N \l_@@_collaboration_plot_target_tl
11661   \str_if_eq:VVTF
11662     \l_@@_collaboration_plot_edge_left_tl
11663     \l_@@_collaboration_plot_focus_tl
11664     {
11665       \tl_set_eq:NN \l_@@_collaboration_plot_target_tl
11666         \l_@@_collaboration_plot_edge_right_tl
11667     }
11668     {
11669       \str_if_eq:VVT
11670         \l_@@_collaboration_plot_edge_right_tl
11671         \l_@@_collaboration_plot_focus_tl
11672       {
11673         \tl_set_eq:NN \l_@@_collaboration_plot_target_tl
11674           \l_@@_collaboration_plot_edge_left_tl
11675       }
```

```
11676     }
11677     \tl_if_blank:VF \l_@@_collaboration_plot_target_tl
11678     {
11679       \@@_collaboration_plot_set_color:n
11680         { \tl_use:N \l_@@_collaboration_plot_target_tl }
11681       \draw[
11682         draw=\tl_use:N \l_@@_collaboration_plot_color_tl,
11683         line~width=.75pt,
11684         opacity=.78,
11685         line~cap=round
11686       ]
11687         (dfg-collaboration-arc-\l_@@_collaboration_plot_focus_tl)
11688         .. controls % chktex 12
11689         ($ (dfg-collaboration-arc-\l_@@_collaboration_plot_focus_tl) % chktex 25
11690            + (0,\dim_use:N \l_@@_collaboration_plot_arc_height_dim) $) % chktex 25 chktex 35
11691         and
11692         ($ (dfg-collaboration-arc-\l_@@_collaboration_plot_target_tl) % chktex 25
11693            + (0,\dim_use:N \l_@@_collaboration_plot_arc_height_dim) $) % chktex 25 chktex 35
11694         ..
11695         (dfg-collaboration-arc-\l_@@_collaboration_plot_target_tl);
11696     }
11697   }
11698
11699   \cs_new_protected:Npn \@@_collaboration_plot_arc_node:n #1
11700   {
11701     \@@_collaboration_plot_set_color:n {#1}
11702     \str_if_eq:VVTF
11703       \l_@@_collaboration_plot_focus_tl
11704       \l_@@_collaboration_plot_label_tl
11705     {
11706       \node[
11707         anchor=center,
11708         minimum~width=6mm,
11709         minimum~height=5mm,
11710         inner~sep=0pt,
11711         rounded~corners=1pt,
11712         draw=\tl_use:N \l_@@_collaboration_plot_color_tl!78!black,
11713         fill=\tl_use:N \l_@@_collaboration_plot_color_tl,
11714         text=white,
11715         font=\scriptsize\bfseries
11716       ] at (dfg-collaboration-arc-#1) {#1};
11717     }
11718     {
11719       \prop_if_in:NVTF \l_@@_collaboration_plot_visible_prop
11720         \l_@@_collaboration_plot_label_tl
11721       {
11722         \node[
11723           anchor=center,
11724           minimum~width=6mm,
11725           minimum~height=5mm,
11726           inner~sep=0pt,
11727           rounded~corners=.8pt,
11728           draw=\tl_use:N \l_@@_collaboration_plot_color_tl!48,
11729           fill=\tl_use:N \l_@@_collaboration_plot_color_tl!14,
11730           text=\tl_use:N \l_@@_collaboration_plot_color_tl,
11731           font=\scriptsize\bfseries
11732         ] at (dfg-collaboration-arc-#1) {#1};
11733       }
11734       {
11735         \node[
11736           anchor=center,
11737           minimum~width=6mm,
11738           minimum~height=5mm,
11739           inner~sep=0pt,
11740           rounded~corners=.8pt,
11741           draw=grey!22,
11742           fill=white,
11743           text=grey!70!black,
11744           font=\scriptsize
11745         ] at (dfg-collaboration-arc-#1) {#1};
11746       }
11747     }
11748   }
11749
11750   \cs_new_protected:Npn \@@_collaboration_plot_arc_group:nn #1#2
11751   {
11752     \clist_if_empty:nF {#1}
11753     {
11754       \tl_set:Nx \l_tmpa_tl { \clist_item:nn {#1} { 1 } }
11755       \tl_set:Nx \l_tmpb_tl { \clist_item:nn {#1} { -1 } }
11756       \draw[draw=#2,line~width=.75pt,line~cap=round]
11757         ($ (dfg-collaboration-arc-\l_tmpa_tl) + (-3mm,-3.5mm) $) % chktex 25
11758         --
11759         ($ (dfg-collaboration-arc-\l_tmpb_tl) + (3mm,-3.5mm) $); % chktex 25
11760     }
11761   }
11762   \cs_generate_variant:Nn \@@_collaboration_plot_arc_group:nn { V V }
```

```
\cs_new_protected:Npn \@@_collaboration_plot_arc_picture:
{
  \begin{tikzpicture}
    \@@_collaboration_plot_arc_place_nodes:
    % Equal control-point heights give every symmetric cubic arc the same peak.
    \dim_set:Nn \l_@@_collaboration_plot_arc_height_dim { 12mm }
    \bool_if:NT \l_@@_collaboration_plot_arcs_bool
    {
      \seq_map_inline:Nn \l_@@_collaboration_plot_edges_seq
        { \@@_collaboration_plot_draw_focus_arc:n {##1} }
    }
    \seq_map_inline:Nn \l_@@_collaboration_plot_nodes_seq
      { \@@_collaboration_plot_arc_node:n {##1} }
    \seq_map_inline:Nn \l_@@_collaboration_plot_areas_seq
      {
        \prop_get:NnN \l_@@_collaboration_plot_area_nodes_prop {##1}
          \l_tmpa_tl
        \prop_get:NnN \l_@@_collaboration_plot_area_color_prop {##1}
          \l_@@_collaboration_plot_color_tl
        \@@_collaboration_plot_arc_group:VV
          \l_tmpa_tl \l_@@_collaboration_plot_color_tl
      }
  \end{tikzpicture}
}

\cs_new_protected:Npn \@@_collaboration_plot_picture:
{
  \str_case:VnF \l_@@_collaboration_plot_layout_tl
  {
    { heatmap } { \@@_collaboration_plot_heatmap_picture: }
    { arc }     { \@@_collaboration_plot_arc_picture: }
  }
  { \@@_collaboration_plot_arc_picture: }
}

% Keep both caption variants at the captured body-text width without scaling
% the TikZ plot, even when the surrounding \linewidth is temporarily narrower.
\cs_new_protected:Npn \@@_collaboration_plot_full_width_picture:
{
  \makebox[\dim_use:N \l_@@_collaboration_plot_width_dim][l]
    {\@@_collaboration_plot_picture:}
}

\NewDocumentCommand{\ProjectCollaborationPlot}{ O{} m o }
{%
  \group_begin:
  % Capture the normal body width before hvfloat builds either bottom caption.
  \dim_set:Nn \l_@@_collaboration_plot_width_dim { \textwidth }
  \exp_args:NNV \prop_set_from_keyval:Nn
    \l_@@_collaboration_plot_area_color_override_prop
    \l_@@_custom_collaboration_area_colors_tl
  \exp_args:NNV \seq_set_from_clist:Nn
    \l_@@_collaboration_plot_area_palette_seq
    \l_@@_custom_collaboration_area_palette_tl
  \tl_clear:N \l_@@_collaboration_plot_caption_tl
  \keys_set:nn { dfg / project-collaboration-plot }
  {
    layout = arc,
    arcs = false,
    #1
  }
  \IfNoValueF{#3}
    { \tl_set:Nn \l_@@_collaboration_plot_caption_tl {#3} }
  \tl_set:Nn \l_@@_collaboration_plot_focus_tl {#2}
  \tl_trim_spaces:N \l_@@_collaboration_plot_focus_tl
  \DTLifdbexists{project}
  {
    \@@_collaboration_plot_collect_nodes:
    \prop_if_in:NVTF \l_@@_collaboration_plot_status_prop
      \l_@@_collaboration_plot_focus_tl
    {
      \@@_collaboration_plot_collect_edges:
      \@@_collaboration_plot_prepare:
      \tl_if_blank:VTF \l_@@_collaboration_plot_caption_tl
      {%
        \hvFloat[
          floatPos=h,
          capPos=bottom
        ]
        {figure}
        {\@@_collaboration_plot_full_width_picture:}
        {Collaborations~of~\getProject{#2}.}
        {fig:project-collaboration-#2-
          \tl_use:N \l_@@_collaboration_plot_layout_tl}%
      }
      {%
```

```
        \hvFloat[
          floatPos=h,
          capPos=bottom
        ]
        {figure}
        {\@@_collaboration_plot_full_width_picture:}
        {Collaborations~of~\getProject{#2}.~
          \tl_use:N \l_@@_collaboration_plot_caption_tl}
        {fig:project-collaboration-#2-
          \tl_use:N \l_@@_collaboration_plot_layout_tl}%
      }
    }
    {
      \msg_warning:nne { dfg } { collaboration-focus-not-found }
        { \l_@@_collaboration_plot_focus_tl }
      \textcolor{error}
        {Project~collaboration~plot:~unknown~project~'#2'.}
    }
  }
  { \msg_warning:nnn { dfg } { database-not-found } { project } }
  \group_end:
}

% Since the workpackage is closely related to the Gantt chart we continue with
% that.  \cs{GanttPlot} is the plot dispatcher's target.  The old
% \cs{printGantt} spelling remains as a compatibility wrapper.
\tl_new:N \l_@@_gantt_plot_caption_tl
\keys_define:nn { dfg / gantt-plot }
{
  caption .tl_set:N = \l_@@_gantt_plot_caption_tl,
  unknown .code:n = { \msg_warning:nnn { dfg } { unknown-key } {#1} }
}

\NewDocumentCommand{\GanttPlot}{ O{} m o }
{
  \group_begin:
  \tl_clear:N \l_@@_gantt_plot_caption_tl
  \keys_set:nn { dfg / gantt-plot } {#1}
  \IfNoValueF{#3}
    { \tl_set:Nn \l_@@_gantt_plot_caption_tl {#3} }
  \DTLifdbexists{gantt-#2}
  {
    \tl_if_blank:VTF \l_@@_gantt_plot_caption_tl
    {%
      \hvFloat[
        floatPos=h,
        capPos=bottom
      ]
      {figure}
      {\dfgGanttEnvironment{#2}}
      {Gantt~chart~of~\getProject{#2}.}
      {fig:gantt#2}%
    }
    {%
      \hvFloat[
        floatPos=h,
        capPos=bottom
      ]
      {figure}
      {\dfgGanttEnvironment{#2}}
      {Gantt~chart~of~\getProject{#2}.~
        \tl_use:N \l_@@_gantt_plot_caption_tl}
      {fig:gantt-#2}%
    }
  }
  {
    Database~ \texttt{gantt-#2}~ missing.
    \msg_warning:nne { dfg } { database-not-found }{ gantt-#2 }
  }
  \group_end:
}

% Lowercase plot names accept either an explicit project ID or the active
% project.  An explicit ID may carry a trailing caption; the current-project
% form uses the caption key instead.
\cs_new_protected:Npn \@@_typed_plot:nnnn #1#2#3#4
{
  \IfNoValueTF{#4}
    { #1[#2]{#3} }
    { #1[#2]{#3}[#4] }
}
\cs_new_protected:Npn \@@_typed_plot_current:nnnn #1#2#3#4
  { \@@_typed_plot:nnnn {#2}{#3}{#1}{#4} }

\NewDocumentCommand{\ganttPlot}{ O{} g o }
{
  \IfNoValueTF{#2}
```

```
    {
      \tl_if_blank:VTF \l__tmp_current_project_tl
        { \msg_warning:nnn { dfg } { plot-current-project-missing } { gantt } }
        { \exp_args:NV \@@_typed_plot_current:nnnn
            \l__tmp_current_project_tl {\GanttPlot}{#1}{#3} }
    }
    { \@@_typed_plot:nnnn {\GanttPlot}{#1}{#2}{#3} }
}
\NewDocumentCommand{\collaborationsPlot}{ O{} g o }
{
  \IfNoValueTF{#2}
    {
      \tl_if_blank:VTF \l__tmp_current_project_tl
        { \msg_warning:nnn { dfg } { plot-current-project-missing }
            { collaborations } }
        { \exp_args:NV \@@_typed_plot_current:nnnn
            \l__tmp_current_project_tl {\ProjectCollaborationPlot}{#1}{#3} }
    }
    { \@@_typed_plot:nnnn {\ProjectCollaborationPlot}{#1}{#2}{#3} }
}
% Preserve the earlier singular current-project shorthand.
\cs_new_eq:NN \collaborationPlot \collaborationsPlot

\NewDocumentCommand{\printGantt}{ m o }
{
  \IfNoValueTF{#2}
    { \printPlot{{Gantt}{#1}} }
    { \printPlot[caption={#2}]{{Gantt}{#1}} }
}
```

6.1.26 Gantt chart rendering

The Gantt chart for each project is assembled in two steps:

1. `\dfgGanttContent{`⟨*project*⟩`}`: iterates over the `gantt-\meta{project}` database and generates one `\ganttbar` call per row. Rows where `periodStart = periodEnd` are silently skipped. Row breaks precede each visible bar after the first, so skipped rows and the last bar add no blank bottom row. The bar style is dispatched via `\str_case:VnF`: `crosshatch` uses a pgfgantt pattern fill; `striped` overlays two colours at 45° with `\postaction`, using an explicitly coloured PDF pattern for consistent viewer rendering; `alternative` and `important` use flat fills; the default is a solid `color@gantt@default` bar.
2. `\dfgGanttEnvironment{`⟨*project*⟩`}`: wraps the content in a `pgfgantt ganttchart` environment spanning 16 columns (quarters Q3 of year 0 through Q2 of year 5, matching the standard DFG 5-year period). The chart is scaled with `\resizebox` to fill the full text width. A 0.9-row bar with no top shift leaves a narrow gap between the final bar and the bottom rule.

`\printPlot{{Gantt}{`⟨*project*⟩`}}` is the preferred user-facing command; `\printGantt` remains a compatibility alias. The dispatcher calls `\GanttPlot`, which wraps the chart in an `hvfloat` figure with an appropriate caption. Both `\hvFloat` calls put the caption below the chart; the second one appends optional extra text. Both variants use the strict `h` position so the chart remains directly after the work packages; the following `\FloatBarrier` closes that project-plan block.

Quarter numbering: columns 1–2 correspond to Q3–Q4 of the calendar year preceding the funding start; columns 3–16 are Q1–Q4 of years 1–4; the final 16th column is Q1–Q2 of year 5. Bar start/end values in the Gantt database should use these numbers directly.

```
% Store the stripe colour inside the PDF pattern. Some renderers paint
% stroked, uncoloured patterns black even when their fill colour is set.
% Include the resolved model and components in the pattern cache key, so
% per-chart colour changes produce a new pattern instead of reusing one.
\tl_new:N \l_@@_gantt_stripe_color_model_tl
\tl_new:N \l_@@_gantt_stripe_color_spec_tl
\pgfdeclarepattern{
  name = dfg~gantt~stripes,
  type = colored,
  parameters = {
    \l_@@_gantt_stripe_color_model_tl,
    \l_@@_gantt_stripe_color_spec_tl
  },
  bottom~left = {\pgfpoint{-6pt}{-6pt}},
  top~right = {\pgfpoint{6pt}{6pt}},
  tile~size = {\pgfpoint{12pt}{12pt}},
  tile~transformation = {\pgftransformrotate{45}},
  code = {
    \pgfsetstrokecolor{dfgGanttStripeColor}
```

```
    \pgfsetlinewidth{6pt}
    \pgfpathmoveto{\pgfpoint{-6pt}{0pt}}
    \pgfpathlineto{\pgfpoint{6pt}{0pt}}
    \pgfusepath{stroke}
  },
}
\bool_new:N \l_@@_gantt_first_bar_bool
\NewDocumentCommand{\dfgGanttContent}{m}{%
  % Resolve the active colour model before extracting the cache parameters.
  \colorlet{dfgGanttStripeColor}{color@gantt@stripe-two}
  \extractcolorspecs{dfgGanttStripeColor}
    {\l_@@_gantt_stripe_color_model_tl}{\l_@@_gantt_stripe_color_spec_tl}
  \bool_set_true:N \l_@@_gantt_first_bar_bool
  \DTLforeach*{gantt-#1}{%
   \ganttWP=id,%
   \ganttName=name,%
   \ganttDescription=description,%
   \ganttPeriodStart=periodStart,%
   \ganttPeriodEnd=periodEnd,%
   \ganttPatternbar=style%
  }{\hypersetup{linkcolor=black}%
   % If we want to omit a bar in the plot,
   % you need to set periodStart and periodEnd
   % the same value.
   \ifdefstrequal{\ganttPeriodStart}{\ganttPeriodEnd}{}{%
    % if the values are different then there will be a bar.
  \bool_if:NTF \l_@@_gantt_first_bar_bool
    { \bool_set_false:N \l_@@_gantt_first_bar_bool }
    { \ganttnewline }
  \def\ganttBarPlot{
   \sffamily
   \dfgShowTag{\ganttWP}{id}{\getWP{\ganttWP}}[gantt-#1]
   \ifdefempty{\ganttDescription}{}
     {:~\dfgShowTag{\ganttWP}{description}{\ganttDescription}[gantt-#1]}}

    \str_case:VnF {\ganttPatternbar}
    {%
     {crosshatch}{\ganttbar[bar/.append~ style = {%
         pattern       = {crosshatch},
         pattern~ color = {color@gantt@crosshatch},
         draw           = {color@gantt@frame}
        }]{\ganttBarPlot}{\ganttPeriodStart}{\ganttPeriodEnd}}%

     {striped}{\ganttbar[bar/.append~ style = {%
         fill          = {color@gantt@stripe-one},
         % the pattern needs to be applied after the filled color
              postaction    ={pattern={dfg~gantt~stripes}},
         postaction    = {draw, semithick},
         draw          = {color@gantt@frame}
        }]{\ganttBarPlot}{\ganttPeriodStart}{\ganttPeriodEnd}}%

     {alternative}{\ganttbar[bar/.append~ style = {%
         fill          = {color@gantt@alternative},
         draw          = {color@gantt@frame}
        }]{\ganttBarPlot}{\ganttPeriodStart}{\ganttPeriodEnd}}%

     {important}{\ganttbar[bar/.append~ style = {%
         fill          = {color@gantt@important},
         draw          = {color@gantt@frame}
        }]{\ganttBarPlot}{\ganttPeriodStart}{\ganttPeriodEnd}}%
    }
    { % default
     {}{\ganttbar[bar/.append~ style = {%
         fill          = {color@gantt@default},
         draw          = {color@gantt@frame}
        }]{\ganttBarPlot}{\ganttPeriodStart}{\ganttPeriodEnd}}%
    }
  }%
   \hypersetup{linkcolor=blue}}
}
```

The following macro makes it easier to debug the position of the bars.

```
\newcommand{\dfgGanttShowQs}[2]{%
  \bool_if:NTF \l_@@_class_showtags_bool
  {\leavevmode\vbox to\z@{{%
     \vss\color{color@tags@line}
     \rlap{\vrule\raise .9em%
      \hbox{{%
        \fontsize{4}{0}\selectfont%
        \ttfamily\,#1:#2}}}}}
   \textcolor{color@tags@font}{#1}}
  {#1}
}
\NewDocumentCommand{\dfgGanttEnvironment}{m}{
  % #1 : Project ID
  \resizebox{\linewidth}{!}{
```

```
12069     \begin{ganttchart}[
12070       title~ height=.8,
12071       title/.append~ style={draw=none, fill=tableHeadingsBG},
12072       dfgTitleTopRule/.style={
12073        title/.append~ style={
12074         path~ picture={
12075          \draw[draw=tableOuterLines,line~ width=2pt]
12076           (path~ picture~ bounding~ box.north~ west) --
12077           (path~ picture~ bounding~ box.north~ east);
12078         }}},
12079       dfgTitleBottomRule/.style={
12080        title/.append~ style={
12081         path~ picture={
12082          \draw[draw=tableOuterLines,line~ width=2pt]
12083           (path~ picture~ bounding~ box.south~ west) --
12084           (path~ picture~ bounding~ box.south~ east);
12085         }}},
12086       dfgTitleBottomRightRule/.style={
12087        title/.append~ style={
12088         path~ picture={
12089          \draw[draw=tableOuterLines,line~ width=2pt]
12090           (path~ picture~ bounding~ box.south~ west) --
12091           (path~ picture~ bounding~ box.south~ east);
12092          \draw[draw=tableOuterLines,line~ width=2pt]
12093           (path~ picture~ bounding~ box.north~ east) --
12094           (path~ picture~ bounding~ box.south~ east);
12095         }}},
12096       title~ label~ font={\bfseries\footnotesize\color{tableHeadingsFont}},
12097       canvas/.style={
12098         shape=rectangle,
12099         draw=none,
12100         fill=white,
12101         path~ picture={
12102          \draw[draw=tableOuterLines,line~ width=2pt]
12103           (path~ picture~ bounding~ box.north~ west) --
12104           (path~ picture~ bounding~ box.north~ east);
12105          \draw[draw=tableOuterLines,line~ width=2pt]
12106           (path~ picture~ bounding~ box.south~ west) --
12107           (path~ picture~ bounding~ box.south~ east);
12108         }},
12109       x~ unit = \textwidth/16,
12110       y~ unit~ title=0.5cm,
12111       y~ unit~ chart=.5cm,
12112     vgrid={*1{tableOuterLines,dashed}},
12113       %hgrid,
12114       %   group label font=\bfseries\footnotesize,
12115       %   milestone label font=\scriptsize,
12116       bar~ inline~ label~ node/.style={anchor=west},
12117       bar~ inline~ label~ anchor=west,
12118       bar~ label~ font={\scriptsize\color{tableHeadingsFont}},
12119       %  title label font=\bfseries\scriptsize\color{white},
12120       % bar/.style={rounded~ corners=1mm},
12121     bar~ top~ shift=0,
12122     bar~ height=.9,
12123       inline
12124       % NOTE: {1}{16} = 16 quarters = 5 years (2 + 4 + 4 + 4 + 2 quarters).
12125       % This is hardcoded. To adapt to a different funding period length,
12126       % update the ganttchart range, the \gantttitle blocks below,
12127       % and the fundingYears-related fields in the meta keyset.
12128      ]{1}{16}
12129    \gantttitle[dfgTitleTopRule]{\dfgFundingYear{1}}{2}
12130    \gantttitle[dfgTitleTopRule]{\dfgFundingYear{2}}{4}
12131    \gantttitle[dfgTitleTopRule]{\dfgFundingYear{3}}{4}
12132    \gantttitle[dfgTitleTopRule]{\dfgFundingYear{4}}{4}
12133    \gantttitle[dfgTitleTopRule]{\dfgFundingYear{5}}{2}\\
12134      %====================
12135      % \gantttitle{\crcTMPganttShowQs{Q1}{1}}{1}
12136      % \gantttitle{\crcTMPganttShowQs{Q2}{2}}{1}
12137      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q3}{1}}{1}
12138      \gantttitle[dfgTitleBottomRightRule]{\dfgGanttShowQs{Q4}{2}}{1}
12139      %====================
12140      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q1}{3}}{1}
12141      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q2}{4}}{1}
12142      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q3}{5}}{1}
12143      \gantttitle[dfgTitleBottomRightRule]{\dfgGanttShowQs{Q4}{6}}{1}
12144      % ====================
12145      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q1}{7}}{1}
12146      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q2}{8}}{1}
12147      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q3}{9}}{1}
12148      \gantttitle[dfgTitleBottomRightRule]{\dfgGanttShowQs{Q4}{10}}{1}
12149      % ====================
12150      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q1}{11}}{1}
12151      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q2}{12}}{1}
12152      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q3}{13}}{1}
12153      \gantttitle[dfgTitleBottomRightRule]{\dfgGanttShowQs{Q4}{14}}{1}
12154      % ====================
12155      \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q1}{15}}{1}
```

```
12156       \gantttitle[dfgTitleBottomRule]{\dfgGanttShowQs{Q2}{16}}{1}
12157       %\gantttitle{\dfgGanttShowQs{Q3}{19}}{1}
12158       %\gantttitle{\dfgGanttShowQs{Q4}{20}}{1}
12159       \\
12160       \dfgGanttContent{#1}
12161     \end{ganttchart}%
12162    }
12163  }
```

6.1.27 TOC entry and work-package helpers

`\dfgTOCprojectPart{⟨letter⟩}{⟨title⟩}` adds a bold chapter-level entry to the table of contents coloured in the project colour (`projectA` for letter A, etc.). Call it immediately before the project's `\chapter` command.

```
12164  \NewDocumentCommand{\dfgTOCprojectPart}{m O{Project~area} m}{
12165    % #1: project       (kann leer sein)
12166    % #2: pre-freetext  (default: "Project area")
12167    % #3: free text     (kann leer sein)
12168    \bool_if:NF \l_@@_class_fastcompile_project_bool
12169    {
12170      \addtocentrydefault{chapter}
12171        { \tl_if_blank:nF {#1} { \texorpdfstring{\getProjectColor{#1}#1}{} } }
12172      {
12173        #2\c_space_tl
12174        \tl_if_blank:nF {#1} { \texorpdfstring{\getProjectColor{#1}#1}{#1} ~--~ #3 }
12175      }
12176    }
12177  }
```

6.1.28 CV / Curriculum Vitae

The CV section produces structured curricula vitae in the DFG-prescribed format for each project leader. The same document class is used for both the integrated CV pages inside the main proposal and the standalone `crc-cv.tex` document (identified by `cv=true`).

Structure per PI:

1. `\dfgCVheader{⟨id⟩}` — chapter heading with name, academic title and (if present) a QR code linking to the PI's homepage.
2. `\dfgCVmandatory{⟨id⟩}` — qualification and career table built from all `cv` entries with `cv-type=mandatory`.
3. `\dfgCVoptional{⟨id⟩}` — the five supplementary sections (supplement, activities, supervision, distinction, other), each rendered only when at least one matching entry exists.
4. `\dfgCVpublications{⟨id⟩}` — the ten most important publications printed via biblatex; skippable with `skip10pub=true`.
5. `\dfgCVdataprivacy` — the mandatory GDPR data-privacy statement.

`\dfgCV{⟨id⟩}` is a convenience wrapper that calls all five steps in order and manages the bold-author activations around the publications.

Creating a boolean for detecting optional content (used to conditionally include the privacy statement when optional sections exist).

```
12178  \bool_new:N \l_@@_cvoptional_bool
```

CV commands

`\dfgCVheader` Renders the standardised CV heading for one PI. The chapter title uses a two-line format: the word "Curriculum Vitae" in the alternative font colour, followed by the PI's full name with academic title in bold. If the PI has a URL in the `person` database, a QR code linking to it is placed in the top-right margin on the first CV page via `\AddToHookNext{shipout/background}`. The QR code is omitted when the URL field is empty to avoid rendering an empty code.

```
12179  \bool_if:NT  \l_@@_class_cv_bool {
12180  % multicols for TOC
12181  % https://tex.stackexchange.com/a/314046/98739
12182  \BeforeStartingTOC[toc]{\begin{multicols}{2}}%
12183  \AfterStartingTOC[toc]{\end{multicols}}
12184
```

```
% no page number for chapters in TOC
\RedeclareSectionCommands[
  tocraggedpagenumber,
  toclinefill=\tocpageseparator,
  tocindent=0em,
  tocnumwidth=0em,
  tocpagenumberbox=\tocpagenumberbox,%\tocgobble,% <- added
  tocpagenumberformat=\textsf,
]{chapter}
}

\newcommand\dfgReduceSpace{\vspace*{-1.5\baselineskip}}
\NewDocumentCommand{\dfgCVheader}{ m }{%
  % #1: person-id
  \bool_set_false:N \l_@@_cvoptional_bool

  % Single scan: pull all needed fields at once
  \getEntry*[person]{#1}{
   \theAcademicDegree = academicdegree,
   \theURLValue       = url,
   \theNameValue      = lastname,
   \theFirstNameValue = firstname,
   \CurrentPosition   = position,
   \CurrentPositionText   = positionText
  }%
  \addchap[{
      head={CV
        \texorpdfstring{\dfgShowTag{#1}{academicdegree}{\theAcademicDegree}[person]}
          {\theAcademicDegree}\space
        \texorpdfstring{\dfgShowTag{#1}{firstname}{\theFirstNameValue}[person]}
          {\theFirstNameValue}\space
        \texorpdfstring{\dfgShowTag{#1}{lastname}{\theNameValue}[person]}
          {\theNameValue}},
      tocentry={
        \texorpdfstring{\dfgShowTag{#1}{lastname}{\theNameValue}[person]}
          {\theNameValue},\space
        \texorpdfstring{\dfgShowTag{#1}{firstname}{\theFirstNameValue}[person]}
          {\theFirstNameValue}}}]{%
        {\normalsize\textcolor{color@font@alternative}{Curriculum \space Vitae}\\[.5\baselineskip]
 \dfgShowTag{#1}{academicdegree}{\theAcademicDegree}[person]\\[.25\baselineskip]}
 \bfseries
       \dfgShowTag{#1}{firstname}{\theFirstNameValue}[person]\space
       \dfgShowTag{#1}{lastname}{\theNameValue}[person]
  }%
  %\pdfbookmark{CV~\theNameValue, ~~ \theFirstNameValue}{cv-#1}

  % Printing the QR code only on one page next to the headline.
  \AddToHookNext{shipout/background}{%
    \DTLifnullorempty{\theURLValue}
    {}
    {\put(\pagewidth-\marginparwidth,-.05\paperheight){%
   \dfgShowTag{#1}{url}{\qrcode[link,height=1.5cm]{\theURLValue}}[person]}}%
  }

  \begin{tblr}[
    theme   =main,
    long
   ]{
    colspec={X[t]X[4,t]},
    % $column{2}={font=\bfseries},
    columns={halign=l},
    row{odd}={bg=white},
    row{even}={bg=white},
       row{1}={font=\footnotesize\bfseries\sffamily,
               bg=tableHeadingsBG,
               fg=tableHeadingsFont}
   }
   Personal~ Data                                                  &                                                  \\
 Title                                                           & \dfgShowTag{#1}{academicdegree}{\theAcademicDegree}[p
 First~ name                                                     & \dfgShowTag{#1}{firstname}{\theFirstNameValue}[person
 Name                                                            & \dfgShowTag{#1}{lastname}{\theNameValue}[person] \\
 Current~ position                                               &
\ifdefempty{\CurrentPositionText}
  {\dfgShowTag{#1}{position}{\dfgCategoryName{\CurrentPosition}}[person]}
  {\dfgShowTag{#1}{positionText}{\CurrentPositionText}[person]}
   \\
    {Current~ institution(s)/\\ site(s),~ country} & {\printList{{Institutions}{#1}} }                \\
   ORCiD                                                           & \getORCiD{#1}                                    \\
  \end{tblr}
  \dfgReduceSpace
}
```

The next section is about `Qualifications and Career`.

```
\NewDocumentCommand{\dfgCVmandatory}{ m }{%
  % #1: person-id
    \DTLsortdata{cv-#1}
```

```
        {
         periodStart ={descending=true},
         periodEnd   ={descending=true},
         position    ={ascending=true}
          }

   \group_begin:
   \def\tblrbody{}

   % Build the institution-name prop on first use (shared with ProjectsTable)
   \@@_proj_pi_build:

   % Single scan over person table for the three fields shown above
   \getEntry*[person]{#1}{
    \theDegreeProgramme = degree-programme,
    \theDoctorateYear   = doctorateYear,
    \theDoctorate       = doctorate
   }%

   \DTLforeach*[
    \DTLiseq{\dfgrefpersonid}{#1}
    \and
    \DTLiseq{\dfgtype}{mandatory}
   ]{cv-#1}{%
    \dfgrefpersonid      = ref-person-id,
    \dfgtype             = type,
    \dfgyearstart        = periodStart,
    \dfgyearend          = periodEnd,
    \dfgpositionText     = positionText,
    \dfgposition         = position,
    \dfgrefinstitutionid = ref-institution-id,
    \dfgdescription      = description
   }{%
    % Hash lookup instead of two \DTLgetvalueforkey calls
    \prop_get:NVNTF \g_@@_inst_narrow_prop \dfgrefinstitutionid \theInstAffName
    {}
    { \tl_clear:N \theInstAffName }%

    \protected@edef\tblrbody{\tblrbody
       \ifdefempty{\dfgyearend}
         {since\space\dfgShowTag{#1}{periodStart}{\dfgyearstart}[cv-#1]}
         {\str_if_eq:eeTF { \dfgyearstart } { \dfgyearend }
         { \dfgShowTag{#1}{periodStart}{\dfgyearstart}[cv-#1] }
         { \dfgShowTag{#1}{periodStart}{\dfgyearstart}[cv-#1]--
           \dfgShowTag{#1}{periodEnd}{\dfgyearend}[cv-#1] }}
     \ifdefempty{\dfgposition}
   {}{:\space\dfgShowTag{#1}{position}{\dfgposition}[cv-#1]}
     &
   {\dfgShowTag{#1}{positionText}{\dfgpositionText}[cv-#1]
      \ifdefempty{\theInstAffName}{}{,\space\theInstAffName}%
    \ifdefempty{\dfgdescription}{}
      {\\\textit{\dfgShowTag{#1}{description}{\dfgdescription}[cv-#1]}}%
     }
     \\
    }%
   }

   \begin{tblr}[
     theme  = main,
     expand = \tblrbody,
     long
    ]{
     colspec = {X[.7,t]X[4,t]},
     columns = {halign=l},
     row{odd}={bg=white},
     row{even}={bg=white},
         row{1}={font=\footnotesize\bfseries\sffamily,
                 bg=tableHeadingsBG,
                 fg=tableHeadingsFont}
    }
    \SetCell[c=2]{l}{Qualifications~ and~ Career} &                                    \\
    \SetRow{tableHeadingsBGalt,font=\footnotesize\bfseries,halign=l}%
    Stages                       & Periods~ and~ Details                \\
 {Degree\\programme}              & \dfgShowTag{#1}{degree-programme}{\theDegreeProgramme}[person] \\
 Doctorate                     & \dfgShowTag{#1}{doctorateYear}{\theDoctorateYear}[person]\space
                                      \dfgShowTag{#1}{doctorate}{\theDoctorate}[person] \\
    \SetHline{-}{tableOuterLines,.5pt}%
    % \SetRow{tableHeadingsBGalt}%
    % \SetCell{font=\footnotesize\bfseries,halign=l}%
    % Stages~ of~ academic/professional~ career &                                  \\
    \tblrbody
   \end{tblr}
   \group_end:
   \dfgReduceSpace
 }
```

The next section is about `Supplementary Career Information`, or `Activities in the Research Syste`

or `Supervision of Researchers in Early Career Phases` since they all are technically the same.

```
\NewDocumentCommand{\dfgCVoptional}{ m m m }{%
  \def\tblrbody{}
  \DTLforeach*[
   \DTLiseq{\dfgrefpersonid}{#1}
   \and
   \DTLiseq{\dfgtype}{#2}
  ]{cv-#1}{%
   \dfgrefpersonid      = ref-person-id,
   \dfgtype             = type,
   \dfgyearstart        = periodStart,
   \dfgyearend          = periodEnd,
   \dfgpositionText     = positionText,
   \dfgposition         = position,
   \dfgrefinstitutionid = ref-institution-id,
   \dfgdescription      = description
  }{%
    % datatool represents an absent value in a numeric column by its null
    % sentinel, which expands like zero in this context.  Normalise those
    % sentinels before testing the optional period fields; description-only
    % CV entries must not acquire a spurious ``0'' in the first column.
    \DTLifnullorempty{\dfgyearstart}{\def\dfgyearstart{}}{}
    \DTLifnullorempty{\dfgyearend}{\def\dfgyearend{}}{}
   \bool_lazy_and:nnTF
     { \tl_if_empty_p:V \dfgyearstart }
     { \tl_if_empty_p:V \dfgyearend }
   {%
  \protected@edef\tblrbody{\tblrbody
        \SetCell[c=2]{l,wd=\dimexpr\textwidth-12pt\relax}
          {\dfgShowTag{#1}{description}{\dfgdescription}[cv-#1]} & \\
    }%
   }
   {%
    \protected@edef\tblrbody{\tblrbody
        \ifdefempty{\dfgyearstart}
          {
            \ifdefempty{\dfgyearend}
              {}
              {\dfgShowTag{#1}{periodEnd}{\dfgyearend}[cv-#1]\space}
          }
          {
            \ifdefempty{\dfgyearend}
              {since\space\dfgShowTag{#1}{periodStart}{\dfgyearstart}[cv-#1]\space}
              {
                \str_if_eq:eeTF { \dfgyearstart } { \dfgyearend }
                  { \dfgShowTag{#1}{periodStart}{\dfgyearstart}[cv-#1] }
                  { \dfgShowTag{#1}{periodStart}{\dfgyearstart}[cv-#1]--
                    \dfgShowTag{#1}{periodEnd}{\dfgyearend}[cv-#1] }
                \space
              }
          }
        &
        \dfgShowTag{#1}{description}{\dfgdescription}[cv-#1]
        \\
    }%
   }%
  }%

  \ifdefempty{\tblrbody}{}{%
   \begin{tblr}[
     theme  = main,
     expand = \tblrbody,
     long
       ]{
            %rowsep=3pt,
     width = \textwidth,
     colspec = {X[0.7,t]X[4,t]},
     columns = {halign=l},
     row{odd}={bg=white},
     row{even}={bg=white},
           row{1}={font=\footnotesize\bfseries\sffamily,
                   bg=tableHeadingsBG,
                   fg=tableHeadingsFont}
    }
    \SetCell[c=2]{l}{#3} & \\*
      \tblrbody
   \end{tblr}
   \bool_set_true:N \l_@@_cvoptional_bool
    \dfgReduceSpace
  }%
}
```

For the publications we define a custom bibliography template that is enumerating the items as

required.

```
\newlist{DFGlist}{enumerate}{1}
\setlist[DFGlist]{leftmargin=*}
\setlist[DFGlist,1]{
  % font=\bfseries,
  % first=\bfseries,
  % after=\normalfont,
  % itemsep=.5\baselineskip,
  % wide,
  label={\bfseries\arabic*.},
  nosep
}
```

Further on we apply this to the bibliography environment.

```
\defbibenvironment{DFGlist}
{\begin{DFGlist}\RaggedRight}
{\end{DFGlist}}
{\item}
```

The next section combines everything and wraps it. To have a reference listed in this section you have to write into the

```
\defbibheading{categoryA}{%
  \par
  \begin{tblr}[ theme=main, long ]{
    rowhead = 0,
    colspec = {X[1,t]},
    row{odd}={bg=white},
    row{even}={bg=white},
    %hline{1,2,Z} = {0pt},
            row{1}={font=\footnotesize\bfseries\sffamily,
                    bg=tableHeadingsBG,
                    halign=l,
                    fg=tableHeadingsFont},
    row{2} = {%bg=tableHeadingsBGalt, fg=tableHeadingsFont,
              font=\footnotesize\sffamily, halign=l, valign=t, guard},
   }Scientific~ results \\*
   Category~ A~ --~ Articles~ in~ peer-reviewed~ journals,~
   contributions~ to~ peer-reviewed~ conferences~ or~ to~
   anthology~ volumes,~ and~ book~ publications             \\
  \end{tblr}
  \vspace*{-1.5\baselineskip}
}

\defbibheading{categoryB}{%
  \par
  \begin{tblr}[ theme=main, long ]{
    rowhead = 0,
    colspec = {X[1,t]},
    row{odd}={bg=white},
    row{even}={bg=white},
    %hline{1,2,Z} = {0pt},
            row{1}={font=\footnotesize\bfseries\sffamily,
                    bg=tableHeadingsBG,
                    halign=l,
                    fg=tableHeadingsFont},
    row{2} = {%bg=tableHeadingsBGalt, fg=tableHeadingsFont,
              font=\footnotesize\sffamily, halign=l, valign=t, guard},
   }Scientific~ results \\*
   Category~ B~ --~ Any~ other~ form~ of~ published~ results \\
  \end{tblr}
  \vspace*{-1.5\baselineskip}
}

\NewDocumentCommand{\dfgCVpublications}{ m }{%
  % #1: person-id
  \bool_if:NTF \l_@@_class_skipTENpublications_bool
  {%
   \par
   \group_begin: \ttfamily
   Publication~ section~ for~ \getFullName{#1}~ omitted~
   for~ faster~ compilation.\\
   To~ enable~ the~ 10~ most~ important~ publications,~
   go~ to~ the~ documentclass~ options\\ and~ state~
   '\textbf{skip10pub=false}'.
   \group_end:
  }{%
   \tl_gset:Nn \g_@@_current_pi_tl {#1}
   \newrefcontext[sorting=ydnt]
   \nocite{*}%
   \datatool_if_has_key:nnT { person } { name-alias }
   {
    \DTLgetvalueforkey { \theAliases } { name-alias } { person } { id } { #1 }
```

```
12510      \DTLifnullorempty { \theAliases }
12511      { }
12512      { \exp_args:No \addboldnames { \theAliases } }
12513     }
12514
12515     \renewcommand*{\bibfont}{\footnotesize}%
12516
12517     \printbibliography[
12518      env          = DFGlist,
12519      resetnumbers,
12520      keyword      = {#1-A},
12521      heading      = categoryA
12522     ]
12523     \printbibliography[
12524      env          = DFGlist,
12525      resetnumbers,
12526      keyword      = {#1-B},
12527      heading      = categoryB
12528     ]
12529
12530     \endrefcontext
12531     \resetboldnames
12532     \tl_gclear:N \g_@@_current_pi_tl
12533    }%
12534  }
12535
12536  \tl_new:N \g_@@_current_pi_tl
```

\ifIDequal Conditional macro for use inside `.bib` files. Inside an `addendum` field, write `\ifIDequal{pi-id}{text}`; the text is only emitted when the bibliography is being rendered for that PI.

The current PI is held in `\g_@@_current_pi_tl`, which is set by `\dfgCVpublications` before each `\printbibliography` call.

The implementation deliberately uses classic LaTeX primitives (`\edef`, `\ifx`, `\csname`) rather than expl3, because `\ifIDequal` is invoked from inside a `biblatex`-rendered bibliography where `\ExplSyntaxOff` is in effect. Wrapping the variable lookup in `\csname...\endcsname` makes the name resolve correctly regardless of catcode state.

```
12537  \DeclareFieldFormat{addendum}{\mkbibitalic{#1}}
12538  \ExplSyntaxOff
12539  \protected\long\def\ifIDequal#1#2{%
12540    \if\relax\detokenize{#2}\relax
12541      % #2 is empty: do nothing
12542    \else
12543      \begingroup
12544        \edef\dfg@tempa{#1}%
12545        \edef\dfg@tempb{\csname g_@@_current_pi_tl\endcsname}%
12546        \ifx\dfg@tempa\dfg@tempb
12547          \endgroup \par\textcolor{color@font@alternative}{#2}%
12548        \else
12549          \endgroup
12550        \fi
12551    \fi
12552  }
12553  \ExplSyntaxOn
```

After each section - especially when any content was written for the optional parts, a data privacy is needed. We only need this section if there has been optional text before. We find this out by checking the bool.

```
12554  \ExplSyntaxOff
12555  \NewDocumentCommand{\dfgCVdataprivacy}{ m }{%
12556    \begin{tblr}[%
12557      theme   =main,
12558      long
12559     ]{%
12560      colspec = {
12561        X[-1,t]
12562        X[10,t]
12563       },
12564      rows = {bg=white},
12565      row{2-Z}={font=\scriptsize},
12566          row{1}={font=\footnotesize\bfseries\sffamily,
12567                      halign=l,
12568                      bg=tableHeadingsBG,
12569                      fg=tableHeadingsFont},
12570     }
12571     \SetCell[c=2]{l}
12572     Data protection and consent to the processing of optional data &   \\
```

```
    \SetCell{font=\large}\ding{55}  & I (\getFullName*{#1}) expressly consent
         to the processing of the voluntary
    (optional) information, including "special categories of personal data" in
    connection with the DFG's review and decision-making process regarding my
    proposal. This also includes forwarding my data to the external reviewers,
    committee members and, where applicable, foreign partner organisations who
    are involved in the decision-making process. To the extent that these
    recipients are located in a third country (outside the European Economic
    Area), I additionally consent to them being granted access to my data for
    the above-mentioned purposes, even though a level of data protection
    comparable to EU law may not be guaranteed. For this reason, compliance with
    the data protection principles of EU law is not guaranteed in such cases. In
    this respect, there may be a violation of my fundamental rights and freedoms
    and resulting damages. This may make it more difficult for me to assert my
     rights under the General Data Protection Regulation (e.g.\ information,
    rectification, erasure, compensation) and, if necessary, to enforce these
    rights with the help of authorities or in court.  \\[2pt]
    & I may \emph{revoke} my consent in whole or in part at any time - with effect for the
    future, freely and without giving reasons - vis-à-vis the DFG
    (\href{mailto:postmaster@dfg.de}{postmaster@dfg.de}). The lawfulness of the
    processing carried out up to that point remains unaffected. Insofar as I
    transmit \enquote{special categories of personal data} relating to third
    parties, I confirm that the necessary legitimation under data protection law
     exists (e.g.\ based on consent). \\[2pt]
       & I have taken note of the DFG's Data Protection Notice relating to research
    funding, which I can access at \url{www.dfg.de/privacy_policy} and I will forward it
    to such persons whose data the DFG processes as a result of being mentioned in
    this CV.
   \end{tblr}
 }
\ExplSyntaxOn
```

We wrap everything into one command.

```
\NewDocumentCommand{\dfgCV}{ m }{%
  % #1: person-id
 \bool_if:NF \l_@@_class_fastcompile_project_bool
 {
  \DTLifdbexists{cv-#1}{
  \StartTimer\dfgCVheader{#1}\StopTimer{#1:dfgCVheader}
  \StartTimer\dfgCVmandatory{#1}\StopTimer{#1:dfgCVmandatory}
  \StartTimer\dfgCVoptional{#1}{supplement}{Supplementary~ Career~ Information}\StopTimer{#1:dfgCVoptional|supplement
  \StartTimer\dfgCVoptional{#1}{activities}{Activities~  in~  the~  Research~ System}\StopTimer{#1:dfgCVoptional|acti
  \StartTimer\dfgCVoptional{#1}{supervision}{Supervision~  of~  Researchers~  in~  Early~  Career~  Phases}\StopTimer
  \StartTimer \dfgCVpublications{#1}\StopTimer{#1: dfgCVpublications}
  \StartTimer\dfgCVoptional{#1}{distinction}{Academic~  Distinctions}\StopTimer{#1:dfgCVoptional|distinction}
  \StartTimer\dfgCVoptional{#1}{other}{Other~ Information}\StopTimer{#1:dfgCVoptional|other}

  % We check for the bool at this place because we dont have to enable
  % \LaTeX3-Syntax within the command which would require to put |~| at each space.
  \bool_if:NT \l_@@_cvoptional_bool  {
   \StartTimer \dfgCVdataprivacy{#1} \StopTimer{#1:dfgCVdataprivacy}
  }
 }
 {\msg_warning:nne { dfg } { database-not-found }{ cv-#1 }}
  }
}
```

The CV should be printed automatically for all persons with the role `PI`.

```
\NewDocumentCommand{\dfgCVsection}{ O{person} }{%
  \bool_if:NF \l_@@_class_fastcompile_project_bool
  {
  \DTLifdbexists{#1}{%
   \DTLsort*{lastname,firstname}{#1}%
   \DTLforeach*[%start of conditions
    \DTLisinlist{\theFundingPeriodNumeral}{\period}% 1st condition
    \and%
    \DTLisinlist{PI}{\role}% 2nd condition
   ]{#1}{%
    \id             =id,
    \period         =period,
    \role           =role%
   }{%
 \exp_args:Ne \dfgCV { \id }
   }
}{
   Database \texttt{#1} missing.
   \msg_warning:nne { dfg } { database-not-found }{ #1 }
  }%
  }
}
```

Finally we load the metadata and customization provided by the user. In case a file is missing a message will be displayed to inform the user.

```
\NewDocumentCommand{\checkFile}{m}{%
    \StartTimer
  \file_if_exist:nTF {#1}
  {\ExplSyntaxOff\input{#1}\ExplSyntaxOn}
  {\msg_warning:nne { dfg } { file-not-found }{#1}}%
    \StopTimer{checkFile:#1}
}
```

There are at least some file we need or should consider checking.

```
\bool_if:NF \l_@@_class_documentation_bool
{
\bool_if:NTF \l_@@_class_fastcompile_project_bool
{
  \bool_if:NT \l_@@_class_fastcompile_project_show_images_bool
    { \checkFile{metadata/dfg-metadata.tex} }
}
{
  \checkFile{metadata/dfg-metadata.tex}
  \bool_if:NTF \l_@@_class_cv_bool
  { \checkFile{metadata/dfg-metadata-cv.tex} }
  { \msg_warning:nnn { dfg } { cv-entry-omitted }
    {metadata/dfg-metadata-cv.tex} }
  \bool_if:NF \l_@@_class_cv_bool
  { \checkFile{metadata/dfg-metadata-tables.tex} }
}
}
```

This is a gimmick but quite fun to experiment with. This is taken from https://tex.stackexchange.com/a/732406/98739. Note that the message is written into the `.log`-file, not in the terminal.

```
\def\dfgFinalComment{Well~ done~ coding!~ Happy~ TeXing.}
\directlua{
  luatexbase.add_to_callback('wrapup_run',
  function ()
  f=io.open("\jobname.log", "a")
  f:write("\dfgFinalComment")
  f:close()
  end,
  "final message"
  )
}
```

6.2 Generated files and examples

6.2.1 Main file

In this section we provide a skeleton of files to be used as a starting point for your proposal. You can adjust the content anytime. The following code block leads to a strange behaviour and in a table are ignored, it can be fixed with a minipage around the table.

```
% \DocumentMetadata{
%   pdfstandard=A-2b,
%   lang=en-EN,
%   % tagging=structure,
%   % uncompress
% }
\documentclass[
  showtags         =true,
  showtags-tables  =false,
  showmarginalia   =true,
  % fastcompile    =true,
  % fastcompile-project =true,
  % fastcompile-project-show-images=true,
  % fastcompile-project-show-publications=false,
  % skip10pub      =true,
  % colorscheme    =rgb,
  % final          =true,
  english % <-- needed here
]{dfg-crc}
```

Use `\frontmatter` before the title pages and table of contents. It selects lowercase Roman page numbers and unnumbered chapters. Use `\mainmatter` before General information: it clears the preliminary pages, restores numbered chapters, and restarts Arabic page numbering at 1. When

updating an existing proposal, also copy the page-numbering setup below into its customized proposal drivers: the update targets preserve these files. We use `includeonly` for the case to compile only one, two or n-many chapters. This speeds up the compiling time in case you are working mainly and currently on one specific chapter.

```
%% % \includeonly{
%% %   content/01_general-information,
%% %   content/02_existing-funds-and-requested-funds,
%% %   content/03_project,
%% %   content/04_bylaws-of-the-crc,
%% %   content/05_declarations
%% %}
\begin{document}
% Preliminary pages use lowercase Roman page numbers.
\frontmatter
\dfgIfFastCompileProjectF{\input{content/00_titlepage}}
% Start numbered chapters on Arabic page 1.
\mainmatter
\include{content/01_general-information}
\include{content/02_existing-funds-and-requested-funds}
\include{content/03_project}
\include{content/04_bylaws-of-the-crc}
\include{content/05_declarations}
\include{content/99_backmatter}
\end{document}
```

6.2.2 Complete proposal included in the manual

The documentation build uses the same main file and chapter inputs as the local proposal, including its shared consortium metadata and project files. Final mode removes drafting aids. The makefile compiles this generated wrapper before the manual and publishes its PDF beside the manual.

```
\PassOptionsToClass{final=true}{dfg-crc}
\input{crc-proposal.tex}
```

6.2.3 Shared research-project example

Each compact project file supplies its own ID and status to this reusable example body. All project leaders and affiliations come from shared metadata.

```
\providecommand{\dfgExampleResearchProject}[2]{%
  \printProjectHead{#1}
  \section{Summary}
  This project investigates \getEntry[project]{#1}{name}. Its project leaders
  combine experimental and analytical approaches within the shared research
  programme of \thisCRC.
  \ifstrequal{#2}{C}{%
    \section{Project progress to date}
    The preceding funding period established the experimental models and
    shared datasets. The next phase tests causal mechanisms and validates
    their relevance across the participating projects.
  }{%
    \section{Research rationale}
    Preliminary observations motivate a new testable hypothesis. The project
    establishes its experimental framework and independent validation strategy
    during the requested funding period.
  }
  \projectPublicationsHeading
  \projectNocite{FAIR2016}
  \printProjectBibliography[heading=none]
  \section{Project plan}
  \begin{aims}
    \addAim{1} Establish reproducible experimental and analytical workflows.
    \addAim{2} Validate the findings in complementary models and datasets.
  \end{aims}
  \begin{workpackages}
    \addWP{wp1}{
      name = {Experimental framework},
      description = {Establish models, controls, and quality criteria},
      periodStart = {1}, periodEnd = {7}, style = {default}
    }
    Define the controls, sampling strategy, and analysis workflow for \getAim{1}.
    \addWP{wp2}{
      name = {Validation and integration},
      description = {Validate results and integrate consortium datasets},
      periodStart = {6}, periodEnd = {16}, style = {important}
    }
    Test robustness and share the validated results for \getAim{2}.
```

```
  \end{workpackages}
  \printPlot{{Gantt}{#1}}
  \section{Relevance of sex, gender and/or diversity in the research project}
  Include relevant biological variables in sampling and analysis; document
  the rationale for each model and the limits of generalisation.
  \section{Role within the Collaborative Research Centre}
  Share methods, data, and validation resources with the linked projects.
  \printPlot{{ProjectCollaboration}{#1}}
  \section{Differentiation from other funded projects}
  The proposed work addresses the CRC-specific hypothesis and does not
  duplicate personnel or experiments supported by other grants.
  \section{Project funding}
  \addEntry[funding]{RequestedFunding}{
    ref-project-id = {#1}, funding-type = {staff}, category = {PhD},
    percentage = {65},
    quantity = {year1=1,year2=1,year3=1,year4=1,year5=1}
  }
  \addEntry[funding]{RequestedFunding}{
    ref-project-id = {#1}, funding-type = {costs}, category = {consumables},
    sum = {year1=1000,year2=2000,year3=2000,year4=2000,year5=1000}
  }
  \printTable{{RequestedFunding}{#1}}
  The doctoral researcher performs the experiments and integrates the data.
  Consumables cover sample preparation, assays, and independent validation.
  \printProjectTail
}
```

6.2.4 Complete local project index

```
\chapter{Project details}
\printThumbIndex{3}
\dfgTOCprojectPart{A}{Research Projects}
\input{content/03_project_A01}
\input{content/03_project_A02}
\input{content/03_project_A03}
\input{content/03_project_A04}
\input{content/03_project_A05}
\input{content/03_project_A06}
\dfgTOCprojectPart{B}{Research Projects}
\input{content/03_project_B01}
\input{content/03_project_B02}
\input{content/03_project_B03}
\input{content/03_project_B04}
\input{content/03_project_B05}
\input{content/03_project_B06}
\dfgTOCprojectPart{Q}{Central Projects}
\input{content/03_project_Q01}
\input{content/03_project_Q02}
\input{content/03_project_Q03}
\dfgTOCprojectPart{A}{Former project}
\input{content/03_project_A00}

\input{content/03_project_research-example}
\dfgExampleResearchProject{A03}{C}

\input{content/03_project_research-example}
\dfgExampleResearchProject{A04}{N}

\input{content/03_project_research-example}
\dfgExampleResearchProject{A05}{C}

\input{content/03_project_research-example}
\dfgExampleResearchProject{A06}{N}

\input{content/03_project_research-example}
\dfgExampleResearchProject{B02}{N}

\input{content/03_project_research-example}
\dfgExampleResearchProject{B03}{C}

\input{content/03_project_research-example}
\dfgExampleResearchProject{B04}{N}

\input{content/03_project_research-example}
\dfgExampleResearchProject{B05}{C}

\input{content/03_project_research-example}
\dfgExampleResearchProject{B06}{N}
```

6.2.5 Complete CV appendix included in the manual

Reuse the starter CV and its metadata, suppress drafting aids, and expose destinations for the manual's individual-CV previews. The header wrapper adds an invisible target after the original header has selected its page.

```
\PassOptionsToClass{final=true}{dfg-crc}
```

```
\AddToHook{class/dfg-crc/after}{%
  \NewCommandCopy{\dfgDocOriginalCVHeader}{\dfgCVheader}
  \RenewDocumentCommand{\dfgCVheader}{m}{%
    \dfgDocOriginalCVHeader{#1}%
    \hypertarget{doc-cv:#1}{}%
  }%
}
\input{crc-cv.tex}
```

6.2.6 Bibliography Files

There are at least two bibliography files in the folder `bib/`. One is for the common `bibtex`-entries, the others contain so called `strings`.

Main Bibliography File

```
@Article{FAIR2016,
   Author          = {Wilkinson, Mark D. and others},
   Title           = {The {FAIR} Guiding Principles for scientific data management and stewardship},
   JournalTitle    = {Scientific Data},
   Date            = {2016-12},
   DOI             = {10.1038/sdata.2016.18},
   Keywords        = {{FAIR}},
   langid          = {english},
   Number          = {1},
   Pages           = {160018},
   ShortJournal    = {Sci Data},
   URLDate         = {2021-05-31},
   Volume          = {3}
  }

@Comment{All ExampleCV records below are fictional template examples, not real research outputs.}

@Article{ExampleCVper1,
  Author       = {Sample, Kim},
  Title        = {Illustrative study of microbial metabolites and epithelial signalling},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per1-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper1Methods,
  Author       = {Sample, Kim},
  Title        = {Reproducible methods for microbial metabolites and epithelial signalling},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per1-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper1Data,
  Author       = {Sample, Kim},
  Title        = {Fictional demonstration dataset: microbial metabolites and epithelial signalling},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per1-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper2,
  Author       = {Example, Alex},
  Title        = {Illustrative study of multimodal data integration and reproducible analysis},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per2-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper2Methods,
  Author       = {Example, Alex},
  Title        = {Reproducible methods for multimodal data integration and reproducible analysis},
```

```
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per2-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper2Data,
  Author       = {Example, Alex},
  Title        = {Fictional demonstration dataset: multimodal data integration and reproducible analysis},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per2-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper3,
  Author       = {Meier, Hansi},
  Title        = {Illustrative study of microbial sensing and innate immune memory},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per3-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper3Methods,
  Author       = {Meier, Hansi},
  Title        = {Reproducible methods for microbial sensing and innate immune memory},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per3-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper3Data,
  Author       = {Meier, Hansi},
  Title        = {Fictional demonstration dataset: microbial sensing and innate immune memory},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per3-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper4,
  Author       = {Miller, Kimmi},
  Title        = {Illustrative study of epithelial repair and tissue regeneration},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per4-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper4Methods,
  Author       = {Miller, Kimmi},
  Title        = {Reproducible methods for epithelial repair and tissue regeneration},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per4-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper4Data,
  Author       = {Miller, Kimmi},
  Title        = {Fictional demonstration dataset: epithelial repair and tissue regeneration},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per4-B},
  Note         = {Fictional reference for template demonstration only},
```

```
  LangID       = {english}
}

@Article{ExampleCVper5,
  Author       = {Redata, Friedhelm},
  Title        = {Illustrative study of structured doctoral training and research integrity},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per5-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper5Methods,
  Author       = {Redata, Friedhelm},
  Title        = {Reproducible methods for structured doctoral training and research integrity},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per5-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper5Data,
  Author       = {Redata, Friedhelm},
  Title        = {Fictional demonstration dataset: structured doctoral training and research integrity},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per5-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper6,
  Author       = {Hartmann, Samira},
  Title        = {Illustrative study of microbial metabolic pathways and immune adaptation},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per6-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper6Methods,
  Author       = {Hartmann, Samira},
  Title        = {Reproducible methods for microbial metabolic pathways and immune adaptation},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per6-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper6Data,
  Author       = {Hartmann, Samira},
  Title        = {Fictional demonstration dataset: microbial metabolic pathways and immune adaptation},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per6-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper7,
  Author       = {Lindberg, Noah},
  Title        = {Illustrative study of spatial immune niches and quantitative imaging},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per7-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper7Methods,
  Author       = {Lindberg, Noah},
```

```
  Title        = {Reproducible methods for spatial immune niches and quantitative imaging},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per7-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper7Data,
  Author       = {Lindberg, Noah},
  Title        = {Fictional demonstration dataset: spatial immune niches and quantitative imaging},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per7-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper8,
  Author       = {Winters-Scholz, Alexis-Marie},
  Title        = {Illustrative study of dietary perturbations and mucosal microbial communities},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per8-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper8Methods,
  Author       = {Winters-Scholz, Alexis-Marie},
  Title        = {Reproducible methods for dietary perturbations and mucosal microbial communities},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per8-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper8Data,
  Author       = {Winters-Scholz, Alexis-Marie},
  Title        = {Fictional demonstration dataset: dietary perturbations and mucosal microbial communities},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per8-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper9,
  Author       = {Navarro, Leonie},
  Title        = {Illustrative study of nutrient sensing and predictive metabolic models},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2025},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per9-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Article{ExampleCVper9Methods,
  Author       = {Navarro, Leonie},
  Title        = {Reproducible methods for nutrient sensing and predictive metabolic models},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per9-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper9Data,
  Author       = {Navarro, Leonie},
  Title        = {Fictional demonstration dataset: nutrient sensing and predictive metabolic models},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per9-B},
```

```
13166   Note         = {Fictional reference for template demonstration only},
13167   LangID       = {english}
13168 }
13169
13170 @Article{ExampleCVper10,
13171   Author       = {Okafor, Daniel},
13172   Title        = {Illustrative study of microbial biomarkers and clinical tissue responses},
13173   JournalTitle = {Fictional Journal of Consortium Research},
13174   Date         = {2025},
13175   Volume       = {1},
13176   Pages        = {1--8},
13177   Keywords     = {per10-A},
13178   Note         = {Fictional reference for template demonstration only},
13179   LangID       = {english}
13180 }
13181
13182 @Article{ExampleCVper10Methods,
13183   Author       = {Okafor, Daniel},
13184   Title        = {Reproducible methods for microbial biomarkers and clinical tissue responses},
13185   JournalTitle = {Fictional Journal of Consortium Research},
13186   Date         = {2024},
13187   Volume       = {1},
13188   Pages        = {1--8},
13189   Keywords     = {per10-A},
13190   Note         = {Fictional reference for template demonstration only},
13191   LangID       = {english}
13192 }
13193
13194 @Dataset{ExampleCVper10Data,
13195   Author       = {Okafor, Daniel},
13196   Title        = {Fictional demonstration dataset: microbial biomarkers and clinical tissue responses},
13197   Date         = {2025},
13198   Version      = {1.0},
13199   Publisher    = {Example Research Repository},
13200   Keywords     = {per10-B},
13201   Note         = {Fictional reference for template demonstration only},
13202   LangID       = {english}
13203 }
13204
13205 @Article{ExampleCVper11,
13206   Author       = {Sato, Mira},
13207   Title        = {Illustrative study of inflammatory circuits between intestine and liver},
13208   JournalTitle = {Fictional Journal of Consortium Research},
13209   Date         = {2025},
13210   Volume       = {1},
13211   Pages        = {1--8},
13212   Keywords     = {per11-A},
13213   Note         = {Fictional reference for template demonstration only},
13214   LangID       = {english}
13215 }
13216
13217 @Article{ExampleCVper11Methods,
13218   Author       = {Sato, Mira},
13219   Title        = {Reproducible methods for inflammatory circuits between intestine and liver},
13220   JournalTitle = {Fictional Journal of Consortium Research},
13221   Date         = {2024},
13222   Volume       = {1},
13223   Pages        = {1--8},
13224   Keywords     = {per11-A},
13225   Note         = {Fictional reference for template demonstration only},
13226   LangID       = {english}
13227 }
13228
13229 @Dataset{ExampleCVper11Data,
13230   Author       = {Sato, Mira},
13231   Title        = {Fictional demonstration dataset: inflammatory circuits between intestine and liver},
13232   Date         = {2025},
13233   Version      = {1.0},
13234   Publisher    = {Example Research Repository},
13235   Keywords     = {per11-B},
13236   Note         = {Fictional reference for template demonstration only},
13237   LangID       = {english}
13238 }
13239
13240 @Article{ExampleCVper12,
13241   Author       = {Keller, Robin},
13242   Title        = {Illustrative study of patient-derived models and biomarker translation},
13243   JournalTitle = {Fictional Journal of Consortium Research},
13244   Date         = {2025},
13245   Volume       = {1},
13246   Pages        = {1--8},
13247   Keywords     = {per12-A},
13248   Note         = {Fictional reference for template demonstration only},
13249   LangID       = {english}
13250 }
13251
13252 @Article{ExampleCVper12Methods,
```

```
  Author       = {Keller, Robin},
  Title        = {Reproducible methods for patient-derived models and biomarker translation},
  JournalTitle = {Fictional Journal of Consortium Research},
  Date         = {2024},
  Volume       = {1},
  Pages        = {1--8},
  Keywords     = {per12-A},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}

@Dataset{ExampleCVper12Data,
  Author       = {Keller, Robin},
  Title        = {Fictional demonstration dataset: patient-derived models and biomarker translation},
  Date         = {2025},
  Version      = {1.0},
  Publisher    = {Example Research Repository},
  Keywords     = {per12-B},
  Note         = {Fictional reference for template demonstration only},
  LangID       = {english}
}
```

Bibliography Strings

The `strings` are used for quick and consistent referencing parts repeatedly.

```
@String {equal      = {\textit{*equal contribution}}}
@String {sharedlast = {\textit{*shared last authorship}}}
```

6.2.7 Customization File

In the file `preamble/dfg-customization.tex` you can adjust settings of this template. Further information you find in section 3.2. If you want to create new commands put them in this file as well.

```
% \title{<Title>}
% \subtitle{<Subtitle>}
%% \changeSettings{
%% dfgLocation                 ={<Location>},
%% dfgDate                     ={<date>},
%% dfgTitle                    ={<Proposal title>},
%% InfoTeX                     ={<Compiler information>},
%% font                        =<nn>,
%% crcLogoPath                 =<file path>,
%% affiliationLogoPath         =<file path>,
%% doiAPIMail                  =<contact email for optional DOI API lookup>,
%% projectPIOncePerSection     =true,
%% projectsTableInstitutionAbbreviations=false,
%% salary@prof                 =126600,
%% salary@groupLeader          =105000,
%% salary@postDoc              =90600,
%% salary@medDoc               =112200,
%% salary@PhD                  =83700,
%% salary@research             =69000,
%% salary@nonResearch          =61500,
%% salary@student              =1,
%% salary@assistant            =1,
%% animalHousing@mouse         =0.75,
%% animalHousing@mouseGnotobiotic=3.9,
%% animalHousing@rat           =1.4,
%% thumbindex                  =true,
%% thumbindex@pattern          =<int>,
%% thumbindex@orientation      =outwards, % or inwards
%% thumbindex@topmargin        =auto,
%% thumbindex@bottommargin     =auto,
%% emphasizeAuthors            =true,
%% emphasizeEntries            =false,
%% biblatexstyleCV             =<nn>,
%% biblatexstyleFeatureCV      ={<nn>,<nn>},
%% biblatexstyleProposal       =<nn>,
%% biblatexstyleFeatureProposal={<nn>,<nn>},
%%%%%   color section   %%%%%
%% color@reference@boldAuthor =<nn>,
%% color@reference@PI         =black, % or current / a named color
%% color@reference@WP         =<nn>,
%% color@reference@Aim        =<nn>,
%% color@table@headingsBG      =<nn>,
%% color@table@headingsBGalt   =<nn>,
%% color@table@headingsFont    =<nn>,
%% color@table@rows            =<nn>,
%% color@table@lines           =<nn>,
%% color@table@outerLines      =<nn>,
```

```
%% color@table@noValue         =<nn>,
%% color@table@firstColumn     =<nn>,
%% color@table@link            =<nn>,
%% color@table@contFootHead    =<nn>,
%% color@table@caption         =<nn>,
%% color@gantt@frame           =<nn>,
%% color@gantt@default         =<nn>,
%% color@gantt@crosshatch      =<nn>,
%% color@gantt@important       =<nn>,
%% color@gantt@stripe-one      =<nn>,
%% color@gantt@stripe-two      =<nn>,
%% color@thumbindex@bg         =<nn>,
%% color@thumbindex@font       =<nn>,
%% color@tags@font             =<nn>,
%% color@tags@line             =<nn>,
%% color@font@alternative      =<nn>,
%% color@item@i                =<nn>,
%% }
```

6.2.8 DFG salary and animal-housing defaults

The DFG specifies standardised all-in salary costs (*Personalmittel*) per position category. The salary defaults use the 2027 rates. These constants are used in the requested-funding staff tables to calculate yearly personnel costs from FTE counts.

The values are updated annually and published by the DFG. Their canonical defaults are keys in the `dfg/custom` key family. Override them together with the other proposal settings in `dfg-customization.tex`:

Overriding salary and animal-housing rates

```
%   \changeSettings{
%     salary@postDoc      = 90000,
%     animalHousing@mouse = 0.80,
%   }
%
```

Source: https://www.dfg.de/resource/blob/408716/60-12-2027-de.pdf (DFG personnel rates for 2027, form 60.12, version 01/27).

Use `\getValue` with the setting key, for example `\num{\getValue{animalHousing@mouse}}`, to read these rates in text. The expandable `\dfgSalaryProf`, `\dfgSalaryGroupLeader`, `\dfgSalaryPostDoc`, `\dfgSalaryMedDoc`, `\dfgSalaryPhD`, `\dfgSalaryResearch`, `\dfgSalaryNonResearch`, `\dfgSalaryStudent`, `\dfgSalaryAssistant`, `\dfgMouseHousing`, `\dfgMouseHousingGnotobiotic`, and `\dfgRatHousing` accessors remain available for compatibility. They expand to the corresponding setting.

6.2.9 Shared proposal and documentation metadata

The manual, local proposal, and CV use one fictional consortium: six active A projects, six active B projects, three Q projects, twelve PIs, and one former project. Person/project references are reciprocal; collaboration targets exist. The developer installer extracts these same records into local/metadata/.

```
⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{meta}{
  id                   = {crc}, %<--- id (do not change)
  name                 = {Mechanisms of Host--Microbiome Communication},
  nameAddon            = {From Molecular Interactions to Organ Function},
  number               = {9999},
  fundingPeriodNumeral = {3}, % 1|2|3
  ref-affiliation-id   = {aff1}, % or {aff1,aff2} for two applicants
  fundedSince          = {2019-07-01},
  fundingYears         = {2027|2,2028,2029,2030,2031|1},
  fundingYearsPrevious = {2022|2,2023,2024,2025,2026|1}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{institution}{
  id                 = {ins1},
  name               = {Institute of Microbial Systems Medicine},
  abbreviation       = {IMSM},
```

```
  ref-affiliation-id = {aff1}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{institution}{
  id                = {ins2},
  name              = {Department of Translational Biosciences},
  abbreviation      = {DTB},
  ref-affiliation-id = {aff2}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{affiliation}{
  id           = {aff1},
  name         = {Example University},
  abbreviation = {EU},
  addressStreet = {University Avenue 1},
  addressZIP   = {12345},
  location     = {Example City},
  country      = {Germany}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{affiliation}{
  id           = {aff2},
  name         = {Partner University},
  abbreviation = {PU},
  addressStreet = {Research Road 2},
  addressZIP   = {54321},
  location     = {Partner City},
  country      = {Germany}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-pis⟩
\addEntry{person}{
  id                  = {per1},
  firstname           = {Kim},
  lastname            = {Sample},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {1,2,3},
  role                = {PI},
  gender              = {d},
  ref-institution-id  = {ins1,ins2},
  ref-project-id      = {A01,A02},
  name-alias          = {
    {Kim Sample},
    {Sample, Kim S.}
  },
  orcid               = {0000-0000-0000-0001},
  doctorateYear       = {2000},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Biology, Example University},
  position            = {prof-W3},
  positionText        = {Professor and Director},
  url                 = {https://www.example.org/kim-sample},
  phone               = {+49 000 100-1000},
  email               = {kim.sample@example.org},
  contract-information = {
    C1  = {no},
    C2  = {yes},
    C2a = {2028-12-31},
    C2b = {2029-01-03},
    C3  = {yes},
    C3a = {2054-03-10},
    C4  = {no},
    C4a = {DFG}
  }
}
⟨/meta | doc-table-a | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{person}{
  id                  = {per2},
  period              = {2,3},
  firstname           = {Alex},
  lastname            = {Example},
  academicdegree      = {Prof. Dr. phil.},
  position            = {groupLeader},
  positionText        = {Research group leader},
  role                = {PI,steering-committee-member},
  doctorateYear       = {1999},
  doctorate           = {Dr. phil., Partner University},
  degree-programme    = {Data Science, Partner University},
```

```
13447   gender               = {m},
13448   ref-institution-id   = {ins2},
13449   ref-project-id       = {A01,B01,Q01},
13450   orcid                = {0000-0000-0000-0002},
13451   phone                = {+49 000 200-2000},
13452   email                = {alex.example@example.org},
13453   contract-information = {
13454     C1  = {no},
13455     C2  = {no},
13456     C2a = {},
13457     C2b = {},
13458     C3  = {no},
13459     C3a = {},
13460     C4  = {yes},
13461     C4a = {BMFTR}
13462   }
13463 }
13464 ⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
13465
13466 ⟨*meta | doc-table-b | doc-table-q | doc-table-pis⟩
13467 \addEntry{person}{
13468   id                   = {per3},
13469   period               = {2,3},
13470   firstname            = {Hansi},
13471   lastname             = {Meier},
13472   academicdegree       = {Dr. rer. nat.},
13473   position             = {postDoc},
13474   positionText         = {Senior research scientist},
13475   role                 = {PI,steering-committee-member},
13476   doctorateYear        = {1999},
13477   doctorate            = {Dr. rer. nat., Partner University},
13478   degree-programme     = {Microbiology, Partner University},
13479   gender               = {f},
13480   ref-institution-id   = {ins2},
13481   ref-project-id       = {B01,B03,Q03},
13482   orcid                = {0000-0000-0000-0003},
13483   phone                = {+49 000 200-3000},
13484   email                = {hansi.meier@example.org},
13485   contract-information = {
13486     C1  = {yes},
13487     C2  = {no},
13488     C2a = {},
13489     C2b = {},
13490     C3  = {no},
13491     C3a = {},
13492     C4  = {yes},
13493     C4a = {DFG}
13494   }
13495 }
13496 ⟨/meta | doc-table-b | doc-table-q | doc-table-pis⟩
13497
13498 ⟨*meta | doc-table-b | doc-table-pis⟩
13499 \addEntry{person}{
13500   id                   = {per4},
13501   period               = {2,3},
13502   firstname            = {Kimmi},
13503   lastname             = {Miller},
13504   academicdegree       = {Prof. Dr. rer. nat.},
13505   position             = {prof-W2},
13506   positionText         = {Professor of Molecular Medicine},
13507   role                 = {PI},
13508   doctorateYear        = {1999},
13509   doctorate            = {Dr. rer. nat., Partner University},
13510   degree-programme     = {Molecular Medicine, Partner University},
13511   gender               = {f},
13512   ref-institution-id   = {ins2},
13513   ref-project-id       = {B02,B04},
13514   orcid                = {0000-0000-0000-0004},
13515   phone                = {+49 000 200-4000},
13516   email                = {kimmi.mueller@example.org},
13517   contract-information = {
13518     C1  = {yes},
13519     C2  = {no},
13520     C2a = {},
13521     C2b = {},
13522     C3  = {no},
13523     C3a = {},
13524     C4  = {yes},
13525     C4a = {DFG}
13526   }
13527 }
13528 ⟨/meta | doc-table-b | doc-table-pis⟩
13529
13530 ⟨*meta | doc-table-b | doc-table-q | doc-table-pis⟩
13531 \addEntry{person}{
13532   id                   = {per5},
13533   period               = {3},
```

```
13534   firstname            = {Friedhelm},
13535   lastname             = {Redata},
13536   academicdegree       = {Prof. Dr.-Ing.},
13537   orcid                = {0000-0000-0000-0005},
13538   position             = {prof-W2},
13539   positionText         = {Professor of Research Training},
13540   role                 = {PI,steering-committee-member},
13541   doctorateYear        = {1998},
13542   doctorate            = {Dr.-Ing., Example University},
13543   degree-programme     = {Biomedical Engineering, Example University},
13544   gender               = {m},
13545   ref-institution-id   = {ins1},
13546   ref-project-id       = {B05,Q02},
13547   phone                = {+49 000 100-5000},
13548   email                = {friedhelm.redata@example.org},
13549   contract-information = {
13550     C1  = {yes},
13551     C2  = {yes},
13552     C2a = {2028-12-31},
13553     C2b = {2029-01-03},
13554     C3  = {yes},
13555     C3a = {2054-03-10},
13556     C4  = {no},
13557     C4a = {DFG}
13558   }
13559 }
13560 ⟨/meta | doc-table-b | doc-table-q | doc-table-pis⟩
13561
13562 ⟨*meta | doc-table-a | doc-table-pis⟩
13563 \addEntry{project}{
13564   id                 = {A01},
13565   project-area       = {A},
13566   number             = {A01},
13567   project-status     = {C},
13568   consider-for-lists = {true},
13569   reference-only     = {false},
13570   name               = {Microbial signalling across epithelial barriers},
13571   project-type       = {research},
13572   research-area      = {2.21-02},
13573   ref-person-id      = {per1,per2},
13574   collaborations     = {A02,A04,A06,B01,B03,B06,Q01},
13575   legal-issues       = {
13576       L1  = no,
13577       L1a = no,
13578       L2  = no,
13579       L2a = no,
13580       L3  = no,
13581       L4  = no,
13582       L5  = no,
13583       L5a = no,
13584       L6  = no,
13585       L7  = no,
13586       L7a = no
13587     }
13588 }
13589 ⟨/meta | doc-table-a | doc-table-pis⟩
13590
13591 ⟨*meta | doc-table-a | doc-table-pis⟩
13592 \addEntry{project}{
13593   id                 = {A02},
13594   project-area       = {A},
13595   number             = {A02},
13596   project-status     = {N},
13597   consider-for-lists = {true},
13598   reference-only     = {false},
13599   project-type       = {research},
13600   name               = {Metabolic control of tissue-resident immunity},
13601   research-area      = {2.21-05},
13602   ref-person-id      = {per1,per6},
13603   collaborations     = {A01,A03,B02,Q02},
13604   legal-issues       = {
13605     L1  = no,
13606     L1a = no,
13607     L2  = yes,
13608     L2a = no,
13609     L3  = yes,
13610     L4  = yes,
13611     L5  = no,
13612     L5a = no,
13613     L6  = yes,
13614     L7  = no,
13615     L7a = yes
13616   }
13617 }
13618 ⟨/meta | doc-table-a | doc-table-pis⟩
13619
13620 ⟨*meta⟩
```

```
\addEntry{project}{
  id                = {A00},
  project-area      = {A},
  number            = {A00},
  project-status    = {E},
  consider-for-lists = {false},
  reference-only    = {true},
  project-type      = {research},
  name              = {Former project on microbial community dynamics},
  research-area     = {},
  ref-person-id     = {per4,per10},
  collaborations    = {}
}
```

⟨/meta⟩

⟨∗meta | doc-table-b | doc-table-pis⟩

```
\addEntry{project}{
  id                = {B01},
  project-area      = {B},
  number            = {B01},
  project-status    = {C},
  consider-for-lists = {true},
  reference-only    = {false},
  project-type      = {research},
  research-area     = {2.21-03,2.21-05,2.22-15},
  name              = {Host defence during microbiome perturbation},
  ref-person-id     = {per3,per2},
  collaborations    = {A01,B02,B03,Q02},
  legal-issues      = {
    L1  = no,
    L1a = no,
    L2  = no,
    L2a = no,
    L3  = no,
    L4  = no,
    L5  = no,
    L5a = no,
    L6  = no,
    L7  = no,
    L7a = no
  }
}
```

⟨/meta | doc-table-b | doc-table-pis⟩

⟨∗meta | doc-table-b | doc-table-pis⟩

```
\addEntry{project}{
  id                = {B02},
  project-area      = {B},
  number            = {B02},
  project-status    = {N},
  consider-for-lists = {true},
  reference-only    = {false},
  project-type      = {research},
  research-area     = {2.22-15},
  name              = {Molecular signatures of tissue regeneration},
  ref-person-id     = {per4,per10},
  collaborations    = {A02,B01,B04,Q01},
  legal-issues      = {
      L1  = yes,
      L1a = no,
      L2  = yes,
      L2a = yes,
      L3  = yes,
      L4  = yes,
      L5  = yes,
      L5a = no,
      L6  = yes,
      L7  = yes,
      L7a = no
    }
}
```

⟨/meta | doc-table-b | doc-table-pis⟩

⟨∗meta | doc-table-q | doc-table-pis⟩

```
\addEntry{project}{
  id                = {Q01},
  project-area      = {Q},
  number            = {Q01},
  project-status    = {C},
  consider-for-lists = {true},
  reference-only    = {false},
  project-type      = {service},
  name              = {Integrated imaging and analysis services},
  research-area     = {2.21-02,2.22-15},
  ref-person-id     = {per2,per7},
  collaborations    = {A01,A03,A04,A06,B02,B03,B05},
  legal-issues      = {
```

```
    L1  = no,
    L1a = no,
    L2  = no,
    L2a = no,
    L3  = no,
    L4  = no,
    L5  = no,
    L5a = no,
    L6  = no,
    L7  = no,
    L7a = no
  }
}
⟨/meta | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-q | doc-table-pis⟩
\addEntry{project}{
  id                = {Q02},
  project-area      = {Q},
  number            = {Q02},
  project-status    = {C},
  consider-for-lists = {true},
  reference-only    = {false},
  project-type      = {mgk},
  name              = {Integrated Research Training Group},
  research-area     = {},
  ref-person-id     = {per5,per9},
  collaborations    = {A02,A05,B01,B04,B06}
}
⟨/meta | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-q | doc-table-pis⟩
\addEntry{project}{
  id                = {Q03},
  project-area      = {Q},
  number            = {Q03},
  project-status    = {C},
  consider-for-lists = {true},
  reference-only    = {false},
  project-type      = {admin},
  name              = {Central coordination and data stewardship},
  research-area     = {},
  ref-person-id     = {per3,per12},
  collaborations    = {A01,A02,A03,A04,A05,A06,B01,B02,B03,B04,B05,B06,Q01,Q02}
}
⟨/meta | doc-table-q | doc-table-pis⟩

⟨*meta⟩
\addEntry{person}{
  id                = {spokesperson},
  role              = {spokesperson,steering-committee-member},
  period            = {1,2,3},
  lastname          = {Sample},
  firstname         = {Morgan},
  academicdegree    = {Prof. Dr. med.},
  gender            = {f},
  position          = {prof-W3},
  positionText      = {CRC spokesperson and professor of systems medicine},
  doctorateYear     = {1998},
  doctorate         = {Dr. med., Example University},
  degree-programme  = {Medicine, Example University},
  ref-institution-id = {ins1},
  ref-affiliation-id = {aff1},
  orcid             = {0000-0000-0000-0006},
  phone             = {+49 000 100-6000},
  email             = {morgan.sample@example.org}
}
⟨/meta⟩

⟨*meta⟩
\addEntry{person}{
  id                = {man1},
  role              = {manager},
  period            = {2,3},
  firstname         = {Jamie},
  lastname          = {Coordinator},
  academicdegree    = {Dr. rer. nat.},
  gender            = {d},
  position          = {nonResearch},
  positionText      = {Scientific manager},
  doctorateYear     = {2012},
  doctorate         = {Dr. rer. nat., Partner University},
  degree-programme  = {Life Sciences, Partner University},
  ref-institution-id = {ins2},
  ref-affiliation-id = {aff2},
  orcid             = {0000-0000-0000-0007},
  phone             = {+49 000 200-7000},
```

```
  email              = {jamie.coordinator@example.org}
}
⟨/meta⟩

⟨*meta⟩
\addEntry{person}{
  id                 = {rector},
  role               = {rector},
  period             = {3},
  lastname           = {Rector},
  firstname          = {Robin},
  academicdegree     = {Prof. Dr.},
  gender             = {d},
  position           = {prof},
  positionText       = {Rector of Example University},
  ref-affiliation-id = {aff1},
  phone              = {+49 000 100-8000},
  email              = {rector@example.org}
}
⟨/meta⟩

⟨*meta⟩
\addEntry{person}{
  id                 = {rector2},
  role               = {rector},
  period             = {3},
  lastname           = {President},
  firstname          = {Casey},
  academicdegree     = {Prof. Dr.},
  gender             = {d},
  position           = {prof},
  positionText       = {President of Partner University},
  ref-affiliation-id = {aff2},
  phone              = {+49 000 200-8000},
  email              = {president@partner.example.org}
}
⟨/meta⟩

⟨*meta | doc-table-a | doc-table-q | doc-table-pis⟩
\addEntry{affiliation}{
  id                  = {aff3},
  name                = {Northbridge University},
  abbreviation        = {NU},
  addressStreet       = {Science Avenue 3},
  addressZIP          = {34567},
  location            = {Northbridge},
  country             = {Germany}
}
⟨/meta | doc-table-a | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{affiliation}{
  id                  = {aff4},
  name                = {Westhaven Research Centre},
  abbreviation        = {WRC},
  addressStreet       = {Science Avenue 3},
  addressZIP          = {34567},
  location            = {Westhaven},
  country             = {Germany}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-q | doc-table-pis⟩
\addEntry{institution}{
  id                  = {ins3},
  name                = {Centre for Computational Microbiology},
  abbreviation        = {CCM},
  ref-affiliation-id  = {aff3}
}
⟨/meta | doc-table-a | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-q | doc-table-pis⟩
\addEntry{institution}{
  id                  = {ins4},
  name                = {Department of Experimental Immunology},
  abbreviation        = {DEI},
  ref-affiliation-id  = {aff3}
}
⟨/meta | doc-table-a | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{institution}{
  id                  = {ins5},
  name                = {Institute of Metabolic Medicine},
  abbreviation        = {IMM},
  ref-affiliation-id  = {aff4}
}
```

```
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩
\addEntry{institution}{
  id                  = {ins6},
  name                = {Centre for Clinical Translation},
  abbreviation        = {CCT},
  ref-affiliation-id  = {aff4}
}
⟨/meta | doc-table-a | doc-table-b | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-pis⟩
\addEntry{person}{
  id                  = {per6},
  orcid               = {0000-0000-0000-0016},
  firstname           = {Samira},
  lastname            = {Hartmann},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI,steering-committee-member},
  gender              = {f},
  ref-institution-id  = {ins3},
  ref-project-id      = {A02,A03},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0006},
  email               = {samira@example.org}
}
⟨/meta | doc-table-a | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-q | doc-table-pis⟩
\addEntry{person}{
  id                  = {per7},
  firstname           = {Noah},
  lastname            = {Lindberg},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {m},
  ref-institution-id  = {ins3,ins4},
  ref-project-id      = {A03,A04,Q01},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0007},
  email               = {noah@example.org}
}
⟨/meta | doc-table-a | doc-table-q | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-pis⟩
\addEntry{person}{
  id                  = {per8},
  firstname           = {Alexis-Marie},
  lastname            = {Winters-Scholz},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {d},
  ref-institution-id  = {ins4},
  ref-project-id      = {A04,A05},
  position            = {prof-W2},
  positionText        = {Professor of Systems Medicine},
  doctorateYear       = {2010},
  doctorate           = {Dr. rer. nat., Example University},
  degree-programme    = {Life Sciences, Example University},
  phone               = {+49 000 300-0008},
  email               = {alexis-marie@example.org}
}
⟨/meta | doc-table-a | doc-table-pis⟩

⟨*meta | doc-table-a | doc-table-q | doc-table-pis⟩
\addEntry{person}{
  id                  = {per9},
  firstname           = {Leonie},
  lastname            = {Navarro},
  academicdegree      = {Prof. Dr. rer. nat.},
  period              = {3},
  role                = {PI},
  gender              = {f},
  ref-institution-id  = {ins5},
  ref-project-id      = {A05,A06,Q02},
  position            = {prof-W2},
```

```
13969   positionText        = {Professor of Systems Medicine},
13970   doctorateYear       = {2010},
13971   doctorate           = {Dr. rer. nat., Example University},
13972   degree-programme    = {Life Sciences, Example University},
13973   phone               = {+49 000 300-0009},
13974   email               = {leonie@example.org}
13975 }
13976 ⟨/meta | doc-table-a | doc-table-q | doc-table-pis⟩
13977
13978 ⟨*meta | doc-table-a | doc-table-b | doc-table-pis⟩
13979 \addEntry{person}{
13980   id                  = {per10},
13981   firstname           = {Daniel},
13982   lastname            = {Okafor},
13983   academicdegree      = {Prof. Dr. rer. nat.},
13984   period              = {3},
13985   role                = {PI},
13986   gender              = {m},
13987   ref-institution-id  = {ins5,ins6},
13988   ref-project-id      = {A06,B02,B06},
13989   position            = {prof-W2},
13990   positionText        = {Professor of Systems Medicine},
13991   doctorateYear       = {2010},
13992   doctorate           = {Dr. rer. nat., Example University},
13993   degree-programme    = {Life Sciences, Example University},
13994   phone               = {+49 000 300-0010},
13995   email               = {daniel@example.org}
13996 }
13997 ⟨/meta | doc-table-a | doc-table-b | doc-table-pis⟩
13998
13999 ⟨*meta | doc-table-b | doc-table-pis⟩
14000 \addEntry{person}{
14001   id                  = {per11},
14002   firstname           = {Mira},
14003   lastname            = {Sato},
14004   academicdegree      = {Prof. Dr. rer. nat.},
14005   period              = {3},
14006   role                = {PI},
14007   gender              = {f},
14008   ref-institution-id  = {ins6},
14009   ref-project-id      = {B03,B04},
14010   position            = {prof-W2},
14011   positionText        = {Professor of Systems Medicine},
14012   doctorateYear       = {2010},
14013   doctorate           = {Dr. rer. nat., Example University},
14014   degree-programme    = {Life Sciences, Example University},
14015   phone               = {+49 000 300-0011},
14016   email               = {mira@example.org}
14017 }
14018 ⟨/meta | doc-table-b | doc-table-pis⟩
14019
14020 ⟨*meta | doc-table-b | doc-table-q | doc-table-pis⟩
14021 \addEntry{person}{
14022   id                  = {per12},
14023   firstname           = {Robin},
14024   lastname            = {Keller},
14025   academicdegree      = {Prof. Dr. rer. nat.},
14026   period              = {3},
14027   role                = {PI},
14028   gender              = {d},
14029   ref-institution-id  = {ins1,ins6},
14030   ref-project-id      = {B05,B06,Q03},
14031   position            = {prof-W2},
14032   positionText        = {Professor of Systems Medicine},
14033   doctorateYear       = {2010},
14034   doctorate           = {Dr. rer. nat., Example University},
14035   degree-programme    = {Life Sciences, Example University},
14036   phone               = {+49 000 300-0012},
14037   email               = {robin@example.org}
14038 }
14039 ⟨/meta | doc-table-b | doc-table-q | doc-table-pis⟩
14040
14041 ⟨*meta | doc-table-a | doc-table-pis⟩
14042 \addEntry{project}{
14043   id                  = {A03},
14044   project-area        = {A},
14045   number              = {A03},
14046   project-status      = {C},
14047   consider-for-lists  = {true},
14048   reference-only      = {false},
14049   name                = {Microbiome-derived metabolites and immune memory},
14050   project-type        = {research},
14051   research-area       = {2.21-02},
14052   ref-person-id       = {per6,per7},
14053   collaborations      = {A02,A04,B03,Q01}
14054 }
14055 ⟨/meta | doc-table-a | doc-table-pis⟩
```

```
14056
14057 ⟨*meta | doc-table-a | doc-table-pis⟩
14058 \addEntry{project}{
14059   id                 = {A04},
14060   project-area       = {A},
14061   number             = {A04},
14062   project-status     = {N},
14063   consider-for-lists = {true},
14064   reference-only     = {false},
14065   name               = {Spatial niches of host--microbe interactions},
14066   project-type       = {research},
14067   research-area      = {2.21-02},
14068   ref-person-id      = {per7,per8},
14069   collaborations     = {A01,A03,A05,B04,Q01}
14070 }
14071 ⟨/meta | doc-table-a | doc-table-pis⟩
14072
14073 ⟨*meta | doc-table-a | doc-table-pis⟩
14074 \addEntry{project}{
14075   id                 = {A05},
14076   project-area       = {A},
14077   number             = {A05},
14078   project-status     = {C},
14079   consider-for-lists = {true},
14080   reference-only     = {false},
14081   name               = {Dietary control of microbial community dynamics},
14082   project-type       = {research},
14083   research-area      = {2.21-02},
14084   ref-person-id      = {per8,per9},
14085   collaborations     = {A04,A06,B05,Q02}
14086 }
14087 ⟨/meta | doc-table-a | doc-table-pis⟩
14088
14089 ⟨*meta | doc-table-a | doc-table-pis⟩
14090 \addEntry{project}{
14091   id                 = {A06},
14092   project-area       = {A},
14093   number             = {A06},
14094   project-status     = {N},
14095   consider-for-lists = {true},
14096   reference-only     = {false},
14097   name               = {Modelling microbial resilience and recovery},
14098   project-type       = {research},
14099   research-area      = {2.21-02},
14100   ref-person-id      = {per9,per10},
14101   collaborations     = {A01,A05,B06,Q01}
14102 }
14103 ⟨/meta | doc-table-a | doc-table-pis⟩
14104
14105 ⟨*meta | doc-table-b | doc-table-pis⟩
14106 \addEntry{project}{
14107   id                 = {B03},
14108   project-area       = {B},
14109   number             = {B03},
14110   project-status     = {C},
14111   consider-for-lists = {true},
14112   reference-only     = {false},
14113   name               = {Immune circuits linking intestine and liver},
14114   project-type       = {research},
14115   research-area      = {2.21-02},
14116   ref-person-id      = {per3,per11},
14117   collaborations     = {A01,A03,B01,B05,Q01}
14118 }
14119 ⟨/meta | doc-table-b | doc-table-pis⟩
14120
14121 ⟨*meta | doc-table-b | doc-table-pis⟩
14122 \addEntry{project}{
14123   id                 = {B04},
14124   project-area       = {B},
14125   number             = {B04},
14126   project-status     = {N},
14127   consider-for-lists = {true},
14128   reference-only     = {false},
14129   name               = {Cross-organ consequences of inflammatory stress},
14130   project-type       = {research},
14131   research-area      = {2.21-02},
14132   ref-person-id      = {per4,per11},
14133   collaborations     = {A04,B02,B06,Q02}
14134 }
14135 ⟨/meta | doc-table-b | doc-table-pis⟩
14136
14137 ⟨*meta | doc-table-b | doc-table-pis⟩
14138 \addEntry{project}{
14139   id                 = {B05},
14140   project-area       = {B},
14141   number             = {B05},
14142   project-status     = {C},
```

```
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Patient-derived models of metabolic disease},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per5,per12},
  collaborations      = {A05,B03,B06,Q01}
}
⟨/meta | doc-table-b | doc-table-pis⟩

⟨*meta | doc-table-b | doc-table-pis⟩
\addEntry{project}{
  id                  = {B06},
  project-area        = {B},
  number              = {B06},
  project-status      = {N},
  consider-for-lists  = {true},
  reference-only      = {false},
  name                = {Translating microbial biomarkers into interventions},
  project-type        = {research},
  research-area       = {2.21-02},
  ref-person-id       = {per10,per12},
  collaborations      = {A01,A06,B04,B05,Q02}
}
⟨/meta | doc-table-b | doc-table-pis⟩

⟨*!example-data⟩
⟨*meta⟩
\addEntry[funding]{RequestedFunding}{
  ref-project-id={A02}, funding-type={staff}, category={PhD}, percentage={65},
  quantity={year1=1,year2=1,year3=1,year4=1,year5=1}
}
⟨/meta⟩
⟨*meta⟩
\addEntry[funding]{RequestedFunding}{
  ref-project-id={A02}, funding-type={costs}, category={consumables},
  sum={year1=1000,year2=2000,year3=2000,year4=2000,year5=1000}
}
⟨/meta⟩
⟨*meta⟩
\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id={A02}, ref-person-id={per6},
  funding-status={existing}, staff-type={research},
  field-of-research={Microbiology}, commitment={8}, funding-source={State funds},
  description={Supervises experiments and analysis.}
}
⟨/meta⟩
⟨/!example-data⟩


%% Fictional CV fixtures shared by the local CV and the embedded manual PDF.

%% Each PI demonstrates three career stages and all five optional categories.


%% Kim Sample: Microbial systems medicine.

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {mandatory},
  periodStart        = {2013},
  periodEnd          = {},
  positionText       = {Professor and Director},
  position           = {Current appointment},
  description        = {Leads the mechanistic microbiome programme and
    coordinates the shared experimental strategy.},
  ref-institution-id = {ins1}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {mandatory},
  periodStart        = {2005},
  periodEnd          = {2012},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on microbial metabolites
    and epithelial signalling and developed collaborative
    methods.},
  ref-institution-id = {ins1}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {mandatory},
  periodStart        = {2001},
  periodEnd          = {2004},
```

```
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    microbial metabolites and epithelial signalling.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {supplement},
  description        = {Research focus: microbial metabolites and epithelial
    signalling. A visiting fellowship at the Example
    Centre for Interdisciplinary Research supported new
    collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {activities},
  description        = {Chairs the consortium steering committee and an
    annual host--microbiome symposium. Contributes to
    peer review and open teaching materials.}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {supervision},
  description        = {Mentors laboratory and clinical doctoral researchers
    through joint thesis committees. Regular progress
    reviews include research integrity and publication
    planning.}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {distinction},
  description        = {2024: Example Academy fellowship for
    interdisciplinary microbiome research. 2021:
    consortium mentoring recognition for support of
    early-career researchers.}
}

\addEntry[cv]{cv-per1}{
  ref-person-id      = {per1},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Alex Example: Computational systems biology.

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {mandatory},
  periodStart        = {2016},
  periodEnd          = {},
  positionText       = {Research group leader},
  position           = {Current appointment},
  description        = {Develops statistical models linking molecular
    measurements with clinical phenotypes.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {mandatory},
  periodStart        = {2004},
  periodEnd          = {2015},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on multimodal data
    integration and reproducible analysis and developed
    collaborative methods.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {mandatory},
  periodStart        = {2000},
  periodEnd          = {2003},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
```

```
    multimodal data integration and reproducible
    analysis.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {supplement},
  description        = {Research focus: multimodal data integration and
    reproducible analysis. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {activities},
  description        = {Leads the data standards working group and maintains
    the consortium code-review programme. Contributes to
    peer review and open teaching materials.}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {supervision},
  description        = {Supervises doctoral projects in statistics and
    biology, with paired computational and experimental
    mentors. Regular progress reviews include research
    integrity and publication planning.}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {distinction},
  description        = {2024: Partner University award for open research
    software. 2021: consortium mentoring recognition for
    support of early-career researchers.}
}

\addEntry[cv]{cv-per2}{
  ref-person-id      = {per2},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Hansi Meier: Microbiology and host defence.

\addEntry[cv]{cv-per3}{
  ref-person-id      = {per3},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Senior research scientist},
  position           = {Current appointment},
  description        = {Coordinates experimental infection models and
    validation of immune mechanisms.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per3}{
  ref-person-id      = {per3},
  cv-type            = {mandatory},
  periodStart        = {2004},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on microbial sensing and
    innate immune memory and developed collaborative
    methods.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per3}{
  ref-person-id      = {per3},
  cv-type            = {mandatory},
  periodStart        = {2000},
  periodEnd          = {2003},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    microbial sensing and innate immune memory.},
  ref-institution-id = {ins2}
}
```

```
14403
14404 \addEntry[cv]{cv-per3}{
14405   ref-person-id      = {per3},
14406   cv-type            = {supplement},
14407   description        = {Research focus: microbial sensing and innate immune
14408     memory. A visiting fellowship at the Example Centre
14409     for Interdisciplinary Research supported new
14410     collaborations and exchange of methods.}
14411 }
14412
14413 \addEntry[cv]{cv-per3}{
14414   ref-person-id      = {per3},
14415   cv-type            = {activities},
14416   description        = {Organises a practical course on host defence and
14417     serves on the laboratory methods committee.
14418     Contributes to peer review and open teaching
14419     materials.}
14420 }
14421
14422 \addEntry[cv]{cv-per3}{
14423   ref-person-id      = {per3},
14424   cv-type            = {supervision},
14425   description        = {Mentors early-career scientists in experimental
14426     design, biosafety and reproducible laboratory
14427     practice. Regular progress reviews include research
14428     integrity and publication planning.}
14429 }
14430
14431 \addEntry[cv]{cv-per3}{
14432   ref-person-id      = {per3},
14433   cv-type            = {distinction},
14434   description        = {2024: Example Society prize for translational
14435     infection research. 2021: consortium mentoring
14436     recognition for support of early-career researchers.}
14437 }
14438
14439 \addEntry[cv]{cv-per3}{
14440   ref-person-id      = {per3},
14441   cv-type            = {other},
14442   description        = {Makes analysis protocols and training resources
14443     available for reuse. No commercial interests related
14444     to the proposed research. All career details and
14445     distinctions in this CV are fictional examples.}
14446 }
14447
14448
14449 %% Kimmi Miller: Molecular tissue regeneration.
14450
14451 \addEntry[cv]{cv-per4}{
14452   ref-person-id      = {per4},
14453   cv-type            = {mandatory},
14454   periodStart        = {2015},
14455   periodEnd          = {},
14456   positionText       = {Professor of Molecular Medicine},
14457   position           = {Current appointment},
14458   description        = {Connects organoid perturbation experiments with
14459     molecular measurements of tissue recovery.},
14460   ref-institution-id = {ins2}
14461 }
14462
14463 \addEntry[cv]{cv-per4}{
14464   ref-person-id      = {per4},
14465   cv-type            = {mandatory},
14466   periodStart        = {2004},
14467   periodEnd          = {2014},
14468   positionText       = {Independent research group leader},
14469   position           = {Independent research},
14470   description        = {Established a research team on epithelial repair and
14471     tissue regeneration and developed collaborative
14472     methods.},
14473   ref-institution-id = {ins2}
14474 }
14475
14476 \addEntry[cv]{cv-per4}{
14477   ref-person-id      = {per4},
14478   cv-type            = {mandatory},
14479   periodStart        = {2000},
14480   periodEnd          = {2003},
14481   positionText       = {Postdoctoral researcher},
14482   position           = {Postdoctoral training},
14483   description        = {Developed experimental and analytical approaches to
14484     epithelial repair and tissue regeneration.},
14485   ref-institution-id = {ins2}
14486 }
14487
14488 \addEntry[cv]{cv-per4}{
14489   ref-person-id      = {per4},
```

```
  cv-type            = {supplement},
  description        = {Research focus: epithelial repair and tissue
    regeneration. A visiting fellowship at the Example
    Centre for Interdisciplinary Research supported new
    collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per4}{
  ref-person-id      = {per4},
  cv-type            = {activities},
  description        = {Coordinates the organoid methods network and an
    interdisciplinary regeneration seminar. Contributes
    to peer review and open teaching materials.}
}

\addEntry[cv]{cv-per4}{
  ref-person-id      = {per4},
  cv-type            = {supervision},
  description        = {Supervises doctoral researchers in molecular medicine
    and provides mentoring for clinician scientists.
    Regular progress reviews include research integrity
    and publication planning.}
}

\addEntry[cv]{cv-per4}{
  ref-person-id      = {per4},
  cv-type            = {distinction},
  description        = {2024: Partner University award for collaborative
    tissue research. 2021: consortium mentoring
    recognition for support of early-career researchers.}
}

\addEntry[cv]{cv-per4}{
  ref-person-id      = {per4},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Friedhelm Redata: Biomedical research training.

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {mandatory},
  periodStart        = {2014},
  periodEnd          = {},
  positionText       = {Professor of Research Training},
  position           = {Current appointment},
  description        = {Directs the graduate programme and evaluates
    interdisciplinary training formats.},
  ref-institution-id = {ins1}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {mandatory},
  periodStart        = {2003},
  periodEnd          = {2013},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on structured doctoral
    training and research integrity and developed
    collaborative methods.},
  ref-institution-id = {ins1}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {mandatory},
  periodStart        = {1999},
  periodEnd          = {2002},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    structured doctoral training and research integrity.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {supplement},
  description        = {Research focus: structured doctoral training and
    research integrity. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
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```
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {activities},
  description        = {Chairs the graduate curriculum committee and
    coordinates the annual doctoral retreat. Contributes
    to peer review and open teaching materials.}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {supervision},
  description        = {Supports thesis committees, individual development
    plans and peer mentoring across all research areas.
    Regular progress reviews include research integrity
    and publication planning.}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {distinction},
  description        = {2024: Example University prize for doctoral
    education. 2021: consortium mentoring recognition for
    support of early-career researchers.}
}

\addEntry[cv]{cv-per5}{
  ref-person-id      = {per5},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Samira Hartmann: Microbial systems medicine.

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Combines microbial genetics with functional assays of
    host metabolic responses.},
  ref-institution-id = {ins3}
}

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on microbial metabolic
    pathways and immune adaptation and developed
    collaborative methods.},
  ref-institution-id = {ins3}
}

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    microbial metabolic pathways and immune adaptation.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {supplement},
  description        = {Research focus: microbial metabolic pathways and
    immune adaptation. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per6}{
```

```
  ref-person-id      = {per6},
  cv-type            = {activities},
  description        = {Coordinates the pathway-discovery workshop and joint
    protocol reviews across the consortium. Contributes
    to peer review and open teaching materials.}
}

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {supervision},
  description        = {Supervises paired microbiology and immunology
    projects, including laboratory exchanges for doctoral
    researchers. Regular progress reviews include
    research integrity and publication planning.}
}

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {distinction},
  description        = {2024: Example Foundation award for microbial systems
    research. 2021: consortium mentoring recognition for
    support of early-career researchers.}
}

\addEntry[cv]{cv-per6}{
  ref-person-id      = {per6},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Noah Lindberg: Spatial immunology.

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Links spatial profiling to microscopy and
    quantitative descriptions of immune-cell
    neighbourhoods.},
  ref-institution-id = {ins3}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on spatial immune niches
    and quantitative imaging and developed collaborative
    methods.},
  ref-institution-id = {ins3}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    spatial immune niches and quantitative imaging.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {supplement},
  description        = {Research focus: spatial immune niches and
    quantitative imaging. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {activities},
  description        = {Leads the imaging quality-control forum and teaches
```

```
    the annual spatial-analysis course. Contributes to
    peer review and open teaching materials.}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {supervision},
  description        = {Mentors doctoral researchers in image analysis, with
    shared supervision across experimental and
    computational groups. Regular progress reviews
    include research integrity and publication planning.}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {distinction},
  description        = {2024: Example Imaging Society early-career research
    prize. 2021: consortium mentoring recognition for
    support of early-career researchers.}
}

\addEntry[cv]{cv-per7}{
  ref-person-id      = {per7},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Alexis-Marie Winters-Scholz: Host--microbe interactions.

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Develops experimental models of diet-dependent
    microbial colonisation and mucosal responses.},
  ref-institution-id = {ins4}
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on dietary perturbations
    and mucosal microbial communities and developed
    collaborative methods.},
  ref-institution-id = {ins4}
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    dietary perturbations and mucosal microbial
    communities.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {supplement},
  description        = {Research focus: dietary perturbations and mucosal
    microbial communities. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {activities},
  description        = {Coordinates the experimental-models working group and
    a public seminar series on microbiome science.
    Contributes to peer review and open teaching
    materials.}
```

```
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {supervision},
  description        = {Supervises doctoral projects in microbial ecology and
    supports exchange placements between partner
    laboratories. Regular progress reviews include
    research integrity and publication planning.}
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {distinction},
  description        = {2024: Example University award for experimental
    microbiome research. 2021: consortium mentoring
    recognition for support of early-career researchers.}
}

\addEntry[cv]{cv-per8}{
  ref-person-id      = {per8},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Leonie Navarro: Metabolic systems biology.

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Integrates dietary interventions with metabolic
    profiling and predictive modelling.},
  ref-institution-id = {ins5}
}

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on nutrient sensing and
    predictive metabolic models and developed
    collaborative methods.},
  ref-institution-id = {ins5}
}

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    nutrient sensing and predictive metabolic models.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {supplement},
  description        = {Research focus: nutrient sensing and predictive
    metabolic models. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {activities},
  description        = {Organises the metabolic modelling summer school and
    contributes to graduate admissions panels.
    Contributes to peer review and open teaching
    materials.}
}

\addEntry[cv]{cv-per9}{
```

```
  ref-person-id      = {per9},
  cv-type            = {supervision},
  description        = {Mentors doctoral researchers in model validation,
    interdisciplinary communication and career planning.
    Regular progress reviews include research integrity
    and publication planning.}
}

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {distinction},
  description        = {2024: Example Society distinction for integrative
    metabolic research. 2021: consortium mentoring
    recognition for support of early-career researchers.}
}

\addEntry[cv]{cv-per9}{
  ref-person-id      = {per9},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Daniel Okafor: Translational systems medicine.

\addEntry[cv]{cv-per10}{
  ref-person-id      = {per10},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Connects patient-derived measurements with biomarker
    validation and experimental tissue models.},
  ref-institution-id = {ins5}
}

\addEntry[cv]{cv-per10}{
  ref-person-id      = {per10},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on microbial biomarkers
    and clinical tissue responses and developed
    collaborative methods.},
  ref-institution-id = {ins5}
}

\addEntry[cv]{cv-per10}{
  ref-person-id      = {per10},
  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    microbial biomarkers and clinical tissue responses.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per10}{
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  cv-type            = {supplement},
  description        = {Research focus: microbial biomarkers and clinical
    tissue responses. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per10}{
  ref-person-id      = {per10},
  cv-type            = {activities},
  description        = {Coordinates the translational data-analysis workshop
    and the clinical assay harmonisation group.
    Contributes to peer review and open teaching
    materials.}
}

\addEntry[cv]{cv-per10}{
  ref-person-id      = {per10},
  cv-type            = {supervision},
  description        = {Supervises clinician scientists and laboratory
```

```
    researchers in joint translational thesis projects.
    Regular progress reviews include research integrity
    and publication planning.}
}

\addEntry[cv]{cv-per10}{
  ref-person-id      = {per10},
  cv-type            = {distinction},
  description        = {2024: Westhaven Research Centre translational science
    award. 2021: consortium mentoring recognition for
    support of early-career researchers.}
}

\addEntry[cv]{cv-per10}{
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  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Mira Sato: Cross-organ immunology.

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Coordinates tissue sampling and comparative analyses
    of inflammatory responses across organs.},
  ref-institution-id = {ins6}
}

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on inflammatory circuits
    between intestine and liver and developed
    collaborative methods.},
  ref-institution-id = {ins6}
}

\addEntry[cv]{cv-per11}{
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  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    inflammatory circuits between intestine and liver.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {supplement},
  description        = {Research focus: inflammatory circuits between
    intestine and liver. A visiting fellowship at the
    Example Centre for Interdisciplinary Research
    supported new collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {activities},
  description        = {Leads the clinical sample-coordination group and
    teaches a course on cross-organ immunology.
    Contributes to peer review and open teaching
    materials.}
}

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {supervision},
  description        = {Mentors doctoral researchers in clinical cohort
    design, sample governance and mechanistic validation.
    Regular progress reviews include research integrity
    and publication planning.}
```

```
}

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {distinction},
  description        = {2024: Example Clinical Research Society award for
    collaborative immunology. 2021: consortium mentoring
    recognition for support of early-career researchers.}
}

\addEntry[cv]{cv-per11}{
  ref-person-id      = {per11},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}


%% Robin Keller: Integrative systems medicine.

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {mandatory},
  periodStart        = {2018},
  periodEnd          = {},
  positionText       = {Professor of Systems Medicine},
  position           = {Current appointment},
  description        = {Combines patient-derived models with biomarker
    translation and stewardship of shared research data.},
  ref-institution-id = {ins1}
}

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {mandatory},
  periodStart        = {2015},
  periodEnd          = {2017},
  positionText       = {Independent research group leader},
  position           = {Independent research},
  description        = {Established a research team on patient-derived models
    and biomarker translation and developed collaborative
    methods.},
  ref-institution-id = {ins1}
}

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {mandatory},
  periodStart        = {2011},
  periodEnd          = {2014},
  positionText       = {Postdoctoral researcher},
  position           = {Postdoctoral training},
  description        = {Developed experimental and analytical approaches to
    patient-derived models and biomarker translation.},
  ref-institution-id = {ins2}
}

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {supplement},
  description        = {Research focus: patient-derived models and biomarker
    translation. A visiting fellowship at the Example
    Centre for Interdisciplinary Research supported new
    collaborations and exchange of methods.}
}

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {activities},
  description        = {Chairs the consortium reproducibility forum and
    coordinates workshops on reusable clinical research
    data. Contributes to peer review and open teaching
    materials.}
}

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {supervision},
  description        = {Co-supervises doctoral researchers across clinical
    and computational projects, with individual
    career-development plans. Regular progress reviews
    include research integrity and publication planning.}
}

\addEntry[cv]{cv-per12}{
```

```
  ref-person-id      = {per12},
  cv-type            = {distinction},
  description        = {2024: Example University award for reproducible
    translational medicine. 2021: consortium mentoring
    recognition for support of early-career researchers.}
}

\addEntry[cv]{cv-per12}{
  ref-person-id      = {per12},
  cv-type            = {other},
  description        = {Makes analysis protocols and training resources
    available for reuse. No commercial interests related
    to the proposed research. All career details and
    distinctions in this CV are fictional examples.}
}

% \iffalse
⟨/meta-cv⟩
⟨*meta-tables⟩
% \fi
```

6.2.10 Title Page (Proposal specific)

```
\pdfbookmark{Titlepage}{titlepage}
\begin{titlepage}
  \def\tmpLocalDB{meta}%
  \vspace*{4em}
  \begin{center}%
   \large
 \crc\,\dfgCRCNumber\\
    \vspace*{1.5em}
    {\Large\bfseries \dfgCRCName}\\
    \dfgCRCNameAddon\\
 \vspace*{2em}
   \crcLogo[height=2cm]\\
    \vspace*{2em}
    \theCRCCoordinatingAffiliation\\
    \dfgApplicantAffiliationsTF
      { }
      {\theCRCOtherApplicantAffiliations\\}
 \vspace*{2em}
    {\Large\bfseries Funding Proposal}\\[1em]
 for the \dfgFundingPeriodText\space Funding Period\\
 \dfgFundingYear{1}--%
 \dfgFundingYear{2}--%
 \dfgFundingYear{3}--%
 \dfgFundingYear{4}--%
 \dfgFundingYear{5}
    \dfgIfCVT{\par\textcolor{tableHeadingsFont}{Appendix: Research profiles}}
    \dfgIfFinalF{{\par\large\InfoTeX}}
   % \iftoggle{crc@chapter}{%
   %   \par\vspace*{5em}\par
   %   \begin{crcTitleBox}
   %       \centering\Huge%
   %       {\color{rwthRed}%
   %           \ttfamily%
   %           content/\textbf{\dfgChapter}.tex}\\
   %       \Large\dfgCompiled
   %   \end{crcTitleBox}%
   % }{}
   \vfill
   %
   %  \crcLogo[height=1cm]
   \hfill%
   \affiliationLogo[height=1cm]
  \end{center}
  \clearpage
  \checkFile{content/00_titlepage-colophon}
\end{titlepage}
\begin{titlepage}
  % setting the default database for general information
  \def\tmpLocalDB{meta}
  \begin{center}
 Proposal for the \dfgFundingPeriodText\space Funding Period of\\
 \crc\,\dfgCRCNumber

 \enquote{\bfseries\dfgCRCName: \dfgCRCNameAddon}
  \end{center}
  funded since
  \begin{center}
   \getEntry{crc}{fundedSince}
  \end{center}
```

```
   for
   \begin{center}
  \dfgFundingYear{1}--%
  \dfgFundingYear{2}--%
  \dfgFundingYear{3}--%
  \dfgFundingYear{4}--%
  \dfgFundingYear{5}
   \end{center}
   Coordinating university:
   \begin{center}
  \theCRCCoordinatingAffiliation
   \end{center}
   \dfgApplicantAffiliationsTF
     { }
     {
       Other applicant
       \dfgOtherApplicantAffiliationsTF{university}{universities}:
       \begin{center}
         \theCRCOtherApplicantAffiliations
       \end{center}
     }
   \Hrule[
    thickness = 1.5pt,
    above     = 2em,
    below     = 2em]

   \begin{minipage}[t]{.49\linewidth}
    \textbf{Spokesperson of\\ the \crc:}\\[1\baselineskip]
     \getPIsWithRole[raster columns=1]{spokesperson}
   \end{minipage}\hfill%
   \begin{minipage}[t]{.49\linewidth}
    \textbf{Management or office of\\the \crc:}\\[1\baselineskip]
     \getPIsWithRole[raster columns=1]{manager}
   \end{minipage}
   \vfill
   \printSignaturesWithRole{spokesperson}{(Spokesperson~of~\crc)}
   \printSignaturesWithRole{rector}
     {(Rector,~chancellor,~or~president~of~
       \getEntry[affiliation]{\theAffiliationOfRoleID}{name})}
\end{titlepage}
\bgroup
\let\cleardoublepage\clearpage% to get the TOC on two pages (left/right)
\pdfbookmark{\contentsname}{Contents}
\tableofcontents
\dfgIfCVT{\KOMAoptions{open=left}}
\egroup
```

6.2.11 Content Files (Proposal specific)

```
\newrefsection
\chapter{General information}
\printThumbIndex{1}
\section{Key data}
\subsection{Governing bodies of the \crc}
% \subsubsection{Spokesperson of the \thisCRC}
% \getPIsWithRole{spokesperson}

\subsubsection*{Steering Committee of the \thisCRC}

\getPIsWithRole{steering-committee-member}

For the third funding period, the steering committee includes representatives
of both research areas and the Integrated Research Training Group.  The
spokesperson and committee members are elected by the CRC general assembly in
accordance with the bylaws.  The composition is reviewed annually with regard
to scientific breadth, career stage, and equitable participation.

\subsection{Project leaders}
\printTable{ProjectLeaders}

\subsection{Participating institutions}
\printList{ParticipatingInstitutions}

\clearpage
\subsection{Project groups and projects}
\printTable{{Projects}{A}{Microbiota and metabolic pathways}}
\printTable{{Projects}{B}{Immunological circuits}}
\printTable{{Projects}{Q}{Central projects}}
N: New projects started in the \dfgFundingPeriodText\space funding period; %
E: Ending projects not continued in the \dfgFundingPeriodText\space funding
period; %
Research areas according to the German Research Foundation (2024--2028)%
\footnote{\raggedright\url{https://www.dfg.de/resource/blob/331950/fachsystematik-2024-2028-en.pdf}}:
\printList{ResearchAreasFromProjects}


% For the following sections we want to have them
```

```
% numbered until 'subsubsection'
\setcounter{secnumdepth}{4}

\clearpage
\section{Research profile of the \crc} %1.2
\begin{dfgSummaryBox}[title={Spotlights: Moving Forward}]
  \item Resolve causal mechanisms of communication between microbial
    communities, epithelial barriers, and tissue-resident immune cells.
  \item Combine spatial profiling, experimental perturbation, and predictive
    modelling across the participating projects.
  \item Translate shared methods into reproducible services, interoperable
    data, and structured training for researchers in early career phases.
\end{dfgSummaryBox}

%% \FloatBarrier % no images in references list.
\defbibheading{frame-references}[References]{\subsection*{#1}}
\printFrameBibliography[heading=frame-references]


\subsection{Summary of the research programme}%1.2.1

The CRC investigates how microbial and host-derived signals cross epithelial
barriers and influence organ function.  During the previous funding periods,
the consortium established shared cohorts, experimental models, imaging
workflows, and computational methods.  The new programme links these resources
in hypothesis-driven projects that test causality across molecular, cellular,
and tissue scales.

\subsection{Detailed presentation of the research programme}%1.2.2

Project area A examines microbial metabolites and barrier signalling, whereas
project area B focuses on immune circuits and tissue responses.  The central
projects provide imaging, analysis, training, coordination, and research-data
services.  Common samples, harmonised metadata, and reciprocal collaborations
connect the individual work programmes and enable validation across models and
participating institutions.

\subsection{Positioning of the \crc within its general research area}%1.2.3

The programme combines mechanistic microbiome research with systems immunology
and organ physiology.  Its distinguishing feature is the coordinated
validation of candidate signals from discovery cohorts through controlled
experimental systems to tissue-level readouts, supported by shared technology
and transparent data-management standards.

\subsection{National and international cooperation and networking} %1.2.4

The CRC maintains method-oriented collaborations with national infrastructure
networks and international partner laboratories.  Cooperation agreements
define sample and data transfer, authorship, quality assurance, and access to
specialised technologies.  Visiting-researcher and exchange formats are
coordinated through the Integrated Research Training Group.

\section{Research profile of the applicant
  \dfgApplicantAffiliationsTF{university}{universities}} %1.3
\subsection{Strategy and planning}%1.3.1

Example University has identified host--microbiome research as a strategic
priority and supports the CRC through permanent scientific positions, shared
technical platforms, research-data infrastructure, and central administration.
Annual reviews align institutional investments with the scientific milestones
of the consortium.

\subsection{Staff situation} % 1.3.2

The participating units contribute principal investigators, research staff,
technical personnel, data stewards, and administrative support.  Recruitment
planning prioritises timely appointments, transparent selection procedures,
career development, and continuity of essential technical expertise.

\subsection{Research infrastructure} % 1.3.3

The CRC has access to advanced microscopy, flow cytometry, sequencing,
gnotobiotic facilities, secure data storage, and high-performance computing.
Central projects coordinate access, training, maintenance, and quality control
so that methods and datasets remain comparable across all projects.

\section{Support structures} % 1.4
\subsection{Researchers in early career phases} % 1.4.1

Researchers in early career phases receive structured scientific and career
mentoring, transferable-skills training, annual progress reviews, and access to
methods courses and exchange opportunities organised by \getProject{Q02}.

\addEntry[person]{EarlyCareerResearcher}{
  ref-project-id= {A01},
  id              = {KM},
```

```
  lastname      = {Sample},
  firstname     = {Kimmi},
  financed-by   = {DFG},
  category      = {PhD},
  topic         = {Epithelial sensing of microbial metabolites},
  periodStart   = {2025-04},
  periodEnd     = {2029-06}
}
\addEntry[person]{EarlyCareerResearcher}{
  ref-project-id= {B01},
  id              = {KiM},
  lastname      = {Wunderlich},
  firstname     = {Kimmi},
  financed-by   = {DFG},
  category      = {MD},
  topic         = {Immune signatures during microbiome perturbation},
  periodStart   = {2025-04},
  periodEnd     = {2029-06}
}

%% \getEntry[EarlyCareerResearcher]{KM}{lastname}
\printTable{EarlyCareerResearchers}

\subsection{Promotion of equity and diversity} % 1.4.2

Recruitment, supervision, and committee appointments follow transparent
criteria and are monitored by career stage and gender.  Flexible working
arrangements, mentoring, accessibility measures, and targeted career support
are available through the central coordination project.

\addEntry[GenderEquality]{GenderEqualityStaff}{
  position                              = {PhD}, %PhD|PostDoc
  previous-objective-percentage-female = {40},
  status-quo-number-female              = {20},
  status-quo-percentage-female          = {14},
  future-objective-percentage-female    = {40}
}

\addEntry[GenderEquality]{GenderEqualityStaff}{
  position                              = {postDoc}, %PhD|PostDoc
  previous-objective-percentage-female = {22},
  status-quo-number-female              = {34},
  status-quo-percentage-female          = {44},
  future-objective-percentage-female    = {34}
}
% -----

\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {postDoc}, %PhD|PostDoc
  previous-objective-percentage-female = {22},
  future-objective-percentage-female    = {34}
}
\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {groupLeader}, %PhD|PostDoc
  previous-objective-percentage-female = {22},
  future-objective-percentage-female    = {34}
}


\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {prof-W2}, %PhD|PostDoc
  previous-objective-percentage-female = {22},
  future-objective-percentage-female    = {34}
}

\addEntry[GenderEquality]{GenderEqualityProjectLeaders}{
  position                              = {prof-W3}, %PhD|PostDoc
  previous-objective-percentage-female = {22},
  future-objective-percentage-female    = {34}
}

\clearpage
\subsubsection*{A. Objectives for the participation of female researchers}
\printTable{GenderEqualityStaff}

\subsubsection*{B. Project leaders}
\printTable{GenderEqualityProjectLeaders}

\subsection{Knowledge transfer and science communication} % 1.4.3

The CRC communicates results through open research resources, public lectures,
school and patient-group activities, and dialogue formats with clinical and
societal stakeholders.  Transfer opportunities are assessed early, with clear
procedures for intellectual property, responsible communication, and reusable
research outputs.

\section{Risks in international cooperation} %1.5
```

```
15527
15528 International partnerships are reviewed for legal, ethical, export-control,
15529 data-protection, and research-security risks.  Written agreements specify
15530 permitted data and material transfers, access rights, publication principles,
15531 and responsibilities.  The participating universities reassess collaborations
15532 when partners, technologies, or regulatory conditions change.
15533
15534 \section{Considerations on aspects of ecological sustainability in project
15535   planning and implementation} %1.6
15536
15537 The CRC considers ecological sustainability when planning experiments,
15538 procurement, travel, computing, sample storage, and shared infrastructure.
15539 Projects consolidate measurement campaigns, share equipment and reference
15540 materials, select proportionate data-retention and computing strategies, and
15541 prefer lower-emission travel or remote participation where this is compatible
15542 with the scientific objective.  The steering committee reviews these measures
15543 alongside scientific feasibility, research quality, legal obligations, animal
15544 welfare, and the responsible use of resources.
15545
15546
15547 % ok, let's go back to the default (from base.tex)
15548 \setcounter{secnumdepth}{2}
15549
15550 \endrefsection
15551 \resetboldnames
15552
15553 \chapter{Existing funds and requested funds} %2
15554 \printThumbIndex{2}
15555 \section{Existing funds} % 2.1
15556 \subsection{Overview of existing funds for direct costs} %2.1.1
15557
15558 \addEntry[funding]{PreviousExistingFundsDirectCosts}{
15559   year        = 1,
15560   applicant   = 1000,
15561   participant = 2000,
15562   other-funds = 3000
15563 }
15564 \addEntry[funding]{PreviousExistingFundsDirectCosts}{
15565   year        = 2,
15566   applicant   = 1000,
15567   participant = 2000,
15568   other-funds = 3000
15569 }
15570 \addEntry[funding]{PreviousExistingFundsDirectCosts}{
15571   year        = 3,
15572   applicant   = 1000,
15573   participant = 2000,
15574   other-funds = 3000
15575 }
15576 \addEntry[funding]{PreviousExistingFundsDirectCosts}{
15577   year        = 4,
15578   applicant   = 1000,
15579   participant = 2000,
15580   other-funds = 3000
15581 }
15582 \addEntry[funding]{PreviousExistingFundsDirectCosts}{
15583   year        = 5,
15584   applicant   = 1000,
15585   participant = 2000,
15586   other-funds = 3000
15587 }
15588
15589
15590 % New-period values are declared once in the individual project files below.
15591 % The class restores those later declarations from the preceding LaTeX run's
15592 % auxiliary files into the overview database used here.
15593 \printTable{ExistingFundsDirectCosts}
15594
15595
15596 \subsection{Overview of existing staff} % 2.1.2
15597 % Data are reused from project-level RequestedFundingStaff declarations with
15598 % funding-status={existing}.  No separate overview database is required.
15599 \printTable{OverviewExistingStaff}
15600
15601 \subsection{List of existing instrumentation} % 2.1.2
15602 \addEntry[instrumentation]{existingInstrumentation}{
15603   ref-project-id = {A01},
15604   quantity       = {1},
15605   instrument     = {Confocal laser-scanning microscope},
15606   company        = {Example Instruments GmbH},
15607   year           = {2025},
15608   cost           = {250},
15609   financed-by    = {DFG}
15610 }
15611 \addEntry[instrumentation]{existingInstrumentation}{
15612   ref-project-id = {A01,B01},
```

```
15613   quantity       = {2},
15614   instrument     = {Anaerobic workstation},
15615   company        = {Laboratory Systems AG},
15616   year           = {2024},
15617   cost           = {80},
15618   financed-by    = {Institutional funds}
15619 }
15620
15621 \printTable{ExistingInstrumentation}
15622
15623 \section{Previous and requested funds} %2.2
15624 \subsection{Overview} % 2.2.1
15625
15626 \addEntry[funding]{PreviousRequestedFunding}{
15627   year            = 1,
15628   staff           = 1000,
15629   costs           = 2000,
15630   instrumentation = 3000,
15631   fellowships     = 4000,
15632   globalfunds     = 5000
15633 }
15634 \addEntry[funding]{PreviousRequestedFunding}{
15635   year            = 2,
15636   staff           = 1000,
15637   costs           = 2000,
15638   instrumentation = 3000,
15639   fellowships     = 4000,
15640   globalfunds     = 5000
15641 }
15642 \addEntry[funding]{PreviousRequestedFunding}{
15643   year            = 3,
15644   staff           = 1000,
15645   costs           = 2000,
15646   instrumentation = 3000,
15647   fellowships     = 4000,
15648   globalfunds     = 5000
15649 }
15650 \addEntry[funding]{PreviousRequestedFunding}{
15651   year            = 4,
15652   staff           = 1000,
15653   costs           = 2000,
15654   instrumentation = 3000,
15655   fellowships     = 4000,
15656   globalfunds     = 5000
15657 }
15658 \addEntry[funding]{PreviousRequestedFunding}{
15659   year            = 5,
15660   staff           = 1000,
15661   costs           = 2000,
15662   instrumentation = 3000,
15663   fellowships     = 4000,
15664   globalfunds     = 5000
15665 }
15666
15667
15668 \printTable{PreviousAndCurrentRequestedFunding}
15669
15670 \subsection{Overview of funds requested for staff} % 2.2.2
15671 % data comes from all the
15672 % \addEntry[funding]{RequestedFunding}{
15673 %    ...
15674 %   category     = {....},
15675 %   funding-type = {staff},
15676 %    ...
15677 % }
15678 \printTable{OverviewOfFundsRequestedForStaff}
15679
15680 \subsection{Overview of funds requested for instrumentation} % 2.2.3
15681
15682 % data comes from all the
15683 % \addEntry[funding]{RequestedFunding}{
15684 %    ...
15685 %   funding-type = {instrumentation},
15686 %    ...
15687 % }
15688 \printTable{OverviewFundsRequestedInstrumentation}
15689
15690
15691 \section{Upkeep of laboratory animals} % 2.3
15692 \addEntry{UpkeepLaboratoryAnimals}{
15693   ref-project-id   = {A01},
15694   species          = {mouse},
15695   quantity         = {1212},
15696   weeks-kept       = {43},
15697   purchasing-costs = {23},
15698   requested-funds  = {100},
15699   existing-funds   = {232}
```

```
}
\addEntry{UpkeepLaboratoryAnimals}{
  ref-project-id   = {A01,A02},
  species          = {rat},
  quantity         = {2},
  weeks-kept       = {3},
  purchasing-costs = {23.54},
  requested-funds  = {1230},
  existing-funds   = {232}
}
\addEntry{UpkeepLaboratoryAnimals}{
  ref-project-id   = {B01},
  species          = {mouse-gnotobiotic},
  quantity         = {3},
  weeks-kept       = {43},
  purchasing-costs = {23},
  requested-funds  = {201},
  existing-funds   = {132}
}
\printTable{UpkeepLaboratoryAnimals}

```

6.2.12 Complete continuing-project example

This file demonstrates the complete heading and command structure for a continuing research project (`project-status=C`).

```
\printProjectHead{A01}

\section{Summary}
\label{A01:summary}

Summarise the continuing project's central question, progress, planned work,
methods, and contribution to the Collaborative Research Centre.

\section{Project progress to date}
\label{A01:project-progress-to-date}

\subsection{Report and current state of research}%
\label{A01:report-and-current-state-of-research}

Report the work and results from the previous funding periods, relate them to
the current state of research, and explain how they motivate the new plan.

\FloatBarrier

\projectPublicationsHeading

%% Add all project-specific publication keys before printing the bibliography.
\projectNocite{FAIR2016}
\printProjectBibliography[heading=none]

\section{Project plan}
\label{A01:project-plan}

\begin{aims}
  \addAim{A} Define the underlying mechanism.
  \addAim{B} Validate the mechanism experimentally.
  \addAim{C} Integrate the findings across the CRC.
\end{aims}

Refer to one aim as \getAim{A}, or to several as \getAim{A,B,C}.

\begin{workpackages}
  \addWP{wp1}{
    name={Mechanistic analysis},
    description={Define the underlying mechanism},
    periodStart={1},
    periodEnd={8},
    style={default},
  }

  Describe the work, methods, milestones, risks, and expected results of
  \getWP{wp1}.

  \begin{workpackages}
    %% Sub-workpackages deliberately have no name key: their authored text is
    %% the visible list item, while description is used only in the Gantt chart.
    \addWP{wp1a}{
      description={Establish the experimental system},
      periodStart={1},
      periodEnd={4},
      style={alternative},
    }
    Establish and validate the experimental system.
```

```
15779
15780       \addWP{wp1b}{
15781         description={Perform the mechanistic experiments},
15782         periodStart={3},
15783         periodEnd={8},
15784         style={important},
15785       }
15786       Perform the mechanistic experiments and define decision points.
15787     \end{workpackages}
15788
15789     \addWP{wp2}{
15790       name={Experimental validation},
15791       description={Validate and integrate the mechanism},
15792       periodStart={7},
15793       periodEnd={16},
15794       style={striped},
15795     }
15796
15797     Describe the validation strategy, expected results, and contingency plan of
15798     \getWP{wp2}.
15799   \end{workpackages}
15800
15801   \printPlot{{Gantt}{A01}}
15802
15803   \FloatBarrier
15804
15805   \section{Relevance of sex, gender and/or diversity in the research
15806     project}
15807   \label{A01:relevance-of-sex-gender-and-diversity}
15808
15809   Explain whether and how sex, gender, and diversity affect the research
15810   questions, design, analysis, and interpretation.
15811
15812   \section{Role within the Collaborative Research
15813     Centre}
15814   \label{A01:role-within-the-collaborative-research-centre}
15815
15816   Describe the project's contribution to the CRC and its principal
15817   collaborations.
15818
15819   %% The default full-width arc layout shows nodes and area rules without edges.
15820   \printPlot{{ProjectCollaboration}{A01}}
15821   %% To draw the connecting arcs, replace the call above with:
15822   %% \printPlot[arcs=true,caption={Optional additional caption text.}]
15823   %%   {{collaborations}{A01}}
15824   %% Use [layout=heatmap] for the alternative heatmap layout.
15825
15826
15827   \section{Differentiation from other funded
15828     projects}
15829   \label{A01:differentiation-from-other-funded-projects}
15830
15831   Explain how this project differs from, and complements, other funded work of
15832   the project leaders.
15833
15834   \section{Project funding}
15835   \label{A01:project-funding}
15836
15837   \subsection{Previous funding}
15838
15839   Funding of this project within the Collaborative Research Centre started
15840   in July 2019.  Preparatory work was funded by Example University's seed-fund
15841   programme from January 2018 to June 2019 (reference number EU-EXAMPLE-01).
15842
15843   \subsection{Requested funding}
15844
15845   %% Without sum, the class calculates from salary@prof, percentage, and
15846   %% quantity, using half the annual amount in year1 and year5.
15847   \addEntry[funding]{RequestedFunding}{
15848     ref-project-id = {A01},
15849     funding-type   = {staff},
15850     category       = {prof},
15851     percentage     = {100},
15852     quantity = {
15853       year1 = {2},
15854       year2 = {2},
15855       year3 = {2},
15856       year4 = {2},
15857       year5 = {2}
15858     }
15859   }
15860
15861   %% With sum, these concrete yearly amounts override the central salary rate;
15862   %% year1 and year5 below already represent half-years.
15863   \addEntry[funding]{RequestedFunding}{
15864     ref-project-id = {A01},
15865     funding-type   = {staff},
```

```
15866   category       = {PhD},
15867   percentage     = {65},
15868   quantity = {
15869     year1 = {1},
15870     year2 = {1},
15871     year3 = {1},
15872     year4 = {1},
15873     year5 = {1}
15874   },
15875   sum = {
15876     year1 = {26500},
15877     year2 = {53000},
15878     year3 = {53000},
15879     year4 = {53000},
15880     year5 = {26500}
15881   }
15882 }
15883
15884 \addEntry[funding]{RequestedFunding}{
15885   ref-project-id = {A01},
15886   funding-type       = {costs}, %staff, costs, ...
15887   category   = {Laboratory consumables},
15888   sum = {
15889     year1      =  {2112},
15890     year2      =  {67},
15891     year3      =  {9843},
15892     year4      =  {2375},
15893     year5      =  {983}
15894    }}
15895
15896 \addEntry[funding]{RequestedFunding}{
15897   ref-project-id = {A01},
15898   funding-type       = {costs}, %staff, costs, ...
15899   category   = {Sequencing and data analysis},
15900   sum = {
15901     year1      =  {1000},
15902     year2      =  {2000},
15903     year3      =  {3000},
15904     year4      =  {4000},
15905     year5      =  {5000}
15906    }}
15907
15908 \addEntry[funding]{RequestedFunding}{
15909   ref-project-id = {A01},
15910   funding-type       = {instrumentation}, %staff, costs, ...
15911   category           = {Live-cell imaging system},
15912   sum = {
15913     year1      =  {1000},
15914     year2      =  {2000},
15915     year3      =  {3000},
15916     year4      =  {4000},
15917     year5      =  {5000}
15918    }}
15919
15920 \printTable{{RequestedFunding}{A01}}
15921
15922
15923
15924
15925
15926 \subsubsection{Requested funding for staff for the new funding period}
15927 \addEntry[staff]{RequestedFundingStaff}{
15928   ref-project-id     = {A01},
15929   funding-status     = {existing},
15930   staff-type         = {research},
15931   ref-person-id      = {per1},
15932   field-of-research  = {Microbiology and systems medicine},
15933   commitment         = {8},
15934   category           = {prof},
15935   funding-source     = {State funds},
15936   description        = {Leads the experimental programme and supervises the
15937     research staff assigned to the project.}
15938 }
15939
15940 \addEntry[staff]{RequestedFundingStaff}{
15941   ref-project-id    = {A01},
15942   funding-status    = {existing},
15943   staff-type        = {nonResearch},
15944   firstname         = {Alex},
15945   lastname          = {Example},
15946   academic-degree   = {M.Sc.},
15947   positionText      = {Technical assistant},
15948   field-of-research = {Laboratory management},
15949   ref-institution-id = {ins1},
15950   commitment        = {20},
15951   category          = {nonResearch},
15952   funding-source    = {Core support},
```

```
15953   description       = {Maintains shared laboratory workflows and coordinates
15954     sample logistics.}
15955 }
15956
15957 \addEntry[staff]{RequestedFundingStaff}{
15958   ref-project-id     = {A01},
15959   funding-status     = {requested},
15960   staff-type         = {research},
15961   firstname          = {Researcher},
15962   lastname           = {TBA},
15963   academic-degree    = {Dr. rer. nat.},
15964   positionText       = {Postdoctoral researcher},
15965   field-of-research  = {Host--microbiome interactions},
15966   ref-institution-id = {ins2},
15967   category           = {postDoc},
15968   description        = {Performs the planned mechanistic experiments and
15969     integrates molecular and clinical datasets.}
15970 }
15971
15972 \addEntry[staff]{RequestedFundingStaff}{
15973   ref-project-id    = {A01},
15974   funding-status    = {requested},
15975   staff-type        = {nonResearch},
15976   firstname         = {Taylor},
15977   lastname          = {Demo},
15978   academic-degree   = {B.Sc.},
15979   positionText      = {Research assistant},
15980   field-of-research = {Biobanking and data curation},
15981   ref-institution-id = {ins2},
15982   category          = {assistant},
15983   description       = {Supports biobanking, quality control, and structured
15984     curation of the project datasets.}
15985 }
15986
15987 \printTable{{RequestedFundingStaff}{A01}}
15988
15989 \subsubsection{Requested funding for direct costs for the new funding
15990   period}
15991
15992 \addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
15993   ref-project-id = {A01},
15994   ref-affiliation-id = {aff2},
15995   financed-by = {Partner University core support},
15996   sum = {
15997   year1      =  {1200},
15998   year2      =  {500},
15999   year3      =  {21100},
16000   year4      =  {5600},
16001   year5      =  {3400}
16002  }}
16003
16004 \addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
16005   ref-project-id = {A01},
16006   ref-affiliation-id = {aff1},
16007   financed-by = {Example University core support},
16008   sum = {
16009   year1      =  {11100},
16010   year2      =  {53800},
16011   year3      =  {21100},
16012   year4      =  {600},
16013   year5      =  {38300}
16014  }}
16015
16016
16017 \printTable{{RequestedAndExistingFundingDirectCosts}{A01}}
16018
16019 \begin{FundingDirectCostsTable}[Text]
16020   Laboratory consumables and antibodies & EUR & 15000 \\
16021   Single-cell sequencing and spatial profiling & EUR & 12000 \\
16022   Research data storage and analysis & EUR & 3000 \\
16023 \end{FundingDirectCostsTable}
16024
16025
16026 \subsubsection{Requested funding for instrumentation for the new funding
16027   period}
16028
16029
16030 \begin{FundingDirectCostsTable}[Text]
16031   Automated live-cell imaging system & EUR & 95000 \\
16032   Environmental control module & EUR & 18000 \\
16033   Analysis workstation and software & EUR & 7000 \\
16034 \end{FundingDirectCostsTable}
16035
16036
16037 \printProjectTail
```

6.2.13 Complete new-project example

This file demonstrates the complete heading and command structure for a new research project (`project-status=N`). Its scientific-background section is intentionally named differently from the progress section of a continuing project.

```
\printProjectHead{A02}

\section{Summary}
\label{A02:summary}

Summarise the new project's central hypothesis, methods, expected results, and
contribution to the Collaborative Research Centre.

\section{Research rationale}
\label{A02:research-rationale}

\subsection{Current state of research and preliminary work}
\label{A02:current-state-of-research-and-preliminary-work}

Describe the relevant state of research, preliminary work, open questions,
and the scientific rationale for the proposed project.

\FloatBarrier

\projectPublicationsHeading
\projectNocite{FAIR2016}
\printProjectBibliography[heading=none]

\section{Project plan}
\label{A02:project-plan}

\begin{aims}
  \addAim{1} Establish the experimental and analytical framework.
  \addAim{2} Test the central hypothesis and validate the findings.
\end{aims}

\begin{workpackages}
  \addWP{wp1}{
    name={Framework and pilot experiments},
    description={Establish the framework and complete pilot experiments},
    periodStart={1},
    periodEnd={7},
    style={default},
  }

  Describe the methods, milestones, expected results, and risks of
  \getWP{wp1} and its contribution to \getAim{1}.

  \addWP{wp2}{
    name={Hypothesis testing and validation},
    description={Test the hypothesis and validate the findings},
    periodStart={6},
    periodEnd={16},
    style={important},
  }

  Describe the validation strategy and contingency plan for \getWP{wp2} and
  \getAim{2}.
\end{workpackages}

\printPlot{{Gantt}{A02}}

\FloatBarrier

\section{Relevance of sex, gender and/or diversity in the research project}
\label{A02:relevance-of-sex-gender-and-diversity}

Explain whether and how sex, gender, and diversity affect the research
questions, design, analysis, and interpretation.

\section{Role within the Collaborative Research Centre}
\label{A02:role-within-the-collaborative-research-centre}

Describe the project's contribution to the CRC and its principal
collaborations.

\printPlot{{ProjectCollaboration}{A02}}

\section{Differentiation from other funded projects of the project leader(s)}
\label{A02:differentiation-from-other-funded-projects}

Explain how this new project differs from, and complements, other funded work
of the project leaders.

```

```
\section{Project funding}
\label{A02:project-funding}

\subsection{Previous funding}

State that this is a new CRC project and list related previous funding of the
project leaders where applicable.

\subsection{Requested funding}

%% Automatic staff-cost example: sum is intentionally omitted; the class
%% halves the calculated amounts in year1 and year5.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A02},
  funding-type   = {staff},
  category       = {postDoc},
  percentage     = {100},
  quantity = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  }
}

%% Explicit staff-cost example: sum overrides the central salary calculation;
%% the concrete year1 and year5 amounts below already represent half-years.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A02},
  funding-type   = {staff},
  category       = {PhD},
  percentage     = {65},
  quantity = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  },
  sum = {
    year1 = {26500},
    year2 = {53000},
    year3 = {53000},
    year4 = {53000},
    year5 = {26500}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A02},
  funding-type   = {costs},
  category       = {Consumables},
  sum = {
    year1 = {2000},
    year2 = {2000},
    year3 = {2000},
    year4 = {2000},
    year5 = {2000}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {A02},
  funding-type   = {instrumentation},
  category       = {Specialised instrument},
  sum = {
    year1 = {5000},
    year2 = {0},
    year3 = {0},
    year4 = {0},
    year5 = {0}
  }
}

\printTable{{RequestedFunding}{A02}}

\subsubsection{Requested funding for staff}

%% When no person database row is needed, all display data can be supplied
%% directly.  Leave academic-degree empty when it is unknown.
\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id     = {A02},
  funding-status     = {requested},
  staff-type         = {research},
  firstname          = {Researcher},
  lastname           = {TBA},
```

```
16204   academic-degree    = {},
16205   positionText       = {Postdoctoral researcher},
16206   field-of-research  = {Experimental systems medicine},
16207   ref-institution-id = {ins1},
16208   category           = {postDoc},
16209   description        = {Establishes the new experimental platform, performs
16210     the mechanistic studies, and analyses the resulting datasets.}
16211 }
16212
16213 \printTable{{RequestedFundingStaff}{A02}}
16214
16215 Provide the scientific and organisational justification for every requested
16216 staff position.
16217
16218 \subsubsection{Requested funding of direct costs}
16219
16220 \addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
16221   ref-project-id     = {A02},
16222   ref-affiliation-id = {aff1},
16223   financed-by        = {State funds},
16224   sum = {
16225     year1 = {500},
16226     year2 = {500},
16227     year3 = {500},
16228     year4 = {500},
16229     year5 = {500}
16230   }
16231 }
16232
16233 \printTable{{RequestedAndExistingFundingDirectCosts}{A02}}
16234
16235 \begin{FundingDirectCostsTable}[Text]
16236   Consumables for pilot experiments & EUR & 10000 \\
16237   External data analysis & EUR & 5000 \\
16238 \end{FundingDirectCostsTable}
16239
16240 \subsubsection{Requested funding for instrumentation}
16241
16242 \begin{FundingDirectCostsTable}[Text]
16243   Specialised instrument including installation & EUR & 5000 \\
16244 \end{FundingDirectCostsTable}
16245
16246 \printProjectTail
```

6.2.14 Second continuing-project example

This compact but complete file demonstrates another continuing research project and keeps the funding-category examples used by the dummy proposal.

```
16247 \printProjectHead{B01}
16248
16249 \section{Summary}
16250 \label{B01:summary}
16251
16252 Summarise the continuing project and the work proposed for the new funding
16253 period.
16254
16255 \section{Project progress to date}
16256 \label{B01:project-progress-to-date}
16257
16258 \subsection{Report and current state of research}
16259 \label{B01:report-and-current-state-of-research}
16260
16261 Report the previous work, published and unpublished results, and the current
16262 state of research.
16263
16264 \FloatBarrier
16265
16266 \projectPublicationsHeading
16267 \projectNocite{FAIR2016}
16268 \printProjectBibliography[heading=none]
16269
16270 \section{Project plan}
16271 \label{B01:project-plan}
16272
16273 \begin{aims}
16274   \addAim{1} Complete the mechanistic analysis.
16275   \addAim{2} Validate the findings with CRC partners.
16276 \end{aims}
16277
16278 \begin{workpackages}
16279   \addWP{wp1}{
16280     name={Mechanistic analysis},
16281     description={Complete the mechanistic analysis},
```

```
    periodStart={1},
    periodEnd={10},
    style={default},
  }
  Describe \getWP{wp1}, including methods, milestones, and risks.

  \addWP{wp2}{
    name={Collaborative validation},
    description={Validate the findings with CRC partners},
    periodStart={8},
    periodEnd={16},
    style={alternative},
  }
  Describe \getWP{wp2} and its links to \getAim{2}.
\end{workpackages}

\printPlot{{Gantt}{B01}}

\FloatBarrier

\section{Relevance of sex, gender and/or diversity in the research project}
\label{B01:relevance-of-sex-gender-and-diversity}

Explain the relevance of sex, gender, and diversity, or justify why they do
not affect this project's research questions and methods.

\section{Role within the Collaborative Research Centre}
\label{B01:role-within-the-collaborative-research-centre}

Describe the role of the project and its reciprocal collaborations.

\printPlot{{ProjectCollaboration}{B01}}

\section{Differentiation from other funded projects}
\label{B01:differentiation-from-other-funded-projects}

Differentiate the project from other funded work of its project leaders.

\section{Project funding}
\label{B01:project-funding}

\subsection{Previous funding}

State when CRC funding began and list other relevant previous funding.

\subsection{Requested funding}

%% Automatic staff-cost example: sum is intentionally omitted; the class
%% halves the calculated amounts in year1 and year5.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {B01},
  funding-type   = {staff},
  category       = {groupLeader},
  percentage     = {100},
  quantity = {
    year1 = {2},
    year2 = {2},
    year3 = {2},
    year4 = {2},
    year5 = {2}
  }
}

%% Explicit staff-cost example: sum overrides the central salary calculation;
%% the concrete year1 and year5 amounts below already represent half-years.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {B01},
  funding-type   = {staff},
  category       = {postDoc},
  percentage     = {100},
  quantity = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  },
  sum = {
    year1 = {45000},
    year2 = {90000},
    year3 = {90000},
    year4 = {90000},
    year5 = {45000}
  }
}

\addEntry[funding]{RequestedFunding}{
```

```
16369   ref-project-id = {B01},
16370   funding-type       = {globalfunds}, %staff, costs, ...
16371   category   = {Project-wide data stewardship},
16372   sum = {
16373     year1 = {1000},
16374     year2 = {2000},
16375     year3 = {3000},
16376     year4 = {4000},
16377     year5 = {5000}
16378   }
16379 }
16380
16381 \printTable{{RequestedFunding}{B01}}
16382
16383 \subsubsection{Requested funding for staff for the new funding period}
16384
16385 \addEntry[staff]{RequestedFundingStaff}{
16386   ref-project-id     = {B01},
16387   funding-status     = {requested},
16388   staff-type         = {research},
16389   firstname          = {Researcher},
16390   lastname           = {TBA},
16391   academic-degree    = {},
16392   positionText       = {Research group leader},
16393   field-of-research  = {Translational microbiology},
16394   ref-institution-id = {ins2},
16395   category           = {groupLeader},
16396   description        = {Coordinates the experimental programme and the
16397     collaborative validation studies.}
16398 }
16399
16400 \printTable{{RequestedFundingStaff}{B01}}
16401
16402 \subsubsection{Requested funding for direct costs for the new funding period}
16403
16404 \addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
16405   ref-project-id     = {B01},
16406   ref-affiliation-id = {aff2},
16407   financed-by        = {Institutional funds},
16408   sum = {
16409     year1 = {1000},
16410     year2 = {1000},
16411     year3 = {1000},
16412     year4 = {1000},
16413     year5 = {1000}
16414   }
16415 }
16416
16417 \printTable{{RequestedAndExistingFundingDirectCosts}{B01}}
16418
16419 \begin{FundingDirectCostsTable}[Text]
16420   Experimental consumables & EUR & 10000 \\
16421   Data-processing services & EUR & 5000 \\
16422 \end{FundingDirectCostsTable}
16423
16424 \subsubsection{Requested funding for instrumentation for the new funding
16425   period}
16426
16427 \begin{FundingDirectCostsTable}[Text]
16428   No instrumentation requested & EUR & 0 \\
16429 \end{FundingDirectCostsTable}
16430
16431 \printProjectTail
```

6.2.15 Complete ending-project example

Ending projects use a deliberately shorter DFG structure: after the general information printed by `\printProjectHead`, they report project history, published results, and closing funding information. They do not contain a new project plan or a request for another funding period.

```
16432 \printProjectHead{A00}
16433
16434 \section{Project history}
16435 \label{A00:project-history}
16436
16437 \subsection{Report}
16438 \label{A00:report}
16439
16440 \begin{projectReport}
16441 Describe the complete funding history, objectives, methods, results, and
16442 contribution of the ending project.  Citations used only in the report are
16443 printed automatically at the end of this environment.  This dummy citation
16444 also demonstrates the exclusion of a result listed below
16445 \autocite{FAIR2016}.
```

```
\end{projectReport}

\subsection{Published project results}
\label{A00:published-project-results}

%% Replace the example key with the project's results.  Use the literal value
%% none only when a category genuinely produced no result; leaving a value
%% empty keeps a visible drafting instruction.
\printProjectResults{
  quality-assured = {FAIR2016},
  other           = none,
  patents         = none
}

\section{Funding}
\label{A00:funding}

Funding of this project within the Collaborative Research Centre started in
April 2018.  The project will be completed by the end of the current funding
period.  Replace these dates and, where applicable, add previous funding under
another DFG programme and its reference number.

\subsection{Project staff in the ending funding period}
\label{A00:project-staff-in-the-ending-funding-period}

%% For an ending project, funding-status=requested is printed as
%% ``Staff funded with approved grant money'', matching the DFG form.
\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id     = {A00},
  funding-status     = {existing},
  staff-type         = {research},
  ref-person-id      = {per4},
  commitment         = {8},
  funding-source     = {State funds},
  field-of-research  = {Molecular tissue regeneration},
  description        = {Led the project, supervised the experimental work,
    and integrated its results across the CRC throughout the ending funding
    period.}
}

\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id     = {A00},
  funding-status     = {requested},
  staff-type         = {research},
  firstname          = {Researcher},
  lastname           = {TBA},
  academic-degree    = {},
  positionText       = {Postdoctoral researcher},
  field-of-research  = {Regenerative cell biology},
  ref-institution-id = {ins2},
  category           = {postDoc},
  description        = {Completed the final validation experiments and data
    publication from April 2025 until the end of the current funding period.}
}

\printTable{{RequestedFundingStaff}{A00}}

\printProjectTail
```

6.2.16 Complete continuing service-project example

Service projects use the continuing-project headings while the prose and work packages focus on services, resources, access rules, and collaborative use.

```

\printProjectHead{Q01}

\section{Summary}
\label{Q01:summary}

Summarise the service portfolio, its scientific purpose, its users, and the
developments planned for the new funding period.

\section{Project progress to date}
\label{Q01:project-progress-to-date}

\subsection{Report and current state of research}
\label{Q01:report-and-current-state-of-research}

Report completed and ongoing services, usage by CRC projects, technical
developments, quality assurance, and relevant service-project research.

\FloatBarrier

```

```
\projectPublicationsHeading
\projectNocite{FAIR2016}
\printProjectBibliography[heading=none]

\section{Project plan}
\label{Q01:project-plan}

\begin{aims}
  \addAim{1} Maintain and quality-assure the core service portfolio.
  \addAim{2} Extend the portfolio for new collaborative requirements.
\end{aims}

\begin{workpackages}
  \addWP{wp1}{
    name={Core service provision},
    description={Provide and quality-assure core services},
    periodStart={1},
    periodEnd={16},
    style={default},
  }
  Describe access, sample or data handover, turnaround, quality assurance,
  responsibilities, and capacity for \getWP{wp1}.

  \addWP{wp2}{
    name={Service development},
    description={Develop and validate new service modules},
    periodStart={3},
    periodEnd={14},
    style={alternative},
  }
  Describe development milestones and the projects participating in
  \getWP{wp2}.
\end{workpackages}

\printPlot{{Gantt}{Q01}}

\FloatBarrier

\section{Relevance of sex, gender and/or diversity in the research project}
\label{Q01:relevance-of-sex-gender-and-diversity}

Explain relevance for the service methods and the studies supported by them.

\section{Role within the Collaborative Research Centre}
\label{Q01:role-within-the-collaborative-research-centre}

Describe which projects use the services and how priorities, access, and
collaborations are coordinated.

\printPlot{{ProjectCollaboration}{Q01}}

\section{Differentiation from other funded projects}
\label{Q01:differentiation-from-other-funded-projects}

Differentiate the CRC service from existing institutional infrastructure and
other funded service activities.

\section{Project funding}
\label{Q01:project-funding}

\subsection{Previous funding}

State when the service project began and describe relevant previous funding.

\subsection{Requested funding}

%% Automatic staff-cost example: sum is intentionally omitted; the class
%% halves the calculated amounts in year1 and year5.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q01},
  funding-type   = {staff},
  category       = {nonResearch},
  percentage     = {65},
  quantity = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  }
}

%% Explicit staff-cost example: sum overrides the central salary calculation;
%% the concrete year1 and year5 amounts below already represent half-years.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q01},
  funding-type   = {staff},
```

```
16611   category       = {assistant},
16612   percentage     = {50},
16613   quantity = {
16614     year1 = {1},
16615     year2 = {1},
16616     year3 = {1},
16617     year4 = {1},
16618     year5 = {1}
16619   },
16620   sum = {
16621     year1 = {9000},
16622     year2 = {18000},
16623     year3 = {18000},
16624     year4 = {18000},
16625     year5 = {9000}
16626   }
16627 }
16628
16629 \addEntry[funding]{RequestedFunding}{
16630   ref-project-id = {Q01},
16631   funding-type       = {costs}, %staff, costs, ...
16632   category   = {Invitation},
16633   sum = {
16634     year1 = {1000},
16635     year2 = {2000},
16636     year3 = {3000},
16637     year4 = {4000},
16638     year5 = {5000}
16639   }
16640 }
16641
16642 \printTable{{RequestedFunding}{Q01}}
16643
16644 \subsubsection{Requested funding for staff for the new funding period}
16645
16646 \addEntry[staff]{RequestedFundingStaff}{
16647   ref-project-id     = {Q01},
16648   funding-status     = {requested},
16649   staff-type         = {nonResearch},
16650   firstname          = {Service},
16651   lastname           = {Specialist},
16652   academic-degree    = {},
16653   positionText       = {Technical specialist},
16654   field-of-research  = {Core-facility services},
16655   ref-institution-id = {ins1},
16656   category           = {nonResearch},
16657   description        = {Operates the service platform, performs quality
16658     control, and supports participating projects.}
16659 }
16660
16661 \printTable{{RequestedFundingStaff}{Q01}}
16662
16663 \subsubsection{Requested funding for direct costs for the new funding period}
16664
16665 \addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
16666   ref-project-id     = {Q01},
16667   ref-affiliation-id = {aff1},
16668   financed-by        = {Core-facility funds},
16669   sum = {
16670     year1 = {1000},
16671     year2 = {1000},
16672     year3 = {1000},
16673     year4 = {1000},
16674     year5 = {1000}
16675   }
16676 }
16677
16678 \printTable{{RequestedAndExistingFundingDirectCosts}{Q01}}
16679
16680 \begin{FundingDirectCostsTable}[Text]
16681   Service-specific consumables & EUR & 10000 \\
16682   Maintenance and quality assurance & EUR & 5000 \\
16683 \end{FundingDirectCostsTable}
16684
16685 \subsubsection{Requested funding for instrumentation for the new funding
16686   period}
16687
16688 \begin{FundingDirectCostsTable}[Text]
16689   Service-platform upgrade & EUR & 5000 \\
16690 \end{FundingDirectCostsTable}
16691
16692 \printProjectTail
```

6.2.17 Complete integrated research training group example

The MGK/IRTG project type has its own required narrative and funding headings. This file mirrors that structure instead of pretending it is a conventional research project.

```
\printProjectHead{Q02}

\section{Summary}
\label{Q02:summary}

Summarise the objectives, development, and added value of the Integrated
Research Training Group.

\section{Tabular report}
\label{Q02:tabular-report}

Provide the required tabular report of all researchers funded by the
Integrated Research Training Group to date.

\section{Qualification programme}
\label{Q02:qualification-programme}

Describe the qualification programme, workshops, courses, retreats, career
development measures, and other training activities.

\section{Management and supervision}
\label{Q02:management-and-supervision}

Describe management responsibilities, recruitment, supervision arrangements,
mentoring, conflict-resolution processes, and quality assurance.

\section{Environment of the Integrated Research Training Group}
\label{Q02:environment-of-the-integrated-research-training-group}

Describe the scientific, institutional, interdisciplinary, and international
training environment.

\section{Interim assessment and outlook}
\label{Q02:interim-assessment-and-outlook}

Assess progress during the current funding period and describe priorities and
measurable objectives for the next funding period.

\section{Project funding}
\label{Q02:project-funding}

\subsection{Previous funding}

The Integrated Research Training Group has been funded within the
Collaborative Research Centre since April 2018.  Replace this date and add any
relevant funding under another programme.

%% Declare all monetary rows before the existing-funds overview: that table
%% compares existing direct-cost resources with the requested direct costs.
%% Automatic staff-cost example: sum is intentionally omitted; the class
%% halves the calculated amounts in year1 and year5.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q02},
  funding-type   = {staff},
  category       = {PhD},
  percentage     = {65},
  quantity = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  }
}

%% Explicit staff-cost example: sum overrides the central salary calculation;
%% the concrete year1 and year5 amounts below already represent half-years.
\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q02},
  funding-type   = {staff},
  category       = {nonResearch},
  percentage     = {50},
  quantity = {
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  },
  sum = {
    year1 = {15000},
```

```
    year2 = {30000},
    year3 = {30000},
    year4 = {30000},
    year5 = {15000}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q02},
  funding-type   = {costs},
  category       = {Qualification programme},
  sum = {
    year1 = {2000},
    year2 = {2000},
    year3 = {2000},
    year4 = {2000},
    year5 = {2000}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q02},
  funding-type   = {fellowships},
  category       = {Short-term fellowships},
  sum = {
    year1 = {1000},
    year2 = {1000},
    year3 = {1000},
    year4 = {1000},
    year5 = {1000}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q02},
  funding-type   = {globalfunds},
  category       = {Training and networking},
  sum = {
    year1 = {3000},
    year2 = {3000},
    year3 = {3000},
    year4 = {3000},
    year5 = {3000}
  }
}

\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {Q02},
  ref-affiliation-id = {aff2},
  financed-by        = {Graduate-school funds},
  sum = {
    year1 = {1000},
    year2 = {1000},
    year3 = {1000},
    year4 = {1000},
    year5 = {1000}
  }
}

\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id     = {Q02},
  funding-status     = {existing},
  staff-type         = {nonResearch},
  firstname          = {Taylor},
  lastname           = {Trainer},
  academic-degree    = {},
  positionText       = {Training coordinator},
  field-of-research  = {Structured doctoral training},
  ref-institution-id = {ins2},
  commitment         = {20},
  funding-source     = {Partner University core funds},
  description        = {Coordinates recruitment, course administration,
    progress monitoring, mentoring, and quality assurance for the
    qualification programme.}
}

\subsection{Existing funds for the new funding period}

Describe the coordination, training, staff, and direct-cost resources made
available to the Integrated Research Training Group by the participating
institutions.

\subsubsection*{Existing staff}

\printTable{{RequestedFundingStaff}{Q02}}

\printTable{{RequestedAndExistingFundingDirectCosts}{Q02}}
```

```
16861
16862 \subsection{Requested funding}
16863
16864 \printTable{{RequestedFunding}{Q02}}
16865
16866 \subsection{Requested funding for staff}
16867
16868 \begin{FundingDirectCostsTable}[Text]
16869   Doctoral-researcher position: interdisciplinary research and participation
16870     in the complete qualification programme & EUR & 5000 \\
16871 \end{FundingDirectCostsTable}
16872
16873 \subsection{Requested funding for direct costs}
16874
16875 \begin{FundingDirectCostsTable}[Text]
16876   Workshops and transferable-skills courses & EUR & 10000 \\
16877   Visiting researchers and networking & EUR & 5000 \\
16878 \end{FundingDirectCostsTable}
16879
16880 \subsection{Requested funding for fellowships}
16881
16882 \begin{FundingDirectCostsTable}[Text]
16883   Short-term research fellowships: state the type of fellowship, average
16884     number of fellows, average duration, monthly allowance per person, and
16885     justification & EUR & 5000 \\
16886 \end{FundingDirectCostsTable}
16887
16888 \subsection{Requested global funds}
16889
16890 \begin{FundingDirectCostsTable}[Text]
16891   Training and networking measures: explain the purpose, allocation rules,
16892     selection process, and administration & EUR & 15000 \\
16893 \end{FundingDirectCostsTable}
16894
16895 \printProjectTail
```

6.2.18 Complete central administrative-project example

The central administrative project has a dedicated DFG structure. It does not reuse the summary, research-progress, publication, project-plan, or instrumentation headings of a continuing research project.

```
16896 \printProjectHead{Q03}
16897
16898 \section{Documentation of the central activities of the Collaborative
16899   Research Centre}
16900 \label{Q03:documentation-of-central-activities}
16901
16902 Document governance and decision-making, scientific coordination, reporting,
16903 internal communication, data stewardship, publication support, equal-
16904 opportunity measures, early-career support, outreach, and the interfaces with
16905 all applicant and participating institutions.  Distinguish CRC-specific tasks
16906 from permanent institutional duties and state responsibilities, workflows,
16907 results from the current period, and priorities for the new funding period.
16908
16909 \section{Project funding}
16910 \label{Q03:project-funding}
16911
16912 %% The existing-funds overview includes the requested direct-cost total, so
16913 %% all monetary records are declared before either table is printed.
16914 %% Automatic staff-cost example: sum is intentionally omitted.  The class
16915 %% applies the half-year factor to year1 and year5.
16916 \addEntry[funding]{RequestedFunding}{
16917   ref-project-id = {Q03},
16918   funding-type   = {staff},
16919   category       = {nonResearch},
16920   percentage     = {100},
16921   quantity = {
16922     year1 = {1},
16923     year2 = {1},
16924     year3 = {1},
16925     year4 = {1},
16926     year5 = {1}
16927   }
16928 }
16929
16930 %% Explicit staff-cost example: the stated sums override the salary settings;
16931 %% year1 and year5 below already represent half-years.
16932 \addEntry[funding]{RequestedFunding}{
16933   ref-project-id = {Q03},
16934   funding-type   = {staff},
16935   category       = {assistant},
16936   percentage     = {50},
16937   quantity = {
```

```
    year1 = {1},
    year2 = {1},
    year3 = {1},
    year4 = {1},
    year5 = {1}
  },
  sum = {
    year1 = {12500},
    year2 = {25000},
    year3 = {25000},
    year4 = {25000},
    year5 = {12500}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q03},
  funding-type   = {costs},
  category       = {Central coordination and communication},
  sum = {
    year1 = {2000},
    year2 = {2000},
    year3 = {2000},
    year4 = {2000},
    year5 = {2000}
  }
}

\addEntry[funding]{RequestedFunding}{
  ref-project-id = {Q03},
  funding-type   = {globalfunds},
  category       = {Central measures},
  sum = {
    year1 = {2000},
    year2 = {2000},
    year3 = {2000},
    year4 = {2000},
    year5 = {2000}
  }
}

\addEntry[staff]{RequestedFundingStaff}{
  ref-project-id     = {Q03},
  funding-status     = {existing},
  staff-type         = {nonResearch},
  ref-person-id      = {man1},
  field-of-research  = {Research management},
  commitment         = {20},
  funding-source     = {Partner University core funds},
  description        = {Coordinates governance, reporting, internal
    communication, central measures, and interfaces with applicant and
    participating institutions.}
}

\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {Q03},
  ref-affiliation-id = {aff1},
  financed-by        = {Central university funds},
  sum = {
    year1 = {1000},
    year2 = {1000},
    year3 = {1000},
    year4 = {1000},
    year5 = {1000}
  }
}

\subsection{Existing funds for the new funding period}

Describe the central staff, facilities, and direct-cost resources contributed
by the applicant and participating institutions.

\subsubsection*{Existing staff}

\printTable{{RequestedFundingStaff}{Q03}}

\printTable{{RequestedAndExistingFundingDirectCosts}{Q03}}

\subsection{Requested funding for the new funding period}

\printTable{{RequestedFunding}{Q03}}

\subsection{Requested funding for staff}

\begin{FundingDirectCostsTable}[Text]
  Scientific coordination: justify the category, percentage, period, and
    CRC-specific responsibilities & EUR & 5000 \\
```

```
\end{FundingDirectCostsTable}

\subsection{Requested funding for direct costs}

\begin{FundingDirectCostsTable}[Text]
  Governance meetings and scientific exchange & EUR & 10000 \\
  Communication and outreach & EUR & 5000 \\
\end{FundingDirectCostsTable}

\subsection{Requested global funds}

\begin{FundingDirectCostsTable}[Text]
  Central measures: explain the purpose, allocation rules, selection process,
    and administration & EUR & 10000 \\
\end{FundingDirectCostsTable}

\printProjectTail


\chapter{Bylaws of the Collaborative Research Centre} %4
\printThumbIndex{4}
\cohead{Bylaws of the \crc}%
\cehead{Bylaws of the \crc}%

\begin{bylaws}
  \item Name, coordinating
    \dfgApplicantAffiliationsTF{university}{universities}, and purpose of the \crc
  \begin{bylaws}
 \item The \crc (CRC) \enquote{\dfgCRCName\space --\space\dfgCRCNameAddon}
      is a legally dependent institution of \theCRCaffiliation.

    \item The \crc conducts interrelated research on microbial signalling,
      epithelial barriers, immune regulation, and organ function.  It is
      divided into research project areas and central projects.

    \item The \crc promotes interaction with other research institutions,
      supports researchers in early career phases, and fosters responsible
      international cooperation, equity, and equal opportunity.
  \end{bylaws}

  \item Membership and governing bodies
  \begin{bylaws}
    \item Members are the project leaders and persons appointed to central
      scientific or administrative responsibilities by the general assembly.
    \item The governing bodies are the general assembly, the spokesperson, and
      the steering committee.  The general assembly elects the spokesperson
      and steering committee for the applicable funding period.
    \item The steering committee prepares strategic and budgetary decisions,
      monitors milestones, and reports to the general assembly at least once
      per year.
  \end{bylaws}

  \item Decisions and allocation of shared resources
  \begin{bylaws}
    \item Decisions are normally taken by simple majority of members present;
      changes to these bylaws require a two-thirds majority.
    \item Shared funds and infrastructure are allocated through documented,
      transparent procedures based on scientific need, feasibility, and equal
      access across participating projects.
    \item Members disclose conflicts of interest and do not participate in the
      corresponding deliberation or decision.
  \end{bylaws}

  \item Research integrity, equity, and conflict resolution
  \begin{bylaws}
    \item All members follow the applicable rules of good research practice,
      data protection, animal welfare, research ethics, and occupational safety.
    \item The CRC implements equal-opportunity and accessibility measures and
      monitors participation in recruitment, training, and governance.
    \item Concerns may be raised confidentially with the spokesperson, the CRC
      management, or the competent independent offices of the participating
      institutions.
  \end{bylaws}

  \item Research-data governance
  \begin{bylaws}
    \item Projects document data, code, samples, access rights, retention, and
      publication in project-specific data-management plans.
    \item Central data stewardship defines common metadata and quality
      standards, manages controlled access, and supports publication according
      to the FAIR principles and applicable legal requirements.
  \end{bylaws}
\end{bylaws}

\chapter{Declarations} %5
\printThumbIndex{5}
I/We hereby confirm the accuracy of the information contained in the proposal, including the research profiles.
```

```
17110
17111  The \emph{DFG Rules of Procedure for Dealing with Scientific Misconduct}
17112  (\emph{known as VerfOwF})
17113  apply to individuals with a high level of scientific/\-academic responsibility in connection with funding proposals s
       academic responsibility in the project proposal submitted by the \dfgApplicantAffiliationsTF{university}{universities
17114
17115  In submitting this proposal, all project leaders undertake to comply with the
17116  relevant discipline-specific research ethics guidelines, to determine whether
17117  a statement from the responsible ethics committee is required and, where
17118  necessary, to obtain it in good time before the start of funding.  They will
17119  inform the DFG immediately if an ethics committee expresses substantive
17120  concerns or issues a negative statement and will submit the statement if the
17121  DFG requests it.
17122
17123  The DFG takes the protection of your personal data very seriously.
17124  Please note the DFG's data protection notice on research funding, which can be viewed and downloaded at
17125  \url{www.dfg.de/privacy_policy}
17126
17127  Please forward this information to those individuals whose data will be processed by the DFG because they are mention
17128
17129  I/We have read the DFG's data protection notice.
17130
17131  I/We consent to the processing and transfer of special categories of personal
17132  data (including the data of all parties involved) for the purposes stated in the
17133  Privacy Policy if, after careful consideration, I/we provide such data either
17134  now or at a later date.\par
17135  \vfill
17136
17137  \printSignaturesWithRole{spokesperson}{(Spokesperson~of~\crc)}
17138  \printSignaturesWithRole{rector}
17139    {(Rector,~chancellor,~or~president~of~
17140      \getEntry[affiliation]{\theAffiliationOfRoleID}{name})}
```

6.2.19 Backmatter File

The final page suppresses running headers and page numbers and delegates the complete layout to the project-overview table defined by the class.

```
17141  \pagestyle{empty}
17142  \printTable{projectOverview}
```

6.2.20 Project collaboration plot test

This focused test is deliberately independent of the proposal metadata. It contains active, ending, and reference-only projects; duplicate and reverse-only declarations; and calls both supported collaboration layouts.

```
17143  %% Compile from the generated local directory with:
17144  %%   lualatex -interaction=nonstopmode -halt-on-error test-project-collaboration.tex
17145  %%
17146  %% This test is deliberately self-contained.  The image switch is initially
17147  %% false so the fast project option prevents the proposal metadata from loading.
17148  %% Immediately after \begin{document} the image switch is enabled so dummy
17149  %% project entries are retained and the plots below exercise their focused-mode
17150  %% dispatch without disabling the other fast-mode reductions.
17151  \documentclass[
17152    fastcompile-project = true,
17153    fastcompile-project-show-images = false,
17154    english
17155  ]{dfg-crc}
17156
17157  \begin{document}
17158
17159  \ExplSyntaxOn
17160  \bool_set_true:N \l_@@_class_fastcompile_project_show_images_bool
17161  \ExplSyntaxOff
17162
17163  %% #1 is either empty or a complete project-area assignment; #2 is the
17164  %% project id, #3 status (C/N/E), #4 collaborations, and #5 reference-only.
17165  \NewDocumentCommand{\AddDummyProjectEntry}{m m m m m}{
17166    \addEntry{project}{
17167      id                = {#2},
17168      #1
17169      number            = {#2},
17170      name              = {Dummy project #2},
17171      project-status    = {#3},
17172      project-type      = {research},
17173      consider-for-lists = {true},
17174      reference-only    = {#5},
17175      ref-person-id     = {},
17176      collaborations    = {#4},
```

```
17177     research-area      = {},
17178     legal-issues       = {
17179       L1  = {no},
17180       L1a = {},
17181       L2  = {no},
17182       L2a = {},
17183       L3  = {no},
17184       L4  = {no},
17185       L5  = {no},
17186       L5a = {},
17187       L6  = {no},
17188       L7  = {no},
17189       L7a = {}
17190     }
17191   }
17192 }
17193 %% Omitting the optional area leaves the project-area database column absent
17194 %% until the first plot and therefore exercises both schema and ID fallbacks.
17195 \NewDocumentCommand{\AddDummyProject}{o m m m m}{
17196   \IfNoValueTF{#1}
17197     { \AddDummyProjectEntry{}{#2}{#3}{#4}{#5} }
17198     { \AddDummyProjectEntry{project-area={#1},}{#2}{#3}{#4}{#5} }
17199 }
17200
17201 %% Active project area A.  A11 repeats B06 deliberately to test that an
17202 %% undirected relation declared twice is collapsed.
17203 \AddDummyProject{A02}{C}{}{false}
17204 \AddDummyProject{A03}{C}{}{false}
17205 \AddDummyProject{A05}{C}{}{false}
17206 \AddDummyProject{A09}{C}{}{false}
17207 \AddDummyProject{A10}{C}{}{false}
17208 \AddDummyProject{A11}{C}{B06}{false}
17209 \AddDummyProject{A12}{N}{}{false}
17210 \AddDummyProject{A13}{N}{}{false}
17211
17212 %% A04 is ending and A00 is reference-only.  Neither may occur in a plot.
17213 \AddDummyProject{A04}{E}{}{false}
17214 \AddDummyProject{A00}{C}{}{true}
17215
17216 %% Active project area B.  The A04 relation checks ending-project filtering.
17217 \AddDummyProject{B01}{C}{}{false}
17218 \AddDummyProject{B04}{C}{}{false}
17219 \AddDummyProject{B05}{C}{}{false}
17220 \AddDummyProject{B06}{C}
17221   {A00,A04,A11,A12,B01,B04,B05,B07,B10}
17222   {false}
17223 \AddDummyProject{B07}{C}{}{false}
17224 \AddDummyProject{B08}{C}{}{false}
17225 \AddDummyProject{B09}{C}{}{false}
17226 \AddDummyProject{B10}{N}{}{false}
17227
17228 %% Active project area Q.  Q02 declares B06 in the reverse direction only.
17229 \AddDummyProject{Q01}{C}{}{false}
17230 \AddDummyProject{Q02}{C}{B06}{false}
17231 \AddDummyProject{Q03}{C}{}{false}
17232 \AddDummyProject{Q04}{C}{}{false}
17233
17234 %% Focused image mode must retain Gantt rows created through the same data path
17235 %% as ordinary project work packages.
17236 \addEntry[gantt]{gantt-B06}{
17237   id = {wp1},
17238   name = {Dummy work package},
17239   description = {Dummy Gantt bar},
17240   periodStart = {1},
17241   periodEnd = {4},
17242   style = {default}
17243 }
17244
17245 \printPlot{{Gantt}{B06}}
17246
17247 %% The first plot is deliberately called before any project-area key has been
17248 %% entered.  It exercises the missing-column and ID-prefix fallbacks.
17249 \printPlot[layout=heatmap,caption={Many collaborations are shown here.}]
17250   {{collaborations}{B06}}
17251
17252 \clearpage
17253
17254 %% Explicit project areas need not match the project-ID prefix.  The longer
17255 %% service-area label exercises dynamic label measurement, while the options
17256 %% exercise explicit color mappings for otherwise unknown areas.
17257 \AddDummyProject[C]{X01}{C}{B06}{false}
17258 \AddDummyProject[Shared services]{Y01}{N}{}{false}
17259 \ExplSyntaxOn
17260 \@@_project_count_build:
17261 \prop_get:NnN \g_@@_project_count_area_prop { C } \l_tmpa_tl
17262 \int_compare:nNnF { \l_tmpa_tl } = { 1 }
17263   { \PackageError{test}{Expected~one~project~in~dynamic~area~C}{} }
```

```
\prop_get:NnN \g_@@_project_count_area_prop { SHARED~SERVICES } \l_tmpa_tl
\int_compare:nNnF { \l_tmpa_tl } = { 1 }
  { \PackageError{test}{Expected~one~project~in~dynamic~Shared~services~area}{} }
\ExplSyntaxOff
\printPlot[
  layout=heatmap,
  area-colors={C=cyan!70!black,Shared services=orange!80!black}
]{{collaborations}{B07}}

\clearpage

%% Select the low arc layout with arcs explicitly. Every displayed arc reaches the
%% same vertical peak, IDs should sit inside the boxes, and each area rule should
%% span its boxes; additional caption text uses the leading options.
\printPlot[
  layout=arc,
  arcs=true,
  area-palette={orange!80!black,violet!75!black,teal!70!black},
  caption={Dummy metadata; A04 and A00 are intentionally omitted.}
]{{collaborations}{B06}}

\clearpage

%% The same full-width node strip can be shown without connecting arcs.
\printPlot[arcs=false]{{collaborations}{B06}}

\clearpage

%% Turning the image switch off again must restore the fast-mode plot skip.
\ExplSyntaxOn
\bool_set_false:N \l_@@_class_fastcompile_project_show_images_bool
\ExplSyntaxOff
\printPlot{{ProjectCollaboration}{B06}}

\end{document}
```

6.2.21 Reference-path lookup regression test

Exercise the public command against real databases, including fan-out, shared destinations, arbitrary database names, missing data and fast mode. The tag hook records the originating row and protected value of every result.

```
\documentclass[english,final=true]{dfg-crc}
\ExplSyntaxOn
\tl_new:N \g_@@_get_test_seen_tl
\int_new:N \g_@@_get_test_count_int
\NewDocumentCommand{\CheckGet}{m m +m}
{
  \tl_gclear:N \g_@@_get_test_seen_tl
  \par\noindent #1:~#3\par
  \str_if_eq:eeF {\tl_to_str:V \g_@@_get_test_seen_tl}{\tl_to_str:n {#2}}
    {\PackageError{get-test}{Failed~#1:~\tl_to_str:V \g_@@_get_test_seen_tl}
      {Expected~\tl_to_str:n {#2}}}
  \int_gincr:N \g_@@_get_test_count_int
  \typeout{GET~TEST~PASS:~#1}
}
% A custom keyset verifies that traversal does not hard-code the four
% built-in relation types or require the source database to match its keyset.
\keys_define:nn {get-test}
{
  id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{id}{#1},
  name .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{name}{#1},
  code .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{code}{#1},
  ref-person-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-person-id}{#1},
  ref-other-unit-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-other-unit-id}{#1},
  ref-absent-db-id .code:n = \DTLnewdbentry{\l_@@_database_name_tl}{ref-absent-db-id}{#1},
}
\ExplSyntaxOff
\begin{document}
\DTLcleardb{person}
\DTLcleardb{institution}
\DTLcleardb{affiliation}
\DTLcleardb{project}
\addEntry{affiliation}{id={a1},name={University One},country={Germany}}
\addEntry{affiliation}{id={a2},name={University Two},country={France}}
\addEntry{institution}{id={i1},name={\textbf{Alpha Institute}},ref-affiliation-id={a1}}
\addEntry{institution}{id={i2},name={Beta Institute},ref-affiliation-id={a1}}
\addEntry{institution}{id={i3},name={Gamma Institute},ref-affiliation-id={a2}}
\addEntry{person}{id={p1},lastname={Example},ref-institution-id={i1},ref-project-id={P1}}
\addEntry{person}{id={p2},lastname={Other},ref-institution-id={i2,i3,i1,i2}}
\addEntry{person}{id={empty},lastname={Empty},ref-institution-id={}}
\addEntry{person}{id={broken},ref-institution-id={unknown,i3}}
\addEntry{project}{id={P1},name={Project One},ref-person-id={p1,p2}}
```

```
\addEntry[get-test]{lookup-source}{id={r1},code={alternate},ref-person-id={p1,p2},ref-other-unit-id={u1}}
\addEntry[get-test]{other-unit}{id={u1},name={Custom Unit},ref-person-id={p1}}
\addEntry[get-test]{lookup-source}{id={r2},ref-absent-db-id={nowhere}}

\ExplSyntaxOn
\RenewDocumentCommand{\dfgShowTag}{m m m o}
{
  \tl_gput_right:Nn \g_@@_get_test_seen_tl {[#4/#1/#2]=#3;}
  #3
}
\ExplSyntaxOff
\CheckGet{direct}{[person/p1/lastname]=Example;}{\get[person]{p1}{lastname}}
\CheckGet{one-hop}{[institution/i1/name]=\textbf{Alpha Institute};}
  {\get[person]{p1}{institution}{name}}
\CheckGet{explicit-key}{[institution/i1/name]=\textbf{Alpha Institute};}
  {\get[person]{p1}{ref-institution-id}{name}}
\CheckGet{two-hops}{[affiliation/a1/name]=University One;}
  {\get[person]{p1}{institution}{affiliation}{name}}
\CheckGet{explicit-chain}{[affiliation/a1/country]=Germany;}
  {\get[person]{p1}{ref-institution-id}{ref-affiliation-id}{country}}
\CheckGet{institution-root}{[affiliation/a1/name]=University One;}
  {\get[institution]{i2}{affiliation}{name}}
\CheckGet{project-reference}{[project/P1/name]=Project One;}
  {\get[person]{p1}{project}{name}}
\CheckGet{person-reference}{[person/p1/lastname]=Example;[person/p2/lastname]=Other;}
  {\get[project]{P1}{person}{lastname}}
\CheckGet{multiple-ids}{[institution/i2/name]=Beta Institute;[institution/i3/name]=Gamma Institute;[institution/i1/na
  {\get[person]{p2}{institution}{name}}
\CheckGet{shared-affiliation}{[affiliation/a1/name]=University One;[affiliation/a2/name]=University Two;}
  {\get[person]{p2}{institution}{affiliation}{name}}
\CheckGet{three-hops}{[affiliation/a1/name]=University One;[affiliation/a2/name]=University Two;}
  {\get[project]{P1}{person}{institution}{affiliation}{name}}
\CheckGet{named-source}{[person/p1/lastname]=Example;[person/p2/lastname]=Other;}
  {\get[lookup-source]{r1}{person}{lastname}}
\CheckGet{custom-target}{[other-unit/u1/name]=Custom Unit;}
  {\get[lookup-source]{r1}{other-unit}{name}}
\CheckGet{custom-explicit}{[person/p1/lastname]=Example;}
  {\get[lookup-source]{r1}{ref-other-unit-id}{ref-person-id}{lastname}}
\CheckGet{alternative-id}{[person/p1/lastname]=Example;[person/p2/lastname]=Other;}
  {\get[lookup-source]{alternate}{person}[code]{lastname}}
\def\tmpLocalDB{person}
\CheckGet{default-database}{[institution/i1/name]=\textbf{Alpha Institute};}
  {\get{p1}{institution}{name}}
\CheckGet{raw-reference}{[person/p1/ref-institution-id]=i1;}
  {\get[person]{p1}{ref-institution-id}}
\CheckGet{following-group}{[person/p1/lastname]=Example;}
  {\get[person]{p1}{lastname}{\textbf{Following prose.}}}
\CheckGet{field-reference-cycle}{[person/p1/lastname]=Example;[person/p2/lastname]=Other;}
  {\get[person]{p1}{project}{person}{lastname}}
\CheckGet{starred}{}{
  \get*[person]{p1}{\TestLastname=lastname}
  \ifdefstring{\TestLastname}{Example}{}{\PackageError{get-test}{Starred lookup failed}{}}
}
\CheckGet{empty-reference}{}{\get[person]{empty}{institution}{name}}
\CheckGet{missing-root}{}{\get[person]{missing}{institution}{name}}
\CheckGet{missing-leaf}{}{\get[person]{p1}{institution}{absent-field}}
\CheckGet{missing-relation}{}{\get[institution]{i1}{person}{lastname}}
\CheckGet{missing-target-db}{}{\get[lookup-source]{r2}{ref-absent-db-id}{name}}
\CheckGet{missing-source-db}{}{\get[absent-db]{r1}{person}{lastname}}
\CheckGet{broken-branch}{[institution/i3/name]=Gamma Institute;}
  {\get[person]{broken}{institution}{name}}
\CheckGet{after-missing-reset}{[affiliation/a1/name]=University One;}
  {\get[person]{p1}{institution}{affiliation}{name}}
\begingroup
\ExplSyntaxOn
\bool_set_true:N \l_@@_class_fastcompile_project_bool
\RenewDocumentCommand{\dfgFastReference}{m}
  {\tl_gput_right:Ne \g_@@_get_test_seen_tl {fast:#1;}}
\RenewDocumentCommand{\dfgShowTag}{m m m o}
{
  \tl_gput_right:Ne \g_@@_get_test_seen_tl
    {tag:\IfNoValueTF{#4}{}{#4}/#1/#2;}
  #3
}
\ExplSyntaxOff
\CheckGet{fast-chain}{tag:person/p1/institution/affiliation/name;fast:p1;}
  {\get[person]{p1}{institution}{affiliation}{name}}
\addEntry[get-test]{skipped-source}{id={skip},ref-other-unit-id={u1}}
\CheckGet{fast-skipped-custom}
  {tag:skipped-source/skip/other-unit/person/lastname;fast:skip;}
  {\get[skipped-source]{skip}{other-unit}{person}{lastname}}
\CheckGet{fast-skipped-scalar}{tag:skipped-source/skip/name;fast:skip;}
  {\get[skipped-source]{skip}{name}{\textbf{Following prose.}}}
\CheckGet{fast-star}{}{
  \def\TestLastname{stale}
  \get*[person]{p1}{\TestLastname=lastname}
```

```
  \ifdefempty{\TestLastname}{}{\PackageError{get-test}{Fast starred lookup failed}{}}
}
\endgroup
\CheckGet{after-fast-reset}{[person/p1/lastname]=Example;}{\get[person]{p1}{lastname}}
\ExplSyntaxOn
\int_compare:nNnF {\g_@@_get_test_count_int} = {33}
  {\PackageError{get-test}{Not~all~33~cases~ran}{}}
\ExplSyntaxOff
\typeout{REFERENCE LOOKUP TEST PASS: 33 cases}
\end{document}
```

6.2.22 Public-command option regression tests

Small probe renderers check the public dispatchers without depending on a particular table or plot. Database-aware lists retain their real lookup and filtering; only their final cells are replaced with assertions.

```
% The customization file is loaded during class initialization. Probe both
% interfaces there, then verify that later class setup preserves the values.
\AddToHook{file/dfg-customization.tex/before}{%
  \let\DefaultDateBeforeCustomization\dfgDate
  \renewcommand{\dfgLocation}{Direct early location\xspace}%
  \renewcommand{\dfgDate}{Direct early date\xspace}%
  \renewcommand{\dfgTitle}{Direct early title\xspace}%
  \renewcommand{\InfoTeX}{Direct early compiler}%
  \changeSettings{
    dfgLocation={Early location}, dfgDate={Early date},
    dfgTitle={Early title}, InfoTeX={Early compiler}}
}
\documentclass[english,final=true]{dfg-crc}
\ExplSyntaxOn
\int_new:N \g_@@_document_settings_test_count_int
\NewDocumentCommand{\CheckDocumentSetting}{m m m}
{
  \tl_set:No \l_tmpa_tl {#2}
  \tl_if_eq:NnF \l_tmpa_tl {#3}
    {\PackageError{settings-test}{Failed~#1}{Unexpected~document~text.}}
  \int_gincr:N \g_@@_document_settings_test_count_int
}
\ExplSyntaxOff
% Resolve the default date's fast-mode conditional, preserving its spacing
% command. Other checks below deliberately retain deferred metadata lookups.
\ExplSyntaxOn
\group_begin:
\cs_set_eq:NN \xspace \scan_stop:
\cs_set:Npe \ExpandedDefaultDateBeforeCustomization
  {\DefaultDateBeforeCustomization}
\CheckDocumentSetting{generic-date}
  {\ExpandedDefaultDateBeforeCustomization}{<date>\xspace}
\group_end:
\ExplSyntaxOff
\CheckDocumentSetting{early-location}{\dfgLocation}{Early location\xspace}
\CheckDocumentSetting{early-date}{\dfgDate}{Early date\xspace}
\CheckDocumentSetting{early-title}{\dfgTitle}{Early title\xspace}
\CheckDocumentSetting{early-compiler}{\InfoTeX}{Early compiler}
\begingroup
\changeSettings{
  dfgLocation={Runtime location}, dfgDate={Runtime date},
  dfgTitle={\get[meta]{crc}{name}}, InfoTeX={Runtime compiler}}
\CheckDocumentSetting{runtime-location}{\dfgLocation}{Runtime location\xspace}
\CheckDocumentSetting{runtime-date}{\dfgDate}{Runtime date\xspace}
\CheckDocumentSetting{deferred-title}{\dfgTitle}{\get[meta]{crc}{name}\xspace}
\CheckDocumentSetting{runtime-compiler}{\InfoTeX}{Runtime compiler}
\renewcommand{\dfgLocation}{Direct location\xspace}
\renewcommand{\dfgDate}{Direct date\xspace}
\renewcommand{\dfgTitle}{Direct title\xspace}
\renewcommand{\InfoTeX}{Direct compiler}
\CheckDocumentSetting{direct-location}{\dfgLocation}{Direct location\xspace}
\CheckDocumentSetting{direct-date}{\dfgDate}{Direct date\xspace}
\CheckDocumentSetting{direct-title}{\dfgTitle}{Direct title\xspace}
\CheckDocumentSetting{direct-compiler}{\InfoTeX}{Direct compiler}
\endgroup
\CheckDocumentSetting{local-settings-scope}{\dfgDate}{Early date\xspace}
\ExplSyntaxOn
\int_compare:nNnF {\g_@@_document_settings_test_count_int} = {14}
  {\PackageError{settings-test}{Not~all~14~cases~ran}{}}
\typeout{DOCUMENT~SETTINGS~TEST~PASS:~14~cases}

% Exercise the public customization getter through real key assignments.
% All rates below are synthetic, group-local test inputs, not proposal defaults.
\int_new:N \g_@@_get_value_test_count_int
\cs_new_protected:Npn \@@_test_custom_value:nn #1#2
  {
    \tl_set:Ne \l_tmpa_tl {\getValue{#1}}
```

```
17503     \str_if_eq:VnF \l_tmpa_tl {#2}
17504       {\PackageError{settings-test}{Unexpected~getValue~result~for~#1}{}}
17505     \int_gincr:N \g_@@_get_value_test_count_int
17506   }
17507 \clist_map_inline:nn
17508   {font,fontspec,sansfont,sansfontspec,crcLogoPath,affiliationLogoPath,PIcontactStyle,collaborationAreaColors,thumbin
      one,color@gantt@stripe-two,color@table@headingsBG,color@table@headingsBGalt,color@table@headingsFont,color@table@rows
17509   {
17510     \group_begin:
17511     \changeSettings{#1={Getter~probe}}
17512     \@@_test_custom_value:nn {#1}{Getter~probe}
17513     \changeSettings{#1={Second~probe}}
17514     \@@_test_custom_value:nn {#1}{Second~probe}
17515     \group_end:
17516   }
17517 \clist_map_inline:nn
17518   {projectPIOncePerSection,PIReferenceFirstName,projectsTableInstitutionAbbreviations,thumbindex,emphasizeAuthors,emp
17519   {
17520     \group_begin:
17521     \changeSettings{#1={false}}
17522     \@@_test_custom_value:nn {#1}{false}
17523     \changeSettings{#1={true}}
17524     \@@_test_custom_value:nn {#1}{true}
17525     \group_end:
17526   }
17527 \clist_map_inline:nn
17528   {salary@prof,salary@groupLeader,salary@postDoc,salary@medDoc,salary@PhD,salary@research,salary@nonResearch,salary@s
17529   {
17530     \group_begin:
17531     \changeSettings{#1={17.25}}
17532     \@@_test_custom_value:nn {#1}{17.25}
17533     \changeSettings{#1={23.5}}
17534     \@@_test_custom_value:nn {#1}{23.5}
17535     \group_end:
17536   }
17537 \clist_map_inline:nn
17538   {thumbindex@pattern}
17539   {
17540     \group_begin:
17541     \changeSettings{#1={17}}
17542     \@@_test_custom_value:nn {#1}{17}
17543     \changeSettings{#1={23}}
17544     \@@_test_custom_value:nn {#1}{23}
17545     \group_end:
17546   }
17547 \clist_map_inline:nn
17548   {thumbindex@width}
17549   {
17550     \group_begin:
17551     \changeSettings{#1={17pt}}
17552     \@@_test_custom_value:nn {#1}{17.0pt}
17553     \changeSettings{#1={23pt}}
17554     \@@_test_custom_value:nn {#1}{23.0pt}
17555     \group_end:
17556   }
17557
17558 \group_begin:
17559 \cs_set:Npn \xspace {}
17560 \changeSettings{dfgLocation=Here,dfgDate=Today,dfgTitle=Title,InfoTeX=Compiler,
17561   collaborationAreaPalette={red,blue},thumbindex@topmargin=17pt,
17562   thumbindex@bottommargin=23pt,thumbindex@orientation=inwards,
17563   animalHousing@mouse=0.85,salary@postDoc=90000}
17564 \@@_test_custom_value:nn {dfgLocation}{Here}
17565 \@@_test_custom_value:nn {dfgDate}{Today}
17566 \@@_test_custom_value:nn {dfgTitle}{Title}
17567 \@@_test_custom_value:nn {InfoTeX}{Compiler}
17568 \@@_test_custom_value:nn {collaborationAreaPalette}{red,blue}
17569 \changeSettings{collaborationAreaPalette={}}
17570 \@@_test_custom_value:nn {collaborationAreaPalette}{red,blue}
17571 \@@_test_custom_value:nn {thumbindex@topmargin}{17.0pt}
17572 \@@_test_custom_value:nn {thumbindex@bottommargin}{23.0pt}
17573 \@@_test_custom_value:nn {thumbindex@orientation}{inwards}
17574 \changeSettings{thumbindex@topmargin=auto,thumbindex@bottommargin=auto,
17575   thumbindex@orientation=outwards}
17576 \@@_test_custom_value:nn {thumbindex@topmargin}{auto}
17577 \@@_test_custom_value:nn {thumbindex@bottommargin}{auto}
17578 \@@_test_custom_value:nn {thumbindex@orientation}{outwards}
17579 \@@_test_custom_value:nn {~animalHousing@mouse~}{0.85}
17580 \fp_compare:nNnF {2 * \getValue{animalHousing@mouse}} = {1.7}
17581   {\PackageError{settings-test}{getValue~failed~in~arithmetic}{}}
17582 \group_begin:
17583 \changeSettings{animalHousing@mouse=1.25}
17584 \@@_test_custom_value:nn {animalHousing@mouse}{1.25}
17585 \group_end:
17586 \@@_test_custom_value:nn {animalHousing@mouse}{0.85}
17587 \changeSettings{biblatexstyle=shared,biblatexstyleFeature=shared-feature}
17588 \@@_test_custom_value:nn {biblatexstyleCV}{shared}
```

```
\@@_test_custom_value:nn {biblatexstyleProposal}{shared}
\changeSettings{biblatexstyleCV=cv-style,biblatexstyleProposal=proposal-style,
  biblatexstyleFeatureCV=cv-feature,biblatexstyleFeatureProposal=proposal-feature}
\bool_set_false:N \l_@@_class_cv_bool
\@@_test_custom_value:nn {biblatexstyle}{proposal-style}
\@@_test_custom_value:nn {biblatexstyleFeature}{proposal-feature}
\bool_set_true:N \l_@@_class_cv_bool
\@@_test_custom_value:nn {biblatexstyle}{cv-style}
\@@_test_custom_value:nn {biblatexstyleFeature}{cv-feature}
\group_end:
\typeout{CUSTOMIZATION~GETTER~TEST~PASS:~\int_use:N\g_@@_get_value_test_count_int~checks}


\tl_new:N \g_@@_options_test_result_tl
\tl_new:N \l_@@_options_test_mode_tl
\tl_new:N \l_@@_options_test_caption_tl
\dim_new:N \l_@@_options_test_width_dim
\int_new:N \g_@@_options_test_count_int
\keys_define:nn {dfg-options-test}
{
  mode .tl_set:N = \l_@@_options_test_mode_tl,
  caption .tl_set:N = \l_@@_options_test_caption_tl,
}
\cs_new_protected:Npn \@@_options_test_probe:nnn #1#2#3
{
  \tl_set:Nn \l_@@_options_test_mode_tl {empty}
  \tl_clear:N \l_@@_options_test_caption_tl
  \keys_set:nn {dfg-options-test} {#2}
  \tl_gput_right:Ne \g_@@_options_test_result_tl
    {#1|\l_@@_options_test_mode_tl|\l_@@_options_test_caption_tl|#3}
}
\NewDocumentCommand{\OptionProbeList}{O{mode=default}}
  {\@@_options_test_probe:nnn {list}{#1}{}}
\NewDocumentCommand{\OptionProbePlot}{O{mode=default}}
  {\@@_options_test_probe:nnn {plot}{#1}{}}
\NewDocumentCommand{\OptionProbeTable}{O{mode=default}}
{
  \int_compare:nNnF {\value{rowNumber}} = {0}
    {\PackageError{options-test}{Table~row~counter~was~not~reset}{}}
  \@@_options_test_probe:nnn {table}{#1}{}
}
\NewDocumentCommand{\OptionArgumentsList}{O{mode=default} m m}
  {\@@_options_test_probe:nnn {list}{#1}{#2/#3}}
\NewDocumentCommand{\OptionArgumentsPlot}{O{mode=default} m m}
  {\@@_options_test_probe:nnn {plot}{#1}{#2/#3}}
\NewDocumentCommand{\OptionArgumentsTable}{O{mode=default} m m}
  {\@@_options_test_probe:nnn {table}{#1}{#2/#3}}
\NewDocumentCommand{\CheckOptions}{m m +m}
{
  \tl_gclear:N \g_@@_options_test_result_tl
  #3
  \str_if_eq:eeF {\g_@@_options_test_result_tl} {#2}
  {
    \PackageError{options-test}{Failed~#1:~\g_@@_options_test_result_tl}
      {Expected~#2.}
  }
  \int_gincr:N \g_@@_options_test_count_int
  \typeout{OPTIONS~TEST~PASS:~#1}
}
% Keep the real database lookup and option parsing; inspect the resulting
% institution ID and boolean settings at the final formatting boundary.
\cs_set_protected:Npn \_@@_one_institution:n #1
{
  \tl_gput_right:Ne \g_@@_options_test_result_tl
  {
    #1|
    \bool_if:NTF \l_@@_institution_abbreviation_bool {1}{0}
    \bool_if:NTF \l_@@_narrow_bool {1}{0}
    \bool_if:NTF \l_@@_noaddr_bool {1}{0}
    \bool_if:NTF \l_@@_nozip_bool {1}{0}
    \bool_if:NTF \l_@@_noloc_bool {1}{0}
  }
}
% Keep the real PI filtering and raster, but use a minimal box to measure
% the available column width independently of the card's text/padding.
\cs_set_protected:Npn \__getperson_one:n #1
{
  \begin{tcolorbox}[blankest]
    \dim_compare:nNnT
      {\dim_abs:n{\linewidth-\l_@@_options_test_width_dim}} > {.1pt}
      {\PackageError{options-test}{PI~raster~column~width~is~wrong}{}}
    \tl_gput_right:Ne \g_@@_options_test_result_tl
      {#1\bool_if:NT \l_@@_pi_academicdegree_below_bool {|below}}
    PI~#1
  \end{tcolorbox}
}
\NewDocumentCommand{\CheckPIOptions}{m m m +m}
```

```
{
  \dim_set:Nn \l_@@_options_test_width_dim
    {(\linewidth-12pt*\int_eval:n{#2-1})/#2}
  \CheckOptions{#1}{#3}{#4}
}
\ExplSyntaxOff

\begin{document}
\begingroup
\changeSettings{animalHousing@mouse=0.85,salary@postDoc=90000}
Test-only mouse housing: \num{\getValue{animalHousing@mouse}}\,EUR per week.\par
Test-only postdoctoral salary: \num{\getValue{salary@postDoc}}\,EUR per year.\par
\endgroup

\section*{Command option regression checks}
\CheckOptions{list-omitted}{list|default||}{\printList{OptionProbe}}
\CheckOptions{list-empty}{list|empty||}{\printList[]{OptionProbe}}
\CheckOptions{list-grouped}{list|default||}{\printList{{OptionProbe}}}
\CheckOptions{list-leading}{list|leading||}{\printList[mode=leading]{OptionProbe}}
\CheckOptions{list-legacy}{list|legacy||}{\printList{OptionProbe}[mode=legacy]}
\CheckOptions{list-precedence}{list|leading||}
  {\printList[mode=leading]{{OptionProbe}}[mode=legacy]}
\CheckOptions{list-arguments}{list|leading|left,right|A01/two words}
  {\printList[mode=leading,caption={left,right}]{{OptionArguments}{A01}{two words}}}
\CheckOptions{plot-omitted}{plot|default||}{\printPlot{OptionProbe}}
\CheckOptions{plot-empty}{plot|empty||}{\printPlot[]{OptionProbe}}
\CheckOptions{plot-grouped}{plot|default||}{\printPlot{{OptionProbe}}}
\CheckOptions{plot-arguments}{plot|leading|left,right|A01/two words}
  {\printPlot[mode=leading,caption={left,right}]{{OptionArguments}{A01}{two words}}}
\CheckOptions{plot-embedded-legacy}{plot|legacy||A01/second}
  {\printPlot{{OptionArguments}[mode=legacy]{A01}{second}}}
\setcounter{rowNumber}{99}
\CheckOptions{table-omitted-and-counter}{table|default||}{\printTable{OptionProbe}}
\CheckOptions{table-empty}{table|empty||}{\printTable[]{OptionProbe}}
\CheckOptions{table-grouped}{table|default||}{\printTable{{OptionProbe}}}
\CheckOptions{table-arguments}{table|leading|left,right|A01/two words}
  {\printTable[mode=leading,caption={left,right}]{{OptionArguments}{A01}{two words}}}

\DTLcleardb{person}
\addEntry{person}{id={default-person},lastname={Default},role={PI},
  period={1,2,3},ref-institution-id={default-institute}}
\DTLnewdb{option-people}
\DTLnewrow{option-people}
\DTLnewdbentry{option-people}{id}{other-person}
\DTLnewdbentry{option-people}{lastname}{Other}
\DTLnewdbentry{option-people}{role}{PI,spokesperson}
\DTLnewdbentry{option-people}{period}{1,2,3}
\DTLnewdbentry{option-people}{ref-institution-id}{other-institute}

\CheckOptions{institutions-default}{default-institute|00000}
  {\printList{{Institutions}{default-person}}}
\CheckOptions{institutions-flags}{other-institute|11111}
  {\printList[database=option-people,abbreviation,narrow,noaddress,nozip,nolocation]
    {{Institutions}{other-person}}}
\CheckOptions{institutions-legacy-database}{other-institute|11000}
  {\InstitutionsList[option-people]{other-person}[abbreviation,narrow]}
\CheckOptions{institutions-precedence}{other-institute|01000}
  {\InstitutionsList[database=option-people,abbreviation=false,narrow]
    {other-person}[abbreviation]}
\CheckOptions{institutions-dispatch-precedence}{other-institute|01000}
  {\printList[database=option-people,abbreviation=false,narrow]
    {{Institutions}{other-person}}[abbreviation]}
\CheckOptions{institutions-empty}{default-institute|00000}
  {\InstitutionsList[]{default-person}}
\CheckOptions{institutions-reset}{default-institute|00000}
  {\InstitutionsList{default-person}}

\CheckPIOptions{pi-default}{3}{default-person}{\getPIsWithRole{PI}}
\CheckPIOptions{pi-leading}{2}{other-person}
  {\getPIsWithRole[database=option-people,raster columns=2]{spokesperson}}
\CheckPIOptions{pi-legacy}{1}{other-person}
  {\getPIsWithRole[option-people]{PI}[raster columns=1]}
\CheckPIOptions{pi-precedence}{2}{other-person}
  {\getPIsWithRole[database=option-people,raster columns=2]{PI}[raster columns=1]}
\CheckPIOptions{pi-degree-below}{3}{default-person|below}
  {\getPIsWithRole[academicdegree-below]{PI}}
\CheckPIOptions{pi-degree-legacy}{1}{other-person|below}
  {\getPIsWithRole[option-people]{PI}[academicdegree-below,raster columns=1]}
\CheckPIOptions{pi-degree-precedence}{3}{default-person}
  {\getPIsWithRole[academicdegree-below=false]{PI}[academicdegree-below]}
\CheckPIOptions{pi-degree-reset}{3}{default-person}{\getPIsWithRole{PI}}
\CheckPIOptions{pi-empty}{3}{default-person}{\getPIsWithRole[]{PI}}
\CheckPIOptions{pi-reset}{3}{default-person}{\getPIsWithRole{PI}}
\CheckPIOptions{pi-role-filter}{3}{}{\getPIsWithRole{spokesperson}}

\ExplSyntaxOn
\int_compare:nNnF {\g_@@_options_test_count_int} = {34}
```

```
  {\PackageError{options-test}{Not~all~34~cases~ran}{}}
\ExplSyntaxOff
\typeout{COMMAND OPTIONS TEST PASS: 34 cases}

% Render real metadata lookups to check punctuation and option defaults.
\clearpage
\section*{Project PI prose lists}
\addEntry{person}{id={pi-test-a},firstname={Alice},lastname={Able},academicdegree={Dr.}}
\addEntry{person}{id={pi-test-b},firstname={Bob},lastname={Baker},academicdegree={Dr.}}
\addEntry{person}{id={pi-test-c},firstname={Carol},lastname={Clark},academicdegree={Dr.}}
\addEntry{person}{id={pi-test-d},firstname={Dan},lastname={Dover},academicdegree={Dr.}}
\addEntry{project}{id={PI0},ref-person-id={}}
\addEntry{project}{id={PI1},ref-person-id={pi-test-a}}
\addEntry{project}{id={PI2},ref-person-id={pi-test-a,pi-test-b}}
\addEntry{project}{id={PI3},ref-person-id={pi-test-a,pi-test-b,pi-test-c}}
\addEntry{project}{id={PI4},ref-person-id={pi-test-a,pi-test-b,pi-test-c,pi-test-d}}
Zero: [\getPIofProject*{PI0}].\par
One: \getPIofProject*{PI1}.\par
Two: \getPIofProject*{PI2}.\par
Three: \getPIofProject*{PI3}.\par
Four: \getPIofProject*{PI4}.\par
Properties: \getPIofProject*[lastname]{PI3}.\par
Final separator: \getPIofProject*{PI3}[\space und \space].\par
Legacy: \getPIofProject[firstname,lastname]{PI3}[; ].\par
Reset: \getPIofProject{PI2}.\par
Empty properties: [\getPIofProject*[]{PI1}].\par
Following group: \getPIofProject*{PI1}{ followed}.\par
\begingroup
\ExplSyntaxOn
\bool_set_true:N \l_@@_class_fastcompile_project_bool
\ExplSyntaxOff
Fast: \getPIofProject*{PI3}.\par
\endgroup

% Use the same destination helper as the PI table. Keep targets on a separate
% page so a PDF check can distinguish working links from fallback destinations.
\clearpage
\section*{PI table link targets}
\ExplSyntaxOn
\clist_map_inline:nn {pi-test-a,pi-test-b,pi-test-c,pi-test-d}
  {\@@_pi_target:n {#1} PI~target~#1\par}
\ExplSyntaxOff
\end{document}
```

6.2.23 Participating-institution layout regression test

Assert the width and complete, ordered content of the rendered institution groups. Shared PI institutions must appear once, inactive/non-PI institutions must be absent, and options must not leak into the next list.

```
\documentclass[english,final=true]{dfg-crc}

\ExplSyntaxOn
\seq_new:N \g_@@_institutions_test_seen_seq
\dim_new:N \l_@@_institutions_test_width_dim
\NewDocumentCommand{\TestInstitutionName}{m}
{
  \dim_compare:nNnT
    {\dim_abs:n {\linewidth-\l_@@_institutions_test_width_dim}} > {.1pt}
  {
    \PackageError{institutions-test}
      {Wrong~column~width~for~#1:~\the\linewidth}
      {The~list~did~not~honour~its~column~count~or~column~gap.}
  }
  \seq_gput_right:Nn \g_@@_institutions_test_seen_seq {#1}
  Institute~#1
}
\AddToHook{env/multicols/before}
{
  \dim_compare:nNnT {\lastskip} < {\medskipamount}
    {\PackageError{institutions-test}{Missing~space~above~columns}{}}
}
\NewDocumentCommand{\TestInstitutionList}{m m m m +m}
{
  \clearpage
  \section*{#1}
  \seq_gclear:N \g_@@_institutions_test_seen_seq
  \dim_set:Nn \l_@@_institutions_test_width_dim
    {(\linewidth-\dim_eval:n{#3}*\int_eval:n{#2-1})/#2}
  #5
  \tl_set:Ne \l_tmpa_tl
    {\seq_use:Nn \g_@@_institutions_test_seen_seq {,}}
  \tl_if_eq:NnF \l_tmpa_tl {#4}
  {
```

```
17840       \PackageError{institutions-test}
17841         {Unexpected~institutions~in~#1:~\l_tmpa_tl}
17842         {Check~filtering,~duplicate~removal,~sorting,~and~option~scoping.}
17843     }
17844     \typeout{INSTITUTIONS~TEST~PASS:~#1}
17845   }
17846   \ExplSyntaxOff
17847
17848   \newcommand{\TestInstitution}[2]{%
17849     \addEntry{institution}{
17850       id={#1},name={\TestInstitutionName{#1}},ref-affiliation-id={#2}
17851     }%
17852   }
17853
17854   \begin{document}
17855   \DTLcleardb{affiliation}
17856   \DTLcleardb{institution}
17857   \DTLcleardb{person}
17858
17859   \addEntry{affiliation}{id={z},name={Zeta University},location={Zeta City}}
17860   \addEntry{affiliation}{id={a},name={Alpha University},location={Alpha City}}
17861   \addEntry{affiliation}{id={b},name={Beta University},location={Beta City}}
17862   \addEntry{affiliation}{id={unused},name={Unused University},location={Unused City}}
17863   \TestInstitution{Z1}{z}
17864   \TestInstitution{Z2}{z}
17865   \TestInstitution{Z3}{z}
17866   \TestInstitution{Z4}{z}
17867   \TestInstitution{Z5}{z}
17868   \TestInstitution{Z6}{z}
17869   \TestInstitution{A1}{a}
17870   \TestInstitution{B1}{b}
17871   \TestInstitution{INACTIVE}{unused}
17872   \TestInstitution{NONPI}{unused}
17873   \addEntry{person}{
17874     id={active},role={PI},period={1,2,3},
17875     ref-institution-id={Z1,Z2,Z3,Z4,Z5,Z6,A1,B1,Z1}
17876   }
17877   \addEntry{person}{
17878     id={shared},role={PI},period={1,2,3},ref-institution-id={Z1,A1}
17879   }
17880   \addEntry{person}{
17881     id={inactive},role={PI},period={1},ref-institution-id={INACTIVE}
17882   }
17883   \addEntry{person}{
17884     id={manager},role={manager},period={1,2,3},ref-institution-id={NONPI}
17885   }
17886
17887   \TestInstitutionList{flow-two}{2}{12pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17888     {\printList[columns=2]{ParticipatingInstitutions}}
17889   \TestInstitutionList{flow-three-alpha}{3}{16pt}{A1,B1,Z1,Z2,Z3,Z4,Z5,Z6}
17890     {\printList[columns=3,column sep=16pt,order=alpha]{ParticipatingInstitutions}}
17891   \TestInstitutionList{raster-two}{2}{12pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17892     {\printList[raster columns=2]{ParticipatingInstitutions}}
17893   \TestInstitutionList{raster-three}{3}{16pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17894     {\printList[raster columns=3,raster column skip=16pt]{ParticipatingInstitutions}}
17895   \TestInstitutionList{legacy-alpha}{2}{12pt}{A1,B1,Z1,Z2,Z3,Z4,Z5,Z6}
17896     {\printList[columns=2]{{ParticipatingInstitutions}}[alpha]}
17897   \TestInstitutionList{leading-order-wins}{2}{12pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17898     {\printList[columns=2,order=count]{{ParticipatingInstitutions}}[alpha]}
17899   \TestInstitutionList{default-reset}{1}{12pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17900     {\printList{ParticipatingInstitutions}}
17901   \TestInstitutionList{flow-after-text}{2}{12pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17902     {Introductory text.\printList[columns=2]{ParticipatingInstitutions}}
17903   \TestInstitutionList{flow-one}{1}{12pt}{Z1,Z2,Z3,Z4,Z5,Z6,A1,B1}
17904     {\printList[columns=1]{ParticipatingInstitutions}}
17905   \DTLcleardb{person}
17906   \TestInstitutionList{empty-list}{2}{12pt}{}
17907     {\printList[columns=2]{ParticipatingInstitutions}}
17908
17909   \end{document}
```

6.2.24 Nested work-package reference test

This focused test creates repeated child-counter values under three parent work packages. It catches both loss of later references during sorting and ranges that incorrectly cross a parent boundary.

```
17910   %% Compile from the generated local directory with:
17911   %%   latexmk -lualatex -g -halt-on-error test-workpackage-references.tex
17912   \documentclass[
17913     fastcompile-project = true,
17914     fastcompile-project-show-images = false,
17915     english
17916   ]{dfg-crc}
```

```
\begin{document}

\ExplSyntaxOn
\tl_set:Nn \l__tmp_current_project_tl {WPTEST}
\ExplSyntaxOff

\begin{workpackages}
  \addWP{wp1}{name={Parent one}}
  Parent one.
  \begin{workpackages}
    \addWP{wp1a}{} Child 1a.
    \addWP{wp1b}{} Child 1b.
    \addWP{wp1c}{} Child 1c.
    \addWP{wp1d}{} Child 1d.
  \end{workpackages}

  \addWP{wp2}{name={Parent two}}
  Parent two.
  \begin{workpackages}
    \addWP{wp2a}{} Child 2a.
    \addWP{wp2b}{} Child 2b.
  \end{workpackages}

  \addWP{wp3}{name={Parent three}}
  Parent three.
  \begin{workpackages}
    \addWP{wp3a}{} Child 3a.
    \addWP{wp3b}{} Child 3b.
    \addWP{wp3c}{} Child 3c.
  \end{workpackages}
\end{workpackages}

\noindent Range result:
\getWP{wp1b,wp1c,wp1d,wp2b,wp3c}.

\noindent Unordered result:
\getWP{wp2b,wp3c,wp1d}.


\noindent Mixed result:
\getWP{wp1a,wp1b,wp2,wp3a,wp3b}.

\noindent Reversed mixed:
\getWP{wp3b,wp2,wp1a,wp3a,wp1b}.

\noindent Parent and children:
\getWP{wp1,wp1a,wp1b,wp1c,wp2,wp2a,wp2b,wp3}.

\noindent Parent range:
\getWP{wp1,wp2,wp3}.

\end{document}
```

6.2.25 Existing direct-cost overview test

This focused two-pass test prints section 2.1.1 before declaring the project-level rows. On the following run, the auxiliary-file mirror must restore those rows early enough for the table and the assertions.

```
%% Compile from the generated local directory with:
%%   latexmk -lualatex -g -halt-on-error test-existing-funds-direct-costs.tex
\documentclass[english,final=true]{dfg-crc}

\providecommand{\dfgDirectCostsAuxReady}{0}

%% Ending-period values remain explicit because they describe the preceding
%% funding period rather than the project requests below.
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year=1, applicant=4300, participant=1200, other-funds=500
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year=2, applicant=0, participant=0, other-funds=0
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year=3, applicant=0, participant=0, other-funds=0
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year=4, applicant=0, participant=0, other-funds=0
}
\addEntry[funding]{PreviousExistingFundsDirectCosts}{
  year=5, applicant=0, participant=0, other-funds=0
}

```

```
\begin{document}

\section*{Overview of existing funds for direct costs}

%% The second and later passes must see the rows restored from the auxiliary
%% file before the live project declarations below are encountered.
\ifnum\dfgDirectCostsAuxReady=1\relax
  \ExplSyntaxOn
  \cs_set_eq:NN
    \@@_direct_costs_test_show_saved:nnN
    \@@_show_calculated_number:nnN
  \cs_set_protected:Npn
    \@@_show_calculated_number:nnN #1#2#3 { }
  \cs_new_protected:Npn \@@_direct_costs_test_expect:nnn #1#2#3
  {
    \getSumRequestedFundingDirectCosts
      [RequestedAndExistingFundingDirectCostsOverview]{#1}{#2}
    \DTLconverttodecimal {\SumRequestedFundingDirectCosts} \l_tmpa_tl
    \fp_compare:nNnF {\l_tmpa_tl} = {#3}
    {
      \PackageError{direct-costs-test}
        {Expected~#1/#2~sum~#3,~got~\SumRequestedFundingDirectCosts}
        {The~section~2.1.1~overview~classification~is~incorrect.}
    }
  }
  \@@_direct_costs_test_expect:nnn {1}{applicant}{5000}
  \@@_direct_costs_test_expect:nnn {1}{participant}{2100}
  \@@_direct_costs_test_expect:nnn {1}{other-funds}{300}
  \@@_direct_costs_test_expect:nnn {1}{all}{7400}
  \@@_direct_costs_test_expect:nnn {all}{applicant}{25000}
  \@@_direct_costs_test_expect:nnn {all}{participant}{10500}
  \@@_direct_costs_test_expect:nnn {all}{other-funds}{1500}
  \@@_direct_costs_test_expect:nnn {all}{all}{37000}
  \cs_set_eq:NN
    \@@_show_calculated_number:nnN
    \@@_direct_costs_test_show_saved:nnN
  \iow_term:n
    {DIRECT~COSTS~TEST~PASS:~auxiliary~restore,~classification,~and~totals}
  \ExplSyntaxOff
\fi

\printTable{ExistingFundsDirectCosts}

%% These rows intentionally occur after the overview.  aff1 is the applicant
%% affiliation in the generated metadata; aff2 is a participating affiliation.
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {A01},
  ref-affiliation-id = {aff1},
  financed-by        = {Applicant university core support},
  sum = {
    year1 = {5000}, year2 = {5000}, year3 = {5000},
    year4 = {5000}, year5 = {5000}
  }
}
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {B01},
  ref-affiliation-id = {aff2},
  financed-by        = {Participating university core support},
  sum = {
    year1 = {2100}, year2 = {2100}, year3 = {2100},
    year4 = {2100}, year5 = {2100}
  }
}
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {Q01},
  ref-affiliation-id = {other-funds},
  financed-by        = {External grant},
  sum = {
    year1 = {300}, year2 = {300}, year3 = {300},
    year4 = {300}, year5 = {300}
  }
}

%% Project fast mode must update existing-funds totals as well as the rows.
%% Keep this local so the normal-mode auxiliary overview remains unchanged.
\ExplSyntaxOn
\group_begin:
\cs_set_protected:Npn \@@_direct_costs_fast_expect:nnn #1#2#3
{
  \@@_funding_sum_set:Nnnnn \l_tmpa_tl
    {RequestedAndExistingFundingDirectCosts}{#1}{__ALL__}{#2}
  \fp_compare:nNnF {\l_tmpa_tl} = {#3}
  {
    \PackageError{direct-costs-test}
      {Expected~#1/#2~sum~#3,~got~\l_tmpa_tl}
      {Existing~direct-cost~totals~must~match~the~project~rows.}
  }
```

```
}
%% Normal-mode rows above must still be counted exactly once.
\@@_direct_costs_fast_expect:nnn {A01}{sum-year1}{5000}
\bool_set_true:N \l_@@_class_fastcompile_project_bool
\bool_set_true:N \l_@@_class_fastcompile_project_show_funding_tables_bool
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id={FASTTEST}, ref-affiliation-id={aff1}, financed-by={Test},
  sum={year1=1000,year2=1000,year3=1000,year4=1000,year5=1000}
}
\clist_map_inline:nn {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}
  { \@@_direct_costs_fast_expect:nnn {FASTTEST}{#1}{1000} }
\@@_direct_costs_fast_expect:nnn {FASTTEST}
  {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}{5000}
%% A second entry adds to the same project without affecting other projects.
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id={FASTTEST}, ref-affiliation-id={aff1}, financed-by={Test},
  sum={year1=200,year2=200,year3=200,year4=200,year5=200}
}
\@@_direct_costs_fast_expect:nnn {FASTTEST}{sum-year1}{1200}
\@@_direct_costs_fast_expect:nnn {FASTTEST}
  {sum-year1,sum-year2,sum-year3,sum-year4,sum-year5}{6000}
\@@_direct_costs_fast_expect:nnn {A01}{sum-year1}{5000}
\iow_term:n {FAST~DIRECT~COSTS~TEST~PASS:~yearly~and~combined~totals}
\group_end:
\ExplSyntaxOff

\makeatletter
\protected@write\@auxout{}{%
  \string\gdef\string\dfgDirectCostsAuxReady{1}}
\makeatother

\end{document}
```

Starting now a template CV file.

```
\documentclass[english,cv]{dfg-crc}
\begin{document}
\input{content/00_titlepage}
\dfgCVsection
% %% \dfgCV{MM}
\end{document}
```

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A Repository guide

B dfg-crc — Collaborative Research Centre proposal class

Current release: **0.3.0 (2026-10-20)**.

The `dfg-crc` LuaLaTeX class provides the proposal and CV structure for a German Research Foundation (DFG) Collaborative Research Centre (CRC). It uses structured metadata to generate project heads, staff and funding tables, project lists, collaboration plots, Gantt charts, and proposal-wide totals.

This is an independent, unofficial template. It is not published, endorsed, or maintained by the DFG; applicants remain responsible for checking the current DFG forms and programme rules.

The PDF manual combines this introduction with the detailed user and code documentation. Release history is kept separately in `CHANGELOG.md`.

The supplied proposal template is designed for five calendar years and supports one or more applicant universities. The project types `research`, `service`, `mgk`, and `admin` are supported; see the documentation for the corresponding metadata and generated sections.

It is appreciated when in the colophon of your proposal a reference to this template and its author is given.

B.1 Requirements

The class requires LuaLaTeX and a LaTeX kernel dated 2022-06-01 or newer. Biber is required for bibliographies. The default setup uses Noto Serif, Roboto, and the example images supplied by the `mwe` bundle. A full TeX Live[1] installation is recommended and contains the class's runtime dependencies.

Repository: https://gitlab.git.nrw/rpdm/projects-and-services/dfg-crc-template

Issue tracker: https://gitlab.git.nrw/rpdm/projects-and-services/dfg-crc-template/-/issues

B.2 Installation

TeX Live and MiKTeX installations obtained after the first CTAN release can install `dfg-crc` through their normal package managers. To install from the CTAN source archive manually, run:

```
luatex crc-proposal-documentation-and-code.ins
```

This generates `dfg-crc.cls`. Move it into a directory searched by TeX, or keep it beside the proposal's main `.tex` file. The complete generated example proposal is maintained in the repository and can be extracted with `make files`.

B.3 Start the program from the repository

For a first local compile, generate the derived files and then compile the proposal and CV examples from the `local/` folder.

```
make files
cd local
latexmk -lualatex crc-proposal.tex
latexmk -lualatex crc-cv.tex
```

1 https://tug.org/texlive/

The class is designed for LuaLaTeX. Use `lualatex` or `latexmk -lualatex`; do not call `LaTeX` with a capital `L`.

Two-sided pages reserve 30 mm on the binding edge by default: a 25 mm inner margin plus `BCOR=5mm`. Pass a different KOMA `BCOR` class option to adjust the binding correction; `BCOR=8mm` gives 33 mm on the inner edge and a 157 mm text block. The default 160 mm text block leaves a 20 mm outer margin for the rotated thumb index and marginal notes. The 20 mm top and 15 mm bottom margins give the text block 5 mm more height than the previous 20 mm margins; the footer remains included in the page layout. By default, the thumb-index tabs span the text block from top to bottom and divide its current height into equal slots. Set `thumbindex@topmargin` or `thumbindex@bottommargin` to a page-edge distance to override either end. Use `\changeSettings{thumbindex@orientation=inwards}` to turn the labels 180 degrees towards the text block. The default `outwards` points the tops of the letters towards the outer page edge on both left and right pages. Labels sit 2 mm towards the text block from the center of each tab. Numbered section and subsection titles start at the same horizontal position after a shared 12 mm number column. Section, subsection, and subsubsection headings have one baseline of space above them; following text starts on the next line without an additional blank line.

Useful make targets:

> `make files`: extracts `dfg-crc.cls` and the local template files from the documented source. `make update-class TARGET_DIR=/path/to/proposal-worktree`: regenerates and copies only `dfg-crc.cls` into a separate Git worktree. `make update-all TARGET_DIR=/path/to/proposal-worktree`: rebuilds and copies `dfg-crc.cls`, `crc-proposal-documentation-and-code.pdf`, and the standalone `dfg-crc-proposal-example.pdf` and `dfg-crc-cv-example.pdf` into that worktree. `make test`: compiles the generated template test document. `make test-direct-costs`: compiles the focused two-pass regression for DFG table 2.1.1. `make test-workpackage-references`: compiles the focused nested-work-package reference and range-formatting regression. `make test-aim-references`: checks custom aim labels, forward references, plural references, ranges, and list option scope. `make test-bibliography-highlights`: checks frame and project entry highlighting, font sizes, options, and isolation from ordinary lists. `make proposal-example`: compiles the generated local proposal example. `make cv-example`: compiles the generated local CV example. `make check`: runs the test and compiles both examples. `make proposal-doc-example`: builds the complete documentation consortium proposal in final mode as `dfg-crc-proposal-example.pdf`. `make cv-doc-example`: builds the complete twelve-PI CV appendix in final mode as `dfg-crc-cv-example.pdf`. `make doc`: builds both PDFs first, then includes all their pages in the user documentation, followed by the implementation reference. `make ctan-check`: builds the curated CTAN archive and verifies a clean installer extraction and minimal LuaLaTeX compile. `make ctan-validate`: runs the official online CTAN validator without submitting the archive.

For a focused project-text build, select the project with `\includeonly` and use these class options:

```
\documentclass[
  fastcompile-project=true,
  fastcompile-project-show-images=false,
  fastcompile-project-show-publications=false,
  fastcompile-project-show-funding-tables=true,
  english
]{dfg-crc}
```

`fastcompile-project` skips metadata loading and most `\addEntry` execution, calculated project heads, generated tables/lists and plots, embedded graphics, and other front-matter work while retaining authored sections, paragraphs, aims, work packages, labels, and citation commands. Skipped generated tables and lists retain the same visible placeholder and log note as ordinary `fastcompile` builds. Project identifiers in focused fast compilation are red typewriter placeholders such as . Set `fastcompile-project-show-images=true` to load the project

metadata and render images and generated plots encountered in the selected project while retaining the other fast-mode reductions. Set `fastcompile-project-show-publications=true` when the selected project's publication headings, explicit `\nocite` entries, and bibliography should be included. Use `\printProjectBibliography` for project publication lists; plain `\printbibliography` uses biblatex's normal font and bypasses the project-publication switch.

To use footnote-sized text and remove extra space between bibliography entries, add this hook to `preamble/dfg-customization.tex` (outside `\changeSettings`):

```
\AddToHook{package/biblatex/after}{%
  \renewcommand*{\bibfont}{\normalfont\footnotesize}%
  \setlength{\bibitemsep}{0pt}%
}
```

The hook waits until `biblatex` has loaded because the customization file is read earlier. `\bibitemsep` controls the extra gap between entries; change `0pt` to `2pt`, for example, for a small gap. It does not change line spacing within an entry. Without an override, the bibliography style supplies its spacing; the alphabetic styles inherit the standard list gap (4 pt plus 2 pt minus 1 pt in the default 10 pt setup). The CV's custom `DFGlist` already uses `nosep` and is configured separately through its `itemsep` list option.

With `emphasizeEntries=true`, a full project bibliography entry is bold when its `.bib` keywords include the current project ID followed by `-highlight`, for example `B06-highlight`. One such keyword is enough even if two PIs from Project B06 coauthor the entry. An entry may also carry another project's highlight keyword when it should be bold in that project's list. The `B06-project` keyword is optional project metadata; it does not trigger bold type. Other keywords, including CV selectors such as `pabst-A`, can remain in the same field. To use only whole-entry highlighting, set `emphasizeAuthors=false` and `emphasizeEntries=true`.

For the frame-text references in chapter 1, use `\printFrameBibliography` with the same options as `\printbibliography`. With `emphasizeEntries=true`, entries tagged `frametext-highlight` are bold in this list. Project-specific highlight keywords do not affect it, and the normal bibliography font size is preserved. For example:

```
\changeSettings{emphasizeEntries=true}
\defbibnote{frame-references}{%
  \usekomafont{subsection}%
  \bfseries\sffamily References
}
\printFrameBibliography[prenote={frame-references},heading=none]
```

The `.bib` entry may combine keywords, for example `keywords = {pabst-A,B06-highlight,frametext-highlight}`. The keyword selects formatting only; cite the entry or use `\nocite` to include it as usual. Existing chapter 1 files need the command replacement above because template updates preserve edited chapter files.

Wrap optional metadata and front matter with `\dfgIfFastCompileProjectF{...}` so those files are not even read in this mode. With publications disabled, bibliography source files and stale `.bbl` data are also left unread; inline citations display their keys as lightweight drafting markers. The three project-local funding tables remain visible by default. Set `fastcompile-project-show-funding-tables=false` to suppress them too. Focused mode processes their local rows and sums but does not restore or populate proposal-wide overview databases. Compile a document using these options with its own build command; the template Makefile deliberately operates only on the generated test and example documents in `local/`.

B.4 Template validation workflow

Run `make check` to regenerate the template files, compile the test document, and compile both local examples. Run `make all` to perform those checks and build the complete implementation and user documentation. These targets use only files maintained or generated by this repository.

GitLab CI runs the same compile-based checks for branches and merge requests. After a successful default-branch pipeline, GitLab Pages publishes the newest generated `local/crc-proposal.pdf`. The Pages site is rebuilt from an empty `public/` directory each time and therefore exposes only the latest dummy proposal, not an archive of earlier PDFs. CI job artifacts retain the current logs and example outputs temporarily for diagnosing failed pipelines.

B.5 Updating a separate proposal repository

Edit the maintained `.dtx` source, then install the regenerated class into a separately controlled proposal worktree:

```
make update-class TARGET_DIR=/path/to/proposal-worktree
# Or update the class, compiled manual, and complete example PDF:
make update-all TARGET_DIR=/path/to/proposal-worktree
```

Both targets verify that `TARGET_DIR` is the root of a different Git worktree, copy the requested release files, and show their resulting Git status. `update-class` copies only `dfg-crc.cls`; `update-all` first rebuilds the manual and then copies the class, `crc-proposal-documentation-and-code.pdf`, and `dfg-crc-proposal-example.pdf` and `dfg-crc-cv-example.pdf`. Neither target overwrites the proposal repository's README, changelog, or licence. They do not store the destination, inspect its remote, commit, or push. Review and commit the updates from inside that proposal repository.

B.6 Repository boundary

This repository is intentionally limited to the reusable proposal template. Project-specific source, synchronization configuration, credentials, remote identifiers, and private working copies must remain in separately controlled repositories or local worktrees and must not be added here as submodules.

B.7 Structure of the files and folders

The maintained source is `crc-proposal-documentation-and-code.dtx`. Generated files such as `local/dfg-crc.cls`, `local/test-existing-funds-direct-costs.tex`, `local/test-project-collaboration.tex`, `local/test-workpackage-references.tex`, `local/test-participating-institutions.tex`, `local/test-command-options.tex`, `local/test-reference-lookups.tex`, `README.md`, `CHANGELOG.md`, and the local example files are extracted from that source. Edit the `.dtx` for template or documentation changes, then regenerate the derived files with `make files` or by building the documentation.

The manual uses `dfg-crc` itself, with `documentation=true` to load only its explicit fictional fixtures. The DTX driver maintains executable `dfgExample` boxes: code on the left, rendered output on the right. Add `[dfg wide]` for wide results below their code. `make doc` regenerates the fixtures and compiles the examples; incomplete setup fragments remain code-only. This does not read or overwrite the manual scratch file `local/test.tex`.

The manual and local examples share the DTX `meta` records: twelve PIs, six institutions, four affiliations, and fifteen active projects (A01–A06, B01–B06, Q01–Q03), plus a former project. The embedded proposal uses these same metadata in `local/`, with individual files for all fifteen active projects and the former project A00. The local CV uses the same twelve PIs, with matching career records and explicitly fictional publications for each person. Examples use `dfg project=A03` when they need the context normally established by a project heading. Project overviews and the PI roster follow their code boxes at normal size and may span pages; collaboration plots retain the available text width.

`make test-participating-institutions` checks flowing/raster column widths, active-PI filtering, shared-institution deduplication, sorting, spacing above columns, and option scope. `make test-command-options` checks omitted/empty, leading/legacy options, precedence, grouped command specifications, mandatory arguments, custom databases, boolean flags, and per-call defaults for lists, tables, plots, and PI rasters. `make test-reference-lookups` covers generic

reference paths, branching ID lists, destination deduplication, diagnostics, and focused-build placeholders.

The standalone regression document is maintained in `ci/test-template.tex`. `make test` copies it to `local/test-template.tex` before compiling it. `local/test.tex` is reserved for manual experiments and is neither generated nor compiled by the template checks.

B.8 Generated project templates

The generated `local/content/03_project.tex` is a small master file that loads separate, complete project examples. They are intended to be copied and adapted rather than reduced to a list of headings:

`03_project_A01.tex`: continuing research project (`project-status=C`).

`03_project_A02.tex`: new research project (`project-status=N`).

`03_project_B01.tex`: second continuing research-project example.

`03_project_B02.tex`: ending project (`project-status=E`), including the report bibliography, the three published-result categories, and the staff roster for the ending funding period.

`03_project_Q01.tex`: continuing service project.

`03_project_Q02.tex`: continuing Integrated Research Training Group (`project-type=mgk`), including existing funds, fellowships, and global funds.

`03_project_Q03.tex`: central administrative project with its dedicated documentation and funding structure.

Every example starts with `\printProjectHead{<id>}`. That command already generates the general-information heading, project title, project leaders, and the applicable contract and legal-information blocks from metadata. The source files therefore begin with the first status-specific authored heading: `Summary` for active projects and `Project history` for ending projects.

The generated proposal-wide example follows the February 2026 continuation template, including the ecological-sustainability section, separate coordinating and co-applicant universities, the current legal-issues wording, and one rector, chancellor, or president signature per applicant university.

> `bib/`: all bibliography files. `content/`: all files which are part of the actual content of the proposal, e.g. all project textfiles. `figures`: folder contains all figures of the projects. `img`: folder can be used for logo files etc. `metadata`: if you collect all database entries in specific external files, they are placed in this folder (e.g. `dfg-metadata.tex`). `preamble`: all files for the proposal and CV setup, especially `dfg-customization.tex` for customized things or re-newed settings.

B.9 Minimal metadata relationship

Most commands resolve data by following ids through the small databases:

```
person.ref-institution-id -> institution.id
institution.ref-affiliation-id -> affiliation.id
project.collaborations -> project.id
```

For example, a person may list one or more institution ids in `ref-institution-id`. Each institution points to its parent affiliation via `ref-affiliation-id`. Once this relationship exists, commands such as `\printList{{Institutions}{<person-id>}}` can print the institution and affiliation data without further manual wiring. Each institution can also provide an `abbreviation`; generated table cells use this compact label while descriptive lists and legends retain the full `name`.

The same relationship is reused by `RequestedFundingStaff`: when a staff row contains `ref-person-id`, its category and institution are taken from the person's `position` and `ref-institution-id` fields unless the staff row explicitly overrides them. A row without `ref-person-id` must provide `ref-institution-id`; otherwise the class warns because it cannot resolve the institution or assign existing staff to an affiliation in the proposal-wide overview. Existing non-research staff may omit `field-of-research`; that empty table cell is then greyed out as not applicable without a warning. Empty staff positions are omitted without a warning or an empty line. The job descriptions printed below each requested-funding staff table use flush-left number labels and a hanging indent so continuation lines align with the first line of description text. A normal interword space separates the bold name from the description.

Project collaboration plots follow the comma-separated project ids stored in each project's `collaborations` field. Relations are undirected, so declaring the same pair in both project rows is harmless but unnecessary. Ending, reference-only, and list-excluded project rows remain available elsewhere but are omitted from these plots.

B.10 Information retrieval command overview

The following commands can be used directly in proposal text and tables. In the signatures below, square brackets are optional and braces are mandatory. The complete descriptions and examples are in `crc-proposal-documentation-and-code.pdf`.

B.10.1 General database access

Command `\get[database]{id}{field}` prints a metadata field. Add reference steps to retrieve fields from linked rows, for example `\get[person]{MM}{institution}{name}` or `\get[person]{MM}{institution}{affiliation}{name}`. Each step follows `ref-<database>-id`; the full key spelling is also accepted, as in `\get[person]{MM}{ref-institution-id}{name}`. This works from any source database, including named databases populated with a separate keyset. Multiple reference IDs are followed in their stored order and the resulting values are separated by commas. A destination row reached more than once is printed only once; different rows with identical names remain separate. Missing or empty references warn and contribute no value. The terminal field keeps its formatting and `showtags` annotation. Focused fast compilation uses the source ID placeholder, as with `\getEntry`. `\get*` retains the starred multi-field assignment syntax of `\getEntry*`.

Command `\getEntry` with `[database] {id} {field} [id-field]` prints one field from one database row.

Starred command `\getEntry*` with `[database] {id} {assignments} [id-field]` fetches several fields in one lookup and assigns them without printing. For example, fetch person `MM` with `\getEntry*[person]{MM}` and pass the third argument `{\theFirstName=firstname,\theLastName=lastname}`.

Command `\compareEntry` with `[database] {id} {field} {value} {true} {false}` chooses text or code according to a field comparison.

B.10.2 People, institutions, and projects

Commands `\getName` and `\getFullName*` with `{person-id}` print the last name or full name; the star adds the academic degree.

Command `\getPI*` with `{person-id-list} [text]` prints a linked PI reference; the star includes first names. Unstarred references distinguish shared PI surnames with the first given-name initial (K. Schneider / C. Schneider), or the full given name when initials also match. This rule applies across projects and funding periods, including project-first references. Explicit display text bypasses it; starred references retain full given names.

Command `\getORCiD` with `[database] {person-id}` prints a linked, formatted ORCiD from a person row.

Command `\getORCiDvalue` with `[database] {person-id}` stores the unformatted ORCiD in `\theORCiDValue`.

Command `\printList{{Institutions}{<person-id>}}` prints institutions and parent affiliations assigned to a person. Put formatting and database options before the specification, for example `\printList[database=person,abbreviation,narrow]{{Institutions}{MM}}`.

Command `\printList{ParticipatingInstitutions}` lists participating institutions across the proposal, with affiliation names in bold. Use `columns=2` (or `columns=3`, etc.) to flow whole affiliation groups down balanced columns. Short groups follow one another without being padded to the height of a neighbouring group. Groups stay together; reading order is down each column. `column sep=16pt` changes the gap (default: 12 pt). Flowing lists start a new paragraph with a medium vertical skip above the columns. The default layout remains a one-column raster, with 12 pt between columns and 10 pt between rows. For aligned rows, pass `raster columns=2` or another `tcbraster` option through the optional list argument. Sorting defaults to `order=count`; use `order=alpha` for alphabetical affiliation order. The legacy bare `count` and `alpha` options remain supported.

Put all list options before the specification, as with `\printPlot`. Plain and grouped list names are both accepted. Older trailing options remain supported; leading settings take precedence when the same key occurs twice.

```
\printList[columns=2]{ParticipatingInstitutions}
\printList[columns=3,column sep=16pt,order=alpha]{ParticipatingInstitutions}
\printList[order=alpha,raster columns=2]{{ParticipatingInstitutions}}
```

Command `\getPersonInformation` with `[database] {person-id-list}` prints complete formatted PI/contact-information blocks, with a 12 pt column gap in the two-column raster.

Command `\getPIsWithRole` with `[options] {role}` prints formatted current-period people carrying a given role. Put database and raster settings in the leading options, for example `\getPIsWithRole[database=person,raster columns=2]{PI}`. Use `\getPIsWithRole[academicdegree-below]{steering-committee-member}` to put the degree on a separate line below the name within the same PI box. The option applies only to that call; the default (or `academicdegree-below=false`) keeps the degree inline after the name. The default three-column raster has a 12 pt column gap; rows and the following content are separated by 20 pt. Explicit raster options can override that spacing.

PI-box titles are ragged right and top-aligned. All titles within one raster reserve the height of its tallest title, so wrapped names or academic degrees do not misalign the contact information. Recompile until the equal-height layout settles; a clean build can require three LaTeX passes. Phone and email stay individually unbroken, with a one-em gap that permits a line break between them when they do not fit together.

Use `\getPIWithProject*` with `{person-id-list}` to print each PI with their assigned project identifiers in parentheses, without repeating the projects' PI names; the star includes first names. The older `\getPIProject` command remains available but warns that it is deprecated.

Use `\getProjectWithPI*` with `{project-id-list}` to print each project with its assigned PIs; the star includes first names. The older `\getProjectPI` command remains available but warns that it is deprecated. By default, the shared first `\getProject` or `\getProjectPI` reference to a project within one section includes its PIs; later references print only the project. Set `projectPIOncePerSection=false` in `\changeSettings` to repeat the PI names.

Command `\getPIofProject` with `[properties] {project-id} [delimiter]` prints selected PI name properties for one project. The starred form defaults to first and last names, separating earlier names with commas and only the last with "and".

Command `\getProject*` with `{project-id-list} [text]` prints a linked project reference. The star always prints project identifiers only, independently of `projectPIOncePerSection`. Optional `text` replaces the displayed project number while retaining its link target and is best used with a single project. A reference-only project is marked and deliberately left unlinked. Multiple projects are joined with commas and a final `and` in prose, but use compact slash separators

inside table cells, for example A02/B01. With the default projectPIOncePerSection=true, this command shares \getProjectPI's first-reference PI expansion and per-section state.

Command \getProjectCount with [area] prints the active, list-considered project count; omit area for all projects. Any explicit or ID-derived project-area value is accepted.

Command \printPlot{{collaborations}{A01}} prints the default full-width arc layout for A01, with project nodes and unlabelled area rules. Its boxes align with the text edge on both sides of a two-sided page. Its caption always appears below the plot; a final optional argument adds caption text there. Add [arcs=true] before the plot specification to draw the connecting arcs, or use [layout=heatmap] for the heatmap. The optional project-area field creates rows dynamically and falls back to the first character of the project ID. Configure proposal-wide colors with collaborationAreaColors and collaborationAreaPalette in \changeSettings, or use the per-plot area-colors and area-palette options. Gantt charts use the same dispatcher as \printPlot{{gantt}{A01}}. Inside a project, omit the ID: \printPlot{{gantt}} and \printPlot{{collaborations}} use the current project. Options precede the plot specification, for example \printPlot[caption={Additional details}]{{gantt}} or \printPlot[arcs=true]{{collaborations}{B06}}. With an explicit ID, use the same leading caption option: \printPlot[caption={Additional details}]{{gantt}{B06}}. The older Gantt and ProjectCollaboration type names remain available. In project source files, caption={} marks where optional caption text can be added to either plot; left empty, the default caption is still printed.

Command \dfgResearchAreasList with {code} prints a DFG research-area code and its description.

Commands \getAim and \getWP with [project-id] {id-list} cross-references to project aims or work packages; the current project is the default. \getWP places a narrow, unbreakable space after WP or WPs.

The aims environment accepts a custom prefix shared by the numbered items and references:

```
\begin{aims}[label={RDM Aims}]
  \addAim{data} Document research data.
  \addAim{reuse} Publish reusable datasets.
\end{aims}
See \getAim{data,reuse}.
```

This prints RDM Aims 1: and RDM Aims 2:, with RDM Aims 1 and 2 in the reference. \cref and \Cref also use the custom prefix. Its text is used unchanged for single and multiple references; ordinary aims lists retain Aim/Aims. Other list options, such as start=3, remain available. Compile twice after changing labels to update forward references.

Use resume to continue numbering from the previous aims list. Repeat the custom label to keep it in both the items and their cross-references:

```
\begin{aims}[label={RDM Aims},resume]
  \addAim{archive} Archive the datasets for long-term access.
\end{aims}
See \getAim{archive}.
```

Following the two-item list above, this prints RDM Aims 3: and references it as RDM Aims 3. [resume] alone continues the number with the default Aim/Aims prefix. Numbering resumes from the previous aims list, regardless of its label text; custom labels do not create separate counters.

B.10.3 Person statistics

Command \getPositionGenderPhaseAmount with [database] {position} {gender} {period} prints the number of people for one position/gender/period tuple.

Command \getPositionPhaseTotal with [database] {position} {period} prints the all-gender total for a position and funding period.

Command `\getGenderPhaseTotal` with `[database] {gender} {period}` prints the all-position total for a gender and funding period.

Command `\getPositionGenderPhasePercentage` with `[database] {position} {gender} {period}` prints the rounded integer percentage within one position.

Command `\getGenderPhasePercentage` with `[database] {gender} {period}` prints the rounded integer percentage across all counted positions.

The optional database defaults to `person`. Counts include people whose `role` list contains `PI` and whose `period` list contains the requested funding period. The total-row commands sum `postDoc`, `groupLeader`, `prof-W2`, and `prof-W3`. Gender is `f`, `m`, or `d`; percentages are rounded integers without a percent sign and return zero when the denominator is zero.

```
\getGenderPhaseTotal{f}{1}
\getGenderPhasePercentage{f}{1}\%
\getPositionGenderPhaseAmount{prof-W3}{f}{1}
\getPositionGenderPhasePercentage{prof-W3}{f}{1}\%
\getPositionPhaseTotal{prof-W3}{1}
```

Change the last argument to `2` or `3` to query another funding period. These commands are available independently of the current proposal period. Section 1.4.2 B adds the first-period Number and Percentage columns only for proposals with `fundingPeriodNumeral = {3}`. Maintain historical PI records with their appropriate `period` membership to populate these columns. The statistics count people; the share of projects with female leadership requires a separate calculation.

The project-leader table defaults to the female-only DFG layout. For a third-period proposal, periods 1 and 2 each show Number/Percentage; period 3 shows its original objective (from `future-objective-percentage-female`) and current Number/Percentage. The person records' period-3 membership supplies these current counts. Percentages always use all genders as the denominator. Select the expanded f/m/d comparison, including third-period counts and percentages alongside its objective columns, for a single table:

```
\printTable{GenderEqualityProjectLeaders}            % female-only DFG view
\printTable[all-sexes]{GenderEqualityProjectLeaders} % expanded f/m/d view
```

Each call starts in female-only mode; `[all-sexes]` affects only that table. Put options before the table name. Combine `database=MyObjectives,all-sexes` to use a custom objectives database. The prose count and percentage commands are unaffected.

B.10.4 Funding and staff totals

The similarly named funding commands have different scopes: names ending in `RequestedFunding` use the proposal-wide overview by default, while names ending in `RequestedFunds` use project-level funding rows and require a project id.

Command `\getCountRequestedStaff` with `[project] {year} {category}` prints the requested staff quantity for a category, optionally restricted to a project.

Command `\getSumRequestedFunding` with `[database] {year} {type}` prints the proposal-wide total for one funding type and one year; `{all}` as the year totals all five years.

Command `\getTotalSumRequestedFunding` with `[database] {year} {type}` prints the proposal-wide total with `{all}` as the type available for a grand total.

Command `\getSumPreviousRequestedFunding` with `[database] {year} {fields}` prints the previous-period total for one year row, or all rows with `{all}` as the year.

Command `\getSumRequestedFunds` with `[database] {project} {type} {sum-fields}` prints the project total for one funding type and one or several `sum-yearN` fields.

Command `\getTotalSumRequestedFunds` with `[database] {project} {sum-fields}` prints the project total across all funding types for the supplied `sum-yearN` fields.

Command `\getSumRequestedFundsInstrumentation` with `[project] {sum-fields}` prints the instrumentation total, proposal-wide or for the optional project.

Command `\getSumRequestedFundingDirectCosts` with `[database] {year} {group}` prints the direct-cost total for `applicant`, `participant`, `other-funds`, or `all`. Its default database is the auxiliary-file-backed `RequestedAndExistingFundingDirectCostsOverview`, and its displayed result is a full euro amount without decimal places.

Command `\getSumExistingFunds` with `[database] {project} {sum-fields}` prints the existing direct-cost funding for a project and the supplied year fields.

Typical calls are:

```
\getProjectCount[A]                              % active A projects
\getSumRequestedFunding{all}{staff}              % proposal-wide staff total
\getTotalSumRequestedFunding{all}{all}           % proposal-wide grand total
\getSumRequestedFunds{A02}{staff}{sum-year1}     % A02 staff total, year 1
\getTotalSumRequestedFunds{A02}{sum-year1}       % A02 grand total, year 1
```

Funding entries, including requested instrumentation, use EUR as their input unit. Their tables state `Figures in EUR` and display whole euro amounts with thousands separators. The non-aggregated `existingInstrumentation/cost` field instead uses thousands of EUR and its table states `Figures in thousands`; one decimal place is retained when supplied. Explicitly fractional animal-housing rates also retain decimals, while the other monetary columns in that table remain whole euros.

B.10.5 Existing direct-cost support in section 2.1.1

Declare institutional and other existing direct-cost support once, next to the corresponding project table:

```
\addEntry[funding]{RequestedAndExistingFundingDirectCosts}{
  ref-project-id     = {B06},
  ref-affiliation-id = {aff1},
  financed-by        = {Faculty core support},
  sum = {
    year1 = {5000},
    year2 = {5000},
    year3 = {5000},
    year4 = {5000},
    year5 = {5000}
  }
}
```

The sums are stored and displayed in euros. Table 2.1.1 therefore prints `5000` as `5,000`; monetary values are rounded to the nearest euro for display and never acquire a redundant decimal part. An affiliation listed in `meta/crc.ref-affiliation-id` contributes to the applicant-university column; another affiliation contributes to the other-participating-institutions column; and the special value `other-funds` contributes to Other funds. `financed-by` is the human-readable funding-source description and does not select the aggregate column.

The declarations may remain in project files even though section 2.1.1 comes earlier. They are restored into a separate overview database through the LaTeX auxiliary files. A normal multipass build such as `latexmk` is therefore required after adding or changing a row; no duplicate chapter-2 declaration is needed.

B.11 Open-access markers in bibliography entries

The class loads `biblatex-ext-oa` so bibliography entries can show open-access markers in final documents. The DOI API integration is only available with LuaLaTeX. With LuaTeX 1.17.0 or newer, the socket library must also be enabled on the command line, for example with `lualatex --socket`. Provide the contact address required by the API in the customization file:

```
\changeSettings{
  doiAPIMail = researcher@example.org
}
```

The class enables DOI lookups only when `final=true`, `doiAPIMail` is not empty, and LuaSocket is available. Otherwise it retains manual open-access annotations without making network requests.

If the API-based open-access request does not work for a publication, add the manual fallback annotation to the relevant `.bib` entry:

```
options          = {openaccess=auto},
doi+an:default   = {=openaccess},
```

B.12 Troubleshooting

Use `latexmk -lualatex ...` or `lualatex ...`. The command `LaTeX` asks TeX for a format named `LaTeX.fmt` and will usually fail.

If you see `Not allowed in LR mode`, check whether vertical material such as `\vspace`, `\addvspace`, or a paragraph break was inserted inside a table cell that only accepts horizontal material. Prefer the documented commands and options instead of adding raw vertical spacing in table cells.

`Unknown database name` means the requested database has not been loaded or the database name was misspelled. Check the relevant file in `metadata/` and make sure it is input by the document class.

Missing-key or empty-value warnings mean the row exists, but the requested field is empty. This is intentional for fields where the template needs to alert you, such as missing ORCiD values.

If a change has no visible effect, check which `dfg-crc.cls` is being used. Regenerate the class and local examples with `make files`, then compile the relevant example again.

B.13 Maintainer release checklist

1. Update `\dfgVersion`, `\dfgVersionDate`, the README release line, changelog, and `dfg-crc.pkg` in one commit.
2. For CTAN updates after the initial release, set `\update{true}` in `dfg-crc.pkg`. Keep the CTAN path `/macros/luatex/latex/dfg-crc`, and omit `\home` when it duplicates the repository URL.
3. Do not include the generated `dfg-crc.cls` in the CTAN archive. CTAN requires files generated from a documented source and installer to be omitted; the PDF manual and example PDF are documentation exceptions.
4. Confirm that `bossert@itc.rwth-aachen.de` is registered as the second address of CTAN member `lukascbossert`. Keep release information in separate Announcement and Note fields; CTAN explicitly accepted that structure for the initial release.
5. Build the documentation and the CTAN archive, then run the portable installer and minimal-class smoke test:

   ```
   make ctan-check
   ```

6. Run at least one proposal/CV smoke build from `local/`:

   ```
   cd local
   latexmk -lualatex -g -halt-on-error crc-cv.tex
   ```

7. Before a public release, run the complete template check:

```
make check
```

8. Run the official online validator with `make ctan-validate`.
9. Review `git status --short`, `git diff --check`, and the archive file list before committing and tagging the exact release version.

B.14 Support

The current maintainer is Lukas C. Bossert[2].

B.15 License

This material is subject to the LaTeX Project Public License 1.3c or later. The work has the LPPL maintenance status *maintained*. See `LICENCE.md` for the complete notice.

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C Licence

C.1 LICENCE

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This work has the LPPL maintenance status “maintained”.

The Current Maintainer of this work is Lukas C. Bossert.

This work consists of the source file `crc-proposal-documentation-and-code.dtx`, the installer `crc-proposal-documentation-and-code.ins`, the documentation files `README.md`, `CHANGELOG.md`, and `LICENCE.md`, and the derived files `dfg-crc.cls` and `dfg-crc.pdf`.

Bibliography

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